SWAFL Desktop — Complete Technical Manual

Step-by-step use, every control and trigger, every layer and simulation, and the governing equations behind every output

Naziru Halilu

Version 1.1.0

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About this manual

This manual documents **SWAFL Desktop**, the Soil–Water–Atmosphere–Field–Loop decision-support workbench that accompanies the repository *Coupled Water Fluxes, Pore-Scale Dynamics, Precision Irrigation and Mulching Feedback Loops*. It is written for three readers: a farmer or adviser who wants to use the screens, a student who wants to know what each picture means, and a scientist who wants to know exactly which equation produced each number.

How the manual is organised. This is the manual of version 1.1.0. It follows the manual of version 1.0 chapter by chapter, with every picture taken again from version 1.1.0 and every numbered arrow placed on the item it names in the new interface, and it adds the chapters and screens of the new release. The manual of version 1.0 stays in the repository (`docs/SWAFL_Manual.pdf`).

1. Chapters 1–3 install the program and explain the window and every control.
2. Chapters 4–5 explain the clock (dates) and the two computing engines that sit under every layer.
3. Chapters 6–10 document each *layer* of the Layers panel: what it designs or shows, how the picture is generated, and the governing equations.
4. Chapters 11–13 document the three irrigation design layers (surface, drip, sprinkler) in full, including the irrigation schedule and the "When to irrigate" message.
5. Chapters 14–16 document the dashboard, step-by-step workflows, exports and troubleshooting.
6. Chapter 17 states the validity limits.
7. Chapters 18–24 document the additions of version 1.1: running a design and finding the most efficient inputs, weather data (your own file or the online weather of a place), the real field on the map (satellite indices, image analysis of the satellite photo, water sources and the choice of the irrigation system), the crop library and the crop water requirement, the seasonal data, the reports, and every change from version 1.0.
8. Chapter 25 is the screenshot atlas. It shows every screen of the workbench (all layers, every section of the three design layers, the menus, the toolbox, the Properties panel, the Dashboard and the dialogs of version 1.1) from the program itself, and prints under each screenshot every equation that the screen draws on, with its number, source chapter and provenance. A reverse index gives, for every numbered equation, the screens that use it.
9. The appendices give symbols, control value lists, data sets, the provenance ledger, the references and the reference of all numbered equations.

A note on honesty of the equations. SWAFL mixes three kinds of computation, and the manual labels them so that nothing is over-claimed. Every numbered equation carries its label directly beneath it, and the provenance ledger in the appendices lists them all:

Label	Meaning
[Literature]	A published relation implemented as written, cited to its source.
[Data-driven]	Obtained by interpolating or scaling the 10,190 Daisy simulations (Appendix C of the repository), not computed from a closed-form law.

Label	Meaning
[Engineered]	A parameterisation chosen for the workbench (for drawing, for speed, or because no closed form exists). Stated explicitly with its constants so that it can be audited, but not presented as a physical law.

Where the simulation model is named. The program itself shows results only; the simulation model behind them (Daisy, 10,190 runs on ten soils) is described in this manual.

Units. Unless stated, depths are in mm, lengths in m, suction in cm of water (1 cm $\approx$ 0.0980665 kPa), flows in L s^{-1} or m^3 min^{-1} as marked on the screen, times in min or h, and areas in ha.

Chapter 1

Installing and starting SWAFL

1.1 What you receive

SWAFL Desktop is a self-contained Electron program. The simulation engine is one HTML page (`app/engine/engine.html`) with a few script files beside it, which the program serves to itself on a local port, so **no internet connection is needed once it is installed**. The internet is used only if you ask for the online weather of a location (Chapter 19). Four ways of running it are provided.

Route	Who it suits	Steps
A. Build and install on Windows	Most users	Unzip `SWAFL_Desktop.zip`, double-click `BUILD_AND_INSTALL_SWAFL.bat`.
B. Windows program	Users who received `SWAFL.exe`	Double-click `SWAFL.exe`.
C. From source	Developers	`npm install`, then `npm start` (desktop) or `npm run web` (browser).
D. Online, nothing to install	Anyone with a browser	Open the online workbench (link below).

1.2 Route A — build and install, step by step

1. Install **Node.js** (the LTS version) from nodejs.org if it is not already installed. Close and reopen any terminal afterwards.
2. Unzip `SWAFL_Desktop.zip` to a short path outside cloud-synchronised folders, for example `C:\SWAFL_Desktop`. The zip contains a folder `SWAFL_Desktop`; the folder you work in is the one that contains `package.json`.
3. Double-click **`BUILD_AND_INSTALL_SWAFL.bat`**. It installs the packages, builds the program and puts **SWAFL** on the Desktop. The first build downloads Electron and takes a few minutes.
4. Start **SWAFL** from the Desktop. The title bar reads *SWAFL 1.1.0*. If Windows SmartScreen shows "Windows protected your PC", click **More info** and then **Run anyway** (the program is not code-signed).

`START_SWAFL.bat` starts the program from the source folder without building it, and `START_SWAFL_WEB.bat` opens the same program in your browser from your own computer.

1.3 Route D — the online workbench

The same program runs online in any current browser, with nothing to install and no server or Node.js:

SWAFL 1.1.0 online workbench

https://halilunaziru73-creator.github.io/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/Software/SWAFL_Desktop/app/index.html?v=1.1.0

1.4 Route C — from source

```
git clone https://github.com/halilunaziru73-creator/Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops.git
cd Coupled-Water-Fluxes-Pore-Scale-Dynamics-Precision-Irrigation-Mulching-Feedback-Loops/Software/SWAFL_Desktop
npm install
npm start          # desktop window
npm run web        # the same program in a tab of your browser
npm run get-app    # build SWAFL.exe, copy it to the Desktop and start it
```

1.5 Requirements

- Windows 10 or 11 (64-bit) for the built program; any current Chrome, Edge or Firefox for the browser route.
- A screen of at least 1280 × 720 px. The layout is designed for 1600 × 900; panels can be hidden (View menu) or the drawing shown alone (*Focus view*, F9).
- About 400 MB of free memory. The 20-year daily simulation is computed in the program; no server is involved.

1.6 First-run check

After start-up, verify three things: (i) the status bar at the bottom reads *Ready* and ends with *SWAFL 1.1.0*; (ii) the Results tab of the dashboard shows *On* followed by today's date with numbers below; (iii) clicking **Surface irrigation design** in the Layers panel opens the design with its inputs and a *Run the design* box. If the drawing area stays blank, see Chapter 16.

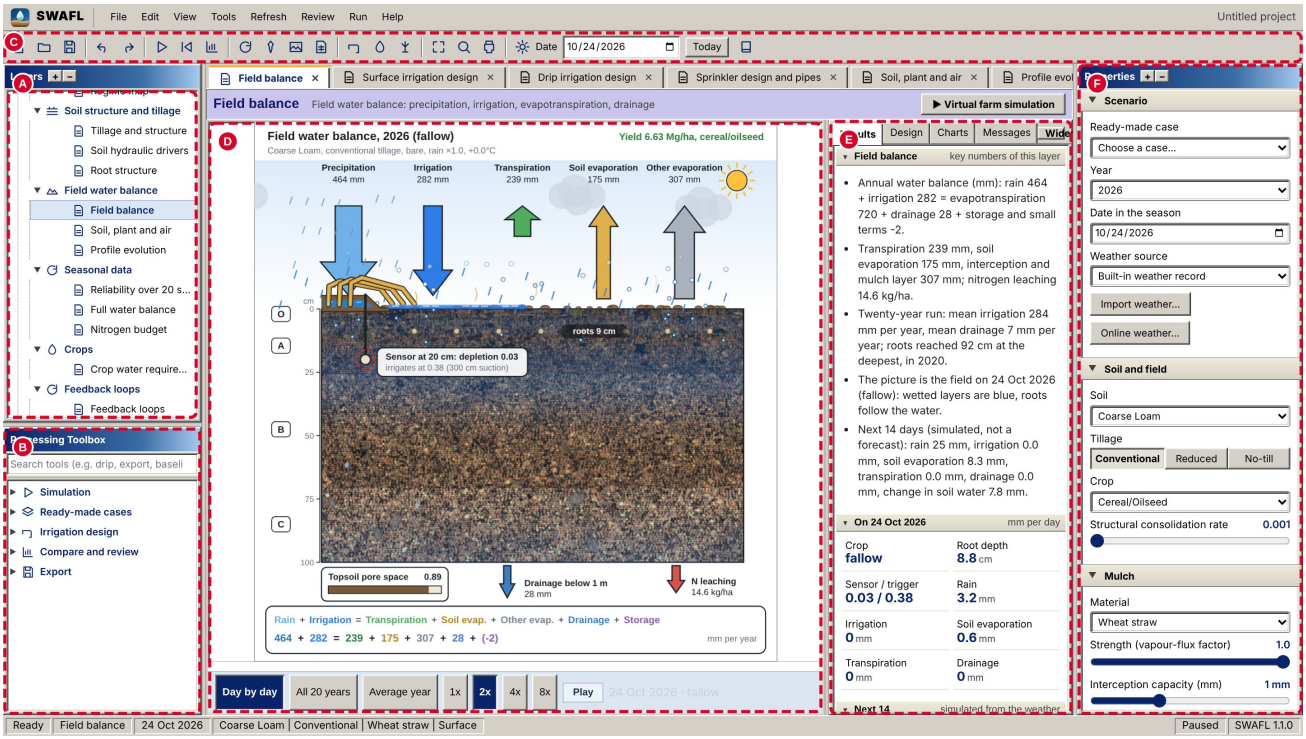


Figure 1.1: The workbench on first start: the Field balance layer for today's date. Left: Layers and Processing Toolbox. Centre: the drawing. Right of the drawing: the dashboard. Far right: Properties.

Chapter 2

The window: a guided tour

2.1 Anatomy

The window has eight regions, named here as they are named in the program.

1. **Menu bar** — File, Edit, View, Tools, Refresh, Review, Run, Help.
2. **Toolbar** — icon buttons for New, Open, Save, Undo, Redo, Play/Pause, Restart, 20-year simulation, the three design layers, fit-to-window, search, the report, and the **Date** control with a **Today** button.
3. **Layers** panel (top left) — the project tree. Each layer is one drawing of the model; each design layer lists its sections.
4. **Processing Toolbox** (bottom left) — a searchable list of ready-to-run actions.
5. **Document tabs and view bar** (above the drawing) — one tab per open layer; the view bar names the layer and gives the buttons of its sections.
6. **Drawing** (centre) — the active layer, with the playback bar and the legend under it.
7. **Dashboard** (right of the drawing) — four tabs: *Results, Design, Charts, Messages*. Every block can be folded and unfolded by clicking its header.
8. **Properties** (far right) — every model input, in collapsible groups.

The **status bar** shows, from left to right: the state (*Ready*), the active layer, the working date, the soil, tillage, mulch and irrigation method, *Paused* or *Running*, and the version (*SWAFL 1.1.0*).

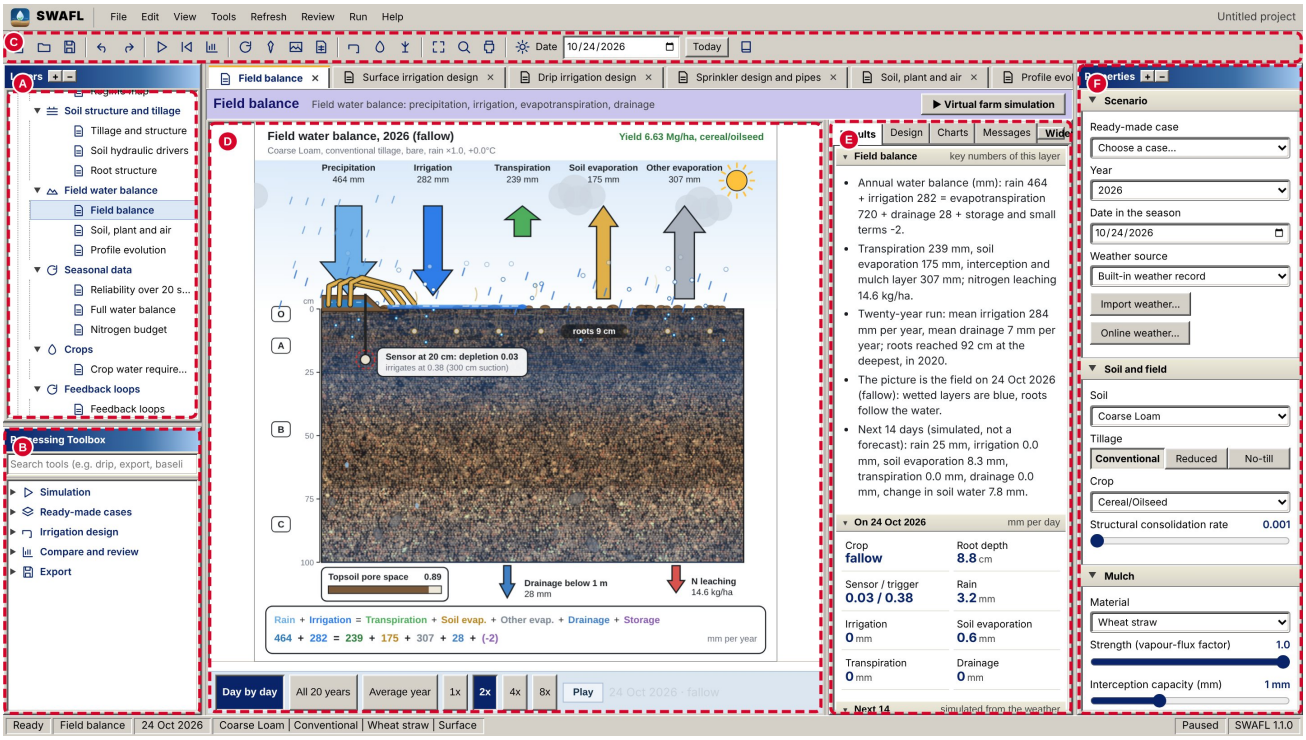


Figure 2.1: The window with the Field balance layer open.

2.2 Menu bar reference

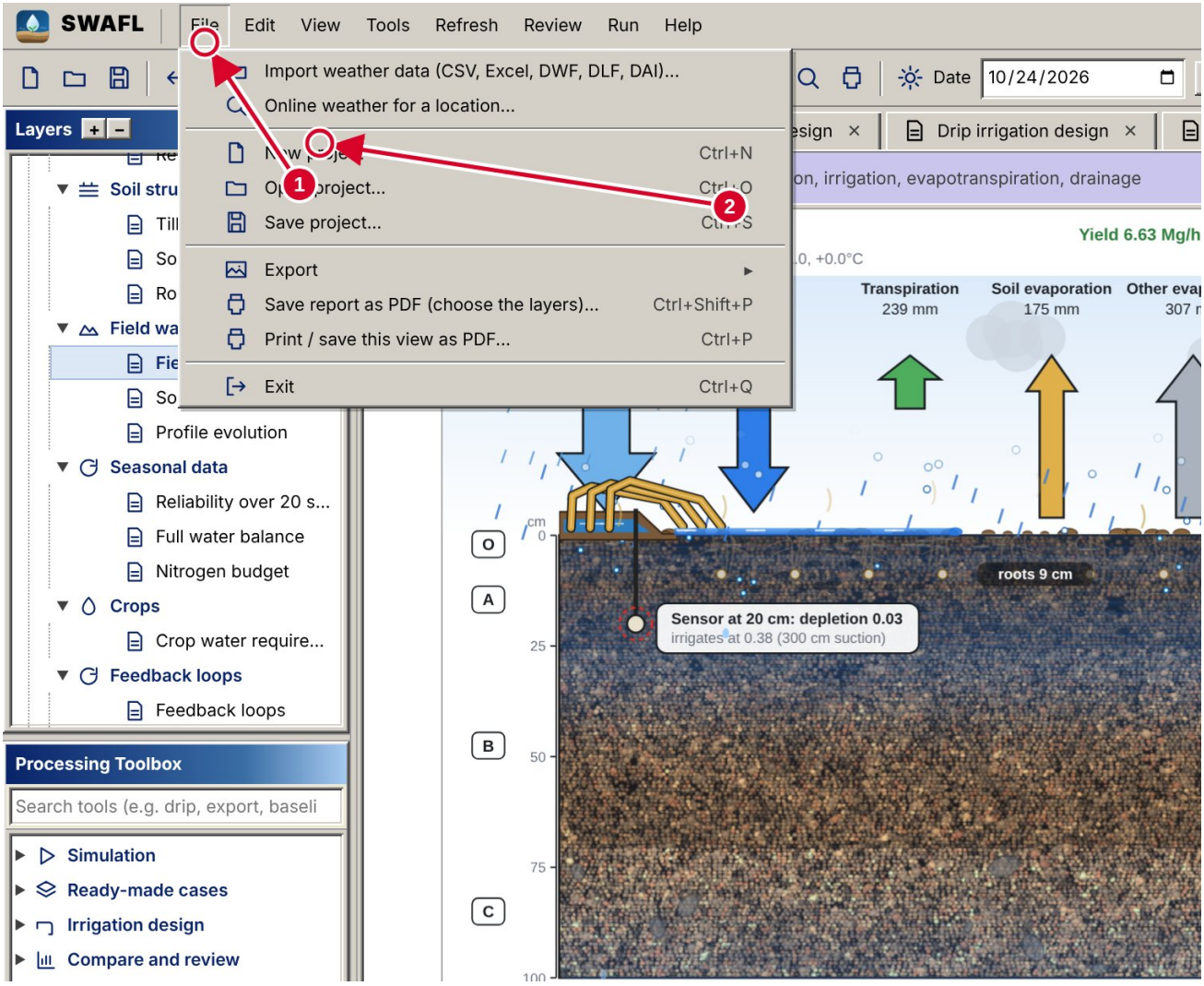


Figure 2.2: The File menu.

Menu	Item	Shortcut	Action
File	Import weather data (CSV, Excel, DWF, DLF, DAI)…		Reads a daily weather file; every layer, the dashboard and the crop water requirement then use it (Chapter 19).
	Online weather for a location…		Gets twenty years of daily weather and a 16-day forecast for a place (Chapter 19).
	New project	Ctrl+N	Resets every control to the defaults (Appendix B).
	Open project… / Save project…	Ctrl+O / Ctrl+S	Loads or saves a `.swafl` file: JSON with the version, the open layer, every control, the year and the date.

Menu	Item	Shortcut	Action
	Export ▸	Ctrl+E	Picture of the window (PNG); yearly results (CSV); design results (CSV).
	Save report as PDF (choose the layers)…	Ctrl+Shift+P	Asks which layers to include (all are ticked to start with) and writes the report (Chapter 23).
	Print / save this view as PDF…	Ctrl+P	Prints the open layer through the system print dialog.
	Exit	Ctrl+Q	Closes the program.
Edit	Undo / Redo	Ctrl+Z / Ctrl+Y	Steps through the history of control changes.
	Reset all settings; Copy settings as text		Defaults for the controls only; a text list of every control value on the clipboard.
	Pin results as baseline / Clear baseline	Ctrl+B	Freezes the 20-year averages for comparison (Chapter 14).
	Find a tool…	Ctrl+F	Moves the cursor to the Toolbox search box.
View	Design layouts ▸		Jumps to the layout drawing of the surface, drip or sprinkler design.
	Field sketch to design…		The field sketch (Chapter 18).
	Layers and Processing Toolbox / Properties / Dashboard		Show or hide each dock.
	Fit drawing to window / Drawing at full width (scroll) / Focus view (hide panels)	F9	Three ways of showing the drawing.
	Expand all layers / Collapse all layers; Full screen	F11	Tree display; full screen.
Tools	Field sketch to design…; Compare the irrigation systems on this field; Find the most efficient design inputs		Chapter 18.
	Import weather data…; Online weather for a location…		Chapter 19.
	Ready-made cases ▸; Irrigation design ▸; Irrigation method ▸; Mulch material ▸		Set the control directly.
	Run 20-year season simulation; Find a tool…		Chapter 5.
Refresh	Refresh display (F6); Reload simulation engine; Clear message log		
Review	Results dashboard; Design results; Yearly charts; Messages; Pin / Clear baseline; Go to top of the drawing		Switches the dashboard tab.
Run	Run / pause the season (F5); Restart from day 1 (Shift+F5); 20-year season simulation; Optimise surface layout; Ready-made cases ▸		

Menu	Item	Shortcut	Action
Help	Manual (PDF) (F1); How to use SWAFL; Troubleshooting (running the app); Keyboard shortcuts; SWAFL on GitHub; Full documentation on GitHub; Report a problem; About SWAFL		

The eight menus are pictured, opened, in the screenshot atlas (Chapter 25).

2.3 The command palette (Ctrl+K)

Press **Ctrl+K** anywhere to open a search box over *every* layer, section, tool, setting and date. Type part of a name (for example `drip`, `trigger`, `mulch`, `2031`) and press **Enter** to run the first match, or use the arrow keys to choose.

2.4 Layers panel

The tree has the project root *SWAFL project* and eight groups, in the order in which a field is usually designed and studied. Click a group to expand it, click a layer to open it as a tab, and click the small triangle beside a layer to list its sections; a section jumps the drawing to that part.

Group	Layers
Irrigation design	Surface irrigation design; Drip irrigation design; Sprinkler design and pipes — each with its sections
Irrigation control	Sensor and trigger; Trigger law
Mulch and evaporation	Mulch against bare soil; Mulch control; Evaporation physics; Fate of saved water; Regime map
Soil structure and tillage	Tillage and structure; Soil hydraulic drivers; Root structure
Field water balance	Field balance; Soil, plant and air; Profile evolution
Seasonal data	Reliability over 20 seasons; Full water balance; Nitrogen budget
Crops	Crop water requirement
Feedback loops	Feedback loops

In version 1.0 every layer carried a short code (S2, V1, N4 …). The codes have gone from the program; this manual uses the layer names. Open layers stay as tabs above the drawing; closing a tab does not lose any setting, because the controls belong to the project, not to the layer.

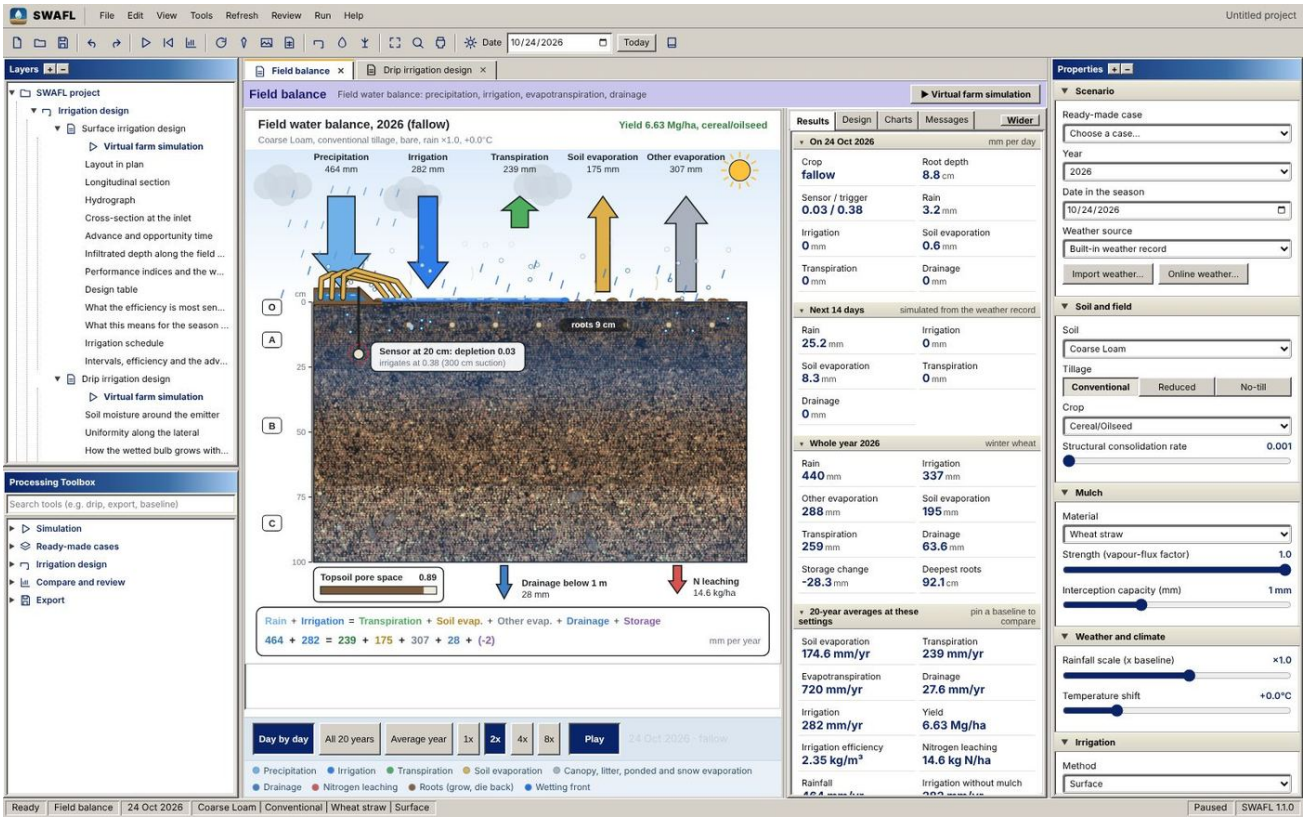


Figure 2.3: The Layers panel with every group expanded.

2.5 The view bar

Above the drawing, the view bar names the open layer. For the design layers it carries one button per section (*Whole design, Layout design, Simulation, ...*).

2.6 Processing Toolbox

Double-click (or select and press Enter) to run an action. Typing in the search box filters all tools by name and by hidden keywords.

Group	Tools
Simulation	Play/pause the season; Restart from day 1; Run 20-year season simulation; Reset all settings
Ready-made cases	The seven cases of Section 3.8
Irrigation design	Open the surface, drip or sprinkler design; Optimise surface layout; set the irrigation method
Compare and review	Pin and clear the baseline; show the Results, Design or Charts tab
Export	PNG of the window; yearly results CSV; design results CSV; save project; print / PDF

2.7 Resizing and remembering the layout

Drag the thin splitters between the docks to resize them. The program remembers the panel widths between sessions. The *Wider* button above the dashboard widens it; **View ▸ Drawing at**

full width lets the drawing keep its design width and scroll; **Focus view** (F9) hides the panels.

2.8 Reading the drawings

- **Hot spots.** Many drawings are clickable. Hovering shows a tooltip; clicking applies a setting (for example, clicking a year bar jumps to that year, clicking a soil on a chart selects that soil).
- **Value boxes.** Every input of the three design layers is a value box. Click it: a small window opens in which you type the exact value, choose its unit (for example m, km or ft; L/s, m^3/h or L/min; ha, m^2 or acre; kPa, bar, m of water or psi) or pick a value from the list of the allowed range. A value outside the range is brought back to its nearest limit.
- **Nothing runs while you type.** The design layers compute only when you press **Run simulation** (Chapter 18); the other layers follow every control at once.
- **Colours.** Water blues, soil browns, crop greens; text is dark on a white sheet; labels sit in white boxes drawn after the pictures, so no arrow covers a label.

Chapter 3

Every control, value by value

All model inputs live in the **Properties** panel (far right). Each group can be collapsed; **+** and **−** in the panel header expand or collapse all. The sliders are *indices* into lists of discrete values, because every value in the lists corresponds to a set of simulations (Chapter 5). The number shown beside a slider is the value the index stands for. The inputs of the three irrigation designs are not in Properties: they are the value boxes inside each design layer (Chapters 11–13 and 18).

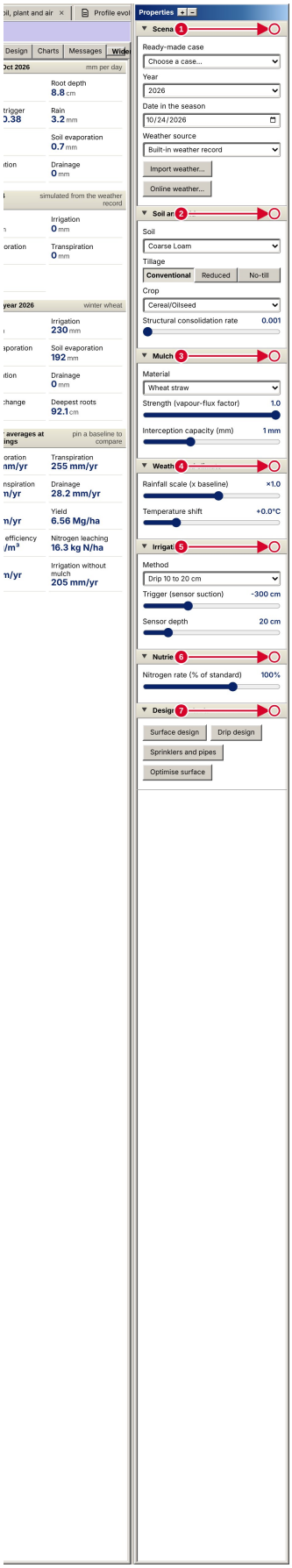


Figure 3.1: The Properties panel, with every group open (the arrows point to the groups; Chapter 25).

3.1 Scenario

Control	Values	Default	What it does
Ready-made case	seven cases (Section 3.8)	—	Replaces many controls at once, as one undoable step.
Year	1980 … 2100	the current year	The calendar year of the working date. Years 1980–1999 use the recorded weather of that year; later years use the typical year of the record, or your own weather when it covers the year (Chapter 4 and Chapter 19).
Date in the season	any calendar date	today	The working date: the day drawn, the “On …” block of the dashboard and the first day of every irrigation schedule.
Weather source	Built-in weather record; Imported weather file; Online weather of a location	built-in	Which daily weather every layer uses. The second and third choices become active only after you have imported a file or fetched the weather of a place (Chapter 19).
Import weather… / Online weather…	buttons		Open the two weather dialogs.

3.2 Soil and field

Control	Values	Default	Effect
Soil	Sand, Loamy Sand, Sandy Loam, Coarse Sand, Coarse Loam, Loam, Silt Loam, Sandy Clay Loam, Clay Loam, Silty Clay Loam	Coarse Loam	Selects the soil’s texture, its van Genuchten parameters (α, n) (Table 5.3), the baseline run, plant-available water per cm (`AWC`) and the trigger anchor. Re-computes every layer and cancels the last run of every design.
Tillage	Conventional / Reduced / No-till	Conventional	Sets the initial pore index p_0 (1.00, 0.72, 0.42) and therefore the starting structure of the topsoil (Section 8.1).
Crop	the simulated cereal/oilseed rotation and the 48 crops of the library, in the groups *Staple food crops* (9), *Cash and oil crops* (8), *Vegetables* (4) and *Trees, vines and palms* (27)	Cereal/Oilseed	Chooses the crop calendar, the crop coefficients, the root depth, the depletion fraction and the yield response (Chapter 21). The cereal rotation is the simulated crop; the library crops are transferred from it with their FAO-56 crop coefficients (Section 21.4).

Control	Values	Default	Effect
Structural consolidation rate	0.001, 0.0025, 0.005, 0.01, 0.02 d^{-1}	0.001	The rate τ_c at which conventionally tilled pores close (Eq. 8.1). In the simulations it changes the results of the five finer soils (loam to silty clay loam, up to about 6 mm a year of drainage and 14 kg N/ha of leaching); in the five coarser soils every rate gives the same result.

3.3 Mulch

Control	Values	Default	Effect
Material	Rice husk, Wheat straw, Wood chips, Sawdust, Grass clippings, Black plastic film	Wheat straw	Sets both sliders below to typical values for the material and the drawn layer thickness (Table 3.1).
Strength (vapour-flux factor) φ	0.1, 0.2, ... 1.0	**1.0 (no mulch)**	The fraction of the bare-soil vapour flux that passes the mulch. 1.0 means bare; 0.1 means the flux is cut by 90 %.
Interception capacity	0.5, 1, 2, 4 mm	1 mm	Water the mulch can hold and lose before it reaches the soil.

Table 3.1. Mulch materials as defined in the engine.

Material	φ	capacity (mm)	drawn thickness (relative)
Rice husk	0.5	2	1.0
Wheat straw	0.4	4	1.35
Wood chips	0.3	2	1.7
Sawdust	0.6	1	0.9
Grass clippings	0.7	4	1.2
Black plastic film	0.1	0.5	0.25

The material only *sets* the sliders. Moving a slider afterwards keeps the material for drawing but uses the slider value for the calculation. Any $\varphi \geq 0.95$ is treated as “no mulch” in the drawings.

3.4 Weather and climate

Control	Values	Default	Effect
Rainfall scale	$×$0.5, 0.6, ... 1.4	$×$1.0	Multiplies the baseline annual rainfall and selects the matching runs. When weather is imported, it is set to the step nearest to the ratio of the imported to the recorded annual rain (Eq. 19.4).

Control	Values	Default	Effect
Temperature shift	−1.0, −0.5, 0, +0.5, … +3.5 °C	0 °C	Selects the warming runs and changes the evaporative demand (Eq. 5.26). When weather is imported, it is set to the step whose demand is nearest the imported reference evapotranspiration (Eq. 19.5).

3.5 Irrigation

Control	Values	Default	Effect
Method	Surface; Overhead; Drip 10–20 cm; Drip 20–30 cm; Drip 40–50 cm; Drip 50–60 cm	Surface	Selects the irrigation run set and, in the day-by-day engine, where each dose is placed (Section 5.2).
Trigger (sensor suction)	−50, −75, −100, −150, −200, −250, **−300**, −350, −400, −500, −600, −700, −800, −1000, −1200, −1500, −2000, −3000, −5000, −10000 cm	−300 cm	Irrigation starts when the soil at the sensor depth is as dry as this suction. A *less negative* number means wetter, more frequent irrigation.
Sensor depth	5, 10, 15, … 100 cm (20 values)	20 cm	Depth of the simulated tensiometer that is compared with the trigger.

Sand and loamy sand. In the simulations of these two soils the sensor-triggered runs at the reference trigger of −300 cm at 20 cm never irrigate, because the coarse soil drains past the sensor before it dries to that suction. A ratio to zero is undefined, so for these soils the irrigation of another trigger, sensor depth or method is added to the reference amount as the difference of the two simulated amounts (Eq. 5.12a). At −50 cm the sand then receives 400 mm a year instead of the reference 278 mm.

Trigger mode (Trigger law layer). The slider above applies one suction to every soil (*common* mode). The **Trigger law** layer offers an *own* mode, in which each soil gets the suction that corresponds to a soil-independent depletion fraction f^* (default 0.45): see Chapter 6.

3.6 Nutrients

Control	Values	Default	Effect
Nitrogen rate	10, 25, 40, 55, 70, 85, **100**, 115, 130, 150 % of the standard rate	100 %	Selects the nitrogen runs; changes yield and nitrogen leaching and, slightly, the maximum rooting depth (Eq. 5.18).

3.7 Ready-made cases

#	Case	Changes
0	Default: Coarse Loam, bare, normal climate	all defaults
1	Dry year on Sand, no mulch	Sand; rain $×$0.6; +1 °C
2	Wheat straw mulch on Loam, drip 10 to 20 cm	Loam; straw; drip 10–20
3	Wet year on Silty Clay Loam, no-till	Silty Clay Loam; no-till; rain $×$1.3
4	Hot year (+3 °C), tomato on Sandy Loam	Sandy Loam; tomato; +3 °C
5	Plastic film, deep drip 40 to 50 cm, Clay Loam	Clay Loam; plastic; drip 40–50
6	Low nitrogen (40 %), cowpea on Silt Loam	Silt Loam; cowpea; N 40 %

Every case first resets the controls to their defaults and then applies its changes, so cases never accumulate.

3.8 Design shortcuts

The last group of Properties opens the three designs (*Surface design, Drip design, Sprinklers and pipes*) and runs **Optimise surface**, which searches the surface design for the inflow and cut-off with the highest application efficiency at the present field.

3.9 Change-by-change: what each control moves

“Run set” means that the response is read from the simulations (Chapter 5); “engine” means that the day-by-day model of Section 5.3 responds as well.

Control changed	Annual outputs (20-year means)	Day-by-day layers and schedules	Design layers
Soil	all (baseline run for that soil)	AWC, (α, n), trigger anchor ⇒ dates, depths	intake (k, A) via Saxton–Rawls and Green–Ampt; Z_n; the sprinkler maximum rate
Tillage, consolidation	drainage, evaporation, yield	pore index in the structure layers	surface storage S_d of the sprinkler design
Crop	yield, transpiration, evaporation (Section 21.4)	crop calendar, K_c, $R_{\max}$, cover	peak ET_c, Z_n, row spacing
Mulch φ, capacity	evaporation, irrigation, drainage, yield	evaporation term	interception loss of the sprinklers
Rain scale	all	rain amount per event	—
Temperature	evaporation, transpiration, irrigation, yield	E_0 and seasonal weight	air temperature in the wind drift (Eq. 13.9)
Method	irrigation, drainage, yield	dose placement	the design that belongs to the method
Trigger	irrigation, yield, drainage	**dates of irrigation**	Z_n (as $f_T \cdot \mathrm{AWC} \cdot R_{\max}$)
Sensor depth	irrigation, yield	**dates of irrigation**	—
Nitrogen	yield, N leaching	$R_{\max}$	—
Weather source	through the nearest rain and temperature steps	**every daily value**	the crop water requirement and the schedules
Date and year	none	**which day is shown; first date in the schedule**	the schedule table

Chapter 4

Dates, years and the clock

4.1 Why the date matters

SWAFL runs on **twenty years of daily weather** arranged on a 365-day calendar without leap days. With the built-in weather these are the twenty recorded years 1980–1999 of the study site (Chapter 5); with imported or online weather they are the last twenty complete years of your own record (Chapter 19). The program maps *any calendar date* you choose onto one of those twenty years. The working date therefore decides (i) which day is drawn, (ii) which of the twenty years acts as the *analogue* for that date, and (iii) the first line of every irrigation schedule.

4.2 The date control

The **Date** box on the toolbar and the **Year** and **Date in the season** controls in Properties ▸ Scenario all set the same working date; the year list runs from 1980 to 2100. **Today** sets the computer's date. When you change the date:

1. With the built-in weather, a year from **1980 to 1999** is its own analogue: that recorded year is drawn.
2. With imported or online weather, a year inside the twenty years of your record is its own analogue in the same way.
3. Any other year (for example 2026 or 2075) takes the **typical** year of the record as its analogue: the median of the twenty years by growing-season rain reaching the soil.
4. The engine sets the field clock to day-of-year d_{oy} of the analogue year, so that 22 October 2029 shows the crop state, soil water and irrigation events of the analogue year on 22 October.

In version 1.0 a *Weather year* control let you choose the typical, the driest or the wettest year for dates outside the record. In version 1.1 the typical year is always used for those dates; to study a dry or a wet year, choose that year inside the record (the *Full water balance* layer of the Seasonal data group shows the rain of each of the twenty years), or import weather.

With Y_0 the calendar year of the chosen date, a the analogue year index $(0, \dots, 19)$ and k the model day number $(0, \dots, 7299)$ of the twenty-year run,

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \quad (4.1)$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

and the calendar label of the first plotted year is $\text{CAL}_0 = Y_0 - a$. The choice of analogue year is

$$a = \begin{cases} \operatorname{argsort}_y \left(P_{\mathrm{thr},y}\right)_{[10]} & \text{typical (median, 11th smallest of 20): years outside the record} \\ Y_0 - Y_{\mathrm{rec}} & Y_{\mathrm{rec}} \le Y_0 \le Y_{\mathrm{rec}} + 19 \end{cases} \tag{4.2}$$

Provenance (Eq. 4.2): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

where $P_{\mathrm{thr},y}$ is the sum of throughfall (rain reaching the soil) in year y (Section 5.2) and Y_{rec} is the first year of the record (1980 for the built-in weather, the first of the twenty years of your weather otherwise).

What this means for you. A date in 2029 is *not* a forecast for 2029 unless the online weather is in use and the date falls in its 16-day forecast. Otherwise it shows how this soil, crop, trigger and method would behave on that day of the year in a typical year. Chapter 17 repeats this as a validity limit.

4.3 The playback bar

Under the drawing, the **playback bar** carries the play button (F5) and, for the field layers, the *Day by day*, *All 20 years* and *Average year* views and the speed buttons. In the field layers the play button animates the field clock: each real second advances 15 model days times the speed setting, and the clock *dwells* for about $3.2/\sqrt{\text{speed}}$ s on every day on which an irrigation, a wetting front of at least 20 cm, or drainage occurs. The clock wraps at the end of the twenty years. **Restart** (Shift+F5) returns to day 1 of the working year. In the three design layers the same button is **Run simulation** (Chapter 18). The year strip that sat under the drawing in version 1.0 has gone: the year is set only in Properties and on the toolbar.

4.4 The 14-day look-ahead

The “Next 14 days” block on the Results tab sums the same simulated daily fluxes from the working date forward. It is labelled *simulated from the weather record* because the rain on those days is the analogue year’s rain. When the online weather is in use and the days fall in its 16-day forecast, the block says *from the weather forecast* instead (Chapter 19).

4.5 Detail year and sensor record

The *Sensor and trigger* and *Profile evolution* layers can show either all twenty years or one year in detail. The suction traces shown there are the study’s **daily suction at 20 cm for 1980–1999**, stored for the Coarse Loam and the Coarse Sand; for every other soil, and for every setting other than the standard one, the trace is moved by the difference between two runs of the daily model, the run at your settings and the run at the standard settings (Eq. 6.4a). The growing season of these records is the 183 days starting 1 April.

Chapter 5

The two computing engines

Every number in SWAFL comes from one of two engines, which are tied together so that they agree on the twenty-year means.

- **Engine 1, the annual meta-model.** It reads 20-year-mean results of the 10,190 Daisy simulations and scales them for the controls. It produces the Dashboard "20-year averages" and every bar, line and arrow in the annual layers. **[Data-driven]**
- **Engine 2, the daily bucket model.** A 40-layer soil-water model run for twenty years of daily weather. It produces dates, depths, schedules, root growth, water flow arrows and the animated field. It is **calibrated** so that its mean irrigation equals Engine 1's, and its fluxes are **scaled** to Engine 1's annual means (Eq. 5.24). **[Engineered, with literature components]**

The design layers (Chapters 11–13) are a third, separate set of *closed-form hydraulic models*; they take their soil, crop, trigger and depth from Engines 1 and 2 (Section 5.3).

5.1 Engine 1 — the Daisy simulations and the meta-model

5.1.1 The simulation campaign

The repository's study ran Daisy 7.1 (Abrahamsen and Hansen, 2000; University of Copenhagen), a one-dimensional soil–plant–atmosphere model that solves Richards' equation with van Genuchten–Mualem hydraulics, coupled to the WEPP dynamic structure model for the effect of tillage and consolidation on pore structure with time. Ten soils (USDA textures from sand to silty clay loam, Ap and Bt horizons) were simulated in **10,190 runs across 17 experiments (E0–E16)**. The families that the workbench reads are:

Experiment	Runs	Varied	Used for
E2	200	Trigger suction	Trigger correction, Eq. (5.11)
E3	200	Sensor depth	Sensor-depth correction, Eq. (5.11)
E4	100	Method (overhead, surface, drip at three depths)	Method correction, Eq. (5.11)
E5	300	Tillage × structural consolidation rate	Baseline **b**, Eqs. (5.4)–(5.10)
E7	200	Mulch vapour-flux factor × interception capacity	Soil evaporation $\bar{E}$, Eq. (5.4)
E11	1,000	Temperature shift × evaporation factor × rain scale	Warming ratios, Eqs. (5.4)–(5.10)
E13	1,000	Nitrogen rate × water strategy × soil	Nitrogen correction, Eq. (5.11)

Experiment	Runs	Varied	Used for
E14	1,000	Mulch factor × capacity × rain scale	Mulch ratios, Eqs. (5.4)–(5.10)

(The other experiments are used by the repository figures and described in the repository README. The *Regime map* and *Fate of saved water* layers are computed from the E14 and E11 grids listed above.)

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

(van Genuchten, 1980), with relative hydraulic conductivity from the Mualem (1976) model,

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Equation (5.2) is used in the Soil, plant and air layer (Section 7.2); Eq. (5.1) is used for all water-retention drawings. The parameters (α, n) used by the engine are *effective* values that pass through the four anchors of the project's trigger table; for the standard textures they match the Carsel and Parrish (1988) values (Table 5.3).

5.1.2 Plant-available water and depletion

Depletion of plant-available water is defined between field capacity (suction 100 cm, pF 2.0) and wilting point (15,849 cm, pF 4.2):

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

$f = 0$ is field capacity and $f = 1$ is wilting point; $f < 0$ (wetter than field capacity) is kept as the *signed* value only in the Trigger law layer. The inverse $h(f)$ is found by bisection on $\log_{10}|h| \in [0, 4.6]$ (60 halvings).

Table 5.3. Effective van Genuchten parameters used by the engine, plant-available water, and the soil's critical suction at the soil-independent depletion $f^* = 0.45$.

Soil	α (cm^{-1})	n	AWC (mm cm^{-1})	h^* at $f^* = 0.45$ (cm)	f at 300 cm
S01 Coarse Loam	0.0961	1.353	1.5	387	0.380
S02 Coarse Sand	0.0536	1.600	0.5	262	0.496
S03 Sand	0.145	2.68	0.6	143	0.842
S04 Loamy Sand	0.124	2.28	0.9	160	0.756
S05 Sandy Loam	0.075	1.89	1.2	195	0.627
S06 Loam	0.036	1.56	1.7	284	0.468
S07 Silt Loam	0.020	1.41	2.0	401	0.367
S08 Sandy Clay Loam	0.059	1.48	1.4	311	0.439
S09 Clay Loam	0.019	1.31	1.6	503	0.314
S10 Silty Clay Loam	0.010	1.23	1.7	729	0.237

(AWC is "typical available water capacity by texture class" in mm of water per cm of soil; it is an engineered per-texture constant, not a measured value of the study soil.)

5.1.3 The meta-model

Let $\mathbf{b} = (E_b, T_b, ET_b, I_b, Y_b, D_b, N_b)$ be the 20-year-mean Daisy results for the chosen soil, tillage and consolidation rate (run set E5), and $\mathbf{b}_{\text{ref}}$ the same for conventional tillage at $\tau_c = 0.005$ d^{-1}. Let φ be the mulch strength, c the mulch capacity, ρ the rainfall scale and ΔT the warming. Define the mulch table lookups $M(\varphi, \rho)$ (E14 at capacity c) and warming table lookups $W(\Delta T, \rho)$ (E11); write

$$\text{now} = M(\varphi,\rho),\ \text{ref} = M(\varphi,1),\ \text{nm} = M(1,\rho),\ \text{bare} = M(1,1), \quad \text{tN} = W(\Delta T,\rho),\ \text{tB} = W(0,\rho),\ \text{tR} = W(0$$

Then the engine computes (the ratio $[x/y]$ is 1 where $|y| < 10^{-6}$):

$$E = \bar{E}(\text{soil},\varphi,c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

$\bar{E}(\text{soil},\varphi,c)$ is the linear interpolation, in φ, of the soil's simulated soil evaporation (run set E7) at capacity c, closed at $\varphi = 1$ by the bare-soil value. Fluxes are combined multiplicatively and drainage/leaching additively because ratios of small numbers (drainage near 0 mm) are unstable.

Four *set-specific corrections* then act in turn on the result. For each family $F \in \{$trigger (E2), sensor depth (E3), method (E4), nitrogen (E13, yield and N_L only)$\}$ with current key κ and baseline key κ_0 (trigger -300 cm; depth 20 cm; method Surface; nitrogen 100 %), with r_κ the soil's row at κ (nearest row if that exact run is absent):

$$Q \leftarrow Q\ \text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E,T,ET,I,Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.$$

(5.11)

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Families whose reference run never irrigates (release 1.1). In the sand and the loamy sand the sensor-triggered runs at the reference trigger (-300 cm at 20 cm, surface method) apply no irrigation at all: the coarse soil drains past the sensor before it dries to that suction. The ratio of Eq. (5.11) is then undefined, and version 1.0 returned 1, so that the trigger, the sensor depth and the method had no effect on the irrigation of these soils. Version 1.1 adds the simulated difference instead,

$$I \leftarrow I + \sum_{F \in \{\text{E2,E3,E4}\}} \max\big(0,\ r_\kappa.I - r_{\kappa_0}.I\big) \quad \text{when } r_{\kappa_0}.I < 1 \text{ mm}, \qquad \Delta S = P + I - ET - D, \tag{5.12a}$$

Provenance (Eq. 5.12a): **[Data-driven]**: differences of Daisy run sets E2–E4; the storage term closes the balance.

so that in the sand a trigger of -50 cm (122 mm more in the runs) gives 400 mm a year instead of 278 mm, -75 cm gives 317 mm and -100 cm 285 mm; drier triggers keep the reference amount, as in the runs.

Finally the model enforces non-negativity, a soil-dependent yield ceiling $Y \le Y_{\max}(\text{soil})$ (the largest simulated yield), rain $P = P_{\text{base}}\rho$, and **water-balance closure**:

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

E_{other} is interception plus evaporation from the mulch layer, and ΔS collects the change in storage and small terms. IE converts Mg ha^{-1} per mm of irrigation to kg of grain per m^3 of irrigation water (1 mm $= 10$ m^3 ha^{-1}).

Crops. The simulated crop is the cereal/oilseed rotation; its yield is shown in Mg ha^{-1}. The 48 crops of the library are transferred from it by the ratio of two FAO-56 daily soil-water balances, one for the library crop and one for the cereal reference, run on the same soil, weather and trigger (Section 21.4, Eqs. 21.5–21.9). Their yield is shown as the relative yield, in per cent of the potential.

5.1.4 What "20-year average at these settings" means

Every Dashboard tile in the last block of the Results tab is the output of Eqs. (5.4)–(5.12) for the current controls. Every slider of Properties is an index into a list of simulated values, so no value ever lies between runs; but two controls changed together (for example, mulch and warming) use the product of ratios above, which assumes the effects are **independent**. The yield ceiling and the closure window (Eq. 5.12) stop independent effects from compounding beyond what the water and the crop allow.

5.2 Engine 2 — the daily bucket model

5.2.1 Layout and state

The soil is $N = 40$ layers of thickness $\Delta z = 2.5$ cm (100 cm total). Each layer holds water w_i (mm) between 0 and the plant-available capacity $W_{\max} = \text{AWC} \cdot \Delta z$. All layers start at $0.72\,W_{\max}$. Relative content $s_i = w_i / W_{\max}$ and depletion $f_i = 1 - s_i$. The run is 20 years of 365 days (7,300 days). The sensor sits in layer $j_s = \text{round}(z_s/\Delta z)$ for sensor depth z_s.

The trigger suction $|h_T|$ is converted to a depletion fraction $f_T = f_{\text{used}}(|h_T|)$ by Eq. (5.3). The sensor condition "soil at suction h_T" therefore becomes the equivalent statement $f_{j_s} \ge f_T$.

5.2.2 Crop calendar and canopy cover

The cereal crop follows the four-year rotation spring barley, winter barley, winter rape, winter wheat with sowing and harvest days (relative to 1 Jan of the year) $(100, 225)$, $(-90, 205)$, $(-125, 200)$, $(-80, 230)$; winter crops are sown in the previous autumn. Legume: sown day 110, harvested 215. Vegetable: sown day 100, harvested 185. Outside the window the field is *fallow*. Canopy cover $c(d) \in [0, 1]$ is

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp(-(d - d_{\text{mid}})/s)}, \quad s = 13 \text{ (vegetable } 8), \tag{5.13}$$

Provenance (Eq. 5.13): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12 \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp(-(d - d_{\text{pk}} + 45)/12)} & \text{otherwise} \end{cases} \tag{5.14}$$

Provenance (Eq. 5.14): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

and after the peak day d_{pk} cover declines linearly to zero at harvest: $c \leftarrow c \max\{0,\, 1 - (d - d_{\text{pk}})/(d_{\text{harv}} - d_{\text{pk}})\}$. The mid day is $d_{\text{sow}} + 50$ (cereal), $d_{\text{sow}} + 40$ (legume), $d_{\text{sow}} + 28$ (vegetable); the peak is harvest -50 (cereal), sow $+65$ (legume), sow $+65$ (vegetable). These logistic curves are **[Engineered]**: they reproduce the shape of a canopy, not a measured leaf-area record.

5.2.3 Rainfall generator

Rain falls on a day with probability 0.28. With $\mu_0 = P/(0.28 \cdot 365)$ the mean wet-day depth of the scaled annual rain P, the depth on a wet day of simulated year y is

$$R = \min\{70,\ \mu_0\, \gamma_y\, (-\ln U)\}, \qquad U \sim \text{Uniform}(0, 1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

an exponential distribution capped at 70 mm, with the random sequence seeded by a hash of the soil name so every run is reproducible. The year factor $\gamma_y = \text{clip}\big((m_{\text{med}}/m_y)^{0.3};\, 0.65,\, 1.45\big)$ makes years wetter or drier according to the dryness m_y of the study's 20-year rainfed suction record at 20 cm, where $m_y = 10^{\overline{\log_{10} h}}$ is the geometric-mean suction over the 183-day growing season of year y, and m_{med} is the median over the 20 years. This is how the generated weather "follows the dryness of the real record".

The canopy and litter catch the first I_c mm of every shower:

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\, R - I_c\}. \tag{5.16}$$

Provenance (Eq. 5.16): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

P_{other} is the interception-and-mulch loss from Eq. (5.12). Only throughfall $T_f > 0.3$ mm enters the soil. The choice of I_c makes the expected interception over the exponential rain distribution equal to the annual interception of Engine 1 (the expected value of $\min(R, I_c)$ for an exponential with mean μ_0 is $\mu_0(1 - e^{-I_c/\mu_0})$, and $e^{-I_c/\mu_0} = r$ gives $\mu_0(1 - r)$ per wet day).

5.2.4 Infiltration, redistribution and drainage

Water enters the top layer and cascades down: any layer exceeding W_{max} passes the excess to the layer below; the excess leaving layer 40 is **drainage** D_d. The depth of the deepest wetted layer is the **wetting front** (cm). Between layers a slow redistribution moves water toward the drier neighbour,

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\;\; w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

shown as *capillary rise* arrows when $q_i > 0.03\,W_{\max}$ upward. Drainage and the wetting front are therefore **tipping-bucket** results, not Richards solutions; that is why the engine scales them to Engine 1 (Eq. 5.24).

5.2.5 Roots

The maximum rooting depth is

$$R_{\max} = \operatorname{clip}\Big(R_c\,(0.85{+}0.3\,\text{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\;25,\;100\Big)\ \text{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

with $R_c = 95$ (cereal), 115 (legume), 45 (vegetable) cm, sand the sand fraction, and N the nitrogen rate in percent. (Drier triggers push roots deeper; nitrogen-rich soils root slightly shallower.) While the canopy is growing, the rooting depth R_d chases water:

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\operatorname{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35 + \bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

where $\bar{s}_{\text{wet}}$ is the mean relative water content of the four layers beneath the root front, and $\kappa = 1.3$ if the top six layers average below 0.35 (dry topsoil) and 1 otherwise. After harvest $R_d \leftarrow \max\{3, 0.965R_d\}$; drought prunes the roots by 0.9 cm per day when the plant-available-water-weighted root fraction is below 0.12 and $R_d > 8$ cm. Root density follows

$$\rho_i \propto \exp\Big(-\frac{z_i}{\lambda R_d + 8}\Big)(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

with $\lambda = 0.32$ (cereal), 0.24 (legume), 0.40 (vegetable) and z_d the drip emitter depth. The bracket is **[Engineered]**: it raises root density up to fourfold around a drip emitter. These root functions are the *modelled* roots drawn in the R1 layer, “not a scan of real ones”.

5.2.6 Transpiration, evaporation

Daily soil evaporation is the annual evaporation E_{yr} of Eq. (5.4) distributed over the year with a sinusoidal weight, reduced by canopy shade,

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65 \cdot \tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

and is taken from the top four layers with weights $(0.55, 0.25, 0.12, 0.08)$, never drawing a layer below $0.05\,W_{\max}$. Crop transpiration demand is the annual transpiration spread by cover, times a **calibration factor** k_t,

$$T_d = k_t\, T_{\text{yr}}\, \frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\text{ yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

and is divided among the layers by *availability-weighted* root density,

$$q_i^T = T_d\, \frac{\rho_i\, a_i}{\sum_j \rho_j a_j}, \qquad a_i = \text{clip}\Big(\frac{s_i - 0.15}{0.6};\, 0,\, 1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

limited so that no layer falls below $0.10\, W_{\text{max}}$. The availability function a_i is a linear stress function between 15 % and 75 % of plant-available water; it is **[Engineered]** in the spirit of the FAO-56 water-stress coefficient (Allen et al., 1998, Eq. 84), which is also linear in depletion.

5.2.7 Irrigation rule and calibration

An irrigation of dose d (mm) is applied on day d if *all* of these hold: canopy cover $c > 0.2$; $95 \le$ doy ≤ 250; at least m days since the last irrigation ($m = 1$, or $m = \text{clip}(\text{round}(110\, d/\bar{I}); 2, 12)$ if $f_T < 0.1$, to space irrigations when the trigger is almost at field capacity); the yearly total would stay within the budget $6\bar{I}$ (practically no cap); the sensor layer is dry enough, $1 - s_{j_s} \ge f_T$; $f_T < 0.995$; and, for drip, the layer at the emitter depth has depletion ≥ 0.12. The dose is $d = \text{clip}(\bar{I}/10; 15, 35)$ mm, and for drip $d_{\text{eff}} = \max\{8, d/2\}$ delivered as small pulses *spread over* ± 3 *layers around the emitter depth*; for other methods the dose enters at the surface. $\bar{I}$ is the annual irrigation of Eq. (5.7).

Calibration of k_t. The trigger decides *when* but the root-zone water use decides *how often*. To make the day-by-day model reproduce the annual irrigation of Engine 1, $k_t \in [0.6, 8]$ is found by eight bisection steps on the first 8 years so that the mean irrigation per year I_{sim} matches $\bar{I}$ within 6 % (if the first guess $k_t = 1$ is already within 6 %, it is kept); the result is then used for all 20 years.

5.2.8 Scaling to the annual model

The daily results are multiplied by constants so that their 20-year means equal Engine 1:

$$S_P = \frac{P}{\overline{P_{\text{sim}}}}, \quad S_{\text{thr}} = \frac{P - P_{\text{other}}}{\overline{T_{f,\text{sim}}}}, \quad S_E = \frac{E}{\overline{E_{\text{sim}}}}, \quad S_T = \frac{T}{\overline{T_{\text{sim}}}}, \quad S_D = \frac{D}{\overline{D_{\text{sim}}}}, \tag{5.24}$$

Provenance (Eq. 5.24): **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

(each set to 1 if the simulated mean is below 10^{-6}), and the daily fluxes displayed on the Dashboard are $R = P_d S_P$, $T_f = \min\{R,\, T_{f,d} S_{\text{thr}}\}$, $E = E_d S_E$, $T = T_d S_T$, $D = D_d S_D$ and irrigation I unscaled. The **storage change of the day** closes the balance:

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

5.2.9 Evaporation under warming (used by the E1 layer)

For the Evaporation physics layer a separate atmosphere factor is used (Section 9.3):

$$E_{0,\text{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\frac{1}{2}[1+\sin(2\pi(\text{doy}-100)/365)]}{0.675}\,\text{clip}\big(1{-}0.12(\rho{-}1);\,0.85,\,1.12\big)\sqrt{\gamma_y}\ \text{m} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

where 4.2 mm d^{-1} is the engine's season-mean bare-soil demand (the seasonal weight is divided by its mean, 0.675), the factor $1 + 0.07\Delta T$ is a 7 % per °C increase (close to the Clausius–Clapeyron rate of the saturation vapour pressure; **[Engineered]** parameter), the cloud factor reduces demand in wetter years, and $\sqrt{\gamma_y}$ links demand to the dryness of the year.

5.3 The design layers and their link to the engines

The three design layers take from the engines:

Quantity	Source	Used by
Soil texture (sand, silt, clay %)	`DATA.texture` (measured for the 10 soils)	Saxton–Rawls intake (Section 11.2)
Trigger depletion f_T	Eq. (5.3) at h_T	Required depth Z_n
Maximum rooting depth $R_{\max}$	Eq. (5.18)	Z_n
Plant-available water per cm AWC	Table 5.3	Z_n
Peak crop use ET_c	engine's peak daily transpiration plus evaporation	Interval T (drip, sprinkler)
Date and irrigation dates	Engine 2	Irrigation schedule tables
Weather source, mulch, interception	Engine 1 and mulch capacity	sprinkler interception loss

The default required depth is therefore

$$Z_n = \text{clip}\big(f_T\cdot\text{AWC}\cdot R_{\max};\,25,\,150\big)\ \text{mm} \qquad (\text{surface; drip and sprinkler use their own limits}) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

which is the net depth that refills the root zone from the trigger depletion back to field capacity ("*from the DSS*" on the screen). Typing a value overrides it (the label then reads *tap to reset*; click the label to return to the computed value).

5.4 Reproducibility

All random numbers in the engine are generated by a deterministic generator seeded by the soil name (a 32-bit hash), so the same controls always give the same dates, depths and animations. There is no external data call; the engine file contains the simulation tables as JSON.

Chapter 6

Irrigation control layers

The two layers of this group answer one question: **when should irrigation start?** *Sensor and trigger* shows what a sensor sees through twenty years; *Trigger law* shows why the same suction means different things in different soils and how to express the trigger so that it works for every soil.

6.1 Sensor and trigger

6.1.1 What it shows

A simulated tensiometer at the **sensor depth** (default 20 cm) records soil-water suction h (cm of water, plotted on a log axis from 1 to 100,000 cm) every day. Two traces are drawn: *irrigated* (blue) and *rainfed* (orange). The red dashed line is the **trigger** you set. Pale bands are growing seasons. Under the trace, red ticks mark irrigated sensor days beyond the trigger, and the bar chart counts, for each year, the growing-season days (of 183) on which the sensor was beyond the trigger, irrigated against rainfed.

6.1.2 How to use it, step by step

1. Open *Irrigation control* ▸ *Sensor and trigger*.
2. Choose **All 20 years** to see the whole record, or **1994 detail** for the one year with full-profile output (the buttons are on the playback bar under the drawing).
3. Move **Trigger (sensor suction)** in Properties; the red line and every count update.
4. Move **Sensor depth**, the soil, the mulch, the method or any other control: both traces follow it (Eq. 6.4a), and the sentence under the drawing names the record and the settings it is drawn for.
5. Press the play button under the drawing to sweep a cursor through the record; the working year is set in Properties ▸ Scenario.
6. Read the text under the plot: *“Irrigated: 85 of 183 days beyond 300 cm, mean suction 255 cm. Rainfed: 168 of 183 days beyond, mean suction 1,945 cm. Irrigation removes 39 % of the days beyond this trigger.”*

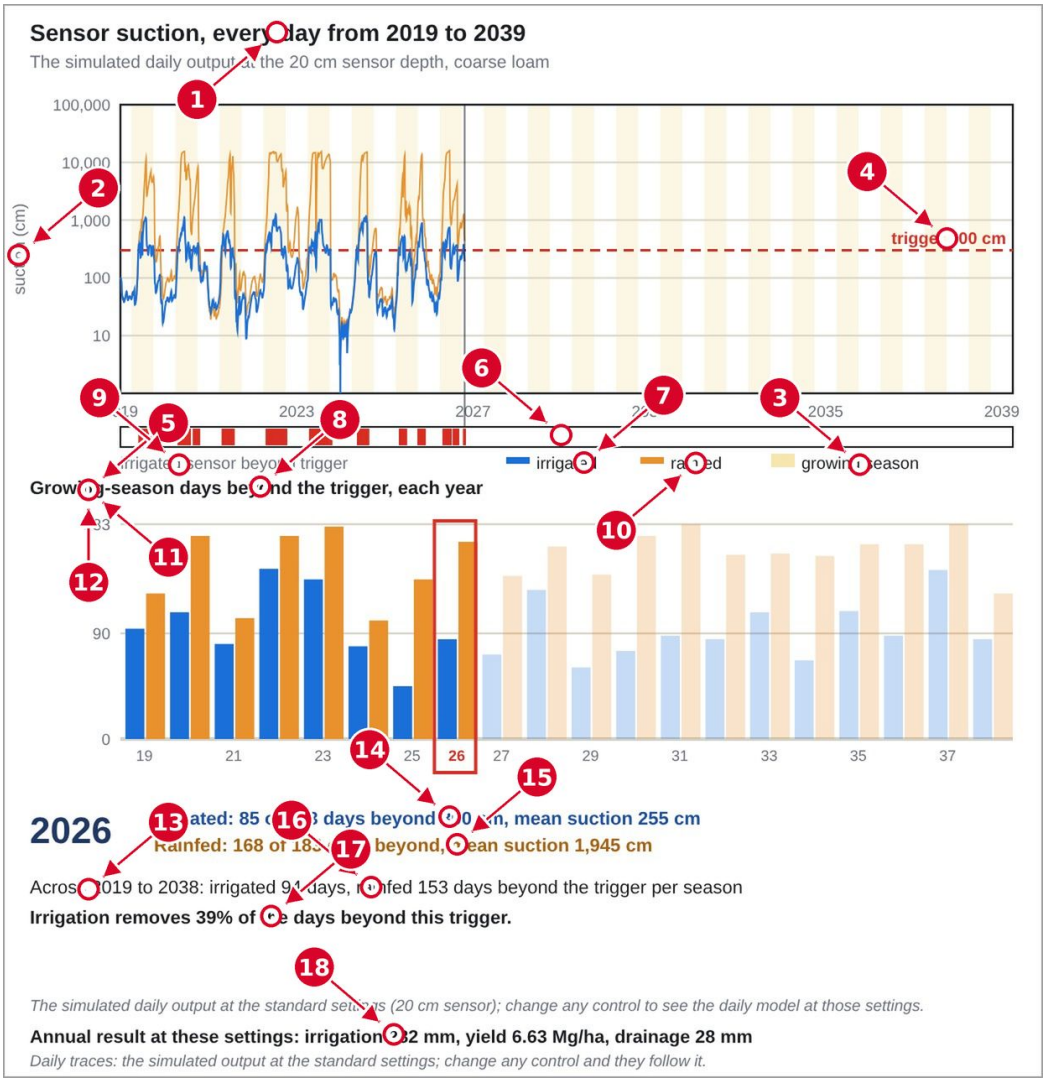


Figure 6.1: Sensor and trigger. Top: 20 years of suction at the sensor, irrigated (blue) and rainfed (orange) with the trigger (red dashed). Bottom: growing-season days beyond the trigger per year. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

6.1.3 How the numbers are generated

- The two suction traces start from the Daisy-simulated daily suction at 20 cm (irrigated with the study's trigger, and rainfed) for the 1980–1999 record, stored in the engine (`D20`) for the Coarse Loam and the Coarse Sand. At the standard settings of those two soils they are drawn as stored. For any other soil, sensor depth or setting the record is moved by the **delta method**: the daily model (Engine 2) is run twice, at your settings and at the standard settings of the same soil and weather, and the difference of the depletion of the sensor layer (smoothed over seven days) is added to the depletion of the stored record, which is then converted to suction in your soil:

$$h_d^{\mathrm{x}} = h_{\mathrm{soil}}\Big(f_{\mathrm{ref}}\big(h_d^{\mathrm{rec}}\big) + \tfrac{1}{7}\sum_{q=-3}^{3}\big[f_{d+q}^{\mathrm{set}} - f_{d+q}^{\mathrm{std}}\big]\Big), \tag{6.4a}$$

Provenance (Eq. 6.4a): **[Data-driven + Engineered]**: the stored simulated record carries the real weather sequence; the change made by your settings is the difference of two runs of the daily

model, converted with the van Genuchten curve of your soil (the inverse of Eq. 5.3, tabulated).

with f_{ref} the depletion of the reference soil whose record is used (the nearer of the two in texture), f^{set} and f^{std} the daily depletion of the sensor layer in the two runs (irrigated or rainfed as the trace), and $h_{\mathrm{soil}}(f)$ the suction of your soil at depletion f. * The count of days beyond the trigger is, for year y,

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.

and the "Irrigation removes X %" statement is

$$X = 100\Big(1 - \frac{\bar{N}_{\mathrm{irr}}}{\bar{N}_{\mathrm{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_{y} N_y. \tag{6.2}$$

Provenance (Eq. 6.2): **[Data-driven]**: counts over the simulated daily series.

- The annual line at the foot ("irrigation 282 mm, yield 6.63 Mg ha^{-1}, drainage 28 mm") is Engine 1 (Eqs. 5.4–5.12) at the current settings.

6.1.4 Reading it

If rainfed days beyond the trigger greatly exceed irrigated days, irrigation is doing its job; if the two bars are similar the trigger is so dry that irrigation rarely starts. A trigger of -50 cm (very wet) produces ticks on almost every day.

6.2 Trigger law

6.2.1 The idea

The same suction does not mean the same dryness in two soils. At -300 cm a sand has lost 84 % of its plant-available water while a silty clay loam has lost only 24 % (Table 5.3, last column). N1 re-expresses the trigger as the **depletion fraction** f of Eq. (5.3), the share of plant-available water already used, and shows that irrigation and yield responses from all ten soils collapse onto a common curve when plotted against f.

[Literature + study result]. The depletion form follows the management-allowed-depletion idea of FAO-56 (Allen et al., 1998, Eq. 83 and Table 22): irrigate when a fixed share p of the total available water is used. The *soil-independent* critical value $f^* = 0.45$ is the study's result (5 % yield loss), and the pooled R^2 rising from 0.48 to 0.75 (irrigation) and 0.23 to 0.70 (yield) is reported in the repository.

6.2.2 How to use it, step by step

1. Open *Irrigation control* ▸ *Trigger law*.
2. The soil column on the left is the soil by depth on the clock's day; the blue gauge beside it shows depletion from FC (0) to WP (1). **Drag the purple trigger marker** on the gauge: the soil's own trigger suction changes, and the right-hand list of ten soils moves in step.
3. Each row on the right is one soil with its trigger suction and the net irrigation depth at that depletion ("143 cm trigger, 278 mm"); the green box gives the relative yield at the soil's trigger.
4. Click a soil name to select it (it becomes the working soil).
5. **Common and own mode.** By default one suction (the Trigger slider) applies to every soil (*common* mode). **Opening the Trigger law layer switches the program to *own* mode**: each soil then uses its own critical suction for the chosen depletion, and the Trigger slider

in Properties shows the nearest value on its 20-value grid (simulated runs exist only on that grid). Own mode stays on until you choose *Reset all settings* or a ready-made case, which restore the common mode.

6. Read the verdict box at the bottom: it states the soil's critical suction, the yield at the current trigger relative to the best trigger, and how many root-zone layers are beyond the trigger on this day.

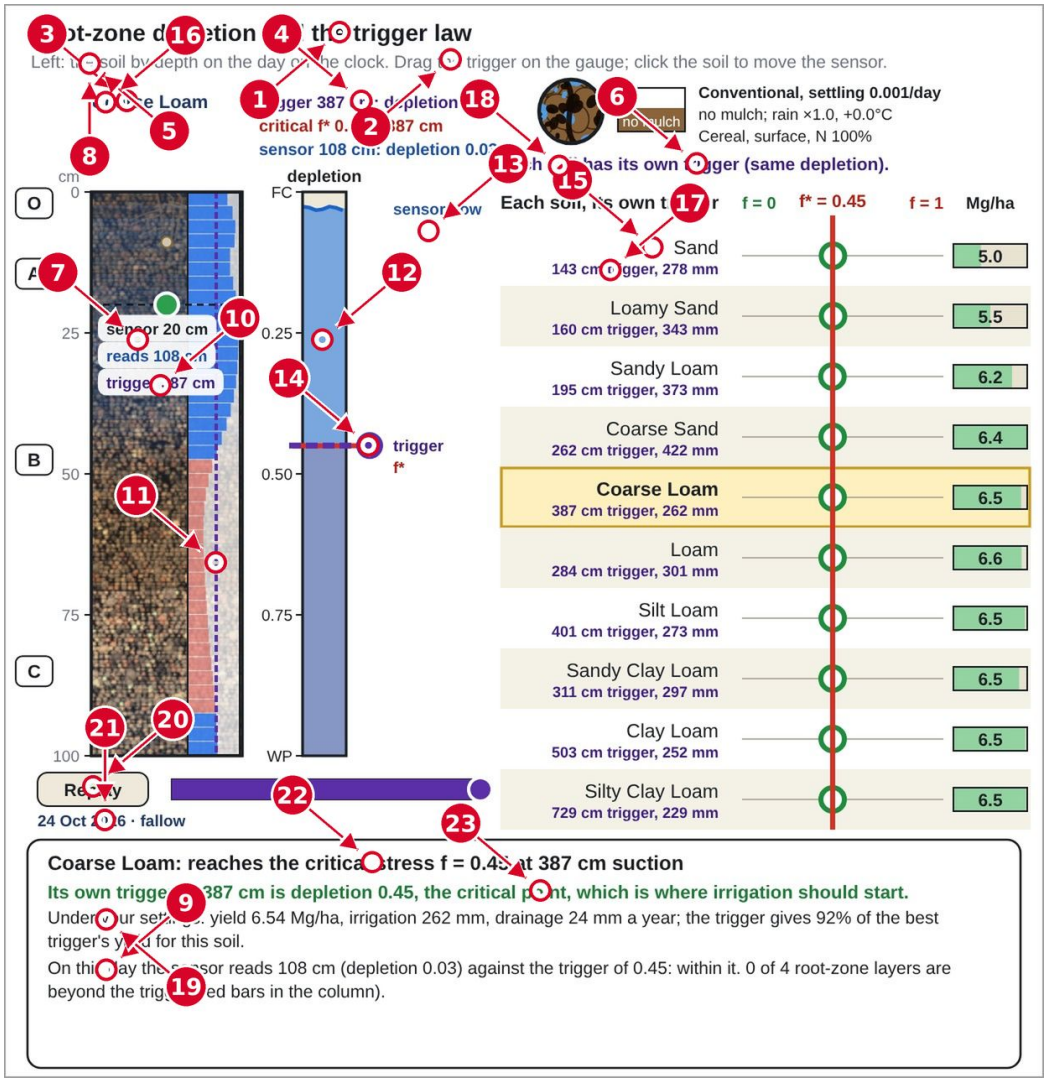


Figure 6.2: Trigger law layer: ten soils on one depletion axis, with the critical $f^* = 0.45$ (red line) and the soil's own trigger. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

6.2.3 Governing relations

The depletion of Eq. (5.3) is evaluated with each soil's effective van Genuchten pair (α, n) (Table 5.3). The suction at which a soil reaches a chosen depletion f is the inverse,

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

Provenance (Eq. 6.3): **[Literature]**: inverse of the van Genuchten curve, solved by bisection.

solved by bisection, giving for $f^* = 0.45$ the soil-specific sensor settings of Table 5.3 (143 cm for sand to 729 cm for silty clay loam). The net irrigation per event shown beside each soil is

$$Z_n = f \cdot \mathrm{AWC} \cdot R_{\max} \quad [\mathrm{mm}] \tag{6.4}$$

Provenance (Eq. 6.4): **[Literature]**: same relation as Eq. (5.27) without the clip.
(the same relation as Eq. 5.27 without the clip). The *relative yield at the trigger* is

$$Y_{\mathrm{rel}} = \frac{Y(h_T)}{\max_h Y(h)}, \tag{6.5}$$

Provenance (Eq. 6.5): **[Data-driven]**: ratio of Daisy run-set E2 yields.
with $Y(h)$ the soil's yield at each simulated trigger (run set E2). **[Data-driven]**.

6.2.4 How fast each soil dries

Each row of the soil list ends with the days from field capacity to the trigger, and the box under the drawing tells how fast the soil in focus dries. Every soil is compared on the same crop demand: the mean evapotranspiration of the peak of the season (days 150 to 220), from the daily model of a well-watered coarse loam at all your other settings. Only the water each soil holds differs:

$$r_s = \frac{ET_c}{\mathrm{TAW}_s}, \qquad \mathrm{TAW}_s = \mathrm{AWC}_s\, z_r, \qquad t_{\mathrm{trig},s} = \frac{f_{\mathrm{trig},s}}{r_s}, \tag{6.6}$$

Provenance (Eq. 6.6): **[Literature]**: the FAO-56 root-zone depletion rate; AWC_s is the available water of the soil (mm per cm), z_r the root depth of the reference run and $f_{\mathrm{trig},s}$ the depletion at the soil's own trigger.

A sand holds about 0.6 mm per cm, so the same demand empties it in about a third of the time of a loam: at the default settings, sand reaches its trigger in about 7 days and coarse loam in about 18. When you press **Play**, the gauge follows the daily model day by day, and its response is tied to the soil: it answers at once in a sand and slowly in a clay (rate $= 0.09 \times 1.5/\mathrm{AWC}_s$ per frame, between 0.04 and 0.35).

Choosing another soil, in the list or in Properties, starts the replay again from the start of the season (day 95), with the gauge set to the new soil.

6.2.5 The core trade-off: f = 0.50 or f = 0.45

Under the explanation, a panel answers the question each grower asks: trigger at a depletion of 0.50 or of 0.45? It has two columns, and each condition is ticked when it applies to your settings.

Choose f = 0.50 if:

- you irrigate a standard field crop (maize, grains, alfalfa): a field crop of the library whose FAO-56 depletion fraction is $p \geq 0.5$;
- your soil is loam or clay (high water-holding capacity): less than 60 % sand and more than 0.9 mm of available water per cm;
- your system can irrigate the entire field quickly: a full round of the design of your irrigation method takes 2 days or less.

Choose f = 0.45 if:

- your irrigation system takes several days to finish a full round: more than 2 days. A round is the time per round of the design divided by the hours a day you irrigate (24 h for surface irrigation).
- your soil is sandy or coarse and dries out very fast: 60 % sand or more, or 0.9 mm of available water per cm or less;

- the crop is in a critical, water-sensitive growth stage such as flowering or pollination: on the date on the clock the crop is at 85 % or more of its full cover.

If any condition of the f = 0.45 column applies, the advice is f = 0.45; otherwise it is f = 0.50. A sentence gives the reasons. Each column has a button that sets the trigger of the soil in focus to that depletion; the button of the trigger in use is green. The time per round is known once the design of your method has been opened; until then the panel says so.

6.2.6 Reading the sensor value

The dashboard tile *Sensor / trigger* reads two numbers: the **current depletion** f at the sensor layer and the **trigger depletion** f_T. When the first reaches the second, irrigation is due. On 18 Oct 2026 the tile reads $0.02 / 0.38$ for the Coarse Loam: the soil is almost at field capacity (0.02), and the trigger is at depletion 0.38 ($= f$ at 300 cm).

Chapter 7

Field water balance layers

7.1 Field balance (the home layer)

7.1.1 What it shows

One picture of the water budget of a field for one day, one year, or the twenty-year average: rain and irrigation arriving, transpiration, soil evaporation and interception leaving upward, drainage leaving downward, and the nitrogen leached with it. The soil profile is drawn to 100 cm with three depth marks A, B and C, the sensor at the chosen depth, the roots as they grow and die back, and the wetting front.

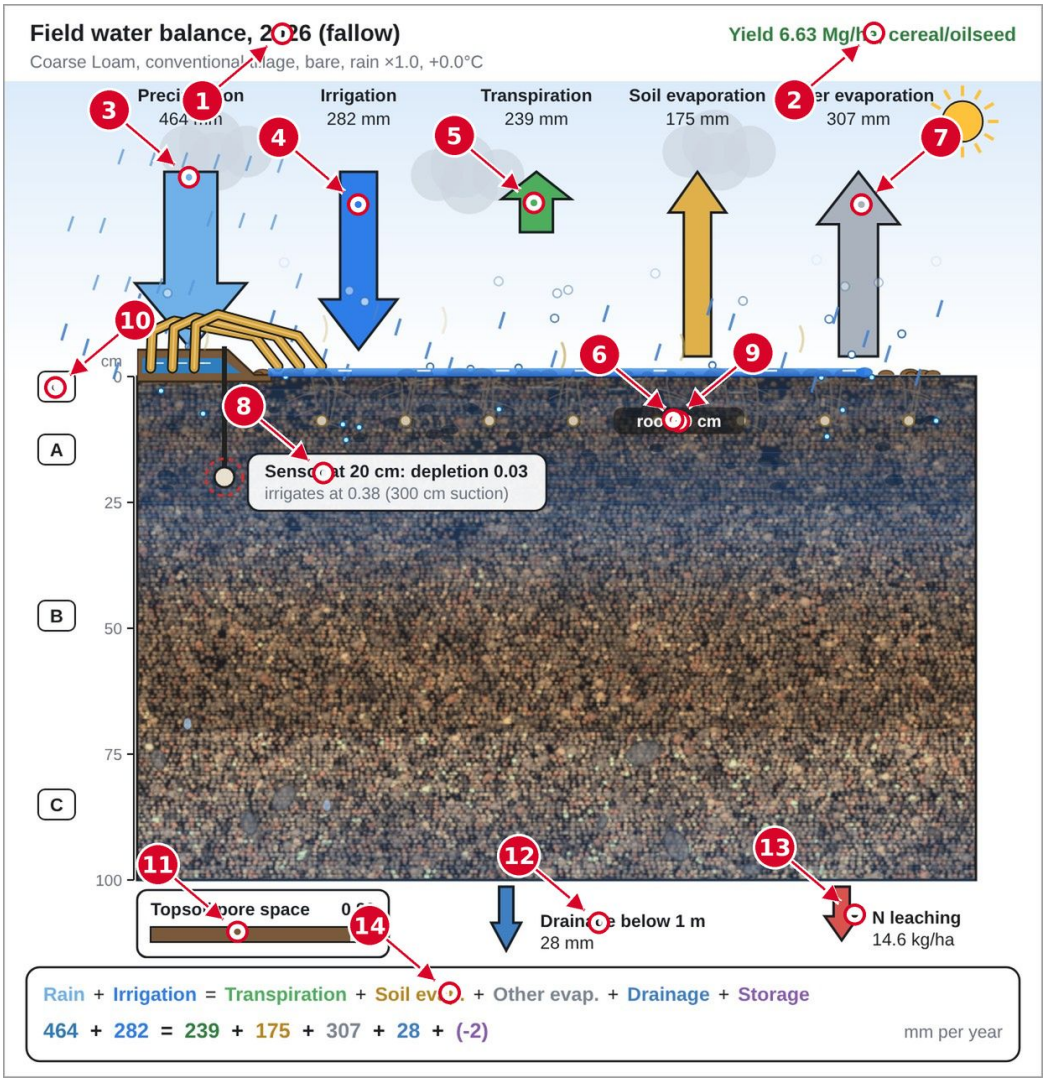


Figure 7.1: Field balance: arrows are the fluxes (mm per year in “Average year” view), the profile shows the root zone, the sensor, and drainage below 1 m. The equation strip under the picture is the closed budget. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

7.1.2 The control bar under the drawing

Button	Meaning
Day by day	The simulated day on the field clock; arrows show that day’s fluxes (T_f, I, T, E, interception, D).
All 20 years	Table and bars of every simulated year (rain, irrigation, interception, evaporation, transpiration, drainage, storage change, crop, deepest roots). Click a year.
Average year	The 20-year means of Engine 1 (Eqs. 5.4–5.12).
◀ year / year ▶	Previous or next simulated year.
1× 2× 4× 8× and speed slider	Play speed.
Play / Pause	Animates the clock.

7.1.3 How each output is generated

On the picture	Generated by
Precipitation, Irrigation, Transpiration, Soil evaporation, Interception, Drainage (mm) in *Average year*	Engine 1: P, I, T, E, E_{other}, D of Eqs. (5.4)–(5.12)
The same in *Day by day*	Engine 2 daily fluxes after scaling, Eqs. (5.15)–(5.25)
Equation strip "Rain + Irrigation = Transpiration + Soil evap. + Interception + Drainage + Storage"	The closure of Eq. (5.12): $P + I = T + E + E_{\text{other}} + D + \Delta S$
Sensor label "depletion 0.02 / irrigates at 0.38"	Engine 2 layer j_s: $f_{j_s} = 1 - s_{j_s}$ against f_T
Roots (depth, density)	Eqs. (5.19)–(5.20)
Wetting front	Depth of the deepest layer reached by the last infiltration event
Topsoil pore space bar	Pore index p of Eq. (8.1) for the chosen tillage and consolidation rate
Drainage below 1 m	D (mm), the water leaving layer 40
N leaching (kg ha^{-1})	N_L of Eq. (5.10)
Yield tag (top right)	$Y \cdot f_{\text{crop}}$ from Eq. (5.8)

7.1.4 Step-by-step: checking a budget

1. Open *Field balance* and choose **Average year**.
2. Read the five numbers over the arrows and the strip below. Check that rain plus irrigation minus the four sinks equals the storage change in brackets.
3. Change **Rainfall scale** to $×$1.3. The rain, drainage and N leaching arrows thicken.
4. Press **Pin results as baseline** (Ctrl+B) *before* each change to see the difference on the Dashboard tiles.
5. Click **Day by day**, Play, and watch irrigation events pause the clock (Section 4.3).

7.2 Soil, plant and air

7.2.1 What it shows

The soil–plant–atmosphere continuum as one water column "from the particles to the sky": a section of the soil with roots, the plant, and the air above; five irrigation layouts (**Furrow, Basin, Border strip, Sprinkler, Drip**) that you can tap, which set the irrigation method; and, on the right, the **water potential along the path** from the air to the deep soil. Two data sources are labelled with badges: *DAISY* (taken from your simulations) and *DERIVED* (computed by the workbench along the path).

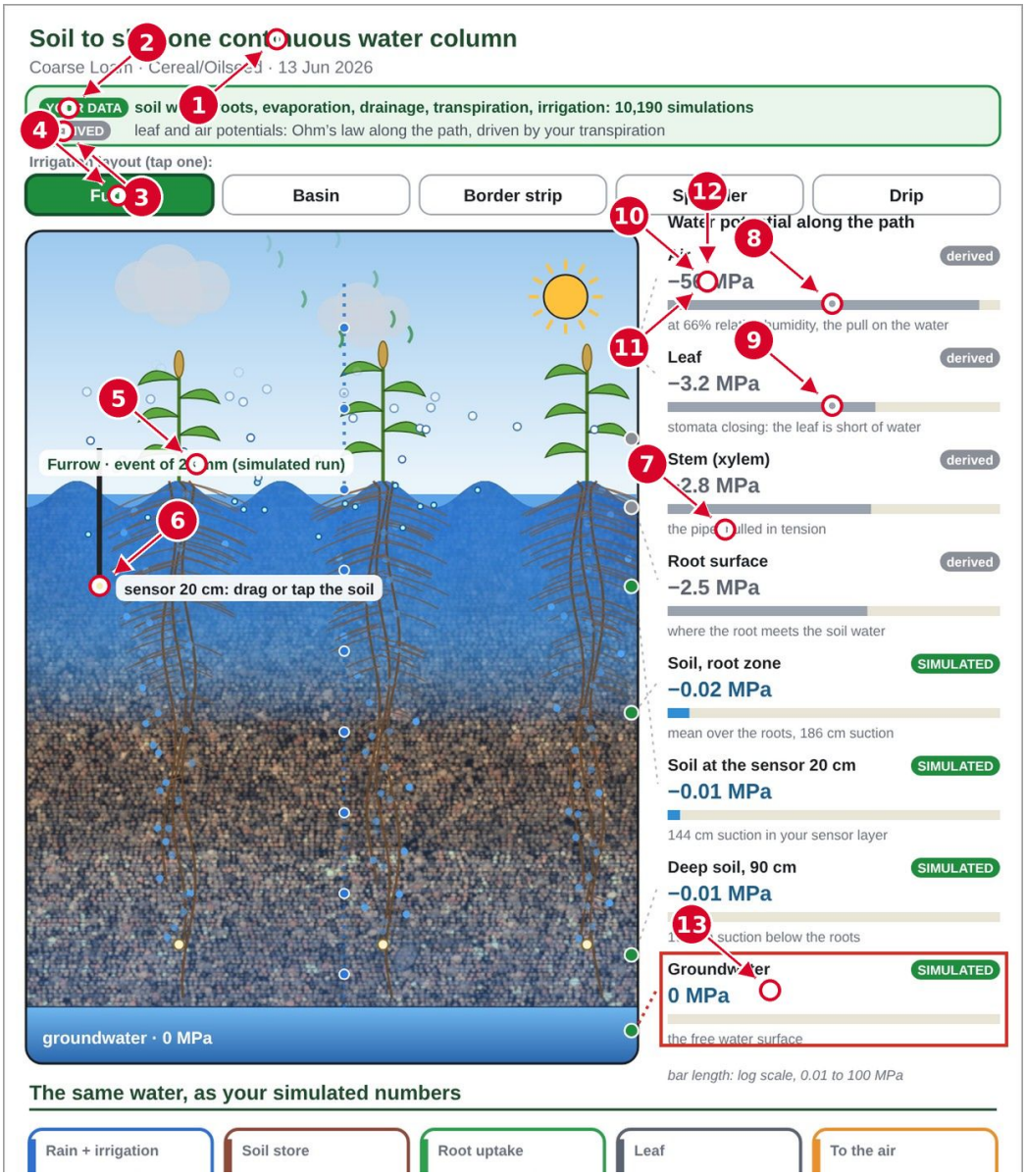


Figure 7.2: Soil, plant and air layer: potentials from the air to the deep soil. *Derived* badges mark the quantities computed along the path; *Simulated* badges mark simulation data. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

7.2.2 Governing equations

Soil. The suction in each layer is the suction that gives that layer's depletion, from the inverse of Eq. (5.3), floored at 30 cm:

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

Provenance (Eq. 7.1): **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

The *root-zone* suction is the root-density-weighted mean, $h_{RZ} = \sum_i \rho_i h_i \big/ \sum_i \rho_i$, and the *sensor* and *deep-soil* values are h_{j_s} and h_{N-3}.

Soil conductivity. The relative conductivity at the root zone uses the Mualem–van Genuchten form, Eq. (5.2), with $S_e = S_e(h_{RZ})$ from Eq. (5.1).

Along the path. The plant is a chain of series resistances, in the spirit of the catenary (Ohm's-law) description of water flow through the soil–plant–atmosphere continuum (van den Honert, 1948). With E the day's transpiration (mm d^{-1}, from Eq. 5.25) and R_{soil}, R_{root}, R_{xyl}, R_{leaf} resistances in MPa per (mm d^{-1}):

$$R_{\text{soil}} = 0.03\ \max\{K_r,\ 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

where $a = 1$ when canopy cover exceeds 0.05 (a crop is present) and $a = 0$ otherwise (fallow). Stomata are drawn as *closing* when $\psi_{\text{leaf}} < -1.6$ MPa.

The air. Water potential of the air above the canopy follows the thermodynamic relation between potential and relative humidity,

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\,\ln \text{RH} \ \approx\ 135.5\,\ln \text{RH}\ \ \text{MPa} \tag{7.5}$$

Provenance (Eq. 7.5): **[Literature]**: Nobel (2009).

(Nobel, 2009), where R is the gas constant, $T \approx 293$ K and $V_w = 18 \times 10^{-6}$ m^3 mol^{-1} the molar volume of water. Relative humidity is $\text{RH} = \text{clip}(0.66 - 0.03\,\Delta T;\ 0.35,\ 0.90)$ with ΔT the temperature shift. At RH = 66 % the air is at -56 MPa, as shown.

What is literature and what is engineered. Eqs. (7.1), (7.5) and (5.2) are physical relations. The resistances in (7.2)–(7.4) are **[Engineered]** values chosen so that a typical transpiring cereal reaches ψ_{leaf} of about -1 to -1.5 MPa and so that stress develops as the soil dries; they are *not* measured resistances, and the layer labels them "derived". **Caution.** Evaluated over mid-season crop days in all ten soils (Paper 3, Section 5.1), the relative conductivity at the root-zone suction is usually below the floor of 0.002 in Eq. (7.2), so R_{soil} sits at its cap (0.92 MPa per mm d^{-1}) and the mean leaf potential is -1.8 to -2.3 MPa in every soil, below the intended range. The leaf, stem and root potentials of this layer therefore illustrate the series-resistance concept and should not be read as plant water status.

7.2.3 Step-by-step

1. Open *Soil, plant and air*.
2. Click a layout. Furrow, basin and border strip set the **Surface** method; Sprinkler sets **Overhead**; Drip sets **Drip 20 to 30 cm**.
3. Drag or click in the soil to move the **sensor depth** (the control snaps to the 5-cm list).
4. Change the **date** to a growing-season day and compare the potentials: with a crop present the leaf falls below -1 MPa as the soil dries; on a fallow date the path is flat (resistance terms are zero).
5. Move the **Trigger** slider and note that the *sensor* suction tile follows the same depletion as the dashboard.

7.3 Profile evolution

7.3.1 What it shows

How the pressure head profile develops through the season in a year, side by side for **irrigated** and **rainfed** soil columns, with the sensor depth shown by a dot. In *All 20 years* mode it draws the pressure head at the sensor through every year and a heat strip of the mean suction of each

growing season (darker is drier). Text states the driest and wettest rainfed season and the share of the 0–60 cm root zone beyond the trigger.

7.3.2 How it is generated

- The **1994** profile is the Daisy-simulated pressure head at every depth, irrigated and rainfed (the only year with full-depth output, as the drawing says).
- In the other years only the 20 cm sensor is available; the columns are shaded only around the sensor depth.
- *Share of the root zone beyond the trigger* is

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

Provenance (Eq. 7.6): **[Engineered]**: indicator defined for the workbench.

the layer-thickness-weighted fraction of the 0–60 cm profile whose suction exceeds the trigger, evaluated for the irrigated and the rainfed profile of 1994 (the year with full-depth output). * The mean suction of a season, used for the heat strip and for the “driest season” statements, is the geometric mean $m_y = 10^{\frac{1}{183}\sum_d \log_{10} h_d}$ over the growing season; the median across years m_{med} also drives the rainfall year factor γ_y (Eq. 5.15).

7.3.3 Step-by-step

1. Open *Profile evolution* and choose **All 20 years**.
2. Click a year in the heat strip to select it; click a column to put the sensor at that depth.
3. Move the trigger: the share beyond the trigger is recomputed.
4. Choose **1994 detail** to see the full profile through the season; press Play to step through it.

Chapter 8

Soil structure and tillage layers

8.1 Tillage and structure

8.1.1 What it shows

A modelled thin section (“microscope”) of the topsoil for the chosen soil and tillage, at three magnifications (10 mm, 2.5 mm, 0.6 mm), with pores filled by water or air according to a suction slider; the same place under the three tillage systems; the relative pore space against time; and the change in drainage and yield caused by tillage at your settings.

8.1.2 Pore index and consolidation

After tillage the loosened topsoil re-settles. The workbench uses a first-order relaxation of the **relative pore space** p (1 = freshly loosened) to a consolidated value $p_c = 0.30$,

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

with t in days since tillage and τ_c the **Structural consolidation rate** control (Table 3, 0.001–0.02 d^{-1}). **No-till** keeps a constant $p = 0.42$ (undisturbed pores). The half-life of loose pores is $\ln 2/\tau_c$: 693 d at 0.001 d^{-1}, 35 d at 0.02 d^{-1}. **[Engineered]**: the exponential form is the standard first-order consolidation relation also used in structure sub-models such as WEPP’s, on which Daisy’s tillage effect is based, but p_0, p_c and the no-till value are *presentation* constants for the drawing; the *hydrological effect* of tillage shown in the bars is read from the Daisy runs (E5), not from Eq. (8.1).

8.1.3 How the section is generated

The section is a random packing of mineral grains and matrix drawn on a raster, with these parameters, all **[Engineered]**:

- Grain-size classes follow the soil’s measured sand, silt and clay percentages (`DATA.texture`).
- **Saturated water content** $\theta_s = \phi$ and **residual content** θ_r are

$$\phi = \text{clip}\big(0.36 + 0.0016\,\text{clay} + 0.0009\,\text{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\text{clay}, \tag{8.2}$$

Provenance (Eq. 8.2): **[Engineered]**: porosity relation with chosen coefficients.

with clay and silt in percent, and the retention curve is the van Genuchten function of Eq. (5.1),

$$\theta(h) = \theta_r + (\phi - \theta_r)\, S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

- **Cracks and aggregates.** The number of cracks is $n_b = \text{round}(2 + 6(1 - f_{\text{till}}))$ with $f_{\text{till}} = 1,\ 0.7,\ 0.4$ for conventional, reduced, no-till; the crack width (mm) is $0.3\ \max\{0,\ p - 0.3\}^{1.3}$.
- **Which pores hold water.** A pore of radius r fills with water at suction h if $r \le r_c(h)$, where the capillary radius follows the Young–Laplace (Jurin) law,

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\, g\, h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

with surface tension $\sigma = 0.0728$ N m^{-1}, contact angle $\gamma \approx 0$ and $g = 9.81$ m s^{-2} (Jurin, 1718; Hillel, 1998). In the drawing a pore is water-filled when its local half-width is at most r_c **or** when it is within a film thickness $\min\{1.6,\ 0.9\,r_c/s + 0.3\}$ pixels of a grain, which draws the water films that survive drying.

- The hover probe states, for a pore, its radius, the suction at which it empties ($h = 1.49/r$) and the present suction.

8.1.4 Controls and steps

1. Open *Soil structure and tillage* ▸ *Tillage and structure*. Press **Play** to run a year of consolidation (the day appears under the picture).
2. Drag the **suction slider** (pF 0–4.2) on the microscope: at low suction the pores are water-filled; as suction rises the large pores empty first.
3. Switch the view: *Natural* (minerals), *Particle sizes* (sand/silt/clay classes), *Water and air*.
4. Change **Tillage** or the **Structural consolidation rate** and watch the curves of $p(t)$ (Eq. 8.1) and the section change.
5. Read the bars *Change in drainage at the current settings* (conventional, reduced and no-till against no-till) and the yields. These come from the Daisy runs for the selected soil: $\Delta D_{\text{conv}} = D_{\text{conv}} - D_{\text{notill}}$, $\Delta D_{\text{red}} = D_{\text{red}} - D_{\text{notill}}$, at the current τ_c.
6. The sentence at the end states whether the consolidation rate changes anything: *"In this soil the consolidation rate changes nothing; tillage is the driver"* or *"Here the consolidation rate matters as well as tillage"*.

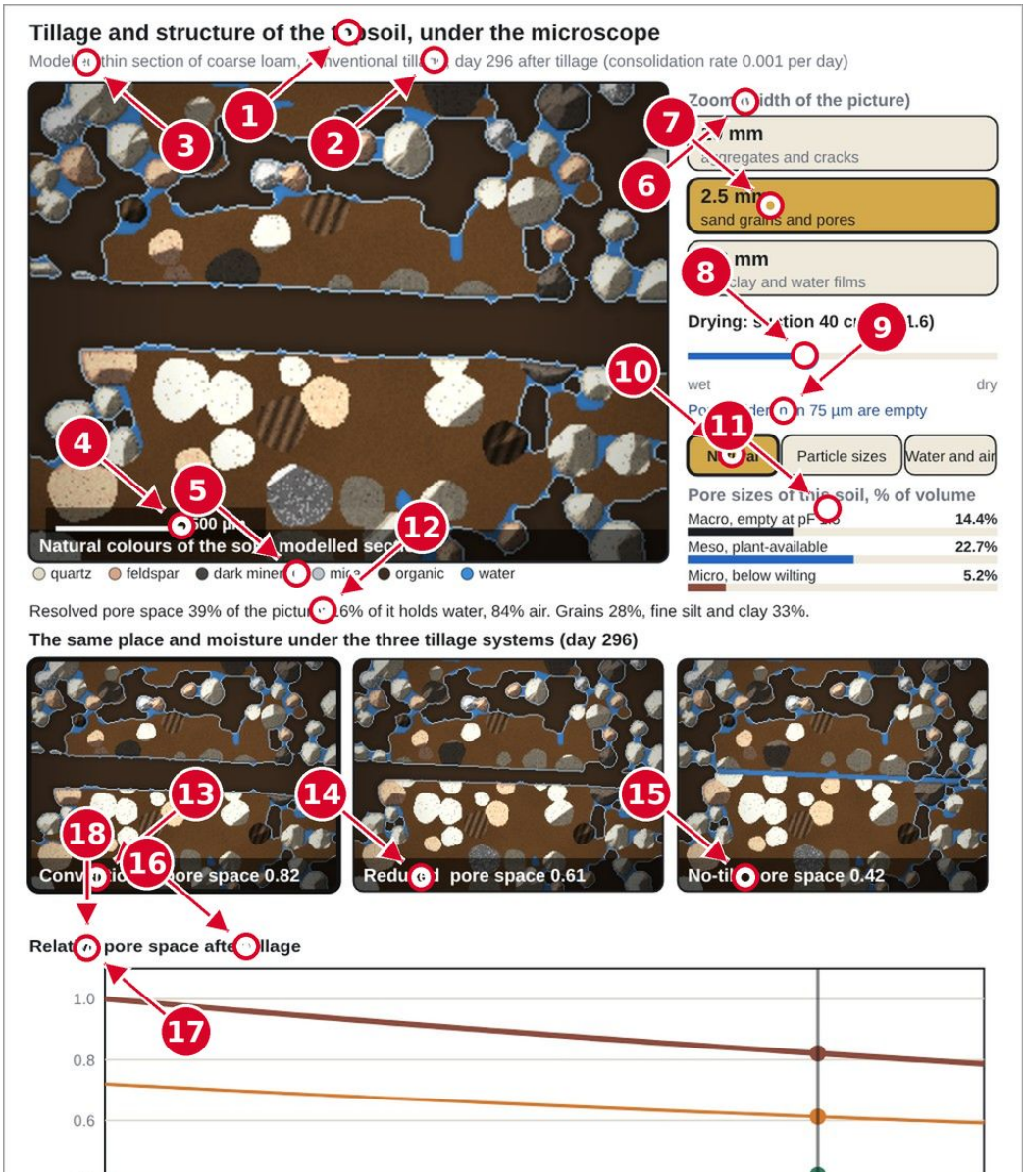


Figure 8.1: Tillage and structure layer: thin section, tillage comparison and the drainage consequences. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

8.2 Soil hydraulic drivers

8.2.1 What it shows

The question *"What drives change in soil hydraulic properties?"* answered for all ten soils together: for each soil sorted by clay content, a blue bar for the **tillage effect** (conventional minus no-till) and an orange bar for the **consolidation effect** (fastest minus slowest rate) on annual drainage, then scatter plots against clay content, and the 1-m soil profiles of the three tillage systems.

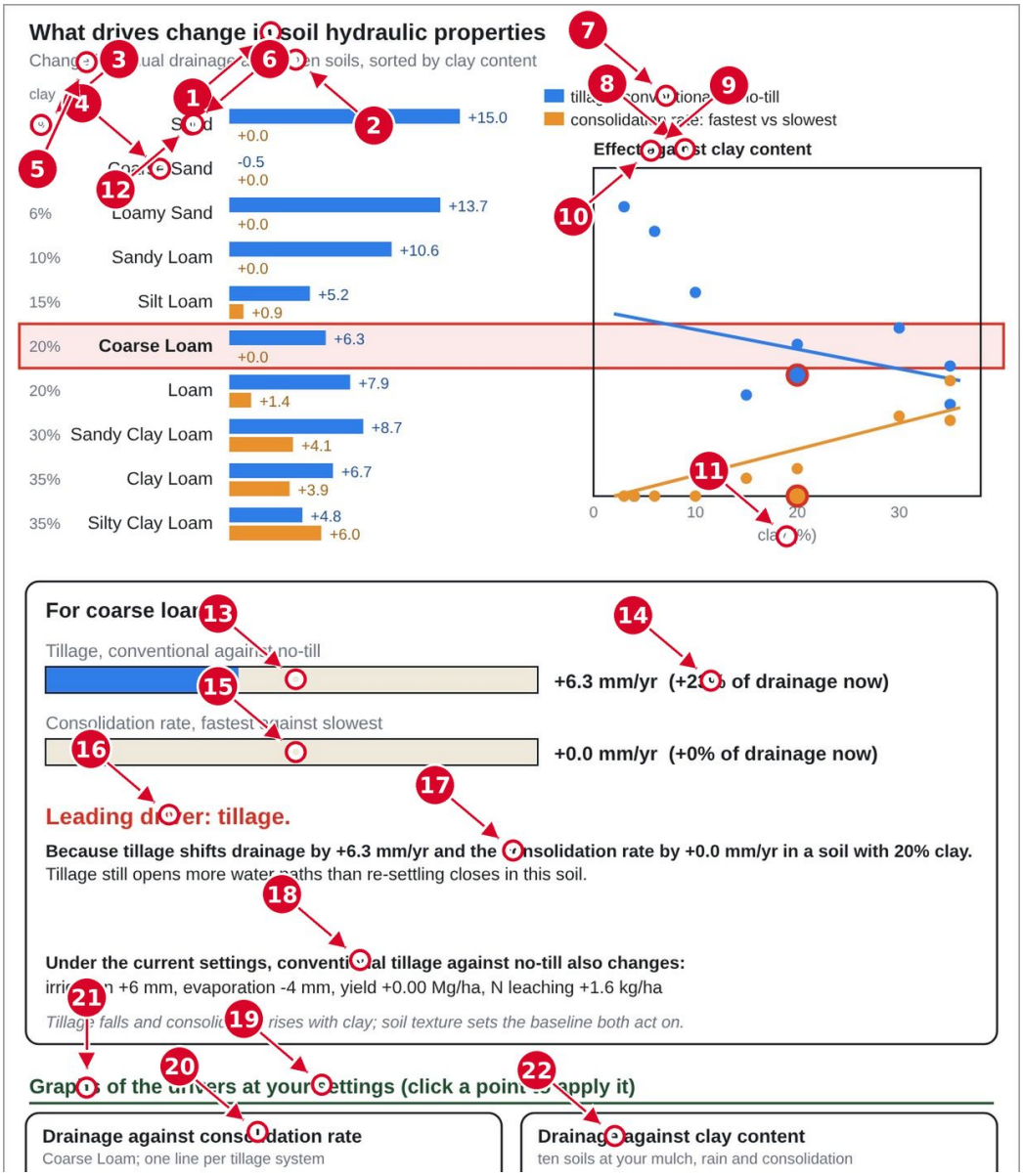


Figure 8.2: Soil hydraulic drivers. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

8.2.2 Generation

For each soil s the Daisy run set E5 gives annual drainage $D_s(\text{tillage}, \tau_c)$. The two bars are

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*), \qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

Provenance (Eq. 8.5): **[Data-driven]**: differences of Daisy run-set E5 results.

with the current settings as starred values; bar length is proportional up to 16 mm yr^{-1}. The lines in the scatter plots are ordinary least-squares fits of each effect on clay content c,

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

Provenance (Eq. 8.6): **[Data-driven]**: ordinary least squares applied to Daisy results.

The headline "Leading driver: tillage" or "consolidation rate" compares Δ_{till} with Δ_{cons} for the selected soil, and the percentage after each bar is $100\,\Delta/D_{\text{now}}$, the share of the present drainage. The last line adds, for conventional against no-till at the current mulch and rain, the changes in irrigation, evaporation, yield and N leaching, obtained by running the meta-model (Eqs. 5.4–5.12) twice (conventional at τ index 2 and no-till) and subtracting.

The lower part of the layer draws, for the three tillage systems, a 48 cm wide, 1 m deep face of the selected soil. The upper 24 cm (the tilled layer) is generated with the pore index p of Eq. (8.1) at day 120 (adjustable), clod and crack density from f_{till}, and the wetness factor $\text{clip}\big((p-0.30)/0.14;\,0.3,\,1.4\big)\cdot\text{clip}(0.75+0.012\,\text{clay};\,0.8,\,1.4)$; below it the subsoil is untouched by tillage. The drainage figures printed under each column are Daisy results (E5).

8.2.3 Steps

1. Open the layer. Click any soil bar or any scatter point to select that soil.
2. Click a point on the *Drainage against consolidation rate* graph to apply a rate and a tillage system at once.
3. Read the verdict, and compare with the profile drawings.

8.3 Root structure

8.3.1 What it shows

Rooting depth in all ten soils at the same day and weather: for each soil a column with the deepest roots (20-year mean, brown dashed), and ticks at the depth that holds 50 % and 90 % of the water taken up (green ticks). Below it, a time view of rooting depth through the 20 years and the daily uptake, and a set of heat maps from the repository's **E15** runs: 1,000 Daisy runs on the Coarse Loam, rooting depth 30–180 cm × ten application amounts × ten irrigation methods, mapped to yield, irrigation, drainage, transpiration, N leached and water-stress days.

8.3.2 Generation

Roots and uptake come from Engine 2 (Eqs. 5.18–5.23). For each soil the engine is run in turn with that soil's parameters (and, in own mode, its own trigger). With per-layer uptake $u_i(d)$ on day d the **depth holding a fraction p of the uptake** is

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k:\ \frac{\sum_{i\le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d\in\text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

Provenance (Eq. 8.7): **[Engineered]**: diagnostic defined for the workbench.

with the layer index counted from 1 (z_p is the *bottom* of layer k_p). It is computed for each of the 20 years from that year's uptake, then averaged over the years; the share of uptake in the top 30 cm is $\sum_{z_i\le 30} U_i \big/ \sum_i U_i$. The model stops roots at 1 m, whereas the Daisy runs reach 120 cm, as the layer notes. The statement "*Across the soils the deepest roots differ by …*" is the range of the 20-year-mean $\max_d R_d$ across soils.

What these roots are. The drawing states it: *"These are modelled roots, not a scan of real ones"*; root density is *relative* (Eq. 5.20), not measured root length. The E15 heat maps are **[Data-driven]** and exist for the Coarse Loam only ("they do not change with the soil you picked").

8.3.3 Steps

1. Open the layer. Click a column to make that soil the working soil.

2. Click or drag on the upper graph to move the day; the lower panel shows the uptake on that day.
3. In the E15 panel choose an output (yield, irrigation, …), click a rooting depth and an application amount in the heat map; the table shows the change against the present setting and tells you the best depth, for example *“Roots at 30 cm leave the crop stressed …”*.

Chapter 9

Mulch and evaporation layers

This group follows the water that mulch keeps in the soil: how mulch lowers evaporation (V4, V6), the physics of drying (E1), where the saved water goes (N2) and when evaporation or drainage dominates (N3).

9.1 Mulch against bare soil

What it shows. Three soil blocks side by side: bare soil, the soil under *your* mulch, and under the strongest mulch (plastic film, $\varphi = 0.1$), with bars for evaporation, irrigation and drainage, the six materials as buttons (click one to apply it), and a verdict.

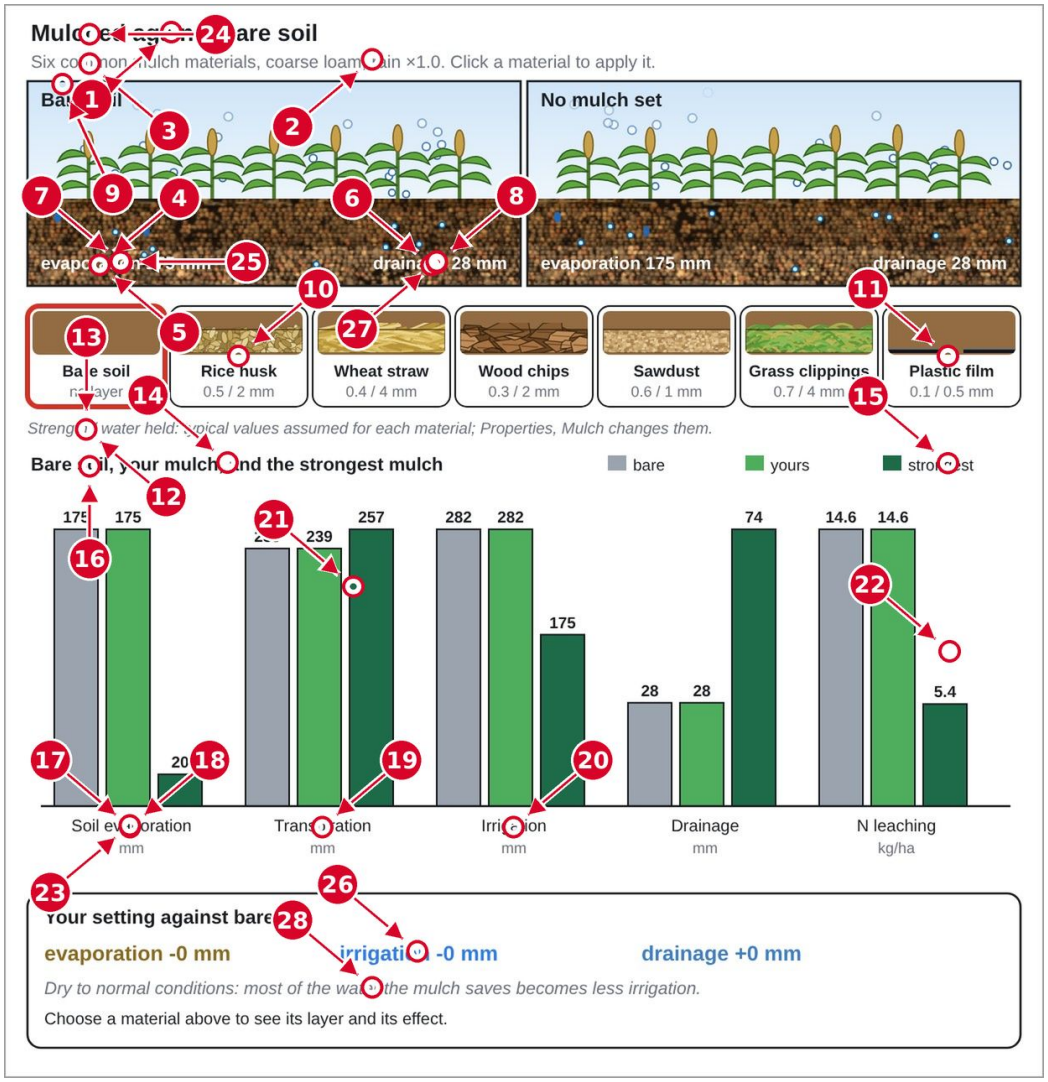


Figure 9.1: Mulch against bare soil. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

How the numbers are generated. All three blocks are Engine 1 evaluated at $\varphi = 1.0$ (bare), at your φ and capacity, and at $\varphi = 0.1$, at the current rainfall. With E_φ, I_φ, D_φ from Eqs. (5.4), (5.7), (5.9),

$$\Delta E = E_{1.0} - E_\varphi, \qquad \Delta I = I_{1.0} - I_\varphi, \qquad \Delta D = D_\varphi - D_{1.0}, \tag{9.1}$$

Provenance (Eq. 9.1): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

are the evaporation saved, the irrigation saved and the extra drainage, in mm yr^{-1}, printed under the blocks. The verdict compares ΔI with ΔD (“*Wet conditions: much of the water the mulch saves now leaves as drainage*” or “*Dry to normal conditions: most of the water the mulch saves becomes less irrigation*”). The block of text for plastic film states that “*plastic film stops vapour almost completely but holds no water, so rain runs to the planting holes*”, which is the capacity of 0.5 mm in Table 3.1.

Steps. (1) Open *Mulch and evaporation* ▸ *Mulch against bare soil*. (2) Click a material. (3) Raise the **Rainfall scale** to $×$1.3 and watch ΔD overtake ΔI. (4) Move the **Strength** slider to see the evaporation saving grow convexly.

9.2 Mulch control

What it shows. Two matrices. The first is **soil evaporation against mulch factor (ten rows, 0.1–1.0) by interception capacity (four columns, 0.5–4 mm)**. The second is **drainage against tillage system (three) by consolidation rate (five)**. Colour darkens with the value; a red outline marks your current setting. The text states what fraction of bare-soil evaporation your mulch removes and whether the soil responds to tillage.

Generation. Each cell of the first matrix is $E(\varphi, c)$ and each cell of the second $D(\text{till}, \tau_c)$, both from Engine 1 at the current soil, rainfall and temperature (Eqs. 5.4 and 5.9). The sentence "*No response here: this soil does not react to tillage or consolidation*" appears when the spread $\max D - \min D$ of the second matrix is below 0.5 mm yr^{-1}. The percentage in the text is $100\,(1 - E_{\varphi,c}/E_{1.0})$.

Steps. Click a cell to apply that mulch factor and capacity or that tillage and rate; the red outline moves; the Dashboard changes at once.

9.3 Evaporation physics

What it shows. A modelled thin section of the topsoil drying after a wetting, with liquid water in the pores (blue), vapour moving up (white), the **drying front** (orange) and, for comparison, the bare-soil curve (grey). To the right: the evaporation rate $e(t)$ and cumulative loss $Q(t)$ for 24 days, split into **stage 1** (the surface stays wet, loss at the atmospheric rate) and **stage 2** (a dry layer grows, loss is limited by the soil).

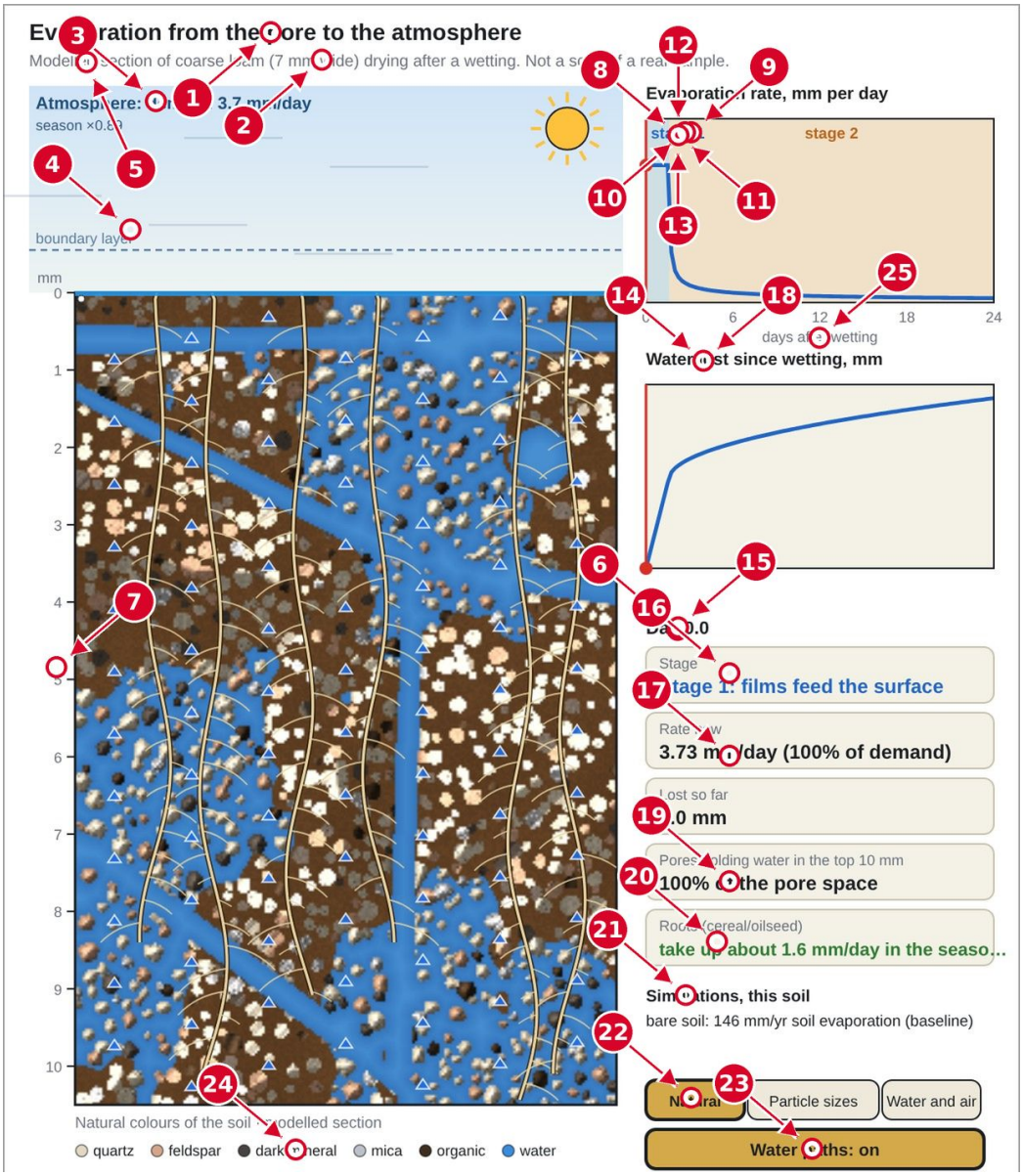


Figure 9.2: Evaporation physics layer. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

Governing relations. The loss is described by the classical two-stage model of soil drying (Philip, 1957; Ritchie, 1972): stage 1 at the atmospheric rate e_0, then stage 2 with a rate falling roughly as $t^{-1/2}$ (cumulative loss $\propto \sqrt{t}$). The implementation in the workbench is

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

Provenance (Eq. 9.2): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

$$Q(t) = q_0 + \begin{cases} e_0\, t, & t \le t_1 \\ (Q_1 - q_0) + 2\, e_0\, \tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

Provenance (Eq. 9.3): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

(the rate is the time derivative of the cumulative loss, so the two are consistent), with the following parameters:

Symbol	Definition	Value or formula
$e_{0,b}$	bare-soil atmospheric demand	Eq. (5.26) (mm d^{-1})
e_0	demand under mulch and canopy	$e_0 = e_{0,b}\,\varphi\,(1 - 0.4\,c)$ with canopy cover c on the date
Q_1	stage-1 storage (mm), water held in the top 40 mm between the wet and the surface-dry state	$Q_1 = \max\{2,\ [\theta(h_i) - \theta(h_s)]\, L_1\, (0.6 + 0.4\rho)\}$, $L_1 = 40$ mm
h_i, h_s	initial and surface-dry suctions (cm)	$h_i = \dfrac{0.25}{\alpha}(1.5 - 0.5\rho)$; $h_s = 4/\alpha$
τ	stage-2 time scale (d)	$\tau = 0.006 + 0.03\,(\text{clay} + \text{silt})/100$
q_0	loss already experienced when the water was placed below the surface (drip)	$q_0 = (1 - w_f)\,0.95\,Q_1$, $w_f = \text{clip}(1 - z_d/50;\ 0.05,\ 1)$, z_d the drip depth in cm; 0 for surface methods
$\Delta\theta$	water lost across the dry layer	$\Delta\theta = \text{clip}(\theta(h_i) - \theta(10^4);\ 0.05,\ 0.4)$

$\theta(h)$ is Eq. (8.3) and α the van Genuchten parameter of Table 5.3; ρ is the rainfall scale. **[Engineered]** — every constant in the table (0.25, 1.5, 0.5, 4, 40 mm, 0.6, 0.4, 0.006, 0.03, 0.95, 50 cm, 0.05–1) is a presentation parameter chosen so that stage 1 lasts a few days in a loam and stage 2 gives the characteristic $\sqrt{t}$ fall; they are **not** calibrated to a laboratory evaporation experiment and the drawing says so ("*the daily rates are a two-stage drying model scaled by your mulch factor*"). The simulated annual evaporation of the workbench (Eq. 5.4) is *Daisy*, not this curve.

The drying front. The depth of the dry layer after stage 1 is

$$\ell(t) = \min\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\}\ \text{mm} \quad (Q > Q_1), \tag{9.4}$$

Provenance (Eq. 9.4): **[Engineered]**: drying-front geometry chosen for drawing.

and the suction profile with depth z is an exponential approach of $\log_{10} h$ with length $\lambda = L_1/3$:

$$\log_{10} h(z) = \log_{10} h_s + \big(\log_{10} h_i - \log_{10} h_s\big)\big(1 - e^{-(z-\ell)/\lambda}\big) \quad (z \ge \ell), \qquad h = 10^5 \text{ cm for } z < \ell. \tag{9.5}$$

Provenance (Eq. 9.5): **[Engineered]**: drying-front geometry chosen for drawing.

Pore picture. Which pores are water-filled at each depth is given by the capillary rule of Eq. (8.4) with the local suction of Eq. (9.5); vapour particles are released from the drying front and rise through air-filled pores. The picture is an **illustration of the mechanism**, not a pore-network simulation.

Steps. (1) Open *Evaporation physics* and press **Play** (24 days). (2) Raise the mulch (lower φ): the grey bare-soil curve stays, your curve lengthens stage 1. (3) Change the soil: stage 1 lasts longest in the sands (largest Q_1, about 3 days for the Sand at the season-mean demand) and is shortest in the silty clay loam (about 1 day), because the window between h_i and h_s scales with $1/\alpha$; in stage 2 the fine soils lose water more slowly (larger τ). (4) Set a drip method: $q_0 > 0$

shortens stage 1 because the surface has less water to lose. (5) Hover over a pore to read its radius and the suction at which it empties.

9.4 Fate of saved water

What it shows. For the soil and mulch you set (a preview of $\varphi = 0.5$ is used if you have not set mulch), four curves against rainfall scale $\times 0.5$–$\times 1.4$: evaporation saved, irrigation saved, extra drainage, extra transpiration. A shaded band marks the rainfall above which drainage exceeds the irrigation saved ("*drainage takes over near* $\times 1.2$"). Below, a bar divides each mm of evaporation saved among less irrigation, more drainage, more transpiration and "storage and other".

Generation. For each rainfall ρ in the ten-value list, with φ^* the mulch factor,

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

Provenance (Eq. 9.6): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

all from Engine 1 at the current soil, tillage and temperature, and the partition at the current rainfall is

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\ f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\ f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\ f_{\text{rest}} = \max\{0,\ 1 - f_I - f_D - f_T\}. \tag{9.7}$$

Provenance (Eq. 9.7): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

The crossover rainfall is the first ρ with $\Delta D \geq \Delta I$ and $\Delta D > 1$ mm. The study reports about 60 % of the saved water going to reduced irrigation in dry years and about 63 % to extra drainage in wet years, with the crossover near $\rho \approx 1.2$ (repository README).

Steps. Choose a material; slide the **Rainfall scale**; the red dashed cursor and the bar move; read the three sentences at the bottom.

9.5 Regime map

What it shows. A map of **mulch factor (rows) against rainfall scale (columns)** coloured by which loss dominates, together with soil columns in which vapour rises (grey) or droplets fall (blue), and a verdict for your settings.

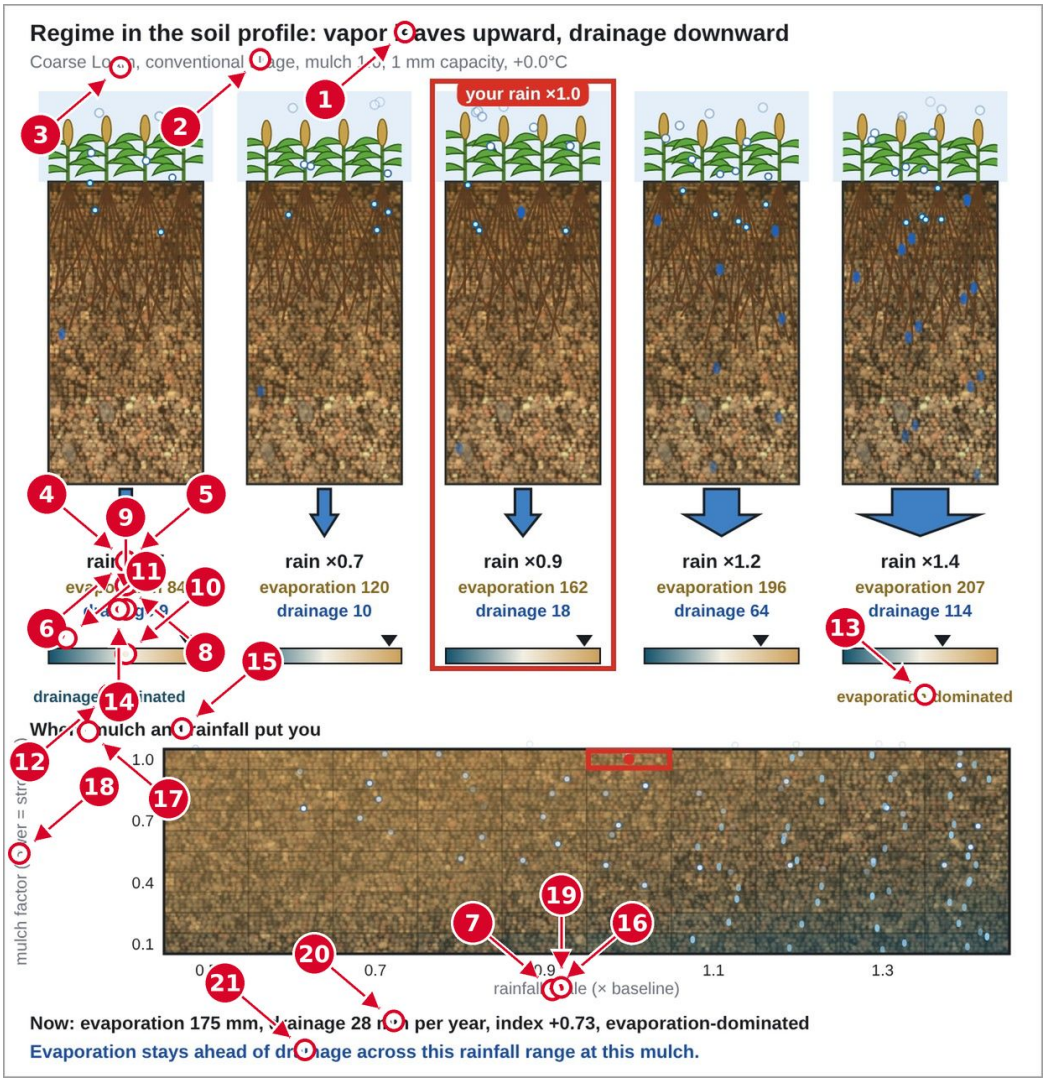


Figure 9.3: Regime map. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

Generation. For each cell the engine takes annual soil evaporation E and drainage D from Engine 1, and forms the **regime index**

$$\mathcal{I} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{I} > +0.05 \\ \text{drainage-dominated} & \mathcal{I} < -0.05 \\ \text{on the boundary} & |\mathcal{I}| \le 0.05 \end{cases} \tag{9.8}$$

Provenance (Eq. 9.8): **[Engineered]**: index defined for the workbench.

so $\mathcal{I} = +1$ is all evaporation and -1 all drainage. The statement "*Climate and mulch act non-additively ... reaching 15 % of irrigation at combined extremes*" is a result of the study (the interaction of mulch and climate on irrigation); it is not recomputed by the layer.

Steps. Click a cell to apply that mulch factor and rainfall; the verdict says *"the main loss is evaporation (...), so mulch will pay off here"* or *"the main loss is drainage ..., so mulch saves less"*.

Chapter 10

Feedback loops

10.1 What it shows

One large picture of a field, from sky to groundwater, on which red arrows join the parts that act on each other: temperature and the crop; rain and soil water; mulch and evaporation; the sensor trigger and irrigation; drainage and nitrogen leaching; nitrogen and the roots; tillage structure and drainage. Each arrow is numbered 1–7, and its **width grows with the size of the effect of one step** of the control behind it. Under the picture a panel lists “*What one step of each control does*”; click a row or an arrow to highlight it.

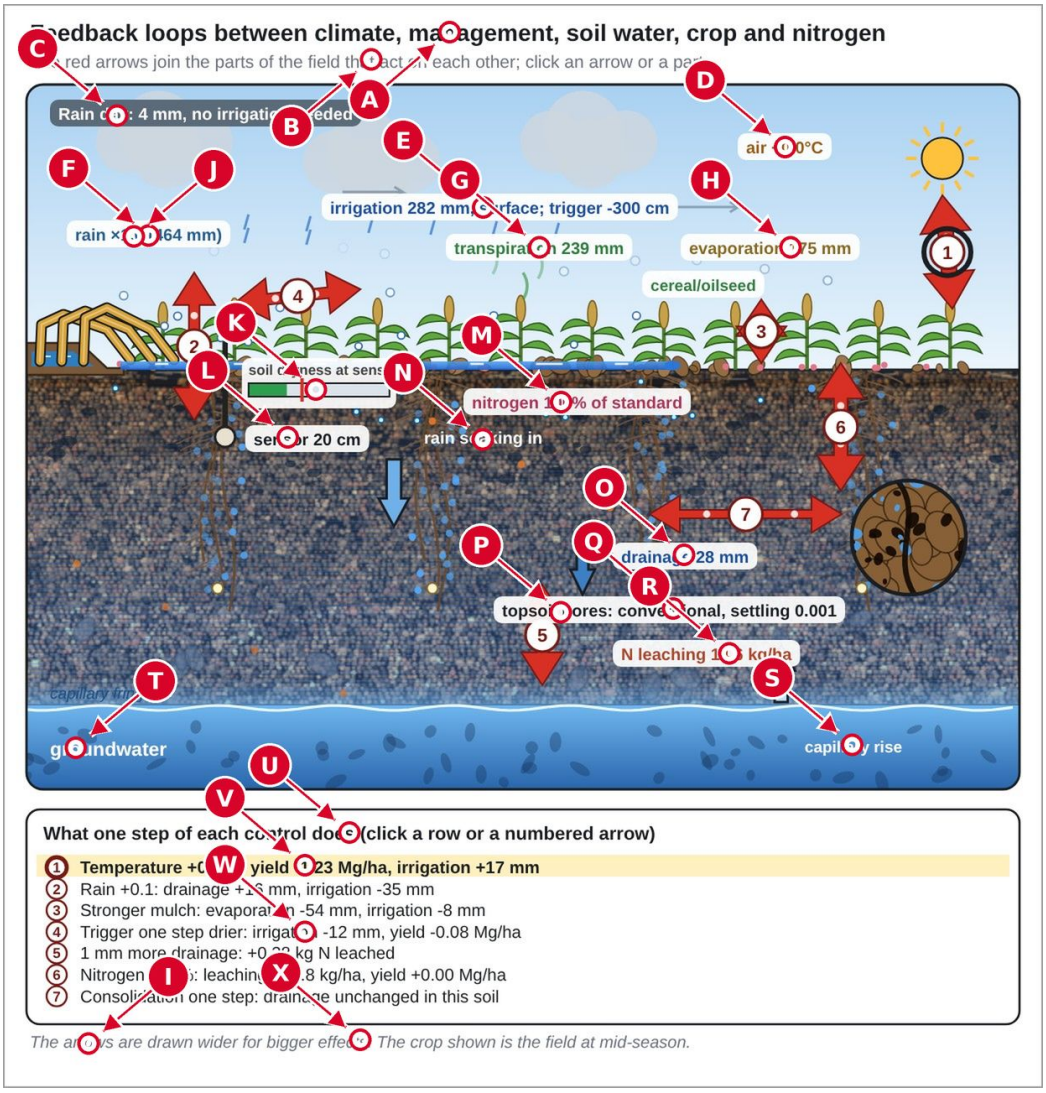


Figure 10.1: Feedback loops. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

The picture is also a set of controls: clicking the clouds cycles the rainfall, the sun the temperature, the mulch layer the material, the sensor the trigger (three grid steps drier each click, wrapping), the soil the tillage, the crop the crop category, the emitter or sprinkler the irrigation method, and the nitrogen band the nitrogen rate.

10.2 How the seven effects are computed

For each control the engine moves **one step** on its list (in the direction that stays inside the list: if the control is at its maximum it steps down) and re-evaluates Engine 1 (Eqs. 5.4–5.12). The change relative to the present result is printed. With m the present result and $m^{(\pm)}$ the result after one step, and f_{crop} the crop factor:

#	Control step	Printed effect	Arrow-width input μ
1	Temperature ± 0.5 °C	$\Delta Y\, f_{\text{crop}}$, ΔI	$8\,\lvert\Delta Y\rvert$
2	Rain ± 0.1 (scale)	ΔD, ΔI	$\lvert\Delta D\rvert / 12$

#	Control step	Printed effect	Arrow-width input μ
3	Stronger mulch (one step lower φ)	ΔE, ΔI	$\lvert\Delta E\rvert/14$
4	Trigger one step drier or wetter	ΔI, $\Delta Y f_{\text{crop}}$	$\lvert\Delta I\rvert/8$
5	One more mm of drainage	$\Delta N_L/\Delta D$ kg N per mm (from the rain step), or "no change" if $\lvert\Delta D\rvert \le 0.5$	$\lvert\Delta D\rvert/10$
6	Nitrogen ± 15 %	ΔN_L, $\Delta Y f_{\text{crop}}$	$\lvert\Delta N_L\rvert/8$
7	Consolidation one step	ΔD (or "unchanged in this soil" if $\lvert\Delta D\rvert < 0.05$)	$\lvert\Delta D\rvert/2$

and the arrow line width in pixels is

$$w_k = 5 + \text{clip}(\mu_k;\, 0,\, 4). \tag{10.1}$$

Provenance (Eq. 10.1): **[Engineered]**: drawing rule.

The sensitivity printed in row 5, the **nitrate cost of drainage**, is the ratio

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D} \quad [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

Provenance (Eq. 10.2): **[Data-driven]**: one-step differences of Daisy results.

which the study reports as roughly 1.06 kg N ha^{-1} per mm of drainage (README). The numbers are **[Data-driven]** one-step differences, not partial derivatives, so a step at the end of a list can be smaller than the step in the middle.

10.3 The sentence under the picture

A status pill reports the day: *"Irrigating now: 28 mm (soil reached the trigger)"*, *"Rain day: 12 mm, no irrigation needed"* or *"Irrigation stopped: soil wet enough"*. The text for the trigger says whether *"the sensor has reached the trigger (depletion f)"*, to what depth *"the water wets the soil"* (the wetting front of the day's irrigation, Section 5.2) and, when that is shallower than the sensor, *"the sensor may stay beyond the trigger"*. If no irrigation is due, it states how many days are free in the next 14 and that rain is needed to keep it that way.

10.4 Steps

1. Open *Feedback loops*. The crop shown is the field at mid-season.
2. Click an arrow (number 4, for example). The matching row is highlighted: a line of the form *"Trigger one step drier: irrigation −X mm, yield −Y Mg/ha"* (the numbers depend on your settings).
3. Click the picture parts to change controls and watch the arrow widths change.
4. Use **Pin baseline** first to compare the 20-year averages for the whole combination.

Chapter 11

Surface irrigation design

The surface-irrigation layer designs a **border strip, furrow or basin** for *your* soil: it derives the soil's intake from texture, solves the advance of the water front, chooses a cut-off time, computes where the water goes, and then searches for a better layout. Everything on this page is a closed-form hydraulic model; no Daisy table is read, except the crop need and the trigger, which come from the DSS (Chapter 5).

The layer is a long scrolling drawing. The sub-section bar above it (*Whole design, Layout design, Simulation, Hydrograph, Advance and opportunity, Results, Design table*) jumps to a section, as do the sub-items of the layer in the Layers panel. The twelve sections are, in order:

#	Section (sub-item)	Purpose
1	Inputs and *Whole design* header	Choose the starting point, the system, intake, cut-off, and the value boxes; then Run simulation
2	Layout in plan: the advance front, shaded by the depth already infiltrated	Plan view of the field with the front
3	Longitudinal section: water running over your soil and soaking in	Side view with Z_0, Z_L, Z_n
4	Hydrograph: inflow, infiltration and runoff through time	Flow partition with time
5	Cross-section at the inlet: the wetted bulb in your soil, and the intake curve	Intake curve $Z(\tau)$ and the soil behind it
6	Advance and opportunity time	$t_x(x)$ and $\tau(x)$
7	Infiltrated depth along the field against the required depth Z_n	$Z(x)$ against Z_n
8	Performance indices and the water balance of one irrigation	E_a, E_s, DP, RO, CU, volumes
9	Design table	Iterations towards $E_a \geq 85\,\%$
10	What the efficiency is most sensitive to	Tornado chart
11	What this means for the season and the farm	Season, sets, hours
12	Irrigation schedule and *Intervals, efficiency and the advice in short*	Dates, gaps, "When to irrigate"

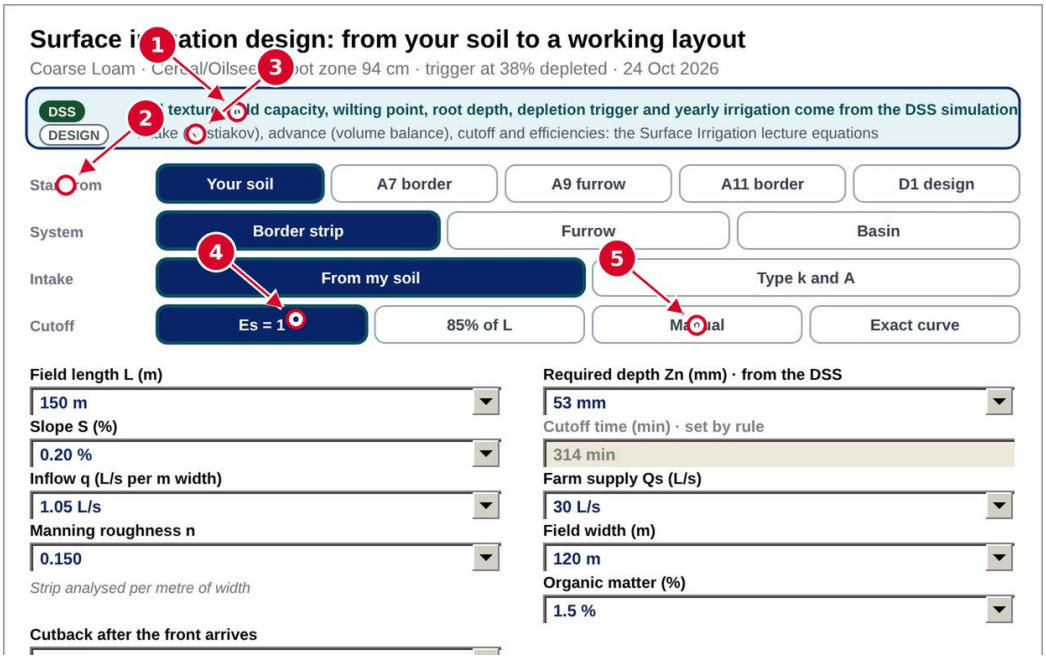


Figure 11.1: Surface irrigation design: the inputs (top) and the layout in plan. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

11.1 Inputs, step by step

1. Open *Irrigation design* ▸ *Surface irrigation design*. The header states the soil, crop, root-zone depth, trigger depletion and date (e.g. *"Coarse Loam · Cereal/Oilseed · root zone 94 cm · trigger at 38 % depleted · 18 Oct 2026"*). The green *DSS* badge lists what is taken from the simulation; the *DESIGN* badge what the design chooses.
2. **Start from** one of five rows:

Chip	Meaning
Your soil	Intake, required depth, cut-off and best inflow are all derived from your soil (default).
A7 border	Lecture worked example: border, $L = 200$ m, $S = 0.20$ %, $q = 2.5$ L s^{-1} m^{-1}, $n = 0.15$, $k = 2$ mm min^{-A}, $A = 0.8$, $Z_n = 90$ mm, given advance $t_L = 120$ min, cut-off 170 min.
A9 furrow	Furrow, $L = 226$ m, $S = 0.30$ %, $q = 2.0$ L s^{-1}, $n = 0.04$, $w = 0.7$ m, $k = 4$, $A = 0.7$, $Z_n = 80$ mm, $\sigma_z = 0.83$.
A11 border	Border, $L = 240$ m, $S = 0.20$ %, $q = 1.667$, $n = 0.15$, $k = 7$, $A = 0.5$, $Z_n = 80$ mm, advance from field data $t_x = x^{1.35} q^{-1.5}/300$ (a power law, Eq. 11.12), cut-off at 85 % of L.
D1 design	Furrow, $L = 240$ m, $S = 0.30$ %, $q = 1.667$, $n = 0.04$, $w = 1$ m, $k = 2$, $A = 0.7$, $Z_n = 80$ mm, approximate advance (Eq. 11.13), $\sigma_z = 0.8154$.

The four worked examples reproduce the course exercises so that the program can be checked against hand calculation; choosing one switches the intake to *Type k and A*.

3. **System**: *Border strip*, *Furrow* or *Basin*. This resets L, S, q, n to the system defaults of Table 11.1.

4. **Intake**: *From my soil* (Green–Ampt from the soil's texture, fitted to Kostiakov, Section 11.2) or *Type k and A* (your own Kostiakov constants).
5. **Cutoff**: *Es = 1* (cut off when the tail has just received Z_n), *85 % of L* (cut off when the front reaches 85 % of the length plus the storage time), *Manual* (set the time), and the toggle *Exact curve* (draw the exact Kostiakov $Z(x)$ instead of the straight-line profile of the lecture notes).
6. Set the value boxes (Table 11.1): click a box, type the exact value or pick it from the list, choose the unit. Then press **Run simulation**; the results below are computed for these inputs (Chapter 18).

Table 11.1. Surface-design inputs (value boxes): ranges (min, max, step) and system defaults.

Control	Border	Furrow	Basin	Notes
Field length L (m)	50–400, 5 (default 150)	60–400, 5 (180)	20–200, 5 (80)	
Slope S (%)	0.02–1, 0.01 (0.20)	0.05–1.5, 0.01 (0.30)	0.02–0.2, 0.01 (0.05)	
Inflow q (L s^{-1})	0.2–20 per m width, 0.05 (6)	0.2–4 per furrow, 0.05 (1.2)	5–60 per m width, 0.5 (25)	In *Your soil* mode q is set automatically (Section 11.9) until you move it
Manning n	0.02–0.25, 0.005 (0.15)	(0.04)	(0.15)	
Furrow spacing w (m)	—	0.5–1.5, 0.05 (0.75 in *Your soil*)	—	$w = 1$ m is used for border and basin (a strip 1 m wide is analysed)
Cutback after the front arrives	0.2–1.0, 0.05 (1 = none)	same	same	Fraction of q after t_L
Required depth Z_n (mm)	20–150, 1	same	same	Default *from the DSS* (Eq. 5.27); tap the label to reset
Farm supply Q_s (L s^{-1})	5–120, 1 (30)	same	same	Farm-level flow
Field width W_f (m)	30–400, 5 (120)	same	same	
Organic matter (%)	0.5–5, 0.1 (1.5)	same	same	Enters Saxton–Rawls
Kostiakov k (mm min^{-A})	0.2–10, 0.05	same	same	Visible in *Type k and A* mode
Kostiakov A	0.2–0.95, 0.01	same	same	Visible in *Type k and A* mode

11.2 Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)

Step 1: soil water properties (Saxton and Rawls, 2006). With sand fraction S, clay fraction C (both 0–1) and organic matter OM (%), the 1500-kPa and 33-kPa water contents are

$$\begin{aligned}
\theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031,\\
\theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02),\\
\theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299,\\
\theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015),
\end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

and saturation and saturated conductivity (mm h^{-1}) follow from

$$\begin{aligned}
\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\
\theta_{(S-33)} &= \theta_{(S-33),t} + \big(0.636\,\theta_{(S-33),t} - 0.107\big), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\
B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.
\end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

These are the published pedotransfer equations of Saxton and Rawls (2006), implemented as written (field capacity $\theta_{fc} = \theta_{33}$, wilting point $\theta_{wp} = \theta_{1500}$). They are regressions on a large US soil database and carry its scatter; the layer uses them for design, not for prediction at a particular site.

Step 2: wetting-front suction. The effective suction at the wetting front (mm) is taken from the van Genuchten pair (α, n) of Table 5.3, $m = 1 - 1/n$,

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

This is a rational-polynomial fit of the Rawls-type wetting-front suction as a function of m; it is an **[Engineered]** approximation (the code comment calls it the "Lisle/Rawls form") and should be read as giving the right order of magnitude (tens of mm for loams, e.g. 16 mm for the Coarse Loam), not a measured ψ_f.

Step 3: available storage in the wetting zone. The soil starts at the trigger depletion f_T and is wetted to 95 % of saturation,

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Step 4: Green–Ampt infiltration (Green and Ampt, 1911). Cumulative infiltration $F(t)$ (mm) solves

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

by Newton iteration from the start value $\max\{K_s t/60,\ \sqrt{2\psi_f \Delta\theta K_s t/60}\}$.

Step 5: Kostiakov fit (Kostiakov, 1932). The design uses the two-parameter empirical intake law

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

and k, A are found by least squares on the log–log plot of $F(\tau)/1000$ at 25 times spaced geometrically between 10 and 600 min:

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

with coefficient of determination $R^2 = \left(\sum (x-\bar{x})(y-\bar{y})\right)^2 \big/ \left(\sum (x-\bar{x})^2 \sum (y-\bar{y})^2\right)$, shown on the layer (e.g. *"Ks 13.1 mm/h · ψf 16 mm · Δθ 0.20 · fit R² 0.997"*). This conversion of a physically based law to a power law is what the lecture method requires; it is an approximation whose quality is reported by R^2.

Step 6: required depth. The net depth to refill the root zone is Eq. (5.27): $Z_n = \text{clip}(f_T\,\text{AWC}\,R_{\text{max}};\,25,\,150)$ mm unless you type another value.

11.3 Inlet flow area (Manning)

The flow cross-section at the inlet, A_0 (m^2), is obtained from Manning's equation (Manning, 1891), in SI units,

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

- **Furrow.** A V-shaped section with side slope $z = 1$ (H:V), depth limit $D = 0.25$ m: $A = zy^2$, $P = 2y\sqrt{1+z^2}$. Eq. (11.8) is solved for the flow depth y by 60 bisection steps for $Q = q$ (m^3 s^{-1}) and the slope S (with $S \geq 10^{-4}$); $A_0 = zy^2$. If $y > D$ the layer warns that the furrow overtops.
- **Border and basin.** A wide sheet: $A \approx y$ per unit width, $R_h \approx y$, so $y_0 = \left(q\,n/\sqrt{S}\right)^{0.6}$ with q in m^2 s^{-1}, and $A_0 = y_0$ (m^2 per m width).
- **D1 preset.** $A_0 = \left(q^2 n^2/792\right)^{0.37313}$ with q in m^3 min^{-1}, the empirical furrow relation of the lecture's worked example. **[Engineered, course-specific]**

11.4 Advance (volume balance)

Definition. For an inflow Q (m^3 min^{-1} per analysed width) the position $x(t)$ of the front is obtained from a volume balance of the water on the surface and in the soil (Walker and Skogerboe, 1987; Elliott and Walker, 1982):

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^A\,w\,x \;\Longrightarrow\; x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^A\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

The code also accepts an optional steady-intake term, $Z = k\tau^A + f_0\tau$ (Kostiakov–Lewis), in which Eq. (11.9) gains the term $f_0\,t\,w/(1+r)$ in the denominator, with r the iterated advance exponent (release 1.1; release 1.0 used a fixed $r = 0.55$); the interface does not expose f_0 (it is zero), and it exists so that the FAO Irrigation Paper 45 examples, which use it, can be reproduced. Here $\sigma_y = 0.8$ is the surface-storage shape factor, w the width analysed (furrow spacing, or 1 m for borders and basins), and σ_z the subsurface-storage shape factor

$$\sigma_z = \frac{A + r\,(1-A) + 1}{(1+A)(1+r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

where r is the exponent of the power-law advance $x = p\,t^{r}$, found by iterating (10 times, starting from $\sigma_z = 0.8$) from the times t_L and $t_{L/2}$ at which the front reaches L and $L/2$,

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

with each time found by bisection on Eq. (11.9). Eq. (11.10) is the standard shape factor of the volume-balance method for the Kostiakov intake law (Elliott and Walker, 1982). The advance time to position x is then $t_x(x) = x^{-1}(t)$, solved on a 300-point grid.

Alternatives (set by the worked examples; the first two are Eq. 11.12):

- *Given advance:* $t_x(x) = t_L\,(x/L)^{1/r}$ with $r = 0.55$ unless set (A7).
- *Field advance* (A11): $t_x(x) = c\,q^{-q_e}\,x^{x_e}$ with $c = 1/300$, $q_e = 1.5$, $x_e = 1.35$.

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes. * *Approximate* (D1, lecture formula):

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

with t_V the volume-balance solution of Eq. (11.9). **[Engineered]**: this is the closed-form approximation of the course notes, not a published general law.

11.5 Cut-off, opportunity time and the infiltrated depth

The **time to infiltrate the required depth** follows from inverting Eq. (11.6),

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

The **cut-off time** t_{co} is

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

A set lasts at most 48 hours. When the inflow is too small for the soil (a sand with a small inflow), the advance can need days or even years. The inflow then stops at $t_{co} = 2880$ min, and the water front has reached $x_{\max}$ (from Eq. 11.9 at that time). The part of the run beyond $x_{\max}$ gets no water, and the deficit shows it. The design says so and suggest raising the inflow or shortening the run. *Provenance:* **[Engineered]**: a practical limit on the length of a set.

The **opportunity time** at distance x is the time the water stays on the soil, $\tau(x) = t_{co} - t_x(x)$ for $t_{co} > t_x$, and the **infiltrated depth** is

$$Z(x) = k\,\big[t_{co} - t_x(x)\big]^{A}, \quad Z_0 = k\,t_{co}^{A}\ \text{(inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^{A}\ \text{(tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

The *straight-line profile* of the lecture notes replaces $Z(x)$ by the line between Z_0 and Z_L (and, for the 85 %-of-L rule, a broken line through $(0.85L,\, Z_n)$); *Exact curve* draws Eq. (11.17). If $t_{co} < t_L$ the front never reaches the end of the field and the layer shows a warning (*the stream does not reach the end*).

11.6 Volumes, efficiencies and uniformity

With w the analysed width, $N = 300$ grid intervals (90 while dragging) and the trapezoidal rule, for $Z_i = Z(x_i)$:

$$\begin{aligned}
&V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b < 1:\ Q\,t_L + c_b\,Q\,(t_{co} - t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w,\\
&V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx,\\
&V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx,\\
&V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0).
\end{aligned} \tag{11.18}$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

The performance indices (Walker and Skogerboe, 1987; Burt et al., 1997) are

$$E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N}|Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

E_a is the **application efficiency**, E_s the **storage (requirement) efficiency**, and CU the uniformity coefficient of Christiansen (1942) evaluated on the infiltrated depths. Because $V_{\text{app}} = V_{\text{inf}} + V_{\text{ro}}$ and $V_{\text{inf}} = V_{\text{st}} + V_{\text{dp}}$, the three fractions satisfy $E_a + \text{DP} + \text{RO} = 1$, which the layer prints as the closed water balance of one irrigation. The deficit V_{df} is the missing part of Z_n. The distance at which Z crosses Z_n, if it does, is $X = L\,(Z_0 - Z_n)/(Z_0 - Z_L)$.

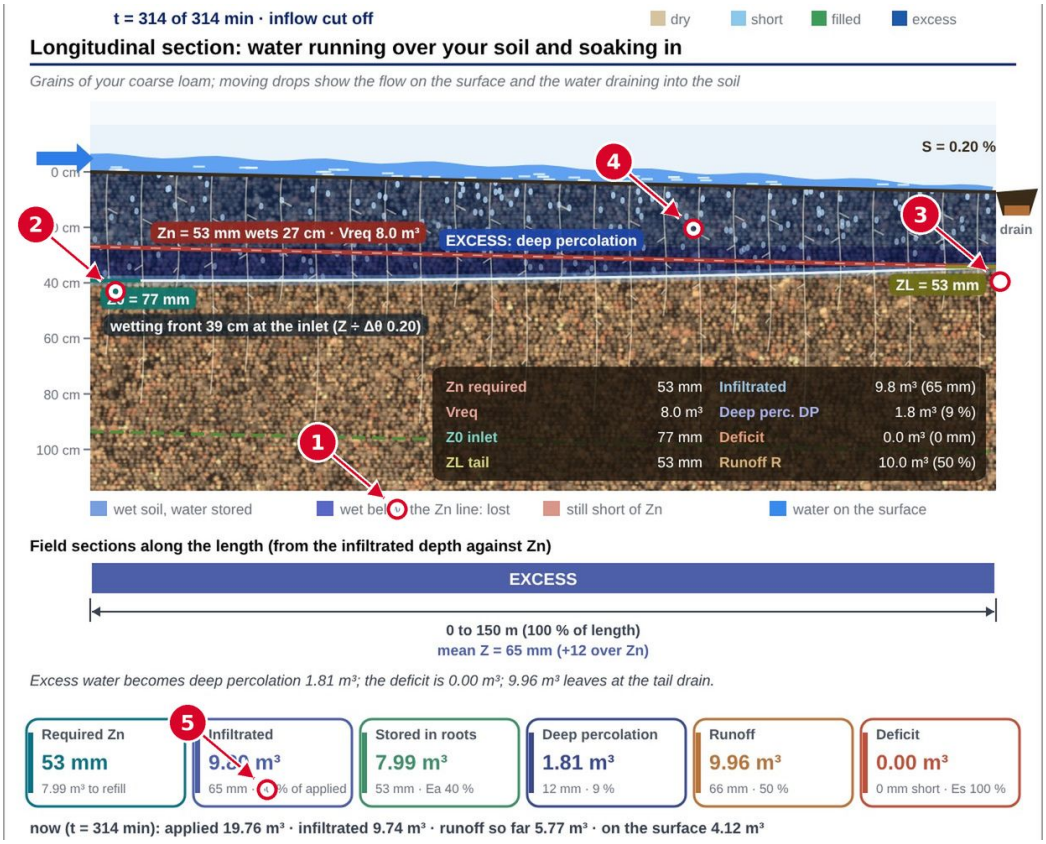


Figure 11.2: Surface design: longitudinal section with Z_0, Z_L, Z_n and the water balance of one irrigation. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

11.7 Hydrograph

The hydrograph (Section 4 of the layer) divides the inflow q into infiltration $I(t)$, runoff $R(t)$ and surface storage. It is built from the same advance and intake:

$$V_c(t) = w \sum_i k \left[\min(t, t_{co}) - \bar{t}_{x,i}\right]_+^A \Delta x \cdot s_c, \qquad s_c = \frac{V_{\inf}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

Provenance (Eq. 11.20): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

where $\bar{t}_{x,i}$ is the advance time at the mid-point of interval i and the factor s_c makes the cumulative infiltration at cut-off equal to $V_{\inf}$ of Eq. (11.18). While $t_L \le t \le t_{co}$ the runoff is $R(t) = \max\{0,\ q - I(t)\}$. After cut-off the surface water *recedes*: with $V_s = \max\{0,\ V_{\text{app}} - V_{\inf} - \int R\,dt\}$ left on the surface, a triangular recession over $t_r = \max\{4,\ \min(0.3\,t_{co},\ 0.25\max(t_L, 10))\}$ min drains it at the tail,

$$R(t) = \frac{2V_s}{t_r}\left(1 - \frac{t - t_{co}}{t_r}\right), \quad t_{co} < t < t_{co} + t_r. \tag{11.21}$$

Provenance (Eq. 11.21): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

The recession shape is **[Engineered]** (the course does not specify it); it conserves the volume V_s. Surface storage at time t is $V_{\text{sto}}(t) = q\,\min(t, t_{co}) - V_c(t) - \int_0^t R\,dt'$.

11.8 Cross-section and intake curve

The *Cross-section at the inlet* section draws, at the inlet, the wetted shape in a soil section generated from the soil's texture, and the intake curve $Z(\tau)$ of Eq. (11.6) beside the Green–Ampt curve $F(\tau)$ of Eq. (11.5), so that the fit can be inspected by eye.

11.9 Best inflow ("Your soil" mode)

In *Your soil* mode, q is chosen automatically from 25 equally spaced values between $q_{\min}$ and $q_{\max}$ of the system as the one that maximises $E_a\,E_s$. For furrows the upper limit is the non-erosive stream

$$q_{\max} = \min\Big\{ q_{\max}^{\text{sys}},\ \frac{0.04}{S}\cdot\frac{1000}{60} \Big\}\ \text{L s}^{-1} \quad (S \text{ in \%}), \tag{11.22}$$

Provenance (Eq. 11.22): **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

about $0.67/S$ L s^{-1}, in line with the classical Criddle et al. (1956) limit $Q_{\max} \approx 0.63/S$. The choice is stored in the inflow box, which becomes manual as soon as you change it.

11.10 The design table

The *Design table* automates one step at a time towards a good design. The **target** is $E_a \geq 85\,\%$ with $E_s \geq 80\,\%$ (the root zone stays at least 80 % refilled). The score of a design is

$$\text{score} = \min(E_a, 1) - 3\,\max(0,\,0.8 - E_s) - 3\,[c_b < 1]\,\max(0,\,0.05 - \text{RO}) - 5\,\max(0,\,E_a - 1). \tag{11.23}$$

Provenance (Eq. 11.23): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

The search tries, in this order, ten single-variable moves — **cut-off, inflow, cutback, length, cut-off, cutback, inflow, length, cut-off, cutback** — and, for each move, evaluates a candidate list and keeps the best candidate **only if the score improves by more than 0.003**:

Move	Candidates
Cut-off	$t_{co} \times$ $\{0.5, 0.58, 0.66, 0.74, 0.8, 0.85, 0.9, 0.95, 1, 1.05, 1.1, 1.2\}$
Inflow	12 values spaced geometrically from $q_{\min}$ to $q_{\max}$
Cutback	$c_b \in \{1, 0.85, 0.7, 0.6, 0.5, 0.4, 0.3, 0.25, 0.2\}$
Length	11 values spaced geometrically from $\max(L_{\min}, 30)$ to $\min(L_{\max}, 1.5L)$, rounded to 5 m

For every candidate other than a pure cut-off move, six cut-off times are also tried (0.62 to 1.02 times the "$E_s = 1$" cut-off). The search stops when the target is met or after the ten moves. Each accepted row lists q, L, cutback, cut-off, E_a, E_s, DP and RO; **tap a row (green when it meets the target) to apply it** to the sliders. If the target cannot be reached the layer says so and suggests a shorter field, tailwater reuse, or drip or sprinklers. The same search is run by the toolbox item *Optimise surface layout (best design)*.

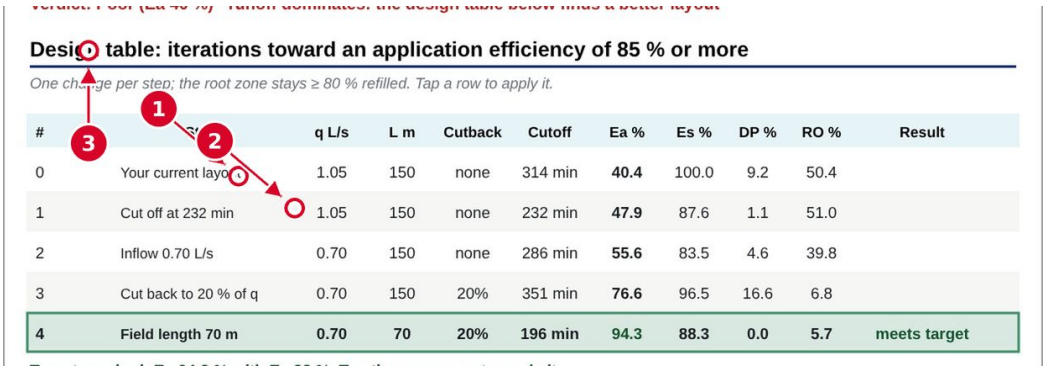


Figure 11.3: Surface design table: iterations toward $E_a \geq 85\,\%$, then the tornado chart. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

11.11 Sensitivity (tornado chart)

Each of seven inputs is moved up and down while the others are held, and the change in E_a is drawn as a bar (green raises E_a, red lowers it):

Input	Intake k	Intake A	Inflow q	Length L	Required depth Z_n	Roughness n	Slope S
Perturbation	± 20 %	± 10 %	± 20 %	± 20 %	± 20 %	± 30 %	± 30 %

The bars are sorted by the larger absolute change, and the printed value is $\max(|\Delta E_a^-|, |\Delta E_a^+|)$ in percentage points (*pts*). The test is local: it is the change around the *present* layout, not a global importance.

11.12 Season and farm

Using the irrigation record of Engine 2 for the year of the working date, with $\sum I$ the net irrigation of that year (mm):

$$n_{\text{ev}} = \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g,$$
$$N_{0M} = \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\left(1, \frac{N_{\text{units}}}{N_{0M}}\right), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}), \tag{11.24}$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

and the seasonal pumping time is $n_{\text{ev}} T_{\text{tot}}$ hours. A depth of Z_{season} mm equals $10\, Z_{\text{season}}$ m^3 per hectare. N_{0M} is the number of furrows or strips that can run at once from the farm supply Q_s.

11.13 Irrigation schedule and “When to irrigate”

The last two sections are common to all three design layers, so they are described once, here.

Schedule table. The next eight irrigation dates after the working date are listed from the DSS (Engine 2, Section 5.2): the sensor in layer j_s reaching the trigger depletion f_T. For each date the table gives

Column	Definition
#	order, soonest first
Date	calendar date of the analogue year (Chapter 4)
In	*today*, *tomorrow*, or *n days*
Net mm	the DSS irrigation depth that day, I_k
Pumped mm	gross depth I_k/E_a with the design's application efficiency (min. $E_a = 0.05$)
Run time	time of one set or one round of the design
Gap to next	days to the following irrigation

Intervals. Over the year of the working date, with irrigation days $d_1 < d_2 < \ldots$ (irrigations above 0.5 mm), the gaps $g_j = d_{j+1} - d_j$ give the mean, the shortest and the longest interval.

Advice message. The message *When to irrigate* is generated from these numbers: the *next irrigation date* and its net and pumped depth; *"Then irrigate about every $\bar{g}$ days (between $g_{\min}$ and $g_{\max}$ days, shorter in hot weather and longer after rain)"*; the efficiencies of the design (E_a, E_s) and the time of one round; and the rule *"If a rain above 10 mm is forecast, delay the irrigation by a day or two"*. If no irrigation is triggered, it says to *keep the sensor at z_s cm and irrigate when it reaches depletion f_T*.

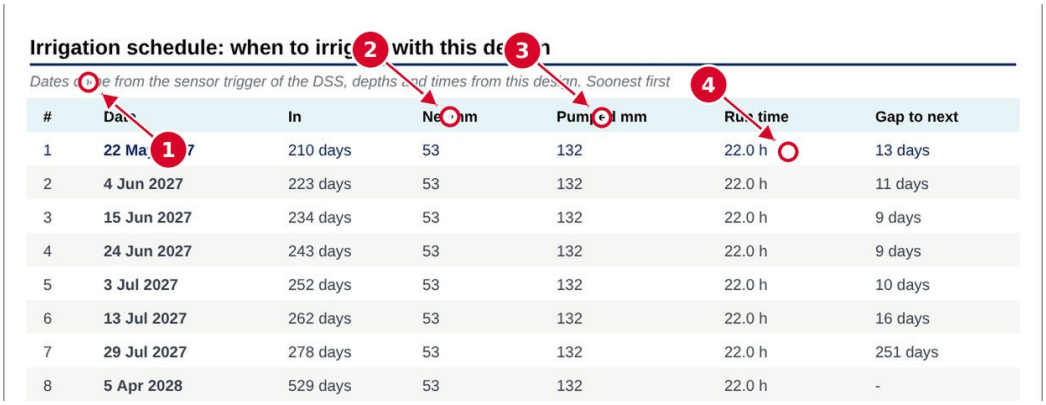


Figure 11.4: The irrigation schedule and the "When to irrigate" message at the end of the surface design. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

Reading it. The first row is today's decision. If the *Gap to next* is short (2–4 days) the trigger is wet or the soil store is small; if it is long (10+ days) the soil store is large or the weather cool. Move the **Date** to see how the schedule changes with the season, and move the **Trigger** or **Sensor depth** to see the dates shift.

Chapter 12

Drip irrigation design

The drip layer designs the **emitter, its placement, the lateral and the pipe network** for the soil, crop and trigger set in the rest of the program. It answers four questions in order: how big a wet bulb does one emitter make, how much of the water it delivers stays in the root zone, how uneven is the delivery along the lateral, and what pump and pipes does the farm need. Everything is a closed-form hydraulic or geometric model, recomputed at every slider move.

The layer's sections, in order: control bar and inputs; soil moisture around the driest, average and wettest emitter (Run simulation animates the wetting); where the water goes; uniformity along the lateral; growth of the wetted bulb with time; result tiles; layout and infrastructure; irrigation schedule; the design table; the sensitivity chart; season and farm; and the "When to irrigate" advice. The last two are shared with the other design layers and are described at the end of Chapter 11.

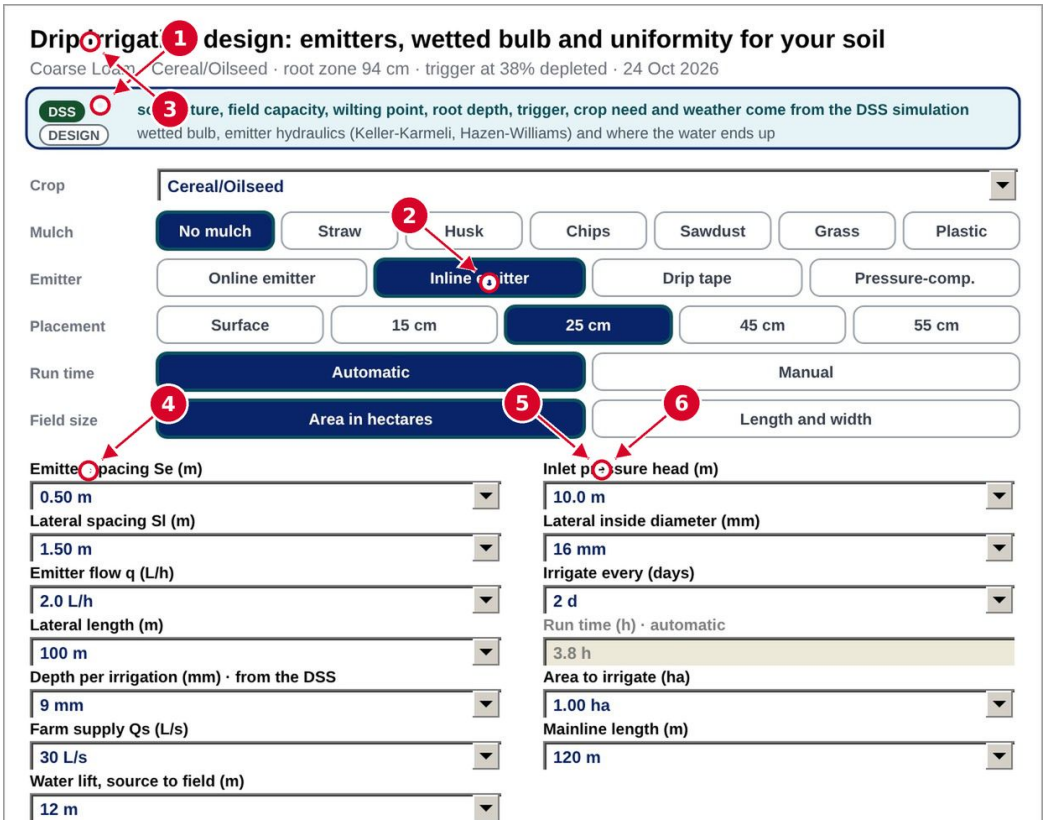


Figure 12.1: Drip design: inputs and the soil moisture around the driest, average and wettest emitter. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

12.1 Inputs, step by step

1. Open *Irrigation design* ▸ *Drip irrigation design*. The crop, soil, root zone, trigger and date come from the DSS (the green badge); the choices below are the design (the amber badge).
2. **Emitter** (a row of four chips). Choosing one sets the nominal flow, spacing, lateral diameter and inlet head of Table 12.1.
3. **Placement**: *Surface* (design only), or a buried line at 15, 25, 45 or 55 cm. A buried choice also sets the explorer's irrigation method to the matching Daisy run (M04, M06, M09 or M10), so that the rest of the program shows the same depth.
4. **Run time**: *Automatic* (time to deliver the depth at the driest emitters, Eq. 12.17) or *Manual*.
5. Set the value boxes of Table 12.2, then press **Run simulation**. The crop, mulch and irrigation method rows at the top are the same controls as the Properties panel.

Table 12.1. Emitter types (defaults are set when the chip is chosen; c_v manufacturer coefficient of variation, x flow exponent, n_e emitters per plant or point, $H_{\min}$ compensation threshold).

Emitter	q (L h^{-1})	S_e (m)	D (mm)	H_0 (m)	c_v	x	n_e	$H_{\min}$ (m)
Online emitter	4	0.75	16	10	0.07	0.5	1	—
Inline emitter	2	0.50	16	10	0.04	0.5	1	—
Drip tape	1	0.20	16	10	0.08	0.6	2	—
Pressure-compensating	2	0.60	16	15	0.03	0.05	1	8

Table 12.2. Drip inputs (value boxes) (range, step, default).

Control	Range, step	Default	Meaning
Emitter spacing S_e (m)	0.1–1.5, 0.05	by emitter	Along the lateral
Lateral spacing S_l (m)	0.5–3, 0.05	1.5 (1.0 for cereal, legume; 1.5 for vegetable, reset when the crop changes)	Between laterals
Emitter flow q (L h^{-1})	0.5–8, 0.1	by emitter	Nominal flow at H_0
Lateral length L_L (m)	20–300, 5	100	
Inlet pressure head H_0 (m)	4–30, 0.5	10 (15 for PC)	
Lateral inside diameter D (mm)	10–32, 1	16	
Irrigate every (days)	1–10, 1	2	Sets the default depth
Run time (h)	0.25–48, 0.25	automatic	Greyed while *Automatic*
Depth per irrigation Z_n (mm)	3–80, 1	from the DSS	Tap the label to reset
Area to irrigate A_f (ha)	0.2–10, 0.1	1	
Farm supply Q_s (L s^{-1})	2–120, 1	30	
Mainline length L_m (m)	20–600, 10	120	
Water lift, source to field (m)	0–60, 1	12	

Depth per irrigation. By default the depth is the crop need over the interval, capped by what the root zone can hold at the trigger,

$$Z_n = \min\!\left\{ Z_n^{\mathrm{DSS}},\ \max\!\left(4,\ \mathrm{ET}_{\mathrm{pk}}\,\Delta t\right)\right\}\ \text{mm}, \tag{12.1}$$

Provenance (Eq. 12.1): **[Engineered]**: depth and demand rules chosen for the workbench.

where Δt is the chosen interval, $Z_n^{\mathrm{DSS}} = \mathrm{clip}(f_T\,\mathrm{AWC}\,R_{\mathrm{max}}; 25, 150)$ is Eq. (5.27), and $\mathrm{ET}_{\mathrm{pk}}$ is the peak five-day moving mean of simulated evaporation plus transpiration over the year, bounded to 2.5–9 mm d^{-1},

$$\mathrm{ET}_{\mathrm{pk}} = \mathrm{clip}\!\left\{ \max_{d} \tfrac{1}{5} \sum_{j=0}^{4} (E_{d+j} + T_{d+j});\ 2.5,\ 9 \right\}. \tag{12.2}$$

Provenance (Eq. 12.2): **[Engineered]**: depth and demand rules chosen for the workbench.

The mean evaporative demand used for surface drying, E_0, is the mean of the simulated daily evaporation on days 141–250 of the year (when positive), clipped to 1–5 mm d^{-1}.

12.2 Lateral hydraulics: head loss and flow along the lateral

An emitter lateral of length L_L carries $N = \mathrm{round}(L_L/S_e)$ intervals and $N+1$ emitters, indexed $j = 0 \ldots N$ from the inlet. The friction head loss in the segment feeding emitter j is the Hazen–Williams equation (Hazen and Williams, 1905), in SI form with a roughness coefficient $C = 140$ for plastic pipe,

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \ \ (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

The head at emitter j is $h_j = h_{j-1} - h_{f,j-1}$ with $h_0 = H_0$ (floored at 0.5 m). Emitter discharge depends on the local head through the emitter equation (Keller and Karmeli, 1974),

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad \text{(non-compensating; } h_j/H_0 \geq 0.01\text{)}, \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.

with $x = 0.5$ for a fully turbulent orifice or long-path emitter and $x = 0.6$ for the drip tape of Table 12.1. The pressure-compensating (PC) emitter is modelled as

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \geq H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.

i.e. a nearly flat discharge above $H_{\min} = 8$ m (a 4 % change in flow per 100 % change in head) and the orifice law below it. The flows depend on the heads and the heads on the flows, so the pair (12.3)–(12.4) is **iterated to convergence** from the uniform start $q_j = q$, $h_j = H_0$ (the first three passes are plain substitution, then the flows are averaged with the previous pass; the iteration stops when the largest relative change in any emitter flow is below 10^{-7}, at most 500 passes). Version 1 used four passes, which is sufficient only when the head loss is small; the converged solver agrees with an independent solution to 3×10^{-7} for head losses up to 60 % of the inlet head. When the friction loss exceeds 20 % of the inlet head, or the iteration fails to converge, the layer prints a red warning, because uniformity then falls and the design is outside the usual rule of thumb. The mean and minimum discharge are then

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600}\ \ (\mathrm{L\,s^{-1}}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.

and the loss along the whole lateral is $h_f = H_0 - h_N$, with $H_{\text{end}} = h_N$ the head at the far end (Section “Uniformity along the lateral”).

Emission uniformity. The design emission uniformity follows Keller and Karmeli (1974),

$$E_u = \text{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

The first factor is the manufacturing variation; the second is the hydraulic variation along the lateral (here $q_{\min}$ replaces the classical quarter-lowest mean, a conservative choice). The *combined* relative standard deviation of the delivery, which drives the spread of bulbs below, is

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Reading the lateral plot: the solid line is q_j/q in per cent against distance, the dashed line is the head as a percentage of H_0. A lateral that is too long or too narrow shows a steep fall of both. For a non-compensating emitter a 10 % loss of head costs about 5 % of flow ($x = 0.5$); the PC emitter stays flat until the head drops below $H_{\min}$.

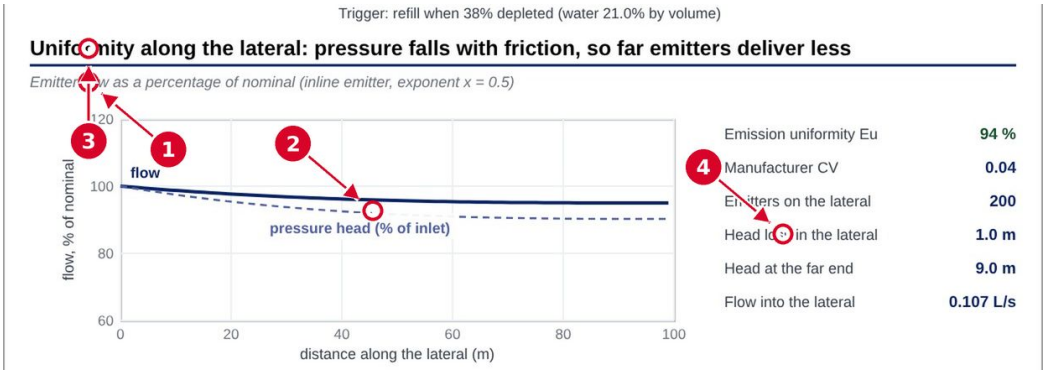


Figure 12.2: Drip design: uniformity along the lateral and the growth of the wetted bulb with run time. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

12.3 The wetted bulb

Water from an emitter spreads into soil that is initially at the trigger water content and is refilled up to field capacity. The layer models the wetted volume around a *buried* point source as a **half-ellipsoid** below the emitter and a short one above it (an **[Engineered]** simplification of the wetted-volume shapes used in drip design; the aspect ratio is derived from the soil, not read from a table).

Water to be stored per unit soil volume.

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Shape. With emitter depth z_d (mm), horizontal radius r (mm) at the emitter level, lower semi-axis $a\,r$ and upper semi-axis $u\,r$ with $u = 0.6$, the wetted radius at depth z is

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Aspect ratio. The shape factor a (depth over radius) increases when gravity is strong relative to capillarity, i.e. for a coarse soil and a low flow. It is evaluated from the ratio of the saturated conductivity to the radial capillary spreading rate $q/(2\pi\lambda^2)$ (the form is a scaling argument and the constants are **[Engineered]**),

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

with K_s in mm h^{-1}, q in L h^{-1} and ψ_f the wetting-front suction of Eq. (11.3) in mm. The result is a bulb that is round ($a = 1$) for a fine soil and elongated downwards (up to $a = 1.9$) for a sand.

Radius. For an emitter that delivers V litres, the wetted volume is $V_w = V \cdot 10^6/\Delta\theta$ mm^3 (1 L $= 10^6$ mm^3). The radius r is found by bisection (38 steps between 5 and 3500 mm) so that the volume of revolution of Eq. (12.10), integrated in 56 layers,

$$V_w = \int \pi\, \rho(z)^2\, dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

equals V_w. When the volume is *line-like*, the cross-section instead of the volume is conserved: for drip tape ($S_e < 0.34$ m) or when neighbouring bulbs overlap ($2r > S_e$ with $S_e < 1.2$ m), the water is spread along the line, $A_w = V \cdot 10^6/(S_e \cdot 1000)/\Delta\theta$ mm^2 per unit length of lateral and the profile area is $2\int \rho\, dz$. The panel is then labelled "line (bulbs merge)" rather than "point". The wetted share of the soil surface is

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\, S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Where the water goes. Of the V litres, the part that falls inside the bulb below the root-zone depth $R_{\max}$ is lost to deep percolation, a fraction B_{below}/V_w of V:

$$V_{\text{dp}}^{(1)} = V\, \frac{\int_{z>R_{\max}} \pi\rho^2\, dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

and the part evaporated from the wet surface patch is limited to half the delivery,

$$V_{\text{ev}} = \min\{0.5\, V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\text{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

(for a line source the area is $2\rho(0)S_e \cdot 10^{-6}$ m^2 per emitter). The factor 2 corresponds to about two days of evaporation from a wet surface patch of mean demand E_0; this is an **[Engineered]** simplification of the two-stage evaporation of Chapter 9. For a surface line ($z_d = 0$) the bulb is half-spherical and the evaporating patch is largest, which is why surface drip stores less than buried drip in the table. The remainder is available to the roots, $V_{\text{root}} = V - V_{\text{dp}}^{(1)} - V_{\text{ev}}$.

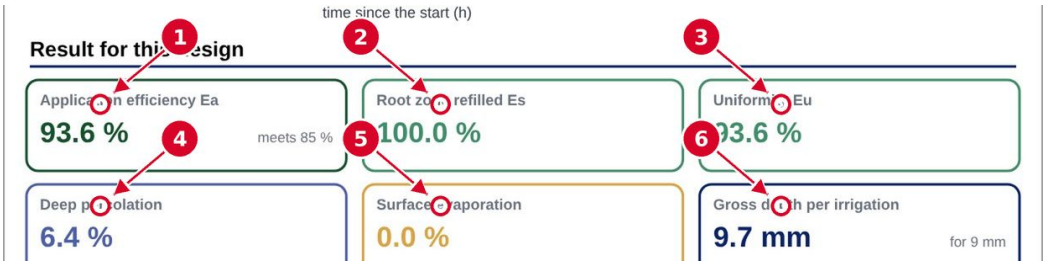


Figure 12.3: Drip design: result tiles and the water partition of one irrigation. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

12.4 Spread between emitters and the efficiencies

Emitters do not all deliver the same volume. The layer evaluates five representative emitters, at standard-normal quantiles $z \in \{-1.28, -0.52, 0, 0.52, 1.28\}$ (the 10th, 30th, 50th, 70th and 90th

percentiles, equal weights), each delivering

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1+\sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

where t is the run time (h). For each, the bulb, deep percolation and evaporation are computed with Eqs. (12.10)–(12.15). The water that the root zone can accept is limited by the need,

$$V_{\text{req}} = Z_n\,S_e\,S_l\ \ (\text{L}), \qquad V_{\text{st},i} = \min\{V_{\text{root},i},\ V_{\text{req}}\}, \qquad V_{\text{ex},i} = V_{\text{root},i} - V_{\text{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Because 1 mm over 1 m^2 is 1 L, V_{req} is the depth needed over the area $S_e S_l$ served by one emitter. **Automatic run time** is the time at which the *driest* emitters deliver the need,

$$t = \text{clip}\Big\{\frac{V_{\text{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \text{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

The efficiencies are means over the five quantiles ($\langle\cdot\rangle$):

$$E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}}, \quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}}\rangle}{\langle V\rangle}, \quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle}, \quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

so that $E_a + \text{DP} + \text{EV} = 1$. Reading them: E_a is how much of the pumped water ends up stored in the root zone; E_s is how well the root zone is refilled (a value below 85 % turns the "Root zone refilled" line red and says *short*); E_u is the lateral uniformity; Z_{gross} is the depth to pump per irrigation. The layer colours $E_a \geq 85$ % green, 65–85 % amber and below 65 % red; $E_u \geq 90$ % is green and 80–90 % amber.

12.5 Layout and infrastructure

The farm side of the design uses the area A_f and the supply Q_s.

Laterals and zones.

$$W_f = \frac{10^4 A_f}{L_L}, \quad N_L = \Big\lfloor \frac{W_f}{S_l} \Big\rfloor, \quad n_{\text{zone}} = \Big\lceil \frac{N_L\,Q_{\text{lat}}}{Q_s} \Big\rceil, \quad n_l = \Big\lceil \frac{N_L}{n_{\text{zone}}} \Big\rceil, \quad Q_z = n_l\,Q_{\text{lat}}, \tag{12.20}$$

Provenance (Eq. 12.20): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

where W_f is the width of the field perpendicular to the laterals, n_{zone} the number of zones that must run in turn so that Q_s is not exceeded, and Q_z (L s^{-1}) the flow of one zone. The total number of emitters is $N_L(\text{round}(L_L/S_e)+1)$.

Pipe sizing. Submain length $L_{sm} = n_l S_l$, mainline length L_m (value box). The pipe diameter is the smallest of the commercial series

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\} \text{ mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

for which the mean velocity $v = 4Q_z/(\pi D^2) \le 1.5$ m s^{-1} (a common design limit that keeps friction and water-hammer moderate; see Keller and Bliesner, 1990, for pipe design practice). The submain loss is 0.4 of the full-length Hazen–Williams loss, Eq. (12.3) with $Q_j = Q_z$ (the Christiansen (1942) reduction factor F for a pipe with many equally spaced outlets is $F \to 1/2.852 = 0.35$ for $m = 1.852$, so 0.4 is slightly conservative), and the mainline loss is the full value:

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852}D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852}D_m^{4.871}}. \tag{12.22}$$

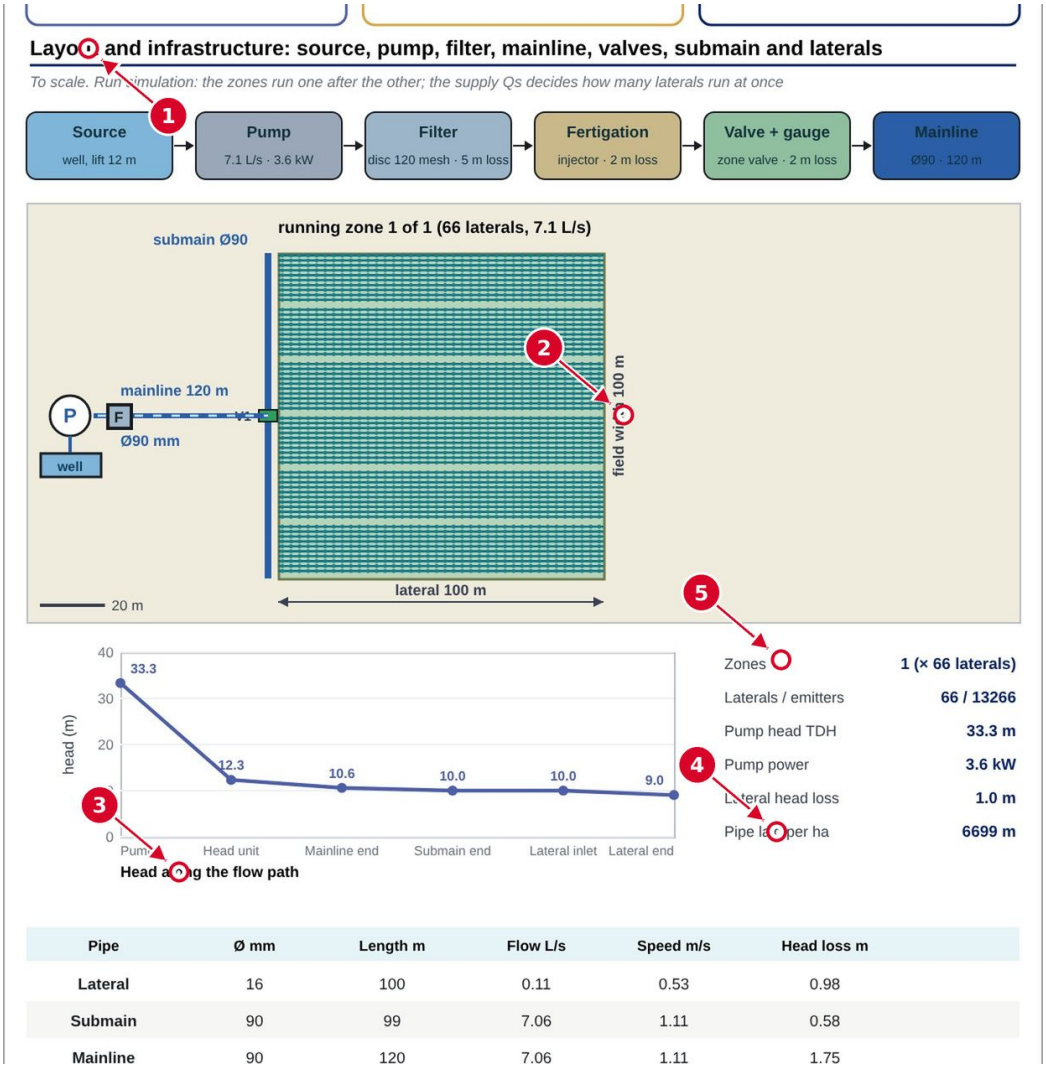
Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Pump. The total dynamic head and the shaft power are

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\text{ m}, \qquad P = \frac{\rho g\, Q_z\,\text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

The constant 9 m covers the filter, valves and fittings, and $\eta = 0.65$ is the overall pump-and-motor efficiency; both are **[Engineered]** design allowances, not measured values. The laterals' own loss is already inside H_0.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Figure 12.4: Drip design: layout and infrastructure, to scale, with the zones running in turn. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

The drawing shows the source, pump, filter, mainline, valves, submain and laterals to scale; during the run the zones run one after the other, and the head along the flow path is drawn underneath. The table beside it gives zones (× laterals per zone), laterals and emitters, TDH, pump power, lateral head loss and pipe laid per hectare, $(N_L L_L + L_{sm})/A_f$.

12.6 Season and farm

With y_I the net seasonal irrigation of the DSS for the year of the working date, E_u and E_a as above,

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

The layer prints these as a table (net irrigation need and number of events, gross water per irrigation and per season in mm and $\text{m}^3\ \text{ha}^{-1}$, peak crop need, flow per hectare, sets for A_f with Q_s, hours of pumping and energy). Then the *Irrigation schedule* and the *When to irrigate* advice, described at the end of Chapter 11, close the layer.

12.7 The design table

The optimiser takes the current design and tries one change at a time, in this order: **placement depth, run time, emitter flow, emitter spacing, placement depth, run time, emitter kind, lateral spacing, emitter flow, run time**. For each move the candidates are:

Move	Candidates
Placement depth	Surface, 15, 25, 45, 55 cm
Run time	$t_0 \times \{0.55, 0.65, 0.75, 0.85, 0.95, 1, 1.1, 1.25\}$, with t_0 the automatic run time of the current design
Emitter flow q	0.6, 1, 1.5, 2, 3, 4, 6, 8 L h^{-1} (run time reset to automatic)
Emitter spacing S_e	0.2, 0.3, 0.4, 0.5, 0.6, 0.75, 1, 1.25 m
Emitter kind	the four emitters of Table 12.1, each with its own default q, S_e
Lateral spacing S_l	0.75, 1, 1.25, 1.5, 1.8, 2.2, 2.6 m

The score is

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

so a design is penalised when the root zone is not refilled to 90 %. A move is accepted when its best candidate improves the score by more than 0.002; the search stops when $E_a \geq 85$ % **and** $E_s \geq 85$ %, or after ten moves. One move is performed per screen update, and the search pauses while a slider is being dragged. Each accepted row shows the step, q, S_e, depth, run time, E_a, E_s and DP; a green row meets the target, and **tapping any row applies that design** to the sliders (including the placement depth and the explorer's irrigation method). If the target is not met, the layer reports the best found and suggests a shorter interval or a different emitter.

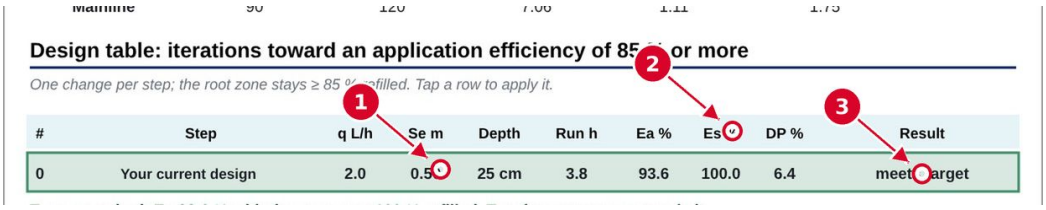


Figure 12.5: Drip design table and the sensitivity chart. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

12.8 Sensitivity

As for the surface layer, each input is moved up and down and the change of E_a is shown as a bar: emitter flow q (± 25 %), emitter spacing (± 25 %), lateral spacing (± 25 %), placement depth (± 30 %, at least 10 cm), depth per irrigation (± 25 %) and run time (± 20 %). The result is a local sensitivity at the present design.

12.9 How to use the layer

- **To choose an emitter for a soil.** Select the soil in the Properties panel, open the layer and compare the four chips with the same run time: the tile of E_a and the line chart of the bulb show which emitter keeps the bulb inside the root zone.
- **To check a lateral length.** Increase L_L until E_u turns amber; the lateral plot shows where the head falls below the pressure at which the emitter holds its flow.
- **To size a pump.** Fix A_f, Q_s, L_m and the lift; read TDH and kW from the layout table.
- **To see the effect of the trigger.** Move *Trigger* in the Properties panel: f_T changes $\Delta\theta$ (Eq. 12.9) and Z_n (Eq. 12.1), and the bulb and the run time adjust.

Chapter 13

Sprinkler irrigation design

The sprinkler layer designs the **nozzle, pressure, spacing and layout** of a sprinkler system for the soil, crop, slope and wind of the case, and then checks the result against the standard course rules (spacing, application rate, uniformity, pipes and pump). It has two parts that are built on the same run: a *simulation* part (the overlapped spray pattern, the wind drift, the soil intake that causes runoff, and the efficiencies) and a *textbook* part (the ten design checks, the pipe and pump sizing, the kinds of sprinkler system).

The layer reads top to bottom: **A** site, **B** sprinkler and layout, **C** operation and **D** farm and pipes (the inputs); the layout in plan; the profile along the lateral; the intake against the application rate; result tiles and the water partition; the textbook checks and system plans; layout and infrastructure; the irrigation schedule; the design table with the sensitivity chart; the season and the advice.

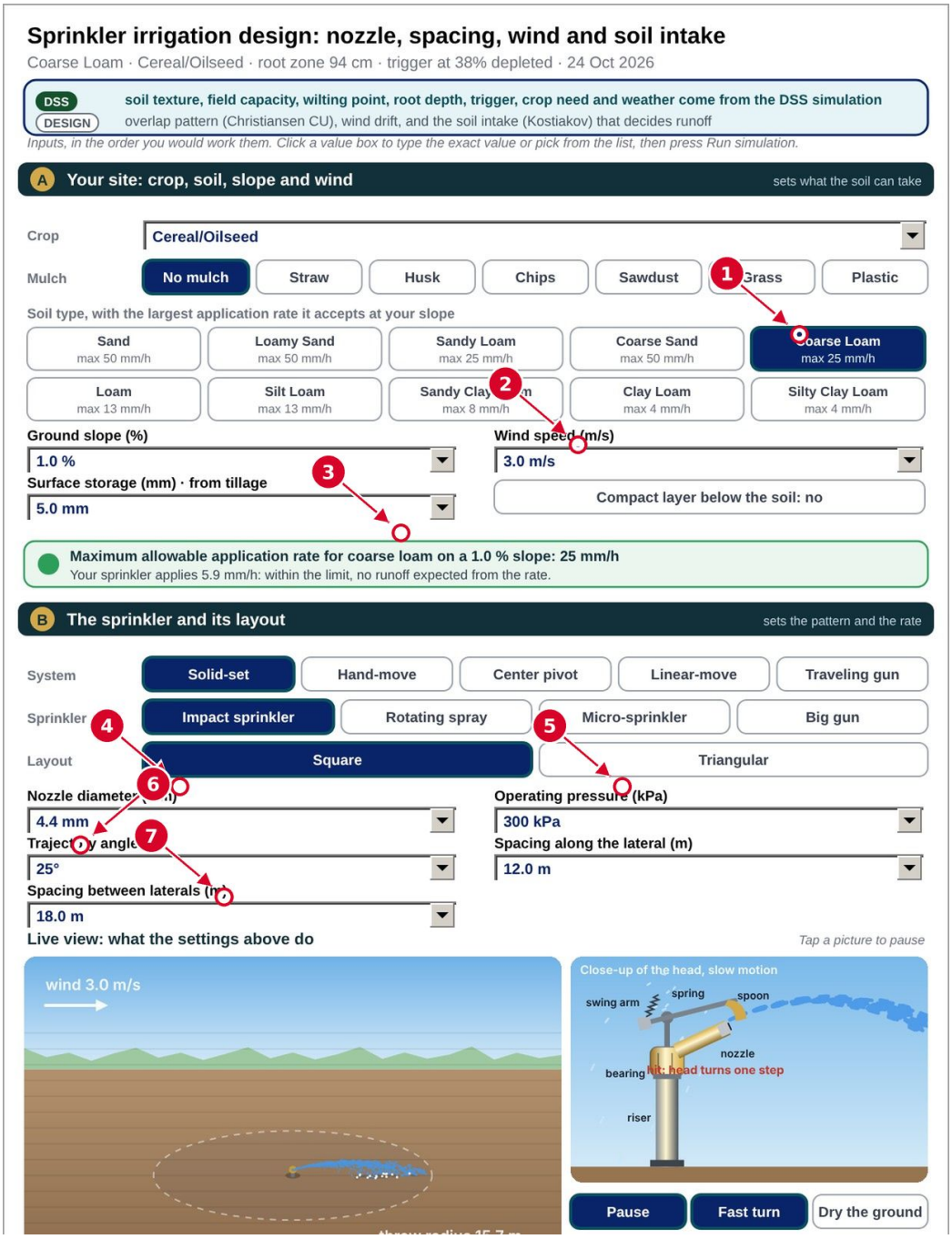


Figure 13.1: Sprinkler design: inputs A to D and the layout in plan with the overlapping spray patterns. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

13.1 Inputs, step by step

A. Your site. Crop, mulch and irrigation method are the same controls as in the Properties panel. The soil row shows each soil with the largest application rate it accepts at the present slope (Table 13.3). Sliders: ground slope S (0–16 %, step 0.5, default 1), wind speed u (0–12 m s^{-1}, step 0.5, default 3), surface storage S_d (0–20 mm, step 0.5; default from tillage, below) and a toggle *Compact layer below the soil* (which lowers the allowed rate, Table 13.3). A banner states

the maximum allowable application rate for the soil and slope and whether the present sprinkler is within it.

B. The sprinkler and its layout. *System* (five chips, see *Kinds of sprinkler system* below), *Sprinkler* (four types, Table 13.1), *Layout* (square or triangular), then value boxes for the nozzle diameter d, operating pressure P, trajectory angle α (8–45°, default 25°), spacing along the lateral S_s and between laterals S_l. Choosing a type resets d, P, S_s, S_l to its defaults and run time to automatic.

Table 13.1. Sprinkler types: ranges (min–max, step) and defaults; c_R range coefficient, m profile exponent, F drift factor. The impact value $c_R = 1.35$ is the Keller and Bliesner (1990) coefficient for impact sprinklers (changed from 1.15 in version 1; the earlier value under-predicted the wetted diameter of their worked example by 14 %). The values for the other three types are **[Engineered]** and have not been verified against published data.

Type	d (mm)	P (kPa)	S_s (m)	S_l (m)	c_R	Profile	F
Impact	2.4–8, 0.1 (4.4)	200–500, 10 (300)	6–30, 1 (12)	6–36, 1 (18)	1.35	$1 - x^{1.3}$	1.0
Rotating spray	1.5–5, 0.1 (3.2)	70–250, 5 (170)	3–15, 0.5 (6)	3–15, 0.5 (6)	0.80	ring	1.6
Micro-sprinkler	0.8–2.5, 0.05 (1.3)	100–250, 5 (180)	1.5–8, 0.25 (3)	1.5–8, 0.25 (3)	0.70	ring-2	0.7
Big gun	10–25, 0.5 (16)	350–700, 10 (500)	20–80, 1 (30)	20–90, 1 (36)	0.95	$1 - x$	1.3

C. How you run it. *Run time* automatic or manual; depth per irrigation Z_n (5–100 mm; default Eq. 5.27, tap to reset); run time T (10–1200 min, step 5); operating hours per day (4–24, default 16).

D. Farm, water supply and pipes. Area A_f (0.5–40 ha, step 0.5, default 5), farm supply Q_s (5–200 L s^{-1}, default 40), lateral length L_L (40–500 m, step 10, default 150), mainline length L_m (20–600 m, default 120) and lift (0–60 m, default 12).

Tillage and surface storage. Unless set by hand, the depression storage S_d follows the tillage of the Properties panel: 5 mm conventional, 3.5 mm reduced, 2 mm none. On a slope the effective storage is reduced,

$$S_d^{\text{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

Provenance (Eq. 13.1): **[Engineered]**: depression storage reduction with slope.

This is an **[Engineered]** relation: storage in depressions decreases as the slope rises (compare the slope dependence of depression storage in Hillel, 1998).

13.2 Nozzle discharge and wetted radius

For a nozzle of diameter d (mm) at operating pressure P (kPa) the pressure head is $h = P/9.81$ m. The discharge follows the orifice equation (Torricelli) with a discharge coefficient of 0.95,

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^4 \;\; (\text{L min}^{-1}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2 \;\text{m}^2, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

and the wetted radius is (an empirical range law of the form used in sprinkler manuals, Keller and Bliesner, 1990, with the coefficient c_R of Table 13.1 chosen per type)

$$R = c_R \sqrt{d\,h}\; f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

with d in mm, h in m and α in degrees (3.1α as an angle in degrees). The angle function peaks near $\alpha = 29^\circ$ and equals 1 at the default 25° ($\sin 77.5^\circ = 0.9763$). **[Engineered]**: the exponent 0.45 and the 3.1α term are chosen to give a mild optimum, not fitted to a data set. The reach R is printed in the plan as a percentage of the spacing along the lateral.

13.3 The spray pattern and the overlap

Radial profile. A single sprinkler deposits water at radius r with a shape $\hat{p}(x)$, $x = r/R_e$, taken from Table 13.1:

$$\hat{p}(x) = \begin{cases} 1 - x^m, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^6), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^5), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

These are **[Engineered]** shapes that reproduce the qualitative patterns published for each class: a single sprinkler with a rapidly falling profile (impact, gun), and a pattern that peaks away from the centre (spray and micro types).

Wind. Wind elongates the pattern downwind and shifts it:

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

where θ is the angle from the wind direction and u is the wind speed in m s^{-1}. The pattern is computed on a 45×45 grid over $\pm 1.4R$ and normalised so that its volume equals the discharge, $\iint p\,dA = q$, with p in mm min^{-1} (1 mm over 1 m^2 is 1 L).

Overlap. The application rate at a point of the repeating cell is the sum over neighbouring sprinklers (square grid $S_s \times S_l$; triangular layout: every second lateral shifted by $S_s/2$, repeating cell $S_s \times 2S_l$),

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\text{mm min}^{-1}), \qquad \bar{a} = \frac{1}{n}\sum_k a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

evaluated on a 14×14 grid of the cell (14×24 for the triangular layout). The mean *gross* application rate is AR $= 60\bar{a}$ mm h^{-1} and the **Christiansen coefficient of uniformity** (Christiansen, 1942) is

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

A CU ≥ 85 % is the usual requirement for a sprinkler system (Burt et al., 1997; Keller and Bliesner, 1990); the tile is green from 84 %. The *Layout in plan* shows the circles of reach, the wind arrow and the depth infiltrated so far (Play animates the time): green is the depth needed, red-brown is short, blue is more than needed, and brown hatching is water running off.

13.4 Wind drift and evaporation loss

The fraction of the water lost between nozzle and soil is

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2+0.5u+0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}\big[1+0.025\,(T_a-20)\big]\Big[0.6+0.4\,\frac{\sin\alpha}{0.4226}\Big];\;0.5,\;45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

where F is the type factor of Table 13.1, P is in kPa, u in m s^{-1} and the air temperature $T_a = 24\,^{\circ}\mathrm{C} + \Delta T$, with ΔT the warming of the Properties panel. The form follows the known trends — loss rises with wind, with pressure (finer droplets), with temperature, and falls with a flatter trajectory — in line with the ranges of 5–15 % usually quoted in textbooks (Keller and Bliesner, 1990); the numerical coefficients are **[Engineered]**. w_d is clipped to 0.5–45 %.

13.5 The soil intake, ponding and runoff

At each cell the water balance is computed with the Kostiakov intake of the soil (Eq. 11.6, from the texture; $K = k \times 1000$ mm min^{-A}). The applied rate after drift is $a_k(1-w_d)$ and for a run of T min the supply is $\mathrm{sup} = a\,T$. Ponding begins when the application rate exceeds the intake rate, which the layer takes as $a = Kt^{A-1}$, i.e. at

$$t_p = \Big(\frac{a}{K}\Big)^{1/(A-1)}, \qquad F = \begin{cases} \mathrm{sup}, & T \le t_p \\ K\,T^A, & T > t_p \end{cases} \qquad R = \mathrm{sup} - F, \quad \mathrm{RO} = \max(0,\, R - S_d^{\mathrm{eff}}), \quad I = \mathrm{sup} - \mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

This is a **simplified point balance** (it uses the cumulative law $F = K\,T^A$ after ponding, and it ignores re-infiltration of water that has run to a lower cell, which makes the layer slightly pessimistic about runoff in a field with surface storage and slightly optimistic on a steep slope). A portion of the infiltrated depth is retained on the crop canopy and mulch and evaporates,

$$I_c = \min\{I,\; I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

where CAP is the mulch storage capacity (mm, Table 3.1), the indicator is 1 when a mulch is present, and $\mathrm{LAI}_{\mathrm{live}}$ is the live leaf-area index of the simulated crop on the working date. The depth that reaches the soil at the cell is $Z_k = I - I_c$.

The intake figure shows the rate the soil accepts, $\dot I(t) = 60\,KA\,t^{A-1}$ mm h^{-1}, the mean applied rate $\overline{AR}(1-w_d)$, the shaded area where the applied rate exceeds the intake (runoff), and the time at which ponding starts. (The plotted curve is the derivative of Kt^A; the point balance uses the ponding condition $a = Kt^{A-1}$, so the two differ by the constant factor A in the intake rate, a documented simplification.)

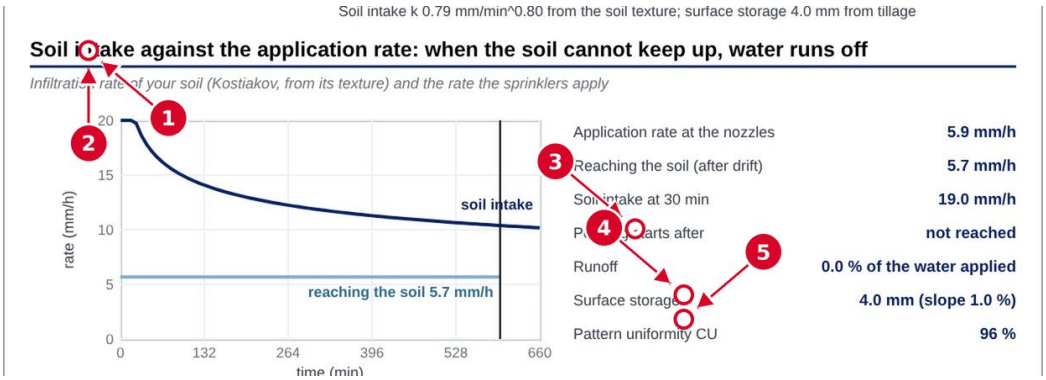


Figure 13.2: Sprinkler design: depth applied and infiltrated along the lateral, and the soil intake against the application rate. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

13.6 Efficiencies and run time

With Z_k the infiltrated depth at cell k and gross $= \bar{a}\,T$ the mean depth discharged:

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

where $\overline{Z}_{25}$ is the mean of the lowest quarter of the Z_k (the distribution uniformity of Merriam and Keller, 1978). The partition is closed, $E_a + \text{DP} + \text{RO} + \text{WD} = 1$. The tiles show E_a (green from 85 %), E_s, CU, runoff, deep percolation and the combined drift, evaporation and interception loss, with a stacked bar of the partition.

Automatic run time. The time T is the smallest value for which the driest quarter of the field receives the required depth,

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\ 2400] \text{ min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

found by 18 bisection steps. If the soil cannot take the water as fast as it is applied, longer runs only raise runoff and the layer shows the run time and low E_a together.

13.7 Maximum allowable application rate

The allowable rate by soil and slope follows the standard table of sprinkler manuals (Keller and Bliesner, 1990, Table of maximum application rates; reproduced here for the textbook check).

Table 13.3. Maximum application rate (mm h^{-1}) by soil type and slope.

Soil group	0–5 %	5–8 %	8–12 %	12–16 %
Deep coarse sandy soils	50	38	25	13
Coarse sandy, compact layer below	38	25	19	10

Soil group	0–5 %	5–8 %	8–12 %	12–16 %
Deep sandy loam	25	20	15	10
Sandy loam, compact layer below	19	13	10	8
Deep silt loam	13	10	8	5
Silt loam, compact layer below	8	6	4	2.5
Clay or clay loam, strong structure	4	2.5	2	1.5

The ten soils of the program are assigned to rows: Coarse Sand, Sand, Loamy Sand $\rightarrow$ 1; Sandy Loam, Coarse Loam $\rightarrow$ 3; Loam, Silt Loam $\rightarrow$ 5; Sandy Clay Loam $\rightarrow$ 6; Clay Loam and Silty Clay Loam $\rightarrow$ 7. The toggle *Compact layer below the soil* moves an odd row to the next (compact) row. The slope selects the column.

13.8 The ten textbook checks

The "Does your design pass?" panel runs the design through the course rules. Let $h_s = P/9.81$ (m), $q_h = 0.06\,q$ (m^3 h^{-1}), $R_w = R$, the wind in km h^{-1} $W = 3.6\,u$, and the wind factor

$$k = \begin{cases} 1, & W \le 5.5 \\ 0.9, & 5.5 < W \le 13 \\ 0.8, & 13 < W \le 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

Provenance (Eq. 13.13): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

The spacing limits, the field layout and the time budget are

$$S_s \le k\,R_w, \qquad S_l \le 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x/S_s \rfloor, \quad N_m = \lfloor L_y/S_l \rfloor, \tag{13.14}$$

Provenance (Eq. 13.14): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}} \ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right) \text{ days}, \tag{13.15}$$

Provenance (Eq. 13.15): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

where $L_x = L_L$ is the lateral length, h_{op} the operating hours per day and Z_{gross} the gross depth per irrigation. For $n = 1 \dots 4$ laterals running at once, the days needed and the inflow are

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\,N\,q_h, \tag{13.16}$$

Provenance (Eq. 13.16): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

and the program picks the smallest n with $N_d \le T_{\text{int}}$ and $Q_{\text{in}} \le 3.6\,Q_s$ (m^3 h^{-1}); if none satisfies both it picks the smallest n meeting the time, else $n = 4$.

Pipes. Friction losses use the Blasius form (Blasius, 1913) for smooth plastic pipe,

$$J = \frac{0.472\,Q^{1.75}}{D^{4.75}} \ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

Provenance (Eq. 13.17): **[Literature]**: Blasius friction law for smooth pipe.

with the Christiansen reduction factor $F = 1/2.75 + 1/(2N)$ for N equally spaced outlets and a 10 % allowance for local losses. The lateral pipe is the smallest of the standard internal diameters $\{46.4,\ 58,\ 63,\ 65.8,\ 75,\ 84.9,\ 100\}$ mm for which

$$h_{f,\mathrm{lat}} = 1.1\, F\, J(Q_L, D)\,(L_x - S_s/2) \le 0.2\, h_s, \qquad Q_L = 1000\, N\, q_h, \tag{13.18}$$

Provenance (Eq. 13.18): **[Literature]**: 20 % rule of Keller and Bliesner (1990).

the classical 20 %-of-operating-head rule that keeps discharge variation along the lateral under about 10 % (Keller and Bliesner, 1990). The mainline (flow $Q_m = n\, Q_L$, length $L_y - S_l/2$) is the smallest diameter not below the lateral's with $J < 0.05$ m m^{-1}.

Manometric head and pump.

$$H_{\mathrm{man}} = z_{\mathrm{lift}} + J\, L_{\mathrm{conv}} + J\, L_{\mathrm{main}} + h_s + 0.75\, h_{f,\mathrm{lat}}, \tag{13.19}$$

Provenance (Eq. 13.19): **[Engineered]**: head and sets bookkeeping.

$$P_{\mathrm{CV}} = \frac{1000\,(Q_m/3.6\times 10^6)\, H_{\mathrm{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\mathrm{kW}} = 0.7355\, P_{\mathrm{CV}}, \tag{13.20}$$

Provenance (Eq. 13.20): **[Literature]**: hydraulic power definition.

with L_{conv} the conveyance length (the mainline length box) and $0.75\, h_{f,\mathrm{lat}}$ the head at the lateral inlet above h_s (the textbook inlet-head rule). The ten checks (spacing along the lateral, spacing between laterals, wind below 16 km h^{-1}, rate below the allowed value, rate below K_s, CU ≥ 85 %, lateral loss ≤ 20 % of h_s, mainline gradient, the whole field within the interval, flow within the farm supply) are shown with a tick or an exclamation point and the numbers; "N of 10 checks pass" closes the panel. The textbook sequence in the layer also plots the class of sprinkler (low, medium or high pressure) from h_s: < 20 m, 20–45 m, > 45 m.

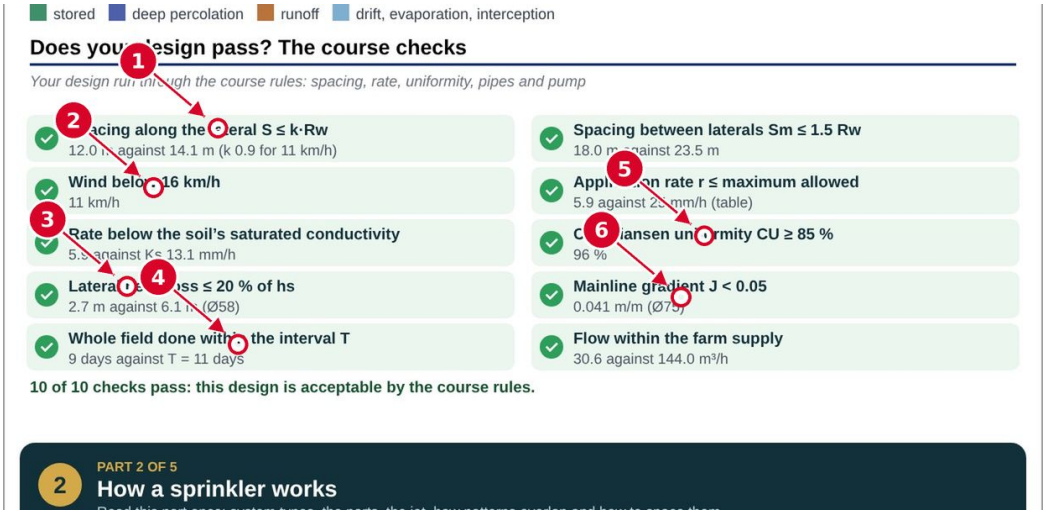


Figure 13.3: Sprinkler design: the ten textbook checks and the kinds of sprinkler system. The numbered arrows are explained in the screenshot atlas (Chapter 25), under the same screen.

13.9 Kinds of sprinkler system

Five systems are offered, each with its efficiency range (Keller and Bliesner, 1990; Burt et al., 1997 report the range of field values):

System	Principle	Typical E_a
Solid-set	Fixed buried laterals; valves open some at a time	70–75 %
Hand-move	The lateral is moved by hand after every set (1–4 sets a day)	70–75 %
Center pivot	A rotating lateral on towers about 30 m apart; the corners stay dry	65–70 %
Linear-move	A lateral travelling in a straight line	65–70 %
Traveling gun	One gun on a hose reel covers a 220° sector, strip by strip	65–70 %

The plan of the selected system is drawn to scale for the present field; the run moves it (the active lateral, the pivot arm, the travelling lateral or the gun strip). A bar shows where the design's E_a lies against the range for the system.

13.10 Layout and infrastructure

The *Layout and infrastructure* block of this layer uses the same hydraulic tools as the drip layer, with these differences. With $N_s = \lfloor L_L / S_s \rfloor$ sprinklers per lateral and $Q_{\text{lat}} = N_s q / 60$ (L s^{-1}), the number of laterals that can run together is $\lfloor Q_s / Q_{\text{lat}} \rfloor$ and the sets

$$N_L = \left\lfloor \frac{10^4 A_f / L_L}{S_l} \right\rfloor, \quad n_l = \min\!\left(N_L,\ \left\lfloor \frac{Q_s}{Q_{\text{lat}}} \right\rfloor\right), \quad n_{\text{set}} = \left\lceil \frac{N_L}{n_l} \right\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.

The lateral loss uses the Hazen–Williams equation (Eq. 12.3, $C = 140$) with the Christiansen factor for N_s outlets and the exponent $m = 1.852$,

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m \, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.

and the lateral pipe is the smallest of the series (12.21) with $h_{f,\text{lat}} \le 0.2\,h$ and $v \le 2.0$ m s^{-1}. The lateral inlet head is $H_0 = h + 1.5 + h_{f,\text{lat}}$ m (1.5 m of riser). The submain, mainline and pump then follow Eqs. (12.22)–(12.23) exactly as for drip, with H_0 from this paragraph. The block's table gives the sets ($\times$ laterals), laterals and sprinklers, TDH, power, lateral loss (≤ 20 %) and pipe laid per hectare.

(Two parallel calculations therefore appear in this layer: the **simulation block** sizes the pipe by Hazen–Williams with $C = 140$, while the **textbook block** uses the Blasius equation and the commercial plastic-pipe sizes of the course. They are separate presentations of the same design question; the numbers will differ by a few per cent, and the textbook block is the one to compare with hand calculations from the course.)

13.11 The design table and sensitivity

The optimiser works like the drip one (Eq. 12.26): score $\min(E_a, 1) - 2.5\max(0, 0.9 - E_s)$, target $E_a \ge 85$ % and $E_s \ge 85$ %, accept a move only if the score rises by more than 0.002. The ten moves are **pressure, spacing along the lateral, run time, nozzle, spacing between laterals, run time, pressure, triangular layout, run time, spacing along the lateral**. For pressure, nozzle and the two spacings the candidates are nine equally spaced values over the range of Table 13.1 (rounded to the step); for the layout the two options; for the run time T_0 $\times$

$\{0.6, 0.7, 0.8, 0.9, 1, 1.1, 1.25\}$ with T_0 the automatic time. Tapping a row applies it. If the target is not met, the layer suggests a lower application rate (a smaller nozzle or wider spacing) or another sprinkler type.

The sensitivity chart moves eight inputs and shows the change of E_a: soil intake k (± 20 %), wind (± 50 %), pressure (± 20 %), nozzle (± 15 %), the two spacings (± 20 %), the depth per irrigation (± 25 %) and the ground slope (± 50 %).

13.12 Season and farm

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60} \text{ h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

The gross water per irrigation is gross $= \bar{a}\, T$ mm, per season n_{ev} gross mm ($= 10\, n_{\text{ev}}$ gross m^3 ha^{-1}). The table also gives one sprinkler's discharge and reach, the sprinklers for A_f, the sets with Q_s, and the pumping hours and energy of the season. The layer closes with the *Irrigation schedule* and the *When to irrigate* message (end of Chapter 11).

13.13 How to use the layer

- **To pick a sprinkler for a soil.** Choose the soil, set the slope and the wind, then compare the types: the banner tells you the largest allowed rate, the intake plot shows where runoff starts, and the checks name the failing rule.
- **To reduce runoff.** Lower the application rate: a smaller nozzle or lower pressure reduces q; a wider spacing spreads it over a larger area; a triangular layout improves CU at the same rate.
- **To reduce drift.** Reduce the pressure, use a flatter angle, or use a micro-sprinkler ($F = 0.7$); the wind arrow and the w_d share of the partition show the effect.
- **To size the system for the farm.** Set the area, farm supply and operating hours; read the sets, the pipe diameters and the pump power from the layout block, and the textbook block for the hand-calculation version.

Chapter 14

The Dashboard: how every number is produced

The Dashboard is the panel to the right of the drawing. It has four tabs (*Results*, *Design*, *Charts*, *Messages*) and the buttons **Wider** (enlarges or shrinks the panel), **Pin** (stores the present 20-year averages as a baseline; Ctrl+B) and **Clear** (removes the baseline). It is laid out like the Properties panel: a white sheet with grey group bars. **Every block can be folded and unfolded on its own** by clicking its bar; the folding is remembered while the program runs, also when the numbers refresh. The dashboard re-reads the engine each time the state, the date or the weather changes, so it always agrees with the drawing.

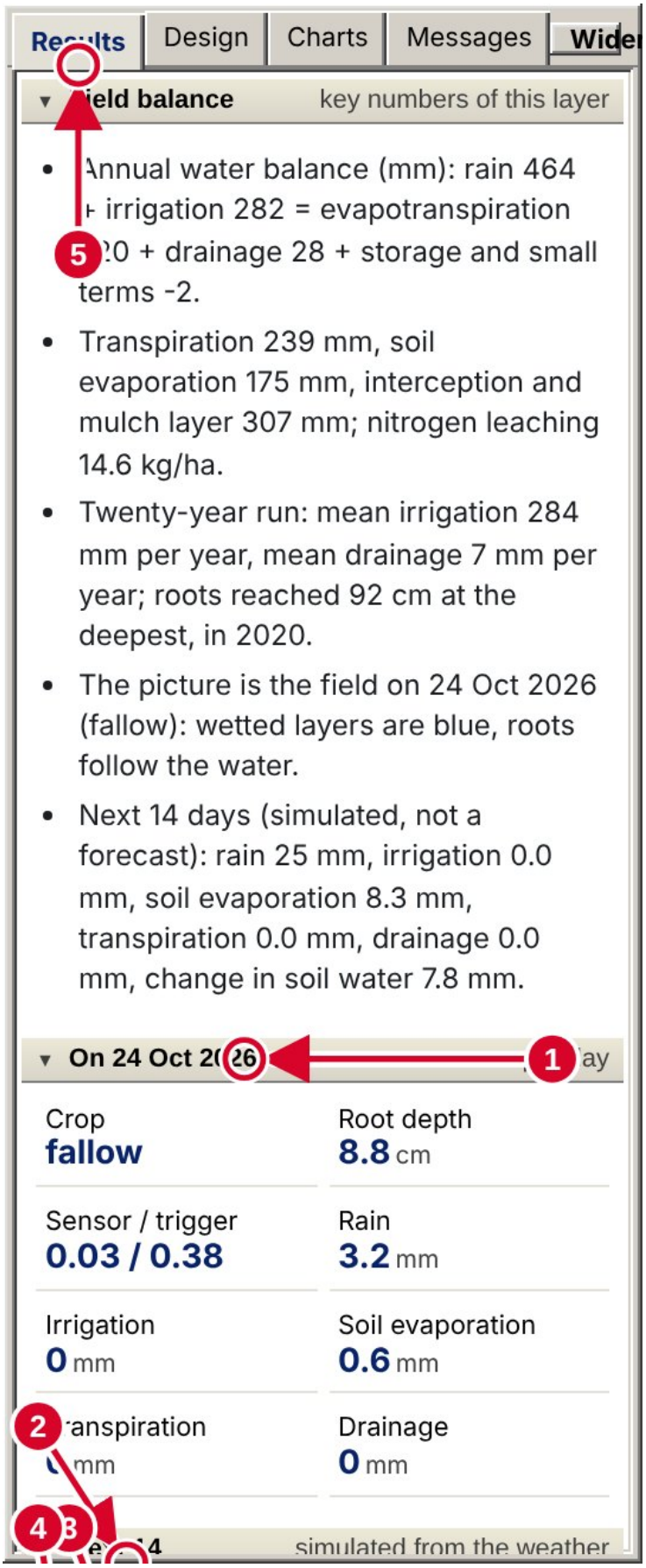


Figure 14.1: The Dashboard, Results tab, with the Field balance layer open.

14.1 Results

The first block of the Results tab carries the name of the open layer and its **key numbers** (the same lines as in the report). For example, on *Soil hydraulic drivers* it shows the change in drainage due to tillage and to the consolidation rate on your soil, and the leading driver. A design layer shows its key numbers after its run.

On *Trigger law* this block shows only the soil you are working on. It lists the trigger (cm) and its depletion, the critical f^*, the sensor reading and its depletion now, the days from field capacity to the trigger, the drying speed, the water held in the root zone, the yield, the irrigation and drainage, the share of the best trigger's yield, and the advice $f = 0.50$ or $f = 0.45$ with its reasons. Click another soil in the drawing to see its numbers. Under the table, the chart of the season shows the depletion at the sensor day by day, the trigger (dashed) and the irrigation days (blue), with both axes named; then the advice, and the two lists *Choose f = 0.50 if* and *Choose f = 0.45 if*, each condition ticked or not with the value that decides it.

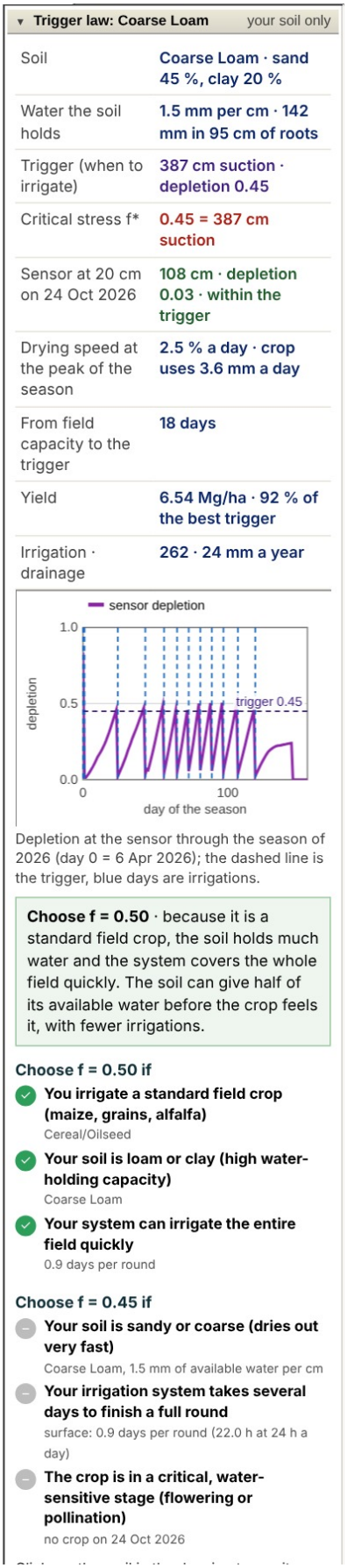


Figure 14.2: The Trigger law block of the Dashboard: the soil, its numbers, the season chart and the advice.

The tab has four blocks.

Block 1: On the working date (mm per day unless stated). Crop, root depth (cm), *Sensor / trigger* (the simulated relative depletion f of the sensor layer and the trigger f_T; irrigation fires when $f \geq f_T$, Eq. 6.1), rain, irrigation, soil evaporation, transpiration and drainage on that day. They are the Engine 2 daily series of Chapter 5, read at the day index k of the working date:

$$\text{Rain} = P_k, \quad \text{Irr} = I_k, \quad \text{Evap} = E_k, \quad \text{Transp} = T_k, \quad \text{Drain} = D_k \ \ (\text{mm d}^{-1}). \tag{14.1}$$

Provenance (Eq. 14.1): **[Engineered]**: bookkeeping of Engine 2 series.

Block 2: Next 14 days. Sums of the same series over the following $N = 14$ days,

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

Provenance (Eq. 14.2): **[Engineered]**: bookkeeping of Engine 2 series.

The bar of the block says *simulated from the weather record* when the days come from the analogue year, and *from the weather forecast* when the online weather is in use and the days fall in its 16-day forecast (Chapter 19).

Block 3: Whole year. For the calendar year of the working date: rain, irrigation, other evaporation (interception, litter, ponded water and snow), soil evaporation, transpiration, drainage, the change in storage and the deepest roots (cm). The water balance of the year closes as

$$\Delta S = P + I - E_{\text{other}} - E - T - D, \tag{14.3}$$

Provenance (Eq. 14.3): **[Literature]**: water-balance identity.

and the tile *Storage change* is ΔS.

Block 4: 20-year averages at these settings. The annual-model outputs of Engine 1 (Chapter 5): soil evaporation, transpiration, evapotranspiration ET $= E + T$, drainage, irrigation, yield (Mg/ha for the simulated cereal, per cent of the potential for a library crop), irrigation efficiency, nitrogen leaching, rainfall, and *Irrigation without mulch* (the same case with no mulch, for the saving).

14.1.1 The pinned baseline

Press **Pin** (or Ctrl+B): the tiles of Block 4 are stored. After any change the tiles show the difference from the pinned values,

$$\Delta_i = x_i - x_i^{\text{pin}}, \tag{14.4}$$

Provenance (Eq. 14.4): **[Engineered]**: bookkeeping.

with ▲ (increase) or ▼ (decrease), green when the change is in the better direction and red when it is worse, and "= no change" when $|\Delta_i| < 10^{-9}$. Lower is better for soil evaporation, drainage, irrigation and nitrogen leaching; higher is better for yield and irrigation efficiency; transpiration, evapotranspiration, rainfall and *Irrigation without mulch* are neutral.

14.2 Design

With a design layer open, the tab shows the results of its **last run**. Before the first run, or after an input changed, it says that the design waits for a run and carries the two buttons **Run simulation** and **Find the most efficient inputs** (Chapter 18). After a run it shows, in folding blocks:

Block	Contents
Key results	Surface: E_a, E_s, deep percolation, runoff, CU, cut-off, fill and advance times, the volumes. Drip: E_a, E_s, E_u, deep percolation, wetted fraction P_w, gross application, irrigation time, volumes per emitter. Sprinkler: E_a, E_s, DU, CU, deep percolation, runoff, gross and net application, set time, discharge, application rate.
Warning	For a surface design whose stream does not reach the end of the field in a practical time.
Most efficient inputs	The best combination found by the search of Section 18.4, with its E_a, E_s, deep percolation and runoff, and **Apply these inputs and run**; or *Already the most efficient* when no combination tried beats your inputs.

The definitions are Eqs. (11.18)–(11.19), (12.19) and (13.11). The Design tab and the *Design results (CSV)* export read the same object, so the numbers agree digit for digit.

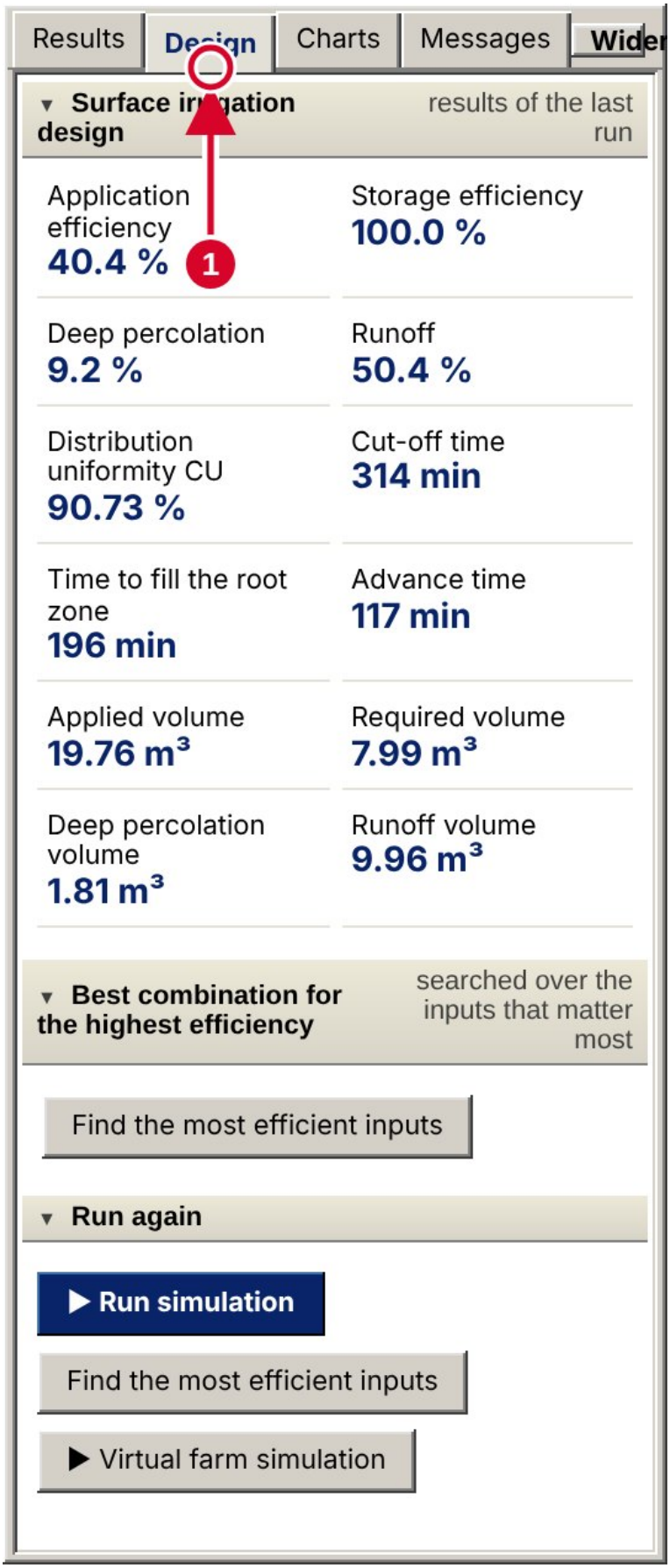


Figure 14.3: The Dashboard, Design tab, after a run of the surface design.

14.3 Charts

On the seasonal layers the Charts tab has one chart for each part of the layer, over the twenty seasons, with the year on the clock marked and the rainfed seasons dashed. Under each chart a line tells how the quantity varies: its mean, the 10 to 90 % range, the lowest and the highest season, and the variation from year to year.

- *Reliability over 20 seasons:* irrigation, drainage below 1 m, grain yield, nitrate leaching, evapotranspiration, days of crop water stress.
- *Full water balance:* rain and irrigation, transpiration and soil evaporation, other evaporation, drainage and drainage through biopores, runoff.
- *Nitrogen budget:* nitrogen supplied, taken up and lost.

With a design layer open and run, the tab draws the charts of the run:

Design	Charts
Surface	Hydrograph: inflow, infiltration and runoff against time, with the advance time t_L and the cut-off t_{co} marked; advance, cut-off and recession along the field; infiltrated depth along the field against the required depth; the times of the irrigation (t_L, t_s to fill the root zone, t_{co}, the recession time t_r and the total); where the water goes.
Drip	Hydrograph at the pump (the zones run one after the other at the zone flow); the times of the irrigation (run time of a zone, the whole field, the interval); pressure head along the lateral; emitter flow along the lateral; where the water goes.
Sprinkler	Hydrograph at the pump (the sets run one after the other); the times of the irrigation; soil intake rate against the application rate; depth caught at each point, sorted (uniformity); where the water goes.

With any other layer open, three charts of the twenty years around the working date (calendar years on the horizontal axis, mm per year): *Rainfall and irrigation*, *Drainage below the root zone* and *Evaporation and transpiration*. They are the annual sums of the Engine 2 daily series,

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

Provenance (Eq. 14.5): **[Engineered]**: bookkeeping of Engine 2 series.

Every chart sits in its own folding block; a chart that is unfolded is drawn again at once. The pump hydrograph of drip and sprinklers is the step series

$$Q_{\text{pump}}(t) = Q_z \quad \text{for } t \in [\,0,\ n_z\, t_{\text{run}}\,), \qquad 0 \text{ afterwards}, \tag{14.6}$$

Provenance (Eq. 14.6): **[Engineered]**: zones of the layout (Eqs. 12.20–12.23 and the sprinkler analogue) run in turn.

with Q_z the flow of one zone or set, n_z the number of zones or sets and t_{run} the run time of one zone (drip) or the set time (sprinklers).

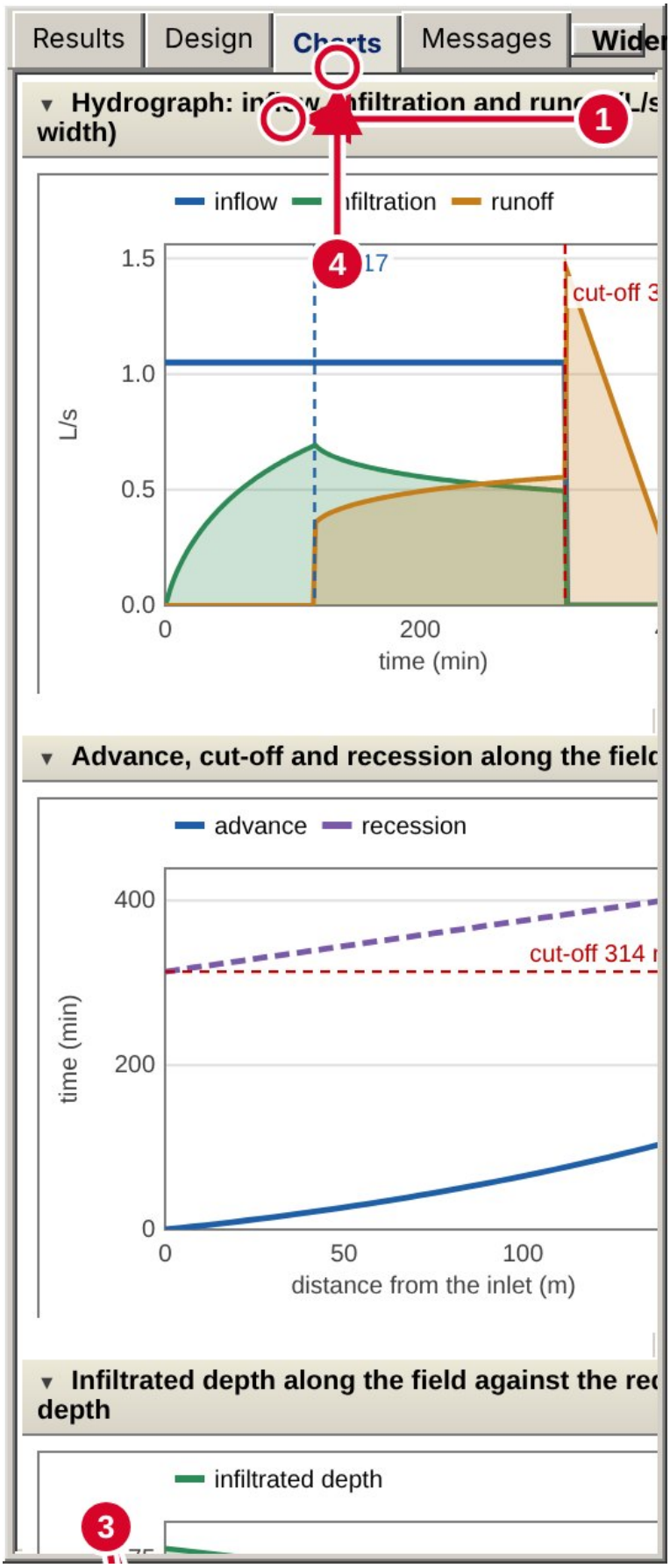


Figure 14.4: The Dashboard, Charts tab, after a run of the drip design.

14.4 Messages

A time-stamped log of what the program did (weather imported, design run, report saved, baseline pinned), newest first, in two folding blocks: *Warnings* (only when there are any) and *All messages*. Use it to find the path of a saved file.

Chapter 15

Worked workflows

Each workflow is a complete sequence of clicks, what to expect, and the equations behind it.

15.1 Workflow 1: How much does straw mulch save?

1. Properties ▸ Scenario ▸ *Ready-made case* ▸ *Default* (Coarse Loam, bare, normal climate).
2. Press **Pin** above the dashboard (Ctrl+B). The tiles of the 20-year block are stored.
3. In Properties ▸ Mulch choose **Wheat straw**; the strength becomes 0.4 and the capacity 4 mm. Read the arrows: soil evaporation falls (green), irrigation falls (green).
4. Open *Mulch and evaporation* ▸ *Fate of saved water* to see where the saved water goes (less irrigation, more drainage, more transpiration).

Why: the strength φ scales the vapour flux through the mulch, and the capacity the water the mulch holds (Chapter 9); *Irrigation without mulch* minus *Irrigation* is the saving on the whole-year balance.

15.2 Workflow 2: When should I irrigate?

1. Set the working date to today and choose the soil and crop.
2. Open *Irrigation design* ▸ *Surface irrigation design* (or drip, or sprinklers), set the inputs and press **Run simulation**. Scroll to the *Irrigation schedule*: the first row is the next irrigation date; the *When to irrigate* message at the end repeats it with the mean interval.
3. Change the **Trigger** in Properties to a drier value (for example $-300 \rightarrow -1000$ cm): the run is cancelled; run again: the interval lengthens and the number of irrigations falls (Eq. 6.3 shifts f_T).
4. Change the **Sensor depth**: a shallow sensor dries first and triggers earlier.

15.3 Workflow 3: Design a border strip for a loam

1. Set the soil to Loam; open *Surface irrigation design*; *Start from: Your soil*; *System: Border*.
2. Type the run length of your field in the *Field length* box (for example 200 m), the width and the slope.
3. Press **Run simulation**. Read E_a and E_s in *Result*, and the hydrograph and the times on the Charts tab.
4. Press **Find the most efficient inputs**. The search tries inflows, cut-off rules, cutbacks and shorter runs (Section 18.4); if it finds a better combination, the Design tab offers **Apply these inputs and run**.
5. Read the sensitivity chart: if the intake constants dominate, the soil intake is the main uncertainty, so measure it in the field.

15.4 Workflow 4: Choose and size a drip system

1. Choose the crop, the soil and the emitter (inline).
2. *Field size*: **Length and width**; type the length and width of the field. The area and the lateral length follow (Eq. 18.1).
3. *Placement*: compare *Surface*, *25 cm* and *45 cm*: each change cancels the run; run each and compare E_a and the losses.
4. Set the farm supply Q_s; read zones, pump head and power in the layout block (Eqs. 12.20–12.23), and the pump hydrograph on the Charts tab.
5. Open *Tools* ▸ *Field on the map...* to read the real field: its indices, its land cover, the water sources around it and the system that suits it (Chapter 20).

15.5 Workflow 5: Check a sprinkler against the course rules

1. Choose the soil and the slope (for example Sandy Loam, 3 %); the banner states the allowed rate (Table 13.3).
2. Choose *Impact sprinkler*; set the nozzle, the pressure and the spacing in their value boxes; press **Run simulation**; read CU and the application rate.
3. Open *Does your design pass?*; fix the red checks in the order the text names; run again.
4. Compare the system chips (solid-set, hand-move, center pivot, linear-move, travelling gun) and their efficiency ranges.

15.6 Workflow 6: My own weather, a warmer future

1. *File* ▸ *Import weather data* and choose your daily weather file, or *File* ▸ *Online weather for a location* (Chapter 19). Every layer now runs on it; Properties shows *Imported weather file* (or *Online weather of a location*) as the weather source.
2. Pin the baseline.
3. Set the temperature shift to +2 °C and the rainfall scale to ×0.8. The arrows show the effect of the change on your own climate.
4. Switch back to *Built-in weather record* in Properties ▸ Scenario at any time: your earlier rainfall and temperature settings return.

15.7 Workflow 7: A report for a client

1. *File* ▸ *Save report as PDF (choose the layers)...*.
2. Untick the groups you do not need (for example *Seasonal data* and *Feedback loops*), keep the irrigation design.
3. **Make the PDF**, choose where to save it. Each layer gets your inputs, its key numbers and its full drawing; long drawings continue on the next pages, cut at their section headings (Chapter 23).

15.8 Workflow 8: Save, share and reproduce

Save the project (*File* ▸ *Save project*) after each case, export the yearly CSV and the design CSV, and note the version shown in the status bar (*SWAFL 1.1.0*). Reopen the `.swafl` file to restore the state exactly.

Chapter 16

Files, exports and troubleshooting

16.1 What can be saved

Item	Menu / toolbox	File	Contents
Report	*File* ▸ *Save report as PDF (choose the layers)...*, Ctrl+Shift+P, toolbar *Report*	`SWAFL_full_report_<soil>_<date>.pdf`	The layers you ticked, each with your inputs, its key numbers and its full drawing (Chapter 23)
Picture of the window	*File* ▸ *Export* ▸ *Picture of the window (PNG)*, Ctrl+E	`SWAFL_<layer>.png`	The whole drawing of the open layer, top to bottom
Yearly results	*File* ▸ *Export* ▸ *Yearly results (CSV)*	`SWAFL_yearly_results.csv`	Two comment lines with soil, tillage, crop, mulch and method; then `Year, Rain_mm, Irrigation_mm, Drainage_mm, Evaporation_mm, Transpiration_mm`
Design results	*File* ▸ *Export* ▸ *Design results (CSV)*	`SWAFL_design_<id>.csv`	Two columns `Quantity,Value`: every number the open design layer computed
Project	*File* ▸ *Save project*	`SWAFL_project.swafl`	JSON: app name and version, the open layer, the state, the year and the date
Field report	*Export the report* in *Tools* ▸ *Field on the map...*	`SWAFL_field_<place>_<date>.pdf` or `.png`	The field on the map, its indices, zones, land cover, water sources and the suitable system, at 96 to 600 dpi, with the coordinates and the author (Chapter 20); also GeoJSON and CSV
Print	*File* ▸ *Print / save this view as PDF*, Ctrl+P	PDF	The open layer through the system print dialog

In the yearly CSV, *Year* is the index 1–20 of the 20-year run starting at the calendar year shown on the Charts tab; the values are the annual sums of Eq. (14.5), rounded to 0.01 mm. In the design CSV the quantity names are the internal names of the model (for example `Ea`, `Es`, `dp`, `ro` for the surface layer; `Eu`, `gross`, `t` for drip; `CU`, `DU`, `AR`, `T` for sprinkler); the efficiencies are fractions (0–1).

The imported weather is not stored in the project file; import it again after opening a project, or keep the file beside the project.

16.2 Troubleshooting

Problem	Fix
“Running scripts is disabled” in PowerShell	Use a Command Prompt terminal, or run `Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned`, or type `cmd /c npm install`.
`npm` is not recognised	Install Node.js, then close and reopen the terminal.
`ENOENT … package.json not found`	You are in the wrong folder; the zip makes a folder inside a folder. `cd` into the one that contains `package.json`.
Install or build is slow or fails inside OneDrive	Copy the project to `C:\SWAFL_Desktop` and run it there.
“Windows protected your PC”	Click *More info* ▸ *Run anyway* (the app is not code-signed).
The window still shows version 1.0	An old `SWAFL.exe` is still on the Desktop or pinned. Delete it and run `BUILD_AND_INSTALL_SWAFL.bat` again: the build removes the old `dist` folder first and the program clears its cache on start.
White or empty window	Wait 10 seconds (the engine loads), then *Refresh* ▸ *Reload simulation engine*. Close any second SWAFL window.
A design shows *Run the design* instead of results	The inputs or a setting changed since the last run. Press **Run simulation**.
The field map says the satellite data or the photo cannot be reached	These need the internet (Chapter 20). Without it the program uses the simulated indices and says so.
A design layer shows “the stream does not reach the end”	The cut-off time is shorter than the advance time: increase the cut-off or the inflow (Eq. 11.16), or shorten the run.
E_a is low and runoff is high (sprinklers)	The application rate is above the soil intake: reduce the nozzle or the pressure, widen the spacing (Chapter 13).
A weather file is refused	Read the message: the file needs a date column (or year, month and day) and a rain column with their names in a header row. A Daisy setup file (`.dai`) holds no weather: the message names the weather file it uses; import that file (Chapter 19).
The online weather is not reachable	The computer is offline or the service does not answer. The current weather stays in use; try again later or import a file.

If a problem remains, open an issue on the project page, and attach the `.swafl` file and the text of the *Messages* tab.

Chapter 17

Validity, limits and honest use

The program is a **teaching and design-exploration tool** built on a research data set. This chapter says what each part can and cannot support.

Engine 1 (the annual meta-model) reproduces, by interpolation, the Daisy 7.1 runs it was fitted to (run sets E2, E3, E4, E5, E7, E11, E13 and E14; the study as a whole has 10,190 runs). Within the ranges of the controls it is an *emulator* of those runs: it is not a new physical model, and it does not extrapolate outside the soils, mulches, climates and irrigation methods that were simulated. It was calibrated to one location's weather record (1980–1999) and its projections to 2100 are **scenarios built from analogue years**, not forecasts.

Engine 2 (the daily bucket model) is a 40-layer, 1-D, daily water balance. It uses closed-form root, evaporation and transpiration functions with a single calibrated coefficient k_t so that its seasonal transpiration matches Engine 1. It does not solve the Richards equation, has no lateral flow, no runoff on slopes and no water table (upward flow is a redistribution between layers only). Its day-to-day irrigation dates are therefore indicative: use them to compare scenarios, not to schedule a real irrigation to the day.

Design layers (surface, drip, sprinkler) are textbook models (Chapters 11–13). They are suitable for teaching, for comparing layouts, and for a first sizing. They are **not a substitute for field evaluation** of intake, advance, pressure and flow, or for a hydraulic design by a qualified engineer. In particular:

- the intake of the soil is derived from texture by pedotransfer functions with their own scatter; measure it in the field for real design;
- the sprinkler pattern is a parametric shape, not a measured catch-can test; the wind-drift loss is a parametric estimate;
- the drip bulb is a geometric simplification of two-dimensional soil water redistribution;
- pump power and pipe sizes use stated allowances (9 m of fittings, 0.65 pump efficiency, $C = 140$) that you should replace with your own;
- the "Engineered" items of the label scheme (Section "About this manual") are modelling choices, not published laws.

Data and reproducibility. The same inputs and version give the same outputs (no random numbers in the design layers; the rainfall generator and all animations of the engine use a deterministic generator seeded by the soil name). Save the `.swafl` file and the version number with any result you report.

Your weather (version 1.1). Imported or online weather drives the daily model and the crop water requirement day by day. The annual model, however, can only move along the simulated steps of rainfall ($×0.5–1.4) and warming (-1$ to +3.5 °C) of the study site (Eqs. 19.4–19.5): a climate far outside those steps (for example a humid tropical climate with 2,000 mm of rain) is represented by the nearest step, and the annual results should then be read as the direction of a change, not as its size. The online forecast covers 16 days; dates after it are the typical year of your record.

The crop library (version 1.1). The 48 library crops are not simulated: they are transferred from the simulated cereal rotation by the ratio of two FAO-56 balances (Section 21.4). The transfer is reliable for the water need and the irrigation (which follow from K_c and ET_0), less so for drainage and nitrogen, and the yield is a relative yield (FAO-33), not a harvest in tonnes.

The real field (version 1.1). Sentinel-2 indices are 10 m pixels of one date and need a scene without cloud over the field. The satellite photo is a mosaic of dates that varies by place and has no near-infrared band, so it gives the visible-band indices and the land cover only; the land cover of Eq. (20.4) is an image analysis by colour and texture, not a trained model, and mistakes dark soil for water or bright soil for roofs now and then. The water sources are those mapped in OpenStreetMap and those seen in the photo; groundwater depth is not known. The DEM is 90 m, so the slope of a small field is approximate. The ranking of the systems is guidance, to be checked on the ground.

Not included. The package does not contain a soil-erosion model, a crop-pest model, an economic optimiser, or a groundwater model.

When you cite results from the program, state the version, the case (soil, crop, mulch, method, trigger, date and weather source) and whether the numbers come from Engine 1, Engine 2 or a design layer.

Chapter 18

Running a design and finding the most efficient inputs

Version 1.1 changes how the three irrigation designs are used. Their inputs are set first, and nothing is computed while you type; the results, charts, design table, schedule and soil-water film appear only when you press **Run simulation**. This chapter describes the run, the value boxes, the field size, the search for the most efficient inputs, the field sketch and the comparison of the three systems.

18.1 The run

When a design layer opens, its inputs are drawn as usual and, below them, the results are replaced by the **Run the design** box:

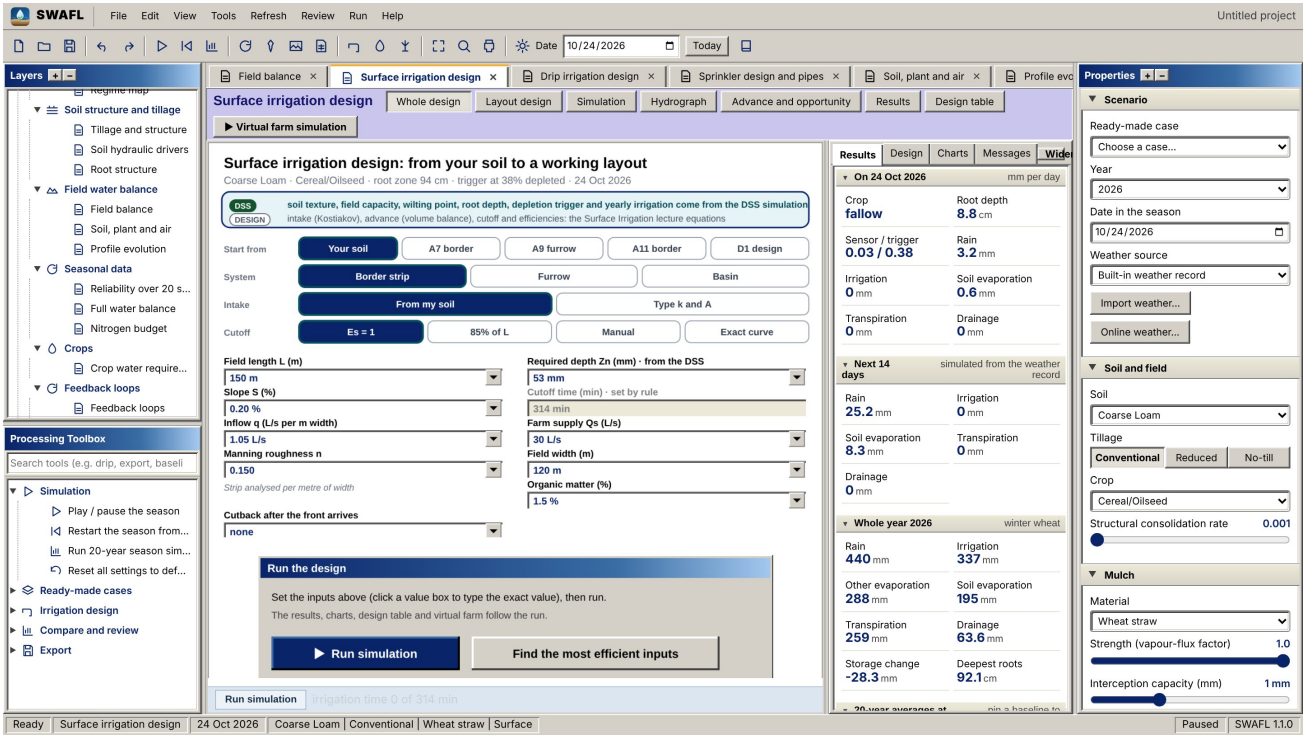


Figure 18.1: The surface design before a run: the inputs are set, the results wait for the run.

- **Run simulation** computes the design with the inputs above it. The run animates the irrigation from its start to its end (the time is shown on the playback bar, for example *irrigation time 0 of 314 min*), and then holds the finished state.
- **Find the most efficient inputs** searches the inputs for the highest application efficiency (Section 18.4), puts the best combination into the value boxes and runs it.
- The play button under the drawing is labelled **Run simulation** in the design layers and does the same.

The last run stays valid until something that changes the result changes: a value box, a chip of the design (system, emitter, placement, sprinkler type, layout, run-time mode, mulch, crop, field size), or a setting of Properties other than the irrigation method (soil, tillage, consolidation, crop, mulch, rain, temperature, trigger, sensor depth, nitrogen, year or date). The box then returns with the reason, for example *The settings in Properties changed: the results wait for a new run*. The Design and Charts tabs of the dashboard follow the same rule.

Formally, a run stores the key κ_{run} of the state with which it was made; the results are shown only while

$$\kappa(\text{state now}) = \kappa_{\text{run}}, \qquad \kappa = \big(\text{every Properties control except the irrigation method, weather mode}\big), \tag{18.0}$$

Provenance (Eq. 18.0): **[Engineered]**: bookkeeping of the run.

and no input of the design has been changed since.

18.2 Value boxes and units

Every input of a design is a value box. Clicking it opens a small window:

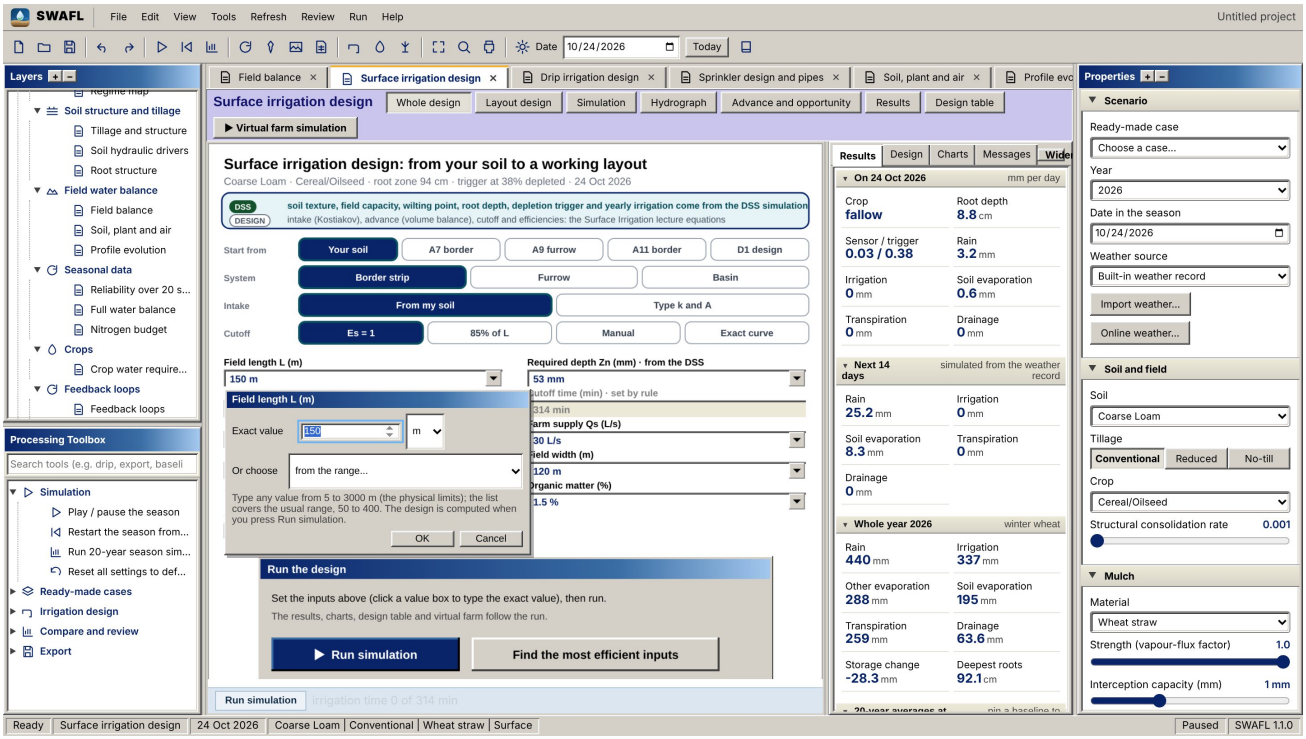


Figure 18.2: The value window of a surface-design input: exact value, unit, list of the allowed range.

- type the exact value, in the unit chosen beside it;

- or pick a value from the list, which runs across the allowed range;
- lengths can be given in m, km or ft; flows in L/s, m³/h or L/min (emitter flow in L/h or L/s); areas in ha, m² or acre; pressures in kPa, bar, m of water or psi; pressure heads in m, kPa, bar or ft; times in min or h; spacings in m or cm.

The value is kept exactly as typed (2.13 stays 2.13); it is not rounded to a step of the list, and the design runs on it. You may type any value up to the physical limits of the system: for example a run of 5 to 3000 m, an inflow up to 300 L/s per m, an emitter flow up to 100 L/h, a pressure up to 1200 kPa. The list covers only the usual range. A value beyond the physical limits is set to the nearest limit, and the window gives both ranges. The value is converted to the unit of the model, $x_{\text{model}} = x_{\text{typed}}\, c_u$, with c_u the factor of the chosen unit (for example 1000 for km, 1/3.6 for m³/h to L/s, 9.80665 for m of water to kPa).

18.3 Field size: an area, or a length and a width

The drip and sprinkler designs need the area to irrigate and the length of the laterals. The row **Field size** offers two ways of giving them:

- **Area in hectares** — type the area; the lateral length is its own value box.
- **Length and width** — type the length and the width of the field in metres. The area and the lateral length then follow, and the lateral-length box is greyed out with the note *from the field length*:

$$A_f = \frac{L\,W}{10^4}\ \ [\text{ha}], \qquad L_{\text{lat}} = \text{clip}\Big(\frac{L}{\lceil L/L_{\max}\rceil};\ L_{\min},\ L_{\max}\Big), \tag{18.1}$$

Provenance (Eq. 18.1): **[Engineered]**: geometry; laterals longer than $L_{\max}$ are split into equal runs from a central submain.

with $L_{\min} = 20$ m and $L_{\max} = 300$ m for drip laterals and $L_{\min} = 40$ m and $L_{\max} = 500$ m for sprinkler laterals. The number of laterals follows from the width, $n_l = \lfloor W_f / S_l \rfloor$ with $W_f = A_f \cdot 10^4 / L_{\text{lat}}$ (Eq. 12.20 and its sprinkler analogue in Chapter 13), and with it the zones, the pipes and the pump.

The surface design is given by its run length and width directly (*Field length L*, *Field width*), and its area is their product.

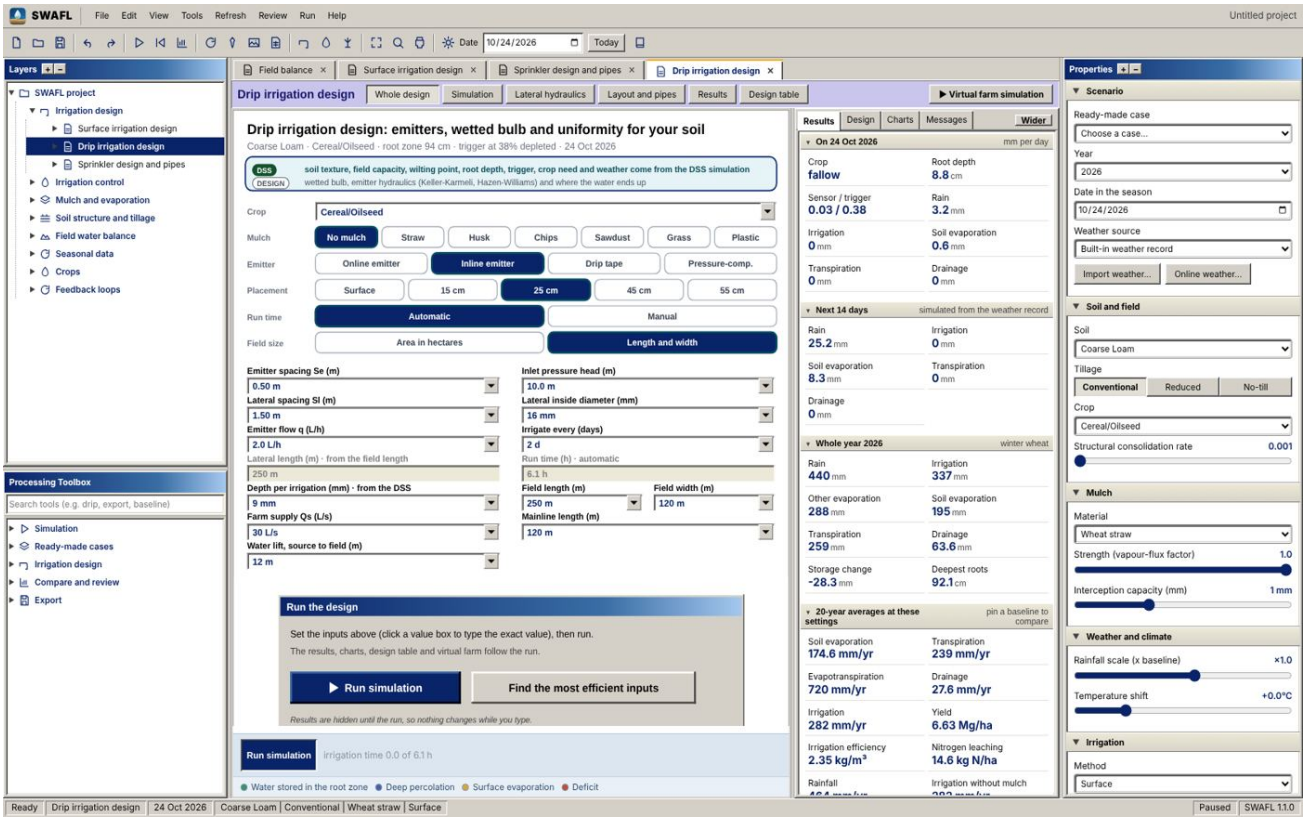


Figure 18.3: The drip design with the field given by its length and width.

18.4 The search for the most efficient inputs

Find the most efficient inputs (in the Run box, on the Design tab, or Tools ▸ *Find the most efficient design inputs*) tries a grid of the inputs that matter most and keeps the one with the highest score

$$s = E_a - 3\,\max\!\left(0,\; 0.95 - E_s\right) + \begin{cases} 0 & \text{surface} \\ 0.05\,E_u & \text{drip} \\ 0.05\,\mathrm{CU} - 2\,R_o & \text{sprinkler} \end{cases} \tag{18.2}$$

Provenance (Eq. 18.2): **[Engineered]**: the application efficiency, with a penalty when the root zone is refilled to less than 95 % (an efficient design that leaves the crop short is not efficient) and a small reward for uniformity.

Design	Grid tried
Surface	11 inflows spaced geometrically across the allowed range, $q_i = q_{\min}(q_{\max}/q_{\min})^{i/10}$; the cut-off rules *when the tail is filled* and *at 85 % of the length*; cutback to 100, 75 or 50 % of the inflow; and the run length L/n for $n = 1$ to 4 (the field split into n runs, each at least the shortest length of the system).
Drip	emitter flow 1, 1.6, 2, 3, 4 or 8 L/h; emitter spacing 0.3, 0.5, 0.75 or 1 m; interval 1 to 4 days.
Sprinkler	five nozzle diameters and four pressures across the range of the sprinkler type; for each, a square spacing of 0.4 to 0.65 times the wetted diameter.

Your present inputs take part in the search. A combination is recommended only when its score is higher than the score of your present inputs; otherwise the Design tab says *Already the most efficient: your present inputs* and offers nothing to apply. The search takes about 0.1–3.5 s and stops after that time.

Applying a combination writes its values into the value boxes, so you can see and change each of them, and runs the design.

18.5 The field sketch

View or *Tools* ▸ *Field sketch to design…* opens a window with a grid on the left and the questions on the right.

On the grid. A red crosshair follows the mouse and gives its position in metres.

- **Field corners:** click to place a corner. Click near a side to add a corner on it, drag a corner to move it, and double-click a corner to remove it.
- **Side lengths:** each side shows its length just outside the field. Click a length to type it exactly; the next corner moves along the side, and the grid changes its scale if the field no longer fits.
- **Water source:** in this mode, click where the water is and drag the blue marker to move it.
- **Inlet:** in this mode, drag the yellow square along the edge of the field to where the pipe or the canal enters.
- **Grid square:** sets the scale, from 1 to 100 m per square.
- **Hand, zoom in and out, Fit:** the hand button moves the drawing; the right or middle mouse button, or Shift and drag, also move it in any mode. **+** and **−**, or the mouse wheel over the grid, zoom in and out around the cursor (10 % to 1200 %). **Fit** shows the whole field. The scale bar and the north arrow stay on the screen and follow the zoom.
- **Locate on the map:** opens the field on the satellite map (Chapter 20), to find the farm, demarcate it on the image and compute its vegetation and water indices.

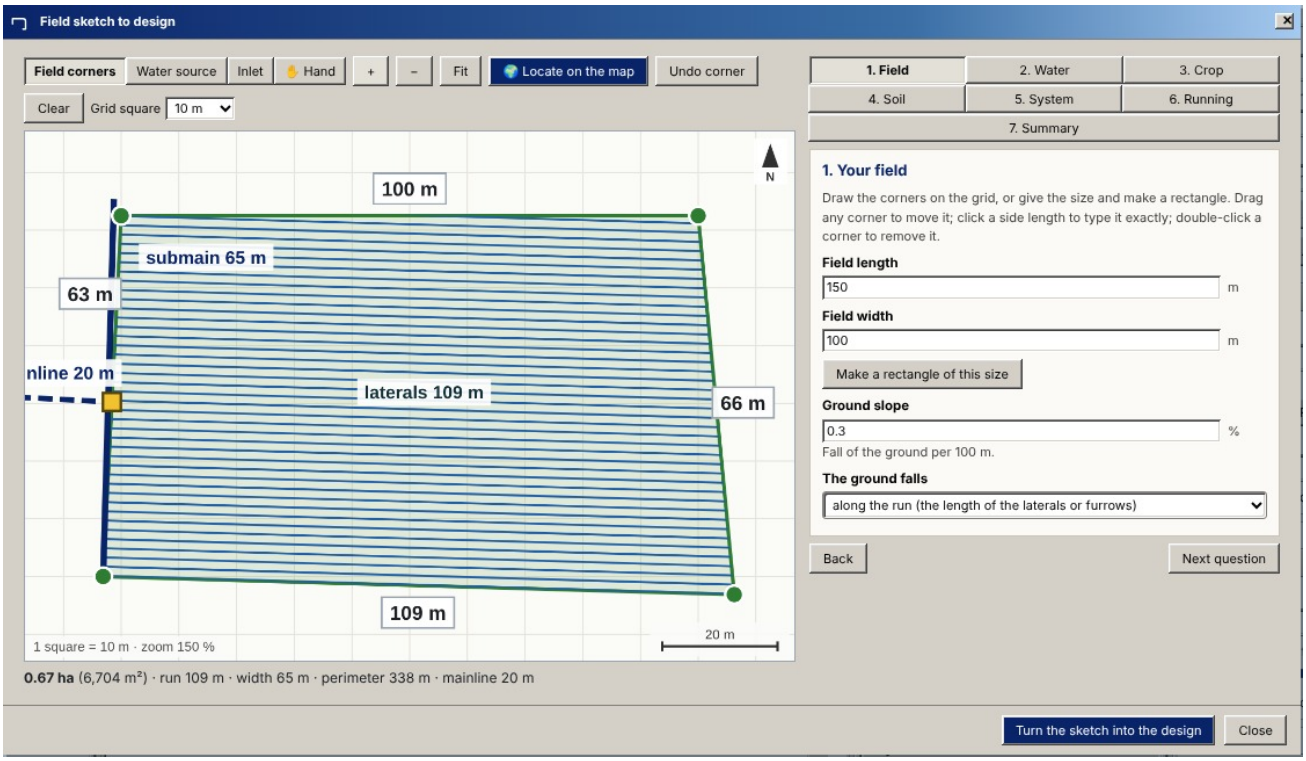


Figure 18.4: The sketch zoomed in, with the hand, zoom and map buttons.

While you draw, the window draws the layout of the system you choose:

- furrows, borders or basins;
- drip laterals at the row spacing;
- sprinkler laterals with the sprinklers and their wetted circles;
- the submain or head ditch along the inlet end;
- the mainline from the source to the inlet, with its length.

The questions, in seven steps (*Next question* and *Back* move through them; the step buttons jump):

1. *Field:* length and width (*Make a rectangle of this size*), ground slope, and the direction in which the ground falls.
2. *Water:* where the water comes from (well, borehole, canal, river, pond or tank), the lift to the field, the flow the source can give, and the hours a day you can irrigate.
3. *Crop:* the crop, the row spacing and the plant spacing.
4. *Soil:* the soil, and whether there is a compact layer below the topsoil.
5. *System:* border strips, furrows, basins, drip or sprinklers; whether the laterals run along the field or across it; for drip, the depth of the laterals and the emitter; for sprinklers, the system, the sprinkler and the two spacings.
6. *Running:* how often to irrigate (drip), and whether the farmer or an automatic controller opens the valves.
7. *Summary:* everything that will be set.

The sketch gives

$$A = \tfrac{1}{2}\Big|\sum_i (x_i y_{i+1} - x_{i+1} y_i)\Big|, \qquad L_{\text{run}} = \max_{\text{along the longest side}} - \min, \qquad W = \text{extent across it}, \tag{18.3}$$

Provenance (Eq. 18.3): **[Literature]**: shoelace formula and projection on the direction of the longest side.

and the mainline length from the source to the inlet. Here L_{run} is measured along the direction of the laterals: along the longest side, or across it.

Turn the sketch into the design sets the crop and the soil in Properties and fills the chosen design:

- *Surface:* the system, the run length, the field width, the slope and the farm supply, and the furrow spacing (from the rows).
- *Drip:* the lateral length, the lateral spacing (the rows), the emitter spacing (the plants), the field length and width, the supply, the interval, the placement and the emitter.
- *Sprinklers:* the area, the slope, the supply, the two spacings, the sprinkler, the system and the lateral length.

It also sets the mainline length, the lift, the source, the hours a day, the compact layer and the operator of the valves. Then it opens the design, which waits for **Run simulation**.

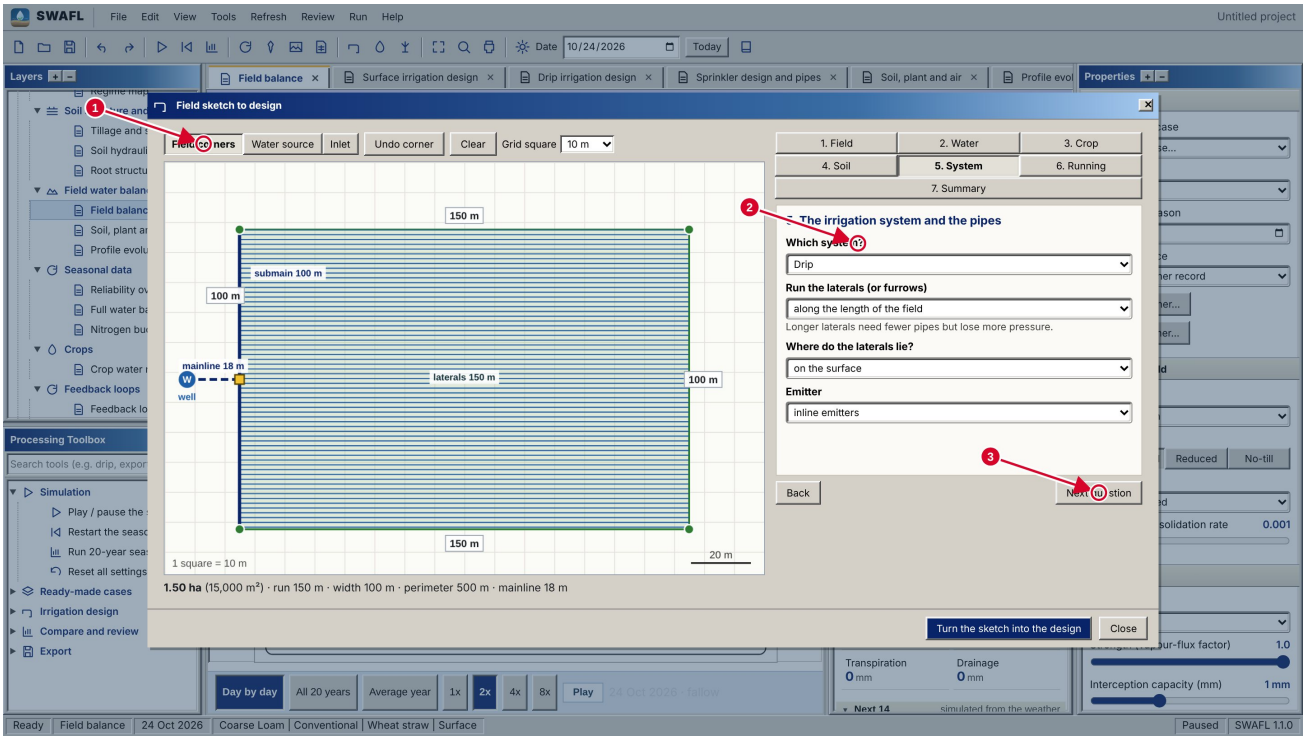


Figure 18.5: The field sketch.

18.6 The soil water in the days ahead

Each design layer (surface, drip and sprinklers) ends with a film of the soil water: **Soil water in the days ahead (DSS)**. It plays day by day; click it to pause or play.

The season of your crop. The crop chosen in Properties is the crop of the decision. When the working date falls in its season, the film runs from the working date to 30 days after the harvest. When the working date has no crop (before sowing or after harvest), the film starts at the **next sowing of that crop** and runs through its whole season, and a yellow note above the film says so; the schedule under the film follows the same season. With the online weather, whose record ends in the present year, the next season is predicted with the weather of the last season of the record, its dates moved on by one year. The crops of the library are sown on their sowing date (Properties), which may be any day of the year; their season may run past 31 December, and the daily model irrigates them whenever they are in the field.

The weather of the decision. The banner at the top of every design has a **WEATHER** line: the weather in use (the recorded weather of Taastrup, Denmark, 1980 to 1999; a weather file; or the online weather of a place with the 16-day forecast) and the button **Online weather of the farm...** (or *Change the weather...*), which opens the online-weather window (Chapter 19). The weather of your farm calibrates everything that follows: the soil water of every day, the irrigation dates and depths, the film, the schedule and the planting advice.

- **Left: the profile of the day.** The water content of each layer of the real soil, 0 to 100 cm, with lines at the wilting point, the trigger and field capacity, the depth of the roots and the sensor. The frame and the label below take the colour of the phase of the day.
- **Right: the days, coloured by phase.** The band at the top colours every day: irrigation day (blue), rain over 5 mm (light blue), comfortable (green), nearing the trigger (yellow), beyond the trigger, i.e. stress (orange, also shaded across the chart), near the wilting point (red), and no crop (grey). The chart also shows:
  - the water held in the root zone in mm (left axis), against the full level (field capacity over the roots of that day) and the trigger level of that day; both levels rise and fall with

the roots, so the axis changes with the soil and the crop;
  - the water left at the sensor, in % of the available water (right axis);
  - rain as ticks at the top and drainage as ticks at the bottom;
  - a dashed red line for what would happen **without irrigation from the day shown**: the root zone dries on with the weather of the record until it reaches the wilting point. It starts again from each day of the film.
- **Below: the analysis.**
  - The day in words: the date, the crop, the phase, the water left, and the sensor against the trigger.
  - The next irrigation after the day shown, with its date, the days to it and the depth of this design, and the date the wilting point would be reached without irrigation from that day. Both are counted again for every day of the film.
  - For the whole span: the number of irrigations, the rain, the evapotranspiration, the drainage, and the share of days in each phase.

Every part of the film is labelled.

- **The profile** has its depth axis (cm) and its water-content axis (% by volume, with the values at the wilting point and at field capacity). Labels mark the wilting point (WP), the trigger and field capacity (FC).
- **The chart** has its axis of the water in the root zone (mm, left), of the water at the sensor (% of available, right) and its axis of dates (the first of each month, with the year at January). The day shown is marked by a line with its date and its water above the chart; arrows mark each irrigation; *next irrigation* names the date and the days to it, and the cross of the wilting point names its date and the days to it without irrigation. These labels move with the film.
- **The colours** are distinct: irrigation dark blue, rain light blue, comfortable green, nearing the trigger yellow, stress orange, near wilting red, no crop grey. The root zone line is dark blue, the sensor purple, the trigger black dashed, the dry-down without irrigation red dotted, the rain ticks blue and the drainage ticks teal.

In the Dashboard, the block *Soil water now: the film of this design* follows the film day by day: the date and day of the film, the crop and phase, the water in the root zone (mm and %), the sensor against the trigger, the next irrigation (date, days to it, depths) and the wilting point without irrigation (date, days to it).

When to plant. Under the film, the chart *When to plant …* gives, for every sowing week of the year, the irrigation the crop needs in a season (bars, mm) and its rain-fed yield (red line, % of potential), from the FAO-56 root-zone balance of the crop library over the 20 years of the weather in use, with your soil, trigger and method. Weeks too cold or too hot for the crop are grey: warm-season crops need a minimum air temperature at sowing and over the season (maize 10 and 14 °C, soybean 10 and 15 °C, tomato 12 and 15 °C, pepper 14 and 17 °C, sorghum 14 and 18 °C, rice, millet, cotton and cassava 15 and 20 °C) and no frost; cool-season crops (wheat, chickpea, mustard, potato, onion, cabbage) a season mean between 5 and 24 °C. The temperatures are those of the weather in use when it has them (online weather, or a file with temperatures), otherwise the typical climate of Taastrup, Denmark, the site of the recorded weather. The best week is the feasible week with the least irrigation (within 10 %) and the highest rain-fed yield; the best window is the run of feasible weeks within 15 % of its need. The text gives the window, the need, the rain-fed yield and the need at your own sowing date; **Sow on … (best)** sets the sowing date in Properties. When no week is warm enough, the warmest window is shown and the text says so. For the simulated cereal rotation, whose dates are fixed, the advice is for wheat from the library (soybean for the legume, tomato for the vegetable) with a button to use that crop. Trees, vines and palms stand all year and have no sowing date.

The Dashboard block *Soil water now* also gives the planting window and the weather in use.

Everything follows the settings. Change the trigger, the sensor depth, the soil, the crop, the mulch, the rain or the weather and the film is computed again.

The phase of a day is set in this order: an irrigation day; a day with no crop; near wilting when the root zone has lost 85 % of its available water; beyond the trigger when the depletion at the

sensor reaches the trigger f_T; a rain day (over 5 mm); nearing the trigger when the depletion at the sensor reaches $0.8\,f_T$; otherwise comfortable.

The water left in the root zone on day d is the mean over the layers within the roots, $\bar{s}(d)$, where $s = (\theta - \theta_{wp})/(\theta_{fc} - \theta_{wp})$. Without the irrigation the depletion follows the FAO-56 root-zone balance:

$$D_r(d{+}1) = \mathrm{clip}\big(D_r(d) + ET(d)\,[0.35{+}0.65\,K_s] - 0.8\,P(d);\ 0,\ \mathrm{TAW}\big), \qquad K_s = \min\Big(1,\ \frac{\mathrm{TAW} - D_r}{0.5\,\mathrm{TAW}}\Big), \tag{18.4}$$

Provenance (Eq. 18.4): **[Literature]** with **[Engineered]** constants: FAO-56 water stress coefficient with $p = 0.5$. $\mathrm{TAW} = \mathrm{AWC}\,z_r$, and ET and P are the evapotranspiration and rain of the daily model. The wilting point is reached when $D_r \geq 0.97\,\mathrm{TAW}$.

The schedule under the film takes its dates from the daily model and its depths and times from the design.

18.7 Comparing the three systems

Tools ▸ Compare the irrigation systems on this field computes the surface, drip and sprinkler designs with the inputs they have now, on the same soil, crop and trigger, and lists their application efficiency, root zone filled, deep percolation, runoff or evaporation and the time per irrigation side by side.

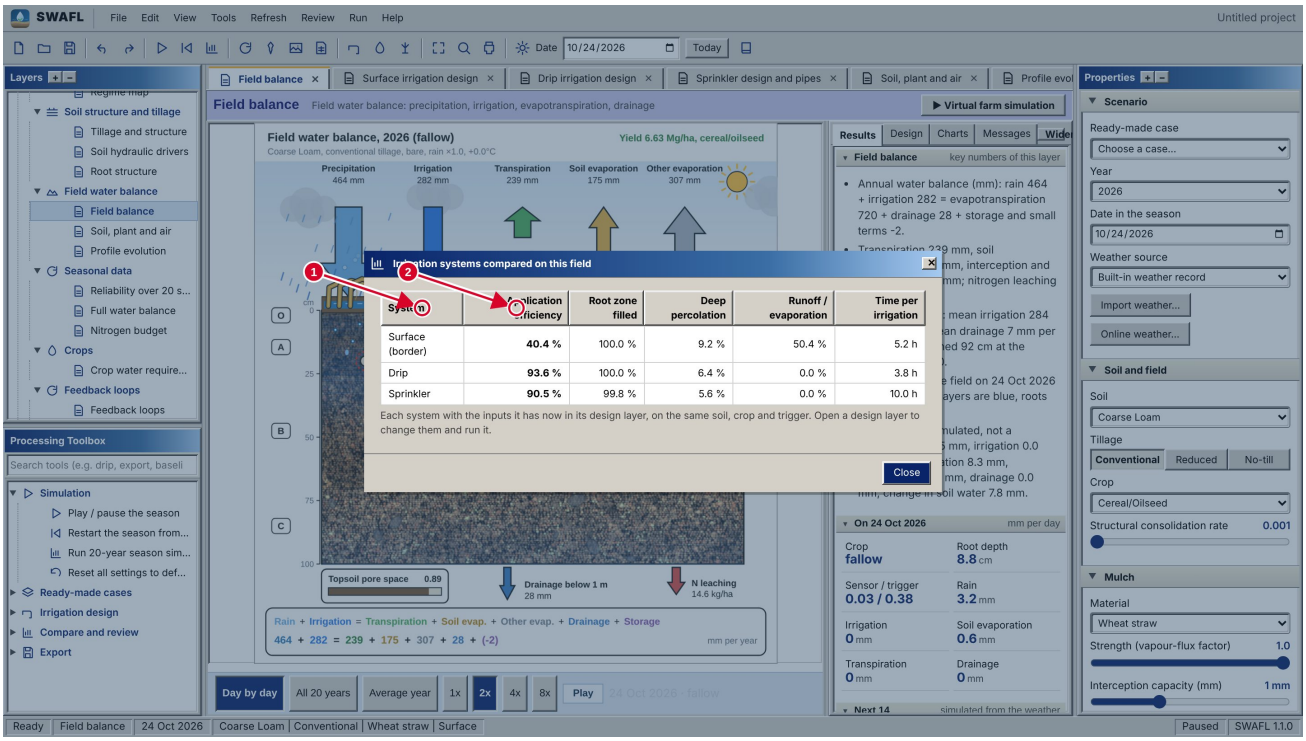


Figure 18.6: The comparison of the three systems.

Chapter 19

Weather data: your own file, or the weather of a place

Without weather data of your own, SWAFL runs on the twenty recorded years of its study site. Version 1.1 lets every layer run on **your weather**: a daily weather file, or the measured daily weather of any place on Earth with its 16-day forecast. The change is seen at once in every layer, the dashboard, the schedules and the crop water requirement.

19.1 Where to start

- **File ▸ Import weather data (CSV, Excel, DWF, DLF, DAI)…**, Tools ▸ *Import weather data…*, or the **Import weather…** button in Properties ▸ Scenario.
- **File ▸ Online weather for a location…**, Tools ▸ *Online weather for a location…*, or the **Online weather…** button.
- Properties ▸ Scenario ▸ **Weather source** switches between *Built-in weather record*, *Imported weather file* and *Online weather of a location*. The last two can be chosen only after a file has been imported or the weather of a place fetched.

19.2 Importing a file

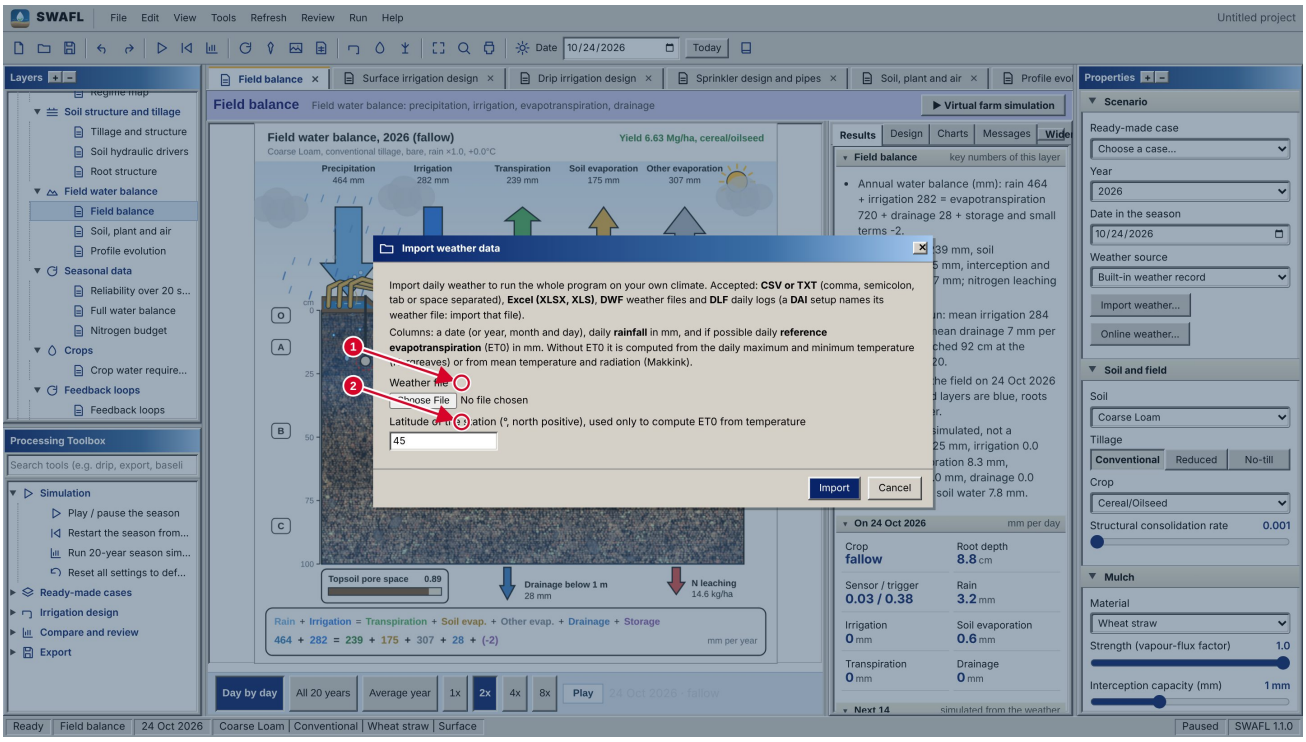


Figure 19.1: The weather import dialog.

Accepted files.

Format	Reading
CSV or TXT	comma, semicolon, tab or space separated; the separator is found from the first lines
Excel (XLSX, XLS, ODS)	the first sheet
DWF (Daisy weather file)	header lines, a line of dashes, the column names, a units line, then the data; hourly values are summed to days
DLF (Daisy daily log)	as DWF
DAI (Daisy setup file)	holds no weather; the program reads the name of the weather file it uses and asks you to import that file

Columns. The columns are found by their names, in any order and with or without units:

Quantity	Names recognised
Date	`date`, `datetime`, `time`; or `year`, `month`, `day`; or `year` and `doy` (day of year)
Rainfall (mm)	`precip…`, `rain…`, `prcp`, `ppt`, `P`; values in inches when the header says `in` or `inch`
Reference evapotranspiration (mm)	`ET0`, `ETo`, `ref…evap…`, `PET`, `potential evap…`
Temperature (°C)	`Tmax`, `Tmin`, `Tmean`, `air temp…`, `Tavg`
Global radiation	`rad`, `srad`, `globrad`, `solar` (MJ m^{-2} d^{-1}, or W m^{-2} when the values exceed 60)

Reference evapotranspiration. When the file has no ET0 column, it is computed. With daily maximum and minimum temperature, by Hargreaves:

$$ET_0 = 0.0023\,(T_{\text{mean}} + 17.8)\,\sqrt{T_{\text{max}} - T_{\text{min}}}\;R_a\;0.408 \quad [\text{mm}\,\text{d}^{-1}], \tag{19.1}$$

Provenance (Eq. 19.1): **[Literature]**: Hargreaves and Samani (1985), FAO-56 Eq. 52; R_a the extraterrestrial radiation (MJ m^{-2} d^{-1}) at the latitude typed in the dialog (FAO-56 Eq. 21).

With daily mean temperature and global radiation R_s, by Makkink:

$$ET_0 = 0.61\,\frac{\Delta}{\Delta+\gamma}\,\frac{R_s}{2.45} - 0.12, \qquad \Delta = \frac{4098\cdot 0.6108\,e^{17.27T/(T+237.3)}}{(T+237.3)^2}, \quad \gamma = 0.066\ \text{kPa}\,\text{K}^{-1}, \tag{19.2}$$

Provenance (Eq. 19.2): **[Literature]**: Makkink (1957), as used in the Netherlands and Denmark.

Without either, the seasonal pattern and the annual demand of the built-in climate are used, and the dialog says so.

19.3 Putting the weather on the twenty-year grid

The program works on twenty years of 365 days. The weather is put on that grid as follows:

1. The **last twenty complete years** of the file are used (a year is complete with at least 300 days). With fewer years, the available years are repeated in turn, and the dialog says how many.
2. Leap days are dropped. A missing day of rain is filled from the same day of another year of the file; a missing day of ET0 from the mean of that day of the year.
3. The mean annual rainfall $\bar{P}$ and reference evapotranspiration $\overline{ET_0}$ are computed.

The daily model (Engine 2) then uses the daily rain P_k of your weather, scaled to the rainfall scale of Properties, and the daily evaporative demand in proportion to your daily ET0:

$$P_k^{\text{model}} = P_k\,\frac{\rho}{\rho_{\text{ref}}}, \qquad e_k = \frac{ET_{0,k}}{\overline{ET_0}/365} \quad (\text{the shape of the demand through the year}), \tag{19.3}$$

Provenance (Eq. 19.3): **[Engineered]**: the daily sequence of your weather replaces the generated rainfall (Eqs. 5.15–5.16) and the seasonal demand weight.

with ρ the rainfall scale of Properties and ρ_{ref} the step chosen for your weather by Eq. (19.4). The annual model (Engine 1) works on the simulated steps of rainfall and warming; it is set to the steps nearest your weather,

$$\rho_{\text{ref}} = \underset{\rho\in\{0.5,\ldots,1.4\}}{\text{nearest}}\ \frac{\bar{P}}{P_{\text{rec}}}, \tag{19.4}$$

Provenance (Eq. 19.4): **[Engineered]**: nearest simulated step; P_{rec} the mean annual rain of the built-in record (464 mm).

$$\Delta T_{\text{ref}} = \underset{\Delta T\in\{-1,\ldots,3.5\}}{\text{nearest}}\ \frac{\overline{ET_0}/E_{\text{rec}} - 1}{0.035}, \tag{19.5}$$

Provenance (Eq. 19.5): **[Engineered]**: the evaporative demand expressed as the warming that gives it (3.5 % per °C, Eq. 5.26); E_{rec} the mean annual reference ET of the built-in record (about 870 mm).

When your weather lies outside the simulated range (for example 1,500 mm of rain, more than 1.4 times the record), the annual results use the nearest step and the dialog says so; the daily layers use your weather as it is.

The crop water requirement (Chapter 21) uses your daily rain and ET0 directly.

The years. With your weather, the twenty years of the grid are the calendar years of your record, so a working date inside them shows that very day of your weather (Eq. 4.2).

After the import a report lists the years used, the days in the file, the mean annual rain and ET0, how ET0 was obtained, the rainfall and temperature steps of the annual model, the columns read, and any notes (repeated years, filled gaps, values outside the simulated range).

19.4 The weather of a place

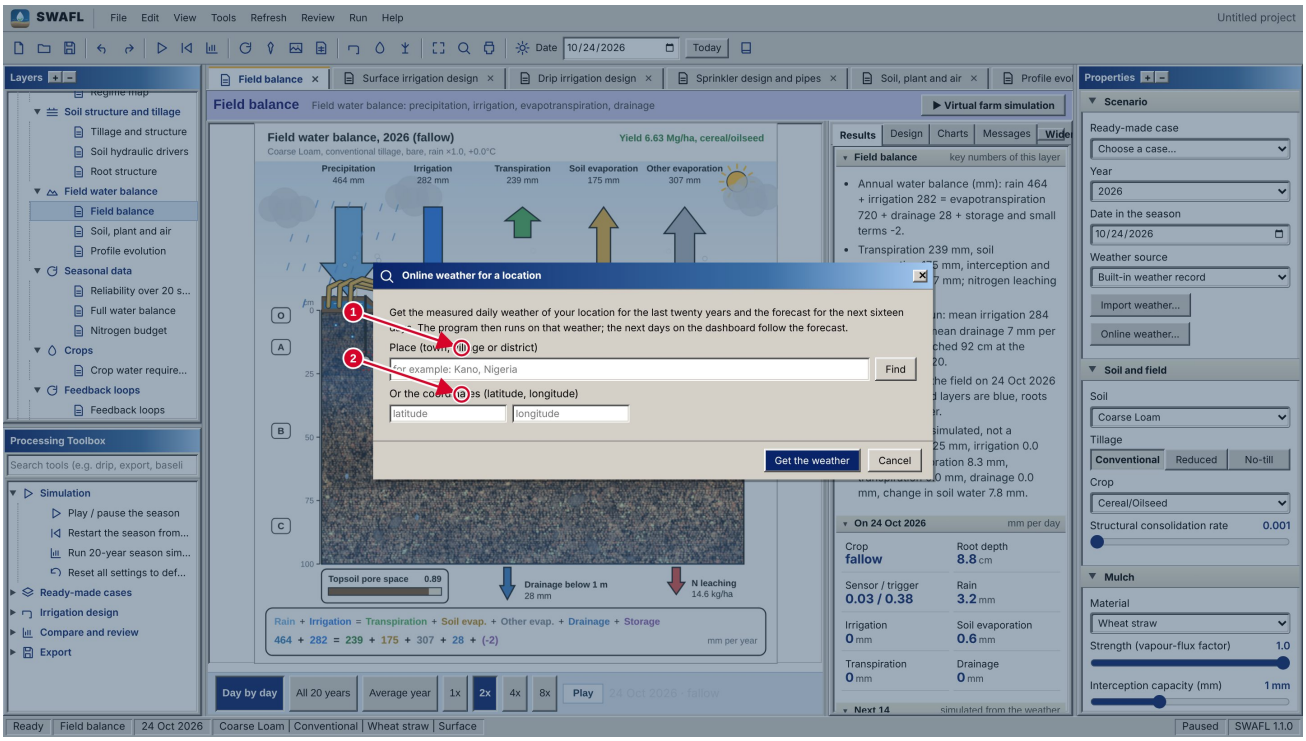


Figure 19.2: The online weather dialog.

Type the name of a town, village or district and press **Find**, then choose the place from the list; or type its latitude and longitude. **Get the weather** fetches:

- the measured daily rain, reference evapotranspiration (FAO Penman–Monteith) and maximum and minimum temperature of the place from 1 January twenty years ago to two days ago;
- the forecast for the next 16 days;
- for the rest of the current year after the forecast, the same days of the most typical year of the record (median rain).

The weather is then put on the grid as above. On the dashboard, the *Next 14 days* block says *from the weather forecast* while the days fall in the forecast.

The online weather needs the internet; everything else does not. If the place search or the weather service cannot be reached, the dialog says so and **the current weather stays in use**; nothing else changes. The service is a free public weather archive that needs no key.

19.5 Switching back

Choose *Built-in weather record* in Properties ▸ Scenario ▸ *Weather source*. The rainfall scale and temperature shift you had before the import come back. The imported file and the online weather stay available in the list until the program is closed.

Chapter 20

The real field: satellite indices, image analysis, water sources and the irrigation system

Tools ▸ *Field on the map…*, or **Locate on the map** in the field sketch, opens the field on the real map. There you find the farm, demarcate the field, and the program reads the real field from open data:

- the **vegetation and water indices** NDVI, EVI, SAVI, NDRE and NDWI from the **Sentinel-2** satellite (Copernicus);
- an **image analysis (computer vision) of the satellite photo** of the field: its land cover (dense and sparse vegetation, bare soil, water, buildings and roads, shadow) and the visible-band indices VARI, GLI, ExG and NGRDI;
- **the site**: the water features around the field (OpenStreetMap), water seen in the photo, the ground elevation and slope (Copernicus DEM), and the soil texture (SoilGrids);
- **suggestions**: where to take the water from, and which irrigation system suits the field, with the reasons.

Everything is computed for the demarcated outline only, after it is drawn, and follows the outline when you change it. All of it needs the internet; without it the program says what it could not reach and uses the simulated indices of SWAFL.

> The pictures of this chapter were made without internet, on a generated test image and test answers of the open-data services, so that every path of the program could be shown. On your computer the real satellite photo and the real data of your field appear in the same places.

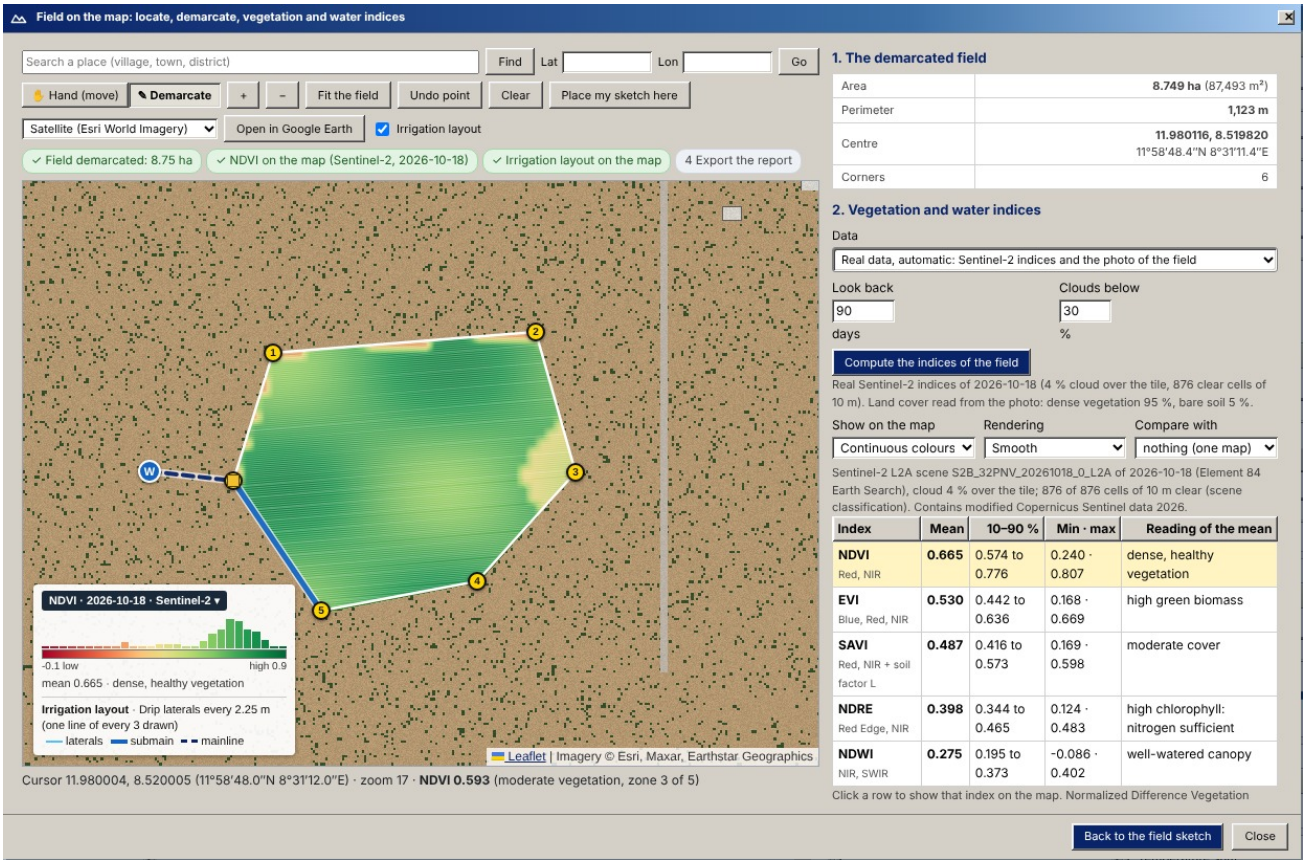


Figure 20.1: The field on the map with its real Sentinel-2 NDVI cut to the outline, the legend on the map (index, date, scale, spread of the field), the irrigation layout and the value under the cursor (test image).

20.1 Finding and demarcating the field

Find the farm. Type a village, town or district and press *Find* (place search of Open-Meteo, the same as for the online weather), or type the latitude and longitude and press *Go*. If the online weather of a place is in use, the map opens there. The background is the satellite image (Esri World Imagery, with place names), the satellite image with roads and names, or the map (OpenStreetMap). **Open in Google Earth** opens Google Earth on the centre of the field, to look at it in 3D.

2. Demarcate the field. In *Demarcate* mode, click the corners of the field on the image. Drag a corner to move it; right-click a corner to remove it; *Undo point* and *Clear* work as in the sketch. The *Hand* moves the map; **+** and **−** or the mouse wheel zoom; *Fit the field* shows the whole field. *Place my sketch here* puts the field of the sketch, at its true size, on the centre of the map. The panel gives the area, the perimeter and the centre, in decimal degrees and in degrees, minutes and seconds (WGS 84). On the sphere of radius $R = 6\,378\,137$ m,

$$A = \frac{R^2}{2}\left|\sum_i (\lambda_{i+1}-\lambda_i)(2+\sin\varphi_i+\sin\varphi_{i+1})\right|, \qquad P = \sum_i 2R\arcsin\sqrt{\sin^2\frac{\Delta\varphi}{2}+\cos\varphi_i\cos\varphi_{i+1}\sin^2\frac{\Delta\lambda}{2}}, \tag{20.1}$$

Provenance (Eq. 20.1): **[Literature]**: spherical polygon area (Chamberlain and Duquette, 2007) and the haversine distance; φ latitude, λ longitude in radians.

The bar of steps above the map shows what is done (with a tick) and what comes next: the field

demarcated, the index on the map, the irrigation layout on the map, the report.

20.2 The indices of the field

Choose the data and press **Compute the indices of the field**:

Data	What it gives
Real data, automatic (the default)	The Sentinel-2 indices of the most recent clear scene, and the image analysis of the satellite photo (land cover). If Sentinel-2 cannot be read, the indices of the photo; if neither, the simulated indices. The message under the button says what was used and what could not be reached.
Sentinel-2 only	NDVI, EVI, SAVI, NDRE and NDWI.
Satellite photo only	The land cover and VARI, GLI, ExG and NGRDI.
Simulated by SWAFL	No internet needed (below).

Sentinel-2 L2A (Copernicus open data). The catalogue of Element 84 Earth Search is searched for the scenes over the field of the last *Look back* days with less than *Clouds below* % cloud over the tile; if it cannot be reached, the catalogue of Microsoft Planetary Computer is used. The most recent scenes are tried in turn until one is clear over the field. The bands B02 (blue), B04 (red), B05 (red edge), B08 (NIR) and B11 (SWIR) are read for the field only (cloud-optimised GeoTIFF, in the UTM zone of the scene), the offset of processing baseline 4 is removed, and cloud, cloud shadow, snow and no-data pixels are left out with the scene classification (SCL). Each cell of 10 m inside the outline gets

$$\mathrm{NDVI}=\frac{\rho_N-\rho_R}{\rho_N+\rho_R},\qquad \mathrm{EVI}=2.5\,\frac{\rho_N-\rho_R}{\rho_N+6\rho_R-7.5\rho_B+1},\quad \mathrm{SAVI}=\frac{(1+L)(\rho_N-\rho_R)}{\rho_N+\rho_R+L},\ L=0.5,$$
$$\mathrm{NDRE}=\frac{\rho_N-\rho_{RE}}{\rho_N+\rho_{RE}},\quad \mathrm{NDWI}=\frac{\rho_N-\rho_{SW}}{\rho_N+\rho_{SW}}. \tag{20.2}$$

Provenance (Eq. 20.2): **[Literature]**: Rouse et al. (1974), Huete et al. (2002), Huete (1988), Barnes et al. (2000), Gao (1996). NDWI after Gao is also called NDMI.

Index	Bands	What it follows
NDVI	Red, NIR	Drought tracking, baseline crop health
EVI	Blue, Red, NIR	Resists saturation in thick canopies (dense crops)
SAVI	Red, NIR, soil factor L	Removes the soil background (dry land, early crop)
NDRE	Red edge, NIR	Deep canopy; nitrogen management, precision farming
NDWI (NDMI)	NIR, SWIR	Water inside the canopy; crop water stress, fire risk

The satellite photo (Esri World Imagery, about 0.3 to 1 m per pixel; the capture date varies by place). The photo of the field is put together from its tiles at the finest zoom that fits about 1700 pixels across; tiles without imagery are left out. Its pixels have only red, green and blue, so it gives the visible-band indices (shadows left out), averaged over the cells of the field:

$$\mathrm{VARI}=\frac{G-R}{G+R-B},\qquad \mathrm{GLI}=\frac{2G-R-B}{2G+R+B},\qquad \mathrm{ExG}=2g{-}r{-}b\ \left(r=\tfrac{R}{R+G+B},\ldots\right),\qquad \mathrm{NGRDI}=\frac{G-R}{G+R}. \tag{20.3}$$

Provenance (Eq. 20.3): **[Literature]**: Gitelson et al. (2002), Louhaichi et al. (2001), Woebbecke et al. (1995), Tucker (1979).

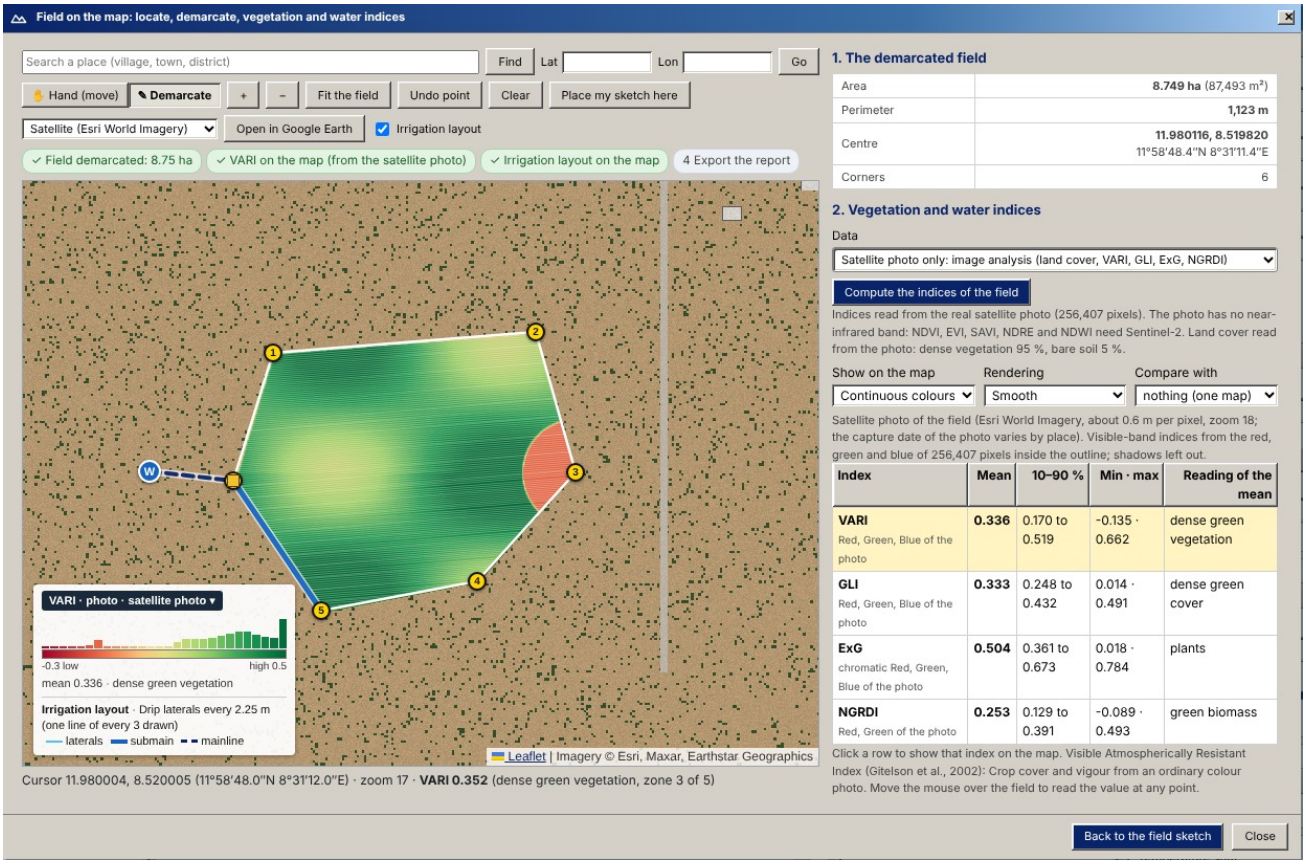


Figure 20.2: The indices read from the real satellite photo: VARI cut to the field, with the table of VARI, GLI, ExG and NGRDI (test image).

The land cover: image analysis of the photo. Every pixel of the photo is described by six features: its brightness L, saturation S, excess green ExG, VARI, blueness $b-(r+g)/2$ and texture (the standard deviation of the brightness over a 5×5 window). The pixels are grouped into seven groups by *k*-means clustering of the standardised features (k-means++ start, 14 iterations), and each group is named from its mean features:

$$\text{class} = \begin{cases} \text{shadow} & L < 0.10 \\ \text{water} & \left(b - \frac{r+g}{2} > 0.025,\ \text{ExG} < 0.03,\ L < 0.55\right) \text{ or } \left(L < 0.2,\ \text{smooth},\ \text{ExG} < 0.02\right) \\ \text{dense vegetation} & \text{ExG} > 0.085 \text{ or VARI} > 0.16 \\ \text{sparse vegetation} & \text{ExG} > 0.035 \text{ or VARI} > 0.05 \\ \text{buildings, roads} & \left(S < 0.13,\ L > 0.45\right) \text{ or } \left(\text{rough},\ S < 0.2\right) \\ \text{bare soil} & \text{otherwise} \end{cases} \tag{20.4}$$

Provenance (Eq. 20.4): **[Engineered]**: classical computer vision (colour indices, texture, *k*-means, rules) and a 3×3 majority filter; not a trained neural network. The shares and areas are those of the pixels inside the outline.

Choose *Land cover read from the photo* under *Show on the map* to see it; the table gives the area and share of each cover. Buildings, roads and water inside the outline are worth leaving out of the irrigated area.

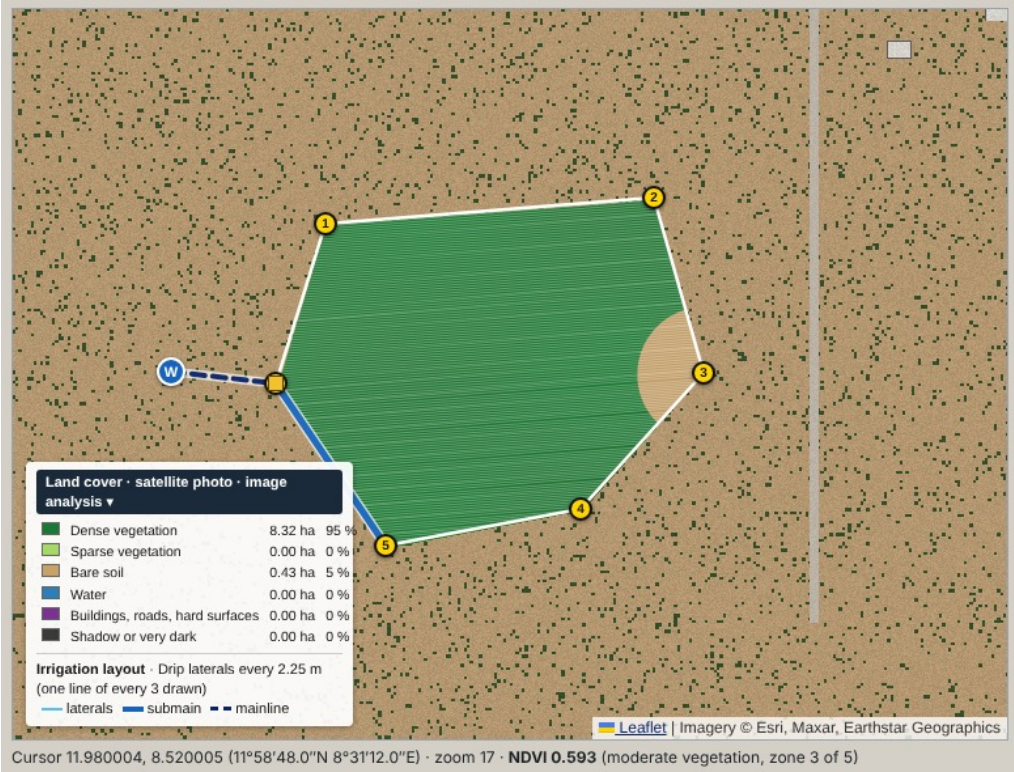


Figure 20.3: The land cover read from the photo, cut to the field, with the area of each cover in the legend (test image).

Simulated by SWAFL (no internet), on the working date or at the peak of the crop: each cell mixes the reflectance of the crop and of the soil by the crop cover c of the daily model, $\rho = c\,\rho_{\text{veg}} + (1 - c)\,\rho_{\text{soil}}$; the soil is darker when wet, the canopy reflects less NIR and more SWIR under stress and less red edge when nitrogen is short; vigour and soil water vary in patches of 25 to 40 m tied to the ground, and the wetting falls from the inlet end to the tail with the uniformity of the design. **[Engineered]**.

The table gives, for each index, the mean, the 10 and 90 % values, the minimum and maximum over the field, and what the mean means. Click a row to show that index on the map.

The index follows your outline. The colours are cut exactly to the outline and laid over the real map. When you add, drag or remove a corner, the simulated indices are computed again at once; satellite and photo data are read again when you press *Compute*.

What the map shows.

- *Continuous colours* or *Management zones (5 classes)*: the zones split the values of the field by natural breaks (Jenks), drawn from the field smoothed over about 9 m so that each zone is a patch you can manage. The table under the indices gives, for each zone, its range, mean, area (ha) and share of the field.
- *Smooth* (interpolated between cells) or *Pixels* (each cell as measured, 10 m for Sentinel-2).
- *Compare with*: a second index side by side; the two maps move and zoom together.
- The legend in the lower left corner of the map gives the index, the date and the source, the colour scale (or the zones with their area), the spread of the field as a histogram and the mean. Click its dark title to fold it.
- Move the mouse over the field: the line under the map gives the coordinates and the value of the index at that point, its reading and its zone (and the value of the second index when two are shown).
- *Irrigation layout* draws the system of the field sketch on the map: the furrows, borders, basins, drip laterals or sprinklers (light lines; one in every few when they are very close),

the submain or head ditch along the edge that holds the inlet (thick blue), the mainline from the water source **W** to the inlet (dashed), and the inlet (yellow square). It appears after *Use this field in the sketch and the designs* (or *Place my sketch here*), and it follows the outline when you move a corner. Choose the system, the spacing and the source in the sketch.

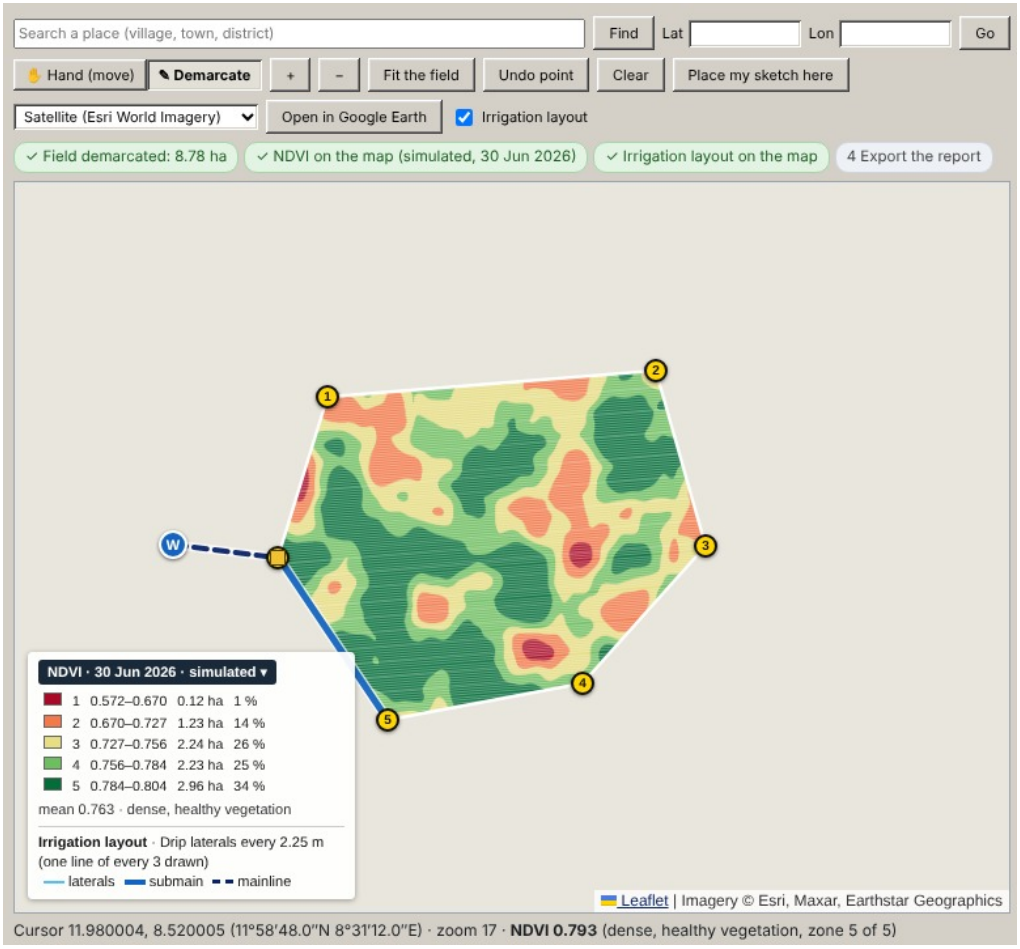


Figure 20.4: Management zones of NDVI (simulated, without internet).

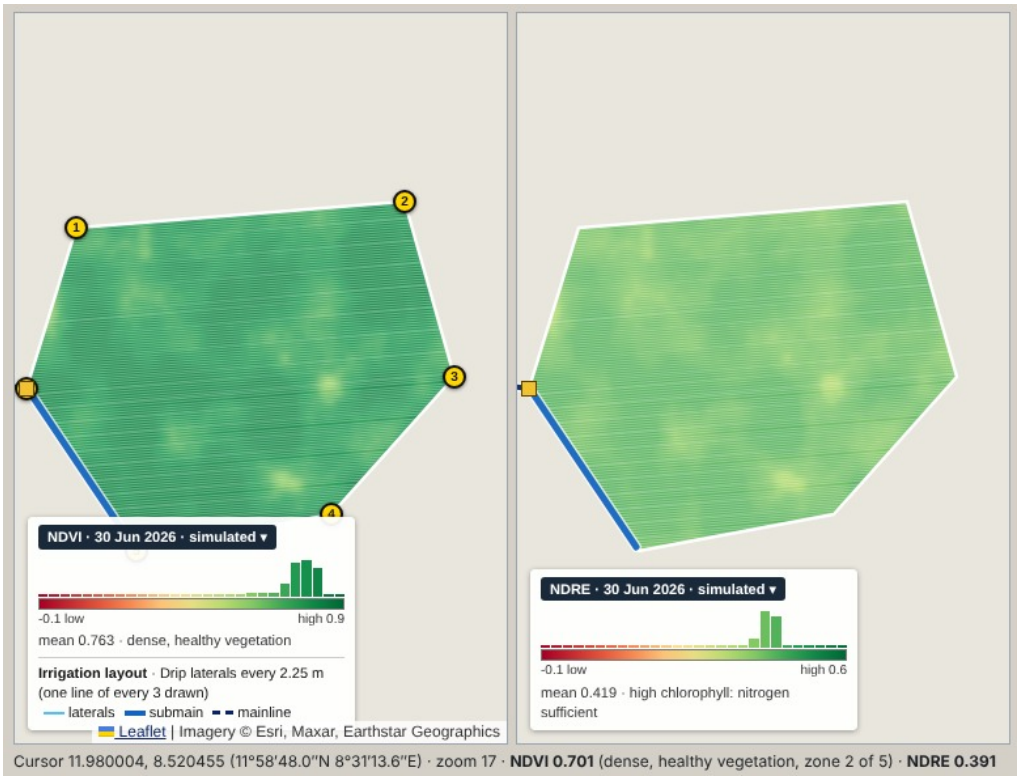


Figure 20.5: Two indices side by side (simulated, without internet).

20.3 The site: water sources and the irrigation system

Analyse the site (real open data) reads, around the demarcated field:

- the **water features** within 3 km from OpenStreetMap (Overpass API): rivers, streams, canals, ditches and drains, lakes, ponds and reservoirs, wells, boreholes, springs and water points;
- **water seen in the photo** within 1.5 km: groups of water pixels of the image analysis (Eq. 20.4) larger than about 600 m², not already mapped;
- the **ground elevation** (Copernicus DEM GLO-90, through Open-Meteo) on a grid over the field, at its corners and at the nearest point of each source;
- the **soil** at the centre of the field from ISRIC SoilGrids (sand, silt, clay and organic carbon, 0 to 60 cm), with its USDA texture class. **Use the nearest SWAFL soil** sets the soil of the project to the soil of the program closest in sand and clay.

The slope of the field is that of the plane fitted to the elevations by least squares,

$$z = a + b\,x + c\,y, \qquad s = 100\sqrt{b^2 + c^2}\ \ [\%], \tag{20.5}$$

Provenance (Eq. 20.5): **[Literature]**: least-squares plane; x, y in metres east and north of the centre. The direction the ground falls to is that of $-(b, c)$.

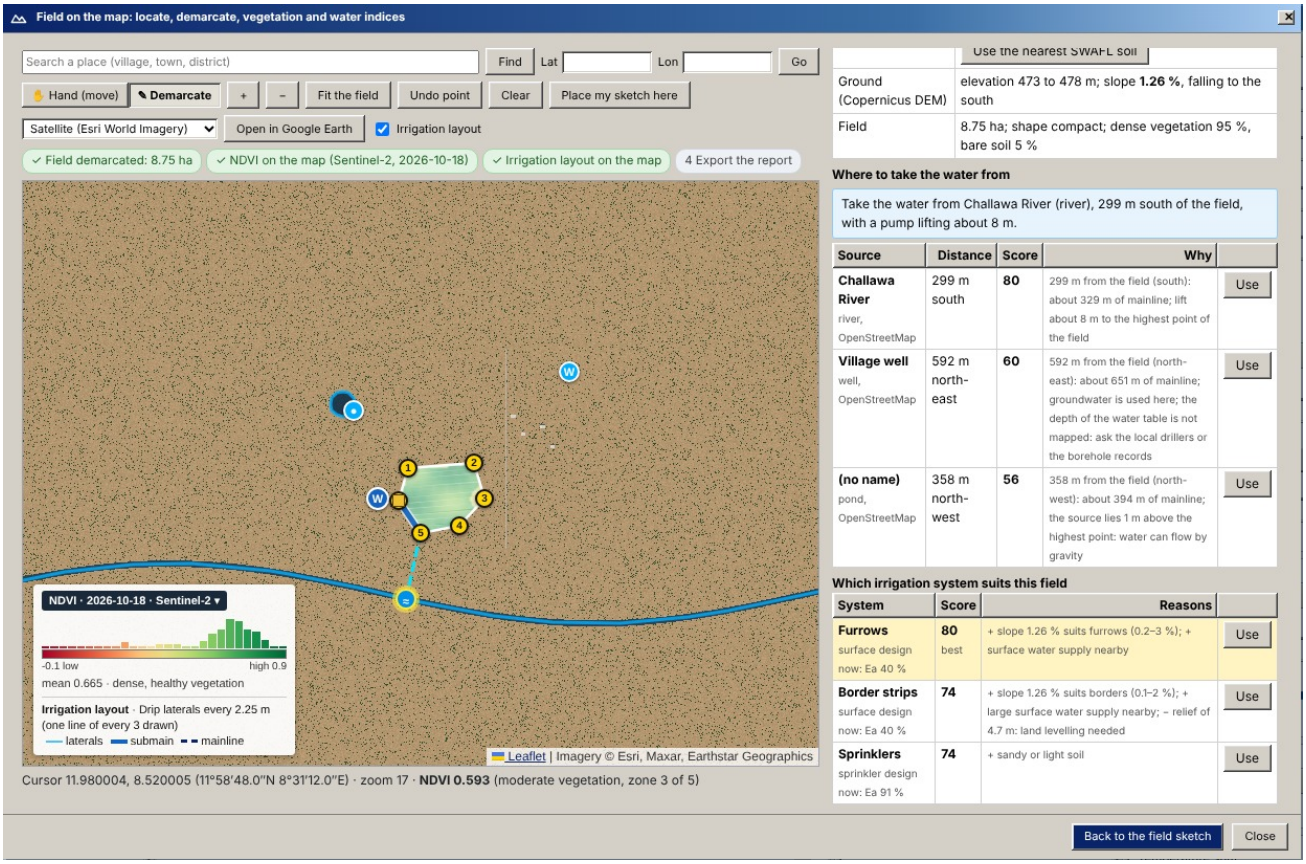


Figure 20.6: The site analysed: the water sources on the map (the best one joined to the field), the soil, the slope, the advice on the water and the ranking of the systems (test data).

Where to take the water from. Each source gets a score

$$\text{score} = B_{\text{type}} - 30\,[\text{seasonal}] - \min\!\left(45,\ d/60\right) - \min\!\left(20,\ 0.8\,\Delta z\right), \tag{20.6}$$

Provenance (Eq. 20.6): **[Engineered]**: B = 92 river, 90 canal, 86 reservoir, 82 lake, 74 borehole, 70 well, 62 pond, 60 spring, 58 stream, 50 water seen in the photo, 42 ditch, 30 drain; d the distance from the field edge (m); Δz the lift from the source to the highest point of the field (m), not counted for wells, whose water table depth is not mapped.

The advice names the best surface source within 1.5 km (with its distance, direction and lift, or *by gravity*); without one, a borehole at the edge of the field when wells are mapped nearby, or a hydrogeological survey first. **Use** puts the source into the field sketch, at its place, with its kind and lift, so the mainline length and the pump head go into the designs.

Which irrigation system suits this field. Each system starts from a base score and gains or loses points for the conditions of the field, after the FAO and USDA guidance for choosing an irrigation system:

System	Gains	Loses
Basins	level ground ($s \le 0.3$ %), clay, large surface supply, rice, orchards	slope over 1 %, sand over 65 %, groundwater only, irregular shape
Border strips	slope 0.1 to 2 %, loam to clay loam, close-growing crops, surface supply	slope over 2 %, sand, groundwater only, irregular shape, relief to level
Furrows	slope 0.2 to 3 %, row crops, surface supply	slope over 3 %, sand, close-growing crops, irregular shape

System	Gains	Loses
Sprinklers	rolling ground (2 to 15 %), light soil, close-growing crops, irregular shape, borehole supply	slope over 20 %, heavy clay, tall trees, level fields with plenty of surface water
Drip	sandy soil, scarce or pumped water, slopes, row crops and orchards	close-growing crops, very large fields, hard surfaces in the outline

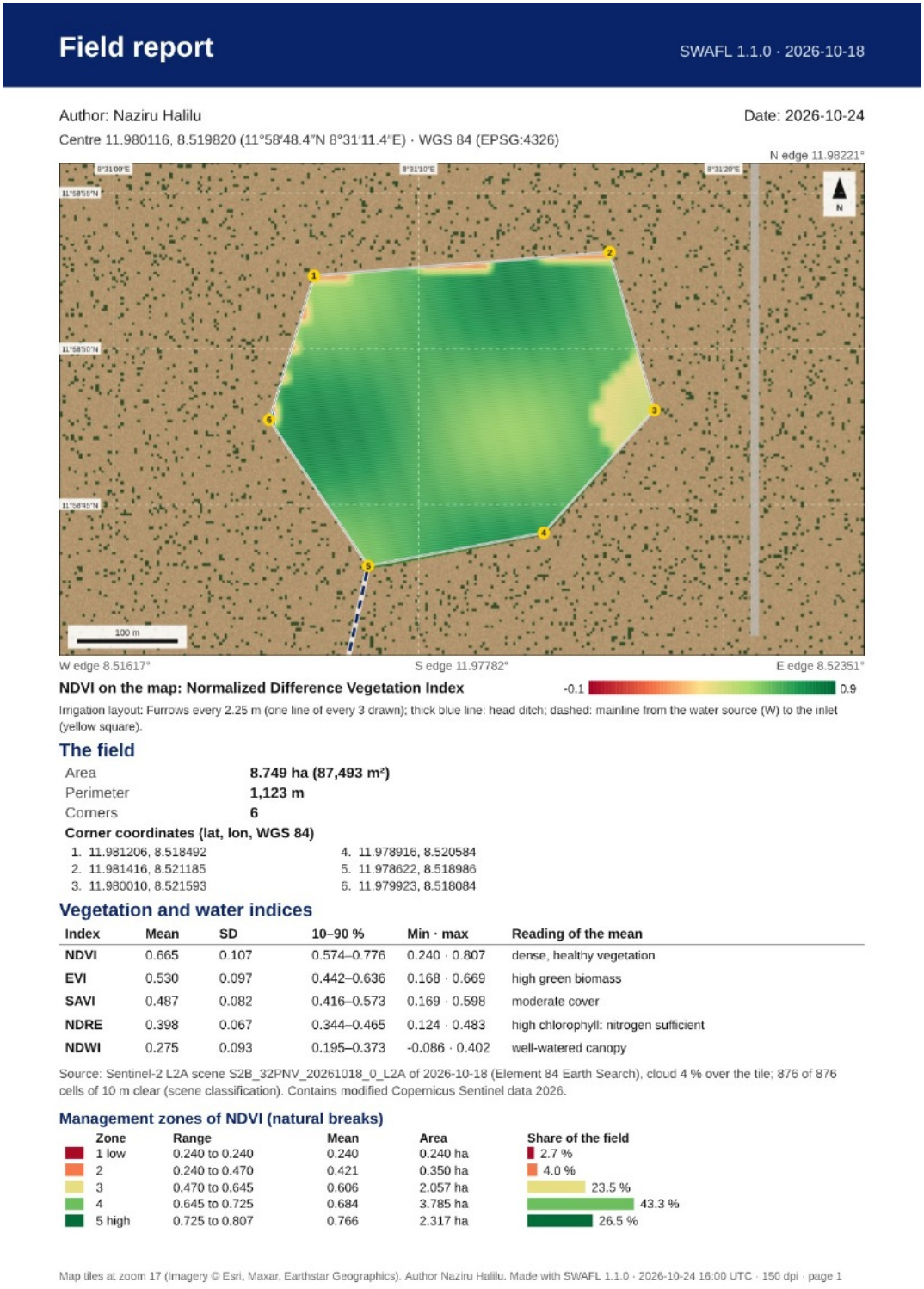
The table lists each system with its score, the reasons, and the application efficiency of the SWAFL design of that system as it stands. **Use** sets the system (and the slope of the site) in the field sketch; then *Turn the sketch into the design* and run it.

20.4 Use and export

Use this field in the sketch and the designs makes the demarcated field the sketch (true lengths, centre coordinates kept), and *Irrigation layout* draws its system on the map. *Online weather of this field* downloads the daily weather of the centre of the field and runs the program on it. The report has a **title, an author and an organisation**, a **resolution of 96, 150, 300 or 600 dpi** and a format:

- **PDF (A4)**: page 1, the map of the chosen index (continuous, zones or land cover) on the satellite photo, cut to the field, with the irrigation layout, a grid of latitude and longitude in degrees, minutes and seconds, north arrow and scale bar, the area, perimeter, centre and corner coordinates, the table of the indices and the zones; page 2, the indices side by side with their spread and what they say together; page 3 (after an image analysis or a site analysis), the land cover, the soil, the slope, the water sources on the map with the advice, and the ranking of the systems. The title, author, coordinates and keywords are also in the properties of the PDF.
- **PNG (A4)**, one image per page, with the resolution written in the file so that it prints at the chosen dpi.
- **GeoJSON** (the field polygon in WGS 84 with the area, the author and the means of the indices) and **CSV** (the corners and every statistic), for a GIS or a spreadsheet.

Every page carries its sources: photo © Esri, Maxar, Earthstar Geographics; map and water features © OpenStreetMap contributors; contains modified Copernicus Sentinel data; Copernicus DEM; ISRIC SoilGrids.



Field report

SWAFL 1.1.0 · 2026-10-18

Author: Naziru Halilu

Date: 2026-10-24

Centre 11.980116, 8.519820 (11°58′48.4″N 8°31′11.4″E) · WGS 84 (EPSG:4326)

W edge 8.51617° S edge 11.97782° E edge 8.52351°

NDVI on the map: Normalized Difference Vegetation Index -0.1 0.9

Irrigation layout: Furrows every 2.25 m (one line of every 3 drawn); thick blue line: head ditch; dashed: mainline from the water source (W) to the inlet (yellow square).

The field

Area	**8.749 ha (87,493 m²)**
Perimeter	**1,123 m**
Corners	**6**

Corner coordinates (lat, lon, WGS 84)

1. 11.981206, 8.518492
2. 11.981416, 8.521185
3. 11.980010, 8.521593
4. 11.978916, 8.520584
5. 11.978622, 8.518986
6. 11.979923, 8.518084

Vegetation and water indices

Index	Mean	SD	10–90 %	Min · max	Reading of the mean
NDVI	0.665	0.107	0.574–0.776	0.240 · 0.807	dense, healthy vegetation
EVI	0.530	0.097	0.442–0.636	0.168 · 0.669	high green biomass
SAVI	0.487	0.082	0.416–0.573	0.169 · 0.598	moderate cover
NDRE	0.398	0.067	0.344–0.465	0.124 · 0.483	high chlorophyll: nitrogen sufficient
NDWI	0.275	0.093	0.195–0.373	-0.086 · 0.402	well-watered canopy

Source: Sentinel-2 L2A scene S2B_32PNV_20261018_0_L2A of 2026-10-18 (Element 84 Earth Search), cloud 4 % over the tile; 876 of 876 cells of 10 m clear (scene classification). Contains modified Copernicus Sentinel data 2026.

Management zones of NDVI (natural breaks)

Zone	Range	Mean	Area	Share of the field
1 low	0.240 to 0.240	0.240	0.240 ha	2.7 %
2	0.240 to 0.470	0.421	0.350 ha	4.0 %
3	0.470 to 0.645	0.606	2.057 ha	23.5 %
4	0.645 to 0.725	0.684	3.785 ha	43.3 %
5 high	0.725 to 0.807	0.766	2.317 ha	26.5 %

Map tiles at zoom 17 (Imagery © Esri, Maxar, Earthstar Geographics). Author Naziru Halilu. Made with SWAFL 1.1.0 · 2026-10-24 16:00 UTC · 150 dpi · page 1

Figure 20.7: The field report, page 1: NDVI of Sentinel-2 on the photo, the coordinates, the indices and the zones (test data).

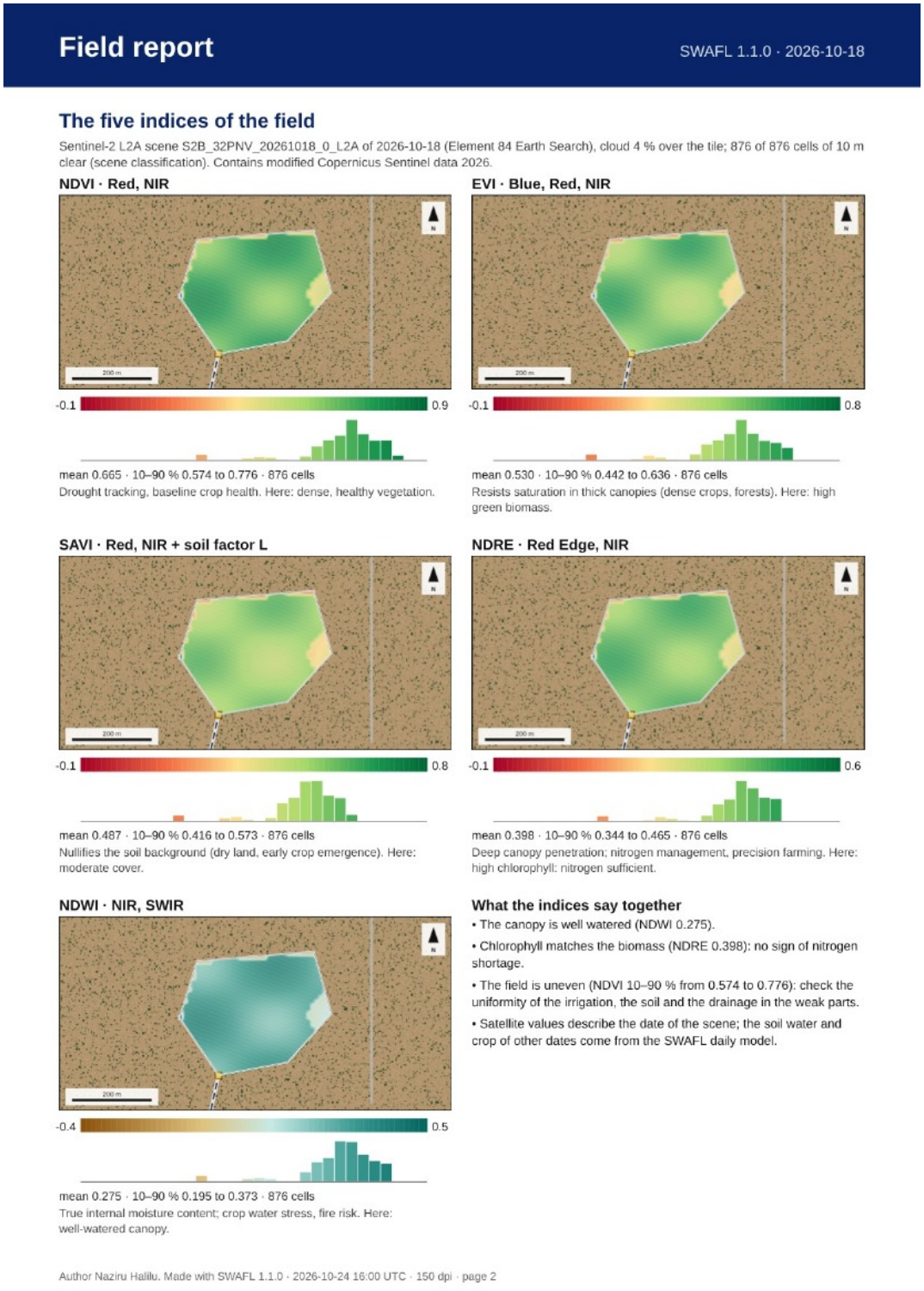
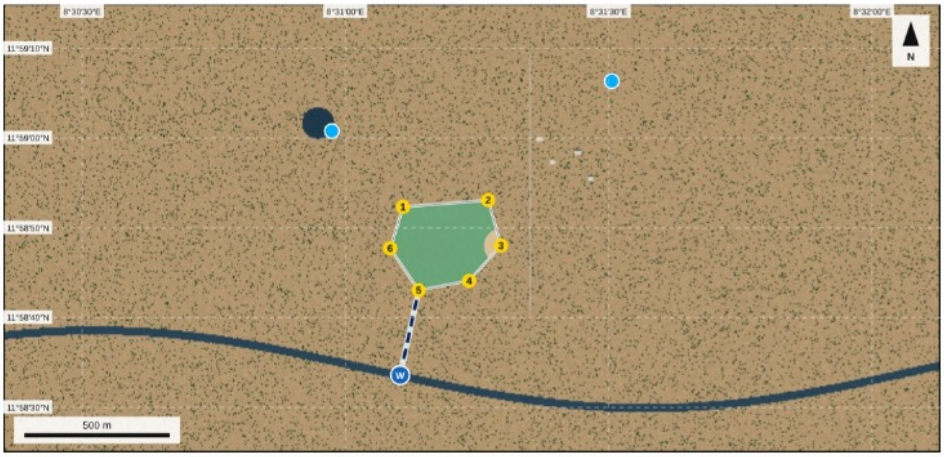


Figure 20.8: The field report, page 2: the five indices.

Field report

SWAFL 1.1.0 · the site

The site: land cover, soil, slope, water sources and the irrigation system



Land cover read from the satellite photo by image analysis (colour and texture of every pixel, k-means clustering, named by rules, majority filter; 0.6 m per pixel):

- Dense vegetation: 95.1 % (8.32 ha)
- Sparse vegetation: 0.0 % (0.00 ha)
- Bare soil: 4.9 % (0.43 ha)
- Water: 0.0 % (0.00 ha)
- Buildings, roads, hard surfaces: 0.0 % (0.00 ha)
- Shadow or very dark: 0.0 % (0.00 ha)

Soil (SoilGrids)	sandy loam: sand 64 %, silt 19 %, clay 17 %
Ground (Copernicus DEM)	elevation 473 to 478 m, slope 1.26 % falling to the south

Where to take the water from

Take the water from Challawa River (river), 299 m south of the field, with a pump lifting about 8 m.

Source	Distance	Score	Why
Challawa River river	299 m south	**80**	299 m from the field (south): about 329 m of mainline; lift about 8 m to the highest point of the field
Village well well	592 m north-east	**60**	592 m from the field (north-east): about 651 m of mainline; groundwater is used here; the depth of the water table is not mapped: ask the local drillers or the borehole records
(no name) pond	358 m north-west	**56**	358 m from the field (north-west): about 394 m of mainline; the source lies 1 m above the highest point: water can flow by gravity

Which irrigation system suits this field

Furrows (best)	**80**	+ slope 1.26 % suits furrows (0.2–3 %); + surface water supply nearby
Border strips	**74**	+ slope 1.26 % suits borders (0.1–2 %); + large surface water supply nearby; − relief of 4.7 m: land levelling needed
Sprinklers	**74**	+ sandy or light soil
Drip	**70**	+ sandy or light soil: drip keeps the water in the root zone
Basin (level basins)	**32**	− slope 1.3 %: basins need level ground; + large surface water supply nearby

Open data: water features © OpenStreetMap contributors (Overpass API); elevation Copernicus DEM GLO-90 (Open-Meteo); soil ISRIC SoilGrids 250 m; photo © Esri, Maxar, Earthstar Geographics. Scores follow the FAO and USDA guidance for choosing an irrigation system; check on the ground.

Map tiles at zoom 15 (Imagery © Esri, Maxar, Earthstar Geographics). Author Naziru Halilu. Made with SWAFL 1.1.0 · 2026-10-24 16:00 UTC · 150 dpi · page 3

Figure 20.9: The field report, page 3: the land cover, the soil, the slope, the water sources and the systems (test data).

Chapter 21

The crop library and the crop water requirement

21.1 The crops

Version 1.1 carries **48 crops** besides the simulated cereal/oilseed rotation. They are chosen in Properties ▸ Soil and field ▸ **Crop**, in four groups: *Staple food crops*, *Cash and oil crops*, *Vegetables*, and *Trees, vines and palms*. For each crop the library holds the crop coefficients $K_{c,\mathrm{ini}}$, $K_{c,\mathrm{mid}}$ and $K_{c,\mathrm{end}}$, the lengths of the four growth stages (initial, development, mid-season, late season), the plant height, the maximum rooting depth, the depletion fraction p, the yield response factor K_y, the ground cover and, for trees and vines, the planting densities or training systems. Field crops take their coefficients and stages from a field study that computed the daily crop water requirement with the FAO Penman–Monteith reference evapotranspiration; trees, vines and berries from the updated FAO-56 coefficients of fruit trees and vines. Rooting depth, p and K_y are the FAO-56 and FAO-33 tabulated values; where a crop is not tabulated the group value is used and the table says *group default*.

The staple and cash crops asked for in this release carry these coefficients:

Crop	$K_{c,\mathrm{ini}}$	$K_{c,\mathrm{mid}}$	$K_{c,\mathrm{end}}$	Note
Cassava	0.30	0.80	0.30	
Yam	0.50	1.10	0.85	stages and rooting depth: group default
Rice	1.05	1.20	0.90	bunded: no runoff in the season
Maize	0.30	1.20	0.60	
Sorghum	0.30	1.00	0.55	
Millet	0.30	1.00	0.30	
Cowpeas	0.40	1.05	0.50	
Groundnut	0.40	1.15	0.60	
Soybean	0.40	1.15	0.50	
Oil palm	0.90	0.95	0.95	perennial
Cocoa	1.00	1.05	1.05	perennial, high canopy shade

The complete library is listed in Table 21.2 at the end of this chapter.

21.2 The Crop water requirement layer

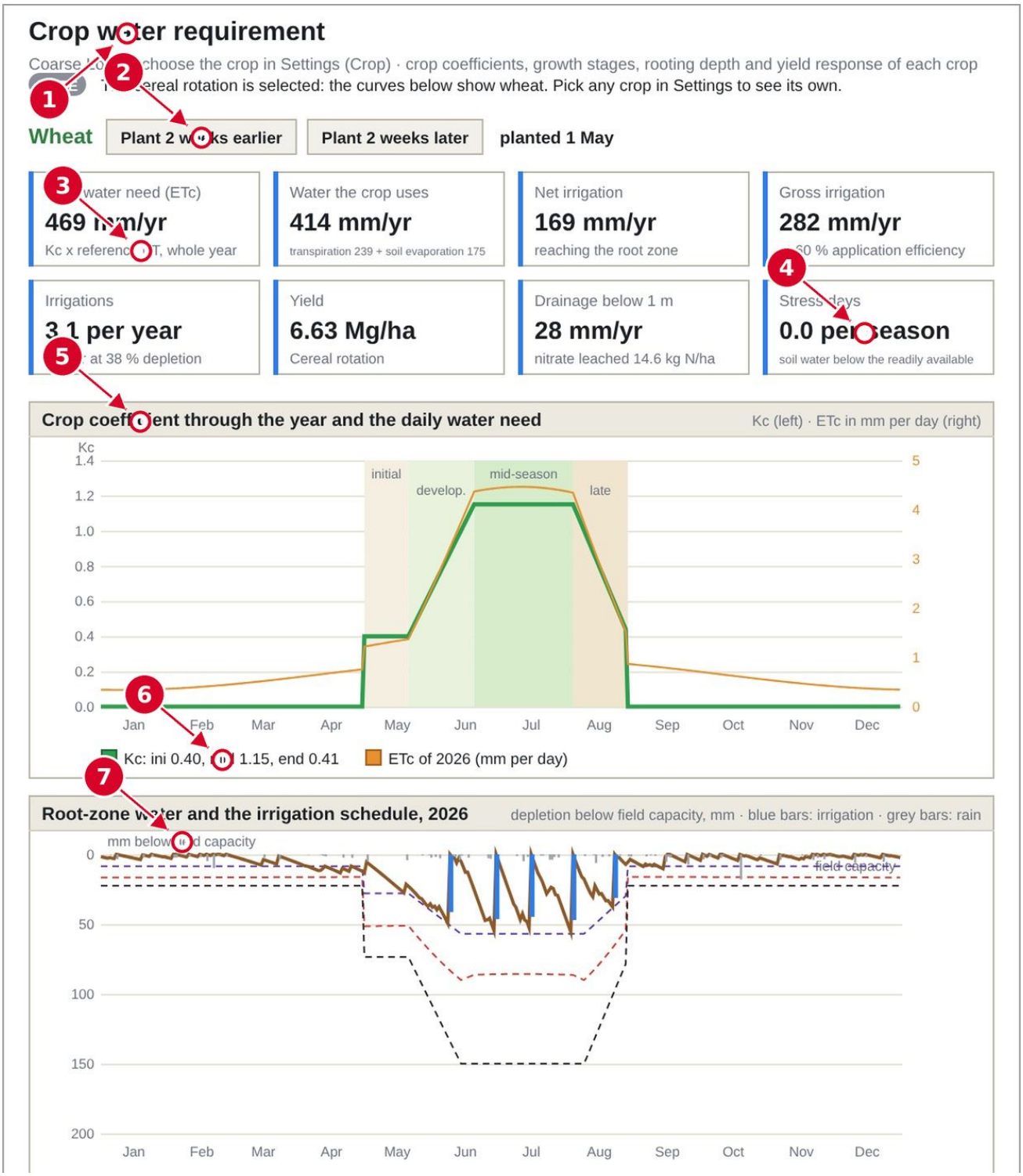


Figure 21.1: The Crop water requirement layer.

The layer *Crops* ▸ *Crop water requirement* shows, for the chosen crop on the soil, weather, trigger and method of the project:

- the **crop coefficient through the year** and the daily water need ET_c of the working year (the label reads *ETc of 2026*, following the year in Properties);
- the **root-zone water and the irrigation schedule** of the working year: the depletion below field capacity against the readily available water and the trigger, with every irrigation of the year;
- buttons to plant two weeks earlier or later (field crops) or to choose the planting density or training system (trees and vines);
- **every crop of the library on this soil**: the yearly crop water need, net irrigation, irrigations per year and relative yield, so that crops can be compared on the same field.

21.3 The daily balance of a crop

The layer computes, for each of the twenty years of the weather in use, the FAO-56 daily balance of the root zone. The crop coefficient follows the four stages,

$$K_c(t) = \begin{cases} K_{c,\mathrm{ini}} & t < L_1 \\ K_{c,\mathrm{ini}} + (K_{c,\mathrm{mid}} - K_{c,\mathrm{ini}})\frac{t-L_1}{L_2} & L_1 \le t < L_1 + L_2 \\ K_{c,\mathrm{mid}} & L_1 + L_2 \le t < L_1 + L_2 + L_3 \\ K_{c,\mathrm{mid}} + (K_{c,\mathrm{end}} - K_{c,\mathrm{mid}})\,\min\!\left(1, \frac{t-L_1-L_2-L_3}{L_4}\right) & \text{after} \end{cases} \tag{21.1}$$

Provenance (Eq. 21.1): **[Literature]**: FAO-56 crop coefficient curve (Allen et al., 1998, Fig. 25); t days after planting; evergreen crops repeat it every year.

and outside the season the bare soil has $K_c = 0.25$. The daily crop water need is

$$ET_{c,d} = K_c(d)\, ET_{0,d}, \tag{21.2}$$

Provenance (Eq. 21.2): **[Literature]**: FAO-56 single crop coefficient approach.

with $ET_{0,d}$ the daily reference evapotranspiration of the weather in use (your own weather when imported; otherwise the annual reference ET of each recorded year shaped by the seasonal weight, adjusted by 3.5 % per °C of the temperature shift). The roots of an annual crop grow with the canopy, perennial roots are at full depth:

$$z_r = \mathrm{clip}\!\left(15 + (R_{\max} - 15)\,\min\!\left(1, \frac{1.15\,K_c}{\max(0.3, K_{c,\mathrm{mid}})}\right);\ 15,\ R_{\max}\right)\ [\mathrm{cm}], \qquad R_{\max} \le 100\ \mathrm{cm}, \tag{21.3}$$

Provenance (Eq. 21.3): **[Engineered]**: root growth tied to the crop coefficient, limited to the 1 m of the model.

The total and readily available water and the depletion D_r of the root zone follow FAO-56,

$$\begin{aligned} &\mathrm{TAW} = \mathrm{AWC}\, z_r, \qquad p_d = \mathrm{clip}\big(p + 0.04\,(5 - K_c\, ET_0);\, 0.1,\, 0.8\big), \qquad \mathrm{RAW} = p_d\,\mathrm{TAW}, \\ &K_s = \begin{cases} 1 & D_r \le \mathrm{RAW} \\ \dfrac{\mathrm{TAW} - D_r}{(1-p_d)\,\mathrm{TAW}} & D_r > \mathrm{RAW} \end{cases}, \qquad ET_a = K_s\, ET_c, \\ &D_{r,d} = D_{r,d-1} - (P_d - RO_d) + ET_{a,d}, \qquad RO_d = 0.3\ \max(0, P_d - 35), \end{aligned} \tag{21.4}$$

Provenance (Eq. 21.4): **[Literature]**: FAO-56 Eqs. 82–85 (water stress coefficient, adjustment of p with the demand) and the root-zone balance Eq. 85; the runoff of heavy rain is **[Engineered]**.

with any negative D_r draining below the roots. Irrigation refills the root zone to field capacity on the day the depletion reaches the trigger of the project, $D_r \geq f_T$ TAW, with f_T the depletion at the trigger suction (Eq. 5.3): the net depth is $I_n = D_r$ and the gross depth $I = I_n / E_{a,m}$, with the method efficiency $E_{a,m}$ = 0.60 (surface), 0.75 (overhead) or 0.90 (drip). The relative yield of a year is

$$Y_{\text{rel}} = \text{clip}\Big(1 - K_y\Big(1 - \frac{\sum ET_a}{\sum ET_c}\Big);\ 0,\ 1\Big), \tag{21.5}$$

Provenance (Eq. 21.5): **[Literature]**: FAO-33 yield response (Doorenbos and Kassam, 1979); $K_y = 1$ when not tabulated, as the table says.

21.4 How a library crop changes every other layer

The annual model (Engine 1) is the result of the simulations of the cereal rotation. A library crop is transferred from it by the ratio of two runs of the balance above on the same soil, weather, trigger and method: one for the crop (c) and one for the cereal reference (r, FAO-56 small grains, K_c 0.30/1.15/0.25). With m the annual-model result of the cereal,

$$I = \begin{cases} I_m \ \text{clip}(I_c / I_r;\ 0,\ 4) & I_r > 5 \text{ mm} \\ I_c & \text{otherwise} \end{cases} \tag{21.6}$$

$$\begin{aligned} T &= \min\Big(0.9\,ET,\ \ T_m \,\text{clip}\Big(\frac{ET_{a,c}}{ET_{c,c}} \cdot \frac{ET_{c,c}}{ET_{c,r}} \cdot \Big(\frac{f_{c,c}}{f_{c,r}}\Big)^{0.3};\ 0.2,\ 3\Big)\Big), \\ E^* &= E_m \,\text{clip}\Big(\frac{1 - 0.7\,f_{c,c}}{1 - 0.7\,f_{c,r}};\ 0.6,\ 1.6\Big), \end{aligned} \tag{21.7}$$

$$D = \max\big(0.2\,D_m,\ D_m + 0.5\,(DP_c - DP_r)\big), \qquad ET = \text{clip}\big(P + I - D - \Delta S_m;\ 50,\ 5000\big), \tag{21.8}$$

$$(E,\ E_{\text{other}}) = (E^*,\ E^*_{\text{other}})\,\frac{ET - T}{E^* + E^*_{\text{other}}}, \qquad N_L = N_{L,m}\,\frac{D + 5}{D_m + 5}, \qquad Y = 100\,Y_{\text{rel},c}\ [\%], \tag{21.9}$$

Provenance (Eqs. 21.6–21.9): **[Engineered]**: transfer by the ratio of two FAO-56 balances; the clips keep the transfer within the range in which the cereal result is informative; the remainder of the balance is shared between soil evaporation and the other evaporation terms in proportion to their targets, so that the balance closes without inflating either.

with f_c the ground cover, $E^*_{\text{other}} = E_{\text{other},m}\,ET/ET_m$ and ΔS_m the storage change of the cereal result. The yield of a library crop is therefore shown as the relative yield in per cent, and the differences of yield in the sensitivity and feedback layers are in *% points*.

21.5 Table 21.2. The crop library

Table 21.2. The 48 crops of the library: crop coefficients, stage lengths (initial/development/mid/late, days), plant height (m), maximum rooting depth (cm), depletion fraction p and yield response factor K_y (— when not tabulated).

Crop	Group	$K_{c,ini}$	$K_{c,mid}$	$K_{c,end}$	Stages (d)	Height (m)	Root (cm)	p	K_y
Rice	field crop	1.05	1.20	0.90	15/30/45/30	0.8	75	0.20	—
Maize (grain)	field crop	0.30	1.20	0.60	15/30/40/20	1.5	135	0.55	1.25
Millet	field crop	0.30	1.00	0.30	15/25/40/25	1.5	150	0.55	—
Sorghum (grain)	field crop	0.30	1.00	0.55	20/35/40/30	1.5	150	0.55	0.90
Cassava	field crop	0.30	0.80	0.30	20/40/90/60	1.0	65	0.35	—
Yam	field crop	0.50	1.10	0.85	30/60/90/60	1.5	60	0.40	—
Cowpea (beans)	field crop	0.40	1.05	0.50	20/30/30/20	0.4	80	0.45	—
Green gram (mung bean)	field crop	0.40	0.96	0.51	17/23/34/16	0.4	75	0.45	—
Soybean	field crop	0.40	1.15	0.50	20/30/45/25	0.6	95	0.50	0.85
Groundnut	field crop	0.40	1.15	0.60	20/35/40/25	0.4	75	0.50	0.70
Cotton	field crop	0.35	1.15	0.50	30/45/65/50	1.1	135	0.65	0.85
Wheat	field crop	0.40	1.15	0.41	20/30/45/25	0.8	165	0.55	1.05
Chickpea	field crop	0.40	1.00	0.40	15/35/45/25	0.4	80	0.50	—
Mustard	field crop	0.35	1.15	0.35	15/35/45/25	1.3	125	0.60	—
Tomato	field crop	0.60	1.15	0.80	30/40/40/25	0.6	100	0.40	1.05
Potato	field crop	0.50	1.15	0.75	25/30/45/30	0.6	50	0.35	1.10
Onion (dry)	field crop	0.70	1.05	0.75	15/25/70/40	0.4	45	0.30	1.10
Cabbage	field crop	0.70	1.05	0.95	40/60/50/15	0.4	65	0.45	0.95
Sweet pepper	field crop	0.60	1.05	0.90	25/35/40/20	0.7	75	0.30	1.10
Table grapes	tree, vine or palm	—	—	—	20/40/120/60	—	150	0.35	—
Wine grapes	tree, vine or palm	—	—	—	20/40/120/60	—	150	0.45	—
Kiwi	tree, vine or palm	—	—	—	20/40/120/60	—	100	0.35	—
Highbush blue-berry	tree, vine or palm	—	—	—	20/50/90/20	—	90	0.50	—
Hop	tree, vine or palm	—	—	—	25/40/80/10	—	110	0.50	—
Citrus	tree, vine or palm	—	—	—	60/90/120/95	—	135	0.50	—
Olive	tree, vine or palm	—	—	—	60/90/120/95	—	145	0.65	—
Apple	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—

Crop	Group	$K_{c,ini}$	$K_{c,mid}$	$K_{c,end}$	Stages (d)	Height (m)	Root (cm)	p	K_y
Pear	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Cherry	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Peach	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Apricot	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Almond	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.40	—
Hazelnut	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Walnut	tree, vine or palm	—	—	—	20/70/90/30	—	205	0.50	—
Pecan	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Pistachio	tree, vine or palm	—	—	—	20/70/90/30	—	125	0.40	—
Fig tree	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Persimmon	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Pomegranate	tree, vine or palm	—	—	—	20/70/90/30	—	150	0.50	—
Avocado	tree, vine or palm	—	—	—	60/90/120/95	—	75	0.70	—
Mango	tree, vine or palm	—	—	—	60/90/120/95	—	120	0.50	—
Date palm	tree, vine or palm	—	—	—	60/90/120/95	—	200	0.50	—
Coconut	tree, vine or palm	—	—	—	60/90/120/95	—	120	0.50	—
Papaya	tree, vine or palm	—	—	—	60/90/120/95	—	90	0.50	—
Guava	tree, vine or palm	—	—	—	60/90/120/95	—	120	0.50	—
Oil palm	tree, vine or palm	0.90	—	—	60/90/120/95	—	90	0.65	—
Cocoa	tree, vine or palm	1.00	—	—	60/90/120/95	—	85	0.30	—
Cherimoya	tree, vine or palm	—	—	—	60/90/120/95	—	120	0.50	—

Chapter 22

Seasonal data: twenty seasons, year by year

The group *Seasonal data* of the Layers panel holds three layers built on the **year-by-year results of the simulations**: the twenty growing seasons of each run, not only their mean. They show how reliable a setting is from one year to the next. In version 1.0 the group was called *Year by year (Daisy data)*; the name of the simulation model has gone from the program and is given here.

22.1 Where the yearly values come from

For each soil the simulations hold the twenty yearly values of rain, irrigation, evaporation, transpiration, drainage, storage change, yield and nitrogen of the reference runs (irrigated and rainfed). A layer reads the run nearest your settings and moves each year by the response of the annual model to your settings,

$$P'_y = \rho\, P_y, \qquad X'_y = X_y \,\mathrm{clip}\Big(\frac{X^{\mathrm{model}}(\text{your settings})}{X^{\mathrm{model}}(\text{reference})};\ 0,\ 6\Big)\ \ (X \in \{I, T, E, E_{\mathrm{other}}, N_L, Y\}),$$

$$D'_y = \begin{cases} D_y \dfrac{D^{\mathrm{model}}}{D^{\mathrm{model}}_{\mathrm{ref}}} & \bar{D} > 5,\ D^{\mathrm{model}}_{\mathrm{ref}} > 5 \\ D_y + \big(D^{\mathrm{model}} - D^{\mathrm{model}}_{\mathrm{ref}}\big) & \text{otherwise} \end{cases}, \qquad ET'_0 = ET_0\,(1 + 0.035\,\Delta T), \quad (22.1)$$

Provenance (Eq. 22.1): **[Data-driven]**: the yearly values of the simulations, moved by the ratio (or, for small drainage, the difference) of Engine 1 between your settings and the reference of the run (your mulch, rain ×1.0, no warming, nitrogen 100 %); for a library crop the yield is the relative yield of Eq. (21.5) year by year and the irrigation follows the ratio of the two crop balances of Section 21.4.

so that every year follows the crop, rain, temperature, nitrogen, mulch, trigger and method you set, while keeping the weather sequence of the record. A sentence under each layer names the run used and the adjustment made. The calendar years on the axes end in the working year.

22.2 Reliability over 20 seasons

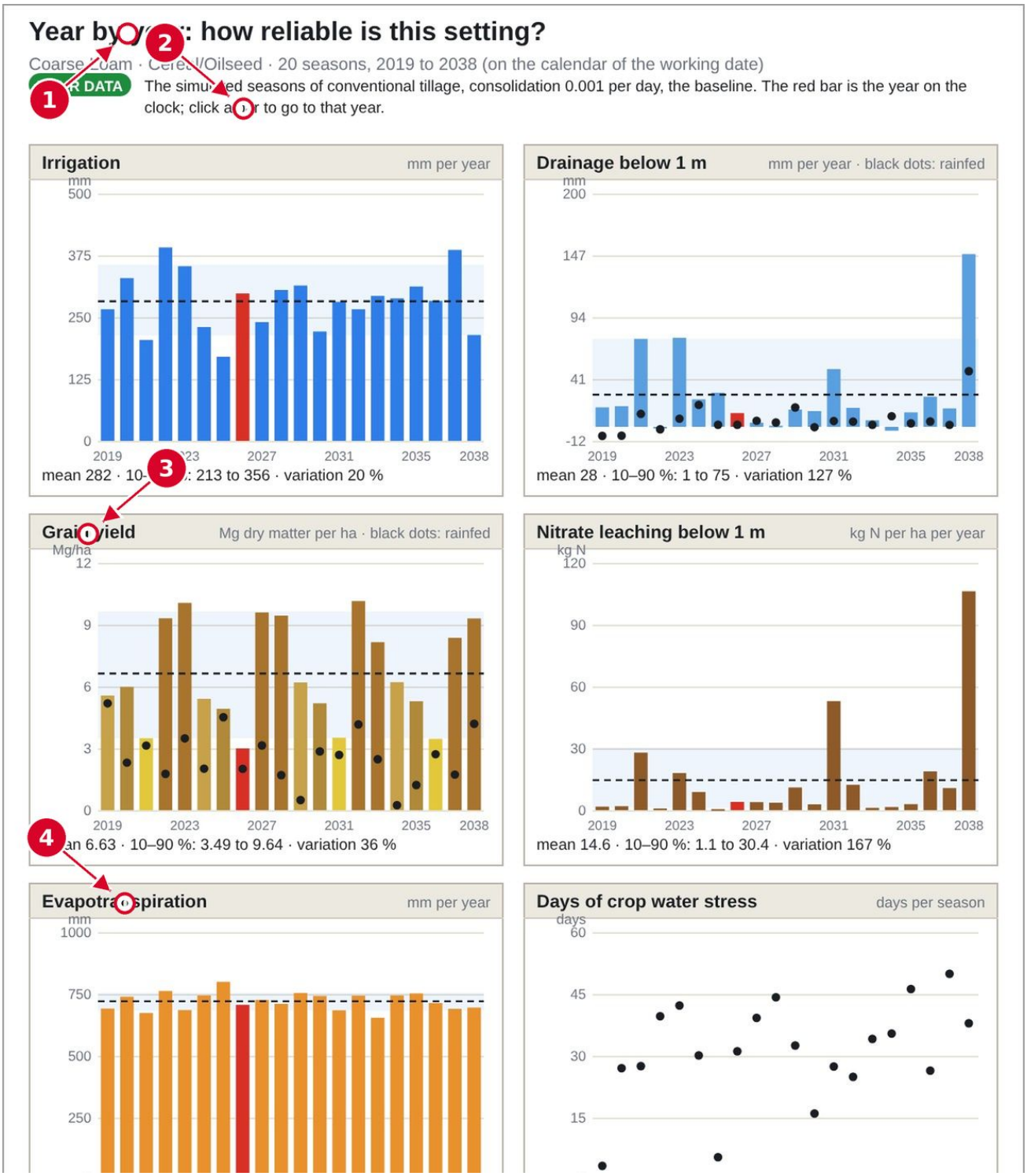


Figure 22.1: Reliability over 20 seasons.

Reliability of this setting over the 20 seasons: the irrigation, drainage, yield (or relative yield) and water stress of each season, with their mean and spread. *How often a given irrigation is exceeded*: the twenty seasonal irrigation amounts sorted into an exceedance curve, so that the

amount needed in four years out of five can be read off. Click a year to set the working year to it.

22.3 Full water balance

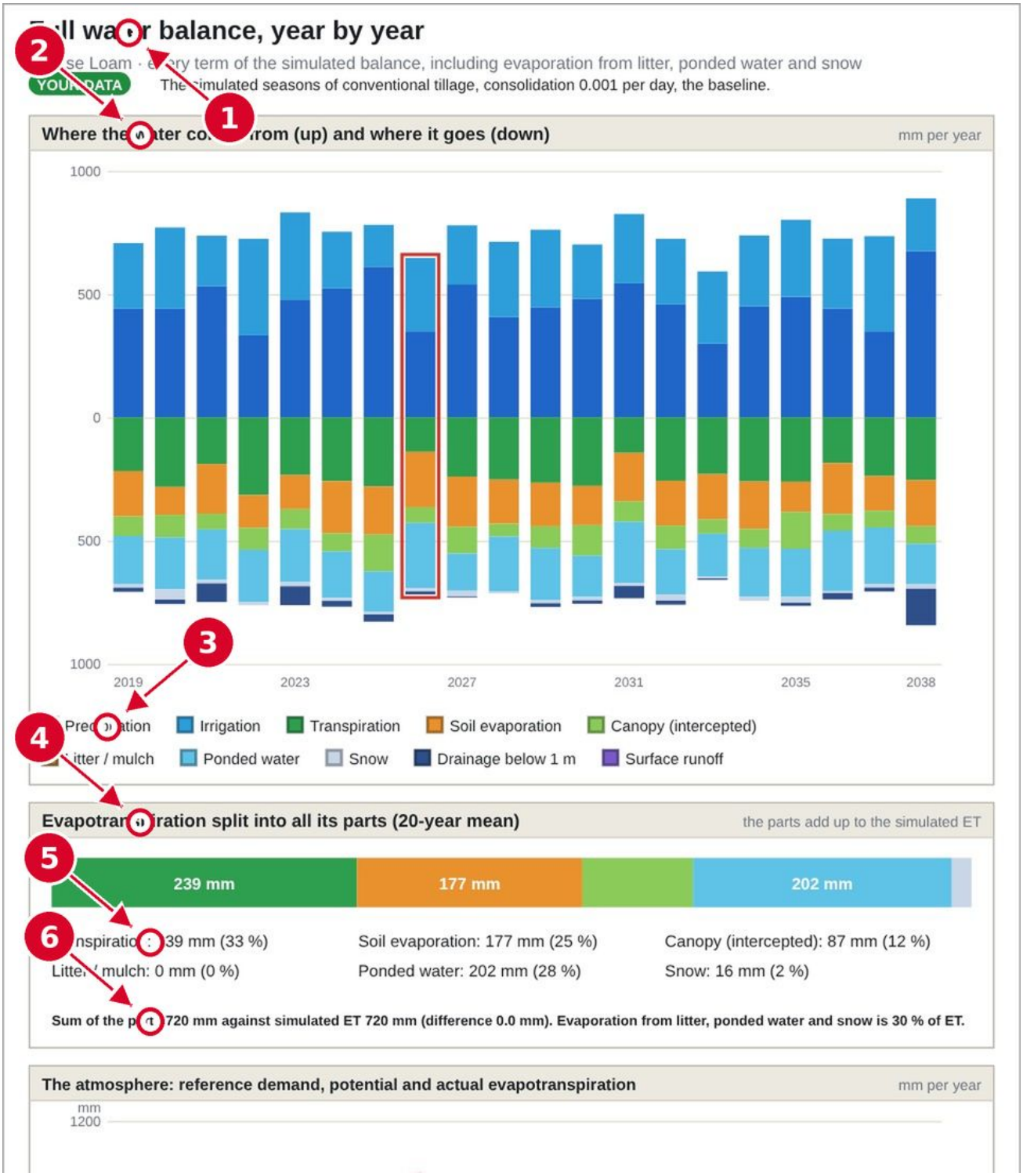


Figure 22.2: Full water balance.

Where the water comes from (up) and where it goes (down): the complete balance of each season, $P + I = E + T + E_{\text{other}} + D + \Delta S$, stacked year by year, so that a dry season (high irrigation, little drainage) and a wet season can be told apart at a glance. *Evapotranspiration split into all its parts*: transpiration, soil evaporation, canopy, litter, ponded and snow evaporation (20-year mean). *The atmosphere*: reference demand, potential and actual evapotranspiration, year by year. *How the water enters and leaves the soil*: infiltration through the soil matrix and the macropores, runoff and drainage.

22.4 Nitrogen budget

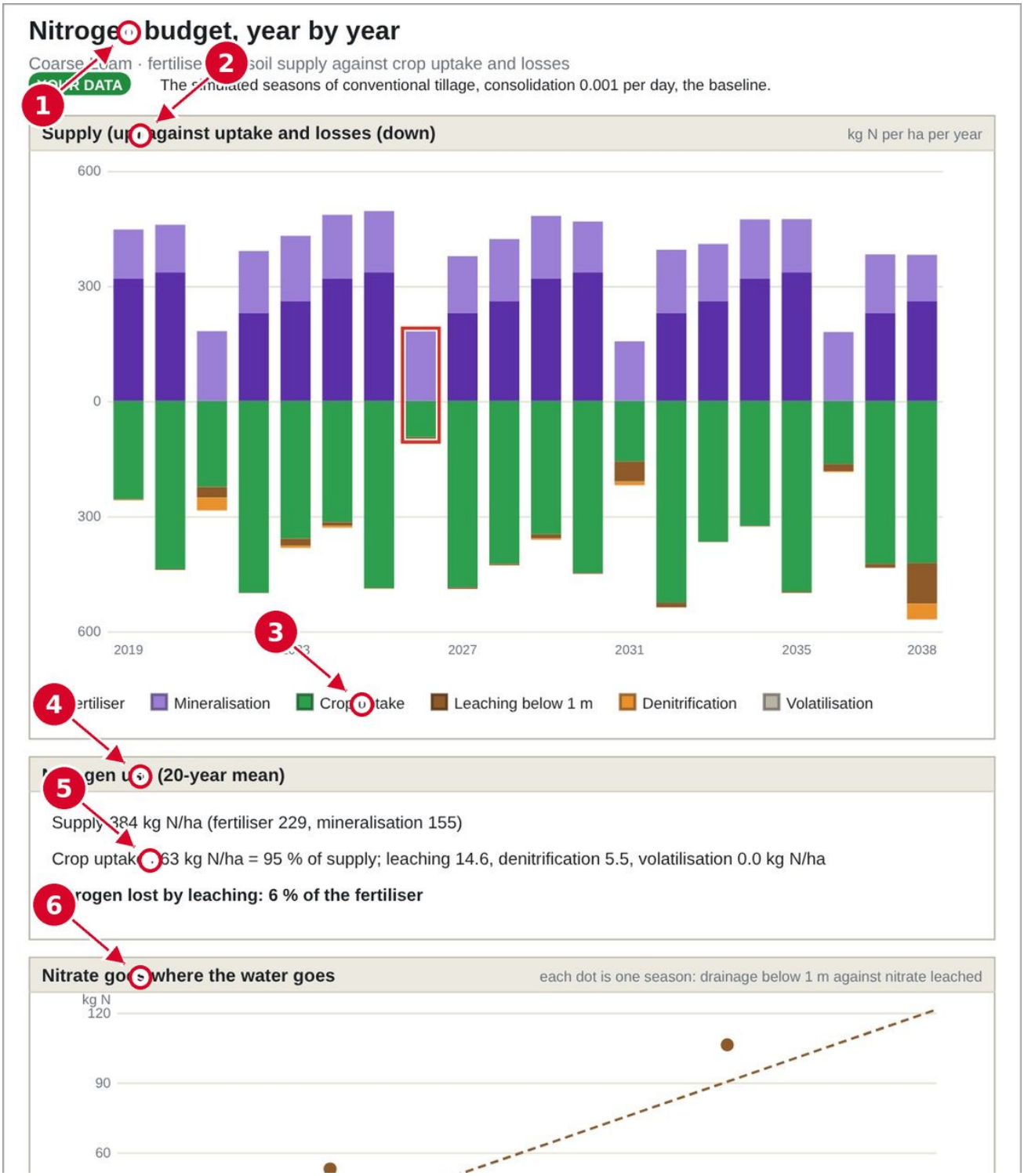


Figure 22.3: Nitrogen budget.

Supply (up) against uptake and losses (down): fertiliser and mineralisation against crop uptake, leaching, denitrification and volatilisation, season by season; *Nitrogen use (20-year mean)*; and *Nitrate goes where the water goes*: the leaching of every season against its drainage, with the

regression line. The slope of that line is printed under the chart: *each extra millimetre of drainage carries … kg N/ha of nitrate*.

Chapter 23

Reports

23.1 Choosing the layers

File ▸ *Save report as PDF (choose the layers)…* (Ctrl+Shift+P, or the *Report* button of the toolbar) opens a window with every layer of the Layers panel, in its groups. **Every layer is ticked to start with.** Untick a layer, or a whole group with its box, or everything with *All layers*; the count at the bottom tells how many layers are ticked. **Make the PDF** builds the report and asks where to save it.

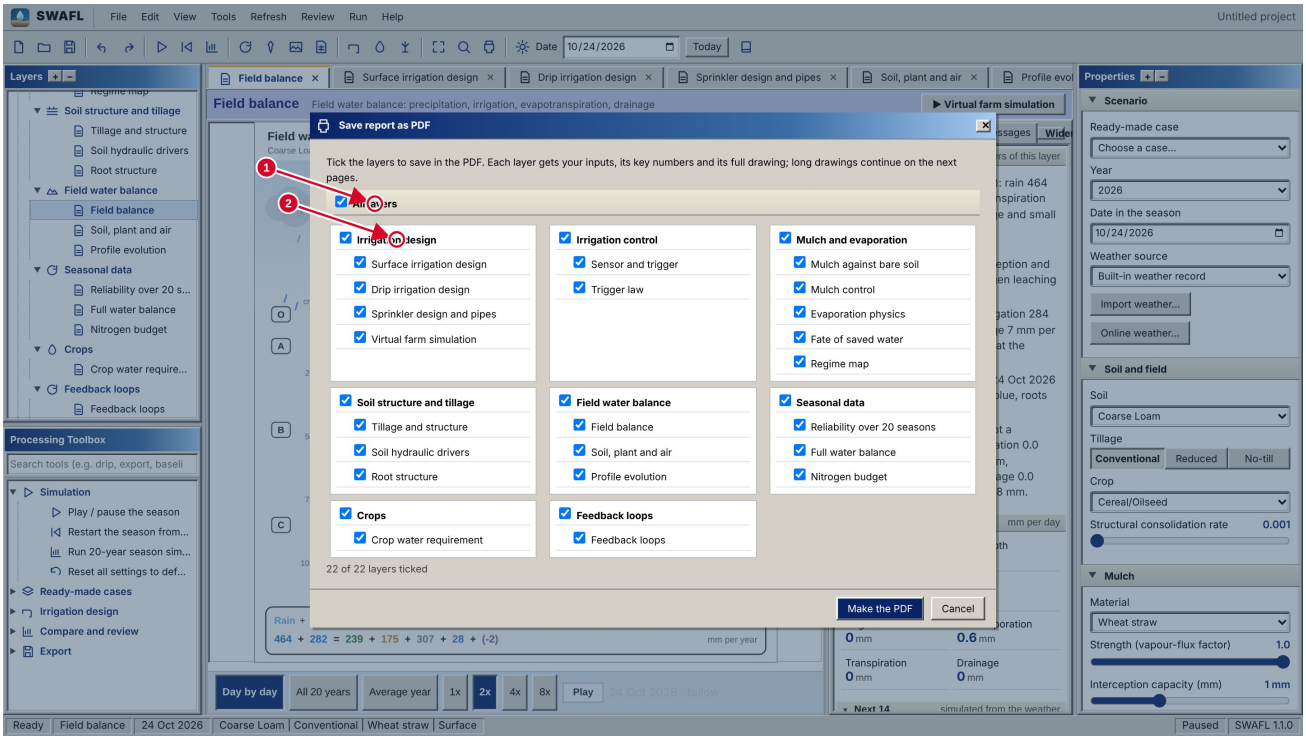


Figure 23.1: The report window: every layer ticked.

23.2 What the report holds

- **The first page**: your inputs (soil, tillage, crop, consolidation rate, mulch, rain, temperature, trigger, sensor depth, method, nitrogen), the results at these inputs (the 20-year averages), the water balance of the year, and the list of the layers included.

- **For each layer**: its name and purpose, the sentence that sums it up, the key numbers at your inputs, and **the whole drawing from top to bottom**. When the key numbers run past the end of the page they continue on the next page. When the drawing is longer than the page it continues on the next pages, cut at its section headings (or, where a layer has none, at a blank band between two parts), so no part of a picture or of its text is hidden or cut through.
- **The design layers** are drawn at the end of the irrigation, so every result is complete.
- **The field balance** appears three times: day by day, all twenty years, and the average year.

A report of every layer has about 60–70 pages and takes a few seconds to build.

23.3 The page and its numbers

The pages are A4. The drawing of a layer is placed at the full text width (515 points) and cut into parts of at most

$$h_{\max} = 748 \text{ pt} \times \frac{W}{515} \ \ [\text{px of the drawing}], \tag{23.1}$$

Provenance (Eq. 23.1): **[Engineered]**: page geometry; W is the width of the drawing in pixels, so a wide drawing is cut into taller parts than a narrow one.

and the first part fills the room left under the key numbers when it is at least 220 points.

Chapter 24

What changed from version 1.0 to version 1.1

24.1 The window

- The program, its title bar, the status bar and the About window show **1.1.0**. The build removes any older program before building, and the program clears its cache on start, so an old version cannot be shown by mistake.
- The **Layers panel** has a new order: Irrigation design, Irrigation control, Mulch and evaporation, Soil structure and tillage, Field water balance, Seasonal data, Crops, Feedback loops. The short codes of the layers (S2, V1, N4, …) are gone; layers are named by their names only.
- The **year strip** and the date bar under the drawings are gone; the year (1980–2100) and the date are set in Properties and on the toolbar.
- The **dashboard** is laid out like the Properties panel and every block folds on its own; the Charts tab draws the hydrograph, the times of the irrigation and where the water goes for each design (Chapter 14).
- **Properties**: the *Weather of years after 1999* selector and the explanatory notes under it, under the consolidation rate and under the design shortcuts have gone; their content is in this manual. *Weather source* chooses the built-in record, an imported file or the online weather.
- The name of the simulation model is no longer shown anywhere in the program; badges that said so now say *Simulated*.
- Labels are drawn after the pictures and moved apart when they would overlap; text near the edge of a drawing is moved inwards instead of being cut; long drawings take their full height at once.
- On screens narrower than 1500 pixels the docks give way so that the drawing keeps its size; the drawing always fits the window, nothing is cut off on the right.
- The selected layer is marked in light blue; the Play/Pause button is small.
- The first block of the Results tab gives the key numbers of the open layer (Chapter 14).
- Layers that play by changing a setting (each soil in turn, rainfall from dry to wet) wait for **Play** in the program, so Properties and the dashboard keep your values.

24.2 The designs

- Inputs are **value boxes** with exact values, units and a list of the allowed range (Section 18.2).
- **Nothing runs while you type**: results appear after **Run simulation** and are withdrawn when an input or a setting changes (Section 18.1).

- **Find the most efficient inputs**, with your present inputs competing (Section 18.4); the surface search also tries shorter runs.
- **Field size** as an area or as a length and a width, for drip and sprinklers (Section 18.3).
- The **field sketch** is a guided drawing: corners you drag, side lengths you type, the source and the inlet, the pipe layout drawn as you choose, and seven steps of questions (field, water, crop, soil, system, running, summary) (Chapter 20). **Compare the irrigation systems** (Section 18.7).
- Each design ends with a **film of the soil water** covering the days ahead and the rest of the season. Every day is coloured by its phase (irrigation, rain, comfortable, nearing the trigger, stress, near wilting, no crop). The film shows the root zone drying towards the trigger and, without irrigation, towards the wilting point, and gives the totals and the share of days in each phase (Chapter 20). The schedule takes its depths and times from the design.
- The seasonal layers have **one dashboard chart per part of the layer**, with how each quantity varies over the twenty seasons (Chapter 14).
- *Trigger law* and *Mulch against bare soil* stand still until **Play**; the season plays at two days a second.
- Values may be typed up to the **physical limits** of the system, beyond the usual range of the list (Section 18.2).
- A surface set lasts at most 48 hours; when the inflow is too small for the soil, the design says so (Chapter 11).
- The surface layer iterates the advance exponent in the steady-intake term (Chapter 11); the slow-advance cases of the textbook example are now within 2 %.

24.3 The data and the models

- **Weather**: import of daily weather files and the online weather of a place, with every layer following (Chapter 19).
- **Crops**: 48 crops with their FAO-56 coefficients; the *Crop water requirement* layer (Chapter 21).
- **Seasonal data**: the twenty seasons of the simulations, following every setting (Chapter 22).
- **Sensor records** follow every setting and every soil by the delta method (Eq. 6.4a).
- **Sand and loamy sand** now respond to the trigger, the sensor depth and the method (Eq. 5.12a).
- The water balance of every crop closes (the limit on evapotranspiration was raised from 2,000 to 5,000 mm).
- Yield is in Mg/ha for the simulated cereal and in per cent of the potential for a library crop, with differences in *% points*.

24.4 Reports

- The report asks which layers to include, all ticked to start with; notes and drawings continue on the next pages; designs are drawn complete (Chapter 23).

24.5 The field on the map

- The field sketch has a hand to move the drawing, zoom in and out (buttons and mouse wheel), *Fit*, and a north arrow and scale bar that follow the zoom.
- *Tools* ▸ *Field on the map…* finds the farm (place search, coordinates, satellite image, Google Earth), demarcates it on the image, computes NDVI, EVI, SAVI, NDRE and NDWI of the demarcated field from Sentinel-2 or simulated by SWAFL, and exports the report (PDF or PNG at 96 to 600 dpi, with coordinates and authorship), GeoJSON and CSV (Chapter 20).
- The virtual farm simulation has been removed; the real field on the map (Chapter 20) takes its place.

- The Trigger law block of the Dashboard shows the current soil as a table, its season chart and the advice with each condition ticked.
- The index map is cut exactly to the demarcated outline on the real map and follows every change of the outline; it shows continuous colours or five management zones with their areas, two indices side by side, the value under the cursor, and the irrigation layout of the sketch.
- The soil-water film of every design follows the season of the crop in Properties (starting at its next sowing when the working date has no crop), gives the advice of when to plant over the whole year, and each design has the weather line with the button for the online weather of the farm.

Chapter 25

Screenshot atlas: every screen with its governing equations

This part of the manual shows every screen of the workbench, one after another, each accompanied by the equations that produce what it displays. The screenshots were taken from the SWAFL workbench itself (the window in `app/index.html`) with the default settings (Coarse Loam, conventional tillage, cereal/oilseed rotation, wheat-straw mulch of strength 1.0, Surface irrigation, typical year). For each screen the first picture is the whole workbench and the second is the drawing area at higher magnification. For the fourteen single layers, whose drawing is longer than the drawing area and is scrolled inside it, the pictures of the whole drawing from top to bottom follow, together with its playback states and with one picture for each control that changes the drawing (every mulch material, every zoom and view of the soil section, every irrigation layout, every day button, and so on). Every picture carries numbered arrows, each ending on the item that its legend entry names. Below the pictures every equation that the screen draws on is printed in full, with its number, the chapter that explains it and its provenance label, so that no number on a screen is without its equation. A reverse index at the end lists, for each of the 134 numbered equations, the screens that use it.

Equation numbers are those of the chapters (for example Eq. 5.4 is the fourth equation of the chapter on the engines). The symbols are defined where the equation first appears; the *Where explained* line gives that place.

25.1 The window and its controls

25.1.1 The welcome dialog

Shown on first start; it carries no equations.

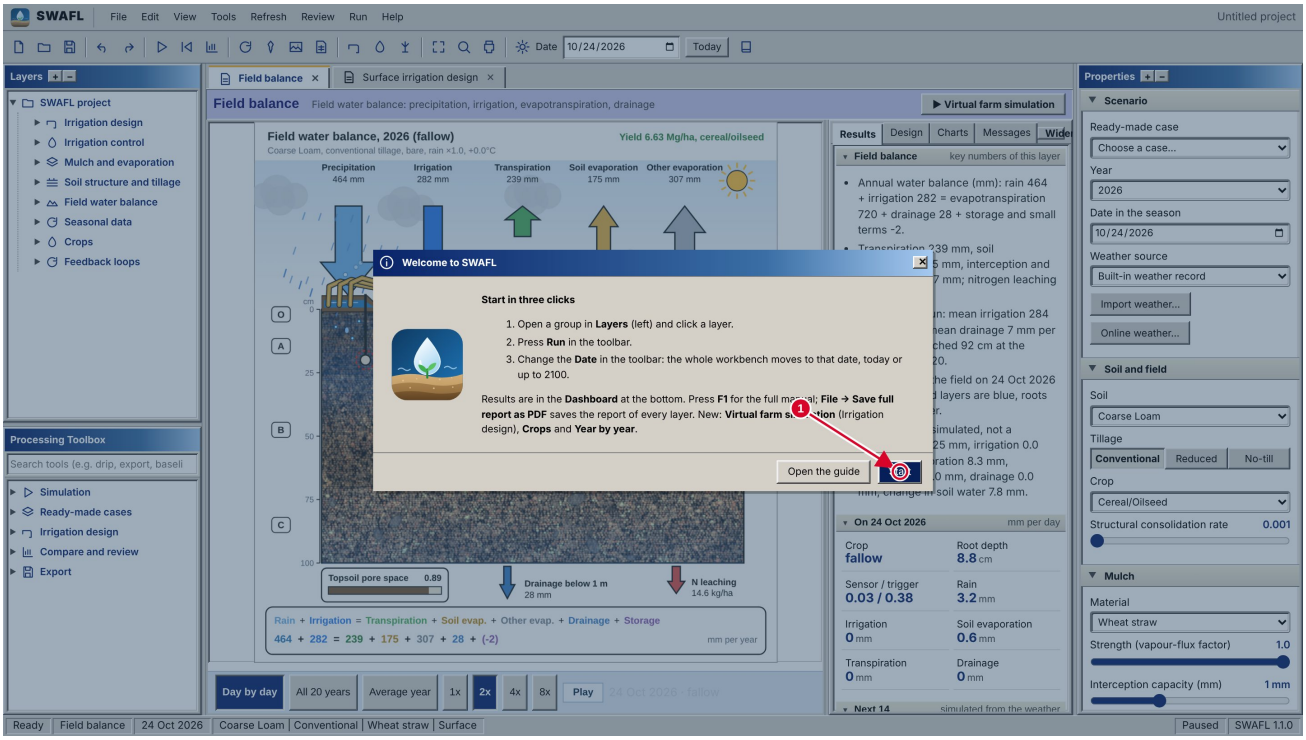


Figure 25.1: The welcome dialog.

What the arrows point to.

- **1** The Start button closes the dialog and opens the workbench; the dialog carries no equation.

25.1.2 The workbench

Layers (left), drawing area (centre), Dashboard (below and right of the drawing) and Properties (right). The date shown in the toolbar sets the whole workbench to that day.

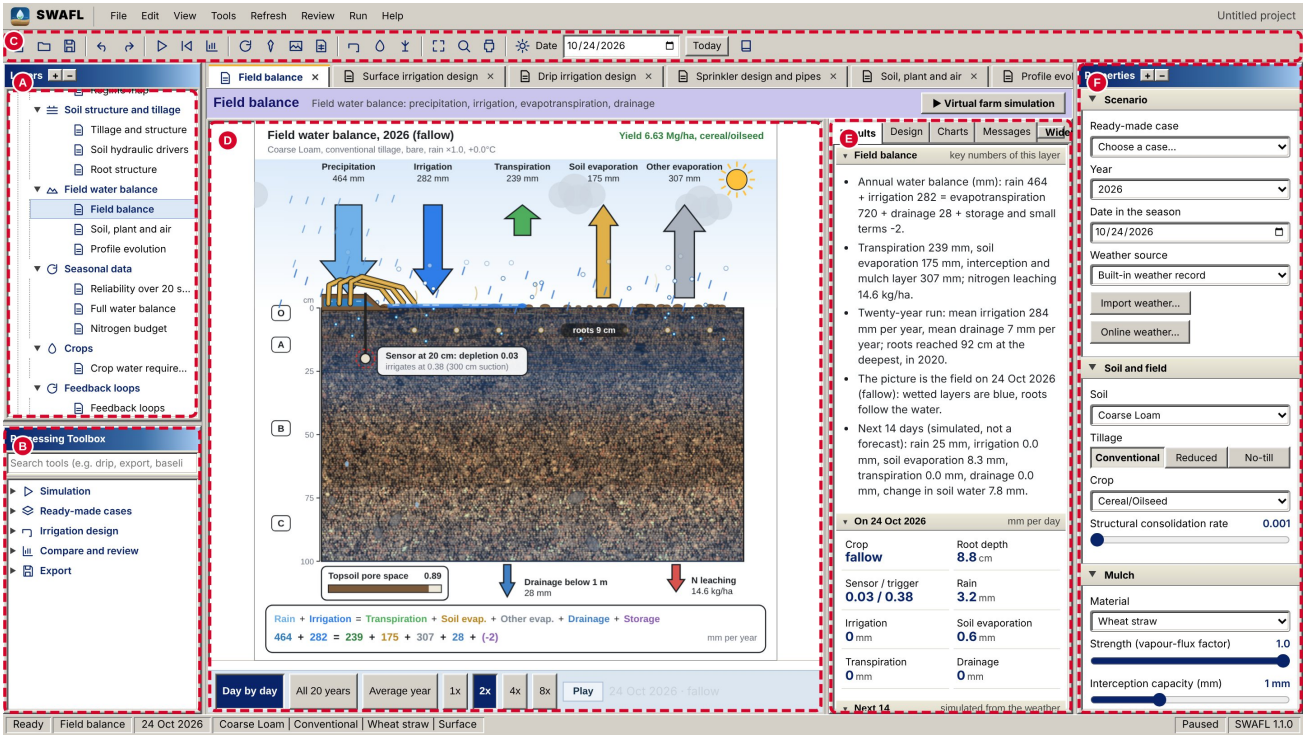


Figure 25.2: The workbench.

What the arrows point to.

- **L** Layers panel: choose a layer or one of its sections. Every choice is one screen of this atlas.
- **X** Processing Toolbox: run, cases, design, compare and export commands.
- **T** Toolbar with the Date control: the date sets the whole workbench to that day (Eqs. 4.1 and 4.2).
- **D** Drawing area: the picture produced by the equations printed under each screen.
- **R** Dashboard: the numerical results of the same equations (Engine 1, Eqs. 5.4 to 5.12; Engine 2, Eqs. 5.13 to 5.25).
- **C** Properties panel: the controls that move the equations (soil, tillage, mulch, weather, irrigation, nutrients).

Equations behind this screen. Eq. 4.1, Eq. 4.2.

Eq. 4.1. Where explained: chapter "Dates", section "The date control".

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \quad (4.1)$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

Eq. 4.2. Where explained: chapter "Dates", section "The date control".

$$a = \begin{cases} \operatorname{argsort}_y \left(P_{\mathrm{thr},y}\right)_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\mathrm{thr},y} & \text{dry} \\ \arg\max_y P_{\mathrm{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

Provenance (Eq. 4.2): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

25.1.3 The File menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

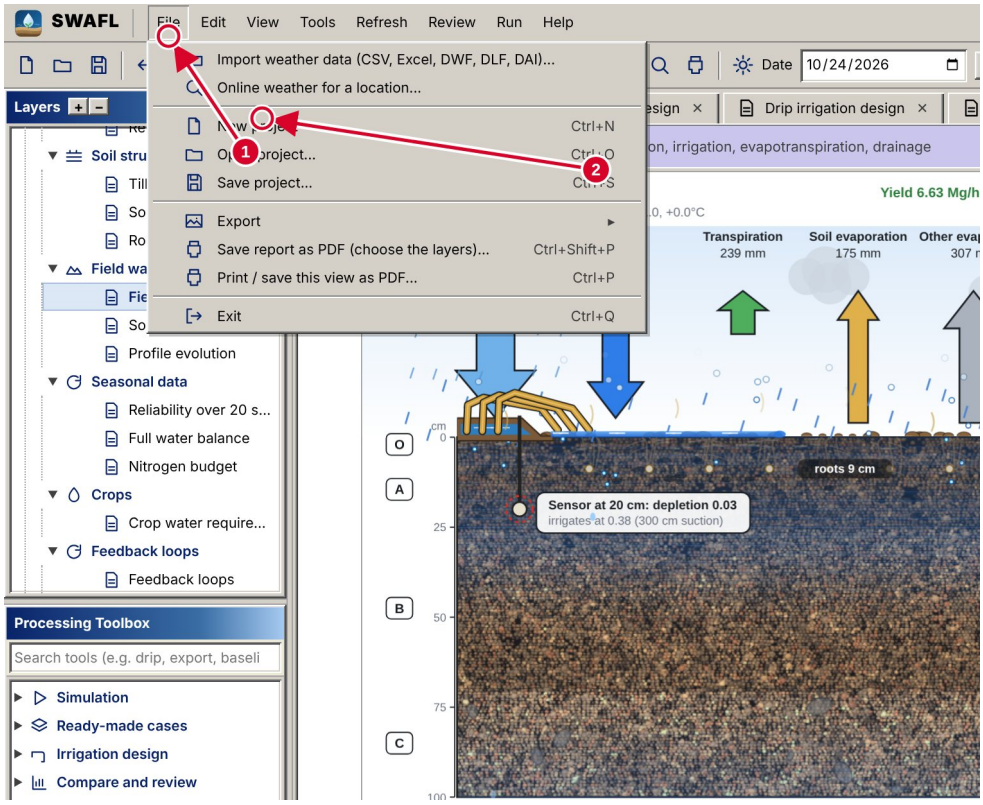


Figure 25.3: The File menu.

What the arrows point to.

- **1** The File menu in the menu bar.
- **2** Its list of commands.

25.1.4 The Edit menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

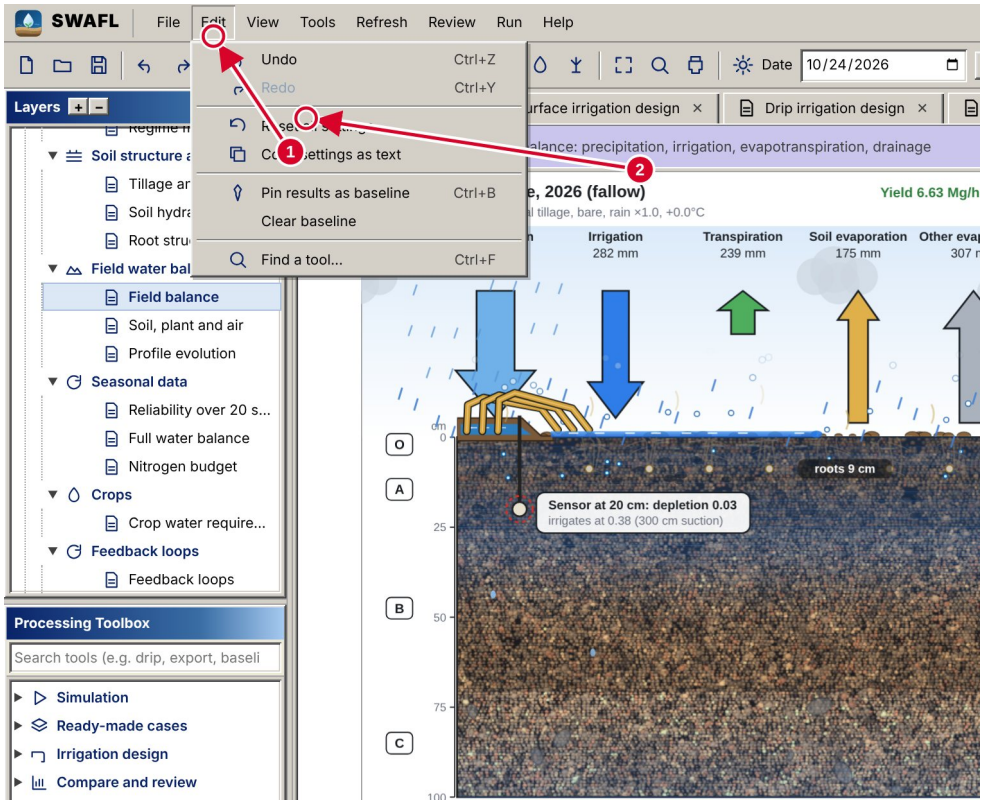


Figure 25.4: The Edit menu.

What the arrows point to.

- **1** The Edit menu in the menu bar.
- **2** Its list of commands.

25.1.5 The View menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

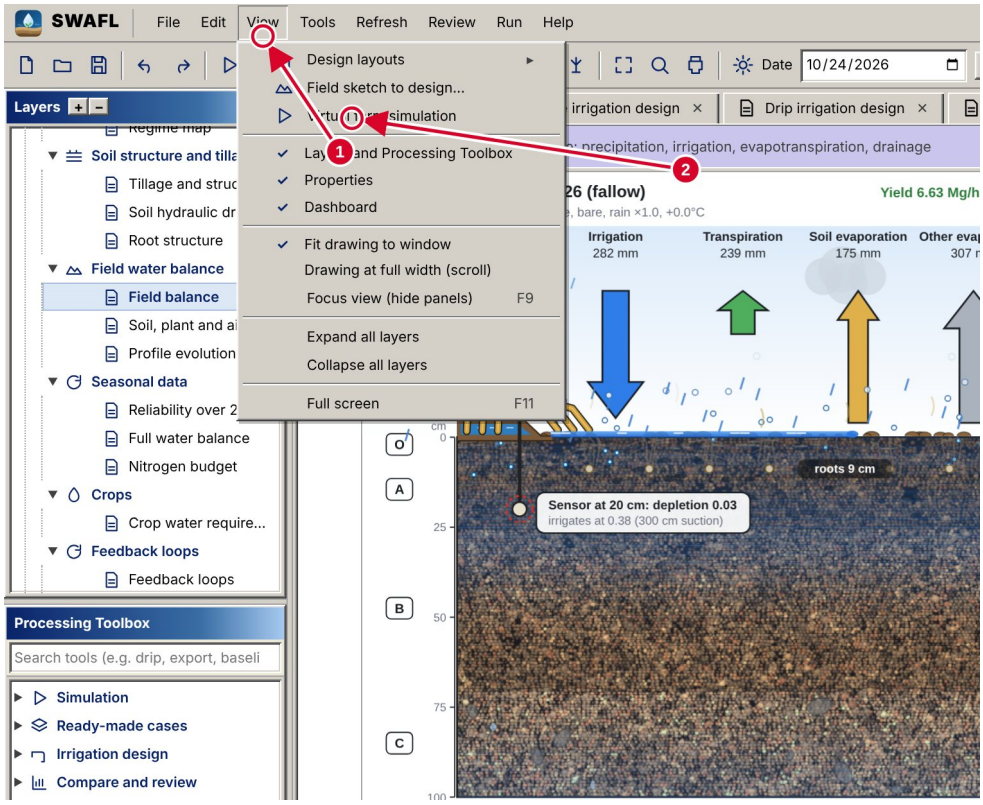


Figure 25.5: The View menu.

What the arrows point to.

- **1** The View menu in the menu bar.
- **2** Its list of commands.

25.1.6 The Tools menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

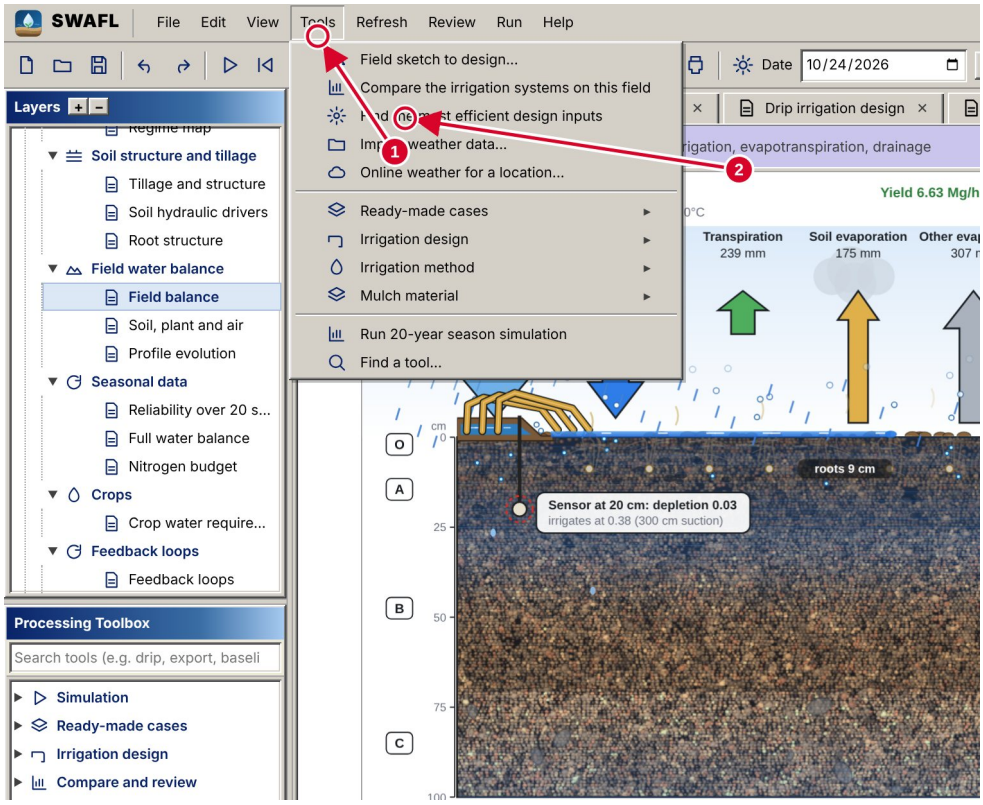


Figure 25.6: The Tools menu.

What the arrows point to.

- **1** The Tools menu in the menu bar.
- **2** Its list of commands.

25.1.7 The Refresh menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

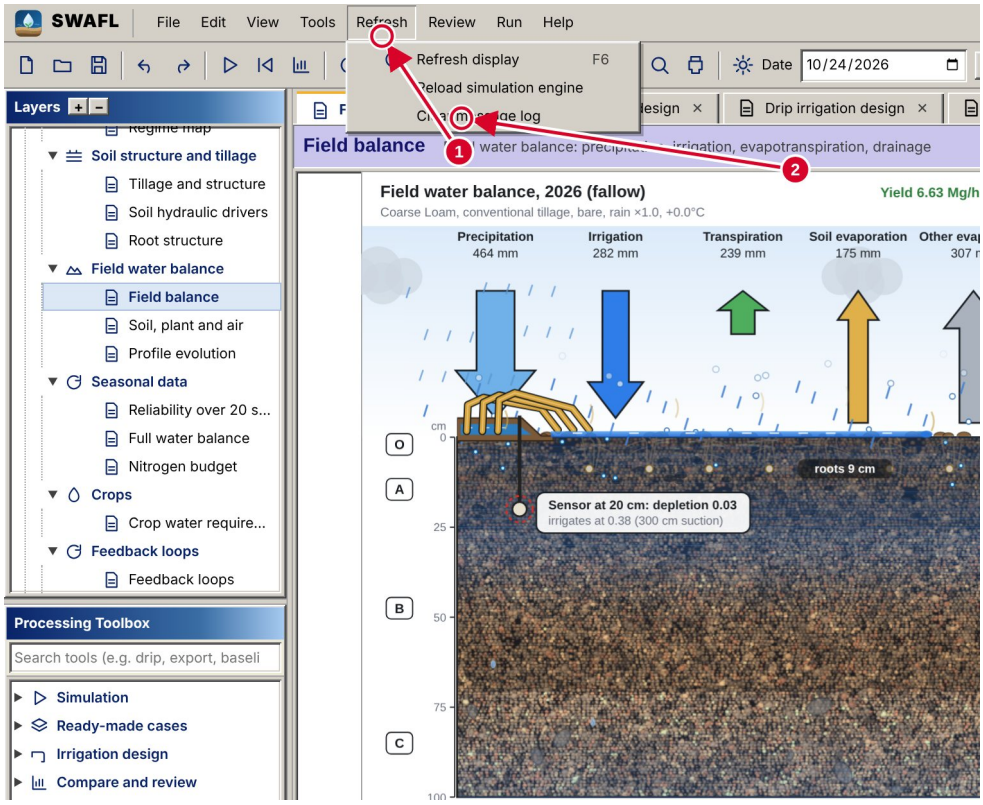


Figure 25.7: The Refresh menu.

What the arrows point to.

- **1** The Refresh menu in the menu bar.
- **2** Its list of commands.

25.1.8 The Review menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

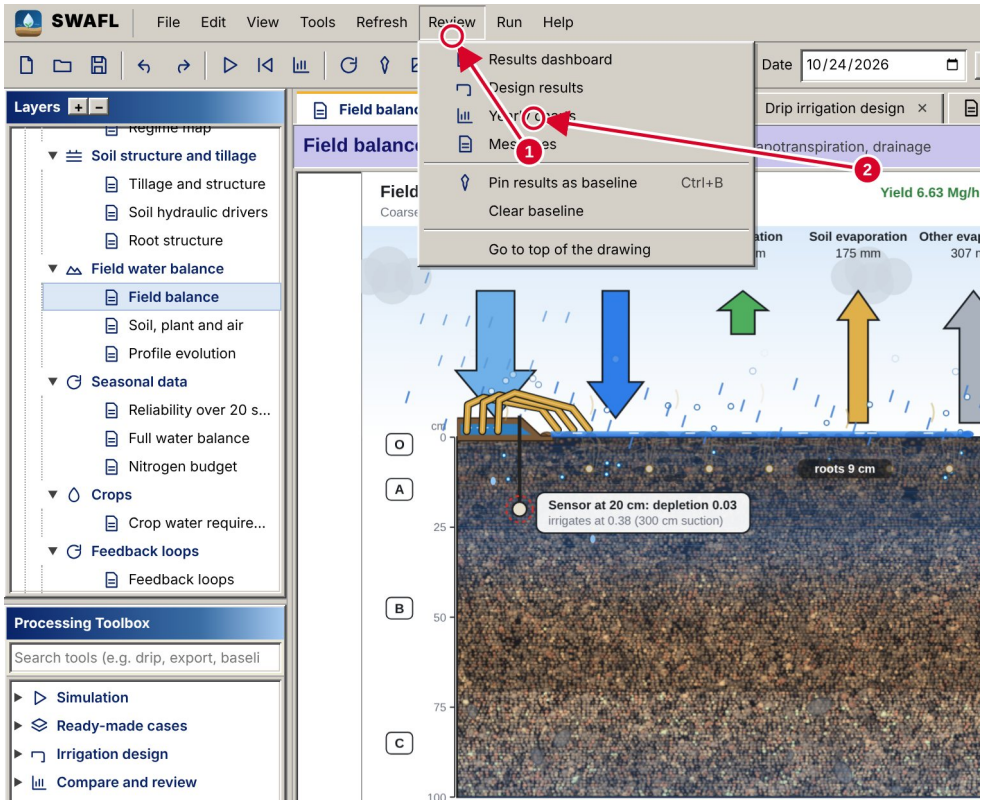


Figure 25.8: The Review menu.

What the arrows point to.

- **1** The Review menu in the menu bar.
- **2** Its list of commands.

25.1.9 The Run menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

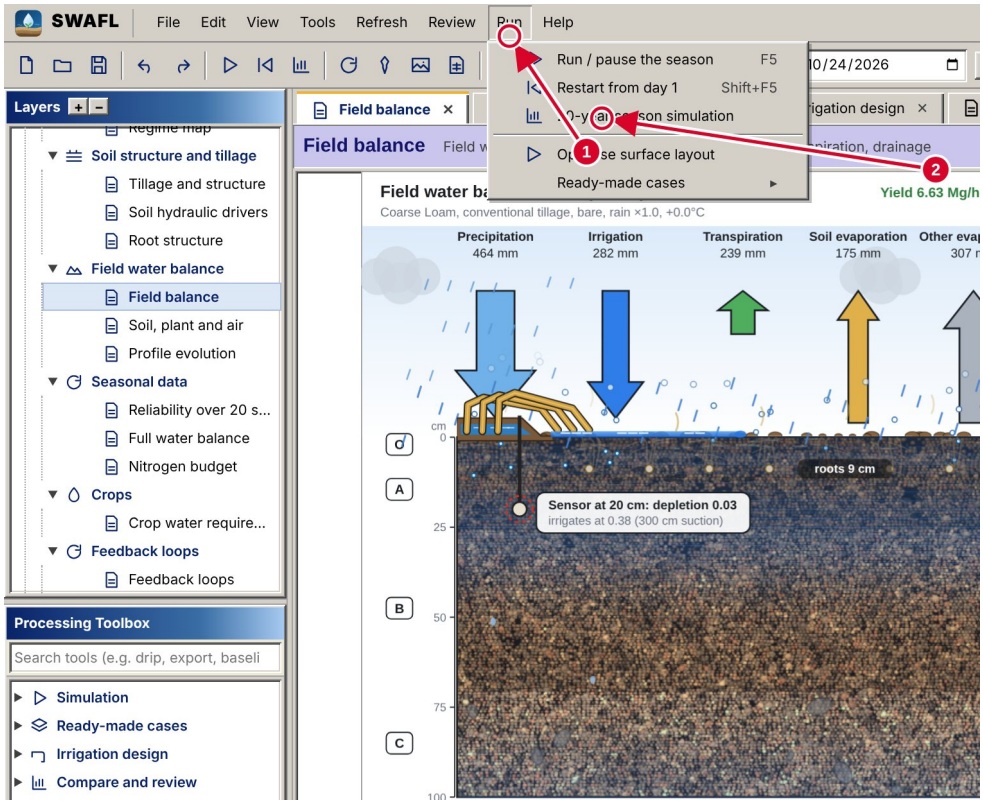


Figure 25.9: The Run menu.

What the arrows point to.

- **1** The Run menu in the menu bar.
- **2** Its list of commands.

25.1.10 The Help menu

Menu commands. Every command is described in the chapter “The window: a guided tour”, section “Menu bar reference”; the commands that run the engines are those of the Run and Refresh menus.

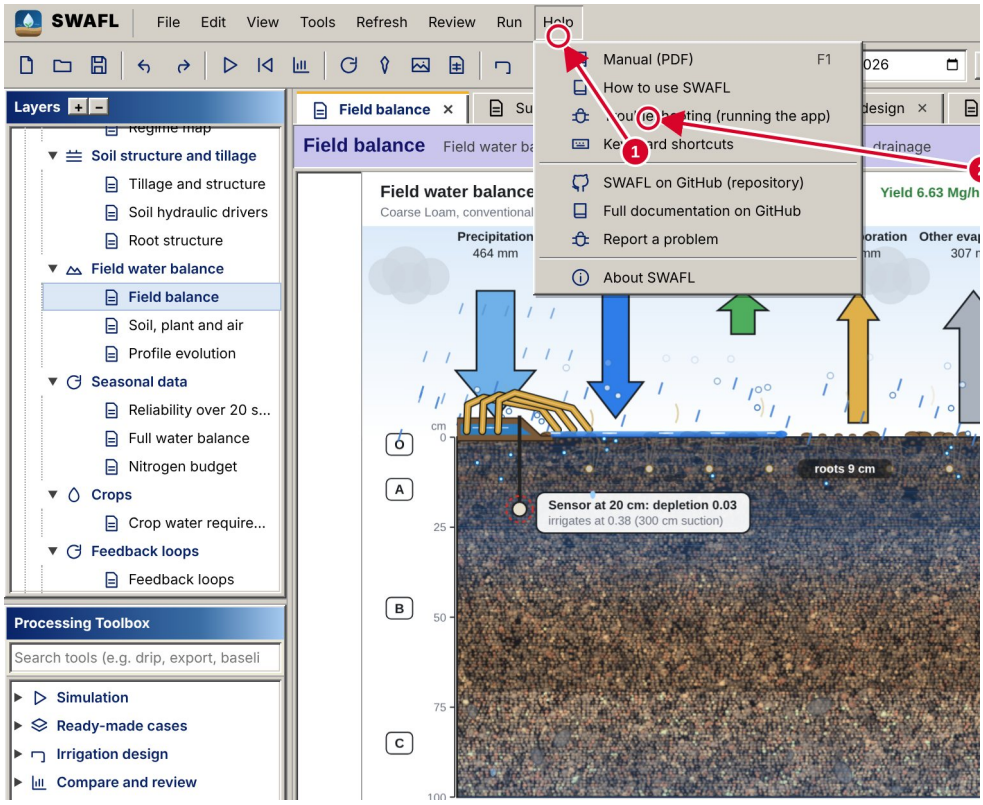


Figure 25.10: The Help menu.

What the arrows point to.

- **1** The Help menu in the menu bar.
- **2** Its list of commands.

25.1.11 The command palette (Ctrl+K)

Searches every layer, control and tool by name.

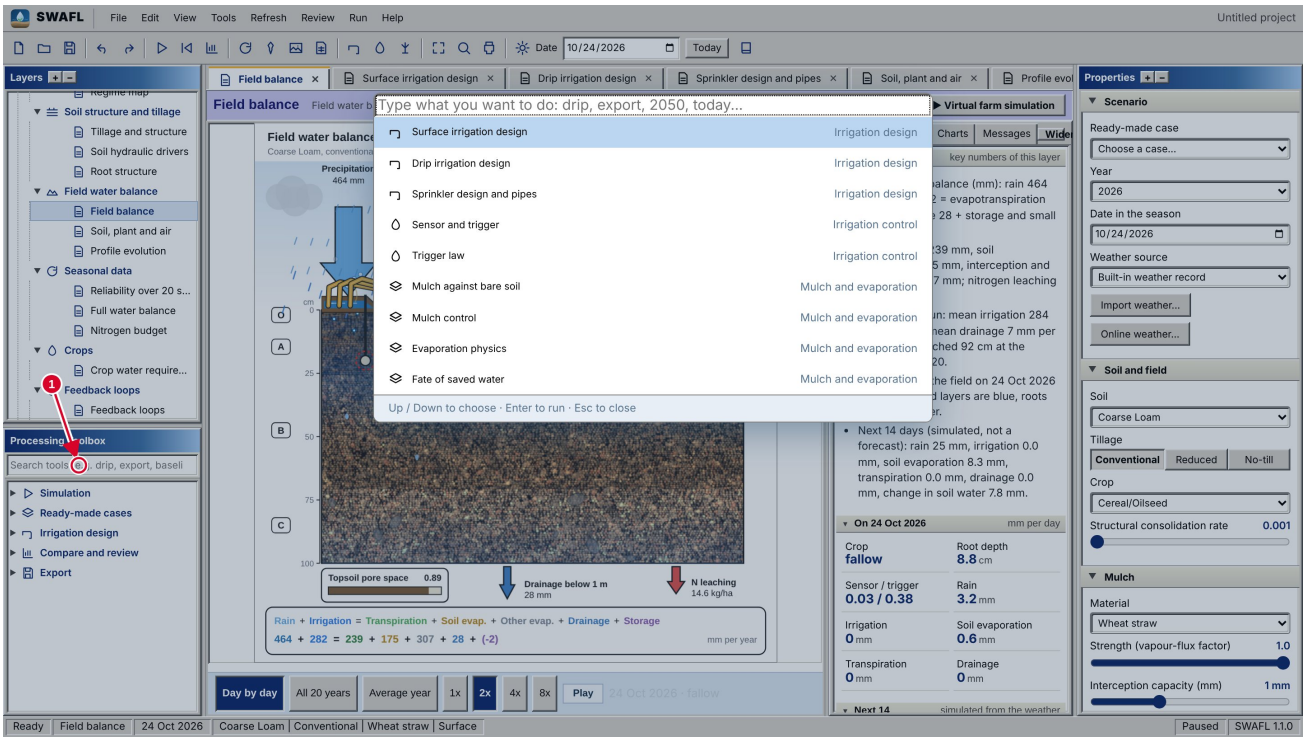


Figure 25.11: The command palette (Ctrl+K).

What the arrows point to.

- **1** Search box of the command palette (Ctrl+K): type a layer, control or tool and press Enter.

25.1.12 The Processing Toolbox

Simulation, ready-made cases, irrigation design, compare and review, export.

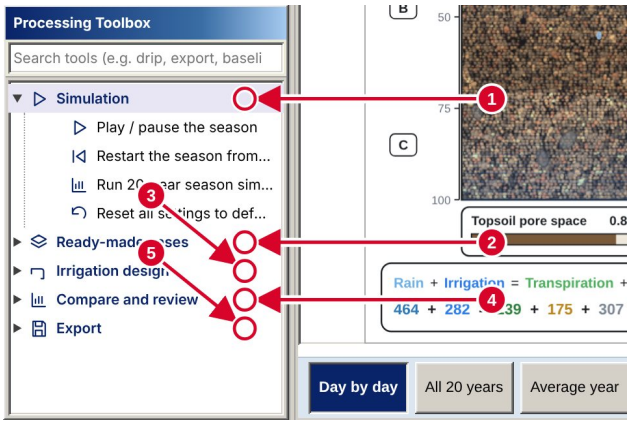


Figure 25.12: The Processing Toolbox.

What the arrows point to.

- **1** Simulation: play or pause the season, restart, 20-year run (Engine 2, Eqs. 5.13 to 5.25).
- **2** Ready-made cases: load a stored combination of soil, tillage, mulch, weather and irrigation.
- **3** Irrigation design: open the surface, drip or sprinkler design layer (Eqs. 11.1 to 13.23).
- **4** Compare and review: pin the current results as the baseline (Eqs. 14.4 and 14.5), clear it, show the dashboard.
- **5** Export: picture, yearly results (CSV), design results (CSV), project, print.

25.1.13 The toolbar

Run, restart, 20-year run, refresh, baseline, picture, focus, search and the Date control (Eqs. 4.1, 4.2).

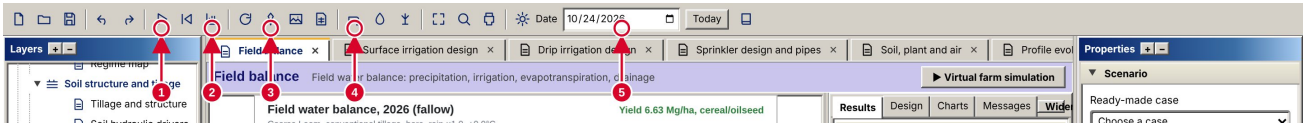


Figure 25.13: The toolbar.

What the arrows point to.

- **1** Run or pause the season (Engine 2).
- **2** Run the 20-year simulation.
- **3** Pin results as baseline (Eq. 14.4).
- **4** Open a design layer: Surface, Drip or Sprinklers.
- **5** Date control: the working date, Eqs. 4.1 and 4.2.

Equations behind this screen. Eq. 4.1, Eq. 4.2.

Eq. 4.1. Where explained: chapter “Dates”, section “The date control”.

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \tag{4.1}$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

Eq. 4.2. Where explained: chapter “Dates”, section “The date control”.

$$a = \begin{cases} \operatorname{argsort}_y \left(P_{\text{thr},y}\right)_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\text{thr},y} & \text{dry} \\ \arg\max_y P_{\text{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

Provenance (Eq. 4.2): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

25.1.14 The status bar

The current layer, date, soil, tillage, mulch and irrigation method.

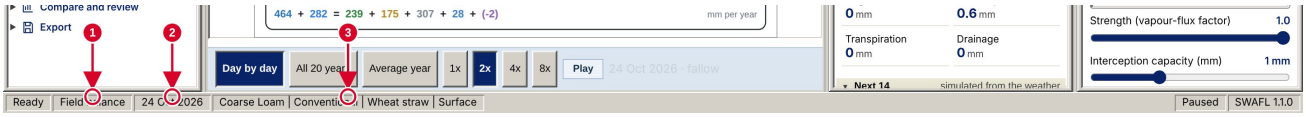


Figure 25.14: The status bar.

What the arrows point to.

- **1** Layer and section on screen.
- **2** Working date.
- **3** Soil, tillage, mulch and irrigation method in force.

25.1.15 The Dashboard, Results tab

Working day, next 14 days, whole year and 20-year averages.

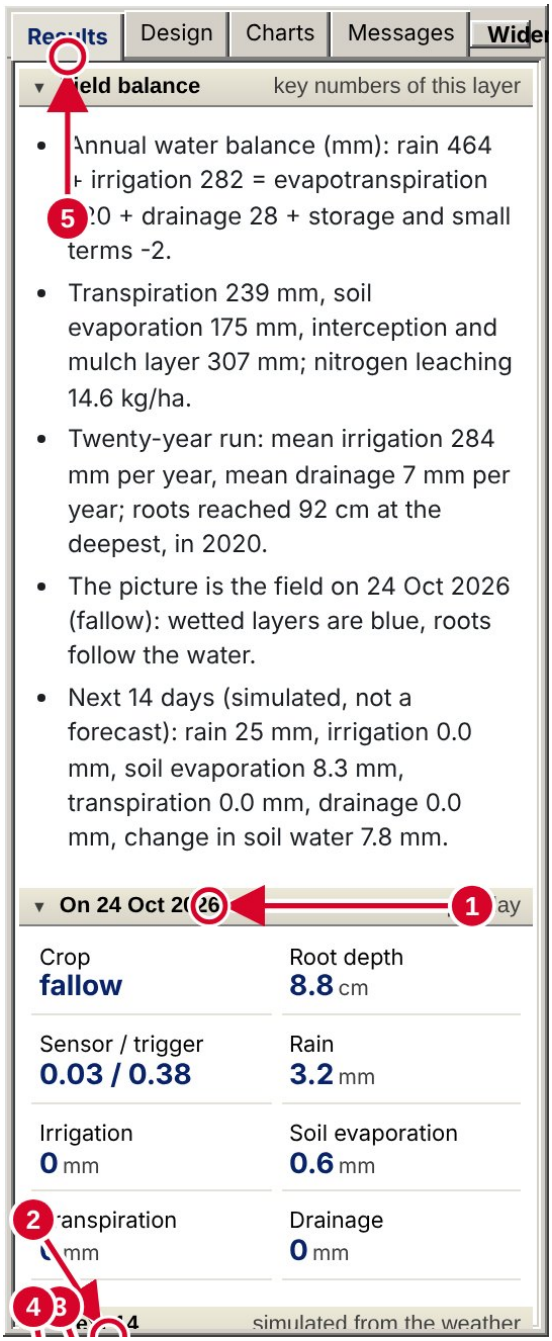


Figure 25.15: The Dashboard, Results tab.

What the arrows point to.

- **1** Working day: crop, root depth, sensor against trigger (Eq. 5.3), rain, irrigation, evaporation, transpiration and drainage of that day, Eqs. 14.1 to 14.3.
- **2** Next 14 days: simulated, not a forecast, from Engine 2.
- **3** Whole year: the annual fluxes of Engine 1, Eqs. 5.4 to 5.12, and the closure of Eq. 5.12.
- **4** Tabs: Results, Design, Charts and Messages.

Equations behind this screen. Eq. 6.1, Eq. 14.1, Eq. 14.2, Eq. 14.3, Eq. 14.4.

Eq. 6.1. Where explained: chapter "Irrigation control layers", section "How the numbers are generated".

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.

Eq. 14.1. Where explained: chapter "The Dashboard", section "Results".

$$\text{Rain} = P_k, \quad \text{Irr} = I_k, \quad \text{Evap} = E_k, \quad \text{Transp} = T_k, \quad \text{Drain} = D_k \ \ (\text{mm d}^{-1}). \tag{14.1}$$

Provenance (Eq. 14.1): **[Engineered]**: bookkeeping of Engine 2 series.

Eq. 14.2. Where explained: chapter "The Dashboard", section "Results".

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

Provenance (Eq. 14.2): **[Engineered]**: bookkeeping of Engine 2 series.

Eq. 14.3. Where explained: chapter "The Dashboard", section "Results".

$$\Delta S = P + I - E_{\text{int}} - E - T - D, \tag{14.3}$$

Provenance (Eq. 14.3): **[Literature]**: water-balance identity.

Eq. 14.4. Where explained: chapter "The Dashboard", section "The pinned baseline".

$$\Delta_i = x_i - x_i^{\text{pin}}, \tag{14.4}$$

Provenance (Eq. 14.4): **[Engineered]**: bookkeeping of Engine 2 series.

25.1.16 The Dashboard, Design tab

Results of the open design layer.

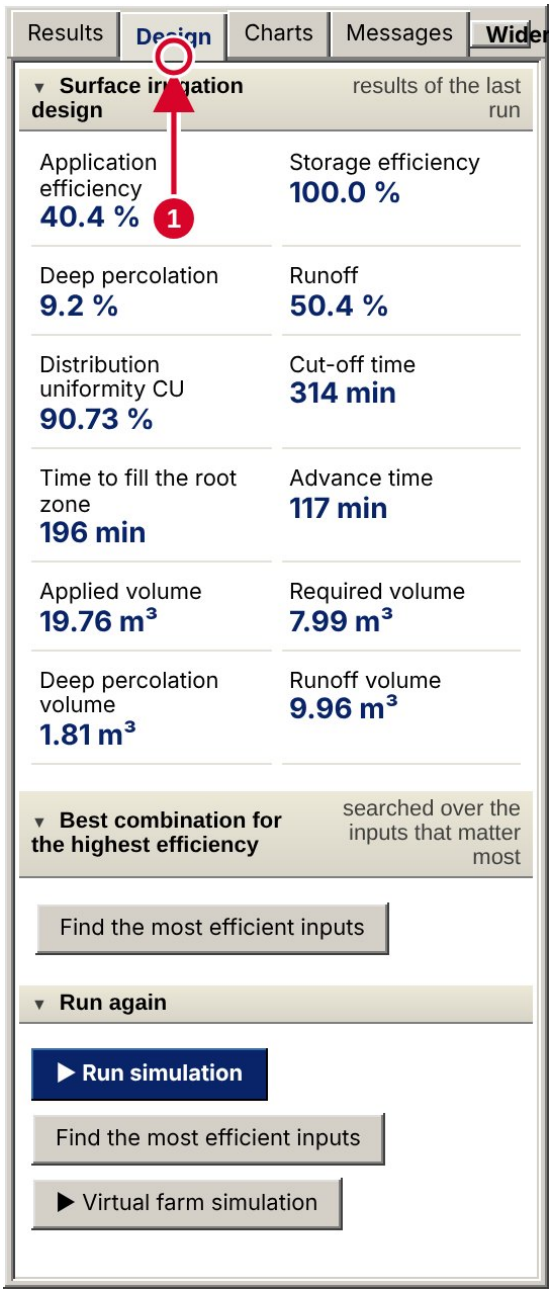


Figure 25.16: The Dashboard, Design tab.

What the arrows point to.

- **1** Design tab with a design layer open: efficiencies, uniformity, times and volumes of the system being designed.
- **2** Tabs.

Equations behind this screen. Eq. 11.18, Eq. 11.19, Eq. 12.19, Eq. 13.11.

Eq. 11.18. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".

$$
\begin{aligned}
&V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b<1:\ Q\,t_L + c_b\,Q\,(t_{co}-t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w,\\
&V_{\text{inf}} = w\int_0^L Z\,dx, && V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx,\\
&V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, && V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx,\\
&V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0).
\end{aligned}
\tag{11.18}
$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.19. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".

$$
E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N}|Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%.
\tag{11.19}
$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$
E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}}, \quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}}\rangle}{\langle V\rangle}, \quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle}, \quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l},
\tag{12.19}
$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 13.11. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".

$$
\begin{aligned}
&E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, && E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},\\
&\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, && \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}},
\end{aligned}
\tag{13.11}
$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

25.1.17 The Dashboard, Charts tab

Yearly charts and the pinned baseline.

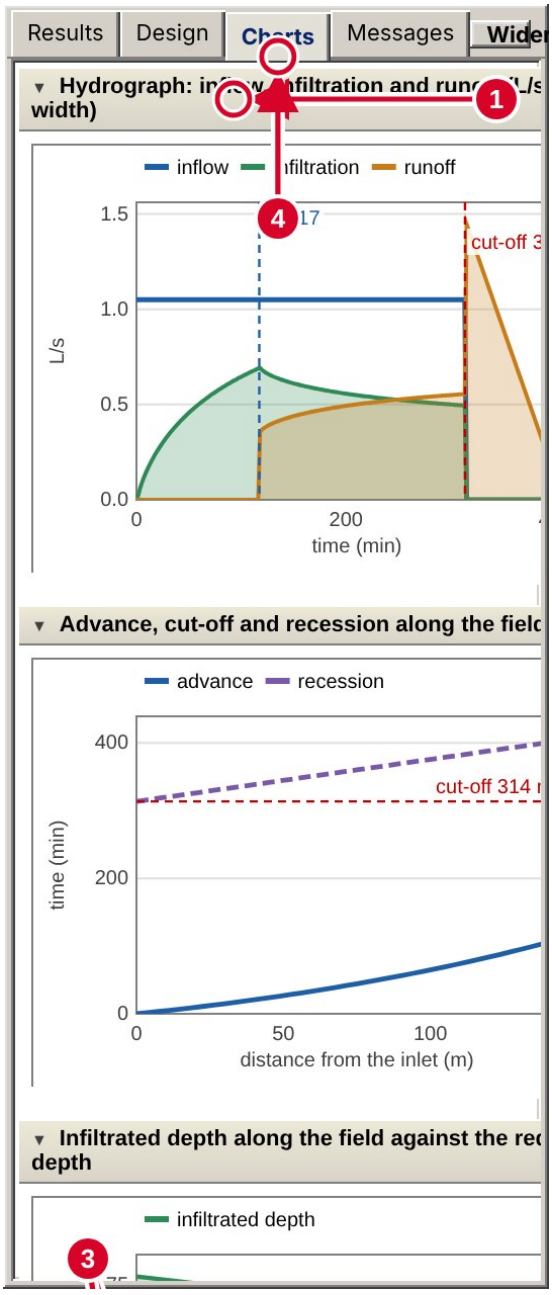


Figure 25.17: The Dashboard, Charts tab.

What the arrows point to.

- **1** Rainfall and irrigation for twenty years (Engine 1 annual values, Eqs. 5.4 to 5.12), with the pinned baseline of Eq. 14.5.
- **2** Drainage below the root zone, D of Eq. 5.9.
- **3** Evaporation and transpiration, E and T of Eqs. 5.4 to 5.12.

Equations behind this screen. Eq. 14.5.

Eq. 14.5. Where explained: chapter “The Dashboard”, section “The pinned baseline”.

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

Provenance (Eq. 14.5): **[Engineered]**: bookkeeping of Engine 2 series.

25.1.18 The Dashboard, Messages tab

Plain-language messages.

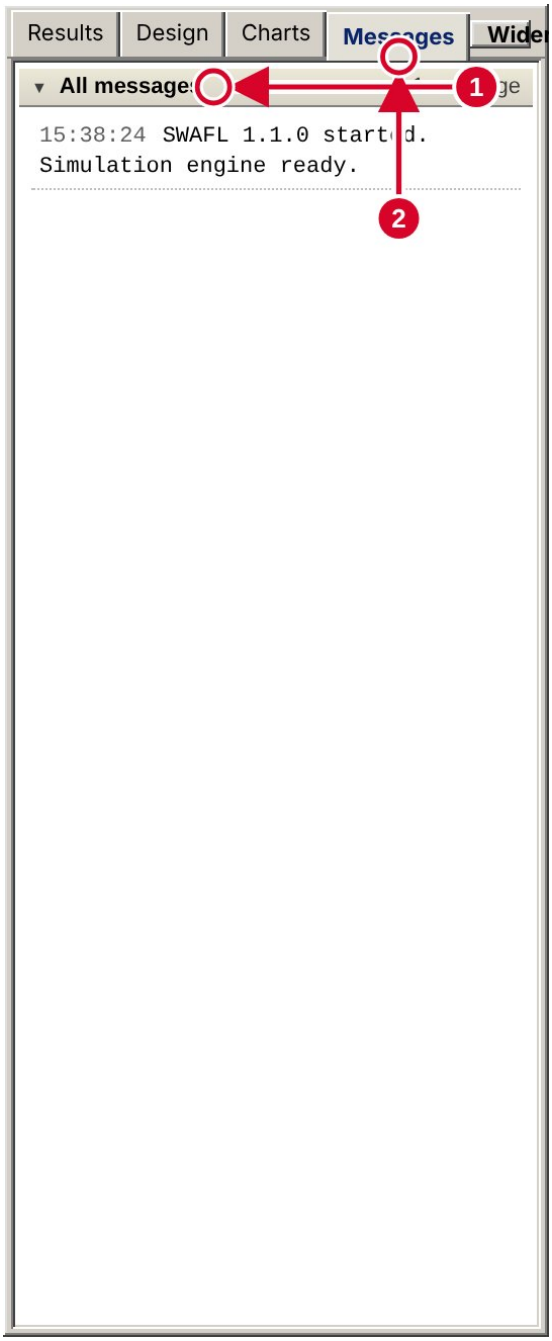


Figure 25.18: The Dashboard, Messages tab.

What the arrows point to.

- **1** Message log: engine start and plain-language notices.

25.1.19 The Properties panel

Every control; the equations are those that the controls move.

What the arrows point to.

- **1** Scenario: ready-made case, weather year and date (Eqs. 4.1 and 4.2).
- **2** Soil and field: soil, tillage, crop and consolidation rate tau_c (Eqs. 5.1, 5.2, 8.1).
- **3** Mulch: material, strength phi and interception capacity c (Eqs. 5.4 to 5.10, 9.1).
- **4** Weather and climate: rainfall scale rho and temperature shift (Eqs. 5.4 to 5.10).
- **5** Irrigation: method, trigger suction h_T and sensor depth (Eqs. 5.3, 5.11, 5.27).
- **6** Nutrients: nitrogen rate (Eqs. 5.10, 5.11).

Equations behind this screen. Eq. 5.18, Eq. 5.26, Eq. 8.1, Eq. 13.9.

Eq. 5.18. Where explained: chapter "The two computing engines", section "Roots".

$$R_{\max} = \mathrm{clip}\Big(R_c\,(0.85{+}0.3\,\mathrm{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \mathrm{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.26. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$E_{0,\mathrm{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\tfrac{1}{2}[1+\sin(2\pi(\mathrm{doy}-100)/365)]}{0.675}\,\mathrm{clip}\big(1{-}0.12(\rho{-}1);\ 0.85,\ 1.12\big)\sqrt{\gamma_y}\ \mathrm{m} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

Eq. 8.1. Where explained: chapter "Soil structure and tillage layers", section "Pore index and consolidation".

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 13.9. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and runoff".

$$t_p = \Big(\frac{a}{K}\Big)^{1/(A-1)}, \qquad F = \begin{cases} \sup, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \sup{-}F, \quad \mathrm{RO} = \max(0, R{-}S_d^{\mathrm{eff}}), \quad I = \sup{-}\mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

25.2 Surface irrigation design

25.2.1 The whole layer: inputs and overview

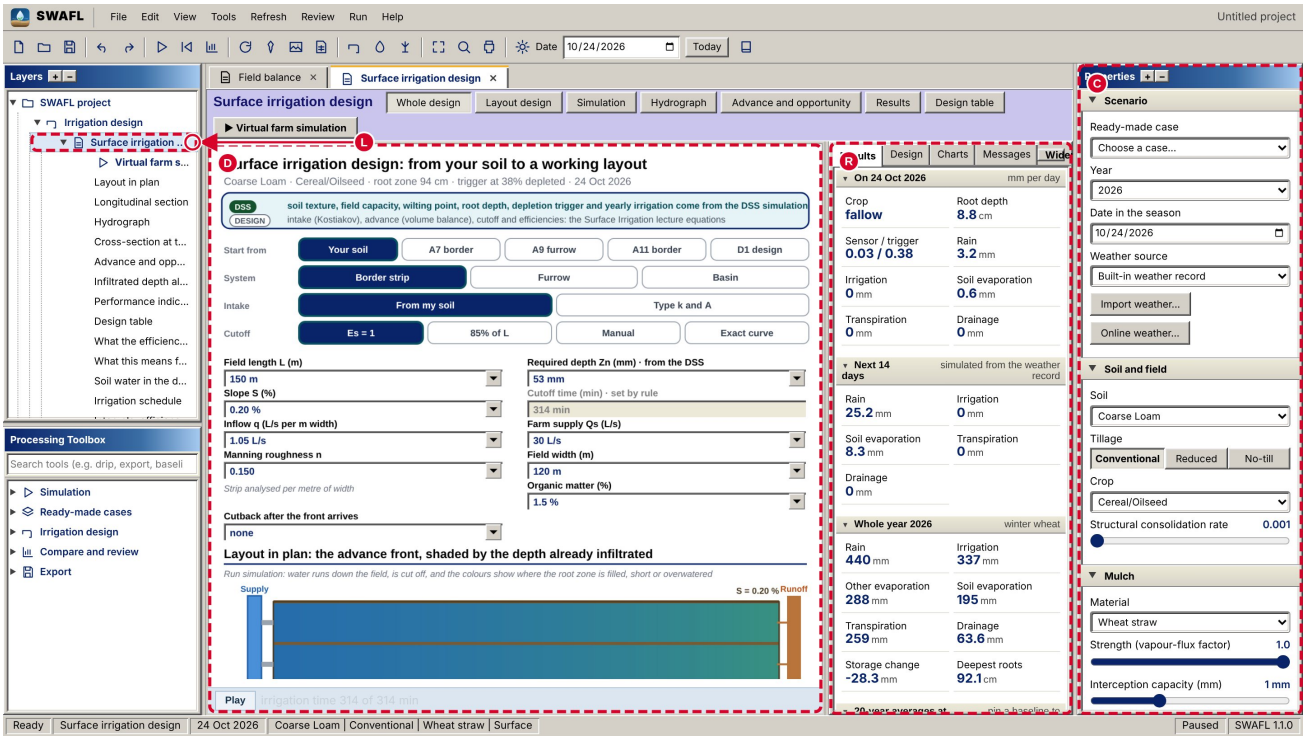


Figure 25.19: Surface irrigation design: the whole workbench.

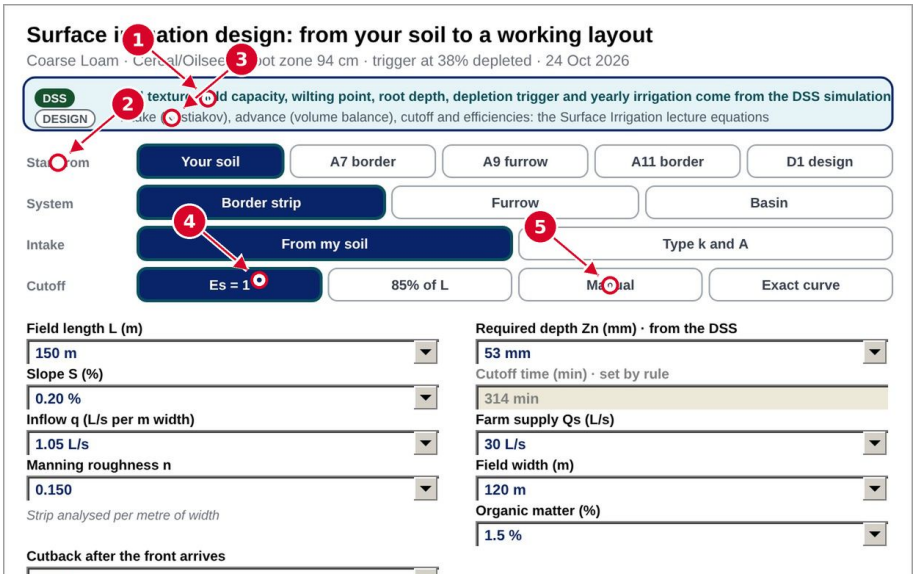


Figure 25.20: Surface irrigation design: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** DSS badge: soil, root zone, trigger and yearly irrigation come from the Engine 1 and 2 results (Eqs. 5.1 to 5.12, 5.27).
- **2** "Start from": in Your soil mode intake, Z_n and the best inflow are derived from the soil, Eqs. 11.1 to 11.7, 5.27 and 11.22.
- **3** Intake "From my soil": Saxton-Rawls, Green-Ampt and Kostiakov, Eqs. 11.1 to 11.7.
- **4** Cut-off "Es = 1": cut-off time t_co, Eq. 11.16.
- **5** Field length L, slope S, inflow q and Manning n: the inputs of Eqs. 11.8 and 11.9.

Equations behind this screen. Eq. 5.27, Eq. 11.1, Eq. 11.2, Eq. 11.3, Eq. 11.4, Eq. 11.5, Eq. 11.6, Eq. 11.7, Eq. 11.8, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13.

Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad (\text{surface; drip and sprinkler use their own limits}) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.1. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\begin{aligned}
\theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031,\\
\theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02),\\
\theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299,\\
\theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015),
\end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.2. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\begin{aligned}
\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\
\theta_{(S-33)} &= \theta_{(S-33),t} + (0.636\,\theta_{(S-33),t} - 0.107), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\
B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.
\end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 11.4. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Eq. 11.5. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$F - \psi_f \,\Delta\theta \,\ln\!\Big(1 + \frac{F}{\psi_f \,\Delta\theta}\Big) = K_s \,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.7. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

Eq. 11.8. Where explained: chapter "Surface irrigation design", section "Inlet flow area (Manning)".

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

25.2.2 Layout in plan: the advance front, shaded by the depth already infiltrated

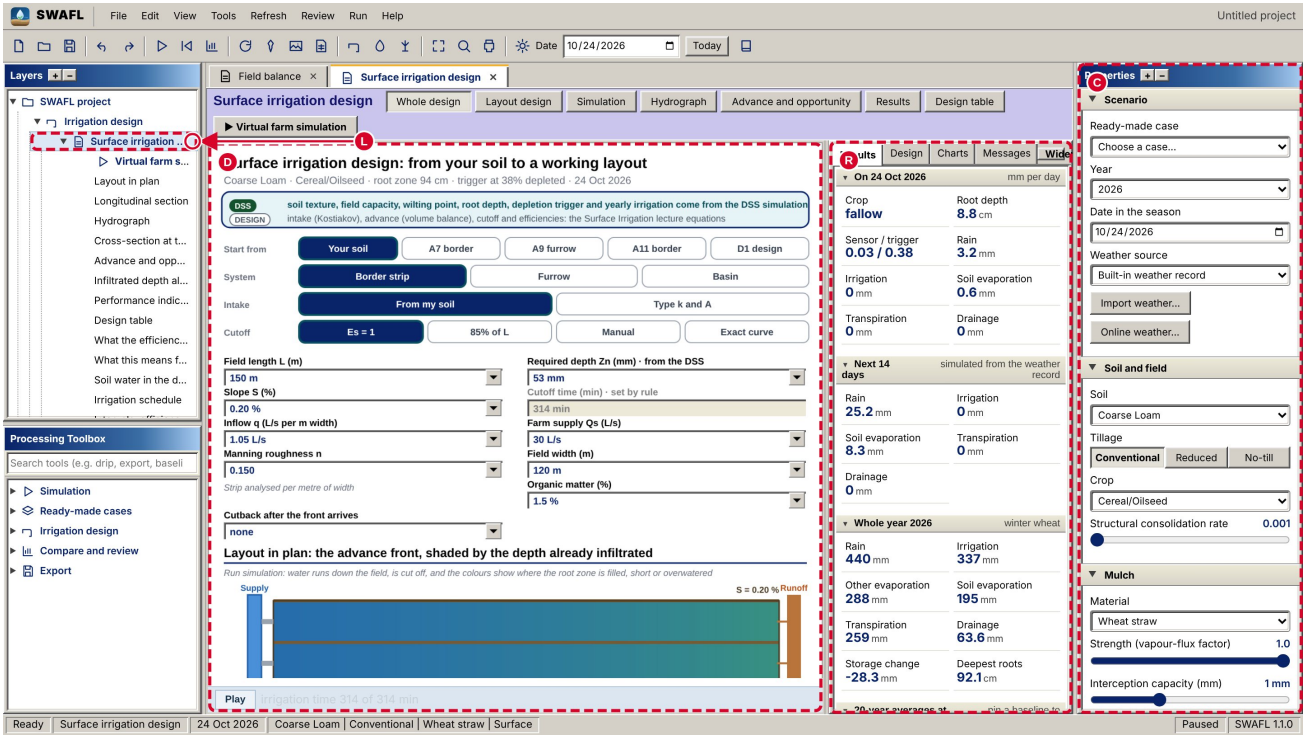


Figure 25.21: Surface irrigation design / Layout in plan: the whole workbench.

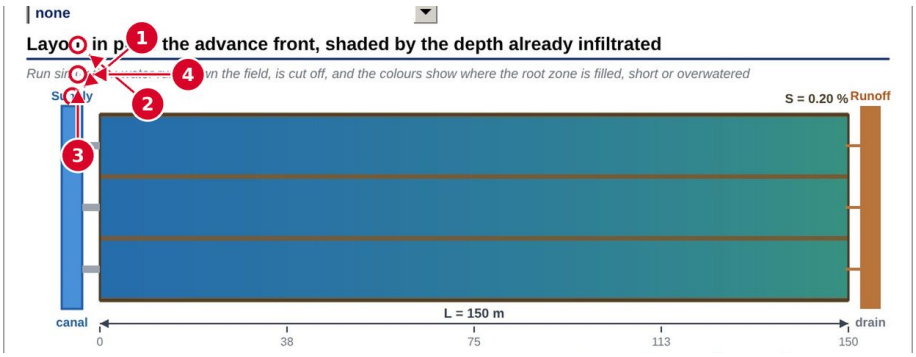


Figure 25.22: Surface irrigation design / Layout in plan: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Supply canal: inflow q enters the strip (flow area A_0 from Manning, Eq. 11.8).
- **2** Strip in plan: front position x(t) from Eq. 11.9; colours show infiltrated depth Z(x) against Z_n, Eq. 11.17.
- **3** Field length L (input).
- **4** Legend dry / short / filled / excess: classes of Z(x) against Z_n, Eq. 11.17.

Equations behind this screen. Eq. 5.27, Eq. 11.8, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13.

Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.8. Where explained: chapter "Surface irrigation design", section "Inlet flow area (Manning)".

$$Q = \frac{1}{n}\, A\, R_h^{2/3}\, S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$Q\,t = \sigma_y\, A_0\, x + \sigma_z\, k\, t^{A}\, w\, x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\, k\, t^{A}\, w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\, q^{-q_e}\, x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\, k\, w\, L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

25.2.3 Longitudinal section: water running over your soil and soaking in

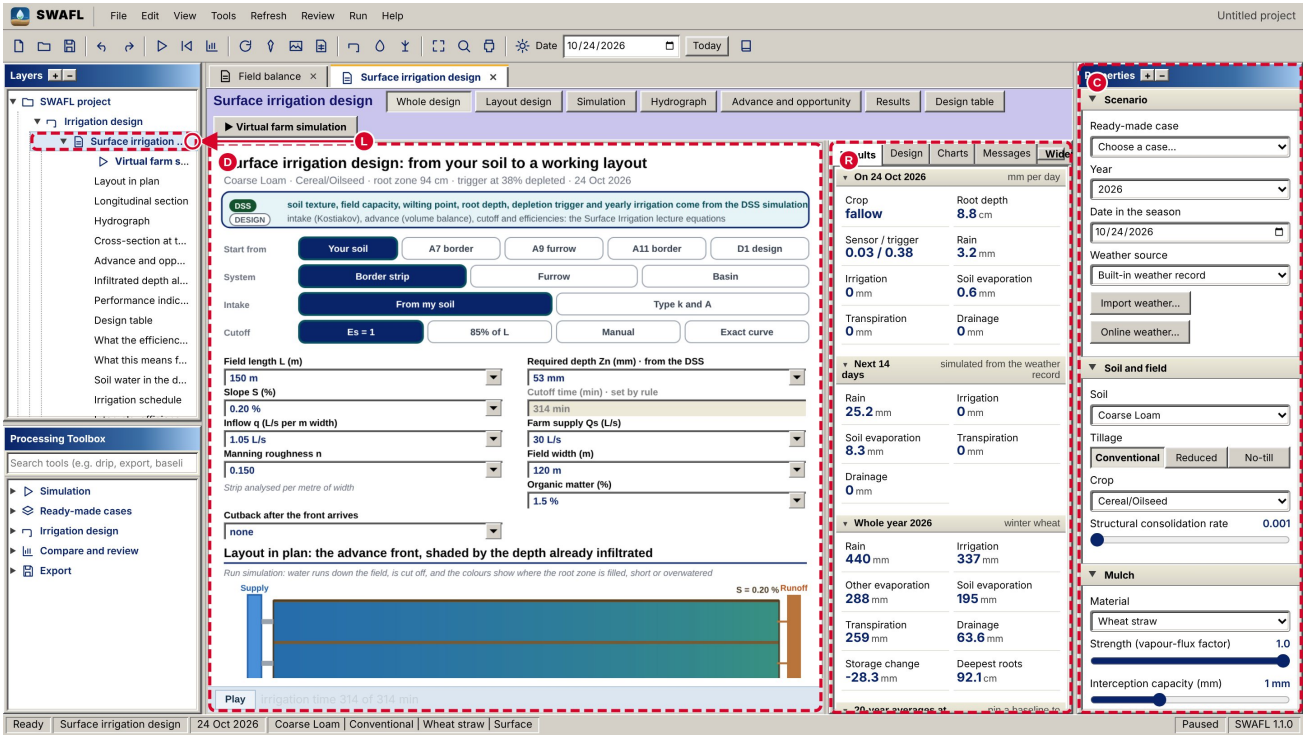


Figure 25.23: Surface irrigation design / Longitudinal section: the whole workbench.

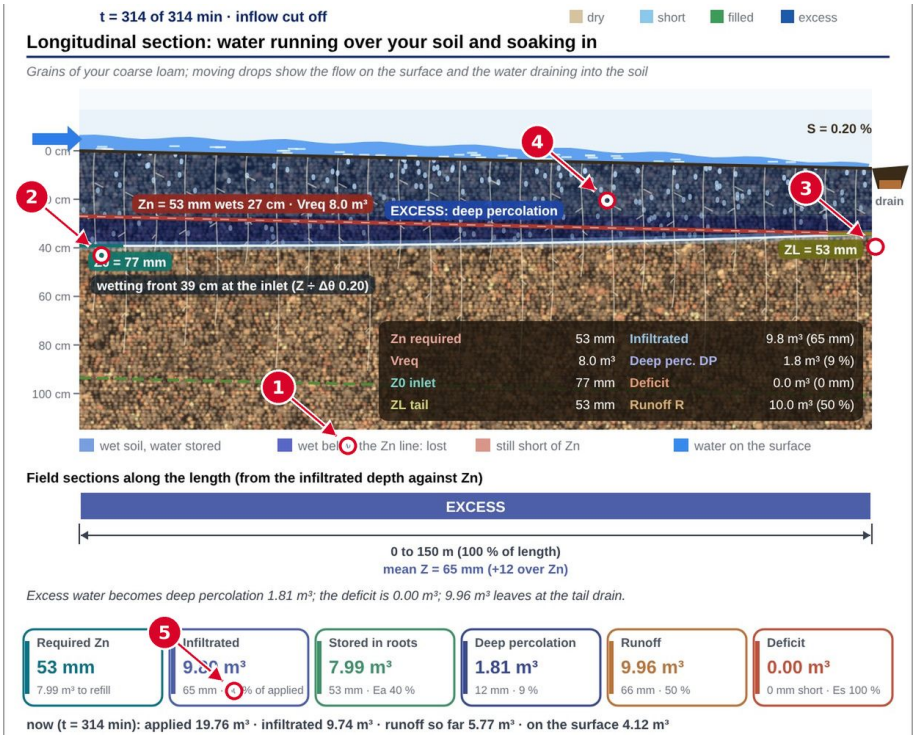


Figure 25.24: Surface irrigation design / Longitudinal section: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Z_n line: required depth from the DSS (Eq. 5.27) drawn as depth of wetted soil.
- **2** Z_0: depth infiltrated at the inlet, Eq. 11.17.
- **3** Z_L: depth infiltrated at the tail, Eq. 11.17.
- **4** EXCESS: water below the Z_n line is lost to deep percolation, V_dp of Eq. 11.18.
- **5** Infiltrated volume V_inf (tile at the foot), Eq. 11.18.

Equations behind this screen. Eq. 11.6, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13, Eq. 11.15, Eq. 11.16, Eq. 11.17.

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$\sigma_z = \frac{A + r\,(1-A) + 1}{(1+A)(1+r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_x(x) = t_L\left(\frac{x}{L}\right)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.15. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

Eq. 11.16. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.17. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$Z(x) = k\,\big[t_{co} - t_x(x)\big]^A, \quad Z_0 = k\,t_{co}^A\ (\text{inlet}), \qquad Z_L = k\,(t_{co} - t_L)^A\ (\text{tail}). \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

25.2.4 Hydrograph: inflow, infiltration and runoff through time

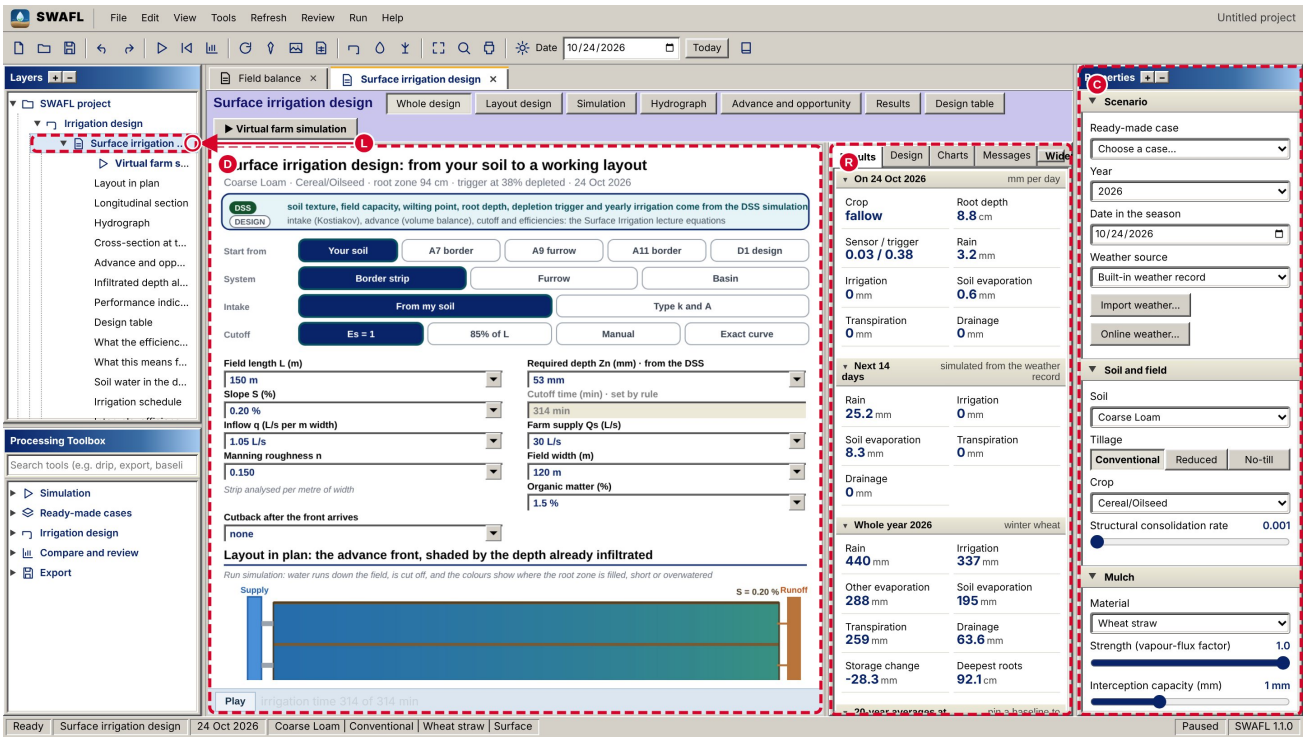


Figure 25.25: Surface irrigation design / Hydrograph: the whole workbench.

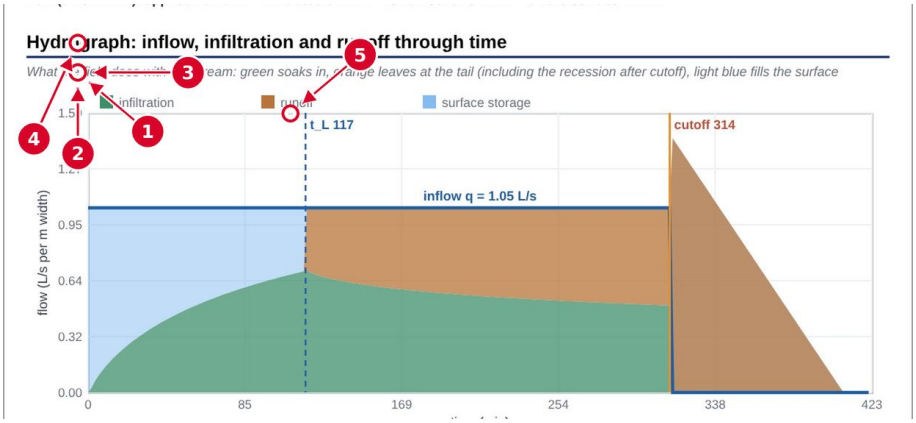


Figure 25.26: Surface irrigation design / Hydrograph: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Green: infiltration I(t): Eq. 11.20.
- **2** Orange: runoff R(t) with its recession after cut-off: Eqs. 11.20 and 11.21.
- **3** Blue: water stored on the surface: Eq. 11.21.
- **4** Cut-off time t_co: Eq. 11.16.
- **5** t_L: advance time to the tail, Eq. 11.9.

Equations behind this screen. Eq. 11.18, Eq. 11.20, Eq. 11.21.

Eq. 11.18. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".

$$V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b<1:\ Q\,t_L + c_b\,Q\,(t_{co}-t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w,$$
$$V_{\text{inf}} = w\int_0^L Z\,dx, \qquad V_{\text{st}} = w\int_0^L \min(Z,Z_n)\,dx,$$
$$V_{\text{dp}} = w\int_0^L \max(Z-Z_n,0)\,dx, \qquad V_{\text{df}} = w\int_0^L \max(Z_n-Z,0)\,dx,$$
$$V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0). \tag{11.18}$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.20. Where explained: chapter "Surface irrigation design", section "Hydrograph".

$$V_c(t) = w\sum_i k\left[\min(t,t_{co}) - \bar{t}_{x,i}\right]_+^A \Delta x\ \cdot s_c, \qquad s_c = \frac{V_{\text{inf}}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

Provenance (Eq. 11.20): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

Eq. 11.21. Where explained: chapter "Surface irrigation design", section "Hydrograph".

$$R(t) = \frac{2V_s}{t_r}\Big(1 - \frac{t-t_{co}}{t_r}\Big), \quad t_{co} < t < t_{co}+t_r. \tag{11.21}$$

Provenance (Eq. 11.21): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

25.2.5 Cross-section at the inlet: the wetted bulb in your soil, and the intake curve

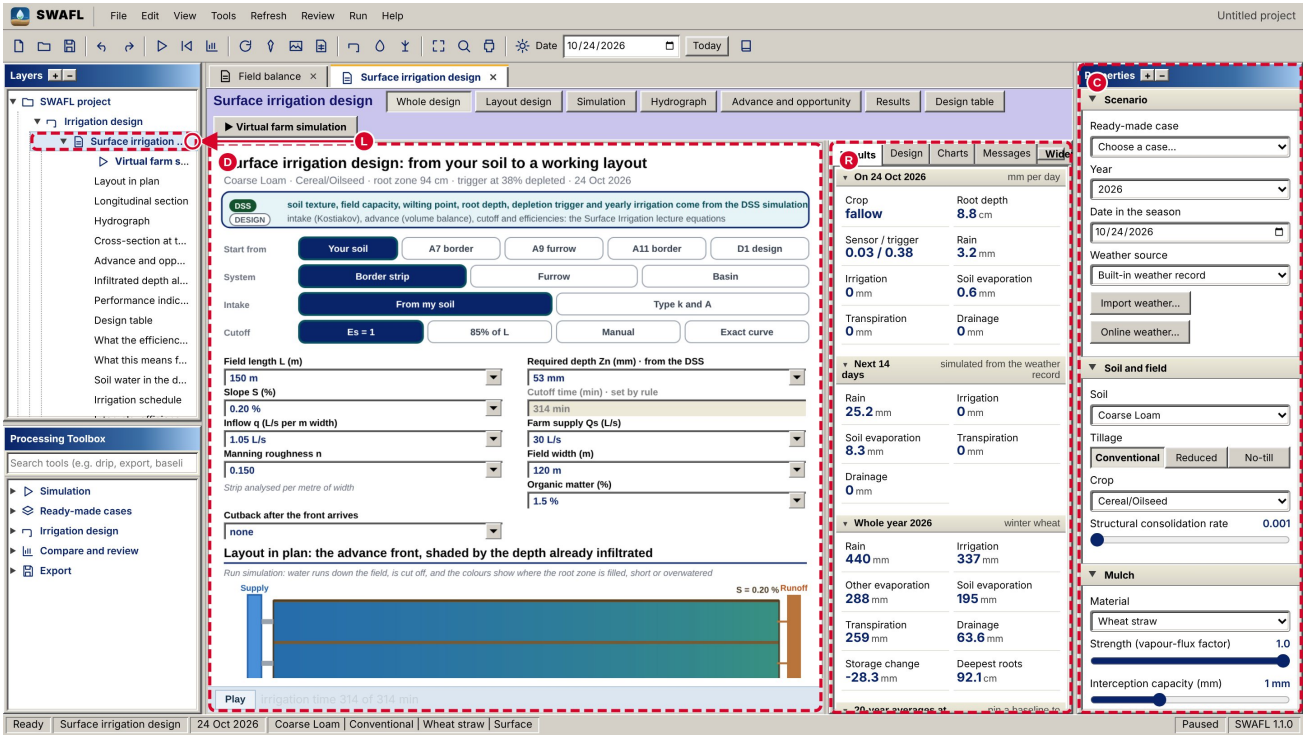


Figure 25.27: Surface irrigation design / Cross-section at the inlet: the whole workbench.

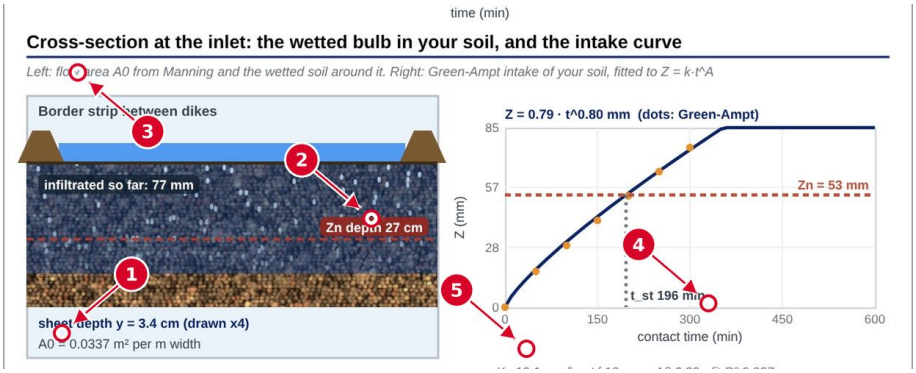


Figure 25.28: Surface irrigation design / Cross-section at the inlet: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sheet depth and flow area A_0 from Manning: Eq. 11.8.
- **2** Z_n depth: the required depth Z_n as a depth of wetted soil (storage of Eq. 11.4).
- **3** Intake curve Z = k tau^A fitted to Green-Ampt: Eqs. 11.5 to 11.7.
- **4** t_st: time to infiltrate the required depth, Eq. 11.15.
- **5** Ks, psi_f and Delta theta of the soil: Eqs. 11.2 to 11.4.

Equations behind this screen. Eq. 5.27, Eq. 11.1, Eq. 11.2, Eq. 11.3, Eq. 11.4, Eq. 11.5, Eq. 11.6, Eq. 11.7.

Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad (\text{surface; drip and sprinkler use their own limits}) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.1. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\begin{aligned}
\theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\mathrm{OM} + 0.005\,S\,\mathrm{OM} - 0.013\,C\,\mathrm{OM} + 0.068\,S\,C + 0.031,\\
\theta_{1500} &= \theta_{1500,t} + \big(0.14\,\theta_{1500,t} - 0.02\big),\\
\theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\mathrm{OM} + 0.006\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} + 0.452\,S\,C + 0.299,\\
\theta_{33} &= \theta_{33,t} + \big(1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015\big),
\end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.2. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\begin{aligned}
\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\
\theta_{(S-33)} &= \theta_{(S-33),t} + \big(0.636\,\theta_{(S-33),t} - 0.107\big), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\
B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.
\end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 11.4. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Eq. 11.5. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

Eq. 11.6. Where explained: chapter “Surface irrigation design”, section “Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)”.

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.7. Where explained: chapter “Surface irrigation design”, section “Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)”.

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

25.2.6 Advance and opportunity time

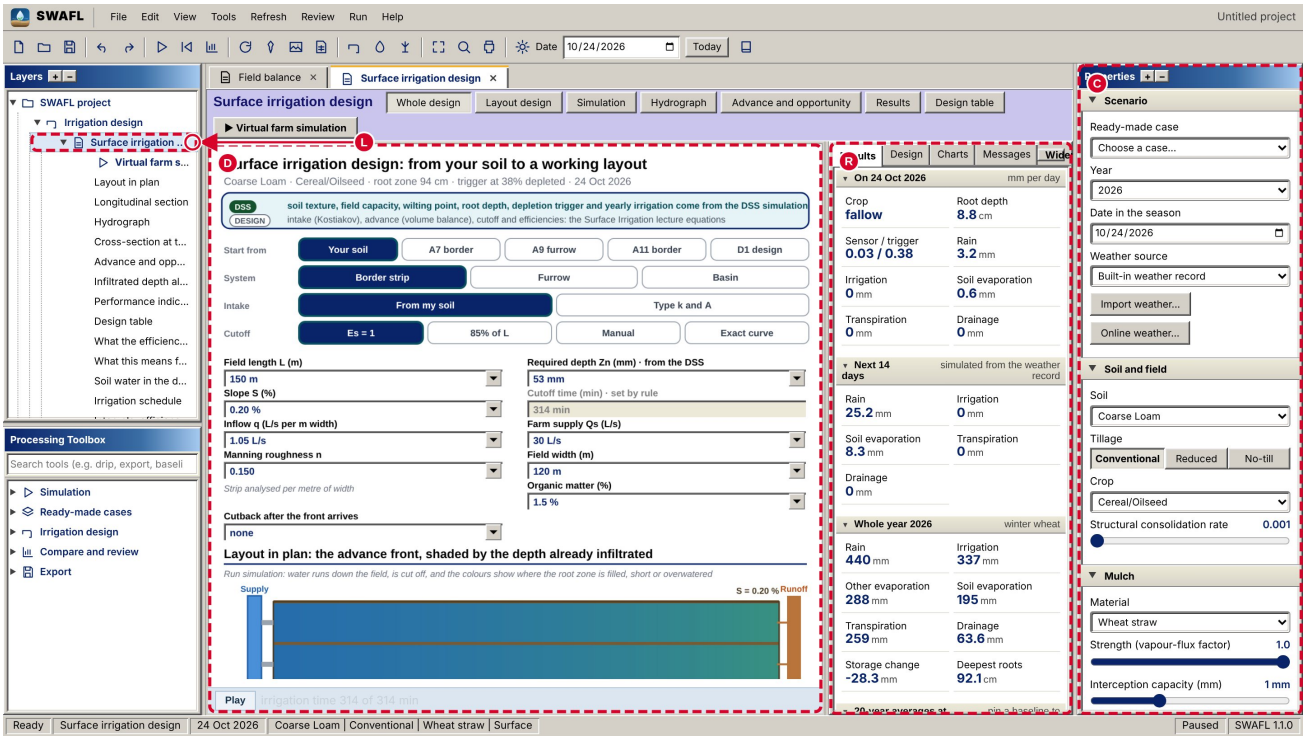


Figure 25.29: Surface irrigation design / Advance and opportunity time: the whole workbench.

Figure 25.30: Surface irrigation design / Advance and opportunity time: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Opportunity time tau = t_co - t_x: Eq. 11.17.
- **2** Advance curve x(t) from the volume balance: Eq. 11.9.
- **3** t_L: time for the front to reach the tail.
- **4** Cut-off 314 min = t_st + t_L: Eqs. 11.15 and 11.16.

Equations behind this screen. Eq. 11.6, Eq. 11.9, Eq. 11.10, Eq. 11.11, Eq. 11.12, Eq. 11.13, Eq. 11.15, Eq. 11.16, Eq. 11.17.

Eq. 11.6. Where explained: chapter “Surface irrigation design”, section “Intake from your soil

(Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.9. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$\sigma_z = \frac{A + r\,(1-A) + 1}{(1+A)(1+r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_x(x) = t_L\Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\,q^{-q_e}\,x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13. Where explained: chapter "Surface irrigation design", section "Advance (volume balance)".

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\,k\,w\,L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1-A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.15. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

Eq. 11.16. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{co} = \begin{cases} t_L + t_{st} & \text{"}E_s = 1\text{": the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{"85 \% of L"} \\ t_{co}^{\text{set}} & \text{"Manual"} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.17. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$Z(x) = k\left[t_{co} - t_x(x)\right]^A, \quad Z_0 = k\,t_{co}^A \text{ (inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \text{ (tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

25.2.7 Infiltrated depth along the field against the required depth Zn

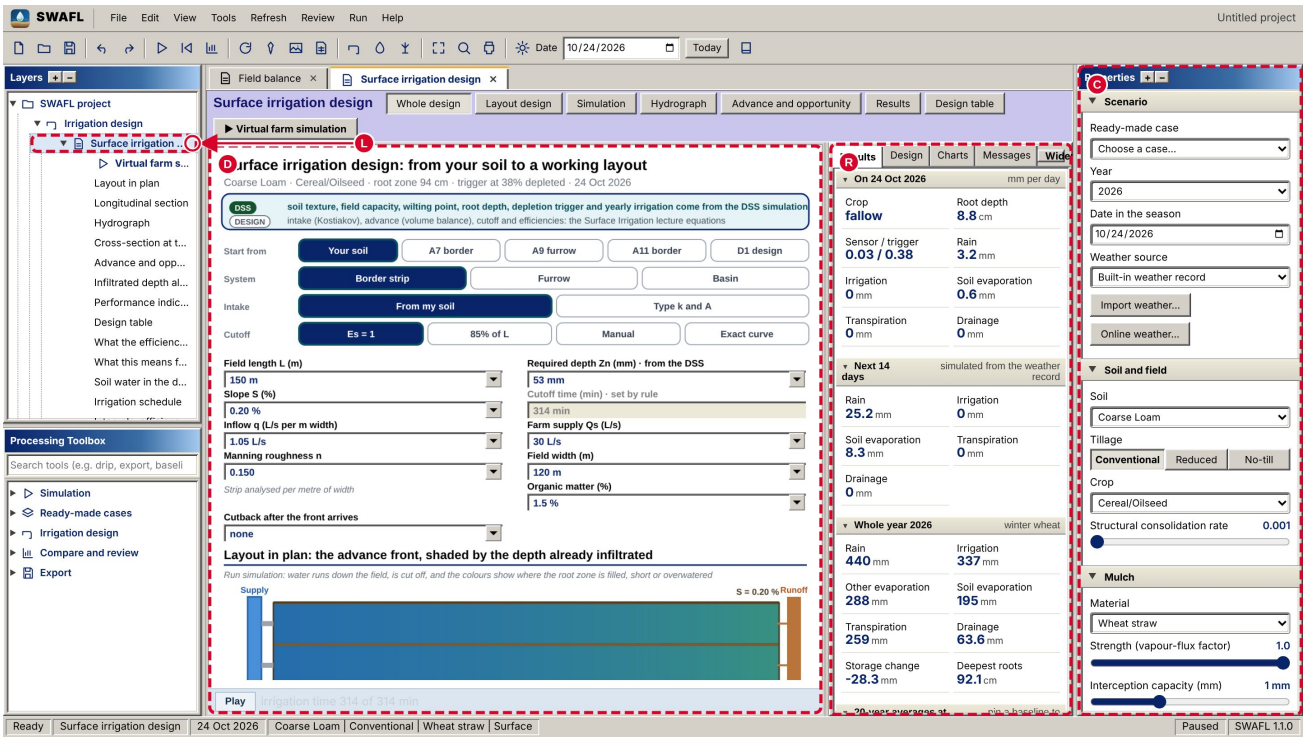


Figure 25.31: Surface irrigation design / Infiltrated depth along the field against the required depth Zn: the whole workbench.

Figure 25.32: Surface irrigation design / Infiltrated depth along the field against the required depth Zn: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Z_0: depth at the inlet, Eq. 11.17.
- **2** Z_L: depth at the tail, Eq. 11.17.
- **3** Z_n: required depth, Eq. 5.27.
- **4** Indigo: depth above Z_n, lost to deep percolation, V_dp of Eq. 11.18.
- **5** Green: water stored in the root zone, V_st of Eq. 11.18.

Equations behind this screen. Eq. 11.6, Eq. 11.15, Eq. 11.16, Eq. 11.17.

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.15. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{st} = \left(\frac{Z_n}{k}\right)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

Eq. 11.16. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.17. Where explained: chapter "Surface irrigation design", section "Cut-off, opportunity time and the infiltrated depth".

$$Z(x) = k\left[t_{co} - t_x(x)\right]^{A}, \quad Z_0 = k\,t_{co}^{A}\ (\text{inlet}), \qquad Z_L = k\,(t_{co} - t_L)^{A}\ (\text{tail}). \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

25.2.8 Performance indices and the water balance of one irrigation

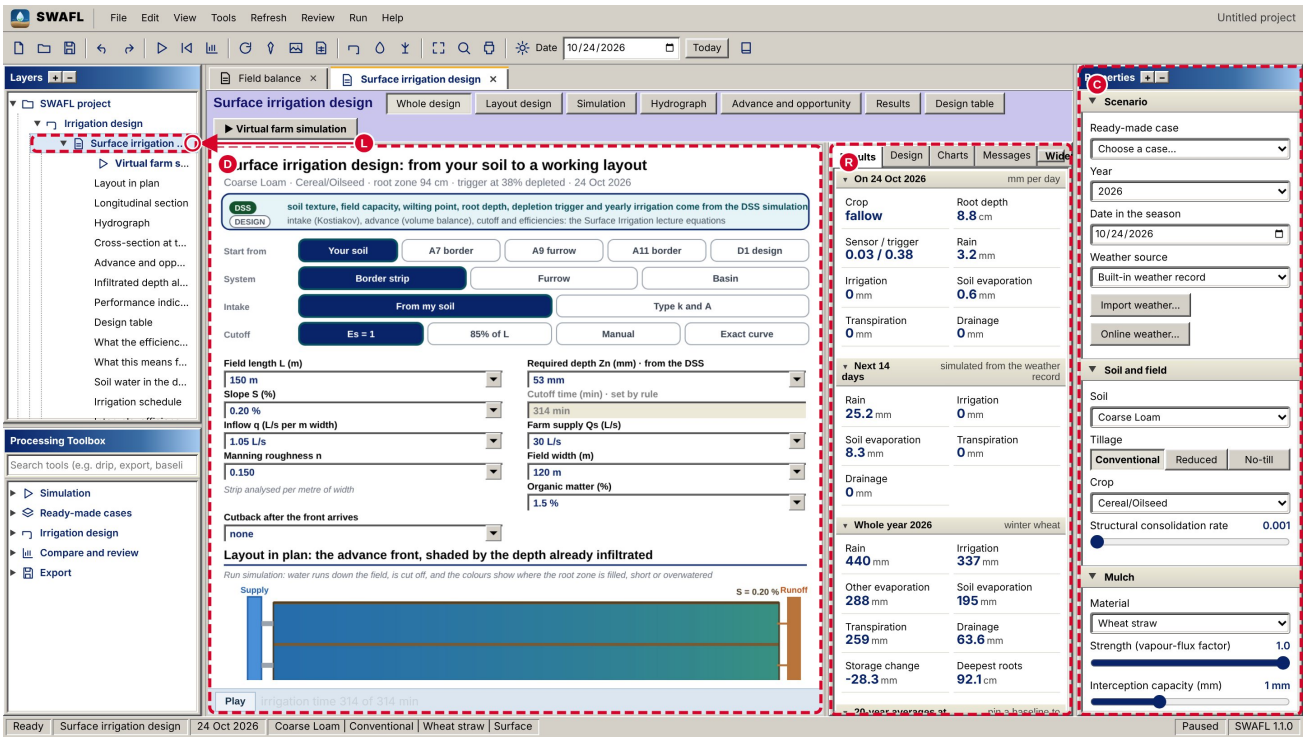


Figure 25.33: Surface irrigation design / Performance indices and the water balance of one irrigation: the whole workbench.

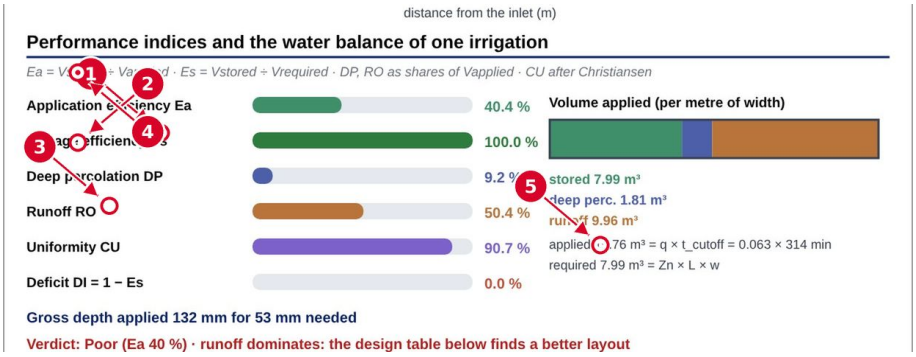


Figure 25.34: Surface irrigation design / Performance indices and the water balance of one irrigation: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Application efficiency E_a = V_st / V_app: Eq. 11.19.
- **2** Storage efficiency E_s = V_st / V_req: Eq. 11.19.
- **3** Deep percolation DP and runoff RO as shares of the applied volume: Eq. 11.19.
- **4** Uniformity CU after Christiansen: Eq. 11.19.
- **5** Volumes applied, stored, percolated and run off: Eq. 11.18.

Equations behind this screen. Eq. 11.18, Eq. 11.19.

Eq. 11.18. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".

$$V_{\text{app}} = Q\,t_{co} \quad \big(\text{with cutback } c_b < 1:\; Q\,t_L + c_b\,Q\,(t_{co} - t_L)\big), \quad V_{\text{req}} = Z_n\,L\,w,$$
$$V_{\text{inf}} = w\int_0^L Z\,dx, \qquad V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx,$$
$$V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, \qquad V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx,$$
$$V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\; 0).$$

(11.18)

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.19. Where explained: chapter "Surface irrigation design", section "Volumes, efficiencies and uniformity".

$$E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N} |Z_i - \bar{Z}|}{\bar{Z}}\Big)\;\%.$$

(11.19)

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

25.2.9 Design table: iterations toward an application efficiency of 85 % or more

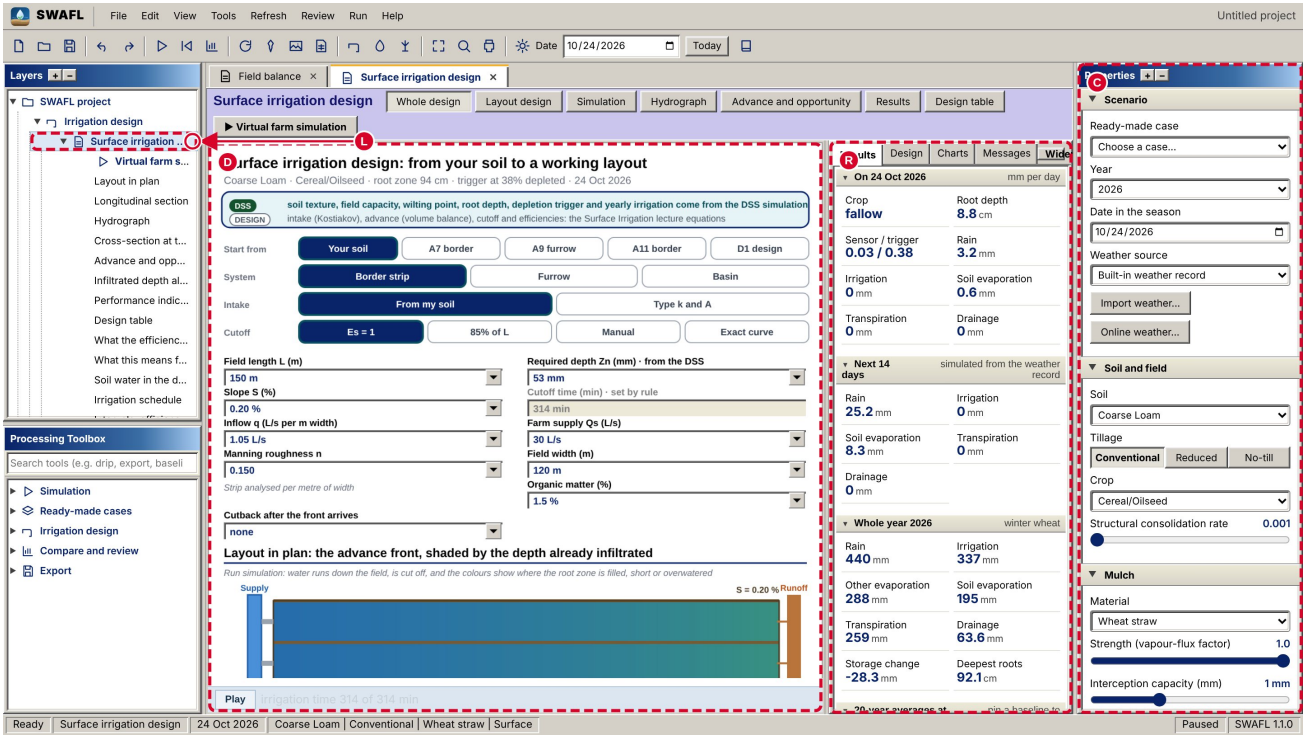


Figure 25.35: Surface irrigation design / Design table: the whole workbench.

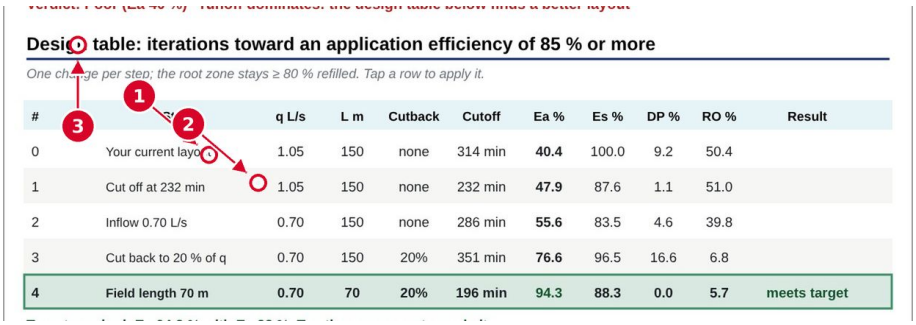


Figure 25.36: Surface irrigation design / Design table: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Row 0: your current layout; its score is Eq. 11.23.
- **2** Candidate moves in cut-off time, Eq. 11.16, then inflow (Eq. 11.22), cutback and length.
- **3** Winning design: E_a of at least 85 % with E_s of at least 80 %, chosen by the score of Eq. 11.23.

Equations behind this screen. Eq. 11.22, Eq. 11.23.

Eq. 11.22. Where explained: chapter “Surface irrigation design”, section “Best inflow (”Your soil” mode)“.

$$q_{\max} = \min\!\left\{ q^{\text{sys}}_{\max},\ \frac{0.04}{S} \cdot \frac{1000}{60} \right\} \ \mathrm{L\,s^{-1}} \quad (S \text{ in \%}), \tag{11.22}$$

Provenance (Eq. 11.22): **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

Eq. 11.23. Where explained: chapter “Surface irrigation design”, section “The design table”.

$$\text{score} = \min(E_a, 1) - 3\max(0,\, 0.8 - E_s) - 3\,[c_b < 1]\ \max(0,\, 0.05 - \text{RO}) - 5\max(0,\, E_a - 1). \tag{11.23}$$

Provenance (Eq. 11.23): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

25.2.10 What the efficiency is most sensitive to

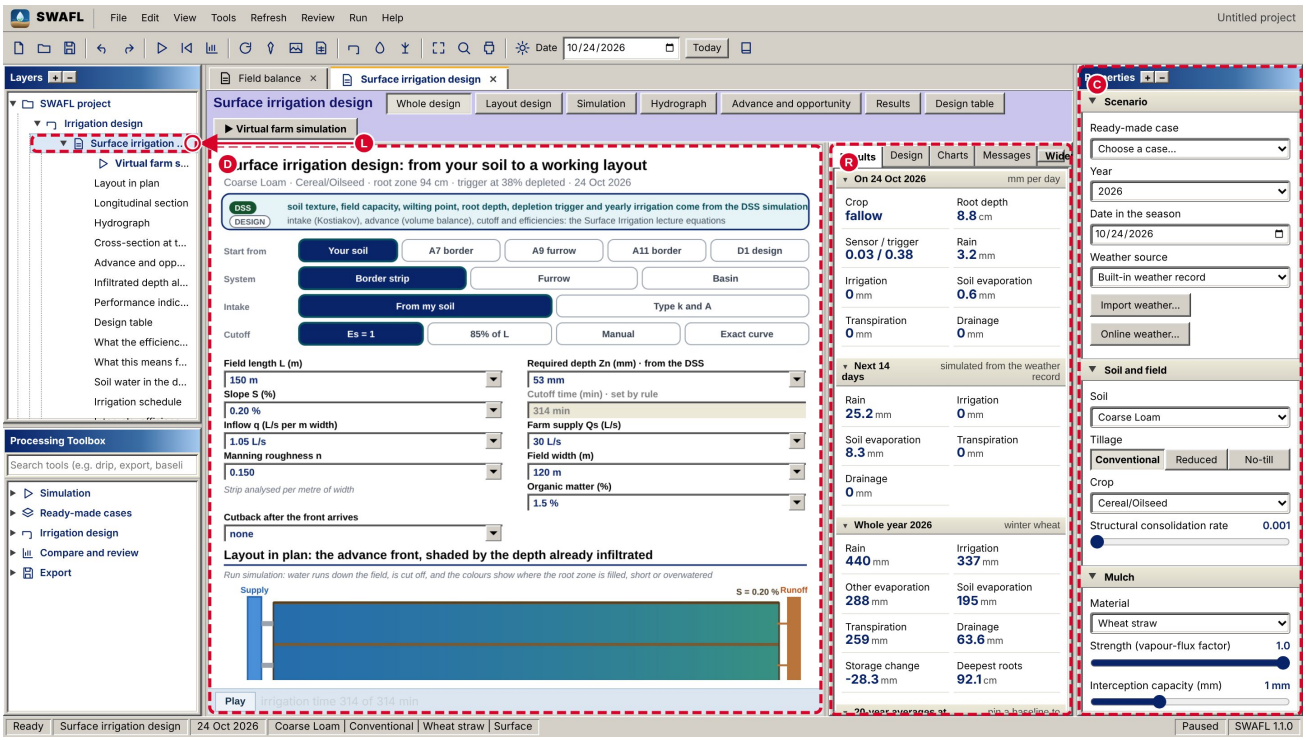


Figure 25.37: Surface irrigation design / What the efficiency is most sensitive to: the whole workbench.

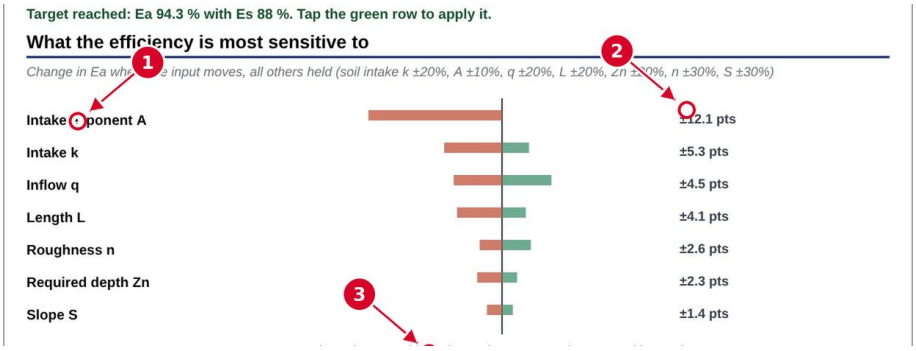


Figure 25.38: Surface irrigation design / What the efficiency is most sensitive to: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Tornado bar: change in E_a (Eq. 11.19) when this input moves by 10 to 30 %.
- **2** Size of the change in percentage points.
- **3** Green raises E_a, red lowers it.

Equations behind this screen. none (the screen shows stored results).

25.2.11 What this means for the season and the farm

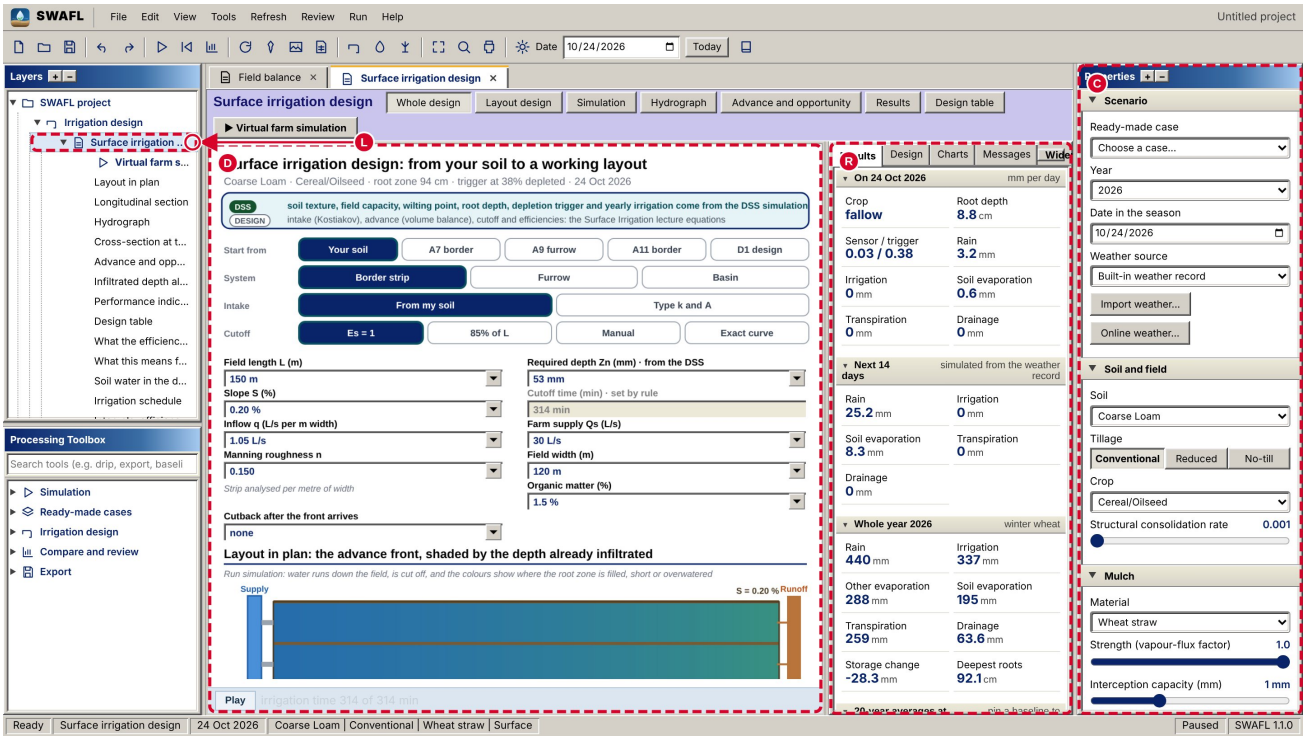


Figure 25.39: Surface irrigation design / What this means for the season and the farm: the whole workbench.

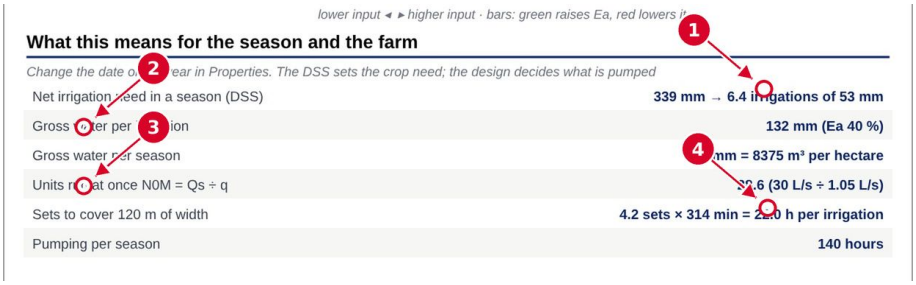


Figure 25.40: Surface irrigation design / What this means for the season and the farm: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Irrigations in a season, n_ev = sum I / Z_n: Eq. 11.24.
- **2** Gross depth per irrigation, Z_g = Z_n / E_a: Eq. 11.24.
- **3** Units run at once, Q_s / q: Eq. 11.24.
- **4** Sets and hours per irrigation: Eq. 11.24.

Equations behind this screen. Eq. 11.24.

Eq. 11.24. Where explained: chapter “Surface irrigation design”, section “Season and farm”.

$$
\begin{aligned}
n_{\text{ev}} &= \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g, \\
N_{0M} &= \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}),
\end{aligned}
\tag{11.24}
$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

25.2.12 Irrigation schedule: when to irrigate with this design

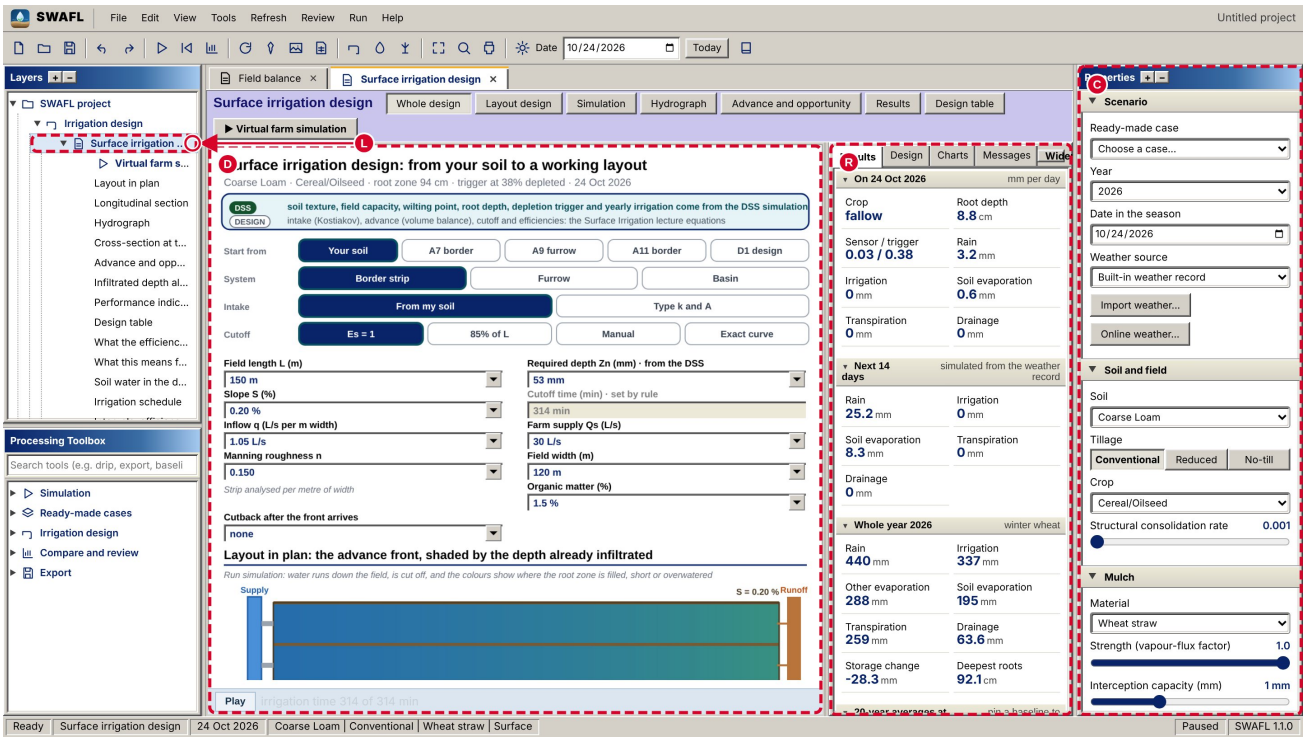


Figure 25.41: Surface irrigation design / Irrigation schedule: the whole workbench.

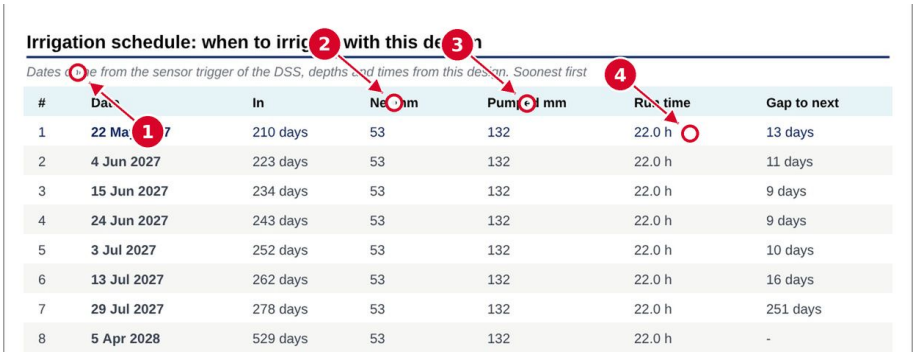


Figure 25.42: Surface irrigation design / Irrigation schedule: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Dates: from the Engine 2 irrigation record of the analogue year (Eqs. 4.1 and 4.2).
- **2** Net mm, I_k: the DSS irrigation depth that day.
- **3** Pumped mm = I_k / E_a: Eq. 11.24.
- **4** Run time of one round: Eq. 11.24.

Equations behind this screen. none (the screen shows stored results).

25.2.13 Intervals, efficiency and the advice in short

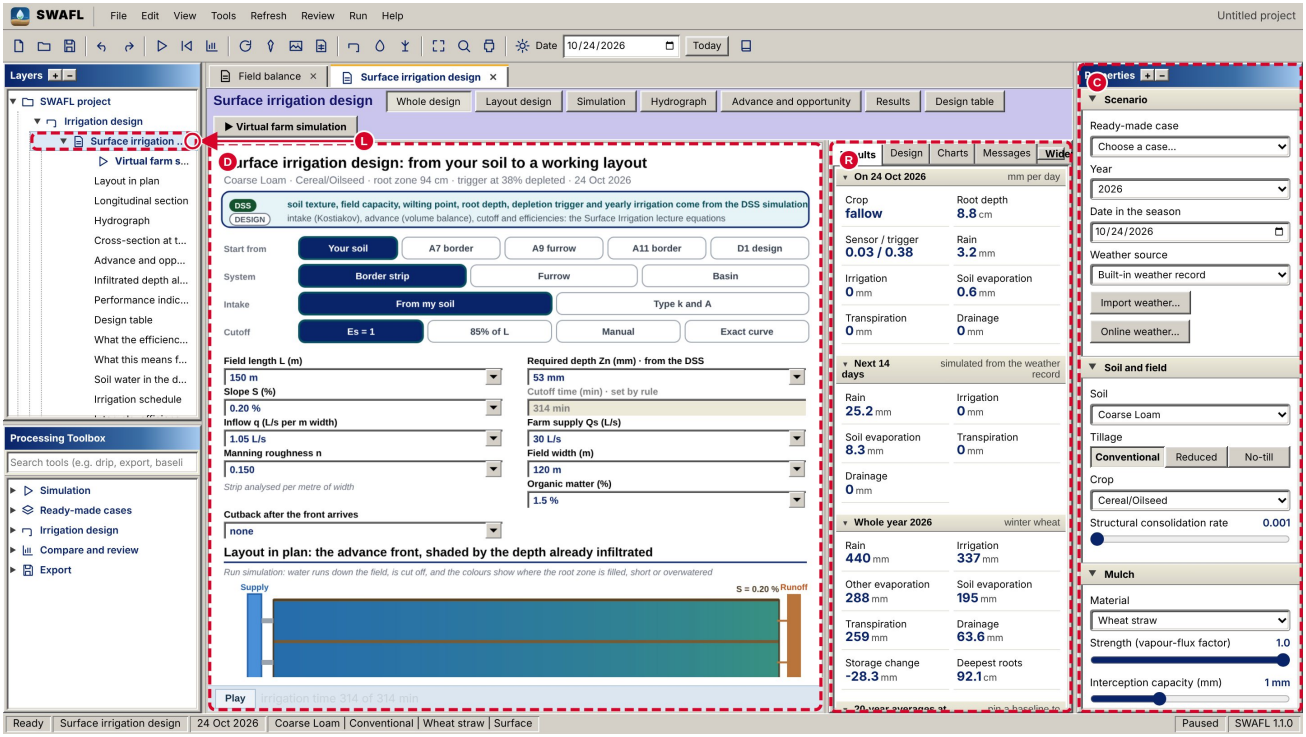


Figure 25.43: Surface irrigation design / Intervals, efficiency and the advice in short: the whole workbench.

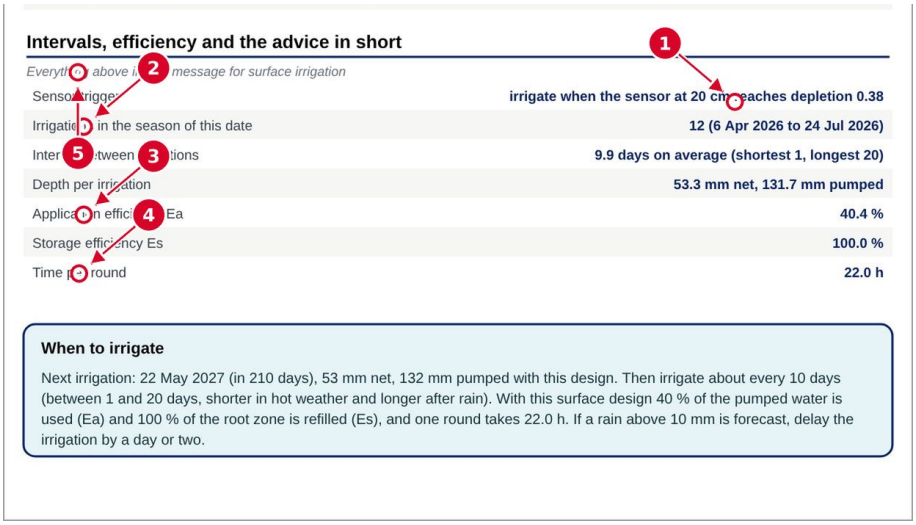


Figure 25.44: Surface irrigation design / Intervals, efficiency and the advice in short: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sensor trigger: depletion f_T, Eqs. 5.3 and 5.27.
- **2** Irrigations in the season: n_ev, Eq. 11.24.

- **3** Application efficiency E_a: Eq. 11.19.
- **4** Time per round: Eq. 11.24.
- **5** “When to irrigate”: a message assembled from the schedule and from E_a and E_s (Eqs. 11.19 and 11.24).

Equations behind this screen. Eq. 11.24.

Eq. 11.24. Where explained: chapter “Surface irrigation design”, section “Season and farm”.

$$
\begin{aligned}
n_{\text{ev}} &= \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g, \\
N_{0M} &= \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}),
\end{aligned}
\tag{11.24}
$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

25.3 Drip irrigation design

25.3.1 The whole layer: inputs and overview

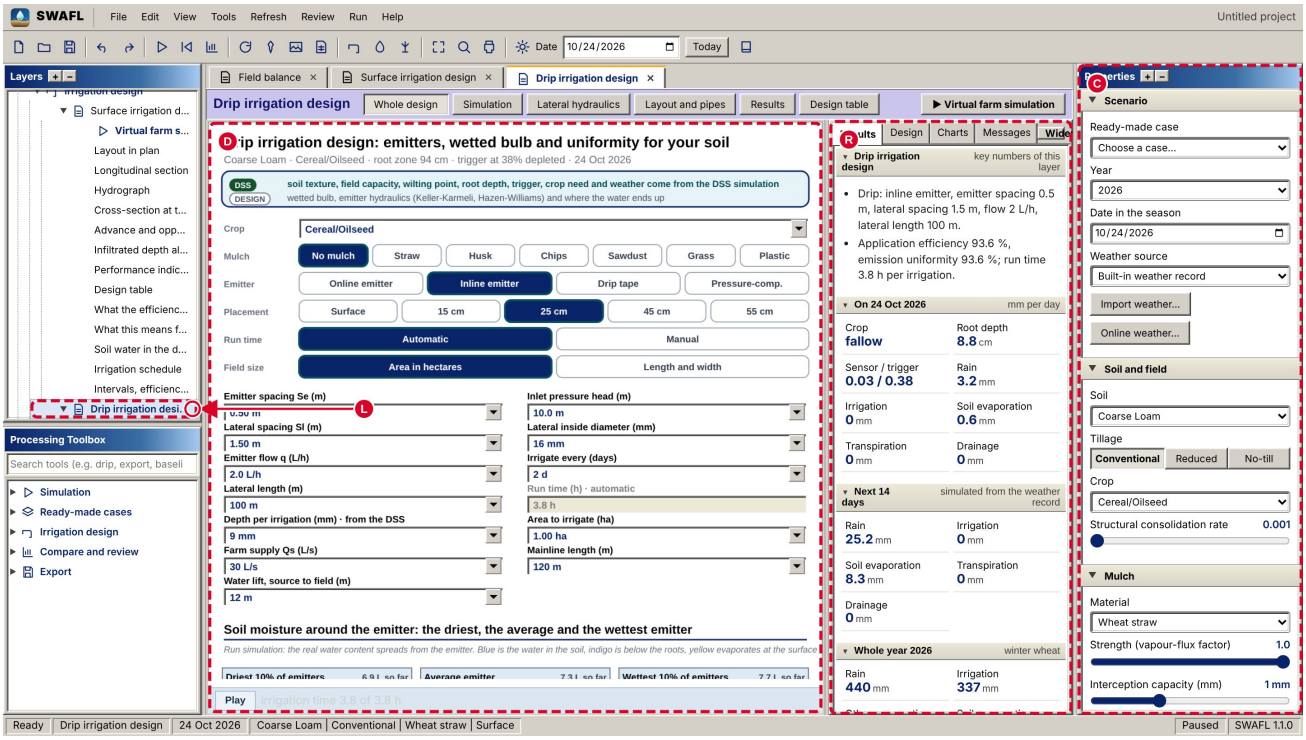


Figure 25.45: Drip irrigation design: the whole workbench.

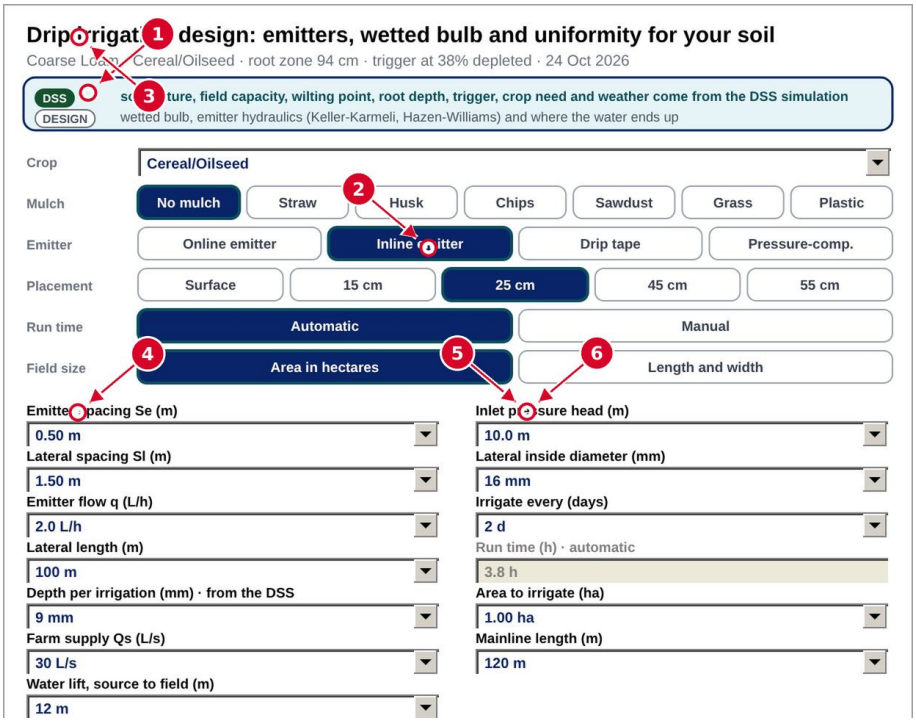


Figure 25.46: Drip irrigation design: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** DSS badge: soil, root depth, trigger and crop need come from Engines 1 and 2.
- **2** Emitter type sets the discharge exponent x: Eqs. 12.4 and 12.5.
- **3** Emitter depth z_d sets the bulb geometry: Eq. 12.10.
- **4** Emitter and lateral spacing, S_e and S_l: area served by one emitter, Eq. 12.17.
- **5** Emitter flow q at the inlet head H_0: Eq. 12.4.
- **6** Inlet pressure head H_0: Eqs. 12.3 and 12.4.

Equations behind this screen. Eq. 5.27, Eq. 11.3, Eq. 12.1, Eq. 12.2, Eq. 12.3, Eq. 12.4, Eq. 12.5, Eq. 12.6, Eq. 12.7, Eq. 12.8, Eq. 12.9, Eq. 12.10, Eq. 12.11, Eq. 12.12, Eq. 12.13, Eq. 12.14, Eq. 12.15, Eq. 12.16, Eq. 12.17, Eq. 12.18, Eq. 12.19.

Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad (\text{surface; drip and sprinkler use their own limits}) \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 12.1. Where explained: chapter "Drip irrigation design", section "Inputs, step by step".

$$Z_n = \min\big\{Z_n^{\mathrm{DSS}},\ \max(4,\ \mathrm{ET}_{\mathrm{pk}}\,\Delta t)\big\}\ \mathrm{mm}, \tag{12.1}$$

Provenance (Eq. 12.1): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.2. Where explained: chapter "Drip irrigation design", section "Inputs, step by step".

$$\mathrm{ET}_{\mathrm{pk}} = \mathrm{clip}\Big\{\max_d \tfrac{1}{5}\sum_{j=0}^{4}(E_{d+j} + T_{d+j});\ 2.5,\ 9\Big\}. \tag{12.2}$$

Provenance (Eq. 12.2): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.3. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$h_{f,j} = \frac{10.67\,S_e\,Q_j^{1.852}}{C^{1.852}\,D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i\ \ (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.4. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating; } h_j/H_0 \geq 0.01), \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.

Eq. 12.5. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \geq H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.

Eq. 12.6. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600}\ \ (\mathrm{L\,s^{-1}}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.

Eq. 12.7. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$E_u = \mathrm{clip}\Big\{\Big(1 - \frac{1.27\,c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

Eq. 12.8. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Eq. 12.9. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\rho(z) = r\,\sqrt{1 - \zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \geq z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \geq 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.11. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.12. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_w = \int \pi\, \rho(z)^2 \, dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.13. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$P_w = \mathrm{clip}\Big\{ \frac{2r}{1000\, S_l};\ 0,\ 1 \Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.14. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{dp}}^{(1)} = V\, \frac{\int_{z>R_{\max}} \pi \rho^2 \, dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{ev}} = \min\big\{ 0.5\, V,\ \pi \rho(0)^2 \cdot 10^{-6} \cdot 2 E_0 \big\} \quad (\mathrm{L},\ \text{area in } \mathrm{m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.16. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_i = \max\big\{ 0.02\, \bar{V},\ \bar{V}\,(1 + \sigma_q z_i) \big\}, \qquad \bar{V} = \bar{q}\, t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.17. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_{\mathrm{req}} = Z_n\, S_e\, S_l \ \ (\mathrm{L}), \qquad V_{\mathrm{st},i} = \min\{ V_{\mathrm{root},i},\ V_{\mathrm{req}} \}, \qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Eq. 12.18. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$t = \mathrm{clip}\Big\{ \frac{V_{\mathrm{req}}}{\bar{q}\, E_u};\ 0.05,\ 96 \Big\} \ \ \mathrm{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$E_a = \frac{\langle V_{\mathrm{st}} \rangle}{\langle V \rangle}, \quad E_s = \frac{\langle V_{\mathrm{st}} \rangle}{V_{\mathrm{req}}}, \quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}} \rangle}{\langle V \rangle}, \quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}} \rangle}{\langle V \rangle}, \quad Z_{\mathrm{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

25.3.2 Soil moisture around the emitter: the driest, the average and the wettest emitter

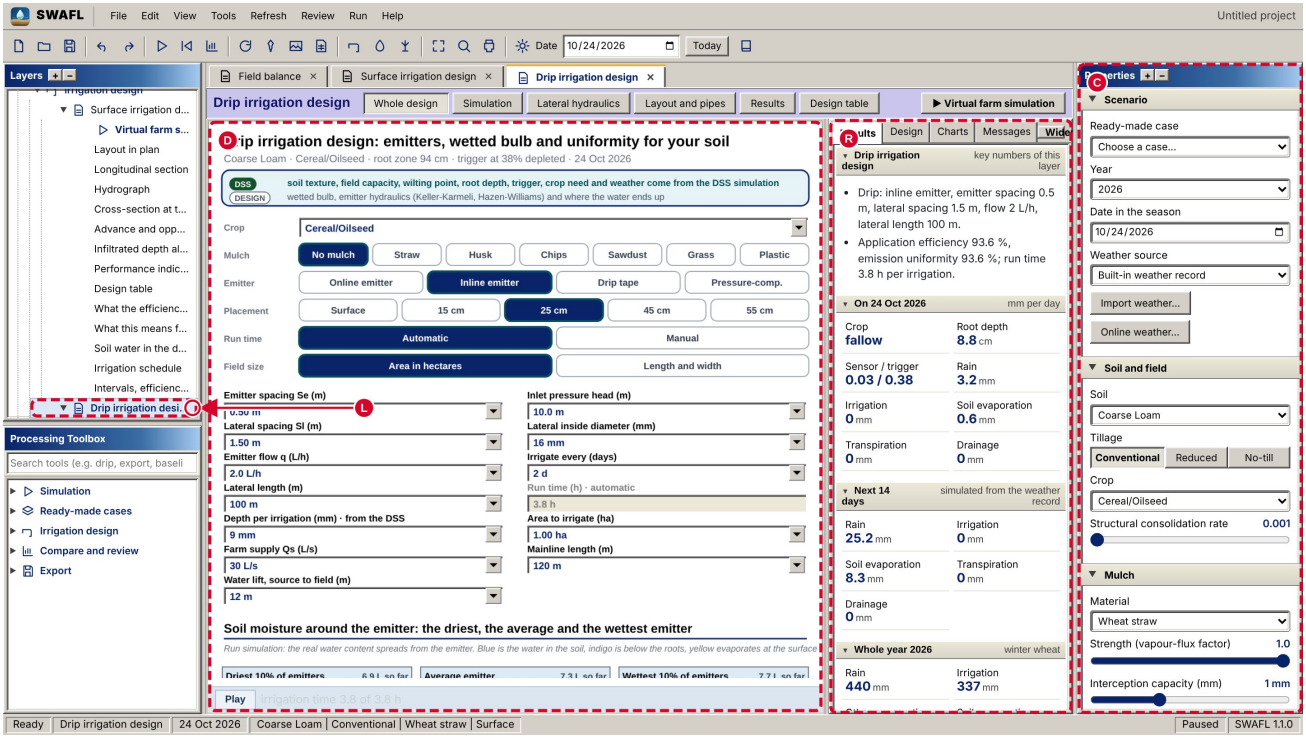


Figure 25.47: Drip irrigation design / Soil moisture around the emitter: the whole workbench.

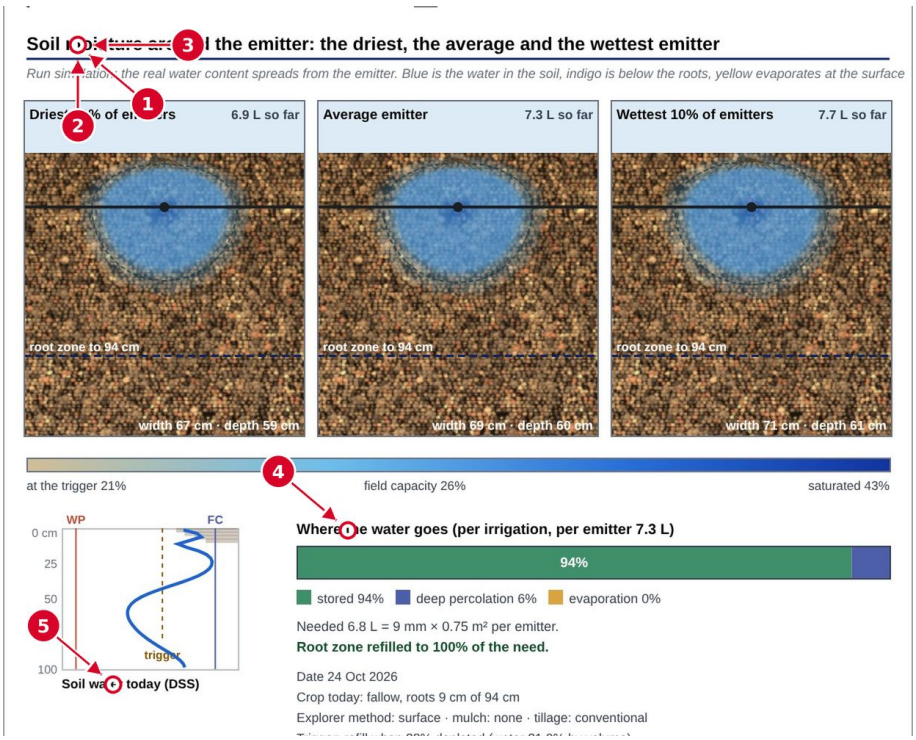


Figure 25.48: Drip irrigation design / Soil moisture around the emitter: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Bulb under the driest 10 % of emitters (low quantile of Eq. 12.16).
- **2** Bulb under the average emitter: Eqs. 12.10 to 12.12.
- **3** Bulb under the wettest 10 % of emitters (high quantile of Eq. 12.16).
- **4** Where the water goes: stored, deep percolation (Eq. 12.14) and evaporation (Eq. 12.15).
- **5** Soil water today from the DSS profile.

Equations behind this screen. Eq. 11.3, Eq. 12.9, Eq. 12.10, Eq. 12.11, Eq. 12.12, Eq. 12.13, Eq. 12.14, Eq. 12.15.

Eq. 11.3. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 12.9. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.11. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.12. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.13. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.14. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi \rho^2\, dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{ev}} = \min\!\left\{0.5\,V,\ \pi \rho(0)^2 \cdot 10^{-6} \cdot 2E_0\right\} \quad (\mathrm{L},\ \text{area in } \mathrm{m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

25.3.3 Uniformity along the lateral: pressure falls with friction, so far emitters deliver less

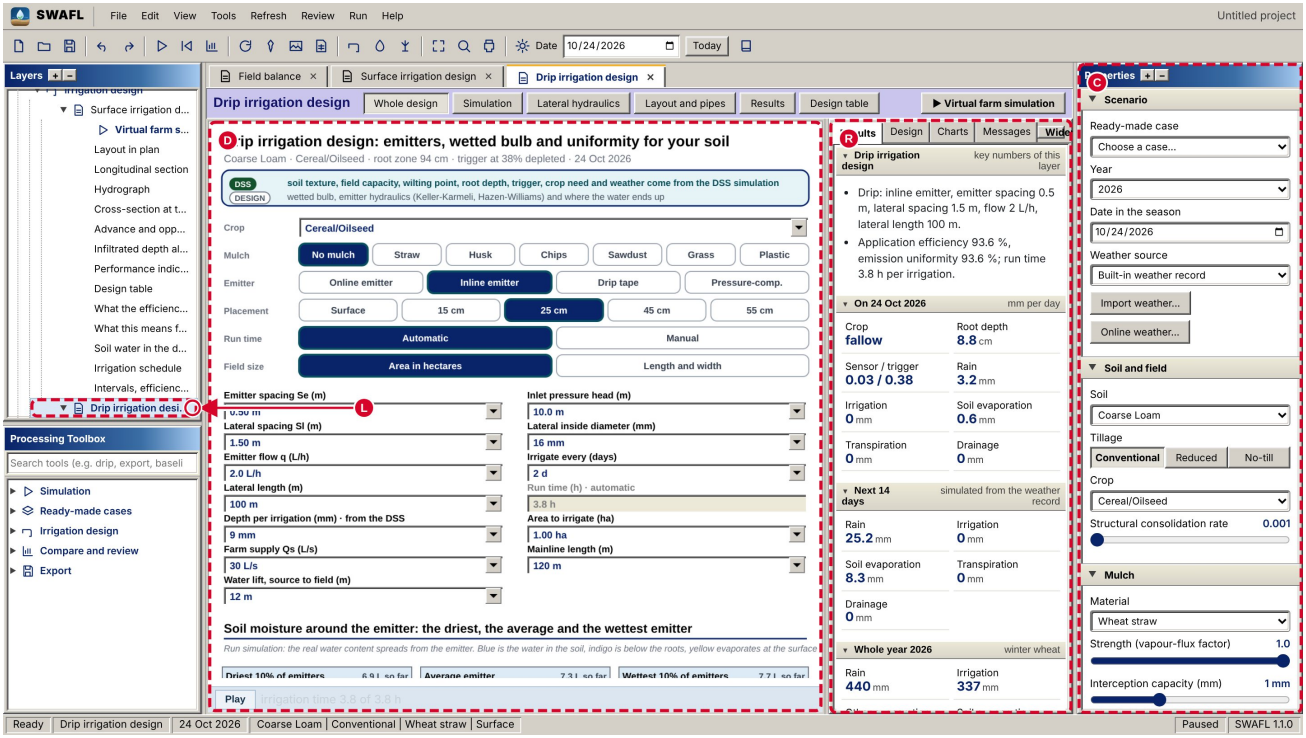


Figure 25.49: Drip irrigation design / Uniformity along the lateral: the whole workbench.

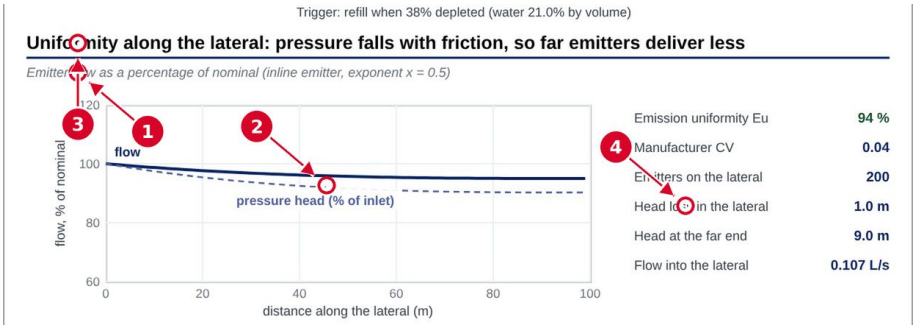


Figure 25.50: Drip irrigation design / Uniformity along the lateral: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Emitter flow q_j along the lateral: Eq. 12.4.
- **2** Pressure head h_j along the lateral, falling with friction: Eq. 12.3.
- **3** Emission uniformity E_u: Eq. 12.7.
- **4** Head loss in the lateral, sum of h_f: Eq. 12.3.

Equations behind this screen. Eq. 12.3, Eq. 12.4, Eq. 12.5, Eq. 12.6, Eq. 12.7, Eq. 12.8, Eq. 12.10, Eq. 12.15, Eq. 12.16, Eq. 12.17, Eq. 12.18, Eq. 12.19.

Eq. 12.3. Where explained: chapter “Drip irrigation design”, section “Lateral hydraulics: head loss and flow along the lateral”.

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.4. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\ h_j/H_0 \geq 0.01), \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.

Eq. 12.5. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \geq H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.

Eq. 12.6. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$\bar{q} = \frac{1}{N+1}\sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600} \;\; (\mathrm{L\,s^{-1}}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.

Eq. 12.7. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$E_u = \mathrm{clip}\Big\{\Big(1 - \frac{1.27\, c_v}{\sqrt{n_e}}\Big)\frac{q_{\min}}{\bar{q}};\ 0.3,\ 1\Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

Eq. 12.8. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\rho(z) = r\,\sqrt{1 - \zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \geq z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \geq 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.16. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1 + \sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.17. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_{\mathrm{req}} = Z_n\,S_e\,S_l\ \ (\mathrm{L}), \qquad V_{\mathrm{st},i} = \min\{V_{\mathrm{root},i},\ V_{\mathrm{req}}\}, \qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Eq. 12.18. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$t = \mathrm{clip}\Big\{\frac{V_{\mathrm{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \ \mathrm{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$E_a = \frac{\langle V_{\mathrm{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\mathrm{st}}\rangle}{V_{\mathrm{req}}}, \quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}}\rangle}{\langle V\rangle}, \quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}}\rangle}{\langle V\rangle}, \quad Z_{\mathrm{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

25.3.4 How the wetted bulb grows with the run time

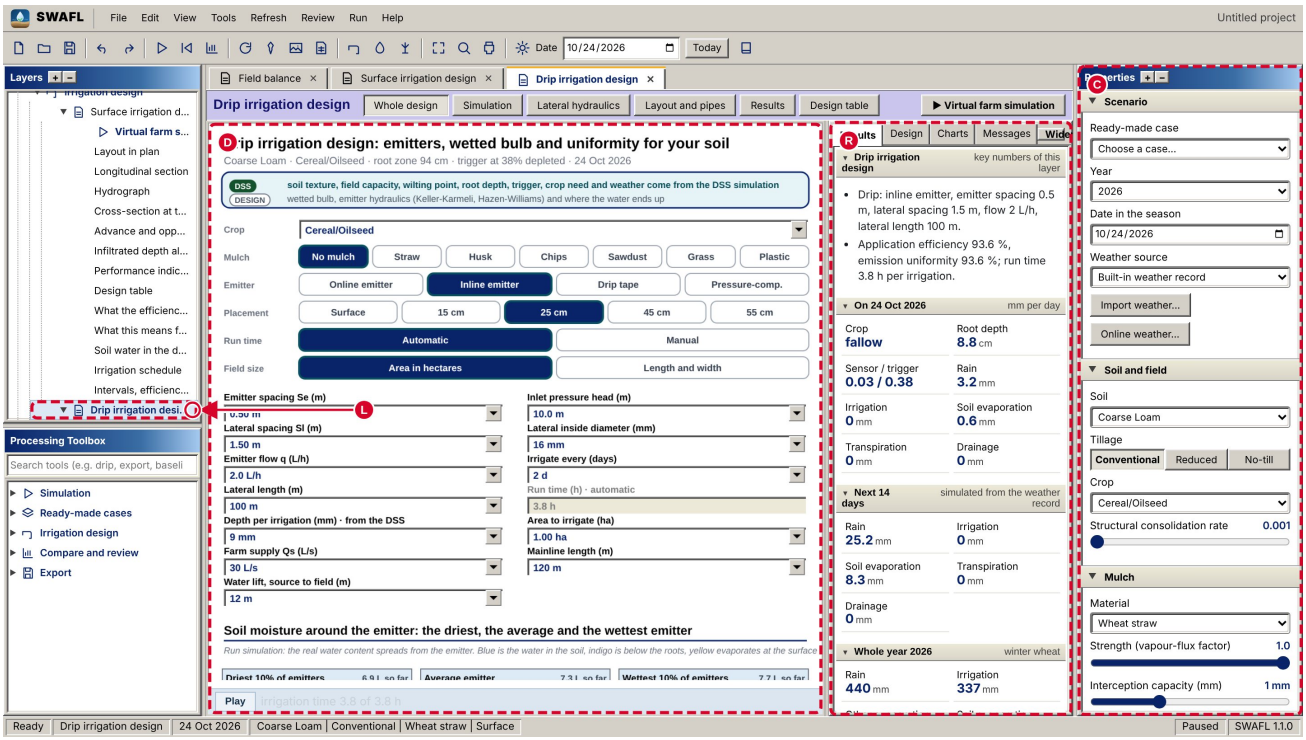


Figure 25.51: Drip irrigation design / How the wetted bulb grows with the run time: the whole workbench.

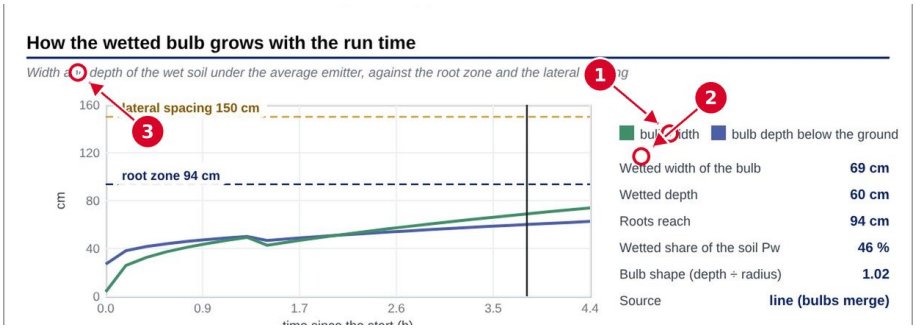


Figure 25.52: Drip irrigation design / How the wetted bulb grows with the run time: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Bulb width (green) and depth (indigo) against time: Eqs. 12.10 to 12.12.
- **2** Root-zone depth R_max, against which the bulb depth gives deep percolation: Eq. 12.14.
- **3** Lateral spacing S_l: the bulbs merge when the width reaches it, P_w of Eq. 12.13.

Equations behind this screen. Eq. 11.3, Eq. 12.9, Eq. 12.10, Eq. 12.11, Eq. 12.12, Eq. 12.13, Eq. 12.14, Eq. 12.15.

Eq. 11.3. Where explained: chapter “Surface irrigation design”, section “Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)”.

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 12.9. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.10. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.11. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$a = \mathrm{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^6/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.12. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.13. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$P_w = \mathrm{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.14. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2\cdot 10^{-6}\cdot 2E_0\} \quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

25.3.5 Result for this design

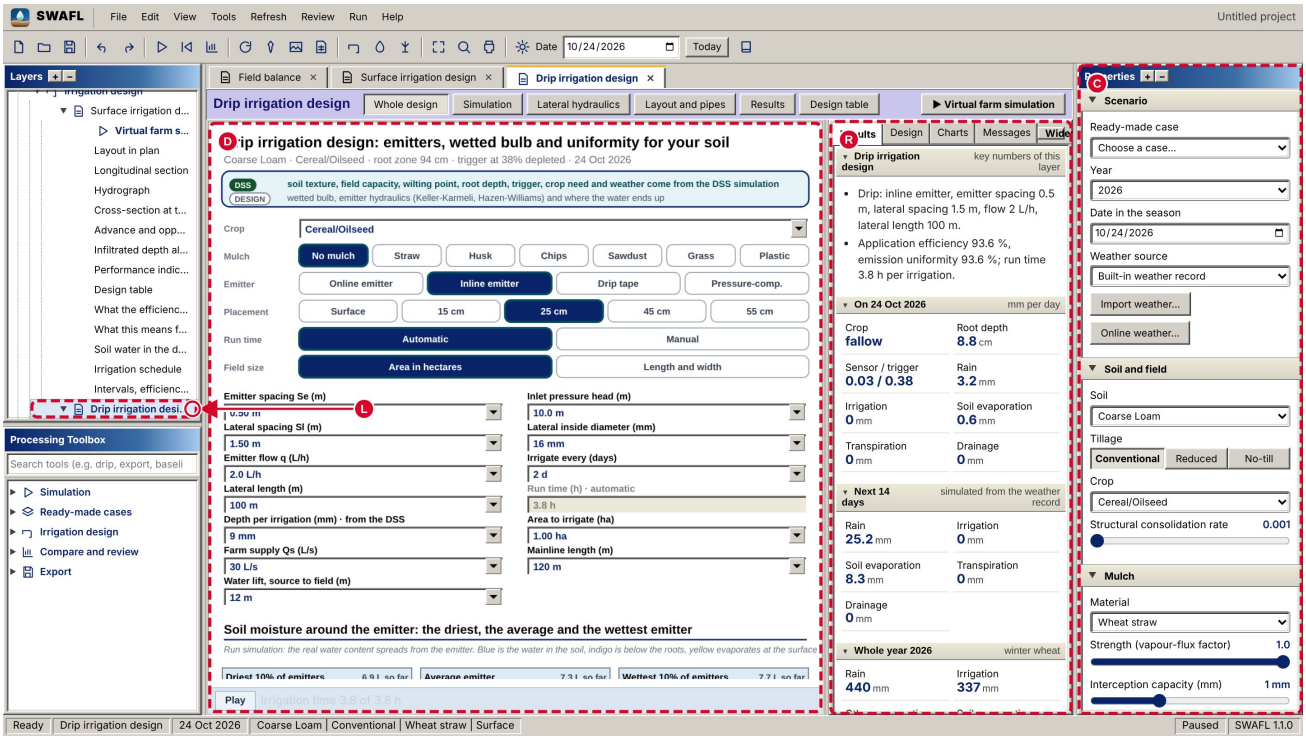


Figure 25.53: Drip irrigation design / Result for this design: the whole workbench.

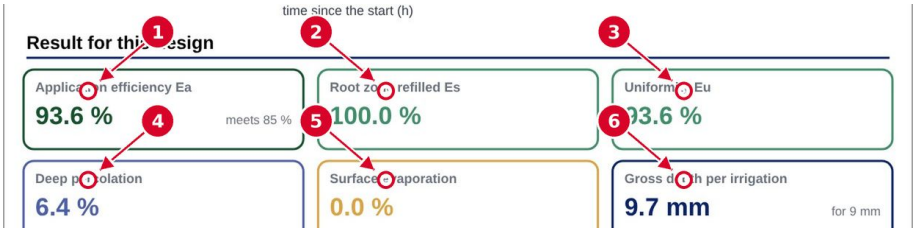


Figure 25.54: Drip irrigation design / Result for this design: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Application efficiency E_a: Eq. 12.19.
- **2** Root-zone refilled E_s: Eq. 12.19.
- **3** Emission uniformity E_u: Eq. 12.7.
- **4** Deep percolation: Eq. 12.14.
- **5** Surface evaporation: Eq. 12.15.
- **6** Gross depth per irrigation: Eqs. 12.17 and 12.18.

Equations behind this screen. Eq. 12.10, Eq. 12.15, Eq. 12.16, Eq. 12.17, Eq. 12.18, Eq. 12.19.

Eq. 12.10. Where explained: chapter “Drip irrigation design”, section “The wetted bulb”.

$$\rho(z) = r\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \geq z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \geq 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15. Where explained: chapter "Drip irrigation design", section "The wetted bulb".

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2 \cdot 10^{-6} \cdot 2E_0\} \quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.16. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1 + \sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.17. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$V_{\mathrm{req}} = Z_n\,S_e\,S_l\ \ (\mathrm{L}), \qquad V_{\mathrm{st},i} = \min\{V_{\mathrm{root},i},\ V_{\mathrm{req}}\}, \qquad V_{\mathrm{ex},i} = V_{\mathrm{root},i} - V_{\mathrm{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Eq. 12.18. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$t = \mathrm{clip}\Big\{\frac{V_{\mathrm{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \ \mathrm{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.19. Where explained: chapter "Drip irrigation design", section "Spread between emitters and the efficiencies".

$$E_a = \frac{\langle V_{\mathrm{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\mathrm{st}}\rangle}{V_{\mathrm{req}}}, \quad \mathrm{DP} = \frac{\langle V_{\mathrm{dp}}^{(1)} + V_{\mathrm{ex}}\rangle}{\langle V\rangle}, \quad \mathrm{EV} = \frac{\langle V_{\mathrm{ev}}\rangle}{\langle V\rangle}, \quad Z_{\mathrm{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

25.3.6 Layout and infrastructure: source, pump, filter, mainline, valves, submain and laterals

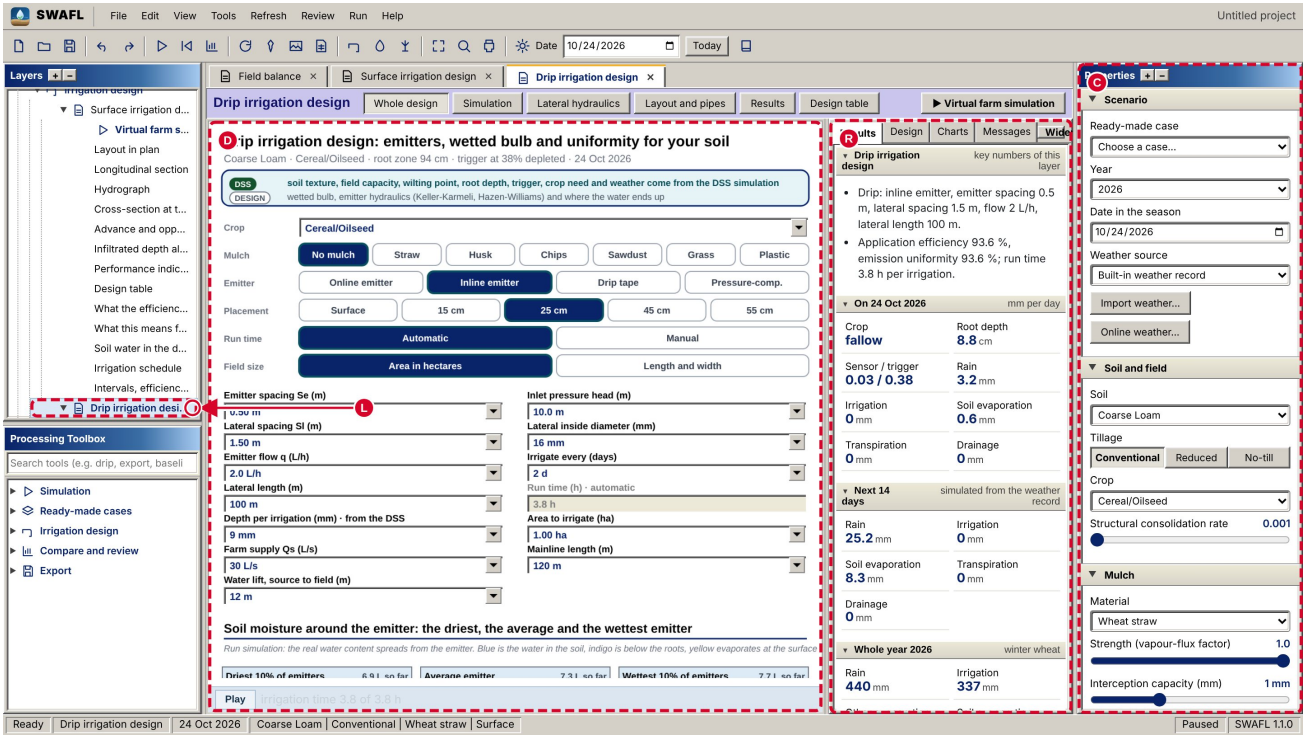
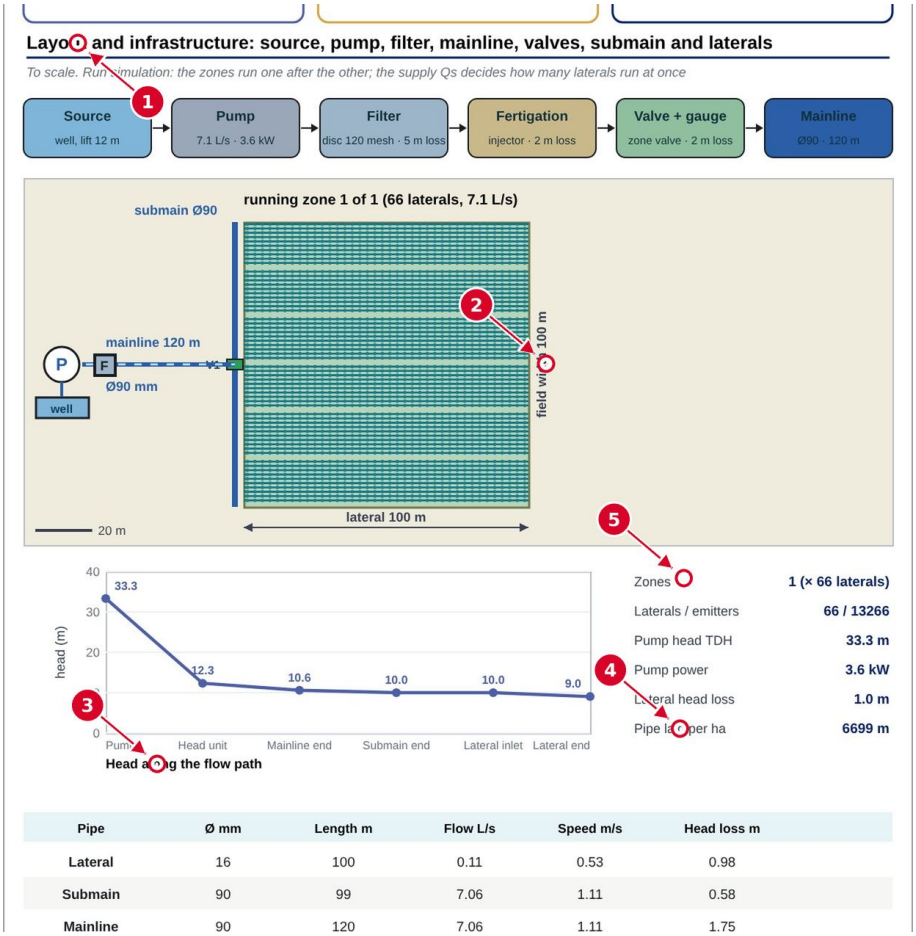


Figure 25.55: Drip irrigation design / Layout and infrastructure: the whole workbench.



Pipe	Ø mm	Length m	Flow L/s	Speed m/s	Head loss m
Lateral	16	100	0.11	0.53	0.98
Submain	90	99	7.06	1.11	0.58
Mainline	90	120	7.06	1.11	1.75

Figure 25.56: Drip irrigation design / Layout and infrastructure: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Source, pump, filter and valves: pump power from the head of Eq. 12.23.
- **2** Laterals across the field, N_L = field width / S_l: Eq. 12.20.
- **3** Head along the flow path: TDH of Eq. 12.23.
- **4** Pipe table: smallest commercial diameter, losses by Eq. 12.22 (Hazen-Williams, Eq. 12.3).
- **5** Zones, laterals and emitters: Eq. 12.20.

Equations behind this screen. Eq. 12.3, Eq. 12.20, Eq. 12.21, Eq. 12.22, Eq. 12.23.

Eq. 12.3. Where explained: chapter “Drip irrigation design”, section “Lateral hydraulics: head loss and flow along the lateral”.

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m.} \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.20. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.

$$W_f = \frac{10^4 A_f}{L_L}, \quad N_L = \left\lfloor \frac{W_f}{S_l} \right\rfloor, \quad n_{\text{zone}} = \left\lceil \frac{N_L\, Q_{\text{lat}}}{Q_s} \right\rceil, \quad n_l = \left\lceil \frac{N_L}{n_{\text{zone}}} \right\rceil, \quad Q_z = n_l\, Q_{\text{lat}}, \tag{12.20}$$

Provenance (Eq. 12.20): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.21. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \text{mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852}D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852}D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \text{m}, \qquad P = \frac{\rho g\, Q_z\, \text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

25.3.7 Design table: iterations toward an application efficiency of 85 % or more

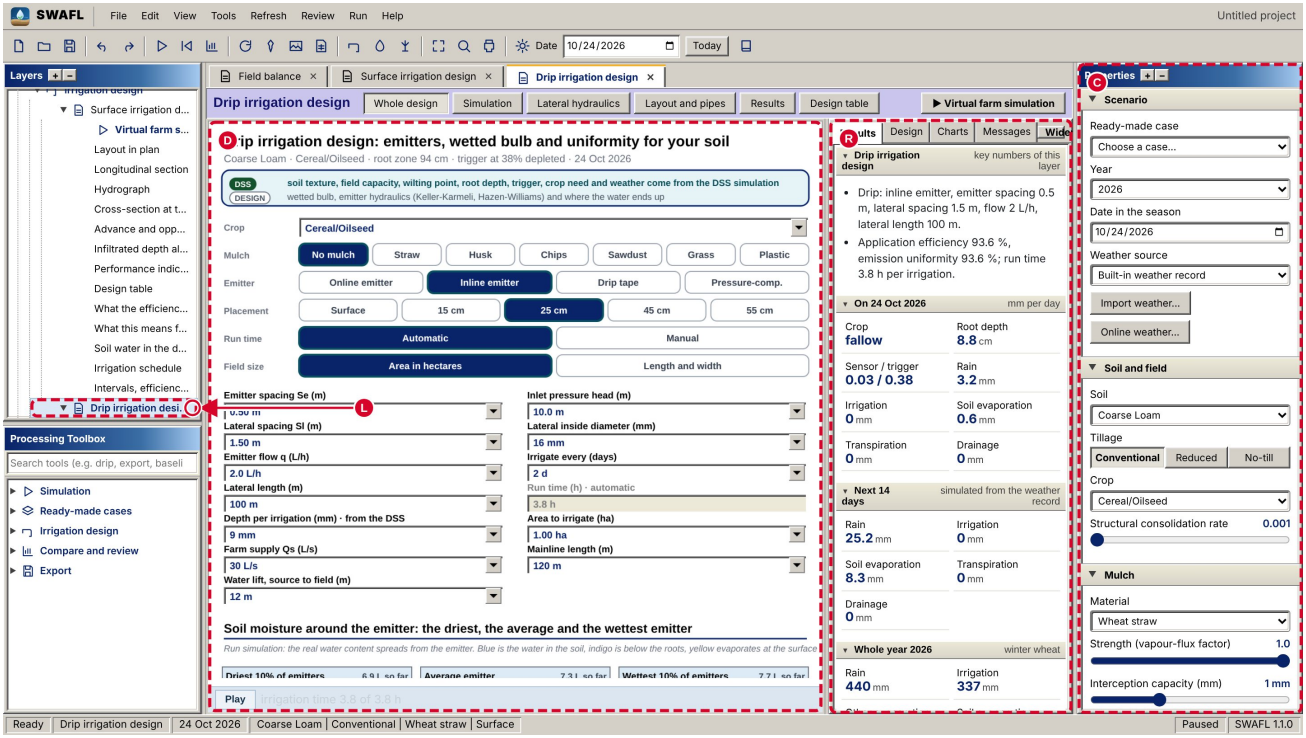


Figure 25.57: Drip irrigation design / Design table: the whole workbench.

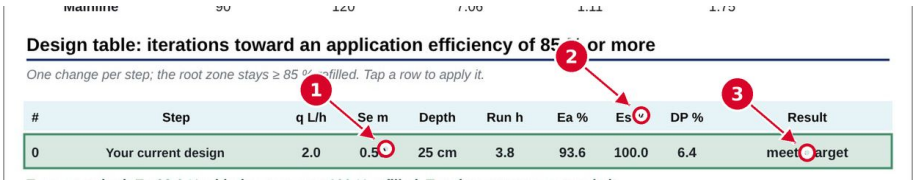


Figure 25.58: Drip irrigation design / Design table: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Row 0: your current design; its score is Eq. 12.26.
- **2** Column E_a: Eq. 12.19.
- **3** Target reached: score = min(E_a, 1) - 2.5 max(0, 0.9 - E_s), Eq. 12.26.

Equations behind this screen. Eq. 12.26.

Eq. 12.26. Where explained: chapter “Drip irrigation design”, section “The design table”.

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

25.3.8 What the efficiency is most sensitive to

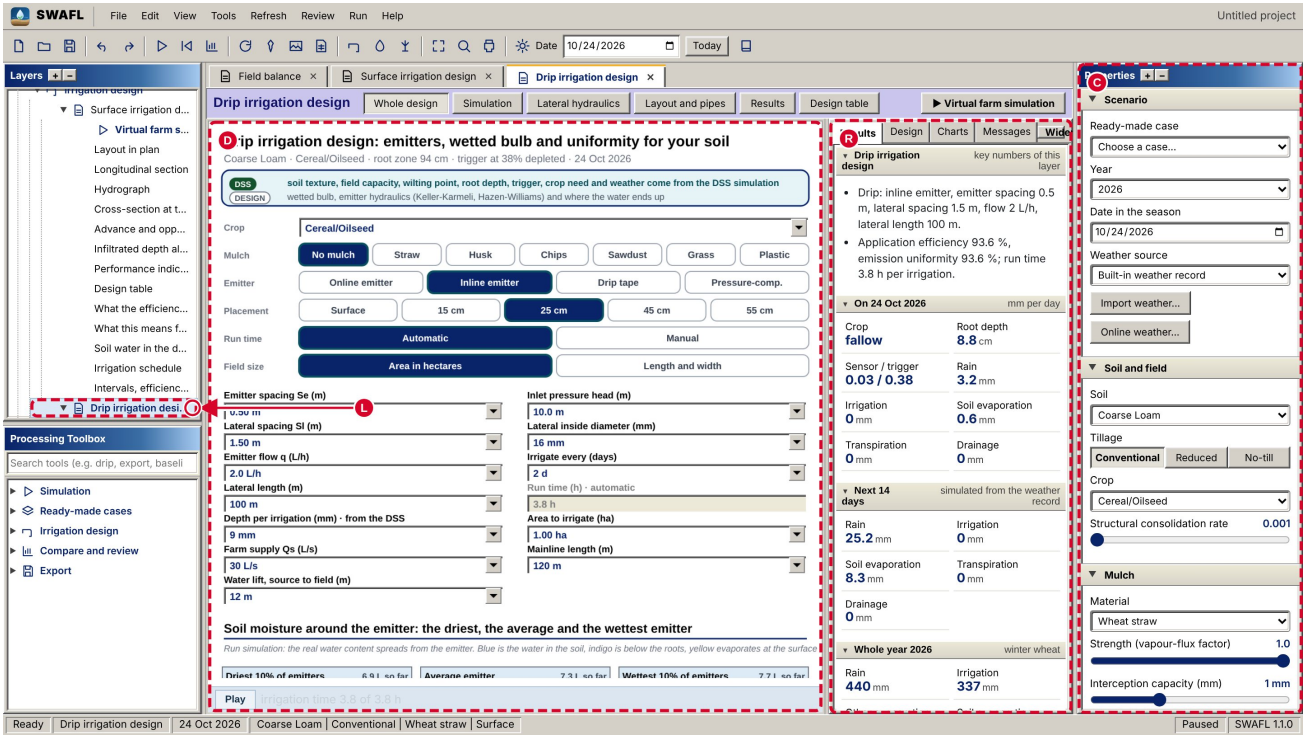


Figure 25.59: Drip irrigation design / What the efficiency is most sensitive to: the whole workbench.

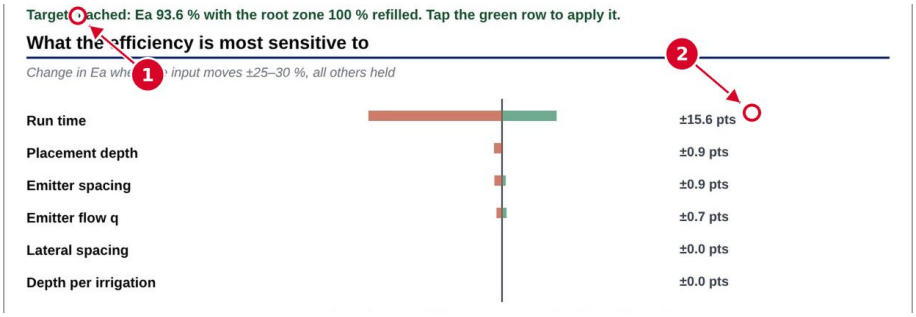


Figure 25.60: Drip irrigation design / What the efficiency is most sensitive to: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Tornado bar: change in E_a (Eq. 12.19) when the run time moves by 25 to 30 %.
- **2** Size of the change in percentage points.

Equations behind this screen. none (the screen shows stored results).

25.3.9 What this means for the season and the farm

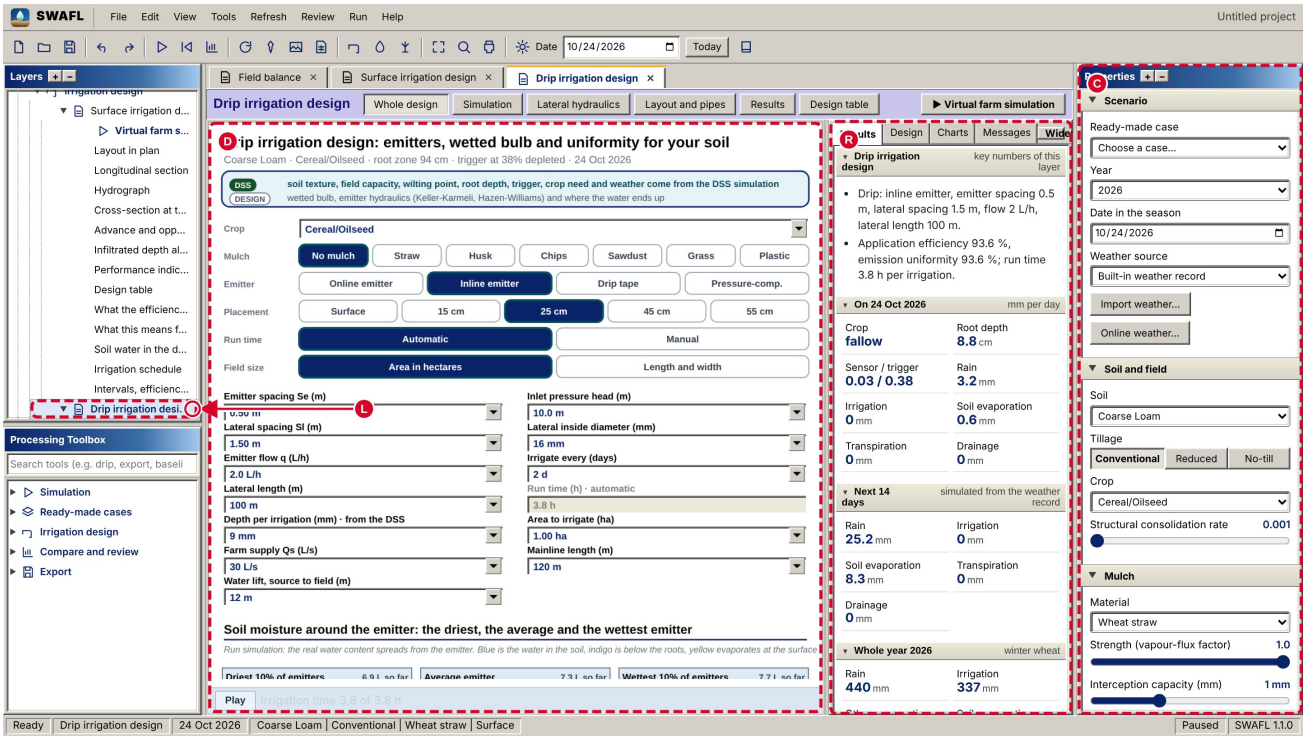


Figure 25.61: Drip irrigation design / What this means for the season and the farm: the whole workbench.

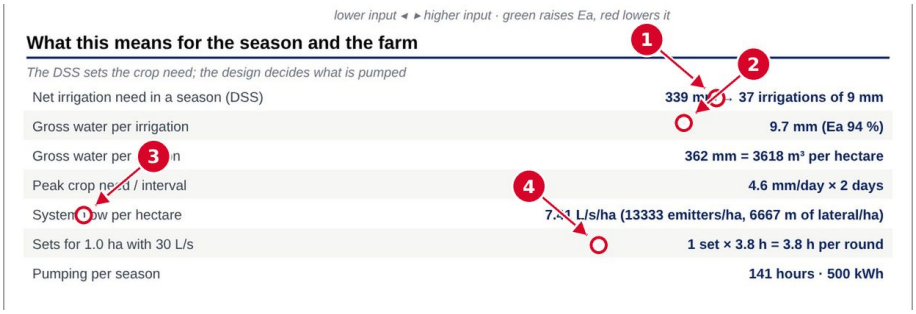


Figure 25.62: Drip irrigation design / What this means for the season and the farm: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Irrigations in a season, n_ev = y_I / Z_n: Eq. 12.24.
- **2** Gross depth per irrigation: Eq. 12.24.
- **3** System flow per hectare: Eq. 12.24.
- **4** Sets and hours: Eq. 12.25.

Equations behind this screen. Eq. 12.24, Eq. 12.25.

Eq. 12.24. Where explained: chapter “Drip irrigation design”, section “Season and farm”.

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\mathrm{L\,s^{-1}\,ha^{-1}}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.25. Where explained: chapter “Drip irrigation design”, section “Season and farm”.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

25.3.10 Irrigation schedule: when to irrigate with this design

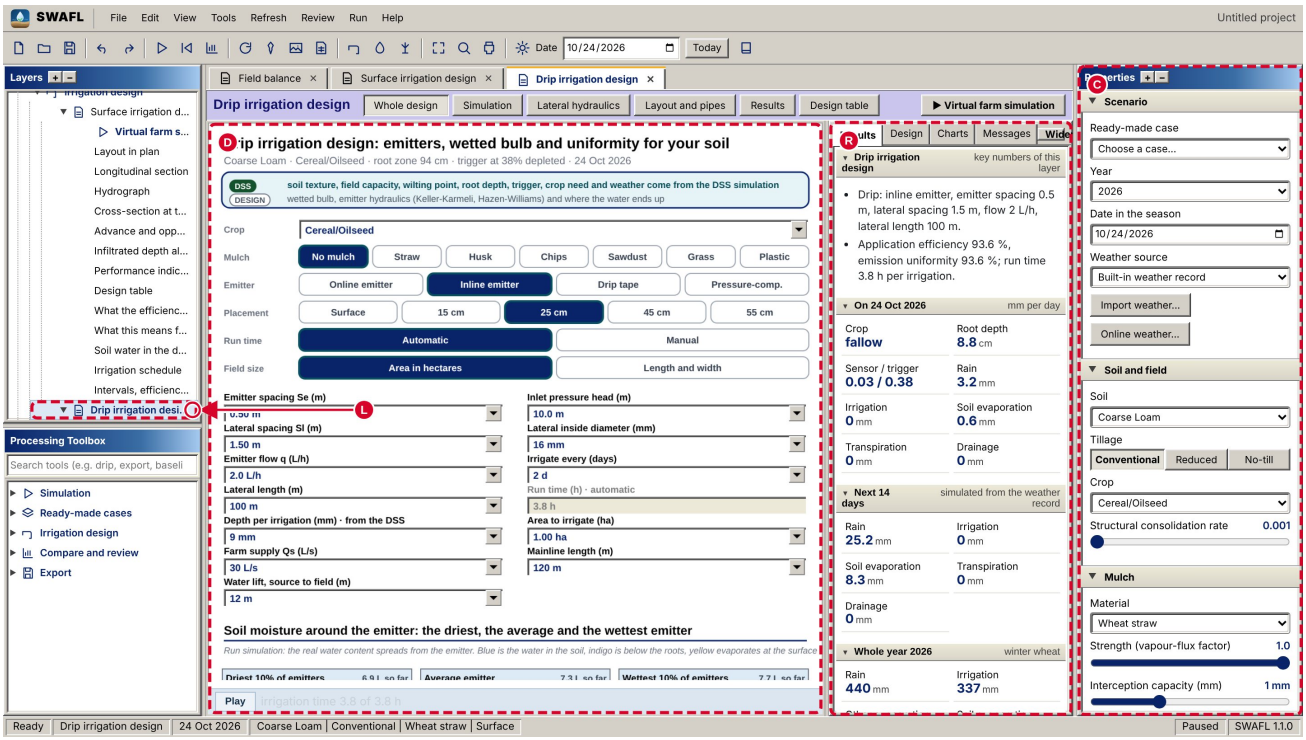


Figure 25.63: Drip irrigation design / Irrigation schedule: the whole workbench.

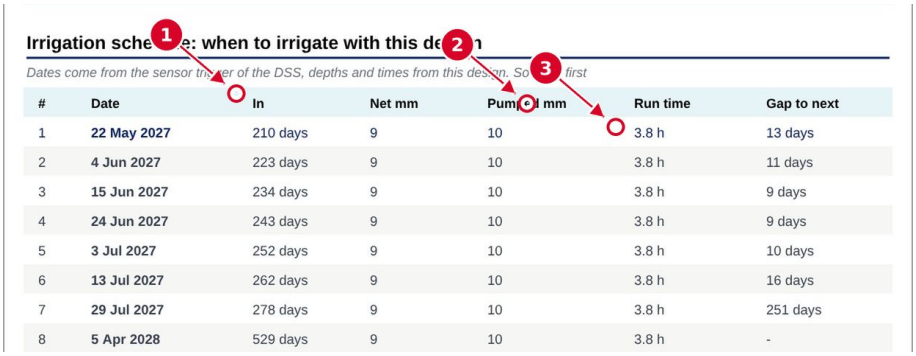


Figure 25.64: Drip irrigation design / Irrigation schedule: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Dates from the Engine 2 irrigation record.
- **2** Pumped mm = I_k / E_a: Eq. 12.24.
- **3** Run time: Eq. 12.18 (automatic run time).

Equations behind this screen. Eq. 12.24, Eq. 12.25.

Eq. 12.24. Where explained: chapter "Drip irrigation design", section "Season and farm".

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.25. Where explained: chapter “Drip irrigation design”, section “Season and farm”.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

25.3.11 Intervals, efficiency and the advice in short

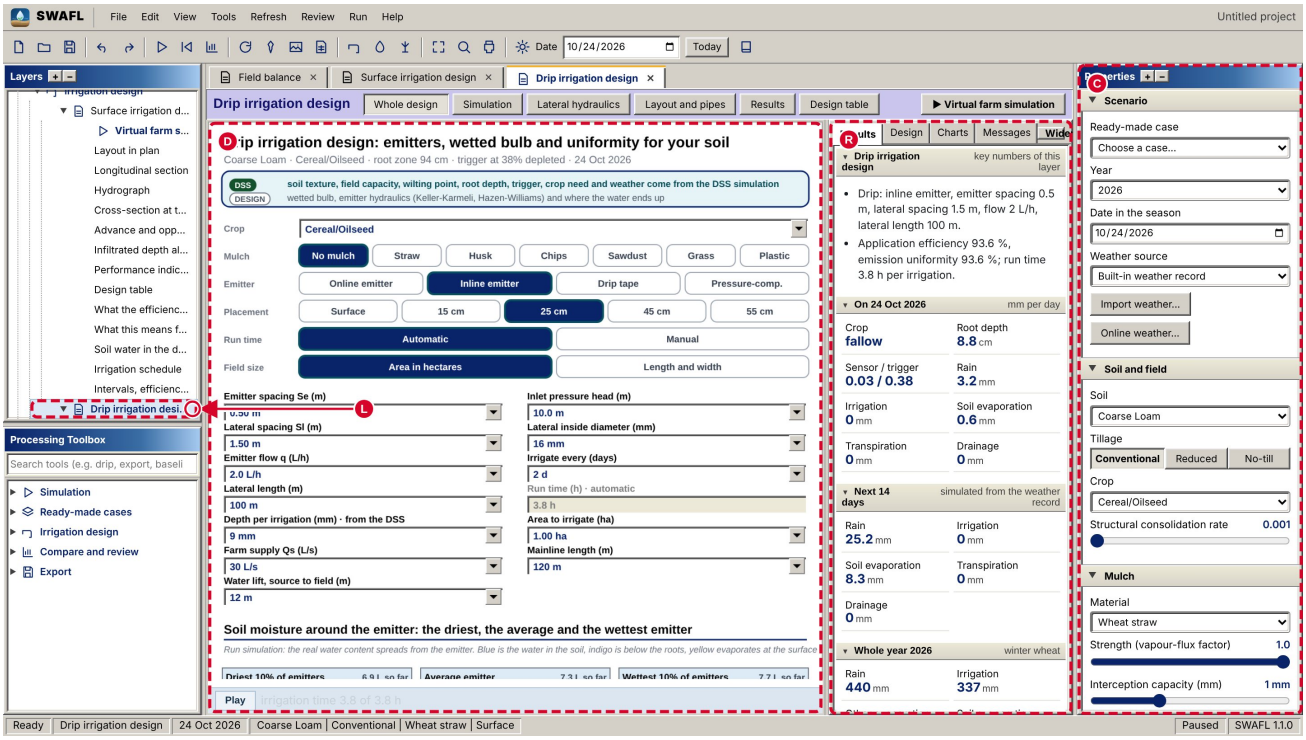


Figure 25.65: Drip irrigation design / Intervals, efficiency and the advice in short: the whole workbench.

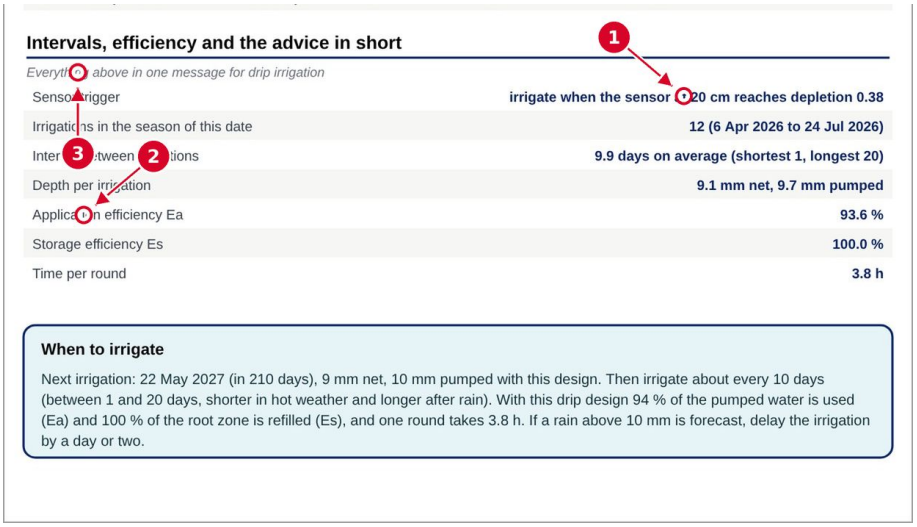


Figure 25.66: Drip irrigation design / Intervals, efficiency and the advice in short: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sensor trigger: Eqs. 5.3 and 5.27.
- **2** Application efficiency E_a: Eq. 12.19.

- **3** “When to irrigate”: message built from the schedule and from E_a and E_s.

Equations behind this screen. Eq. 12.24, Eq. 12.25.

Eq. 12.24. Where explained: chapter “Drip irrigation design”, section “Season and farm”.

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L}\,\text{s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.25. Where explained: chapter “Drip irrigation design”, section “Season and farm”.

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

25.4 Sprinkler design and pipes

25.4.1 The whole layer: inputs and overview

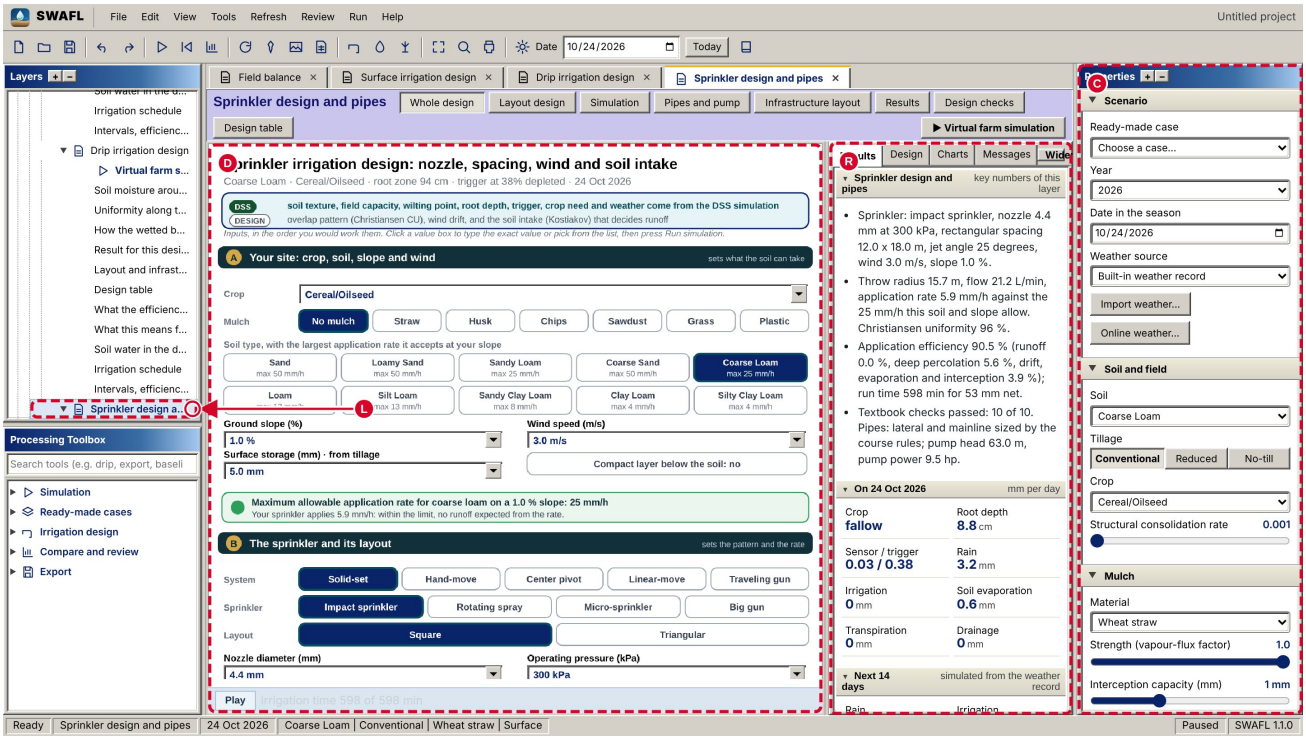


Figure 25.67: Sprinkler design and pipes: the whole workbench.

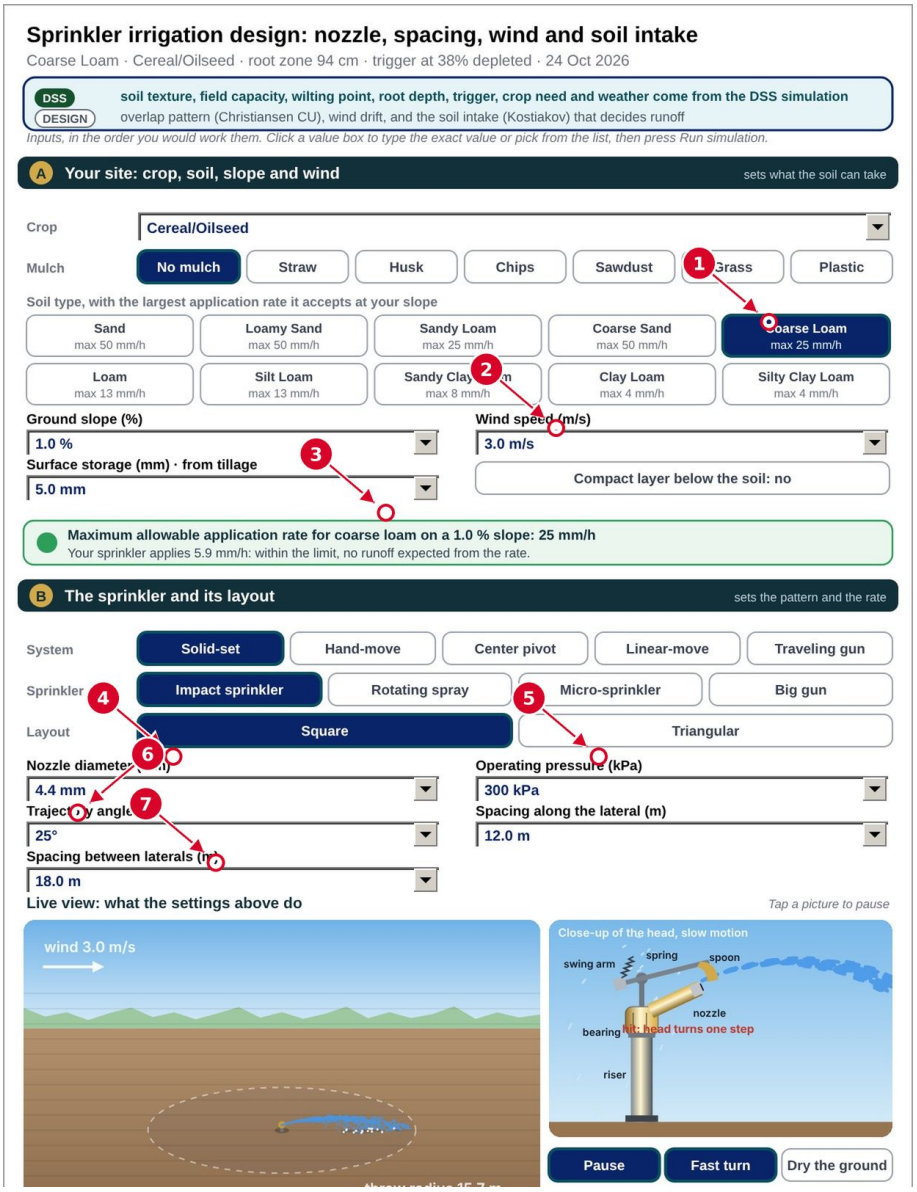


Figure 25.68: Sprinkler design and pipes: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Selected soil: intake K and A and the maximum allowable application rate, Eq. 13.9 and the course table.
- **2** Wind speed u: Eqs. 13.5 and 13.8.
- **3** Banner: application rate against the maximum allowed, Eqs. 13.9 and 13.10.
- **4** Nozzle diameter d: Eqs. 13.2 and 13.3.
- **5** Operating pressure P gives the head h = P / 9.81: Eq. 13.2.
- **6** Trajectory angle alpha: f_alpha in Eq. 13.3.
- **7** Spacing between laterals S_l: Eq. 13.6.

Equations behind this screen. Eq. 5.27, Eq. 11.6, Eq. 13.1, Eq. 13.2, Eq. 13.3, Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7, Eq. 13.9, Eq. 13.10.

Eq. 5.27. Where explained: chapter “The two computing engines”, section “Evaporation under

warming (used by the E1 layer)".

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 13.1. Where explained: chapter "Sprinkler design", section "Inputs, step by step".

$$S_d^{\mathrm{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

Provenance (Eq. 13.1): **[Engineered]**: depression storage reduction with slope.

Eq. 13.2. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^{4}\ \ (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^{2}\ \mathrm{m^2}, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

Eq. 13.3. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".

$$R = c_R\sqrt{d\,h}\ f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\widehat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

Eq. 13.9. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and runoff".

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \sup, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \sup - F, \quad \mathrm{RO} = \max(0, R - S_d^{\mathrm{eff}}), \quad I = \sup - \mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.10. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and runoff".

$$I_c = \min\{I,\; I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

25.4.2 Layout in plan: depth infiltrated so far, from the overlapping spray patterns

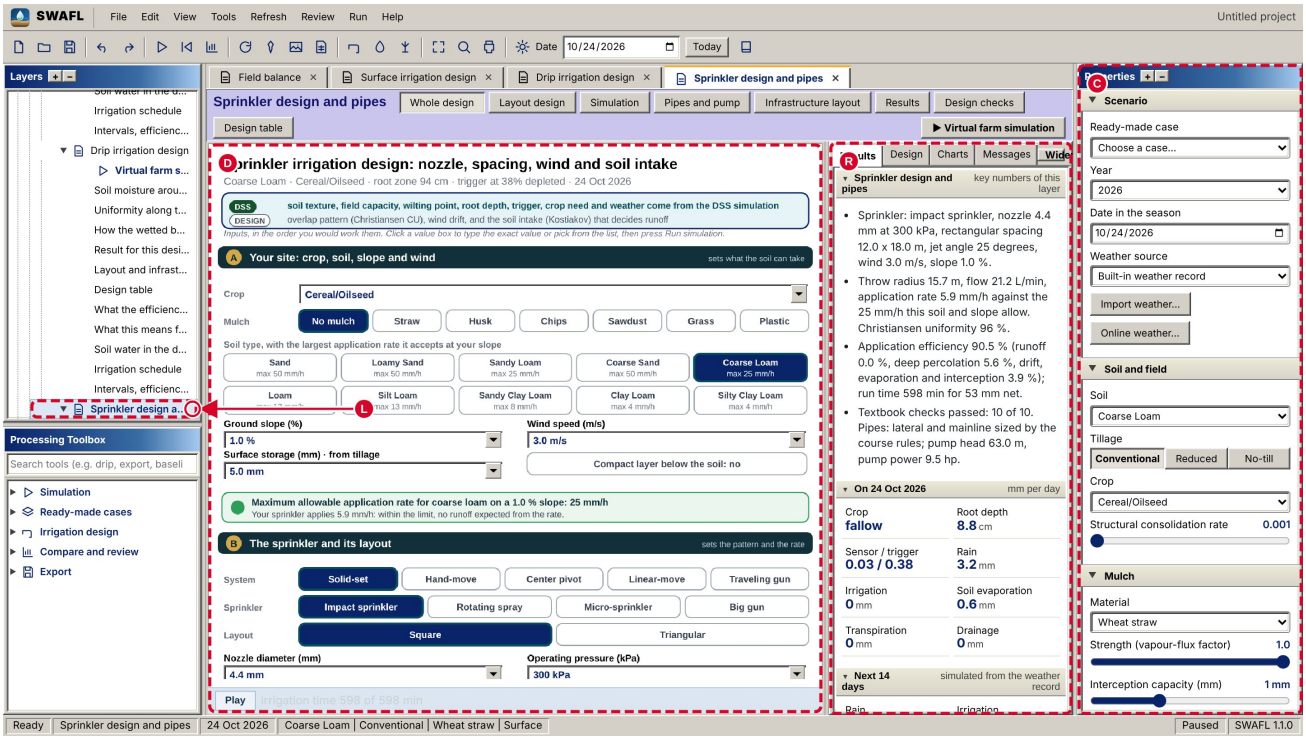


Figure 25.69: Sprinkler design and pipes / Layout in plan: the whole workbench.

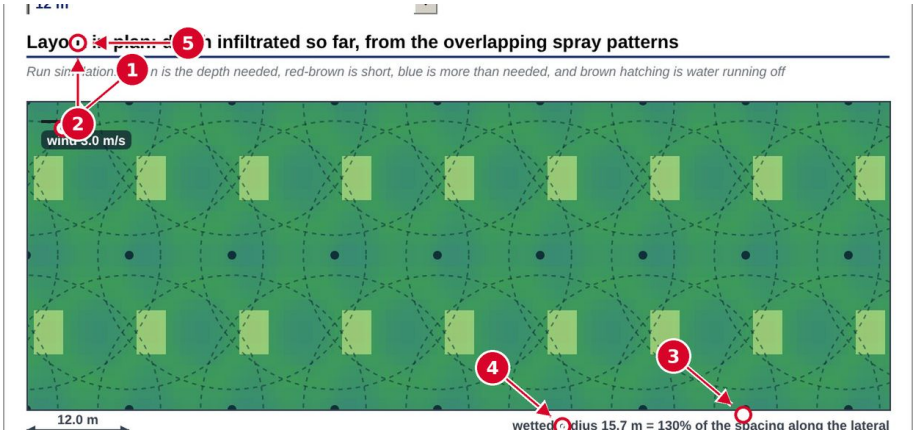


Figure 25.70: Sprinkler design and pipes / Layout in plan: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Wind arrow: wind lengthens and shifts the pattern, Eq. 13.5.
- **2** Overlapping spray patterns: sum over neighbours, Eq. 13.6, with the radial profile of Eq. 13.4.
- **3** Wetted radius R: Eq. 13.3.
- **4** Spacing along the lateral S_s.

- **5** Colour shows infiltrated depth against Z_n: Eq. 13.11.

Equations behind this screen. Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

Eq. 13.4. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$\widehat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$R_{e}(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_{k}, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$\mathrm{CU} = 1 - \frac{\sum_{k} |a_{k} - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

25.4.3 Along the lateral: depth applied, depth infiltrated and where it ends up in the soil

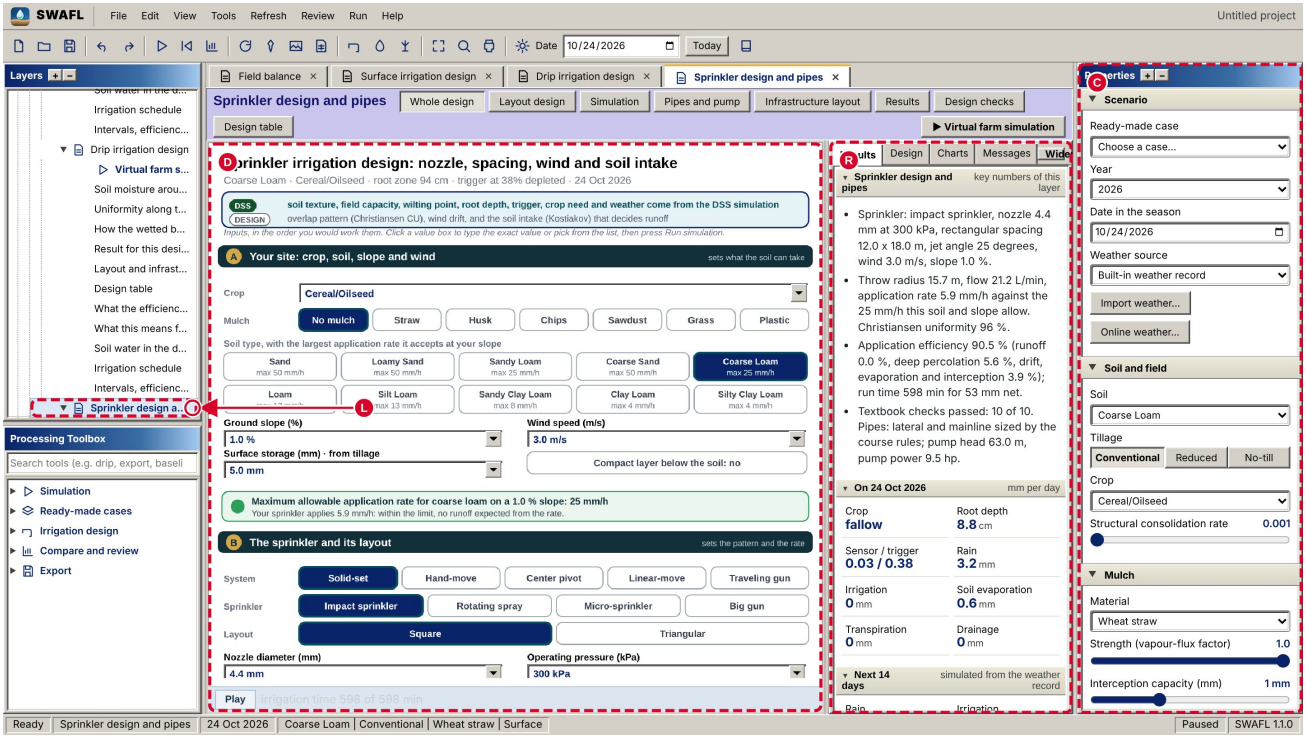


Figure 25.71: Sprinkler design and pipes / Along the lateral: the whole workbench.

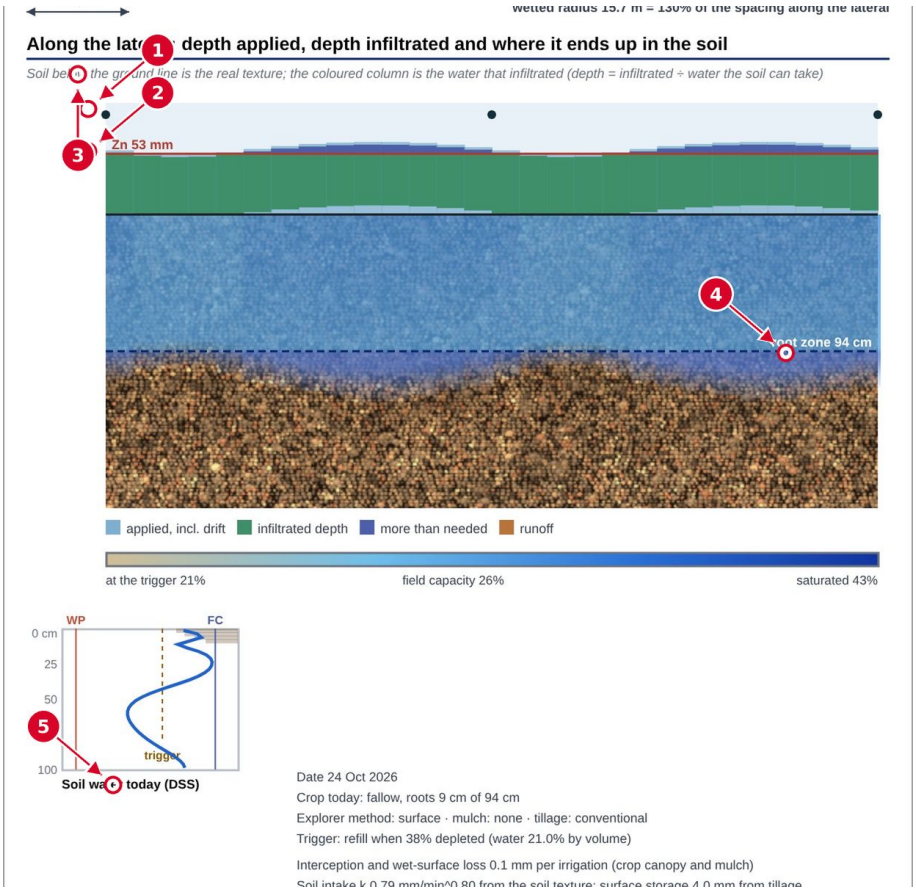


Figure 25.72: Sprinkler design and pipes / Along the lateral: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sprinkler positions.
- **2** Z_n: required depth, Eq. 5.27.
- **3** Infiltrated depth in each cell, Z_k: Eq. 13.11.
- **4** Root-zone depth (94 cm).
- **5** Soil water today from the DSS profile.

Equations behind this screen. Eq. 11.6, Eq. 13.9, Eq. 13.10, Eq. 13.11, Eq. 13.12.

Eq. 11.6. Where explained: chapter "Surface irrigation design", section "Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)".

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 13.9. Where explained: chapter "Sprinkler design", section "The soil intake, ponding and runoff".

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \sup, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \sup - F, \quad \mathrm{RO} = \max(0, R - S_d^{\mathrm{eff}}), \quad I = \sup - \mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.10. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.

$$I_c = \min\{I,\ I_{\text{int}}\}, \qquad I_{\text{int}} = 0.6\,\text{CAP}\,[\text{VFF} < 0.95] + 1.2\,\text{LAI}_{\text{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.11. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.

$$\begin{aligned} E_a &= \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, & E_s &= \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, & \text{DU} &= \frac{\overline{Z}_{25}}{\overline{Z}}, \\ \text{DP} &= \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, & \text{RO} &= \frac{\overline{\text{RO}_k}}{\text{gross}}, & \text{WD} &= w_d + \frac{\overline{I_c}}{\text{gross}}, \end{aligned} \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 13.12. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.

$$T = \min\{T :\ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\ 2400]\ \text{min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

25.4.4 Soil intake against the application rate: when the soil cannot keep up, water runs off

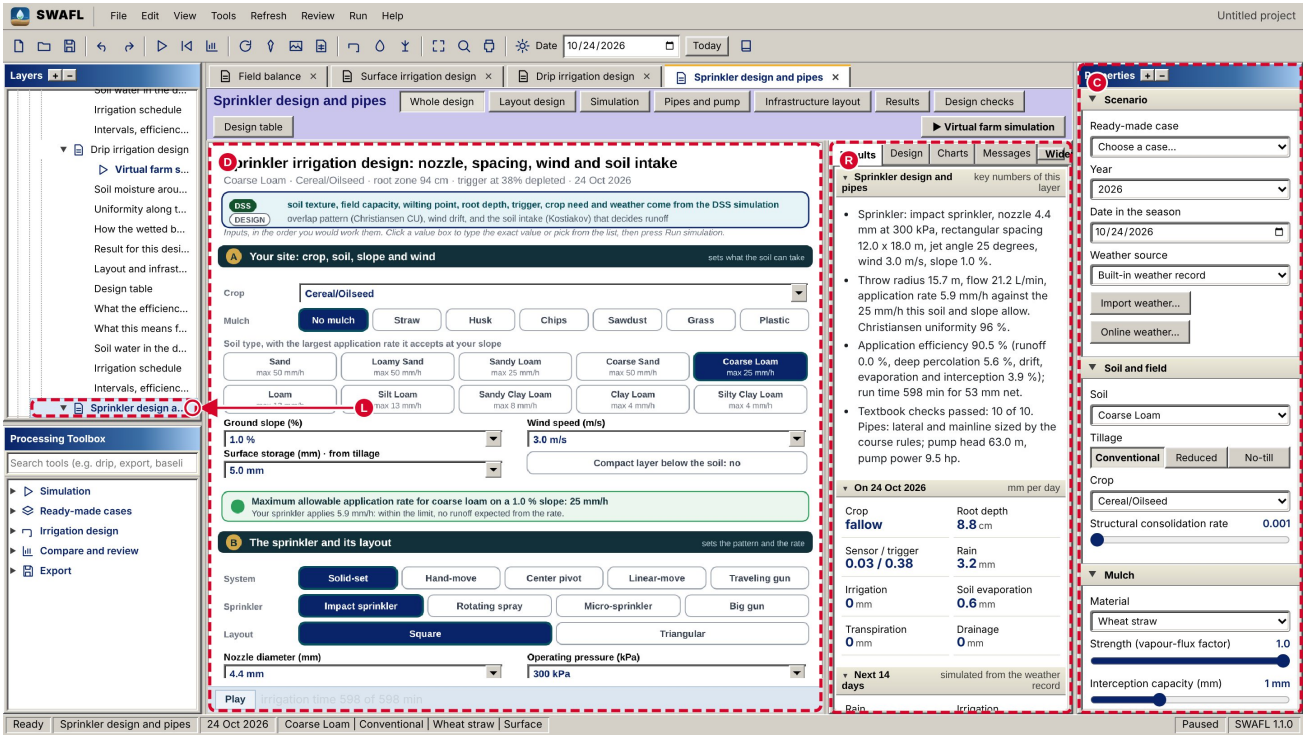


Figure 25.73: Sprinkler design and pipes / Soil intake against the application rate: the whole workbench.

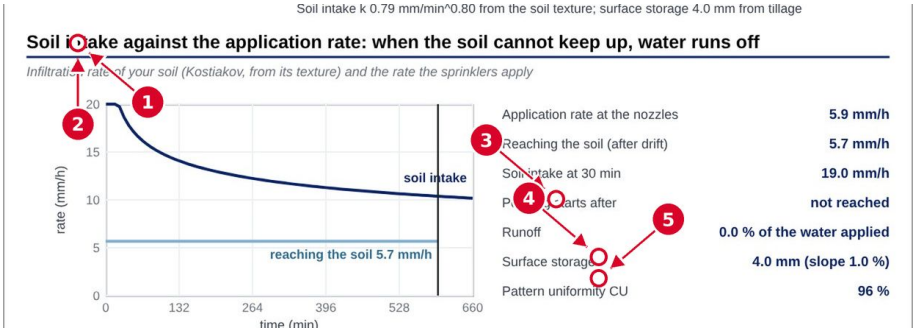


Figure 25.74: Sprinkler design and pipes / Soil intake against the application rate: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Soil intake rate K t^(A-1) (Kostiakov): Eq. 13.9.
- **2** Applied rate a = 60 times the mean depth: Eq. 13.7.
- **3** Ponding starts after t_p: Eq. 13.9.
- **4** Runoff share: Eqs. 13.9 and 13.10.
- **5** Surface storage S_d: Eq. 13.1.

Equations behind this screen. Eq. 11.6, Eq. 13.9, Eq. 13.10.

Eq. 11.6. Where explained: chapter “Surface irrigation design”, section “Intake from your soil (Saxton–Rawls → Green–Ampt → Kostiakov)”.

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 13.9. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.

$$t_p = \left(\frac{a}{K}\right)^{1/(A-1)}, \qquad F = \begin{cases} \text{sup}, & T \le t_p \\ K\,T^{A}, & T > t_p \end{cases} \qquad R = \text{sup} - F, \quad \text{RO} = \max(0, R - S_d^{\text{eff}}), \quad I = \text{sup} - \text{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.10. Where explained: chapter “Sprinkler design”, section “The soil intake, ponding and runoff”.

$$I_c = \min\{I,\ I_{\text{int}}\}, \qquad I_{\text{int}} = 0.6\,\text{CAP}\,[\text{VFF} < 0.95] + 1.2\,\text{LAI}_{\text{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

25.4.5 Result for this design

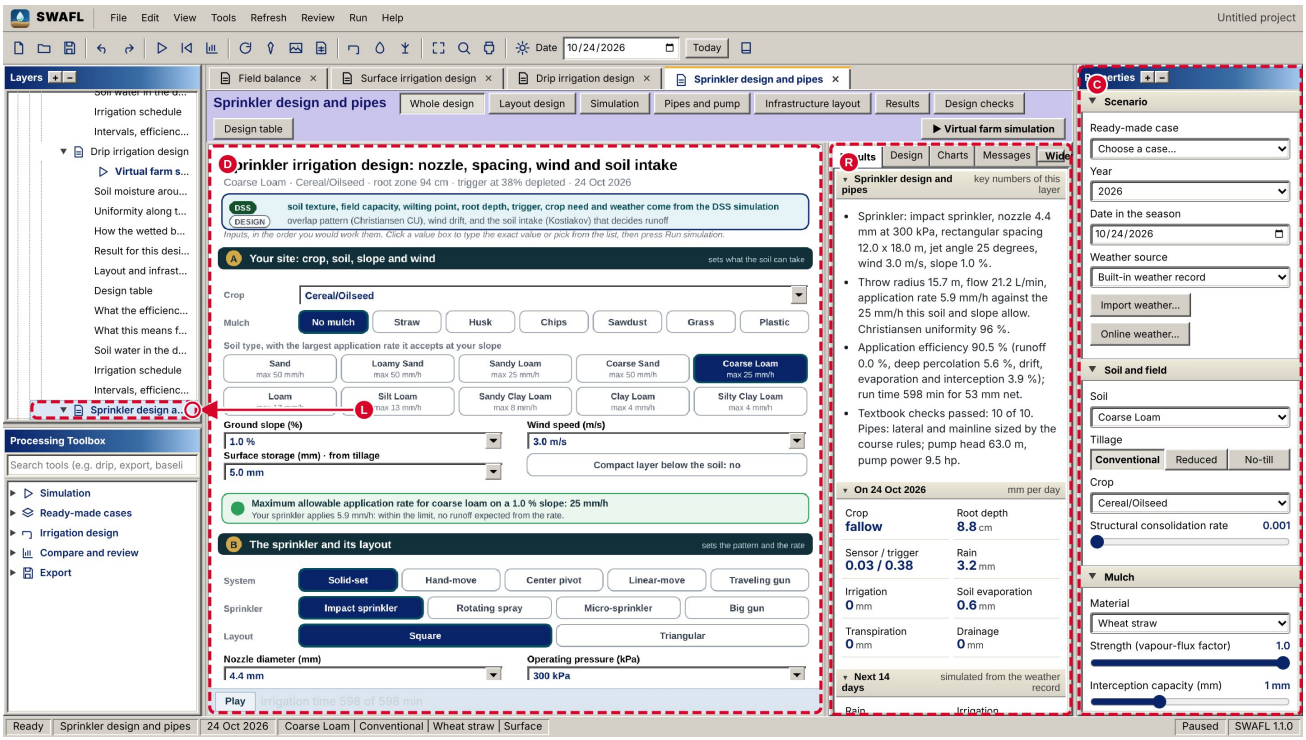


Figure 25.75: Sprinkler design and pipes / Result for this design: the whole workbench.

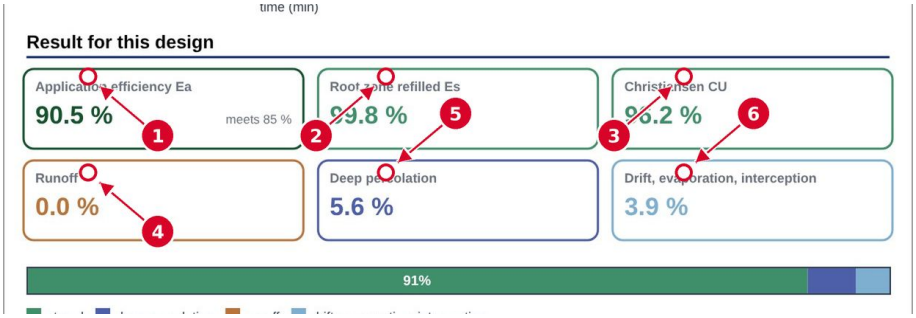


Figure 25.76: Sprinkler design and pipes / Result for this design: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Application efficiency E_a: Eq. 13.11.
- **2** Root-zone refilled E_s: Eq. 13.11.
- **3** Christiansen CU: Eq. 13.7.
- **4** Runoff: Eqs. 13.9 and 13.10.
- **5** Deep percolation: Eq. 13.11.
- **6** Drift, evaporation and interception: Eqs. 13.8 and 13.10.

Equations behind this screen. Eq. 13.11, Eq. 13.12.

Eq. 13.11. Where explained: chapter “Sprinkler design”, section “Efficiencies and run time”.

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 13.12. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\, 2400] \ \text{min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

25.4.6 Does your design pass? The course checks

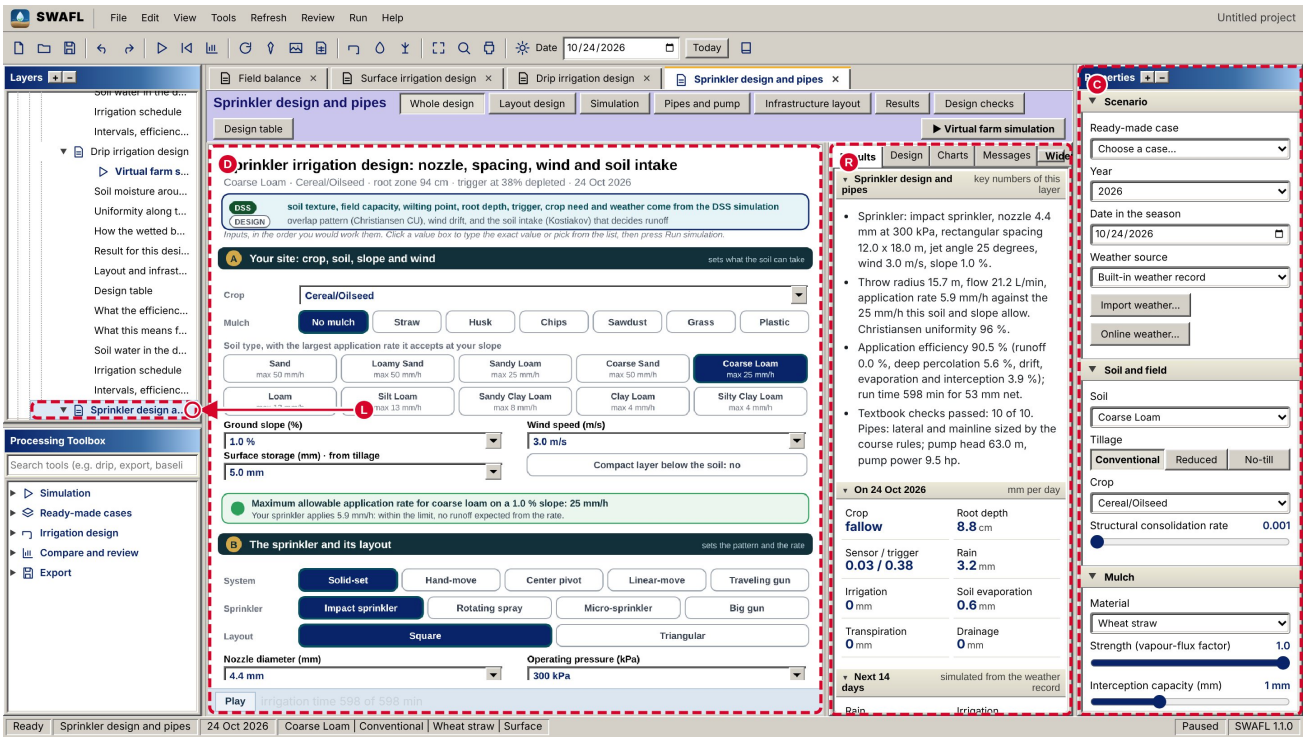


Figure 25.77: Sprinkler design and pipes / Does your design pass? The course checks: the whole workbench.

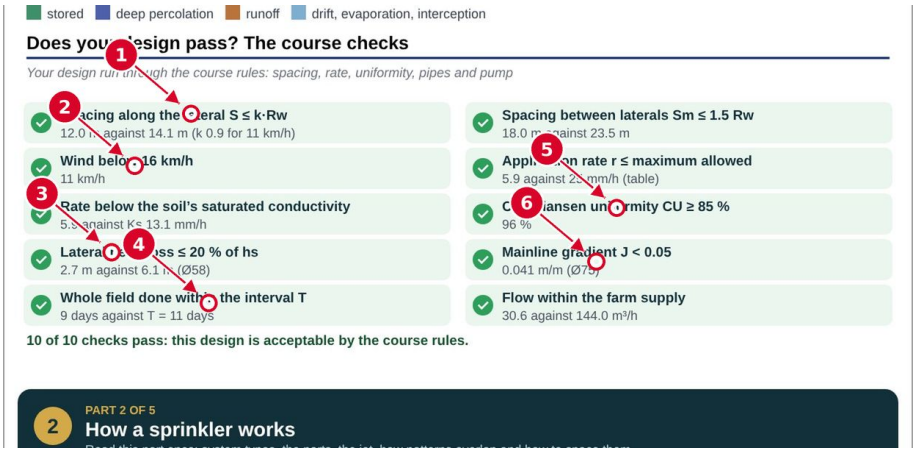


Figure 25.78: Sprinkler design and pipes / Does your design pass? The course checks: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Spacing along the lateral S_s <= k R_w: Eqs. 13.13 and 13.14.
- **2** Wind below 16 km/h: Eq. 13.13.
- **3** Lateral head loss <= 20 % of the head: Eq. 13.18.
- **4** Whole field done within the interval T: Eqs. 13.14 to 13.16.

- **5** Christiansen CU >= 85 %: Eq. 13.7.
- **6** Mainline gradient J < 0.05: Eq. 13.17.

Equations behind this screen. Eq. 13.13, Eq. 13.14, Eq. 13.15, Eq. 13.16, Eq. 13.17, Eq. 13.18, Eq. 13.19, Eq. 13.20.

Eq. 13.13. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$k = \begin{cases} 1, & W \le 5.5 \\ 0.9, & 5.5 < W \le 13 \\ 0.8, & 13 < W \le 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

Provenance (Eq. 13.13): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.14. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$S_s \le k\,R_w, \qquad S_l \le 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x / S_s \rfloor, \quad N_m = \lfloor L_y / S_l \rfloor, \tag{13.14}$$

Provenance (Eq. 13.14): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.15. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}} \ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right) \text{ days}, \tag{13.15}$$

Provenance (Eq. 13.15): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.16. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\, N\, q_h, \tag{13.16}$$

Provenance (Eq. 13.16): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.17. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$J = \frac{0.472\, Q^{1.75}}{D^{4.75}} \ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

Provenance (Eq. 13.17): **[Literature]**: Blasius friction law for smooth pipe.

Eq. 13.18. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$h_{f,\text{lat}} = 1.1\, F\, J(Q_L, D)\,(L_x - S_s/2) \le 0.2\, h_s, \qquad Q_L = 1000\, N\, q_h, \tag{13.18}$$

Provenance (Eq. 13.18): **[Literature]**: 20 % rule of Keller and Bliesner (1990).

Eq. 13.19. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$H_{\text{man}} = z_{\text{lift}} + J\, L_{\text{conv}} + J\, L_{\text{main}} + h_s + 0.75\, h_{f,\text{lat}}, \tag{13.19}$$

Provenance (Eq. 13.19): **[Engineered]**: head and sets bookkeeping.

Eq. 13.20. Where explained: chapter "Sprinkler design", section "The ten textbook checks".

$$P_{\mathrm{CV}} = \frac{1000\,(Q_m/3.6\times10^{6})\,H_{\mathrm{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\mathrm{kW}} = 0.7355\,P_{\mathrm{CV}}, \tag{13.20}$$

Provenance (Eq. 13.20): **[Literature]**: hydraulic power definition.

25.4.7 Kinds of sprinkler system

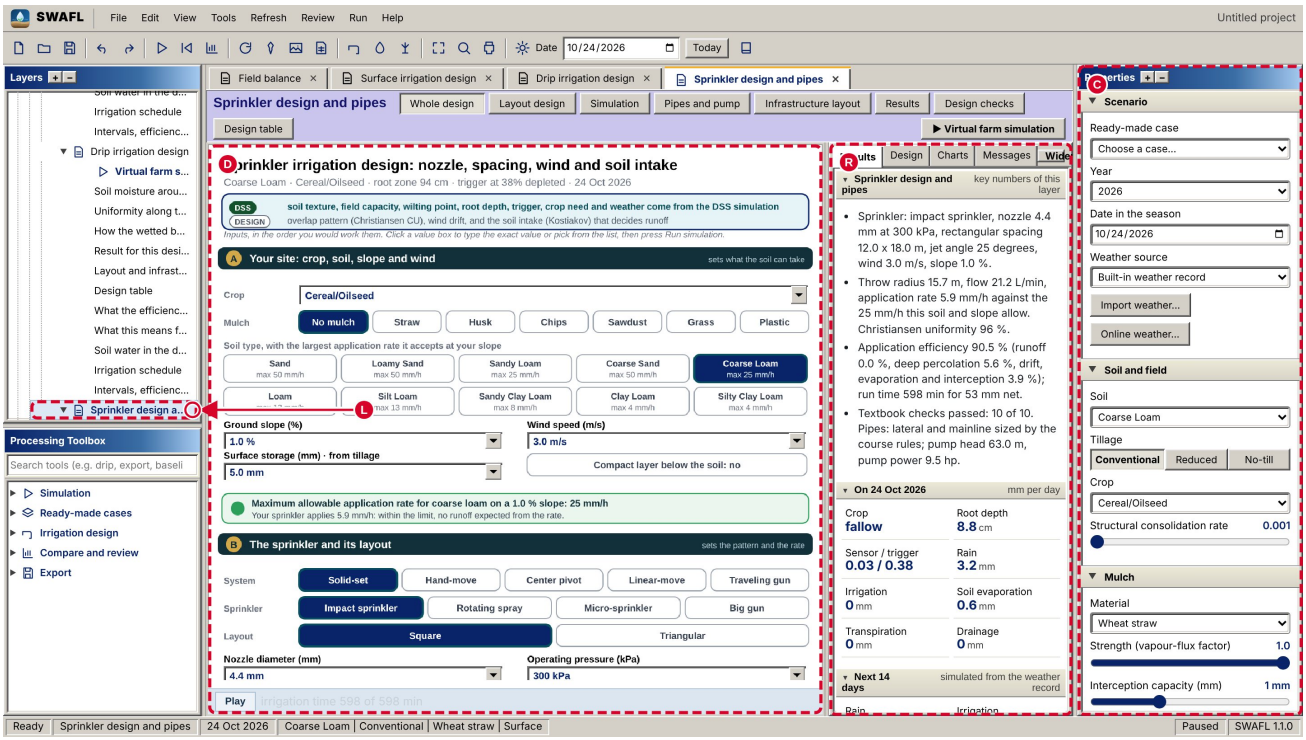


Figure 25.79: Sprinkler design and pipes / Kinds of sprinkler system: the whole workbench.

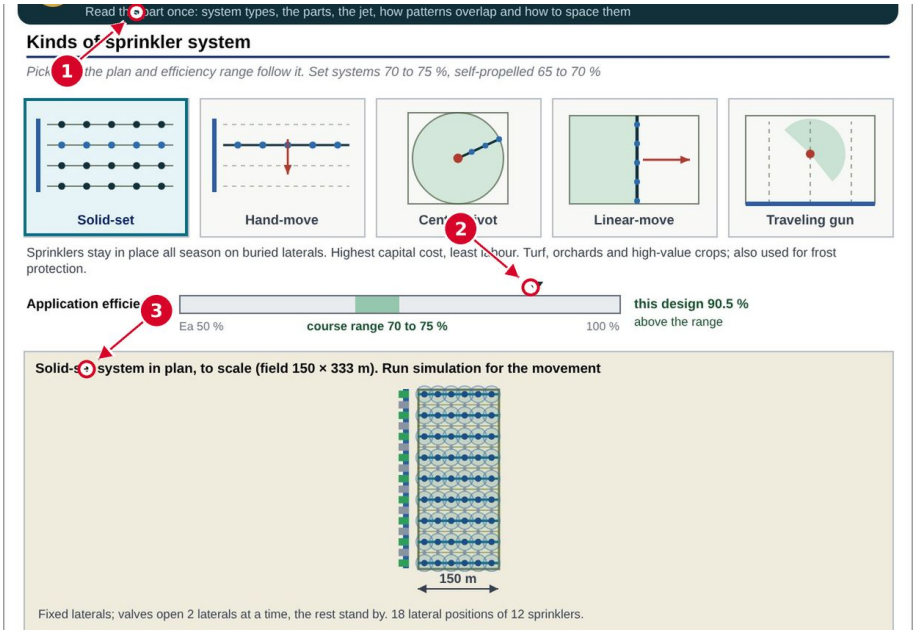


Figure 25.80: Sprinkler design and pipes / Kinds of sprinkler system: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** System type (solid-set selected): sets the efficiency range of the course table.
- **2** This design’s E_a against the course range: Eq. 13.11.
- **3** Field plan to scale: laterals and sets, Eq. 13.21.

Equations behind this screen. none (the screen shows stored results).

25.4.8 The sprinkler: parts and pressure class

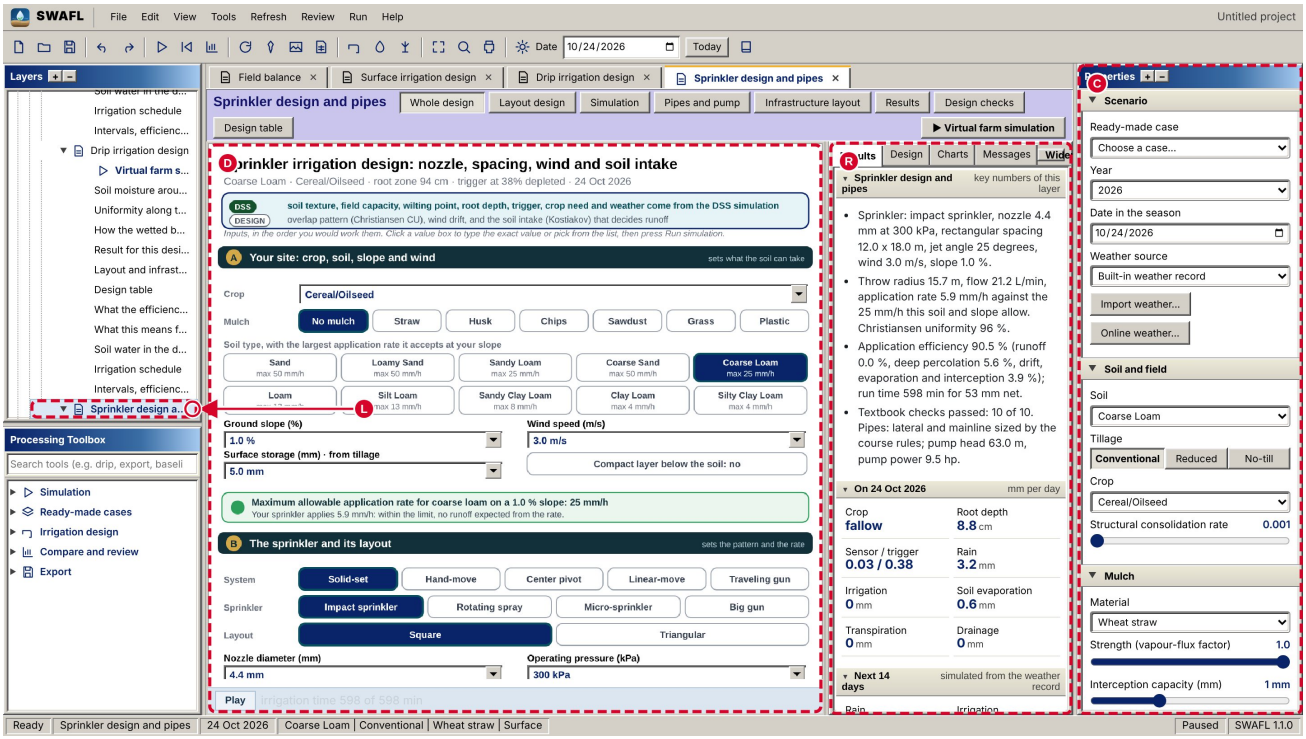


Figure 25.81: Sprinkler design and pipes / The sprinkler: the whole workbench.

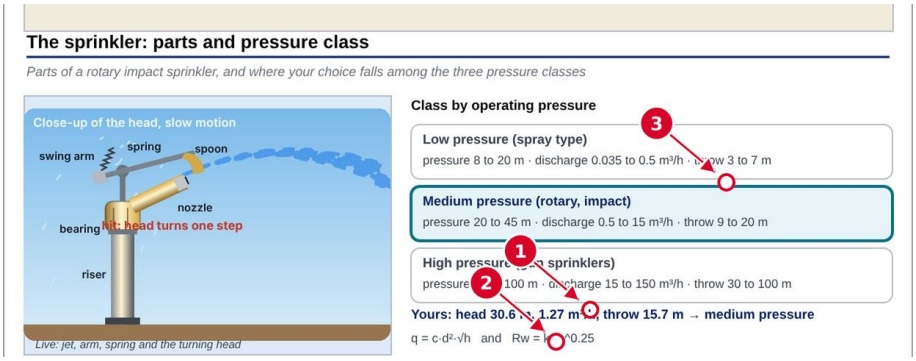


Figure 25.82: Sprinkler design and pipes / The sprinkler: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** “Yours”: head h = P / 9.81 and discharge q (Eq. 13.2), throw R (Eq. 13.3).
- **2** Formulas q = c d^2 sqrt(h) and R = k_r h^0.25: Eqs. 13.2 and 13.3.
- **3** Pressure classes (low, medium, high): table, no equation.

Equations behind this screen. Eq. 13.2, Eq. 13.3.

Eq. 13.2. Where explained: chapter “Sprinkler design”, section “Nozzle discharge and wetted radius”.

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^4 \;\; (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2 \;\mathrm{m^2}, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

Eq. 13.3. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".

$$R = c_R\sqrt{d\,h}\;f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

25.4.9 The jet: trajectory angle, throw radius and droplet size

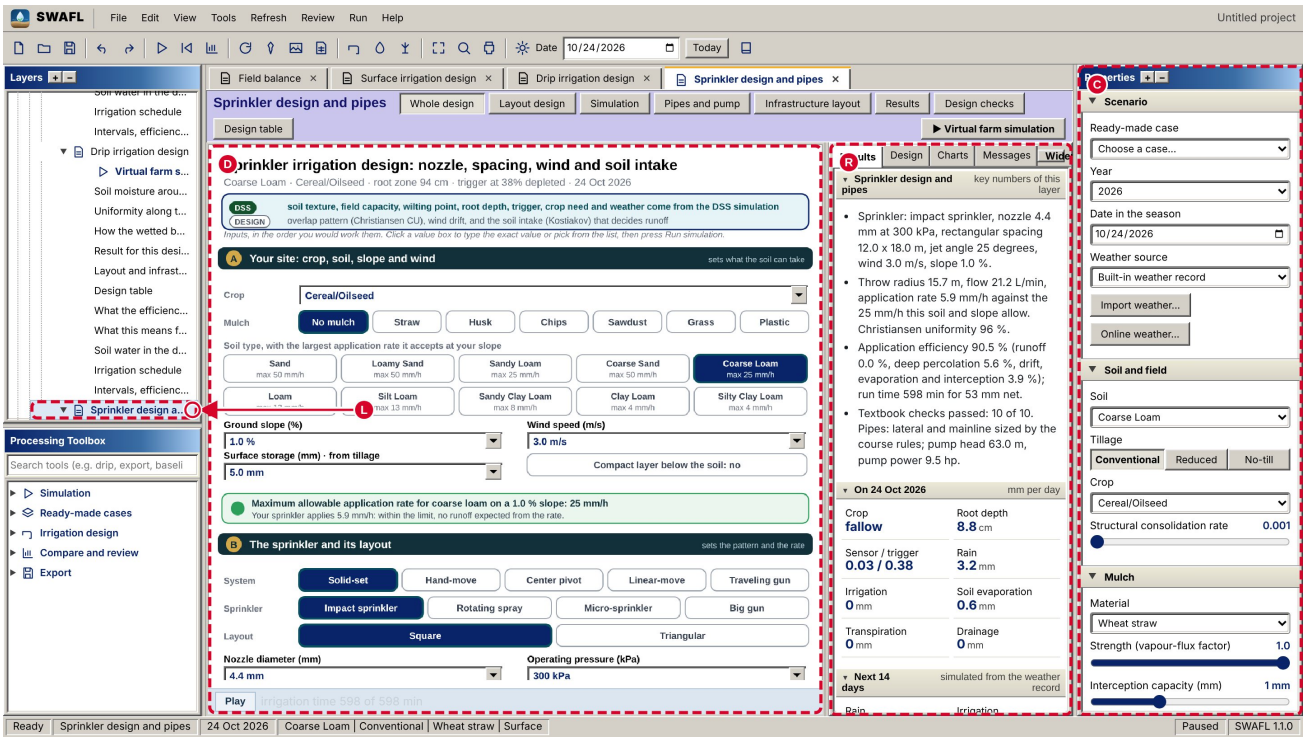


Figure 25.83: Sprinkler design and pipes / The jet: the whole workbench.

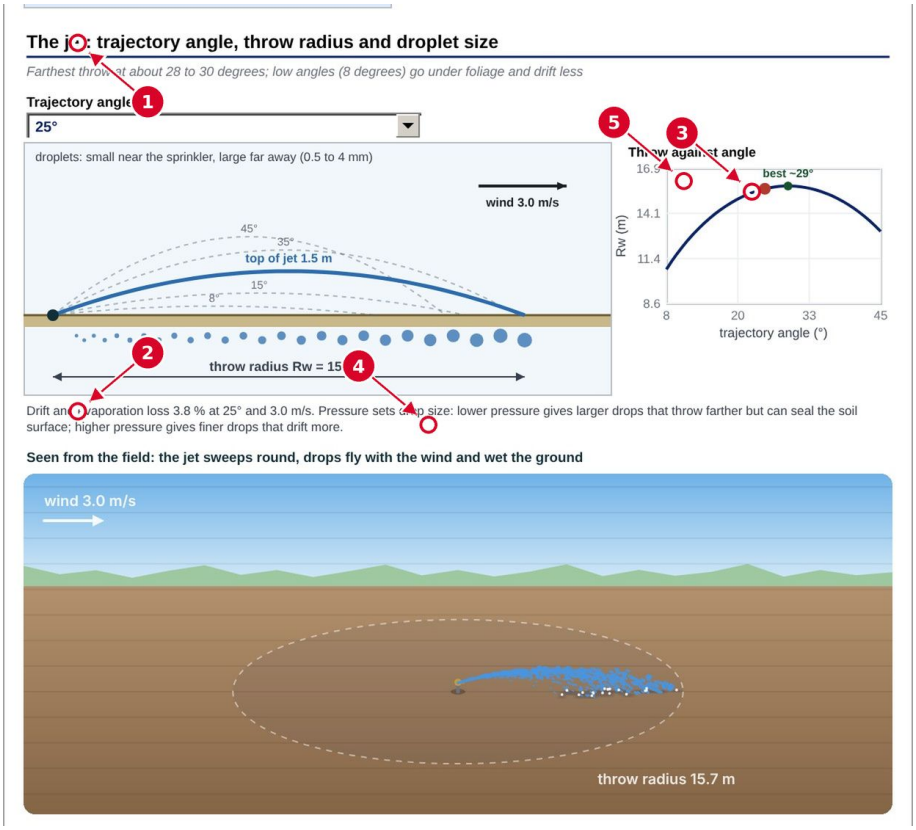


Figure 25.84: Sprinkler design and pipes / The jet: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Trajectory angle alpha: f_alpha(alpha) in Eq. 13.3.
- **2** Throw radius R_w: Eq. 13.3.
- **3** Throw against angle, best near 29 degrees: Eq. 13.3.
- **4** Drift and evaporation loss w_d: Eq. 13.8.
- **5** Wind speed u: Eqs. 13.5 and 13.8.

Equations behind this screen. Eq. 13.2, Eq. 13.3, Eq. 13.8.

Eq. 13.2. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".

$$q = 0.95\, A_n \sqrt{2g\,h} \times 6 \times 10^4 \ \ (\mathrm{L\,min^{-1}}), \qquad A_n = \pi\left(\frac{d}{2000}\right)^2 \ \mathrm{m^2}, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

Eq. 13.3. Where explained: chapter "Sprinkler design", section "Nozzle discharge and wetted radius".

$$R = c_R \sqrt{d\,h}\; f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \left[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\right]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

Eq. 13.8. Where explained: chapter “Sprinkler design”, section “Wind drift and evaporation loss”.

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{\left(1.2 + 0.5u + 0.04u^2\right) F\Big(\frac{P}{250}\Big)^{0.7}\left[1 + 0.025\,(T_a - 20)\right]\Big[0.6 + 0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

25.4.10 The water pattern of one sprinkler, and how overlap evens it out

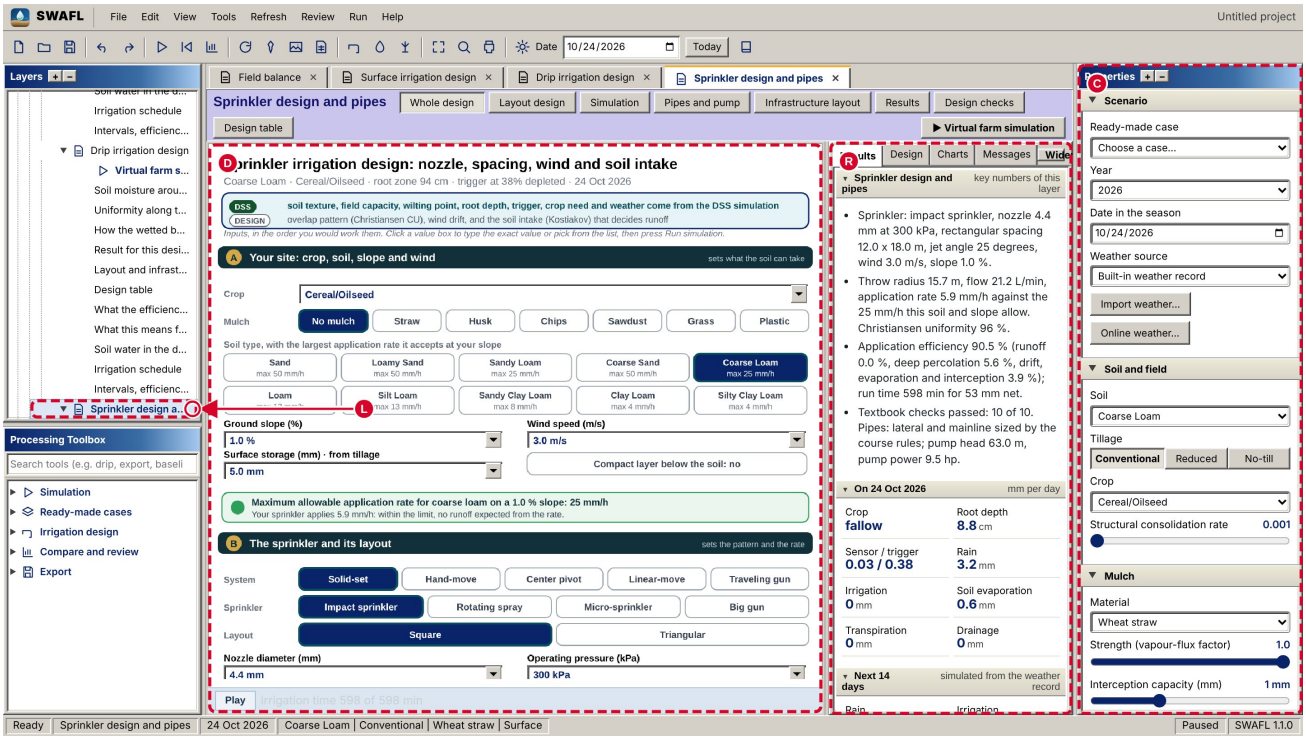


Figure 25.85: Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out: the whole workbench.

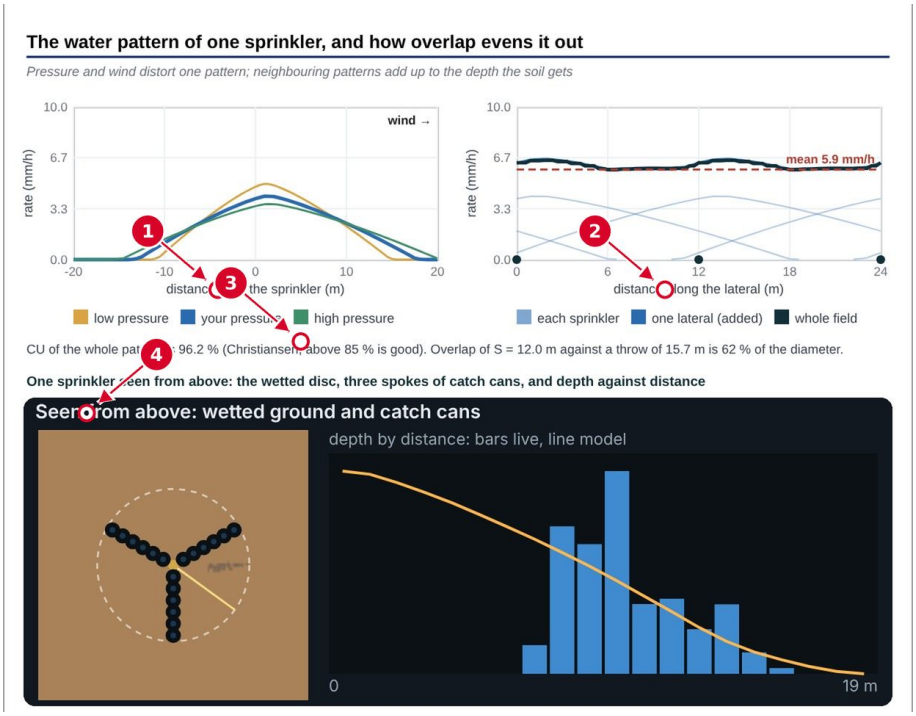


Figure 25.86: Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Radial profile of one sprinkler, p(x): Eq. 13.4, skewed by the wind of Eq. 13.5.
- **2** Sum over neighbouring sprinklers, a(x, y): Eq. 13.6; the dashed line is the mean rate.
- **3** Christiansen coefficient of the whole pattern: Eq. 13.7.
- **4** One sprinkler seen from above with three spokes of catch cans.

Equations behind this screen. Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\widehat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$R_{e}(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$a(x, y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_{k}, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\mathrm{CU} = 1 - \frac{\sum_{k} |a_{k} - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

25.4.11 Layouts: rectangular, square and triangular spacing for this sprinkler

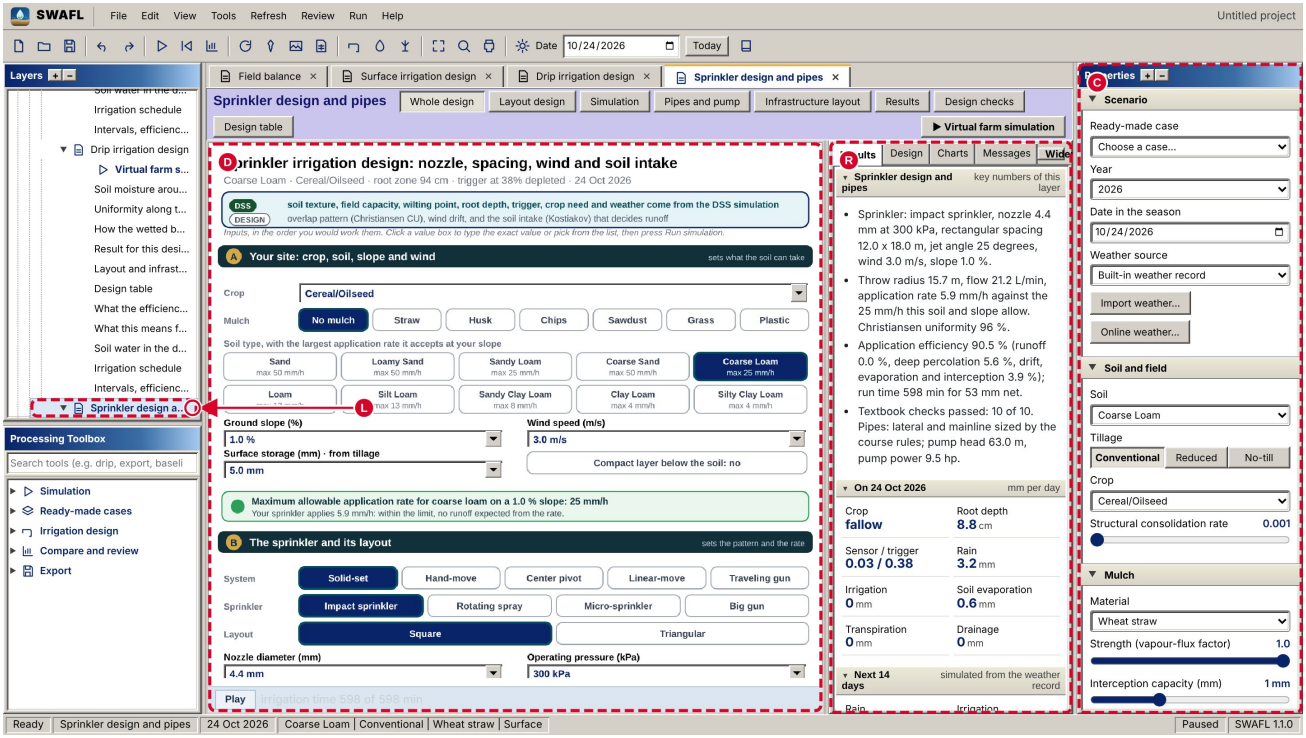


Figure 25.87: Sprinkler design and pipes / Layouts: the whole workbench.

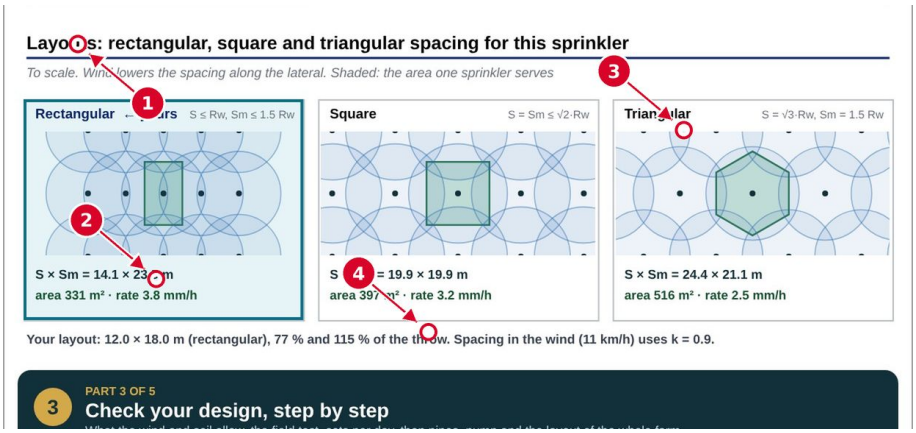


Figure 25.88: Sprinkler design and pipes / Layouts: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Your layout: S_s <= k R_w and S_l <= 1.5 R_w, Eq. 13.14.
- **2** Area per sprinkler and application rate: Eq. 13.7.
- **3** Triangular layout: every second lateral shifted by S_s / 2, Eq. 13.6.
- **4** Wind factor k: Eq. 13.13.

Equations behind this screen. Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

Eq. 13.4. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$\widehat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$R_{e}(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_{k}, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter “Sprinkler design”, section “The spray pattern and the overlap”.

$$\mathrm{CU} = 1 - \frac{\sum_{k} |a_{k} - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

25.4.12 What wind and soil allow

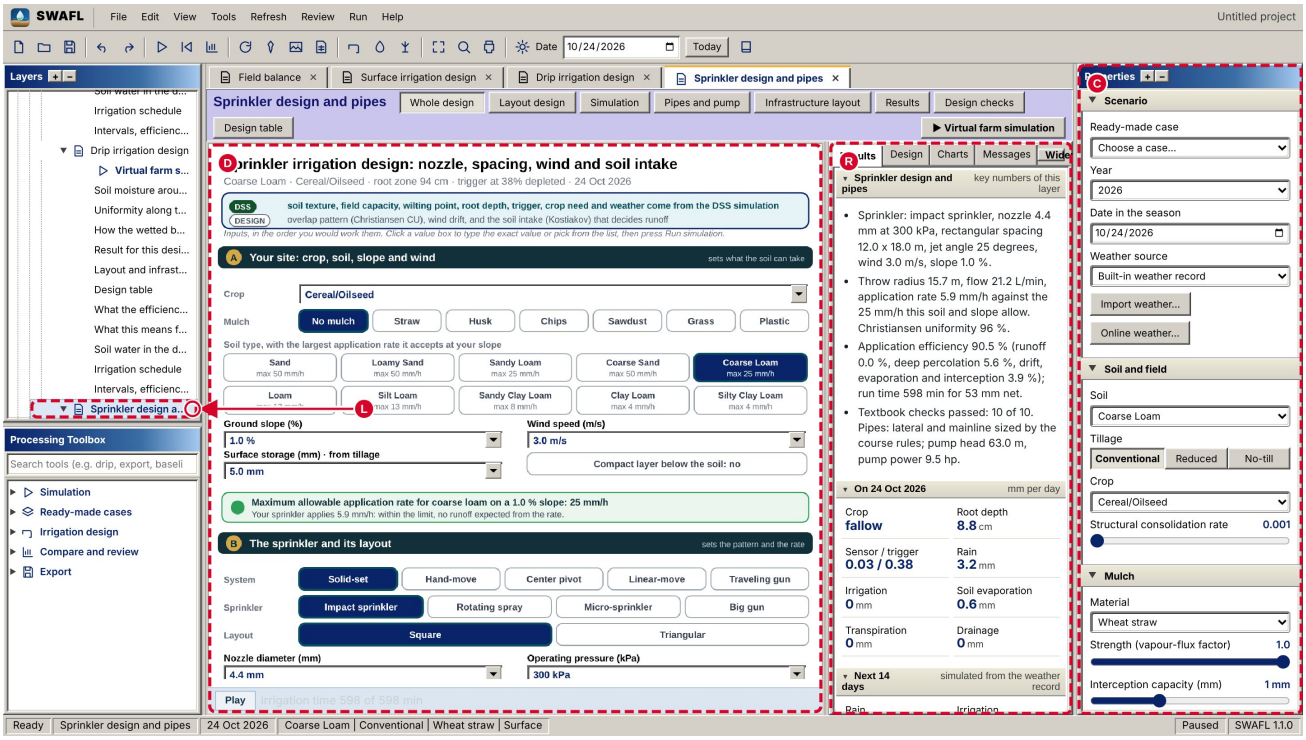


Figure 25.89: Sprinkler design and pipes / What wind and soil allow: the whole workbench.

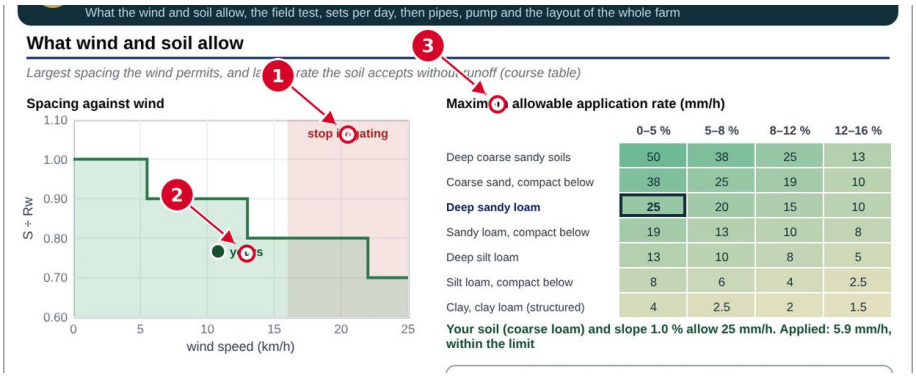


Figure 25.90: Sprinkler design and pipes / What wind and soil allow: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Step line of the wind factor k: Eq. 13.13.
- **2** “Yours”: your wind speed and spacing.
- **3** Maximum allowable application rate by soil and slope (course table), used in Eqs. 13.9 and 13.13.

Equations behind this screen. Eq. 13.8.

Eq. 13.8. Where explained: chapter “Sprinkler design”, section “Wind drift and evaporation loss”.

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2+0.5u+0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}\big[1+0.025\,(T_a-20)\big]\Big[0.6+0.4\,\frac{\sin\alpha}{0.4226}\Big];\;0.5,\;45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

25.4.13 Catch-can test: measuring the uniformity the way it is done in the field

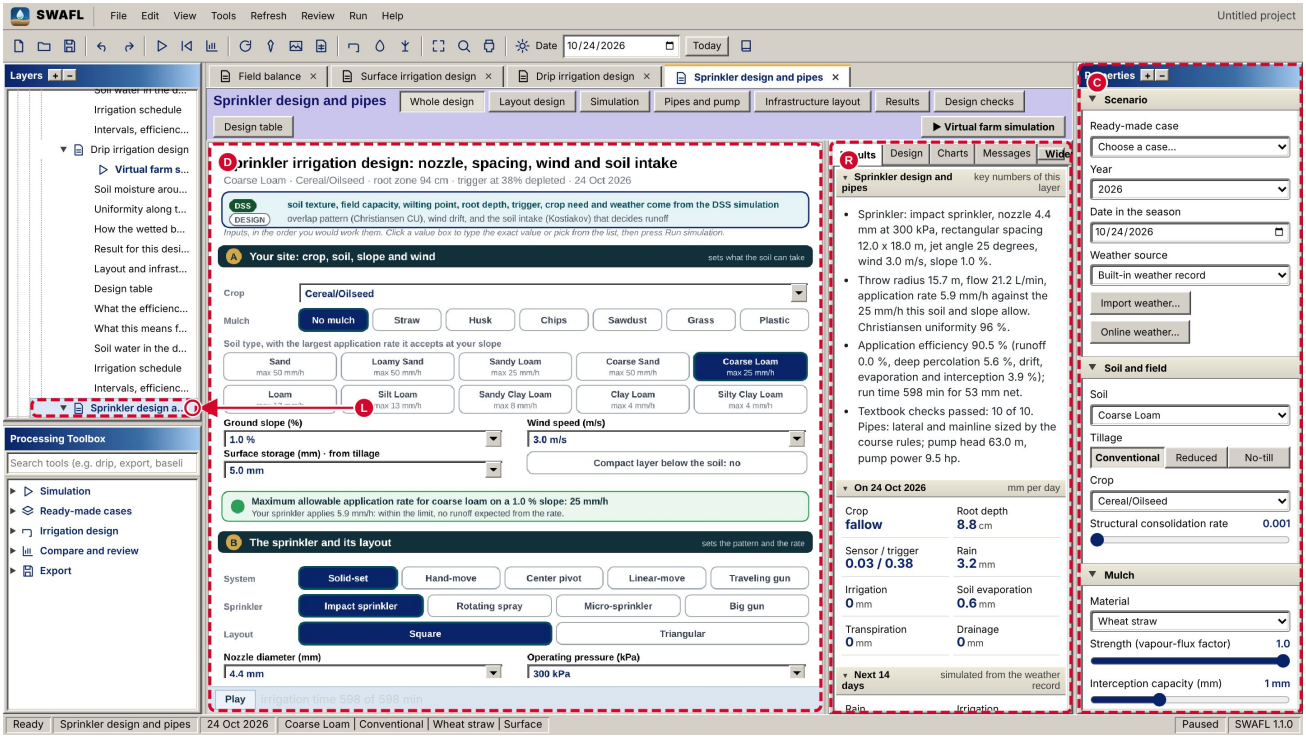


Figure 25.91: Sprinkler design and pipes / Catch-can test: the whole workbench.

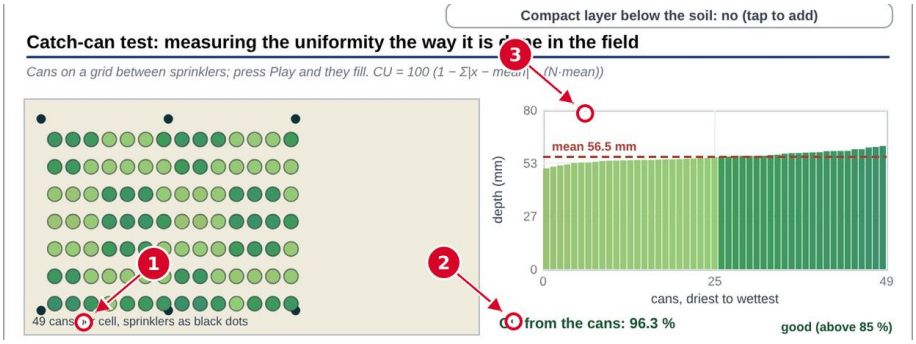


Figure 25.92: Sprinkler design and pipes / Catch-can test: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Catch cans: depth a_k in each can from the pattern of Eq. 13.6.
- **2** CU from the cans = 1 - sum |a_k - mean| / (n mean): Eq. 13.7.
- **3** Cans sorted from driest to wettest, with the mean depth.

Equations behind this screen. Eq. 13.4, Eq. 13.5, Eq. 13.6, Eq. 13.7.

Eq. 13.4. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\widehat{p}(x) = \begin{cases} 1 - x^{m}, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^{6}), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^{5}), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \widehat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7. Where explained: chapter "Sprinkler design", section "The spray pattern and the overlap".

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

25.4.14 Soil water as a reservoir: interval, sets per day and the flow the field needs

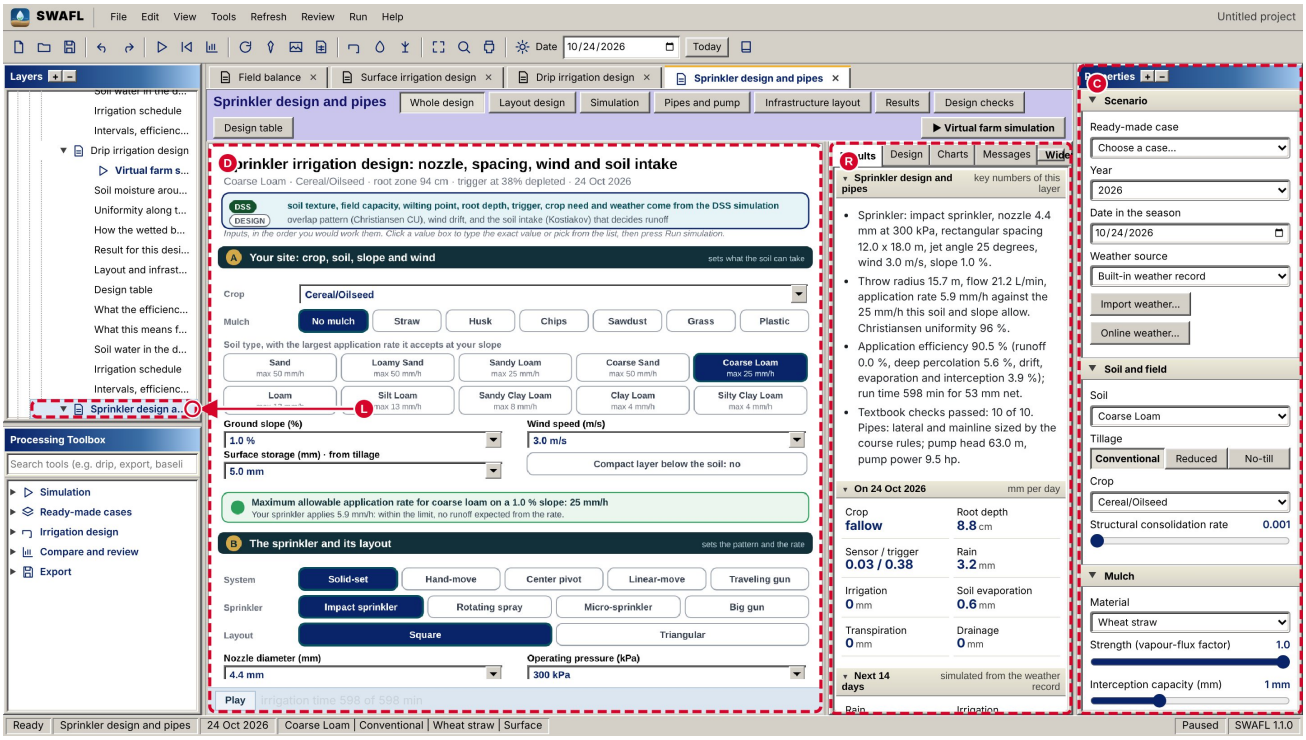


Figure 25.93: Sprinkler design and pipes / Soil water as a reservoir: the whole workbench.

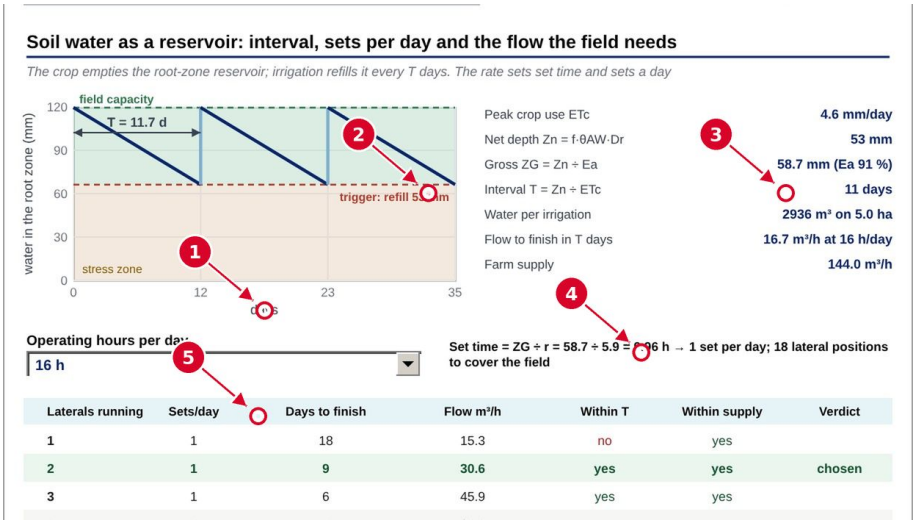


Figure 25.94: Sprinkler design and pipes / Soil water as a reservoir: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Root-zone store drawn down by ET_c and refilled every T days.
- **2** Refill at Z_n from the DSS: Eq. 5.27.
- **3** Gross depth Z_G = Z_n / E_a: Eq. 13.15.

- **4** Set time t = Z_G / r: Eq. 13.15.
- **5** Laterals running together, days to finish and inflow: Eq. 13.16.

Equations behind this screen. Eq. 13.11, Eq. 13.12, Eq. 13.23.

Eq. 13.11. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".

$$E_a = \frac{\overline{\min(Z_k, Z_n)}}{\text{gross}}, \qquad E_s = \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, \quad \text{DU} = \frac{\overline{Z}_{25}}{\overline{Z}},$$
$$\text{DP} = \frac{\overline{\max(Z_k - Z_n, 0)}}{\text{gross}}, \quad \text{RO} = \frac{\overline{\text{RO}_k}}{\text{gross}}, \qquad \text{WD} = w_d + \frac{\overline{I_c}}{\text{gross}}, \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 13.12. Where explained: chapter "Sprinkler design", section "Efficiencies and run time".

$$T = \min\{T : \ \overline{Z}_{25}(T) \geq Z_n\}, \qquad T \in [1,\, 2400] \text{ min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.23. Where explained: chapter "Sprinkler design", section "Season and farm".

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60} \text{ h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

25.4.15 Pipes, pump and the hydraulic grade line (the course sizing)

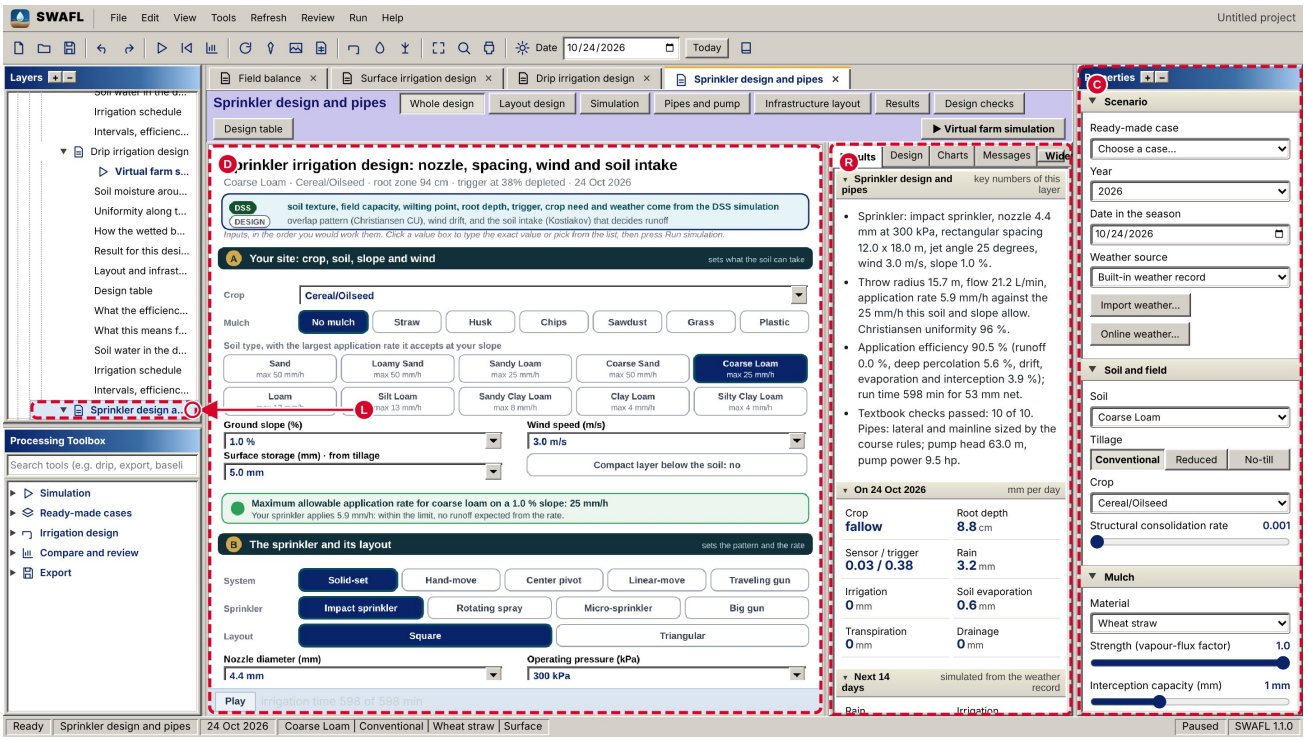


Figure 25.95: Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing): the whole workbench.

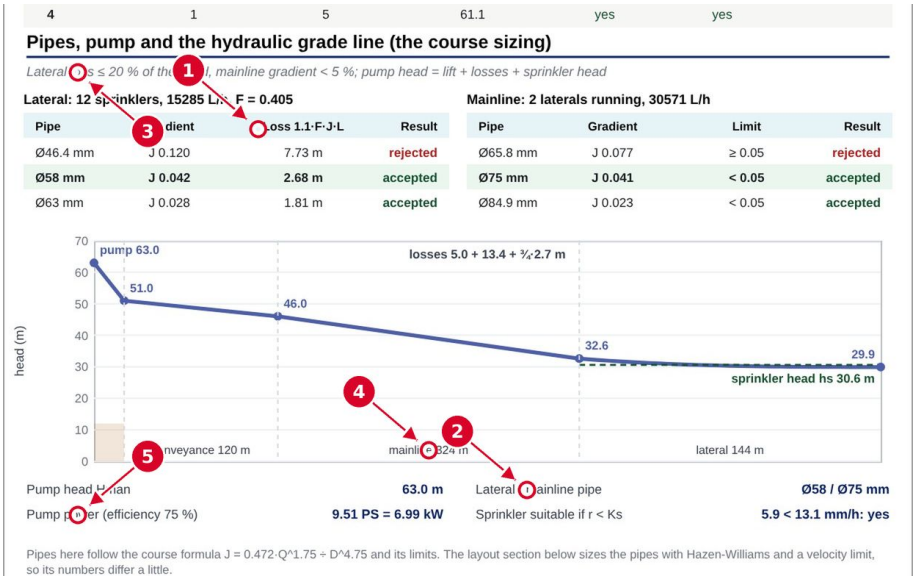


Figure 25.96: Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing): the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Gradient J of the pipe (Blasius form): Eq. 13.17.
- **2** Lateral pipe: smallest standard diameter meeting the loss limit, Eq. 13.18.
- **3** Mainline: gradient below 0.05, Eq. 13.17.
- **4** Head along the flow path to the pump, H_man: Eq. 13.19.
- **5** Pump power: Eq. 13.20.

Equations behind this screen. Eq. 12.3, Eq. 12.21, Eq. 12.22, Eq. 12.23, Eq. 13.21, Eq. 13.22.

Eq. 12.3. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \ \ (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m.} \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.21. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\} \text{ mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$h_{f,sm} = 0.4\, \frac{10.67\, L_{sm}\, Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\, L_m\, Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23. Where explained: chapter "Drip irrigation design", section "Layout and infrastructure".

$$\mathrm{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \mathrm{m}, \qquad P = \frac{\rho g\, Q_z\, \mathrm{TDH}}{\eta} = \frac{9.81\, (Q_z/1000)\, \mathrm{TDH}}{0.65} \ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 13.21. Where explained: chapter "Sprinkler design", section "Layout and infrastructure".

$$N_L = \Big\lfloor \frac{10^4 A_f / L_L}{S_l} \Big\rfloor, \quad n_l = \min\Big(N_L,\ \Big\lfloor \frac{Q_s}{Q_{\text{lat}}} \Big\rfloor\Big), \quad n_{\text{set}} = \Big\lceil \frac{N_L}{n_l} \Big\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.

Eq. 13.22. Where explained: chapter "Sprinkler design", section "Layout and infrastructure".

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m\, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.

25.4.16 Layout and infrastructure: source, pump, filter, mainline, valves, submain and sprinkler laterals

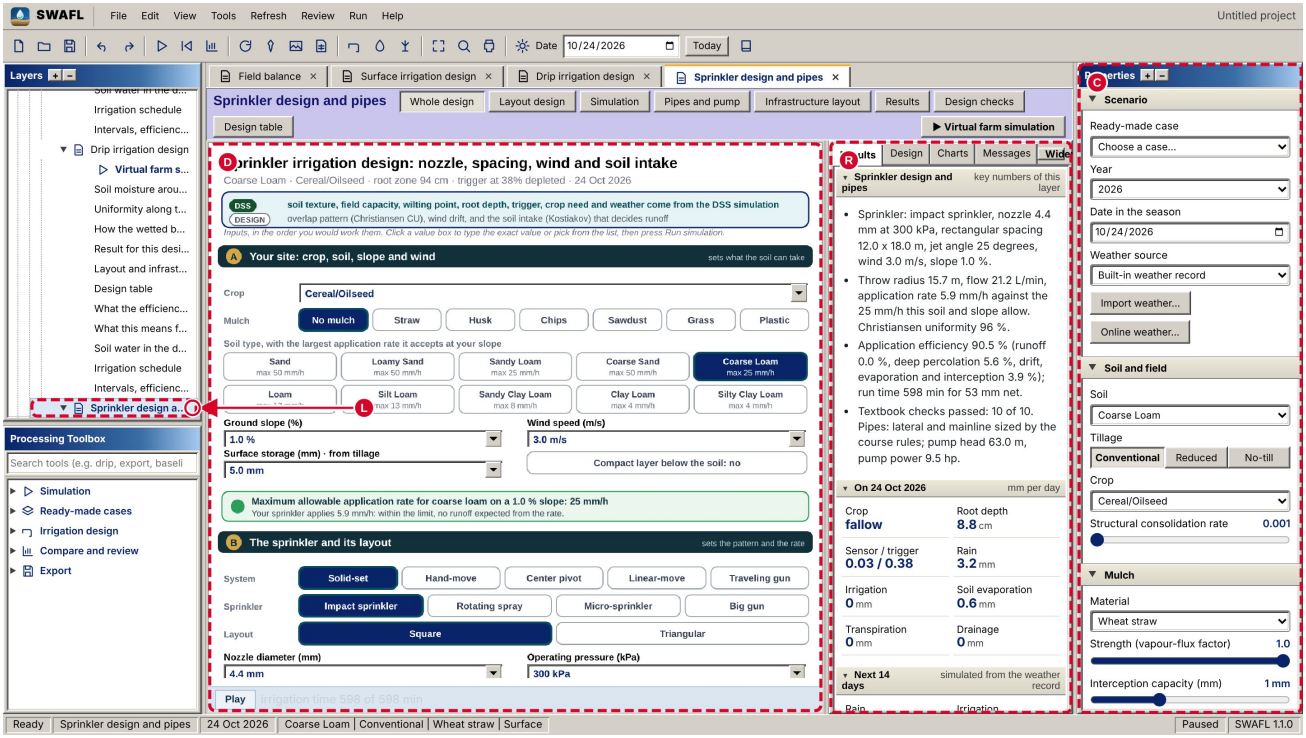


Figure 25.97: Sprinkler design and pipes / Layout and infrastructure: the whole workbench.

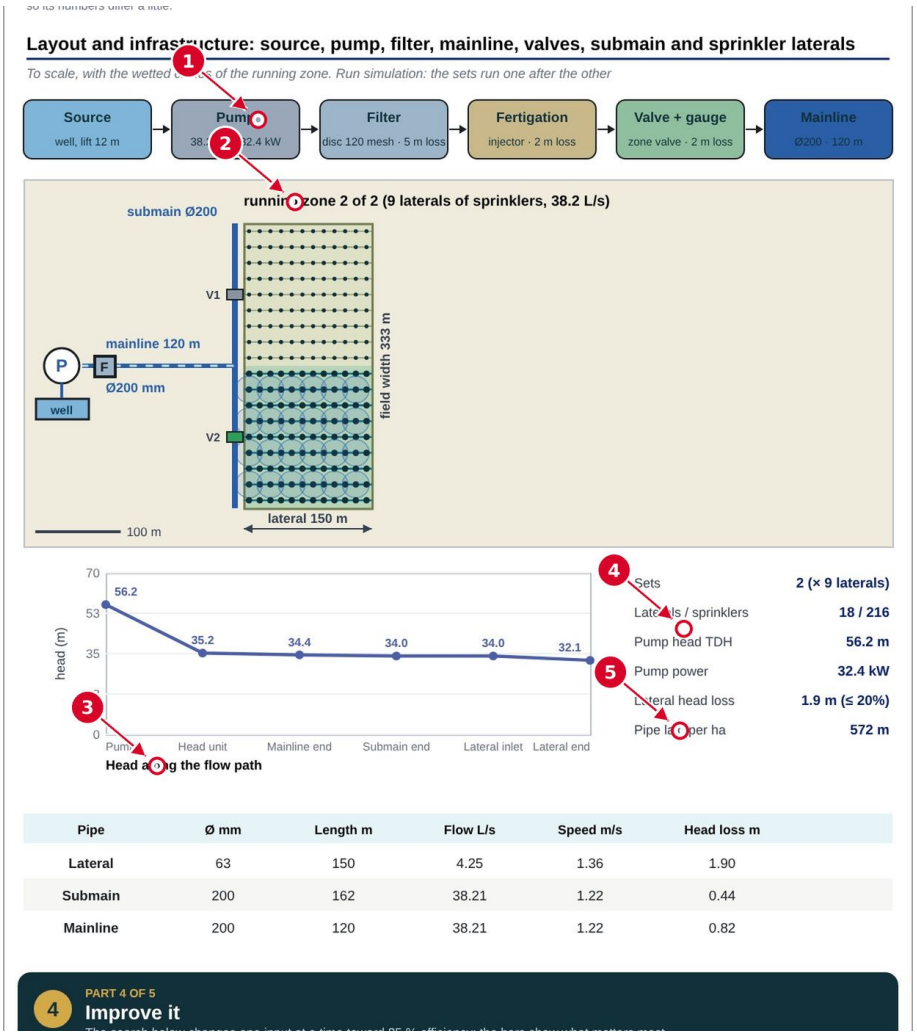


Figure 25.98: Sprinkler design and pipes / Layout and infrastructure: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Source, pump, filter, fertigation and valves in order.
- **2** Laterals, N_L = field length / S_l, and laterals running together: Eq. 13.21.
- **3** Head along the flow path: Eqs. 13.19 and 13.22.
- **4** Lateral head loss with the Christiansen factor F_m: Eq. 13.22.
- **5** Pipe table: Eq. 13.18.

Equations behind this screen. Eq. 12.3, Eq. 12.21, Eq. 12.22, Eq. 12.23, Eq. 13.21, Eq. 13.22.

Eq. 12.3. Where explained: chapter "Drip irrigation design", section "Lateral hydraulics: head loss and flow along the lateral".

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \geq j+1} q_i \;\; (\mathrm{m^3\,s^{-1}}), \quad D \text{ in m.} \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.21. Where explained: chapter "Drip irrigation design", section "Layout and infrastruc-

ture”.

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \text{mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23. Where explained: chapter “Drip irrigation design”, section “Layout and infrastructure”.

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \text{m}, \qquad P = \frac{\rho g\, Q_z\, \text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 13.21. Where explained: chapter “Sprinkler design”, section “Layout and infrastructure”.

$$N_L = \left\lfloor \frac{10^4 A_f / L_L}{S_l} \right\rfloor, \quad n_l = \min\!\left(N_L,\ \left\lfloor \frac{Q_s}{Q_{\text{lat}}} \right\rfloor\right), \quad n_{\text{set}} = \left\lceil \frac{N_L}{n_l} \right\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.

Eq. 13.22. Where explained: chapter “Sprinkler design”, section “Layout and infrastructure”.

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m\, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.

25.4.17 Design table: iterations toward an application efficiency of 85 % or more

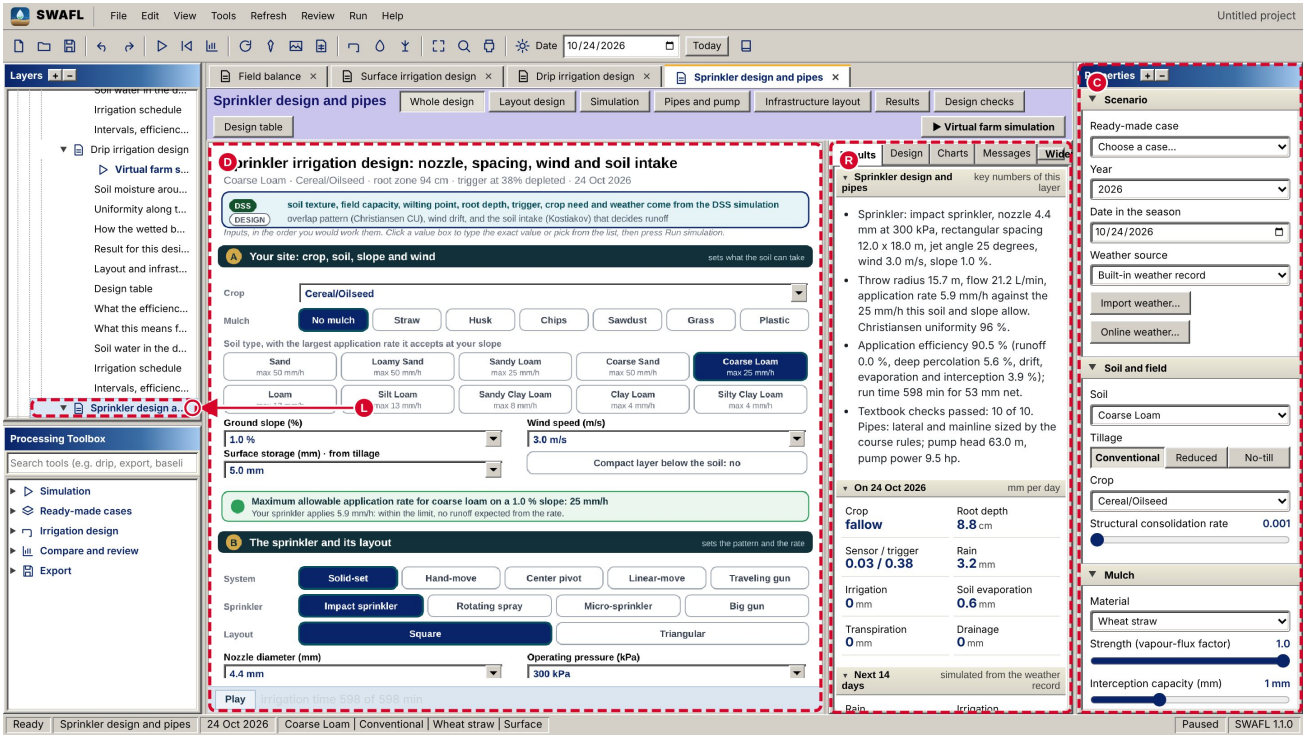


Figure 25.99: Sprinkler design and pipes / Design table: the whole workbench.

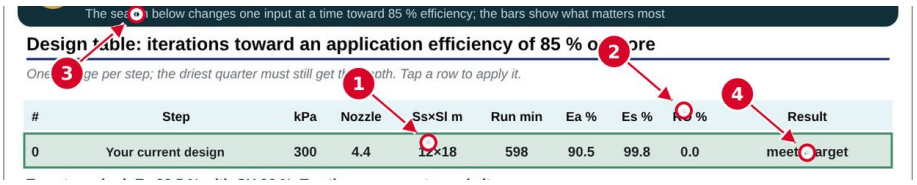


Figure 25.100: Sprinkler design and pipes / Design table: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Row 0: your current design (pressure, nozzle, spacing, run time, E_a, E_s).
- **2** Column E_a: Eq. 13.11.
- **3** Run time T: Eq. 13.12.
- **4** Target reached: E_a of at least 85 % with CU of at least 85 %, Eqs. 13.7 and 13.11.

Equations behind this screen. Eq. 12.26.

Eq. 12.26. Where explained: chapter “Drip irrigation design”, section “The design table”.

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

25.4.18 What the efficiency is most sensitive to

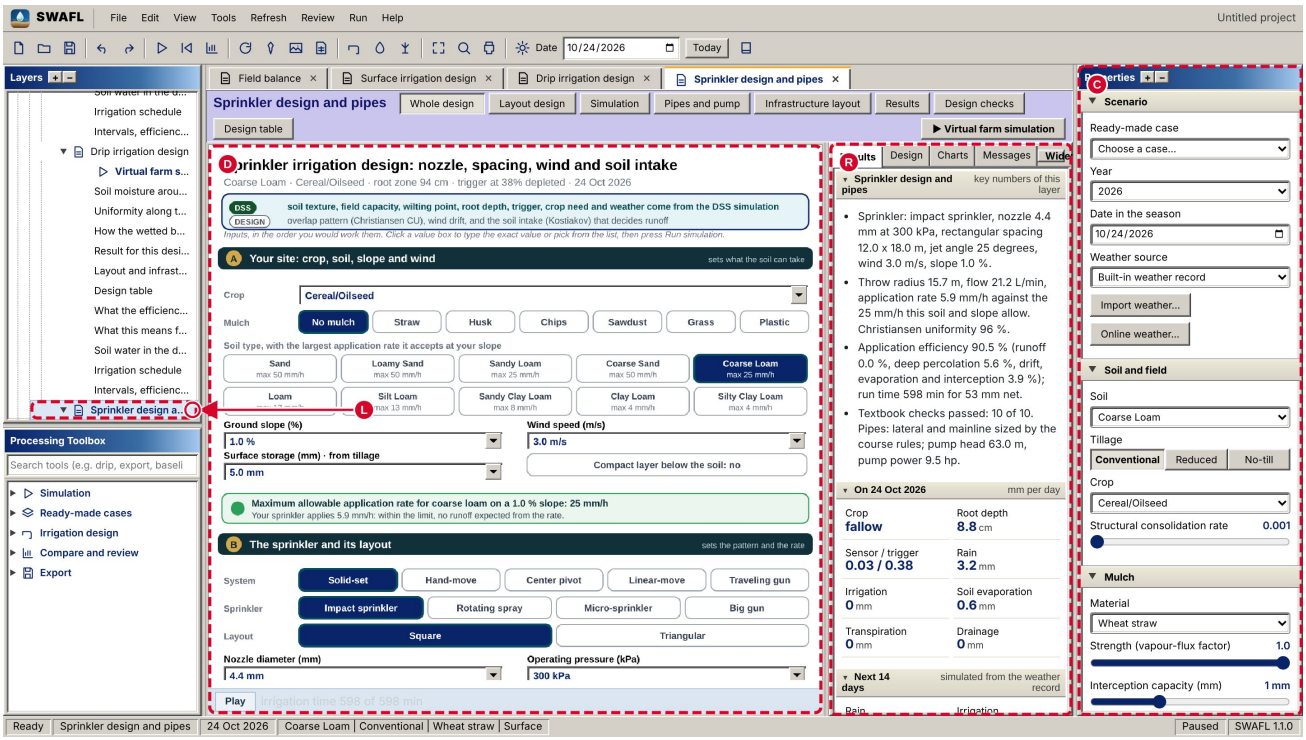


Figure 25.101: Sprinkler design and pipes / What the efficiency is most sensitive to: the whole workbench.

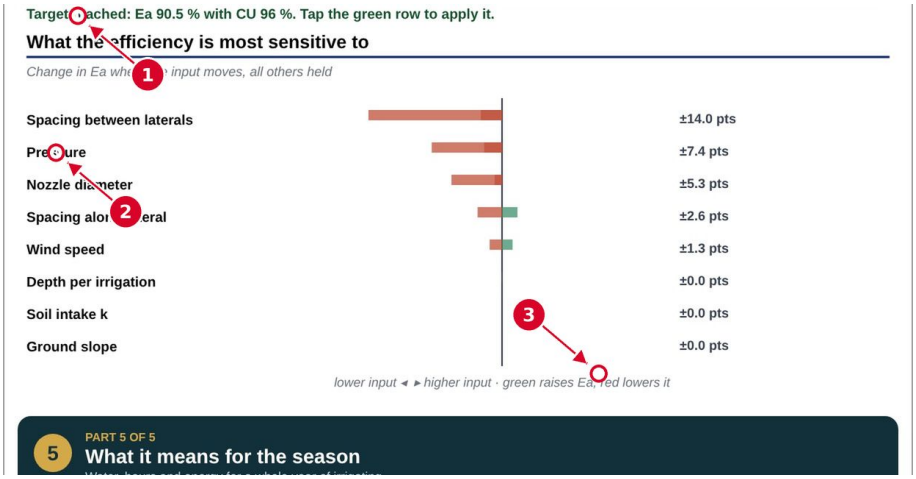


Figure 25.102: Sprinkler design and pipes / What the efficiency is most sensitive to: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Tornado bar: change in E_a (Eq. 13.11) when the input moves.
- **2** Pressure: second most sensitive input.
- **3** Green raises E_a, red lowers it.

Equations behind this screen. Eq. 12.26.

Eq. 12.26. Where explained: chapter “Drip irrigation design”, section “The design table”.

$$\text{score} = \min(E_a, 1) - 2.5\ \max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

25.4.19 What this means for the season and the farm

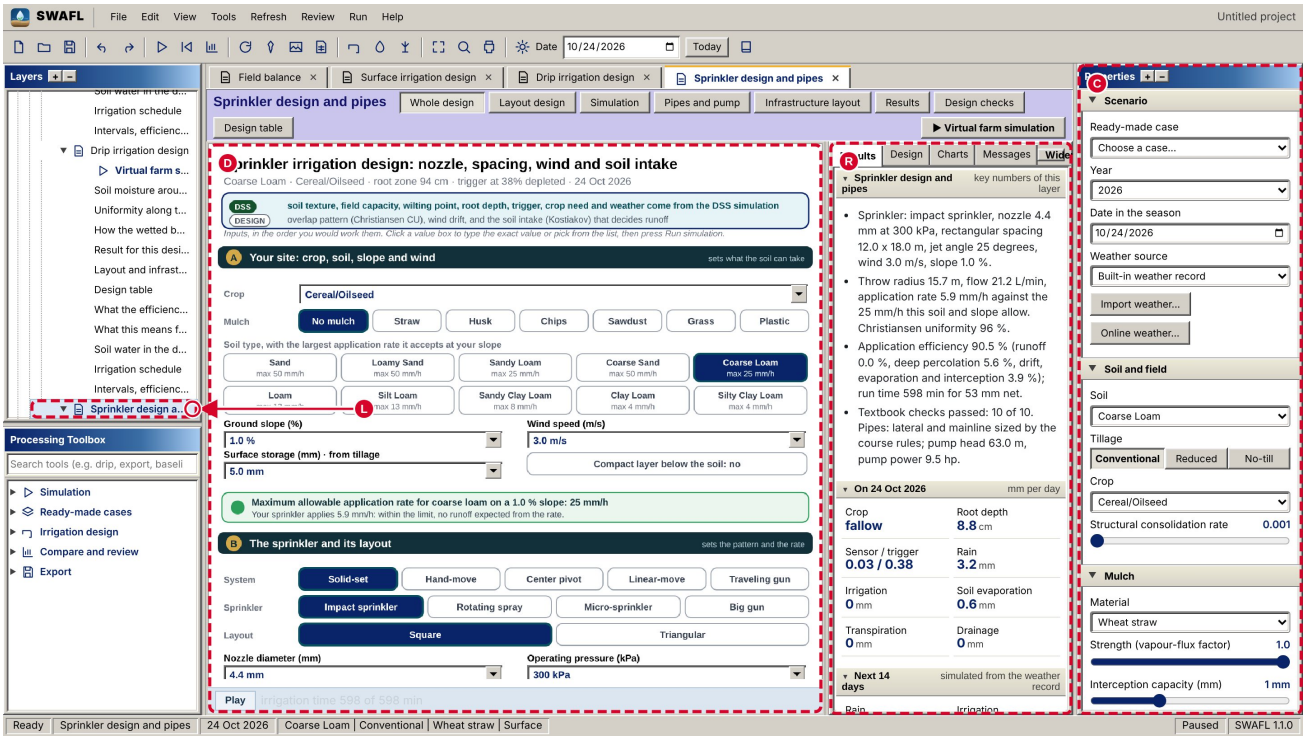


Figure 25.103: Sprinkler design and pipes / What this means for the season and the farm: the whole workbench.

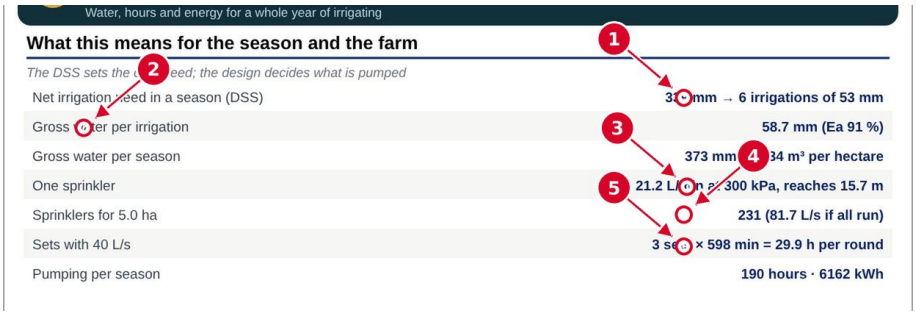


Figure 25.104: Sprinkler design and pipes / What this means for the season and the farm: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Irrigations in a season: Eq. 13.23.
- **2** Gross depth per irrigation: Eq. 13.23.
- **3** One sprinkler discharge q: Eq. 13.2.
- **4** Sprinklers for the farm area, N_s: Eq. 13.21.
- **5** Sets and run time: Eqs. 13.12 and 13.21.

Equations behind this screen. Eq. 13.23.

Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.

$$n_{\mathrm{ev}} = \frac{y_I}{Z_n}, \qquad n_{\mathrm{sp}} = \mathrm{round}\!\left(\frac{10^4 A_f}{S_s S_l}\right), \qquad Q_{\mathrm{tot}} = \frac{n_{\mathrm{sp}}\, q}{60}, \qquad n_{\mathrm{set}} = \left\lceil \frac{Q_{\mathrm{tot}}}{Q_s} \right\rceil, \qquad t_{\mathrm{round}} = \frac{n_{\mathrm{set}}\, T}{60}\ \mathrm{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

25.4.20 Irrigation schedule: when to irrigate with this design

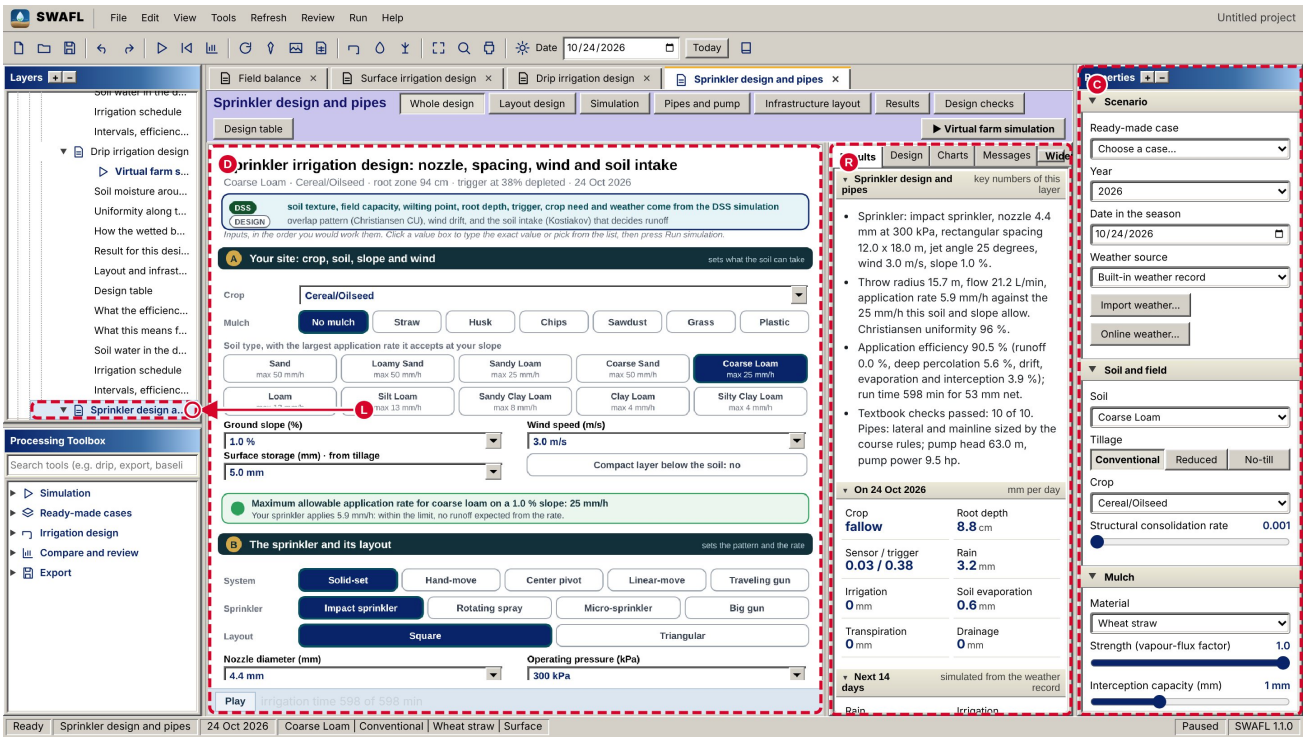


Figure 25.105: Sprinkler design and pipes / Irrigation schedule: the whole workbench.

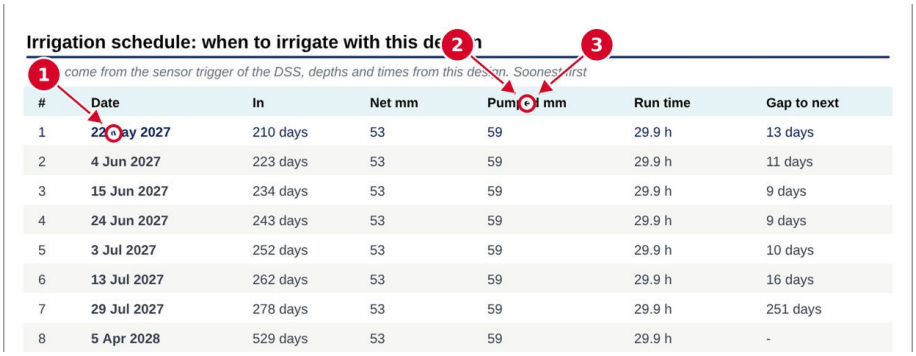


Figure 25.106: Sprinkler design and pipes / Irrigation schedule: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Dates from the Engine 2 irrigation record.
- **2** Pumped mm = I_k / E_a: Eq. 13.23.
- **3** Run time of one round: Eq. 13.12.

Equations behind this screen. Eq. 13.23.

Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.

$$n_{\mathrm{ev}} = \frac{y_I}{Z_n}, \qquad n_{\mathrm{sp}} = \mathrm{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\mathrm{tot}} = \frac{n_{\mathrm{sp}}\, q}{60}, \qquad n_{\mathrm{set}} = \Big\lceil \frac{Q_{\mathrm{tot}}}{Q_s} \Big\rceil, \qquad t_{\mathrm{round}} = \frac{n_{\mathrm{set}} T}{60}\ \mathrm{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

25.4.21 Intervals, efficiency and the advice in short

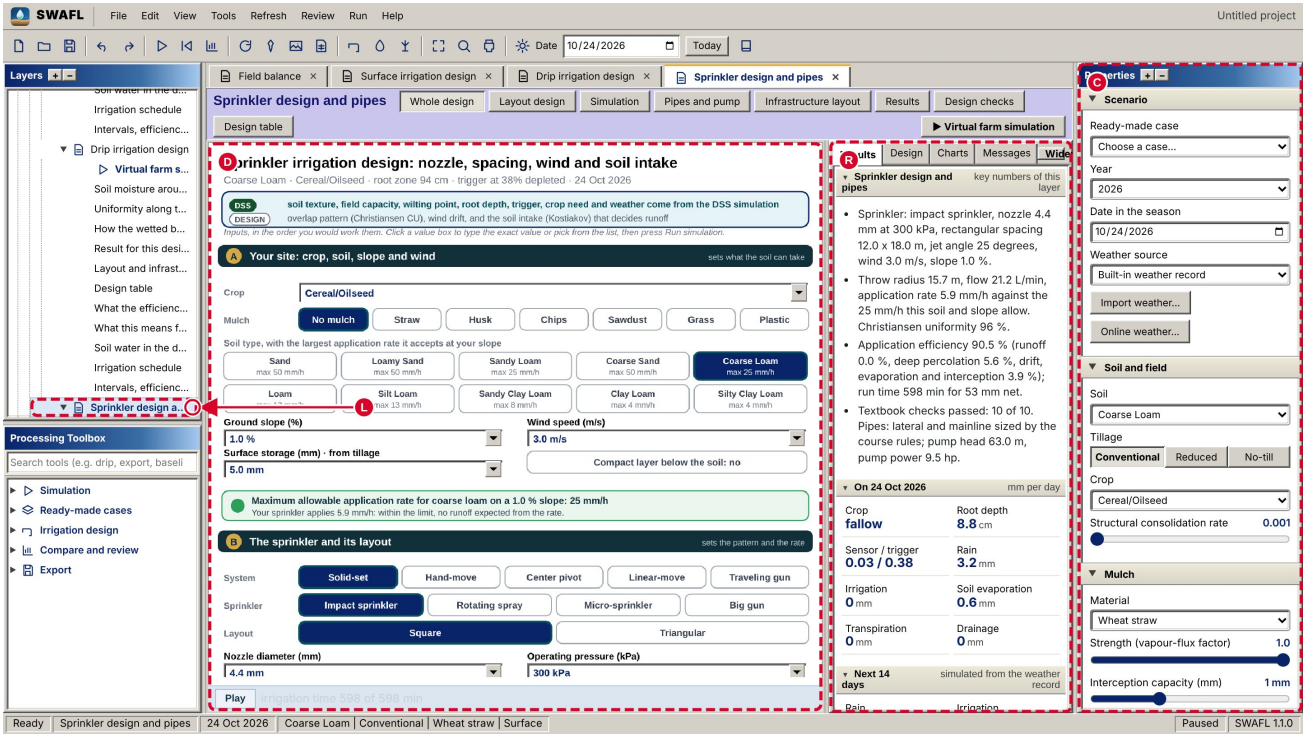


Figure 25.107: Sprinkler design and pipes / Intervals, efficiency and the advice in short: the whole workbench.

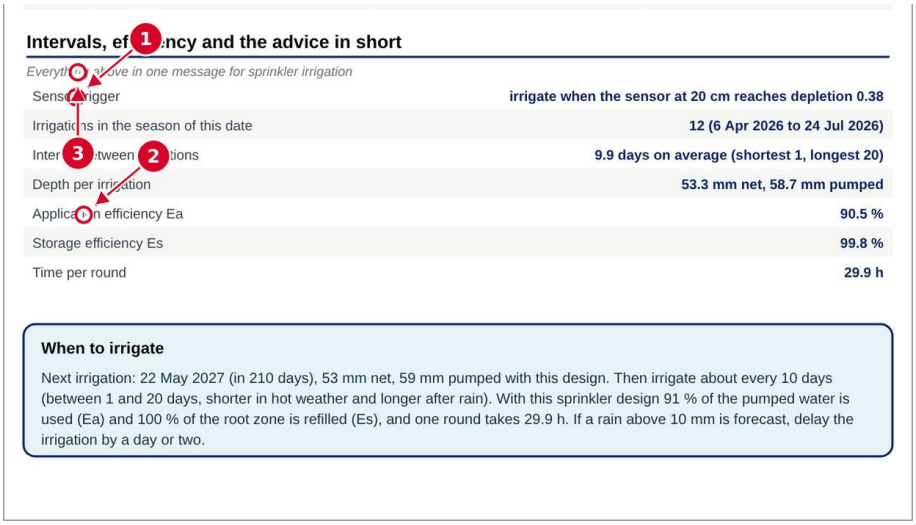


Figure 25.108: Sprinkler design and pipes / Intervals, efficiency and the advice in short: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Sensor trigger: Eqs. 5.3 and 5.27.
- **2** Application efficiency E_a: Eq. 13.11.

- **3** “When to irrigate”: message built from the schedule and from E_a and E_s.

Equations behind this screen. Eq. 13.23.

Eq. 13.23. Where explained: chapter “Sprinkler design”, section “Season and farm”.

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}} T}{60}\ \text{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

25.5 Field balance

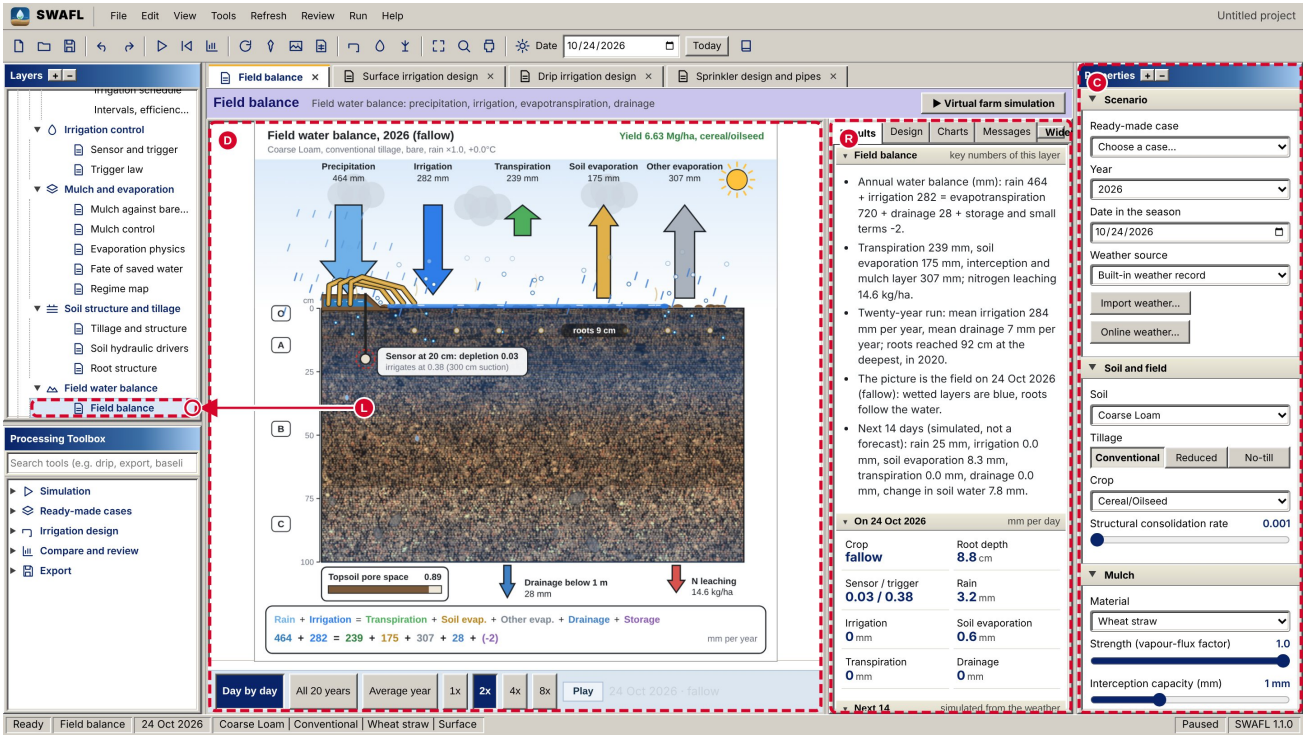


Figure 25.109: Field balance: the whole workbench.

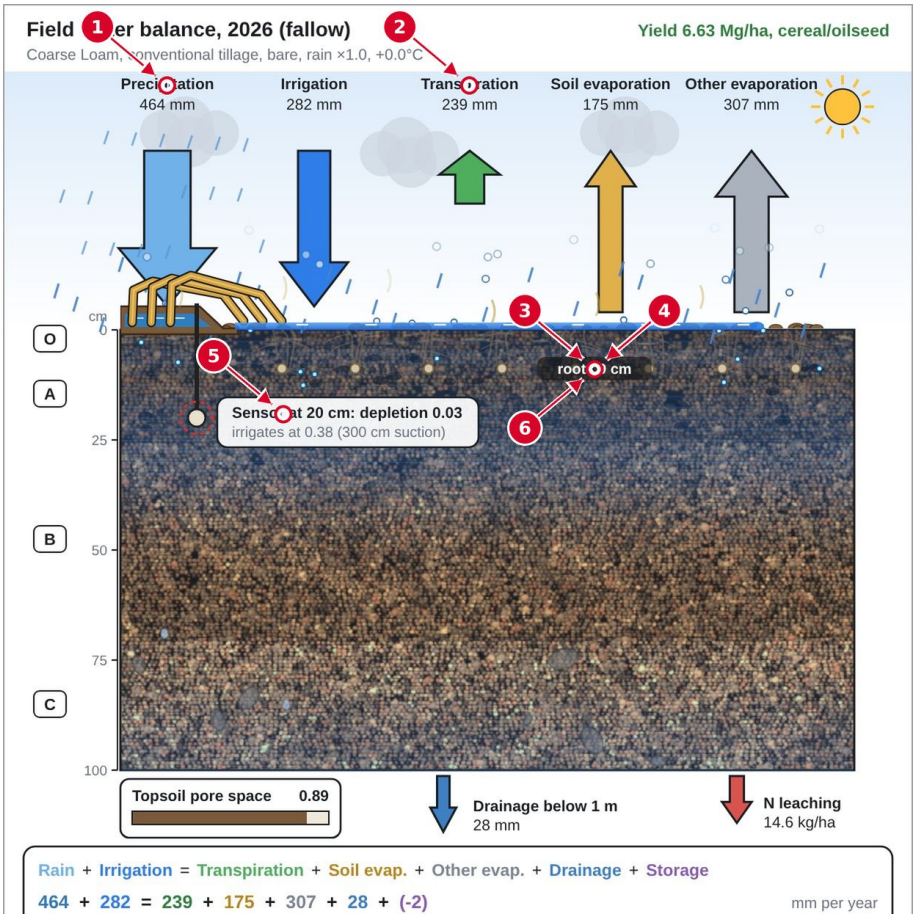


Figure 25.110: Field balance: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Precipitation arrow, P: annual value from Eqs. 5.4 to 5.12 (Average year); daily value from the rainfall generator, Eqs. 5.15 and 5.16 (Day by day).
- **2** Transpiration arrow, T: Eqs. 5.4 to 5.12 (annual) and Eqs. 5.21 to 5.25 (daily).
- **3** Soil evaporation arrow, E: Eqs. 5.4 to 5.12 (annual) and Eqs. 5.21 to 5.26 (daily).
- **4** Interception arrow, E_other: the closure term of Eq. 5.12.
- **5** Sensor tag: depletion f of the sensor layer against the trigger f_T, Eq. 5.3 (Engine 2 layer j_s).
- **6** Roots (depth and density): Eqs. 5.19 and 5.20.

Equations behind this screen. Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 5.13, Eq. 5.14, Eq. 5.15, Eq. 5.16, Eq. 5.17, Eq. 5.18, Eq. 5.19, Eq. 5.20, Eq. 5.21, Eq. 5.22, Eq. 5.23, Eq. 5.24, Eq. 5.25, Eq. 8.1.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".

$$Q \leftarrow Q\,\text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 5.13. Where explained: chapter "The two computing engines", section "Crop calendar and canopy cover".

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp(-(d - d_{\text{mid}})/s)}, \quad s = 13\ (\text{vegetable } 8), \tag{5.13}$$

Provenance (Eq. 5.13): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.14. Where explained: chapter "The two computing engines", section "Crop calendar and canopy cover".

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\ \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp(-(d - d_{\text{pk}} + 45)/12)} & \text{otherwise} \end{cases} \tag{5.14}$$

Provenance (Eq. 5.14): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.15. Where explained: chapter "The two computing engines", section "Rainfall generator".

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0, 1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.16. Where explained: chapter "The two computing engines", section "Rainfall generator".

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

Provenance (Eq. 5.16): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.17. Where explained: chapter "The two computing engines", section "Infiltration, redistribution and drainage".

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

Eq. 5.18. Where explained: chapter "The two computing engines", section "Roots".

$$R_{\max} = \operatorname{clip}\Big(R_c\,(0.85{+}0.3\,\text{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \text{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.19. Where explained: chapter "The two computing engines", section "Roots".

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\operatorname{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35 + \bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.20. Where explained: chapter "The two computing engines", section "Roots".

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.21. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65\cdot\tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.22. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\text{ yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.23. Where explained: chapter "The two computing engines", section "Transpiration, evaporation".

$$q_i^{T} = T_d \,\frac{\rho_i \, a_i}{\sum_j \rho_j a_j}, \qquad a_i = \mathrm{clip}\Big(\frac{s_i - 0.15}{0.6};\, 0,\, 1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.24. Where explained: chapter "The two computing engines", section "Scaling to the annual model".

$$S_P = \frac{P}{\overline{P}_{\mathrm{sim}}}, \quad S_{\mathrm{thr}} = \frac{P - P_{\mathrm{other}}}{\overline{T}_{f,\mathrm{sim}}}, \quad S_E = \frac{E}{\overline{E}_{\mathrm{sim}}}, \quad S_T = \frac{T}{\overline{T}_{\mathrm{sim}}}, \quad S_D = \frac{D}{\overline{D}_{\mathrm{sim}}}, \tag{5.24}$$

Provenance (Eq. 5.24): **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

Eq. 5.25. Where explained: chapter "The two computing engines", section "Scaling to the annual model".

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

Eq. 8.1. Where explained: chapter "Soil structure and tillage layers", section "Pore index and consolidation".

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

25.5.1 The whole layer, part by part

The picture below is the complete drawing of the layer Field balance, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

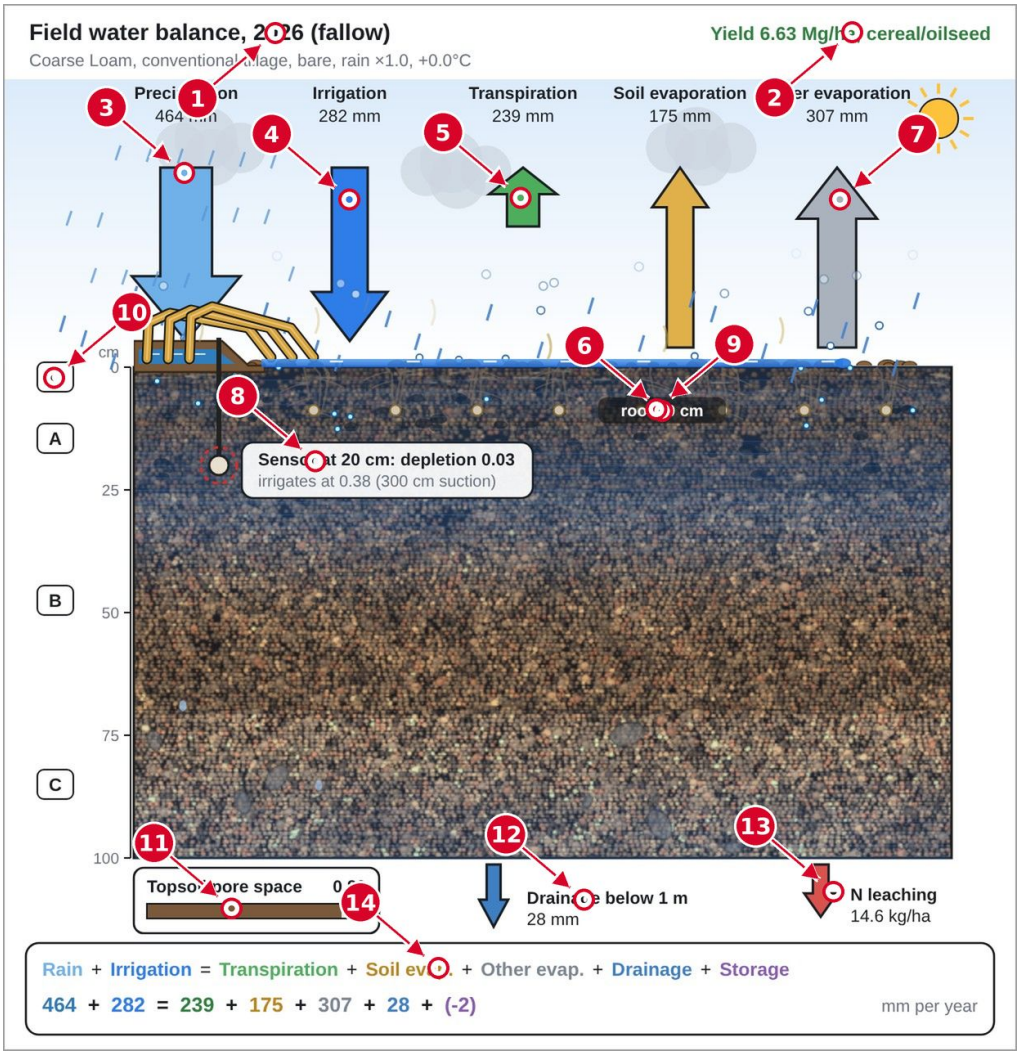


Figure 25.111: Field balance, part 1 of 1.

What the arrows point to.

- **1** The title gives the year, the crop state and the scenario in force (soil, tillage, mulch, rain scaling and warming).
- **2** The yield of the year, with the crop that produced it.
- **3** Precipitation, the wet input of the budget, in mm for the year.
- **4** Irrigation, the applied water decided by the sensor trigger, in mm for the year.
- **5** Transpiration, the water taken up through the roots and released by the leaves.
- **6** Soil evaporation, the water lost directly from the soil surface.
- **7** Interception, the rain held on leaves and mulch and evaporated before reaching the soil.
- **8** The soil sensor at 20 cm, with its depletion and the depletion at which irrigation starts.
- **9** The depth of the roots at this moment of the season.
- **10** The horizons O, A, B and C of the soil profile, with the depth scale in cm.
- **11** The topsoil pore space, the fraction of the topsoil volume that is air and water.
- **12** Drainage below 1 m, the water that leaves the root zone downward.
- **13** Nitrogen leaching carried out of the root zone with the drainage.
- **14** The closed budget strip, in which the inputs equal the outputs plus the change in storage.

25.5.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

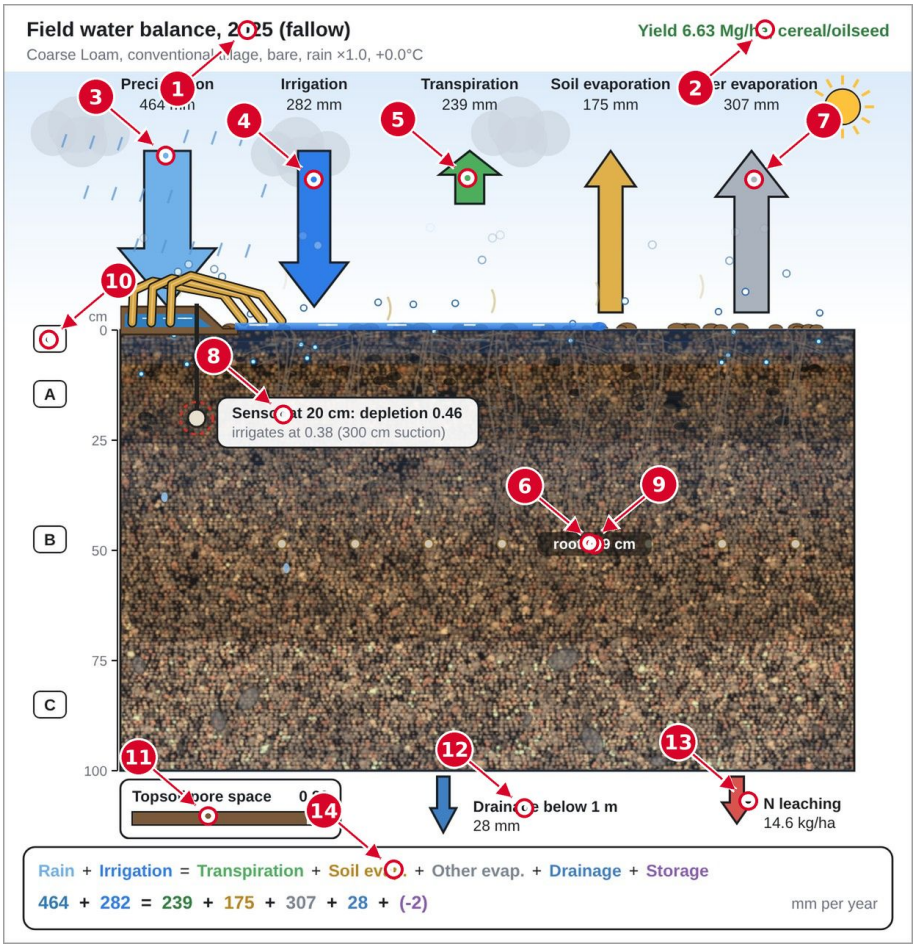


Figure 25.112: Field balance, part 1, playback a third of the way (7 Aug 2025 · fallow).

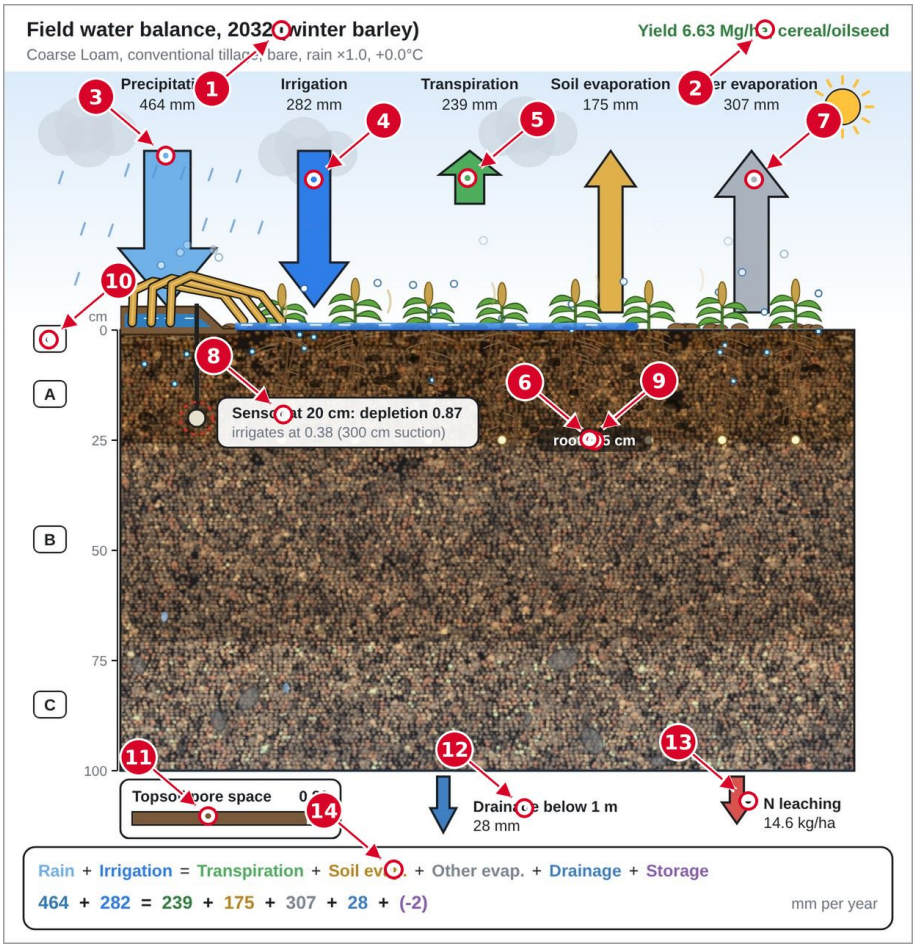


Figure 25.113: Field balance, part 1, playback two thirds of the way (13 Mar 2032 · winter barley).

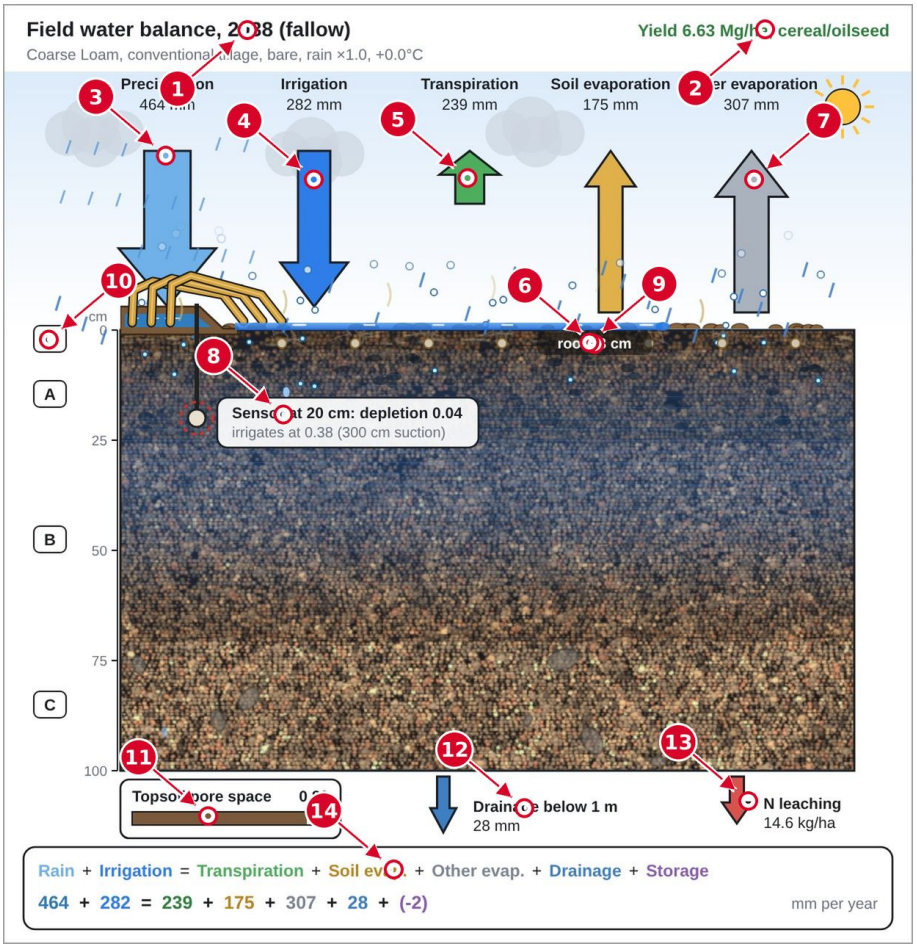


Figure 25.114: Field balance, part 1, playback the end (31 Dec 2038 · fallow).

25.5.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

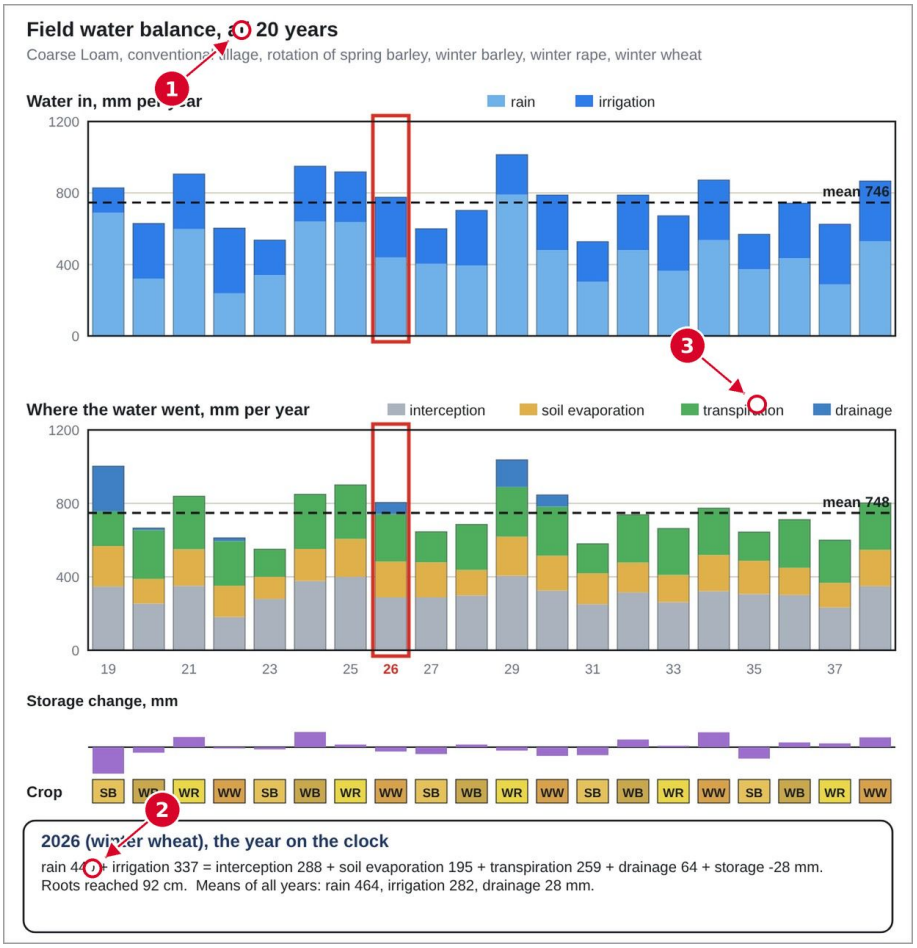


Figure 25.115: Field balance, part 1. The All 20 years button pressed, so the arrows and the budget show the whole twenty-year record in one picture.

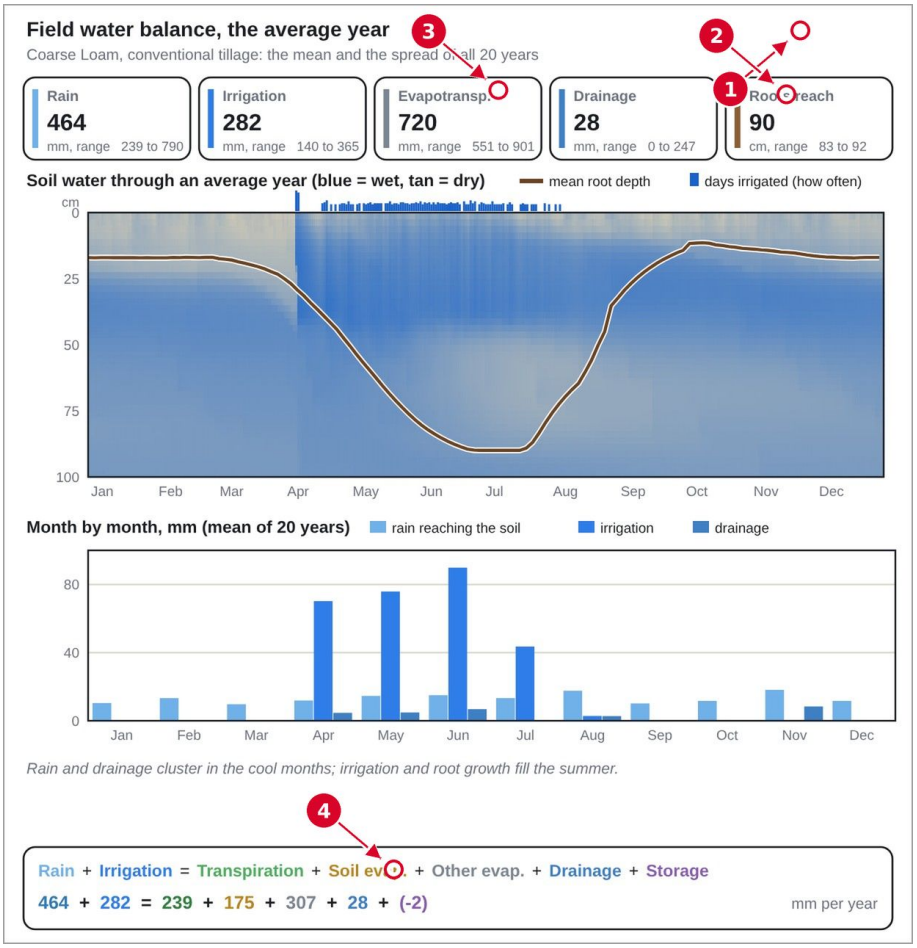


Figure 25.116: Field balance, part 1. The Average year pressed, so every flux is the mean of the twenty simulated years.

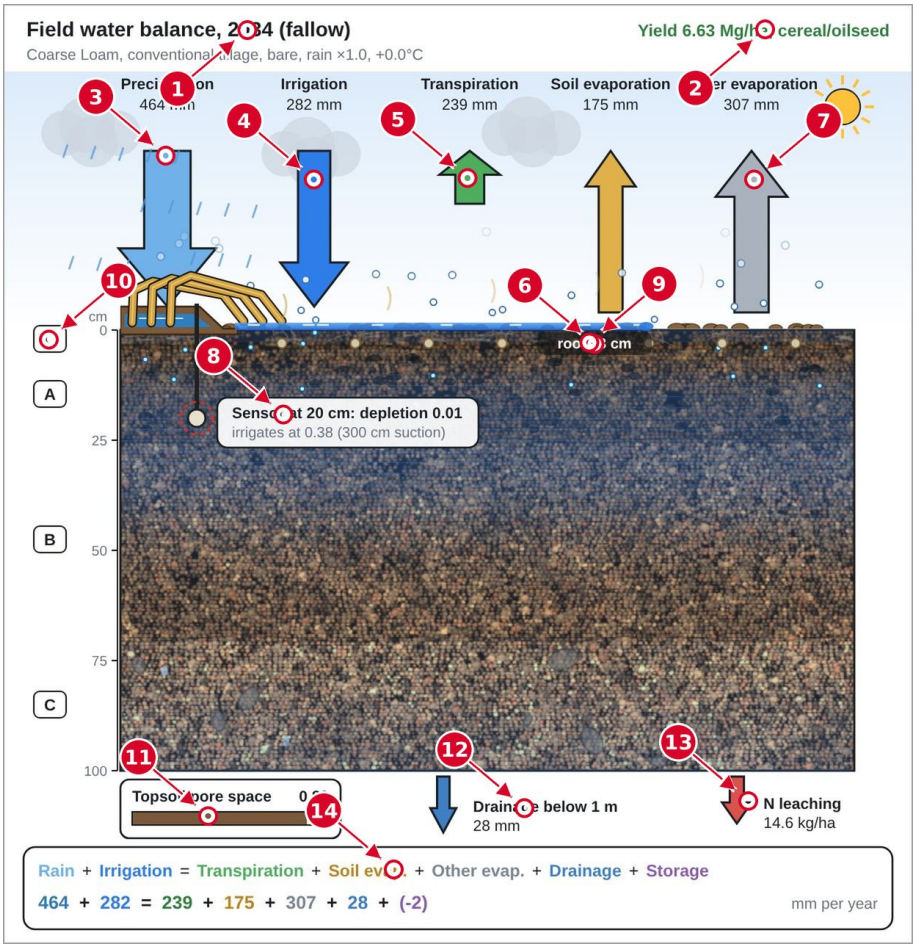


Figure 25.117: Field balance, part 1. The year 2030 pressed, so the picture shows the budget of that single year.

25.6 Soil, plant and air

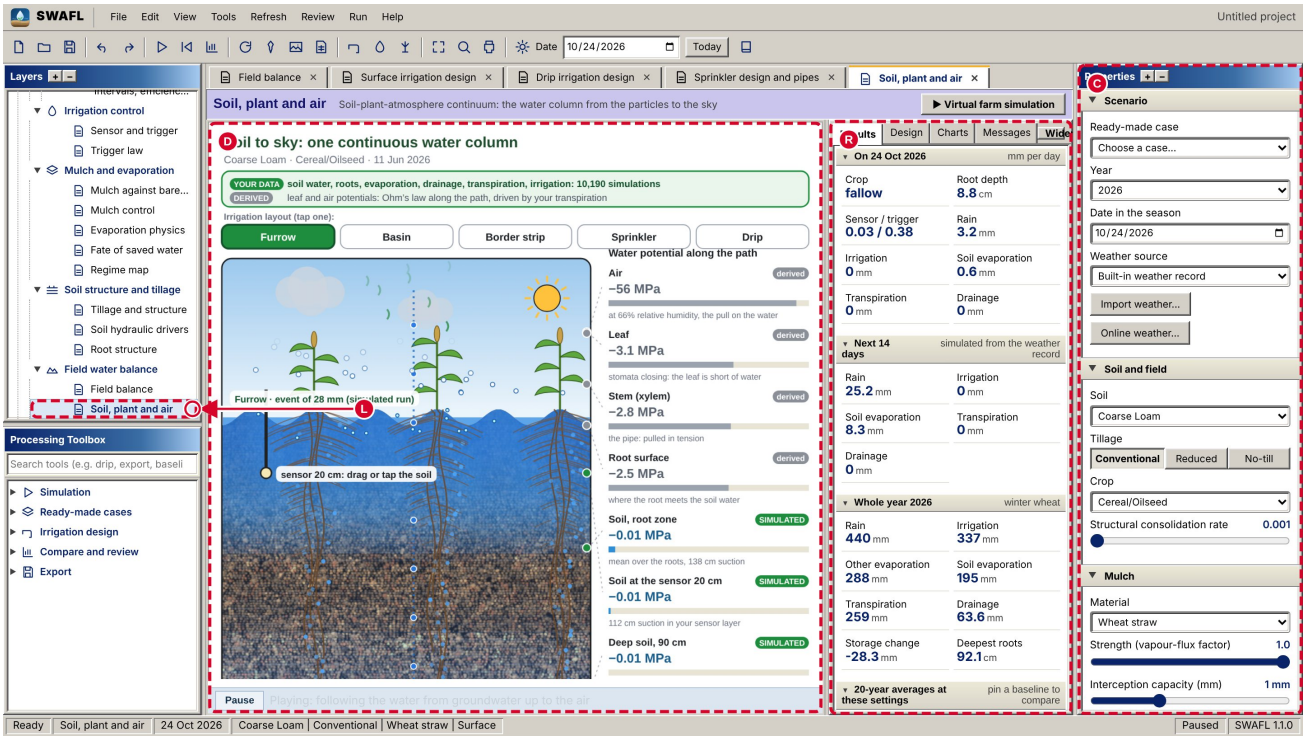


Figure 25.118: Soil, plant and air: the whole workbench.

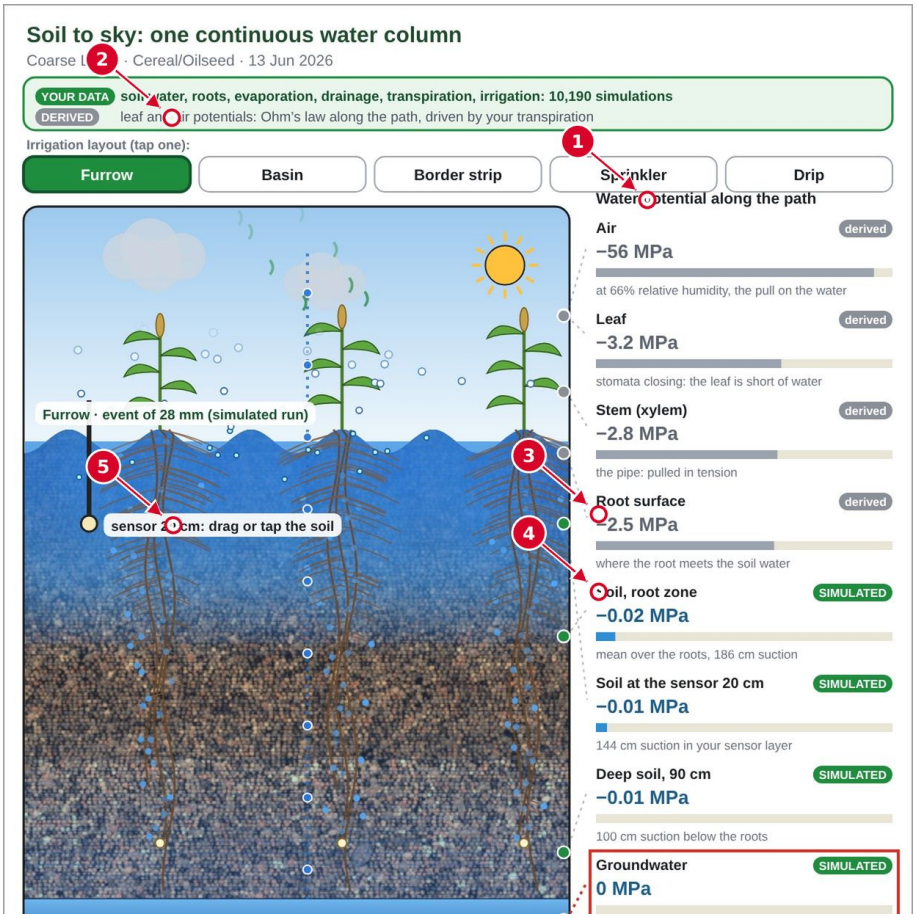


Figure 25.119: Soil, plant and air: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Water potential of the air, psi_air = 135.5 ln RH: Eq. 7.5.
- **2** Leaf potential, psi_leaf: Eq. 7.4.
- **3** Stem and root-surface potentials, psi_stem and psi_root: Eqs. 7.2 and 7.3.
- **4** Root-zone soil potential from the layer suctions: Eq. 7.1 (inverse of Eq. 5.3), with Eqs. 5.1 and 5.2 for the conductivity.
- **5** Sensor at 20 cm: the suction of layer j_s, Eq. 7.1.

Equations behind this screen. Eq. 5.1, Eq. 5.2, Eq. 5.3, Eq. 5.25, Eq. 7.1, Eq. 7.2, Eq. 7.3, Eq. 7.4, Eq. 7.5.

Eq. 5.1. Where explained: chapter "The two computing engines", section "The simulation campaign".

$$S_e(h) = \left[1 + (\alpha|h|)^n\right]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 5.2. Where explained: chapter "The two computing engines", section "The simulation campaign".

$$K_r(S_e) = S_e^{1/2}\left[1 - \left(1 - S_e^{1/m}\right)^m\right]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Eq. 5.3. Where explained: chapter "The two computing engines", section "Plant-available water and depletion".

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.25. Where explained: chapter "The two computing engines", section "Scaling to the annual model".

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

Eq. 7.1. Where explained: chapter "Field water balance layers", section "Governing equations".

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

Provenance (Eq. 7.1): **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

Eq. 7.2. Where explained: chapter "Field water balance layers", section "Governing equations".

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.3. Where explained: chapter "Field water balance layers", section "Governing equations".

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.4. Where explained: chapter "Field water balance layers", section "Governing equations".

$$R_{\text{soil}} = 0.03\ \max\{K_r,\, 0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.5. Where explained: chapter "Field water balance layers", section "Governing equations".

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\,\ln \text{RH} \;\approx\; 135.5\,\ln \text{RH}\ \ \text{MPa} \tag{7.5}$$

Provenance (Eq. 7.5): **[Literature]**: Nobel (2009).

25.6.1 The whole layer, part by part

The picture below is the complete drawing of the layer Soil, plant and air, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

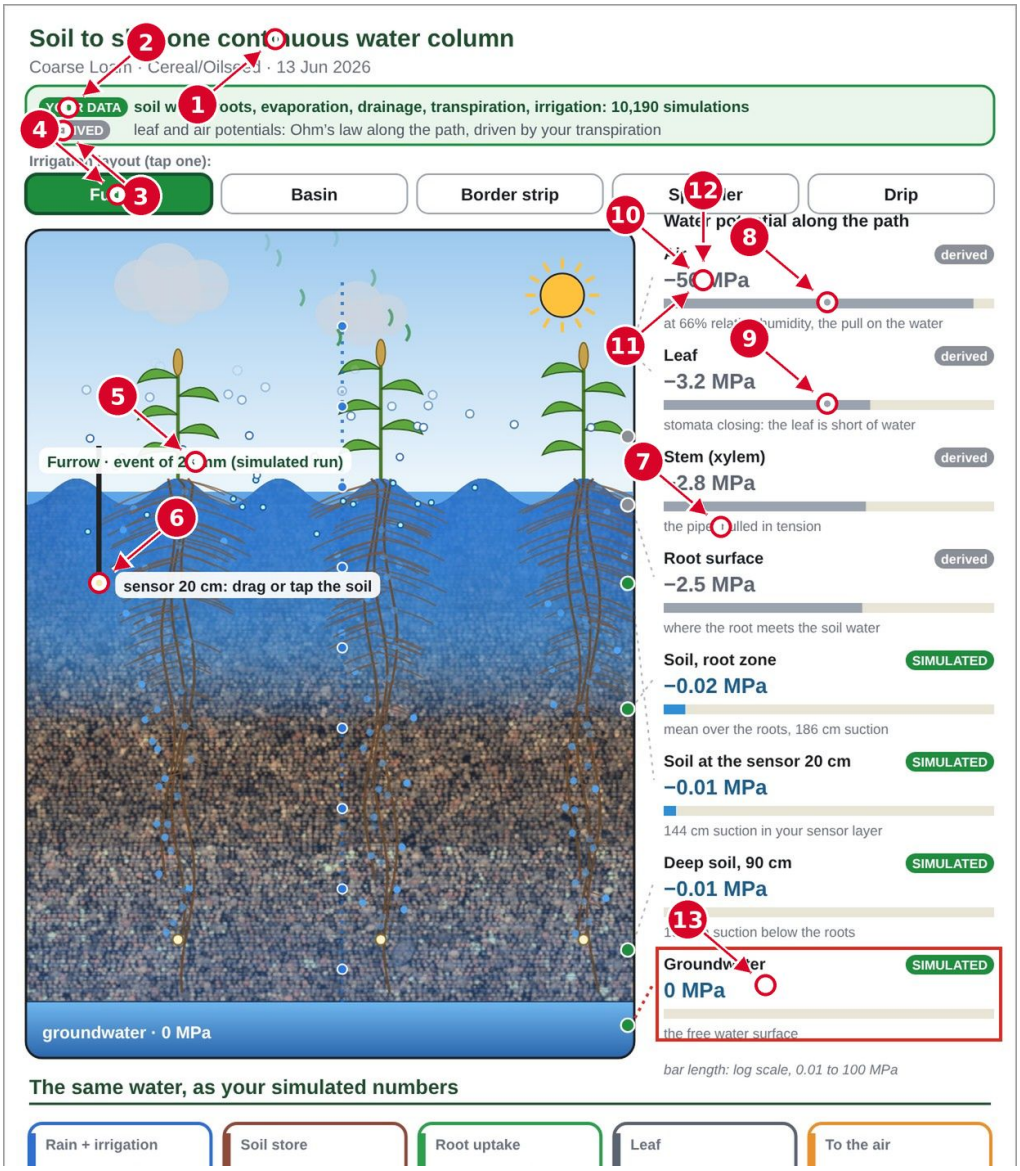


Figure 25.120: Soil, plant and air, part 1 of 3.

What the arrows point to.

- **1** The title and the date of the water column shown, from the groundwater up to the air.
- **2** The SIMULATED badge marks quantities read from the simulations of the ten soils.
- **3** The DERIVED badge marks quantities computed along the path from Ohm's law, the leaf and the air potentials.
- **4** The irrigation layout buttons, here Furrow; the other layouts are Basin, Border strip, Sprinkler and Drip.
- **5** The label of the irrigation event with the depth applied in the simulated run.
- **6** The soil sensor at 20 cm, which can be dragged or tapped to another depth.
- **7** The roots of the crop, with the depth they have reached.

- **8** The potential of the air at the current relative humidity, the strongest pull on the water.
- **9** The potential of the leaf, set by the stomata and the transpiration.
- **10** The potential of the soil in the root zone, mean over the roots.
- **11** The potential of the soil at the sensor depth.
- **12** The potential of the deep soil below the roots.
- **13** The groundwater, the free water surface at zero potential.

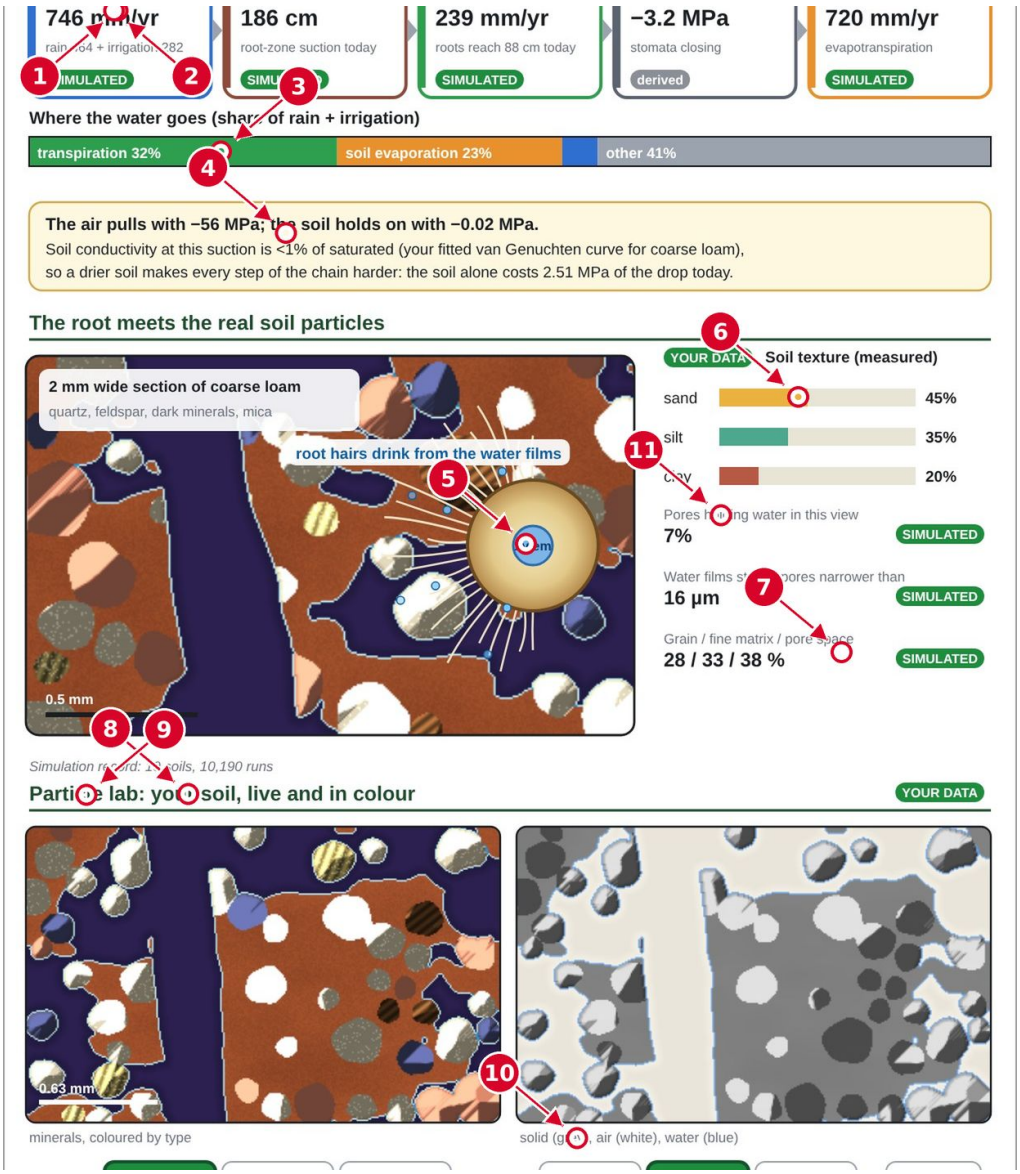


Figure 25.121: Soil, plant and air, part 2 of 3.

What the arrows point to.

- **1** The cards give the same water as numbers, from rain and irrigation to the soil store.
- **2** The root uptake card, the transpiration drawn through the roots in mm per year.
- **3** The partition of rain and irrigation into transpiration, soil evaporation and the other losses.
- **4** The message that compares the pull of the air with the hold of the soil.
- **5** A section of the root surrounded by the real soil particles, with root hairs in the water films.
- **6** The measured texture of the soil, the sand, silt and clay fractions.

- **7** The grain, fine matrix and pore fractions of this section.
- **8** The particle lab, the soil shown live and in colour.
- **9** The left view colours the grains by mineral type.
- **10** The right view shows the solid in grey, the air in white and the water in blue.
- **11** The suction, the water content and the pores holding water at the chosen state.

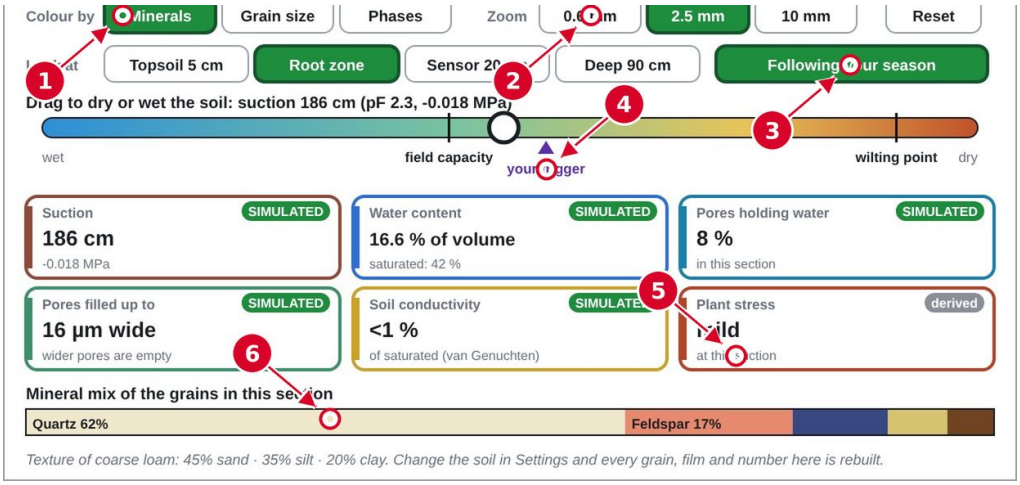


Figure 25.122: Soil, plant and air, part 3 of 3.

What the arrows point to.

- **1** The Colour by buttons choose minerals, grain size or phases.
- **2** The Zoom buttons set the width of the section to 0.6, 2.5 or 10 mm.
- **3** The Look at buttons choose the depth, or follow your season.
- **4** The value box dries or wets the soil, with the field capacity, your trigger and the wilting point marked.
- **5** The plant stress flag, past or inside your trigger at this suction.
- **6** The mineral mix of the grains in this section.

25.6.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

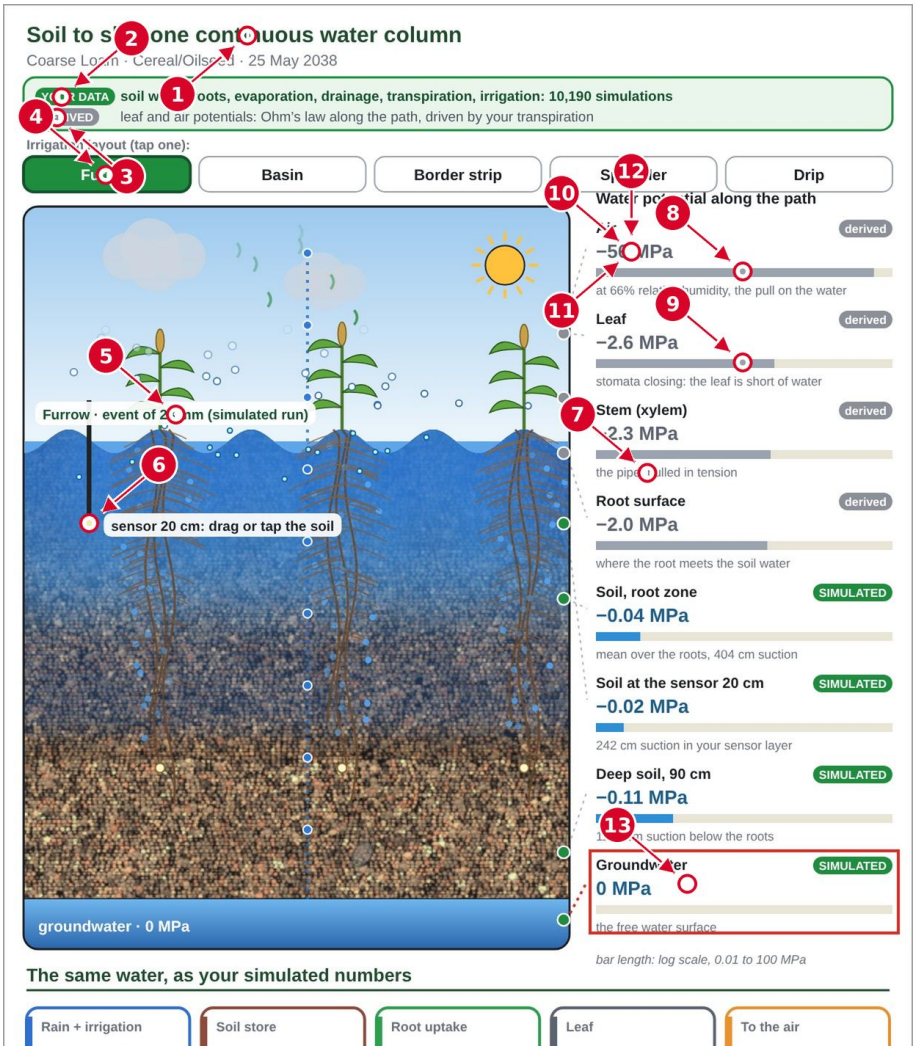


Figure 25.123: Soil, plant and air, part 1, playback a third of the way (Paused: following the water from groundwater up to the air).

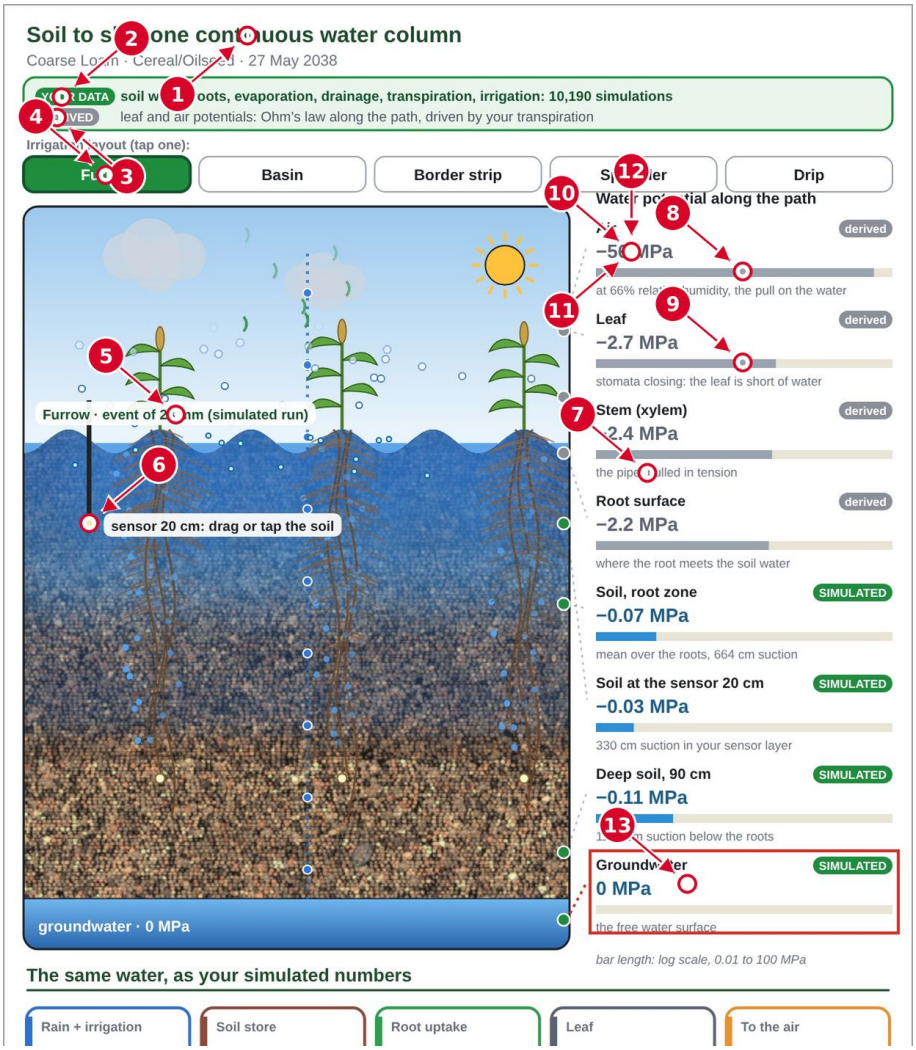


Figure 25.124: Soil, plant and air, part 1, playback two thirds of the way (Paused: following the water from groundwater up to the air).

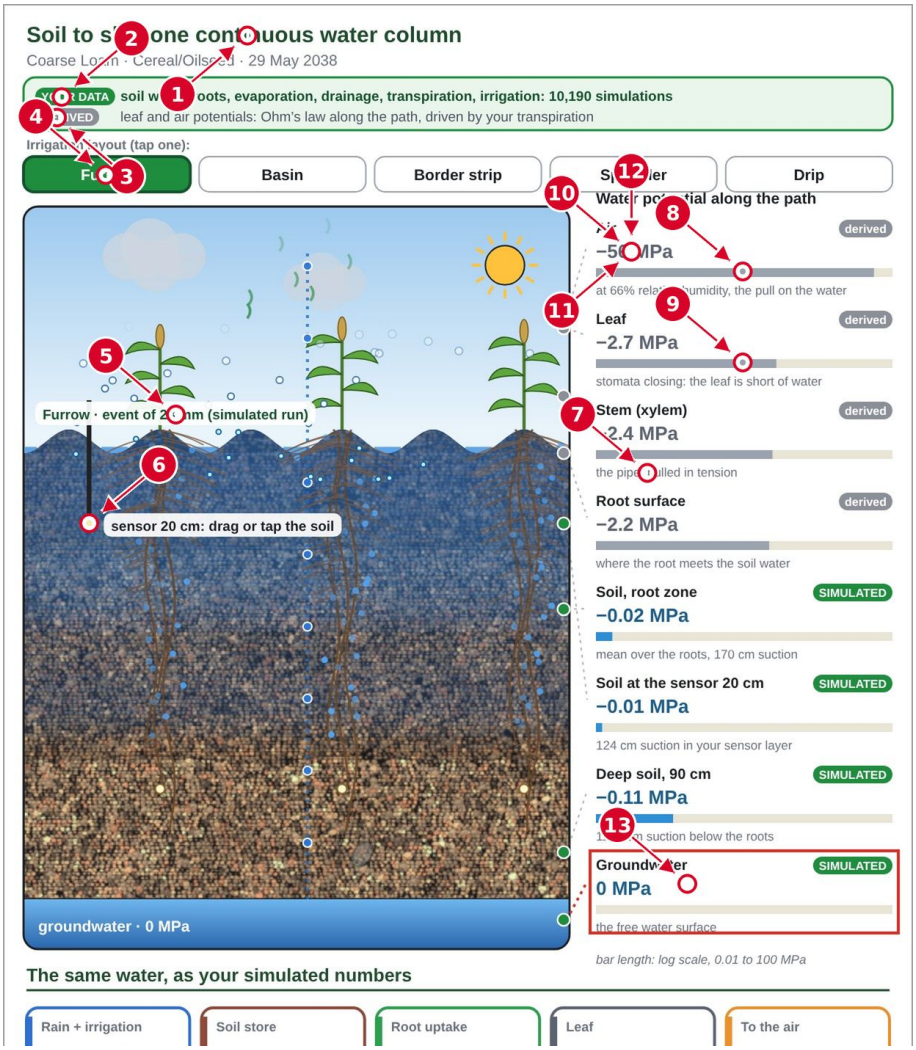


Figure 25.125: Soil, plant and air, part 1, playback the end (Paused: following the water from groundwater up to the air).

25.6.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

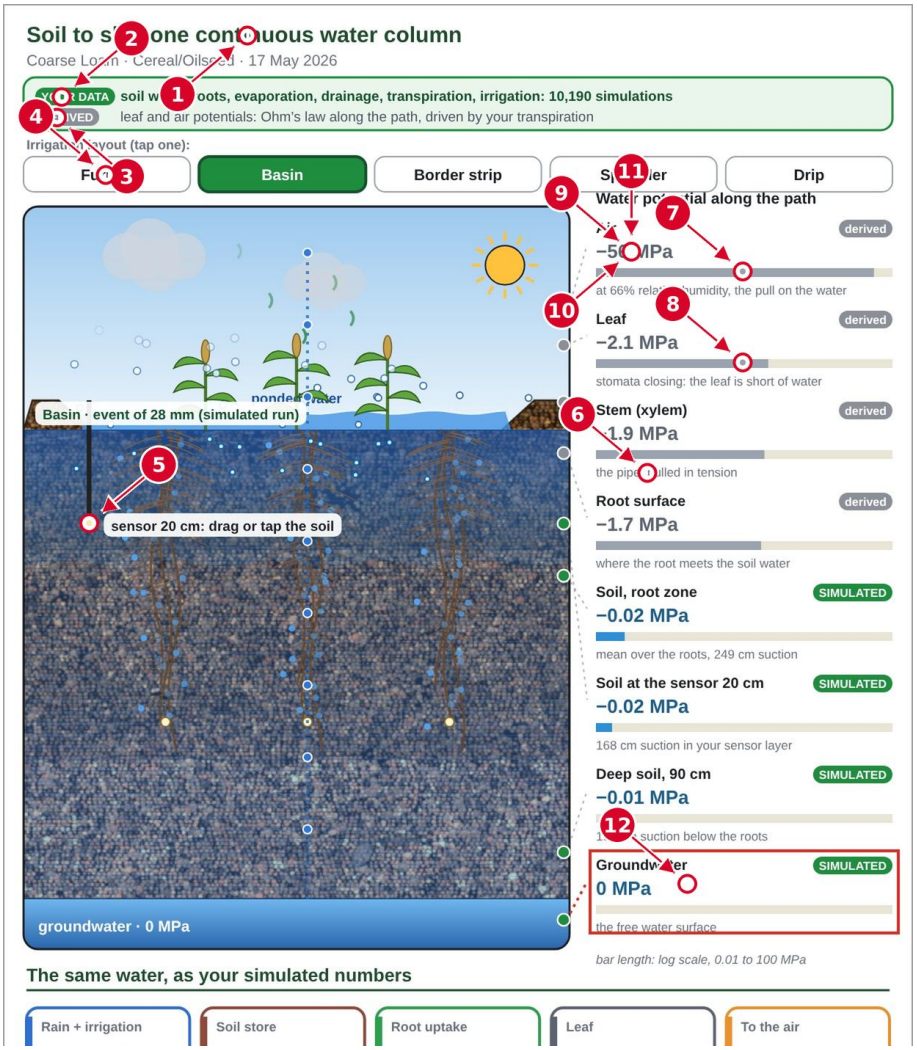


Figure 25.126: Soil, plant and air, part 1. The Basin layout chosen, so the water ponds over the whole level plot.

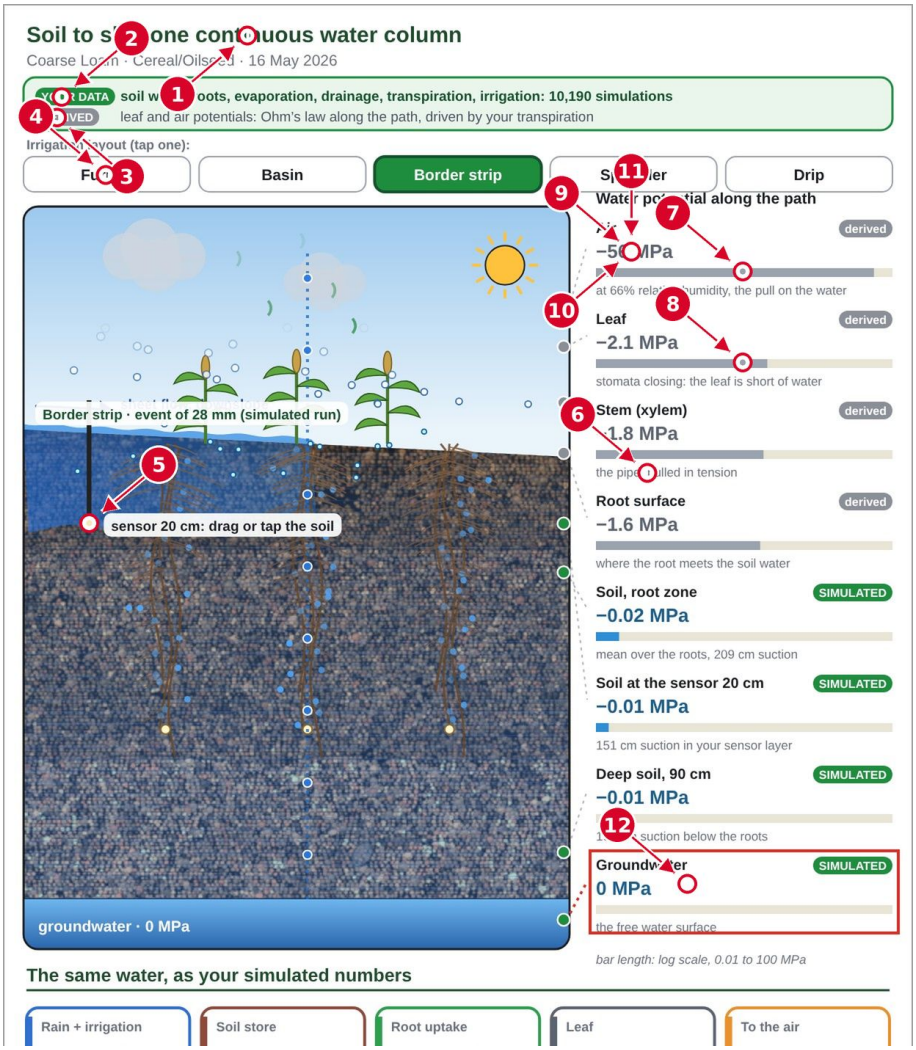


Figure 25.127: Soil, plant and air, part 1. The Border strip layout chosen.

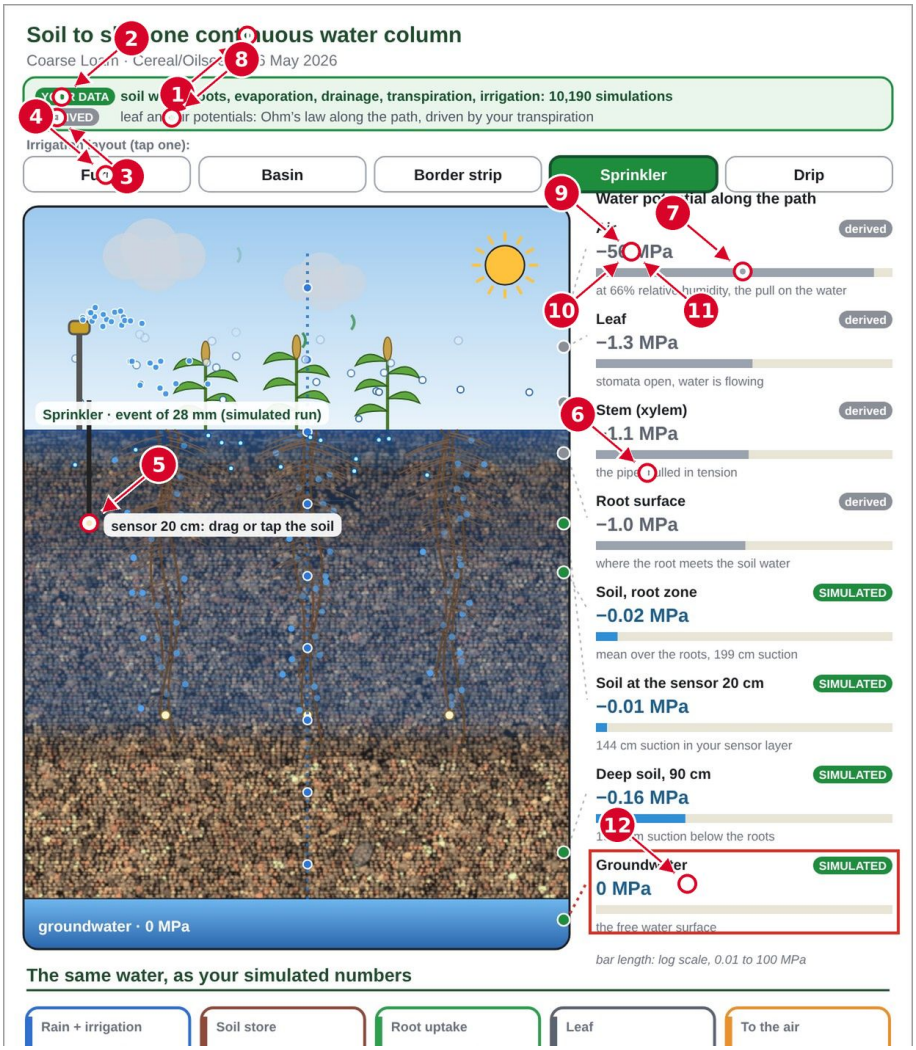


Figure 25.128: Soil, plant and air, part 1. The Sprinkler layout chosen, so the water arrives as spray over the plants.

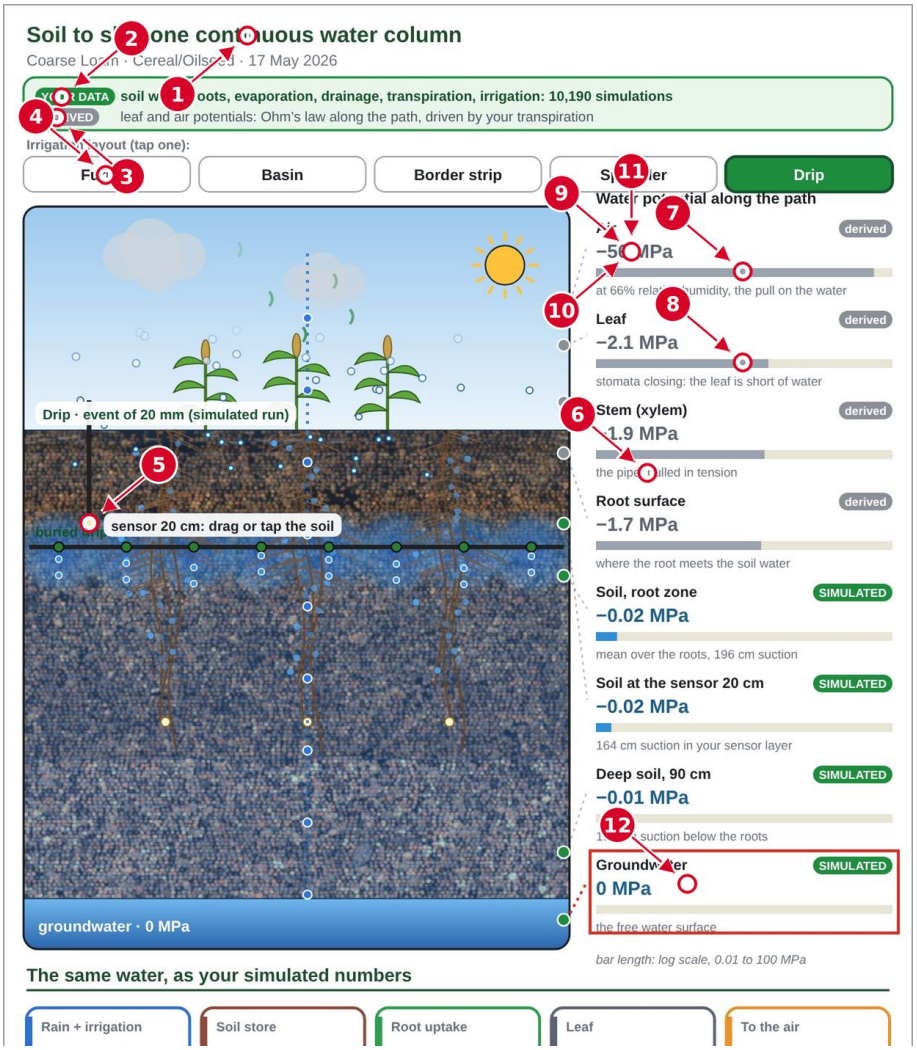


Figure 25.129: Soil, plant and air, part 1. The Drip layout chosen, so the water arrives at the plant row only.

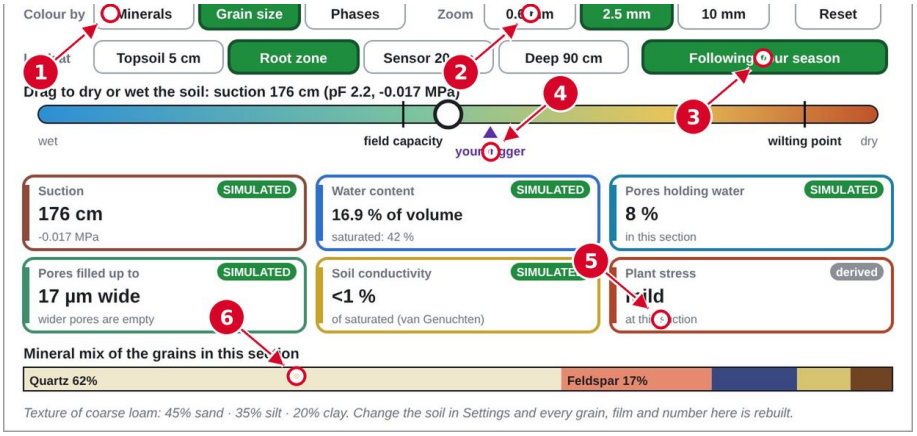


Figure 25.130: Soil, plant and air, part 3. The Colour by Grain size button pressed, so the grains of the section are coloured by their size.

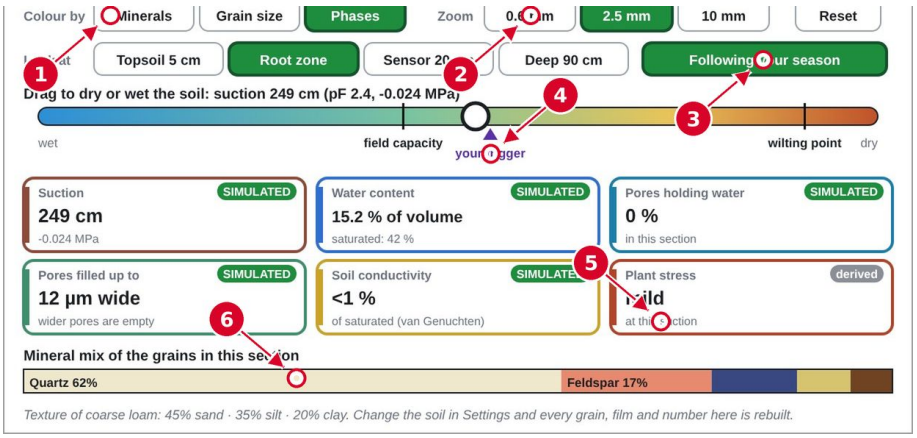


Figure 25.131: Soil, plant and air, part 3. The Colour by Phases button pressed, so the section shows the solid, the air and the water.

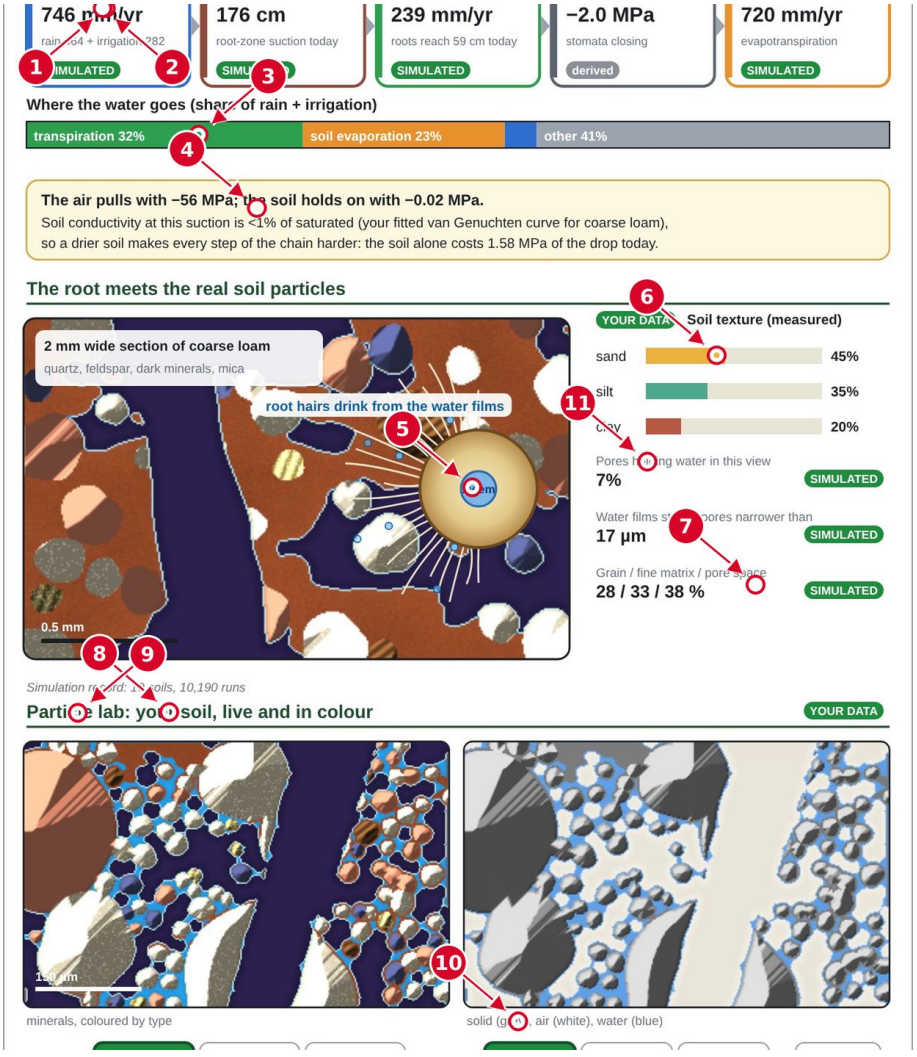


Figure 25.132: Soil, plant and air, part 2. The Zoom 0.6 mm button pressed, so the section shows silt, clay and the water films.

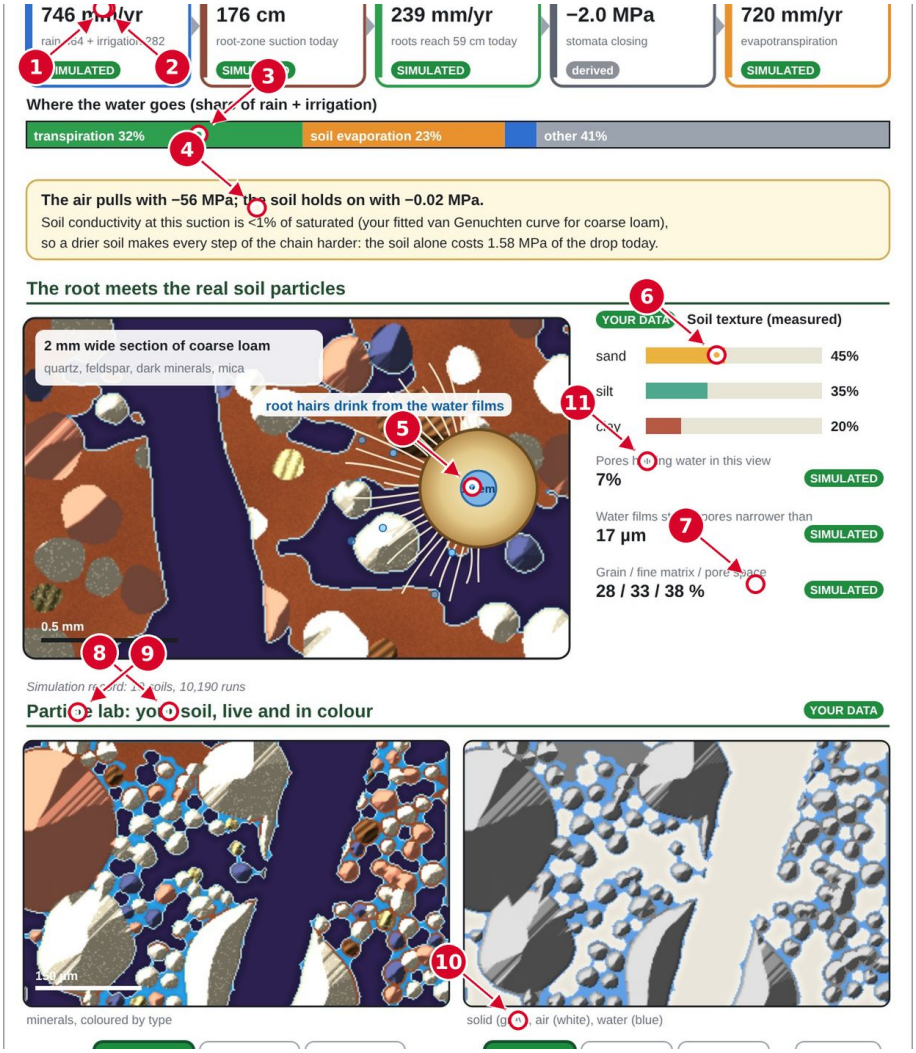


Figure 25.133: Soil, plant and air, part 2. The Zoom 10 mm button pressed, so the section shows aggregates and cracks.

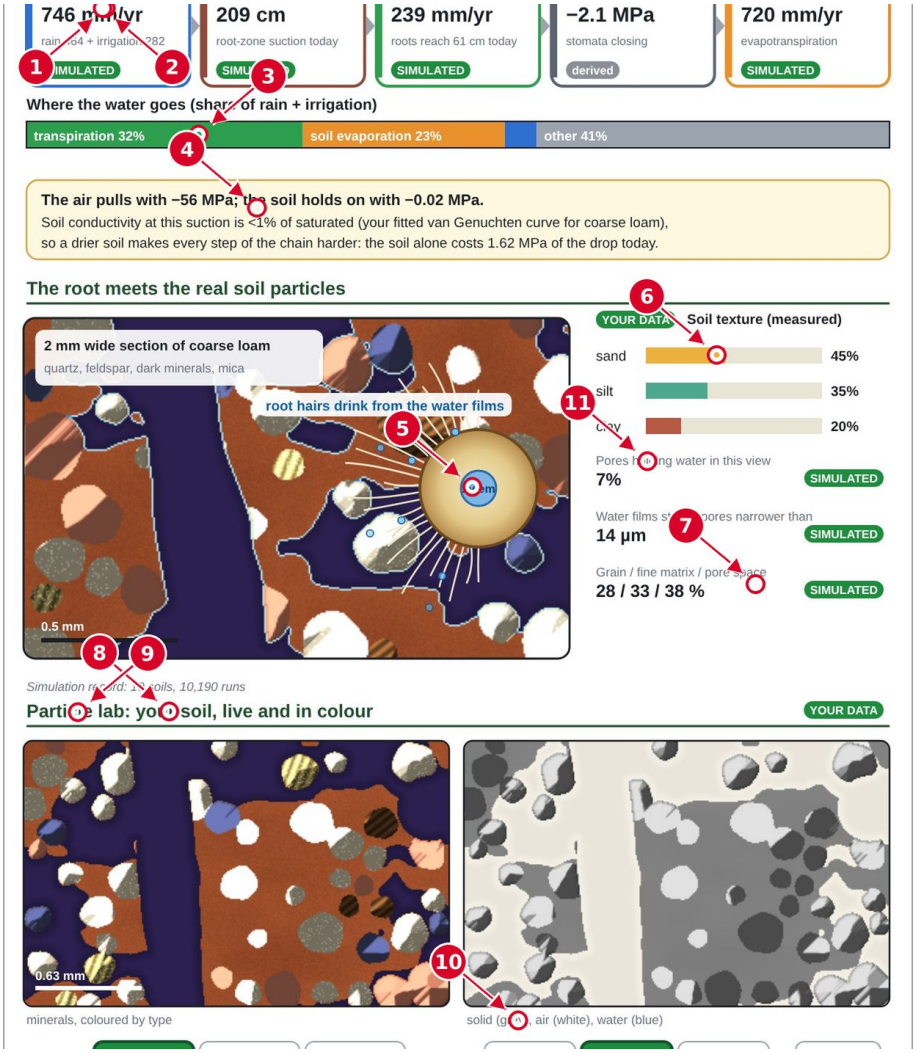


Figure 25.134: Soil, plant and air, part 2. The Look at Topsoil 5 cm button pressed, so the section and the readouts refer to the topsoil.

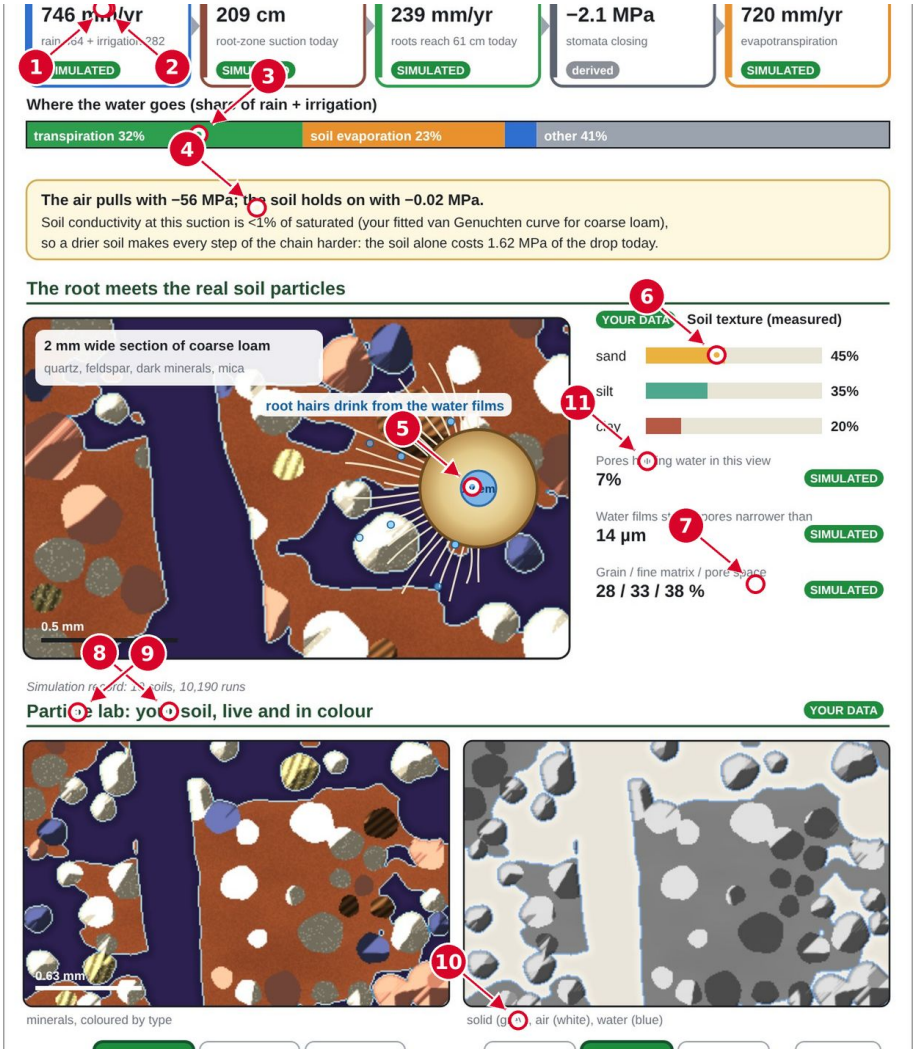


Figure 25.135: Soil, plant and air, part 2. The Look at Sensor 20 cm button pressed.

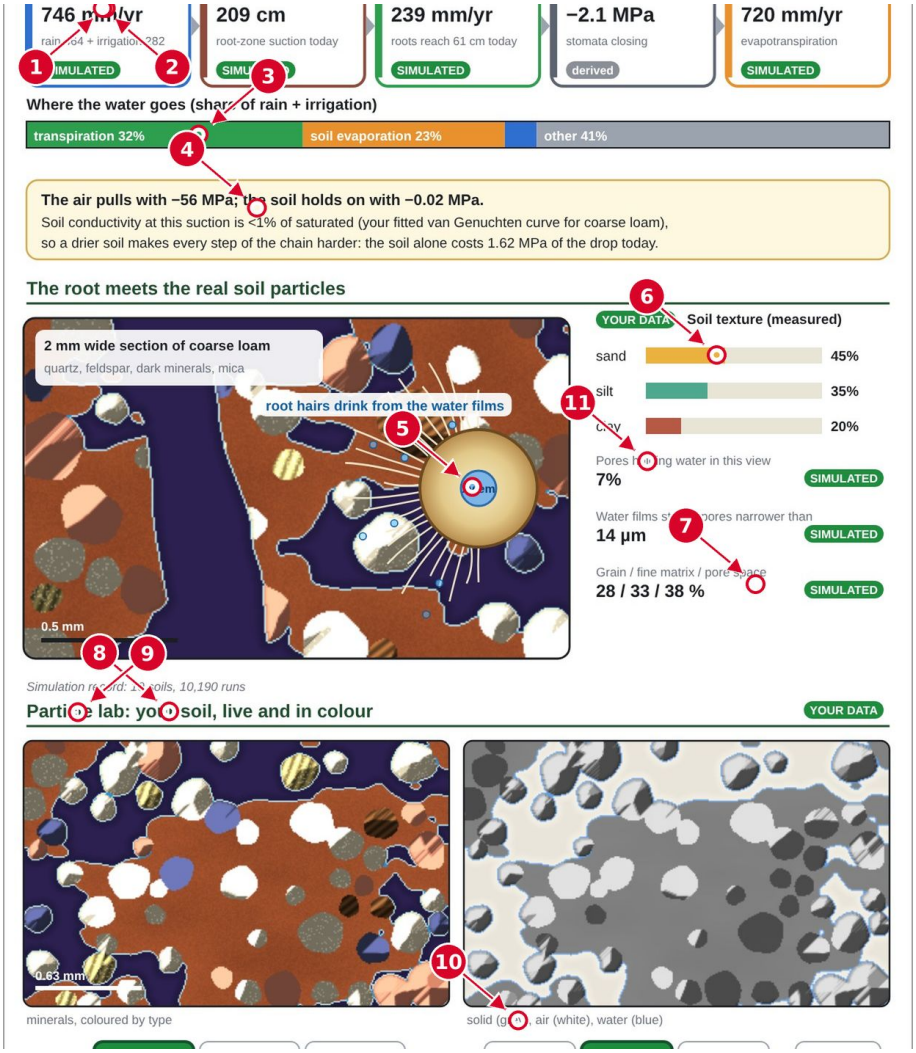


Figure 25.136: Soil, plant and air, part 2. The Look at Deep 90 cm button pressed, so the section and the readouts refer to the deep soil.

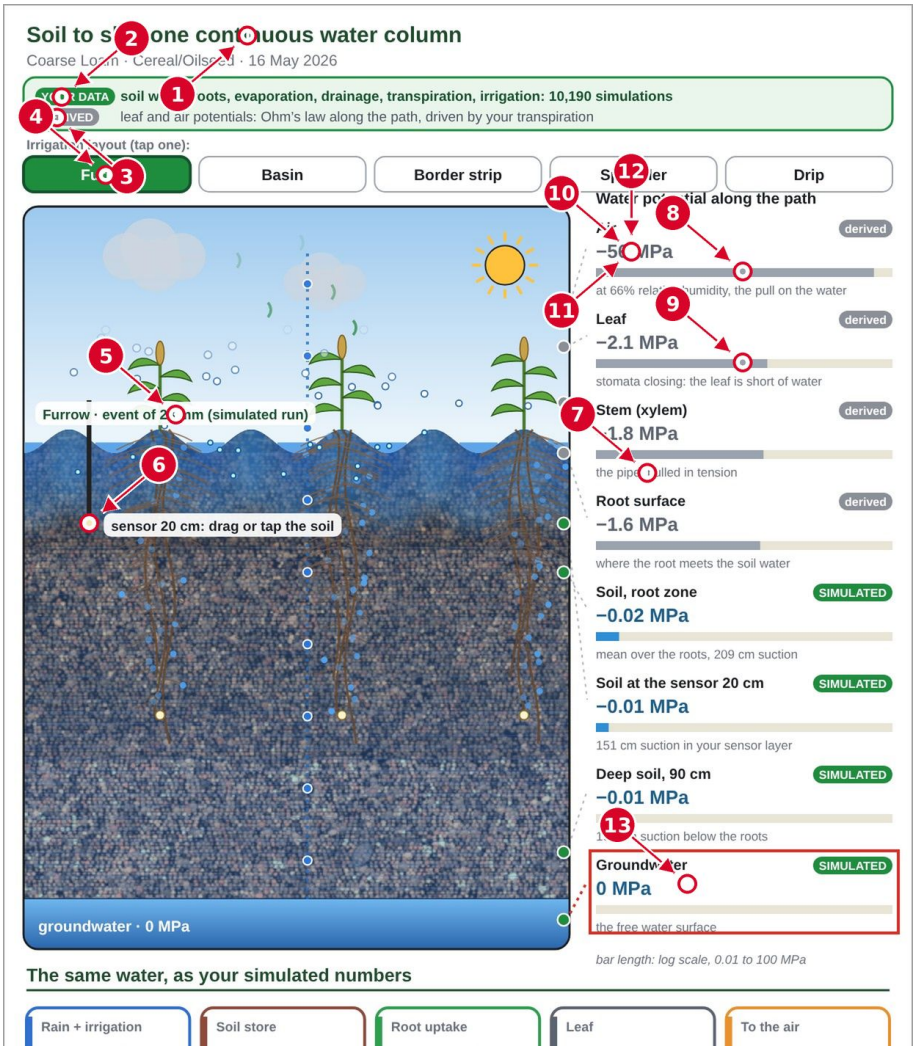


Figure 25.137: Soil, plant and air, part 1. The suction value box dragged to the wet end, so the soil is shown near saturation.

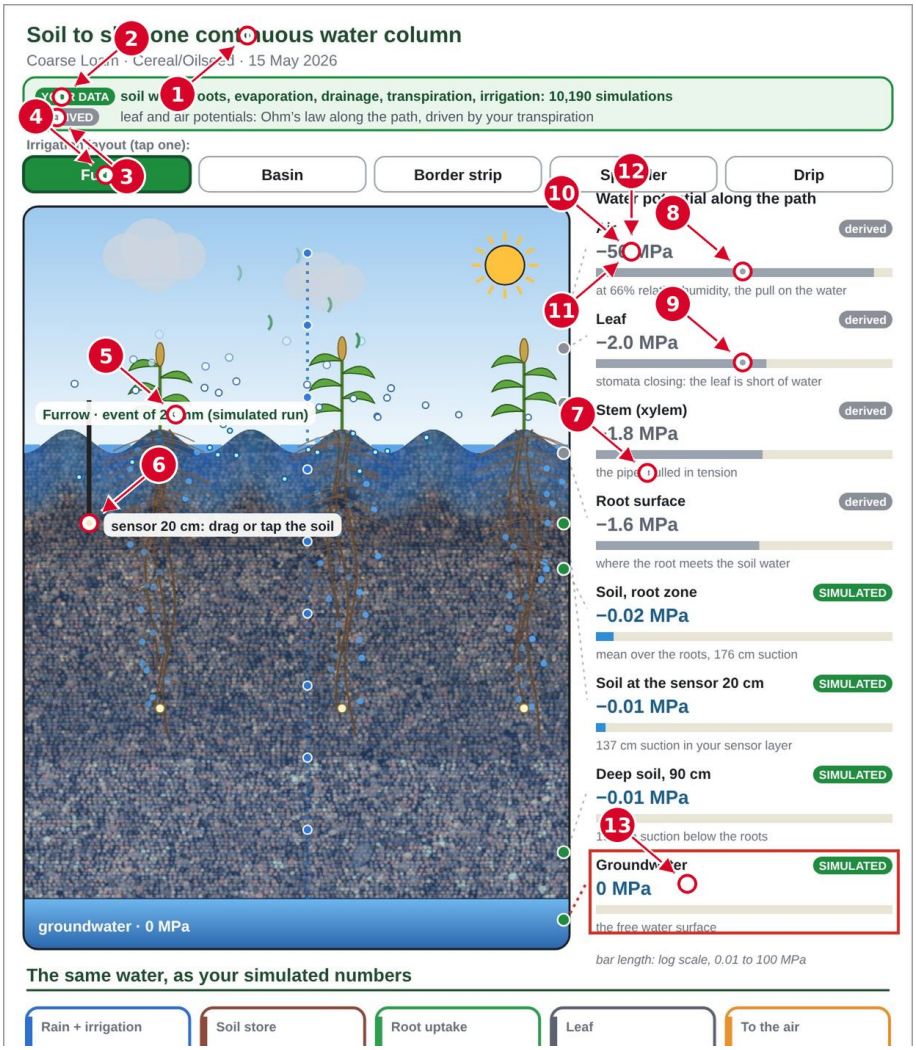


Figure 25.138: Soil, plant and air, part 1. The suction value box dragged to the dry end, so the soil is shown near the wilting point.

25.7 Profile evolution

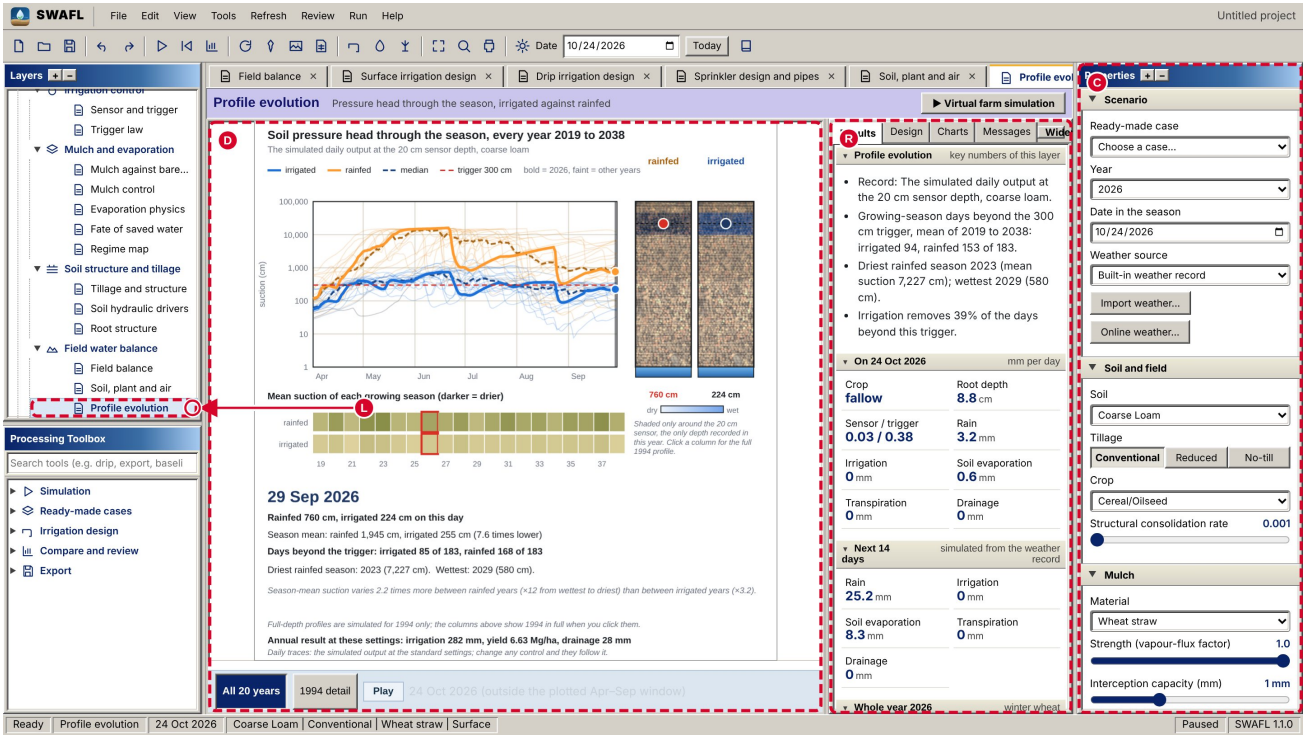


Figure 25.139: Profile evolution: the whole workbench.

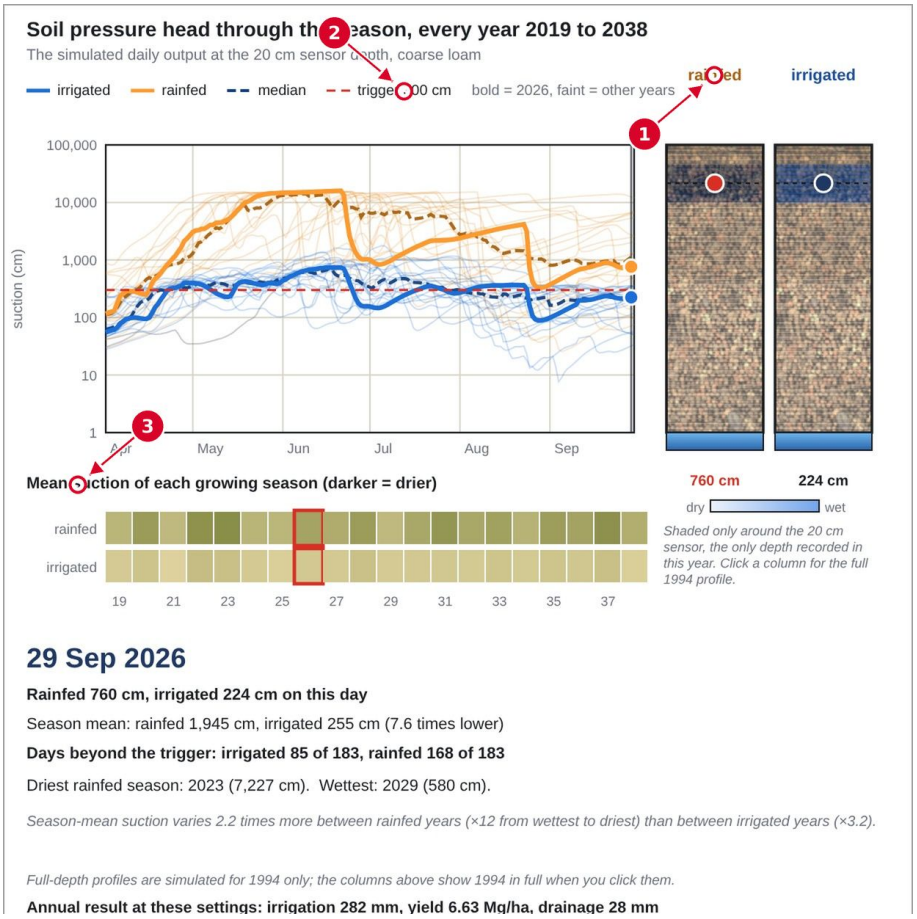


Figure 25.140: Profile evolution: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Daily suction at the 20 cm sensor: stored Daisy series (data-driven, no equation).
- **2** Trigger h_T (dashed red line), set by the value box; the share of the root zone beyond it is Eq. 7.6.
- **3** Heat strip: geometric mean suction of each growing season, text under Eq. 7.6.

Equations behind this screen. Eq. 5.15, Eq. 5.17, Eq. 7.6.

Eq. 5.15. Where explained: chapter "The two computing engines", section "Rainfall generator".

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0,1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.17. Where explained: chapter "The two computing engines", section "Infiltration, redistribution and drainage".

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

Eq. 7.6. Where explained: chapter "Field water balance layers", section "How it is generated".

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

Provenance (Eq. 7.6): **[Engineered]**: indicator defined for the workbench.

25.7.1 The whole layer, part by part

The picture below is the complete drawing of the layer Profile evolution, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

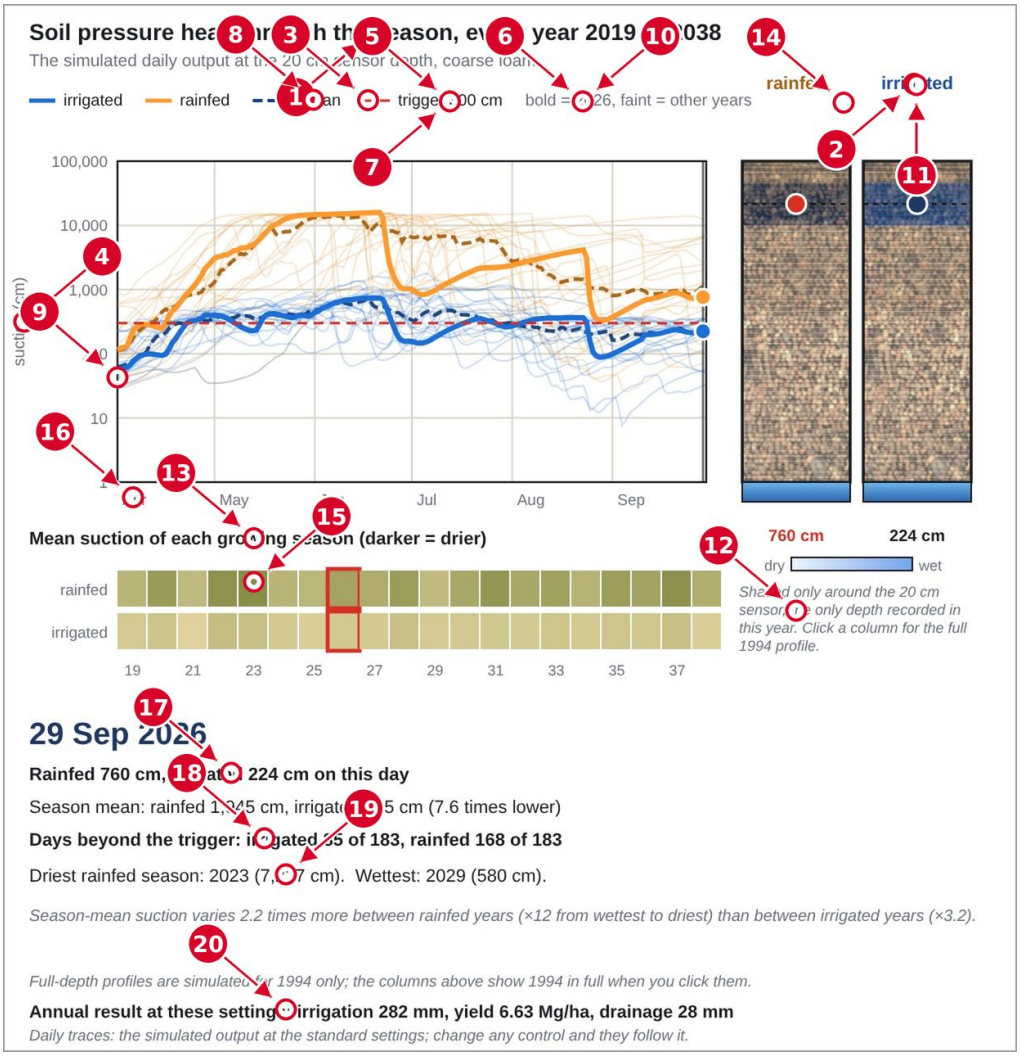


Figure 25.141: Profile evolution, part 1 of 1.

What the arrows point to.

- **1** The title, the daily suction at the 20 cm sensor for every year of the record.
- **2** The legend of the traces, irrigated in blue and rainfed in orange.
- **3** The trigger legend, the dashed red line at the suction that starts an irrigation.
- **4** The suction axis in cm of water on a log scale.
- **5** The bold blue trace, the irrigated season of the year shown.

- **6** The bold orange trace, the rainfed season of the same year.
- **7** The trigger line at 300 cm.
- **8** The faint traces, the other years of the record.
- **9** The marker of the day shown, moving along the year as the animation plays.
- **10** The rainfed soil column, shaded by the suction around the sensor.
- **11** The irrigated soil column.
- **12** The sensor ring at 20 cm, the only depth recorded in this year.
- **13** The strip of the mean suction of every growing season, darker for drier.
- **14** The selected year in the strip; clicking a column selects that year.
- **15** The dark cells are the dry rainfed seasons, far darker than the irrigated row.
- **16** The date of the day shown.
- **17** The suction on this day for the rainfed and the irrigated soil.
- **18** The days beyond the trigger in the growing season, irrigated against rainfed.
- **19** The driest and the wettest rainfed season of the record.
- **20** The annual result at the settings in force, the irrigation, the yield and the drainage.

25.7.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

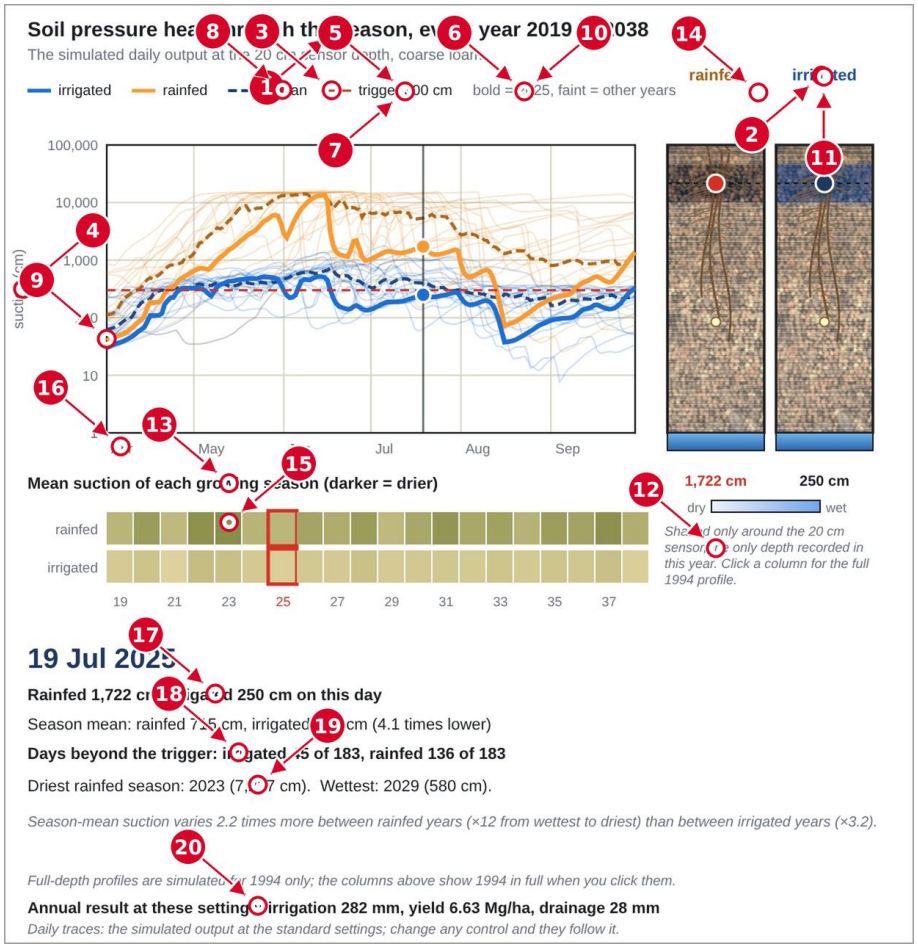


Figure 25.142: Profile evolution, part 1, playback a third of the way (31 Dec 2038 (outside the plotted Apr–Sep window)).

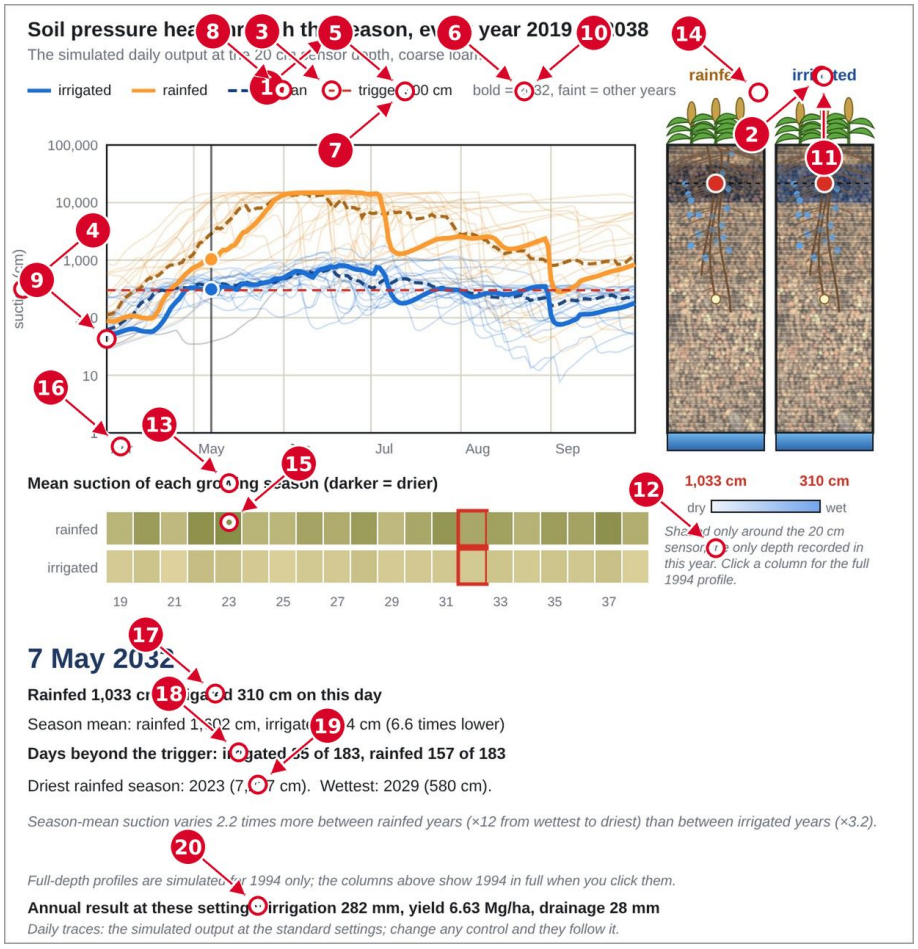


Figure 25.143: Profile evolution, part 1, playback two thirds of the way (31 Dec 2038 (outside the plotted Apr–Sep window)).

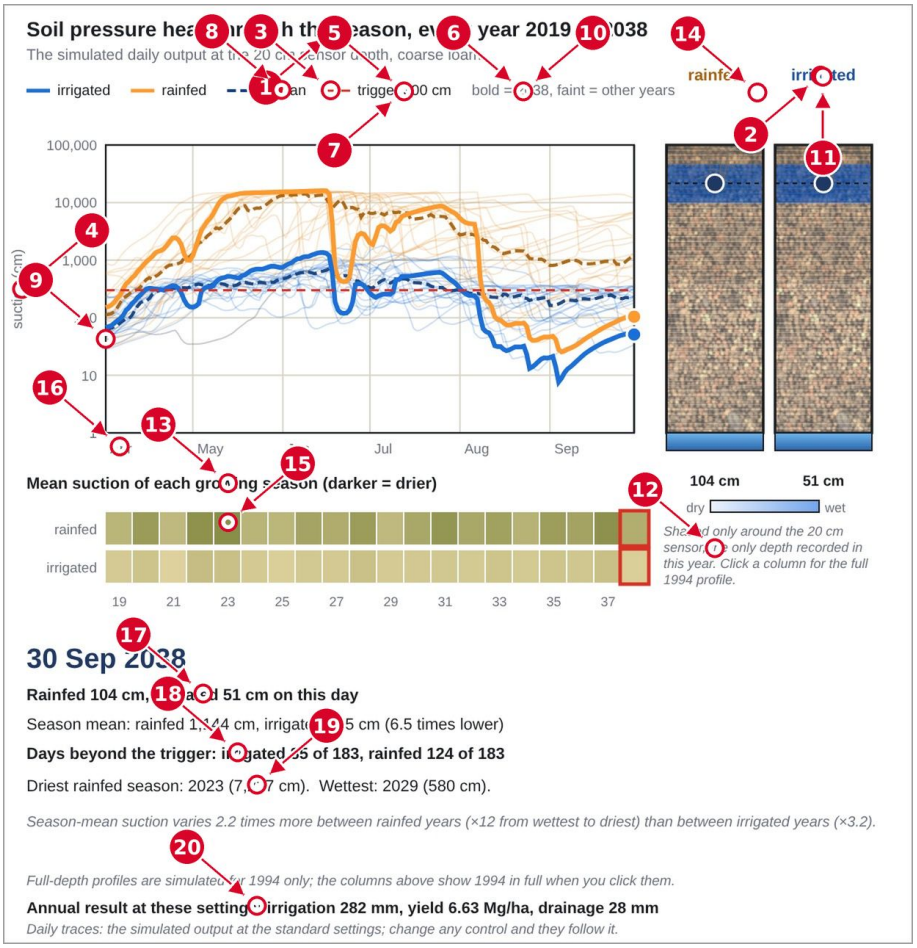


Figure 25.144: Profile evolution, part 1, playback the end (31 Dec 2038 (outside the plotted Apr–Sep window)).

25.7.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

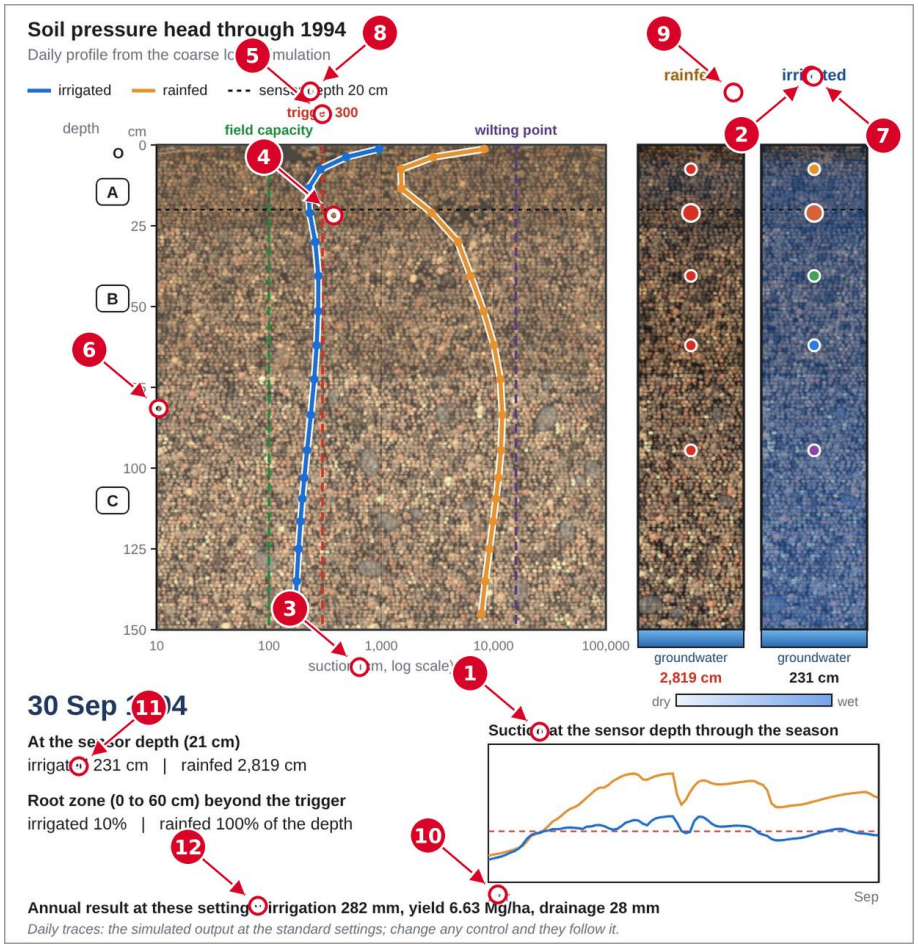


Figure 25.145: Profile evolution, part 1. The 1994 detail button pressed, so the chart shows the daily record of the year 1994 at full depth.

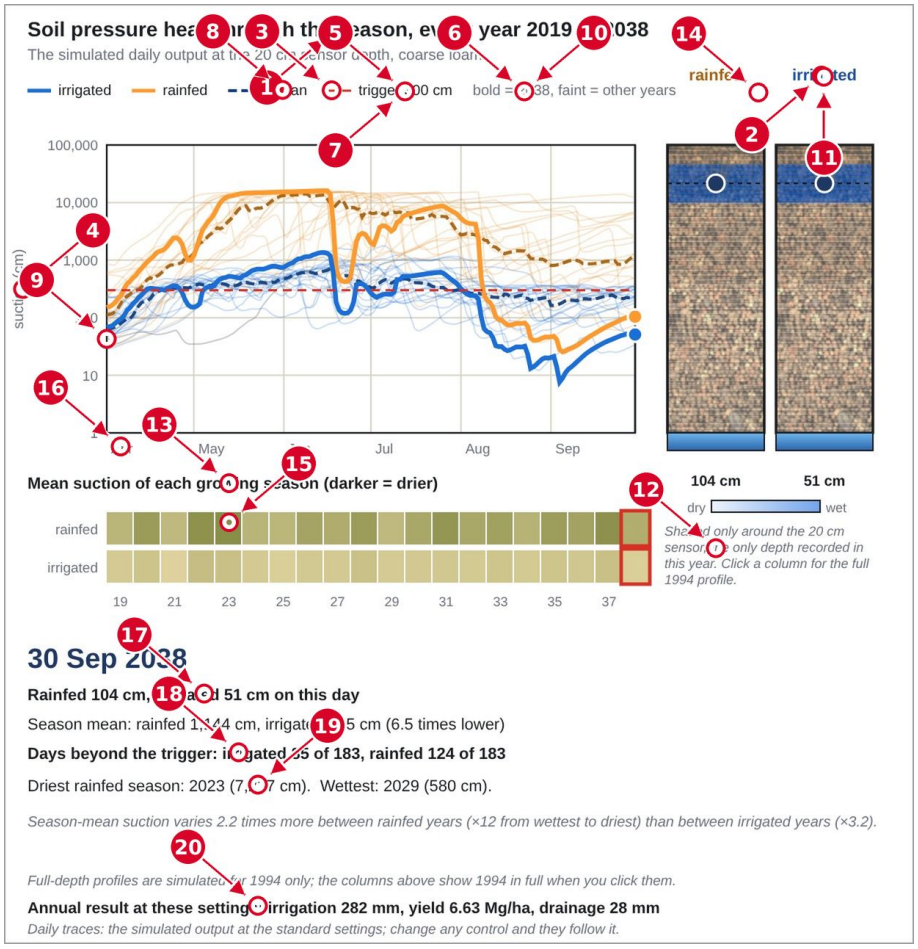


Figure 25.146: Profile evolution, part 1. A column of the season strip clicked, so the chart and the columns show the selected year.

25.8 Sensor and trigger

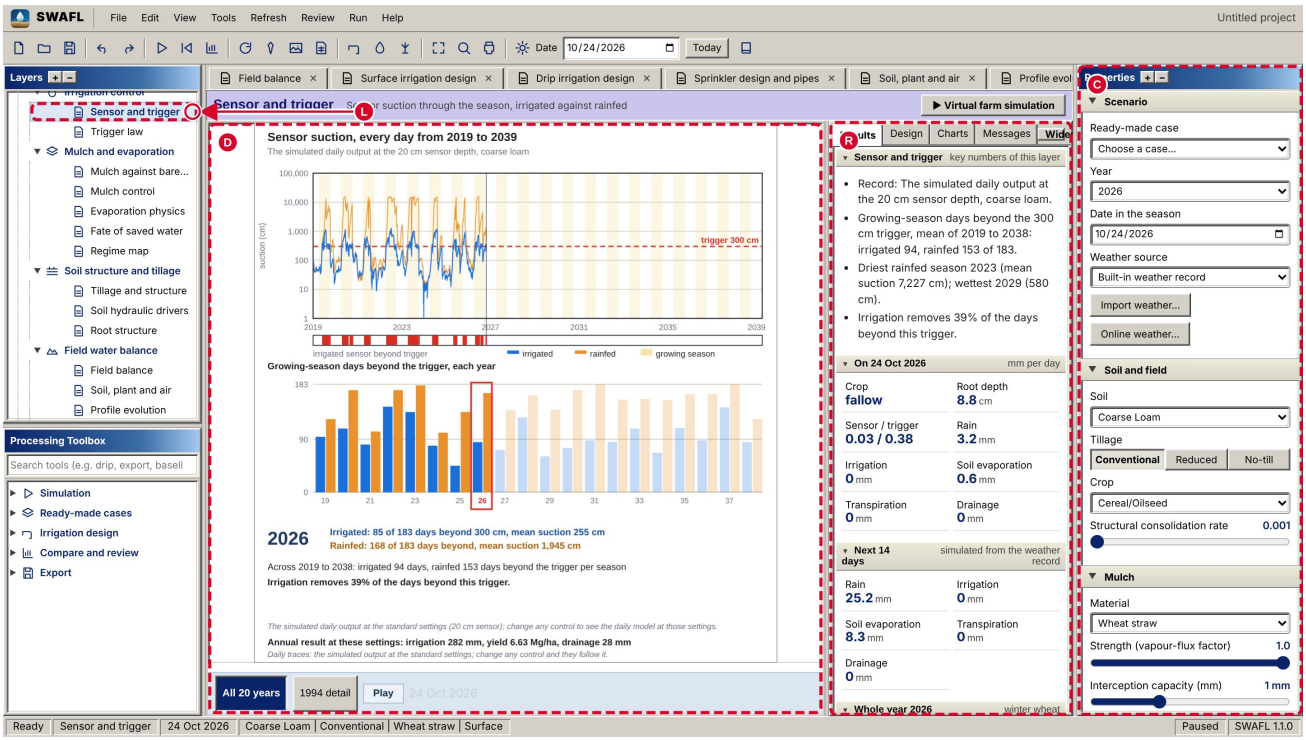


Figure 25.147: Sensor and trigger: the whole workbench.

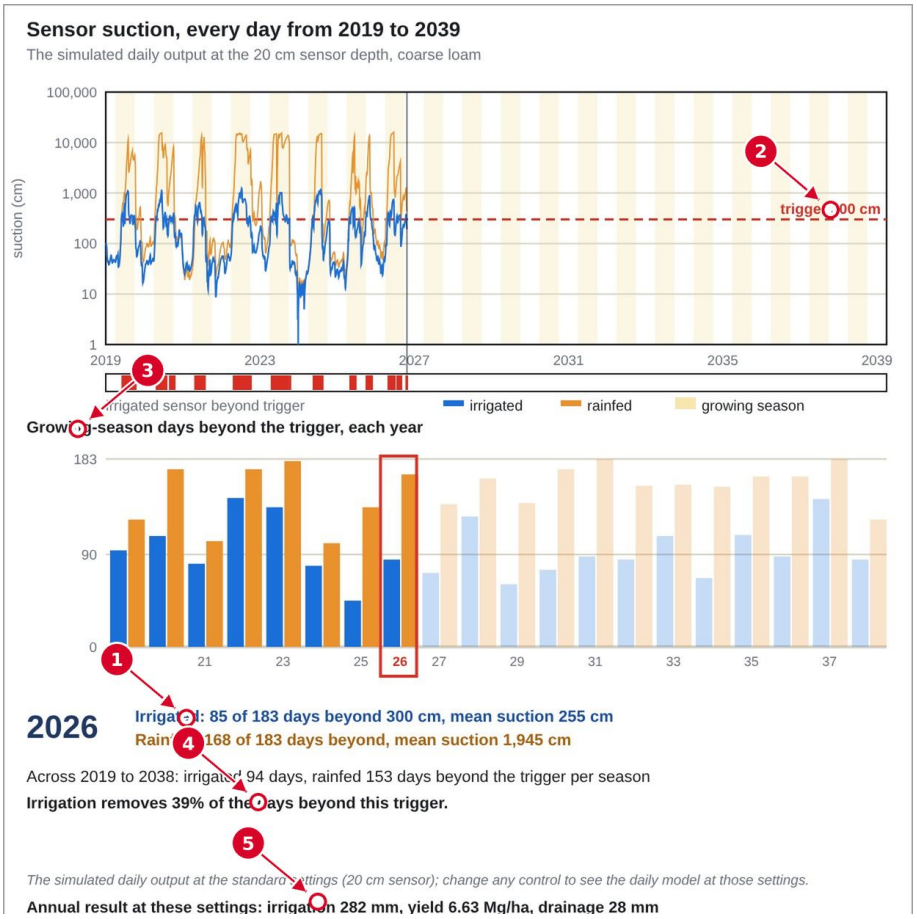


Figure 25.148: Sensor and trigger: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Daily suction at 20 cm, irrigated (blue) and rainfed (orange): stored Daisy series (data-driven).
- **2** Trigger h_T (red dashed line), set in Properties; days beyond it are counted by Eq. 6.1.
- **3** Growing-season days beyond the trigger for each year, N_y(h_T): Eq. 6.1.
- **4** "Irrigation removes 39 % of the days": X = 100(1 - N_irr / N_rainfed), Eq. 6.2.
- **5** Annual result at these settings (irrigation, yield, drainage): Engine 1, Eqs. 5.4 to 5.12.

Equations behind this screen. Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 6.1, Eq. 6.2.

Eq. 5.4. Where explained: chapter "The two computing engines", section "The meta-model".

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".

$$Q \leftarrow Q\,\text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 6.1. Where explained: chapter "Irrigation control layers", section "How the numbers are generated".

$$N_y(h_T) = \#\{\, d \in \text{growing season of year } y : h_d > h_T \,\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.

Eq. 6.2. Where explained: chapter "Irrigation control layers", section "How the numbers are generated".

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_y N_y. \tag{6.2}$$

Provenance (Eq. 6.2): **[Data-driven]**: counts over the simulated daily series.

25.8.1 The whole layer, part by part

The picture below is the complete drawing of the layer Sensor and trigger, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

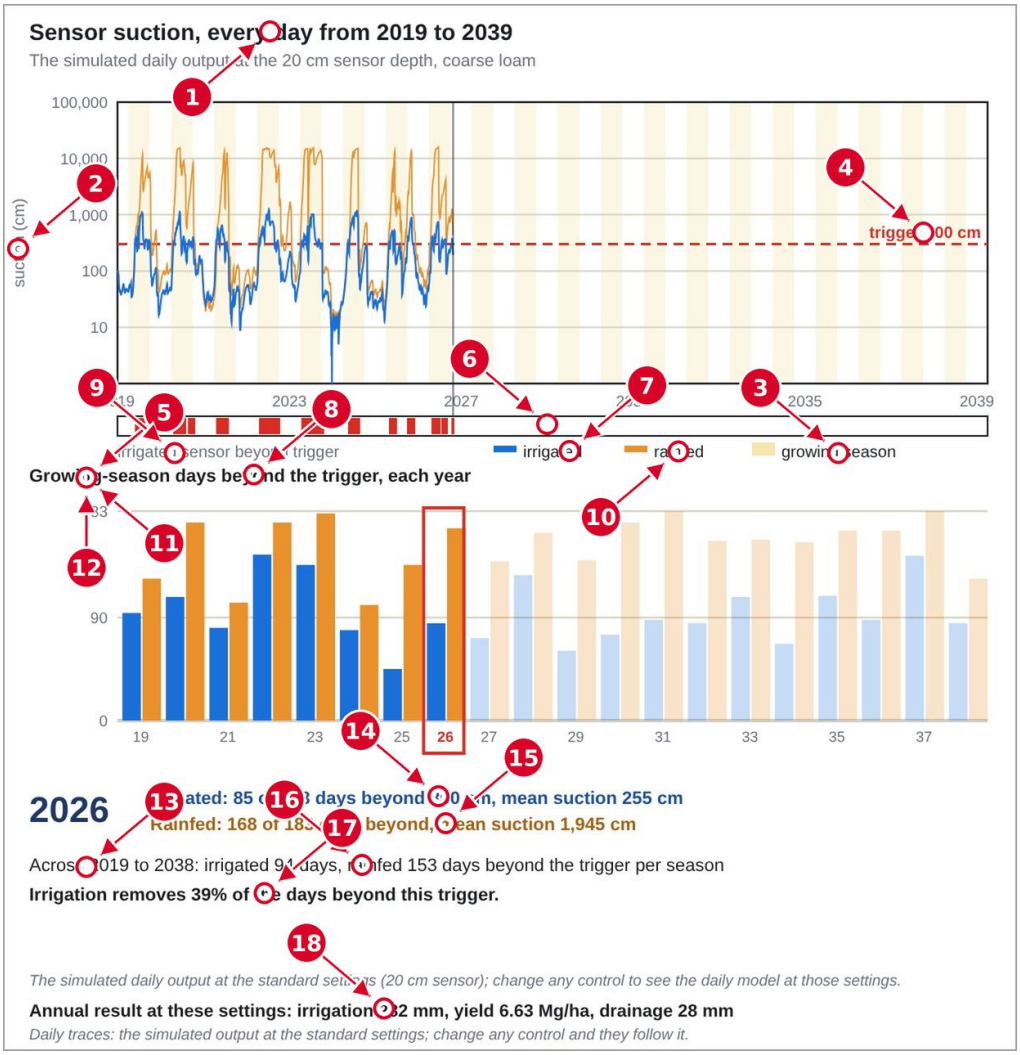


Figure 25.149: Sensor and trigger, part 1 of 1.

What the arrows point to.

- **1** The title, the daily sensor suction for every day of the twenty years.
- **2** The suction axis in cm on a log scale.
- **3** The shaded bands are the growing seasons.
- **4** The trigger line, the suction at which the sensor calls for irrigation.
- **5** The time axis in years.
- **6** The strip that marks the days on which the irrigated sensor is beyond the trigger.
- **7** The legend, irrigated in blue, rainfed in orange and the growing season in tan.
- **8** The chart of the growing-season days beyond the trigger, one pair of bars per year.
- **9** The irrigated bar of the selected year.
- **10** The rainfed bar of the selected year.
- **11** The faint bars are the other years.
- **12** The scale of the days, from 0 to the 183 days of the season.
- **13** The year selected.
- **14** The irrigated days beyond the trigger and the mean suction of the season.
- **15** The rainfed days beyond the trigger and the mean suction of the season.
- **16** The mean over the whole record, irrigated against rainfed.
- **17** The share of the days beyond the trigger that irrigation removes.
- **18** The annual result at the settings in force.

25.8.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

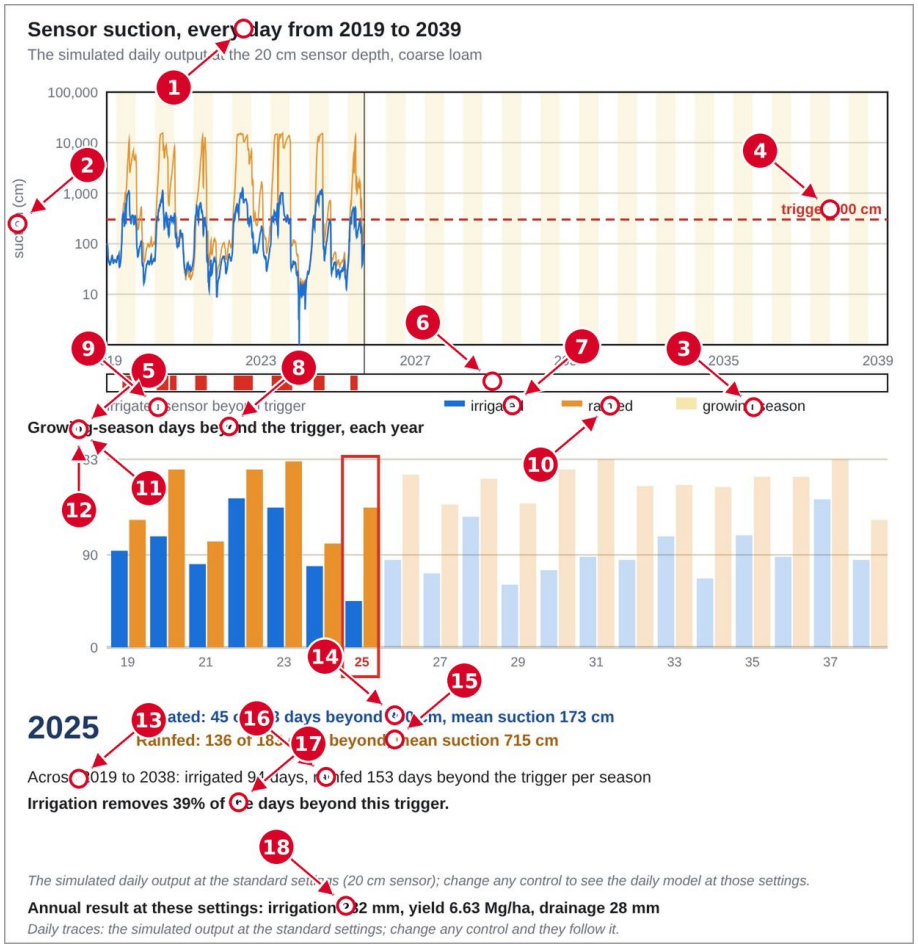


Figure 25.150: Sensor and trigger, part 1, playback a third of the way (7 Sep 2025).

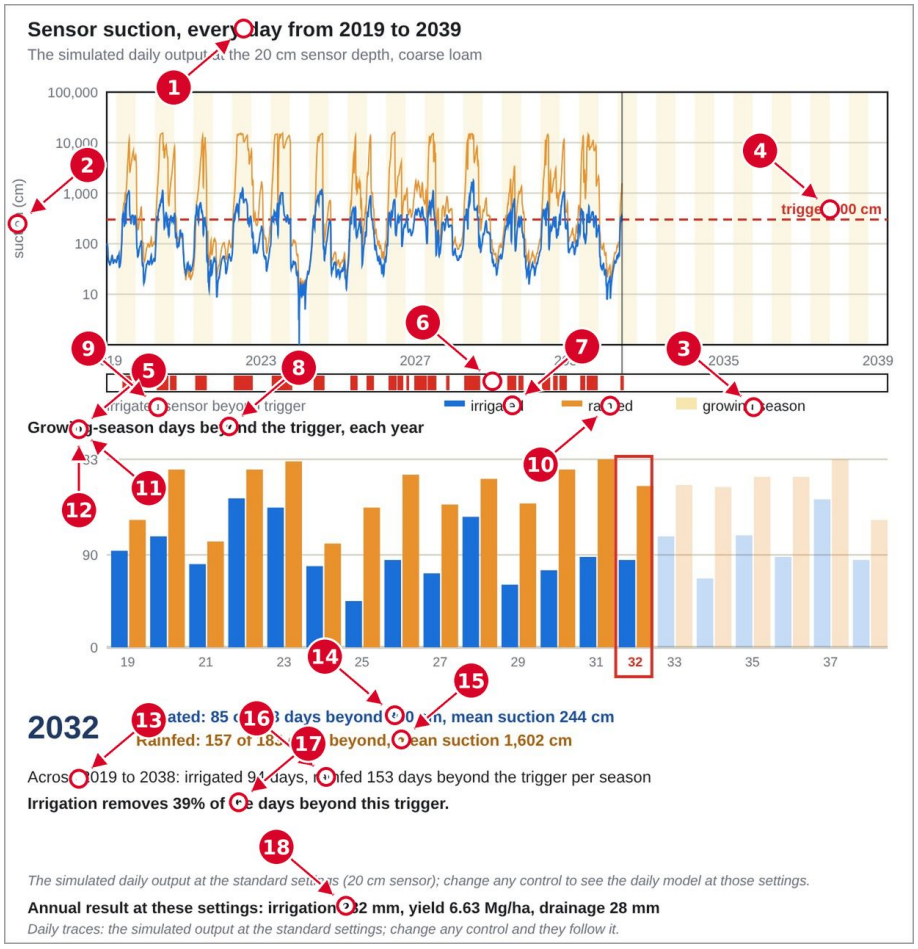


Figure 25.151: Sensor and trigger, part 1, playback two thirds of the way (13 May 2032).

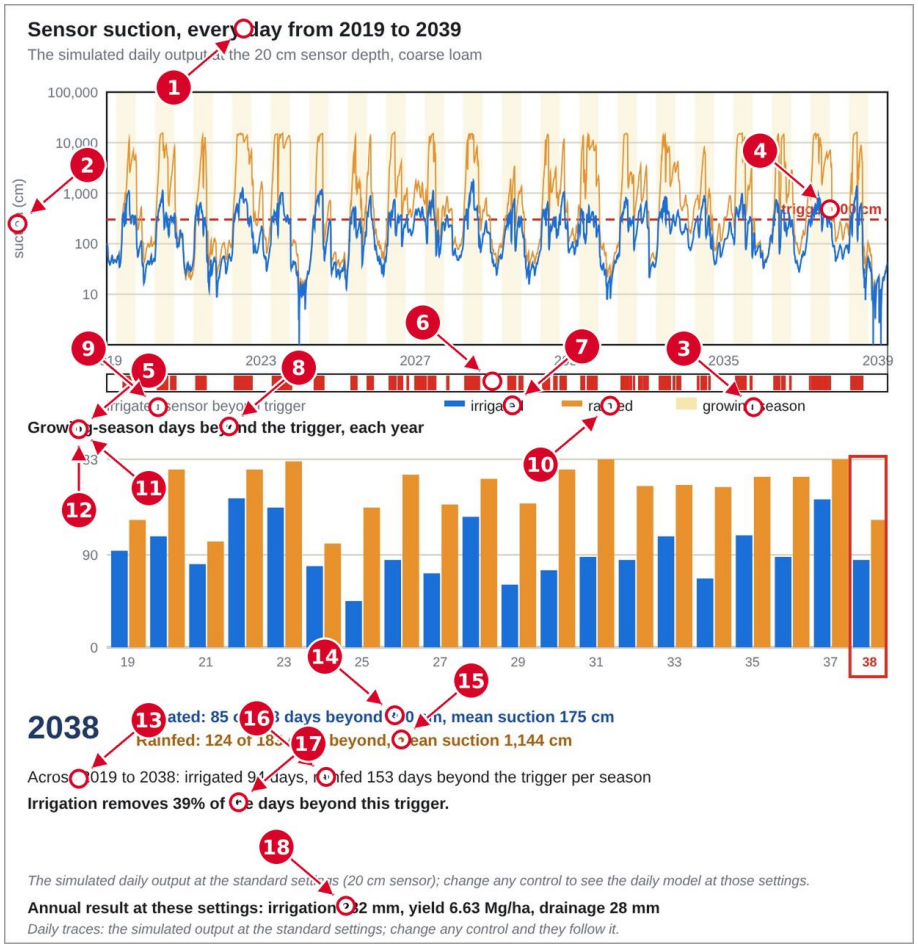


Figure 25.152: Sensor and trigger, part 1, playback the end (1 Apr 2039).

25.8.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

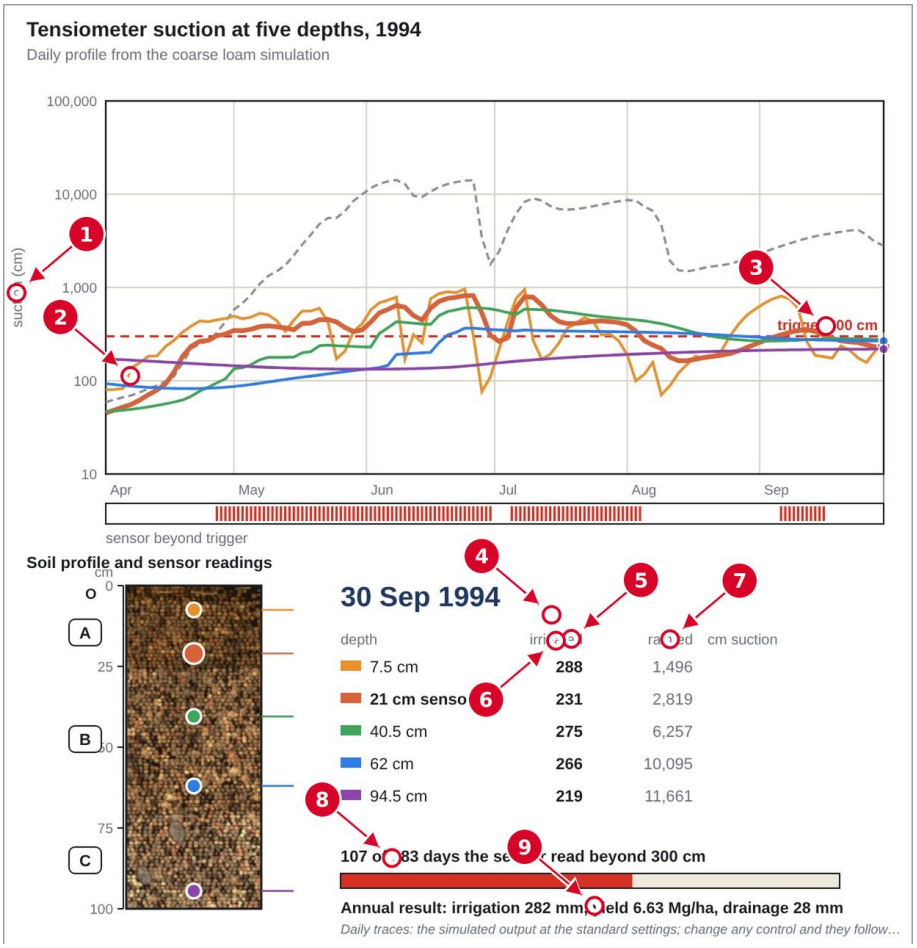


Figure 25.153: Sensor and trigger, part 1. The 1994 detail button pressed, so the chart shows the daily sensor record of the year 1994.

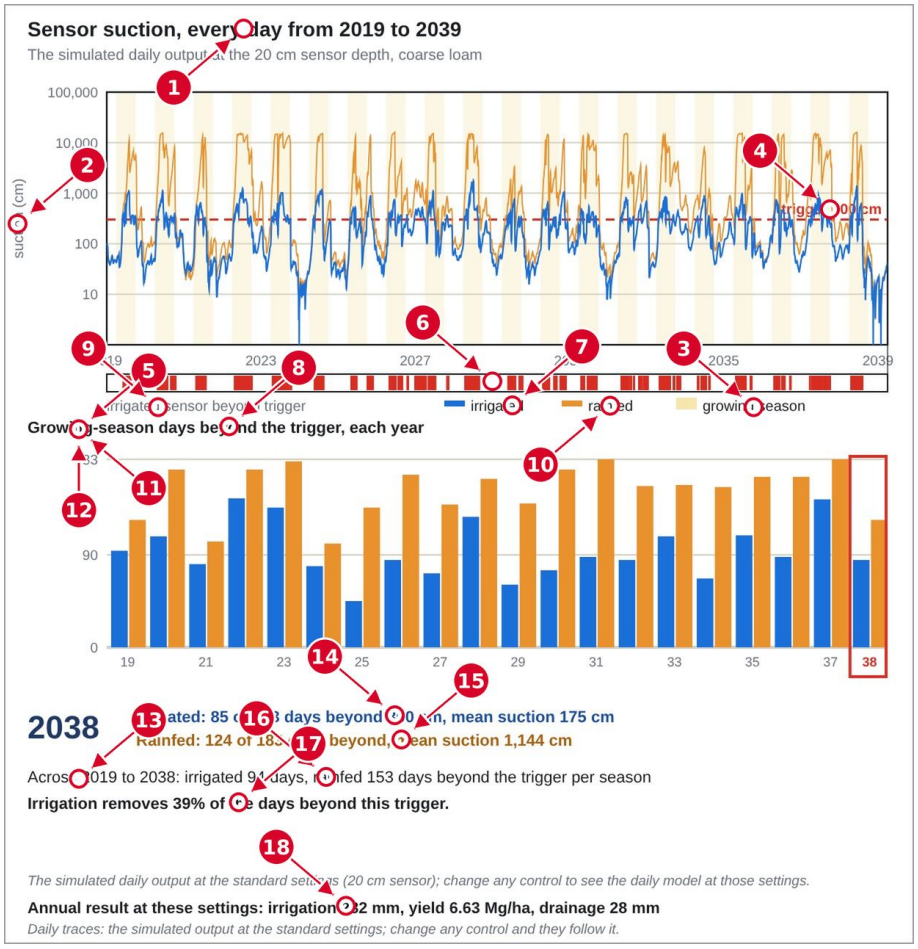


Figure 25.154: Sensor and trigger, part 1. A bar of the lower chart clicked, so the readout below shows the selected year.

25.9 Trigger law

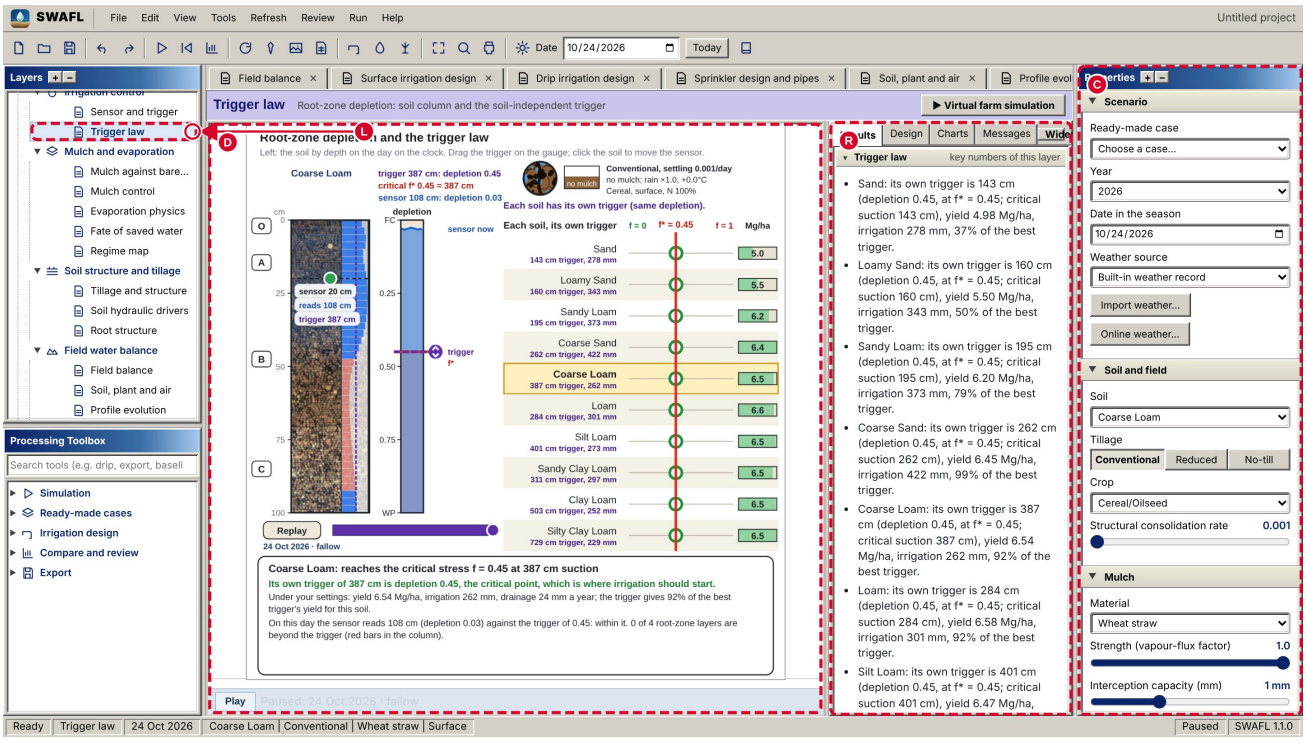


Figure 25.155: Trigger law: the whole workbench.

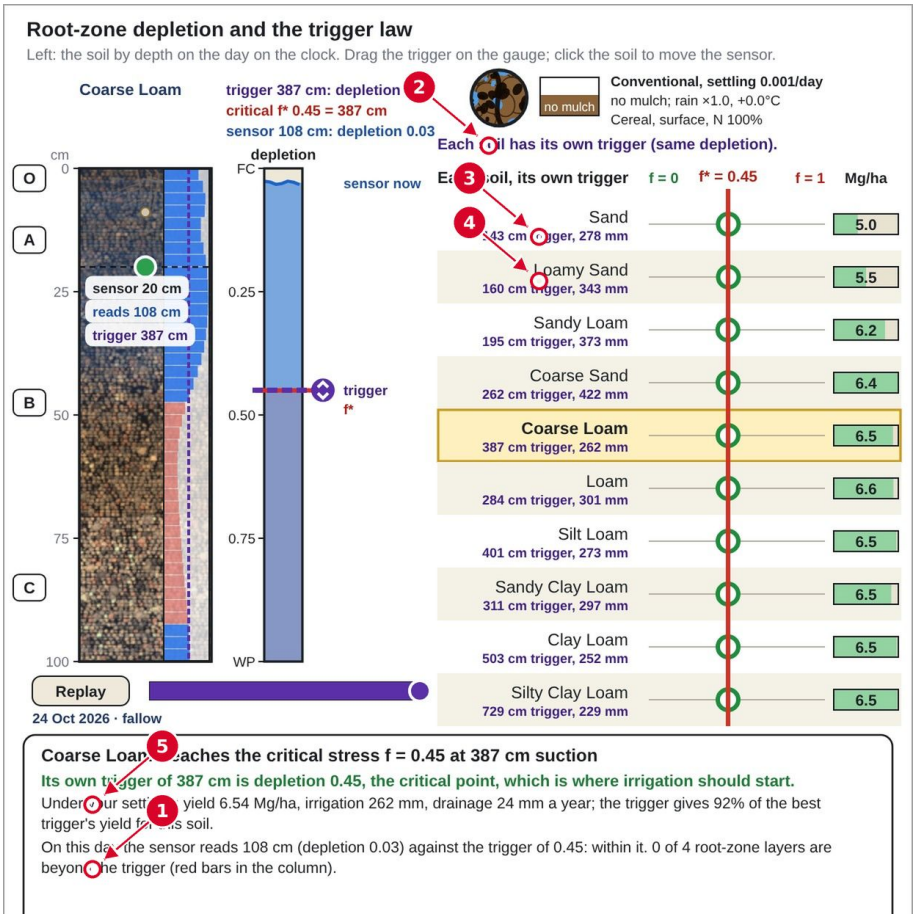


Figure 25.156: Trigger law: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Soil column: each layer coloured by its depletion f, Eq. 5.3 (red = beyond the trigger).
- **2** Critical depletion f* = 0.45, the same for every soil (Eq. 5.3).
- **3** Soil-specific trigger suction h(f*), the inverse of the van Genuchten curve: Eq. 6.3.
- **4** Net irrigation per event Z_n = f AWC R_max: Eq. 6.4.
- **5** Relative yield at the trigger, Y_rel (Eq. 6.5) and the trigger suction (Eq. 6.3) for the selected soil.

Equations behind this screen. Eq. 5.3, Eq. 5.27, Eq. 6.3, Eq. 6.4, Eq. 6.5.

Eq. 5.3. Where explained: chapter "The two computing engines", section "Plant-available water and depletion".

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.27. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$Z_n = \mathrm{clip}\big(f_T \cdot \mathrm{AWC} \cdot R_{\max};\ 25,\ 150\big)\ \mathrm{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 6.3. Where explained: chapter "Irrigation control layers", section "Governing relations".

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

Provenance (Eq. 6.3): **[Literature]**: inverse of the van Genuchten curve, solved by bisection.

Eq. 6.4. Where explained: chapter "Irrigation control layers", section "Governing relations".

$$Z_n = f \cdot \mathrm{AWC} \cdot R_{\max} \quad [\mathrm{mm}] \tag{6.4}$$

Provenance (Eq. 6.4): **[Literature]**: same relation as Eq. (5.27) without the clip.

Eq. 6.5. Where explained: chapter "Irrigation control layers", section "Governing relations".

$$Y_{\mathrm{rel}} = \frac{Y(h_T)}{\max_h Y(h)}, \tag{6.5}$$

Provenance (Eq. 6.5): **[Data-driven]**: ratio of Daisy run-set E2 yields.

25.9.1 The whole layer, part by part

The picture below is the complete drawing of the layer Trigger law, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

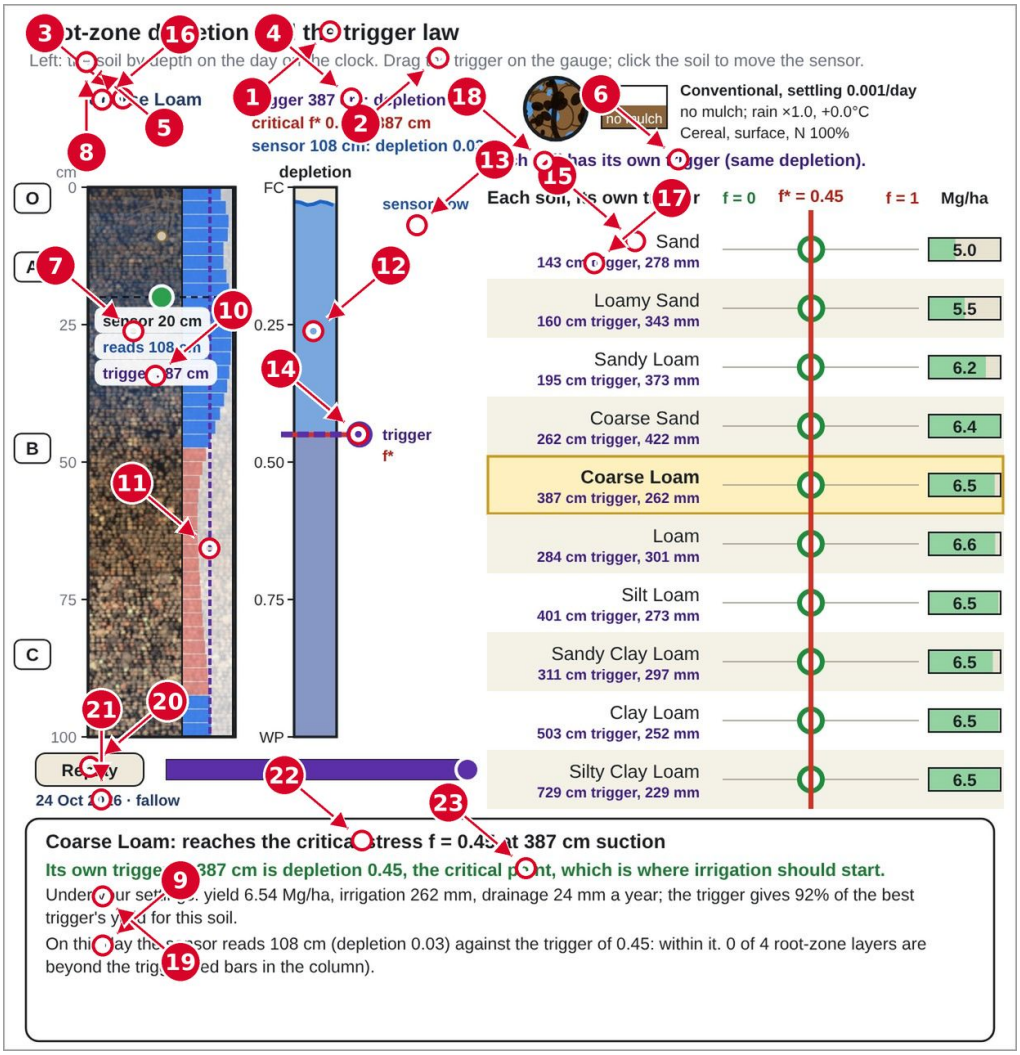


Figure 25.157: Trigger law, part 1 of 1.

What the arrows point to.

- **1** The title, the depletion of the root zone and the trigger law.
- **2** The instruction line, drag the trigger on the gauge and click the soil to move the sensor.
- **3** The soil whose column and gauge are shown.
- **4** The readout of the trigger, the critical depletion f* and the sensor depletion on the day.
- **5** The scenario in force, the tillage, mulch, rain scaling, warming, crop and nitrogen.
- **6** The statement that each soil has its own trigger at the same depletion.
- **7** The soil column by depth with the layers of the root zone.
- **8** The sensor tag, its depth, its reading and the trigger in cm of suction.
- **9** The blue bars are the layers inside the trigger on the day shown.
- **10** The red bars are the layers beyond the trigger.
- **11** The dashed line is the trigger carried down the column.
- **12** The depletion gauge from field capacity (FC) to wilting point (WP).
- **13** The sensor level on the gauge today.
- **14** The handle of the trigger, which is dragged to another depletion.
- **15** The soils ranked in the list, each with its own trigger in cm and its seasonal irrigation.
- **16** The soil selected, highlighted in the list.
- **17** The critical depletion f* as the red line across all ten soils.
- **18** The ring on each soil gives where its own trigger sits between f = 0 and f = 1.

- **19** The simulated yield of each soil at its own trigger, in Mg per ha.
- **20** The Replay button and the time value box that replay the season.
- **21** The date shown on the clock.
- **22** The conclusion, the stress at which the selected soil needs water.
- **23** The sentence that links the soil's own trigger to the critical point.

25.9.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

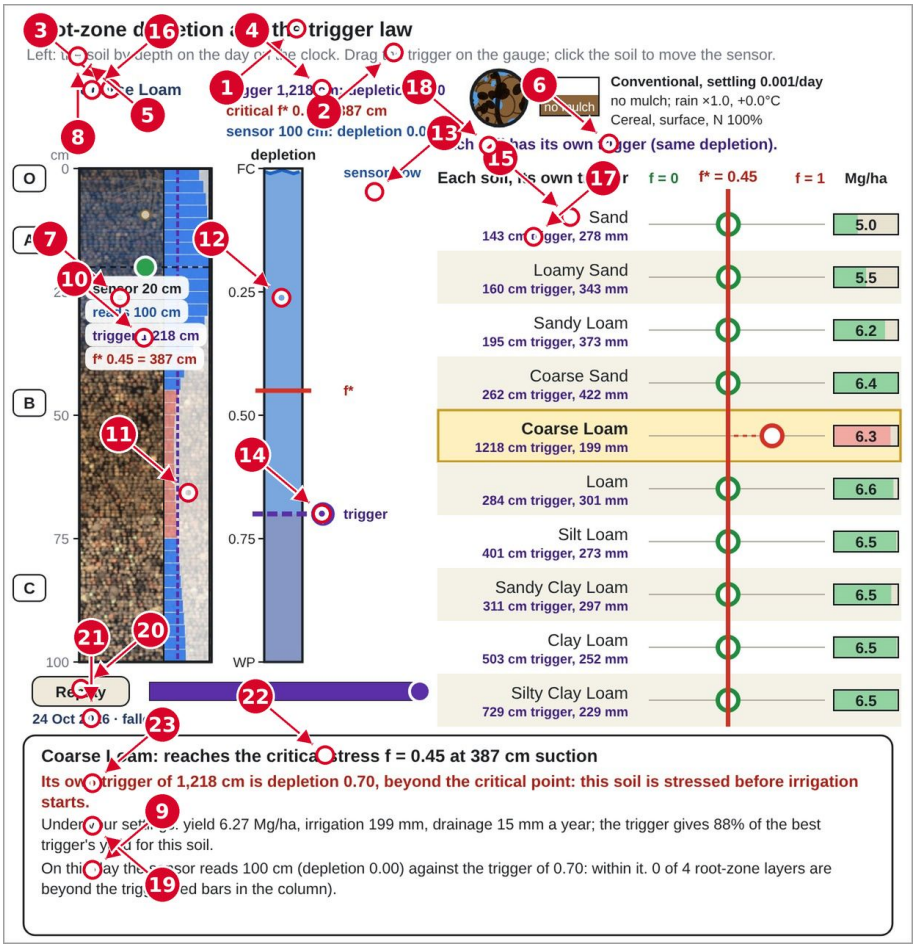


Figure 25.158: Trigger law, part 1. The trigger handle dragged to a deeper depletion on the gauge, so the trigger moves and the number of layers beyond it changes.

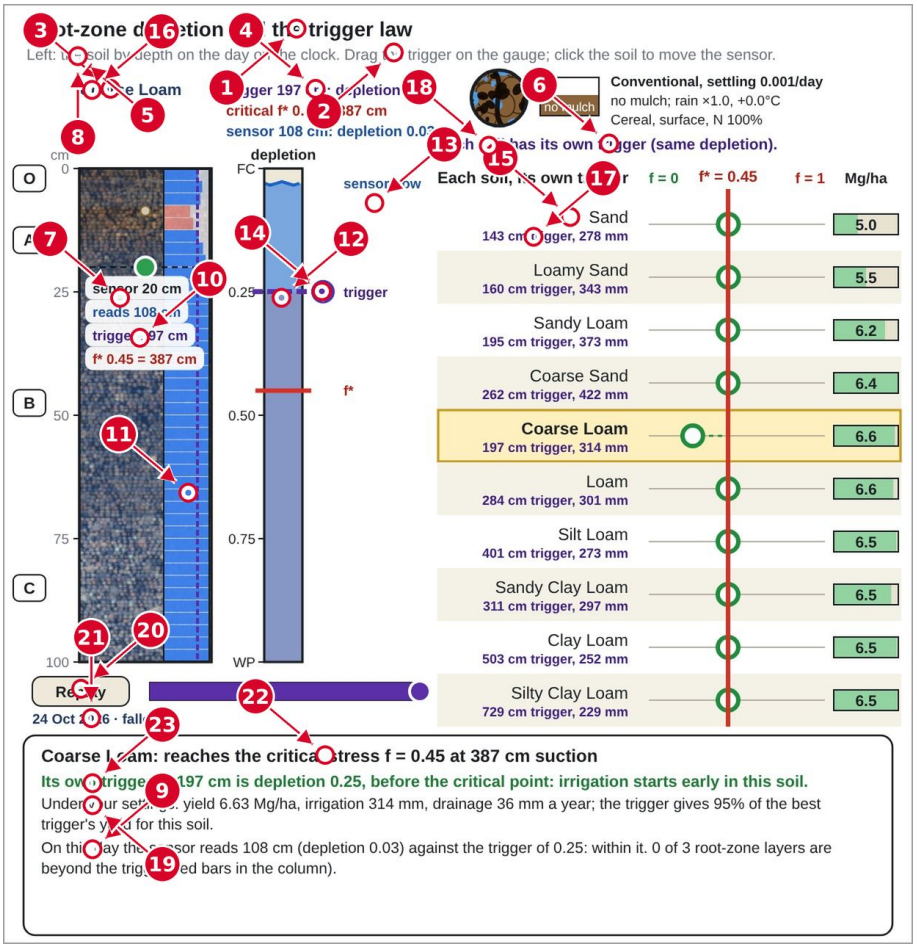


Figure 25.159: Trigger law, part 1. The trigger handle dragged to a shallow depletion on the gauge.

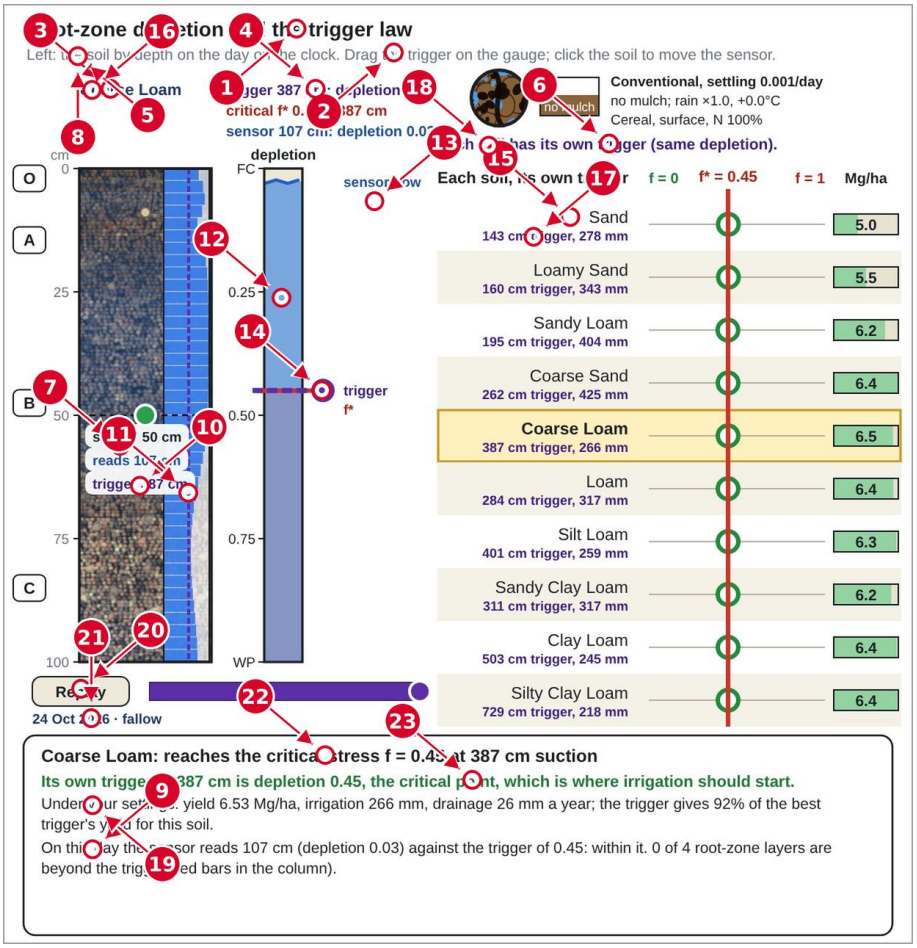


Figure 25.160: Trigger law, part 1. The soil column clicked at 60 cm, so the sensor moves to that depth.

25.10 Tillage and structure

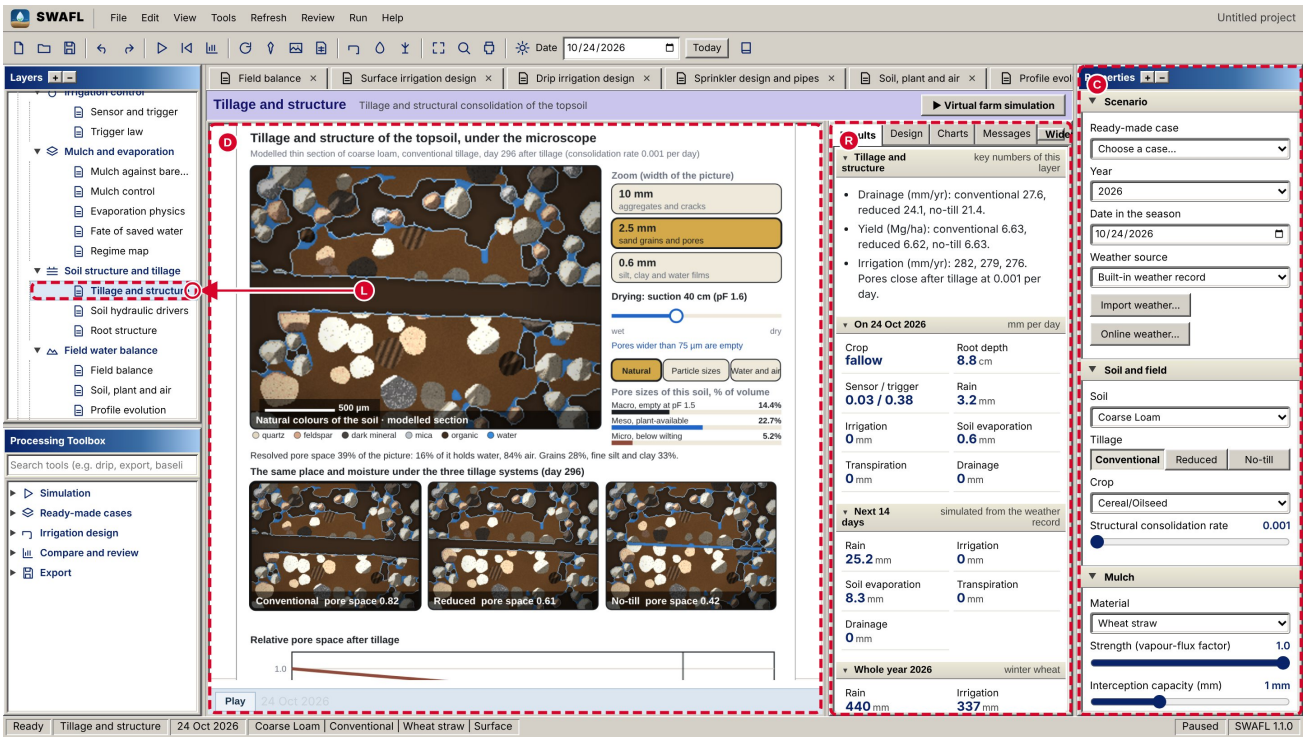


Figure 25.161: Tillage and structure: the whole workbench.

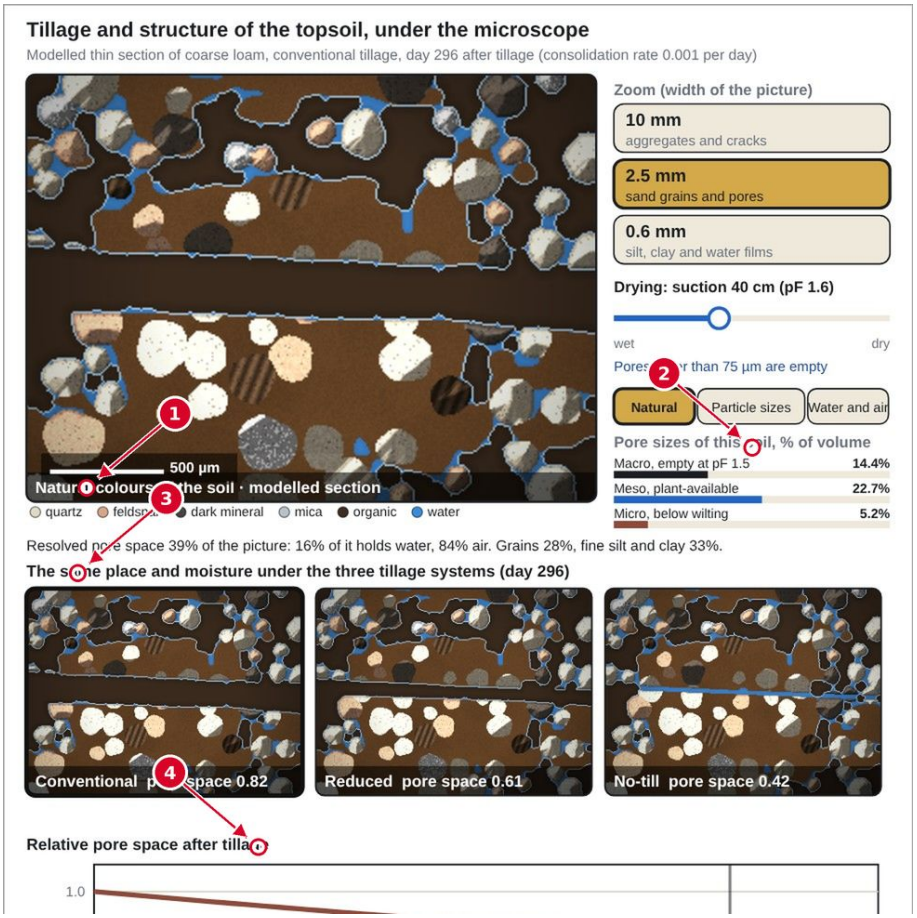


Figure 25.162: Tillage and structure: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Modelled thin section of the topsoil; the share of open pores follows the pore index p(t), Eq. 8.1.
- **2** Pore-size classes from the retention curve, theta(h): Eqs. 8.3 and 8.4 (capillary radius).
- **3** The same place under the three tillage systems; pore space 0.82, 0.61 and 0.42 from Eq. 8.1.
- **4** Relative pore space after tillage, p(t) = p_c + (p_0 - p_c) exp(-tau_c t): Eq. 8.1.

Equations behind this screen. Eq. 5.1, Eq. 8.1, Eq. 8.2, Eq. 8.3, Eq. 8.4.

Eq. 5.1. Where explained: chapter “The two computing engines”, section “The simulation campaign”.

$$S_e(h) = \left[1 + (\alpha|h|)^n\right]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 8.1. Where explained: chapter “Soil structure and tillage layers”, section “Pore index and consolidation”.

$$p(t) = p_c + (p_0 - p_c)\, e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 8.2. Where explained: chapter “Soil structure and tillage layers”, section “How the section is generated”.

$$\phi = \mathrm{clip}\big(0.36 + 0.0016\,\mathrm{clay} + 0.0009\,\mathrm{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\mathrm{clay}, \tag{8.2}$$

Provenance (Eq. 8.2): **[Engineered]**: porosity relation with chosen coefficients.

Eq. 8.3. Where explained: chapter “Soil structure and tillage layers”, section “How the section is generated”.

$$\theta(h) = \theta_r + (\phi - \theta_r)\, S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

Eq. 8.4. Where explained: chapter “Soil structure and tillage layers”, section “How the section is generated”.

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\, g\, h} \;\approx\; \frac{0.149\ \mathrm{cm}^2}{h} \quad (\text{or } r_c[\mathrm{mm}] \approx 1.49/h[\mathrm{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

25.10.1 The whole layer, part by part

The picture below is the complete drawing of the layer Tillage and structure, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

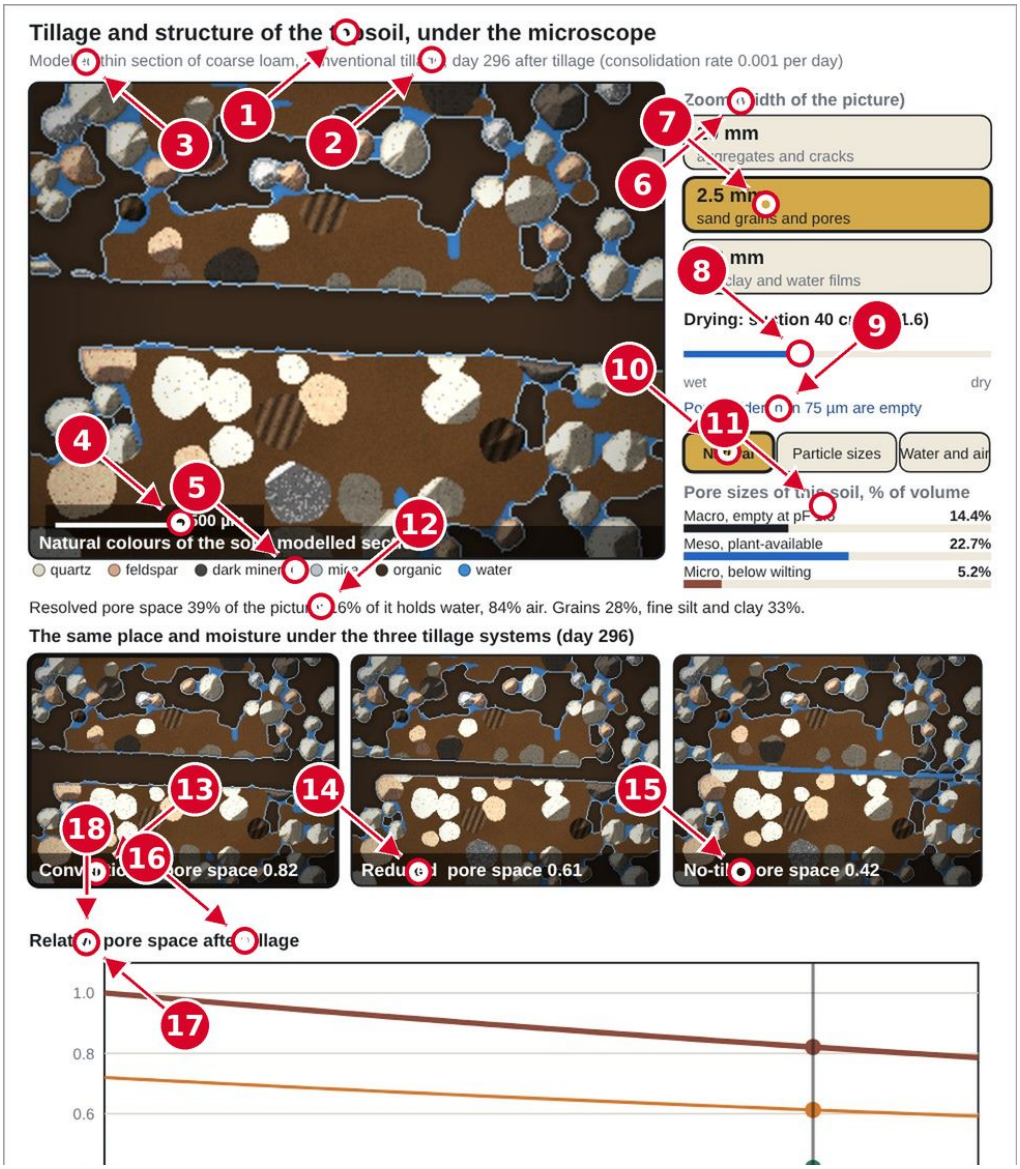


Figure 25.163: Tillage and structure, part 1 of 2.

What the arrows point to.

- **1** The title, tillage and the structure of the topsoil under the microscope.
- **2** The scenario of the section, the soil, the tillage, the day after tillage and the consolidation rate.
- **3** The modelled thin section, with grains, pores and water films drawn at the chosen zoom.
- **4** The scale bar of the section.
- **5** The legend of the colours, quartz, feldspar, dark mineral, mica, organic matter and water.
- **6** The Zoom buttons, 10 mm, 2.5 mm and 0.6 mm wide pictures.
- **7** The zoom selected, here the 2.5 mm picture of sand grains and pores.
- **8** The drying value box, which sets the suction and so which pores hold water.
- **9** The note on the pore width that empties at this suction.
- **10** The view buttons, Natural, Particle sizes and Water and air.
- **11** The pore-size classes of this soil, macro, meso and micro, in percent of the volume.
- **12** The resolved pore space of the picture and the share of it holding water.
- **13** The conventional tillage tile, with its relative pore space of 1.00.
- **14** The reduced tillage tile, with a smaller relative pore space.

- **15** The no-till tile, with the smallest relative pore space.
- **16** The chart of the relative pore space after tillage against the days since tillage.
- **17** The conventional line, which starts at 1.0 and consolidates with time.
- **18** The reduced tillage line.

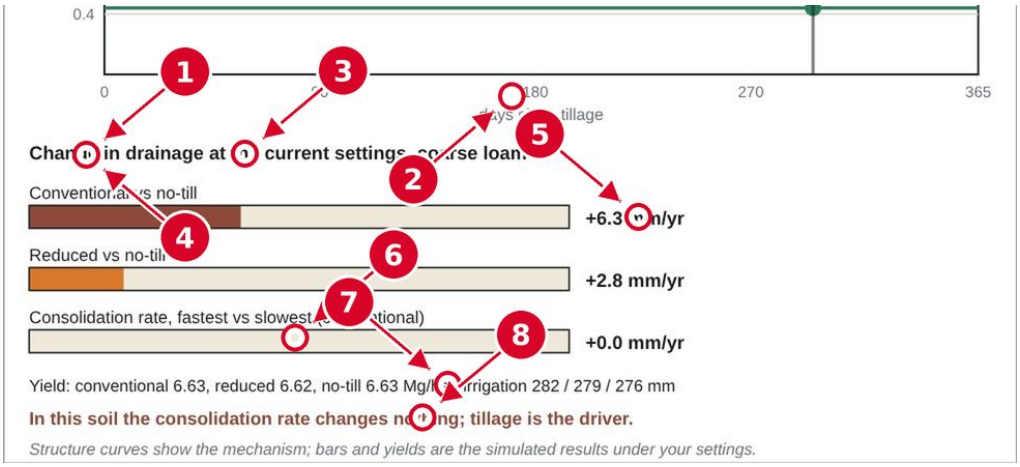


Figure 25.164: Tillage and structure, part 2 of 2.

What the arrows point to.

- **1** The no-till line, flat because the soil is not loosened.
- **2** The time axis in days since tillage.
- **3** The change in drainage at the current settings.
- **4** The bar of drainage for conventional against no-till tillage.
- **5** The value of that difference in mm per year.
- **6** The bar for the consolidation rate, fastest against slowest.
- **7** The yield and the irrigation of the three tillage systems.
- **8** The conclusion, which factor drives the drainage in this soil.

25.10.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

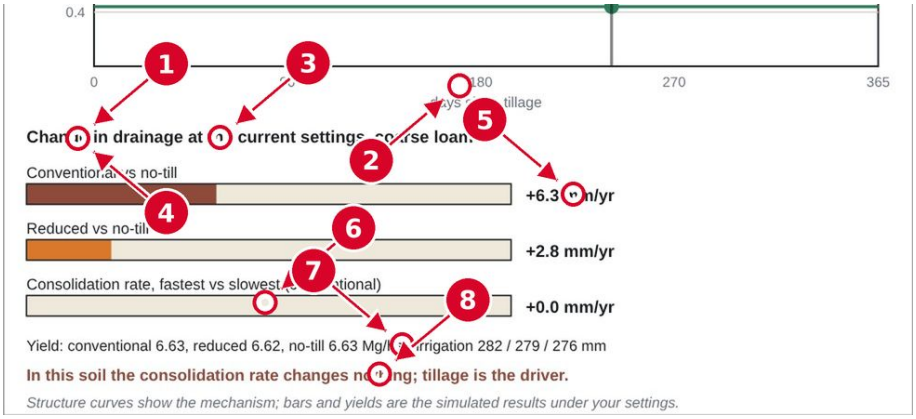


Figure 25.165: Tillage and structure, part 2, playback two thirds of the way (day 241).

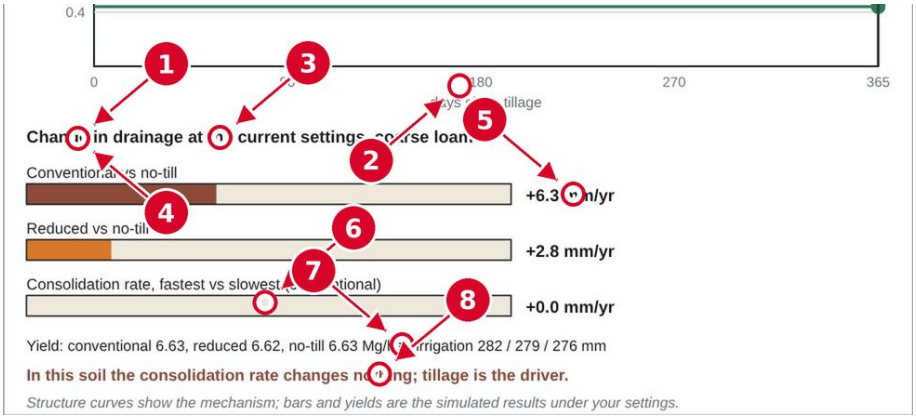


Figure 25.166: Tillage and structure, part 2, playback the end (day 365).

25.10.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

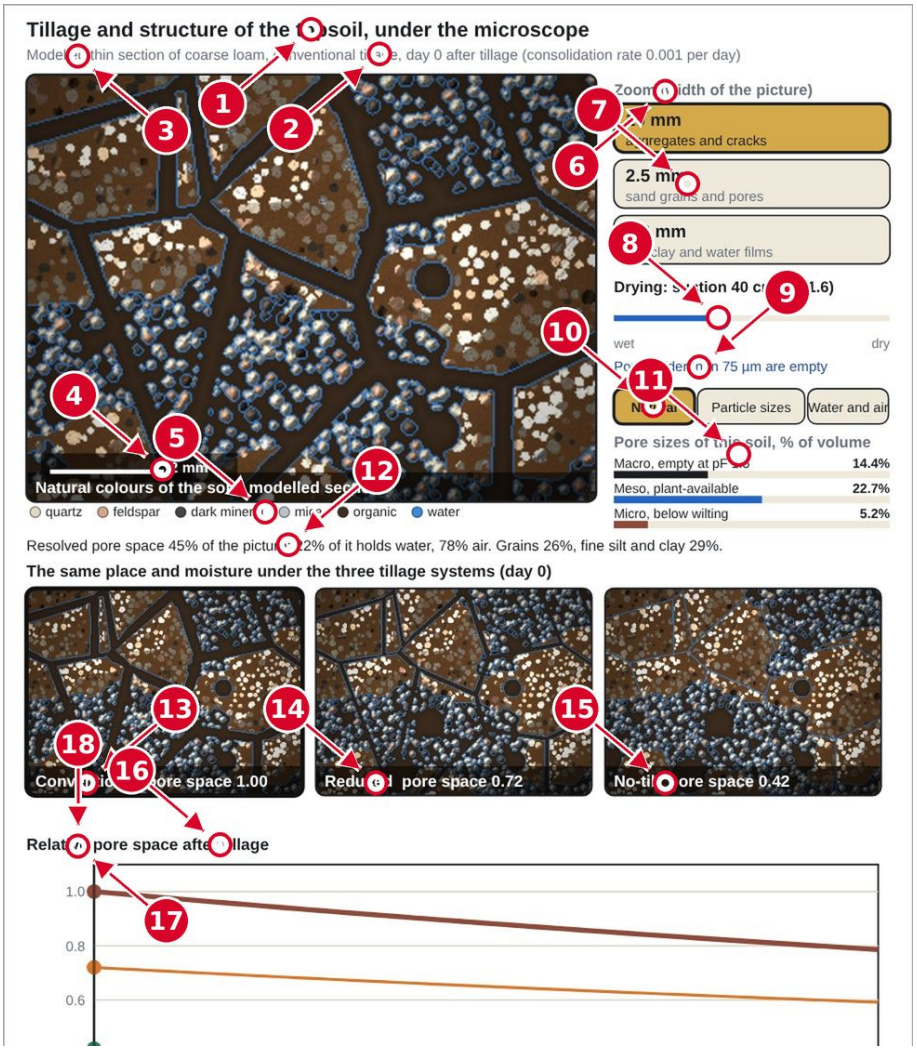


Figure 25.167: Tillage and structure, part 1. The 10 mm zoom chosen, so the picture shows aggregates and cracks.

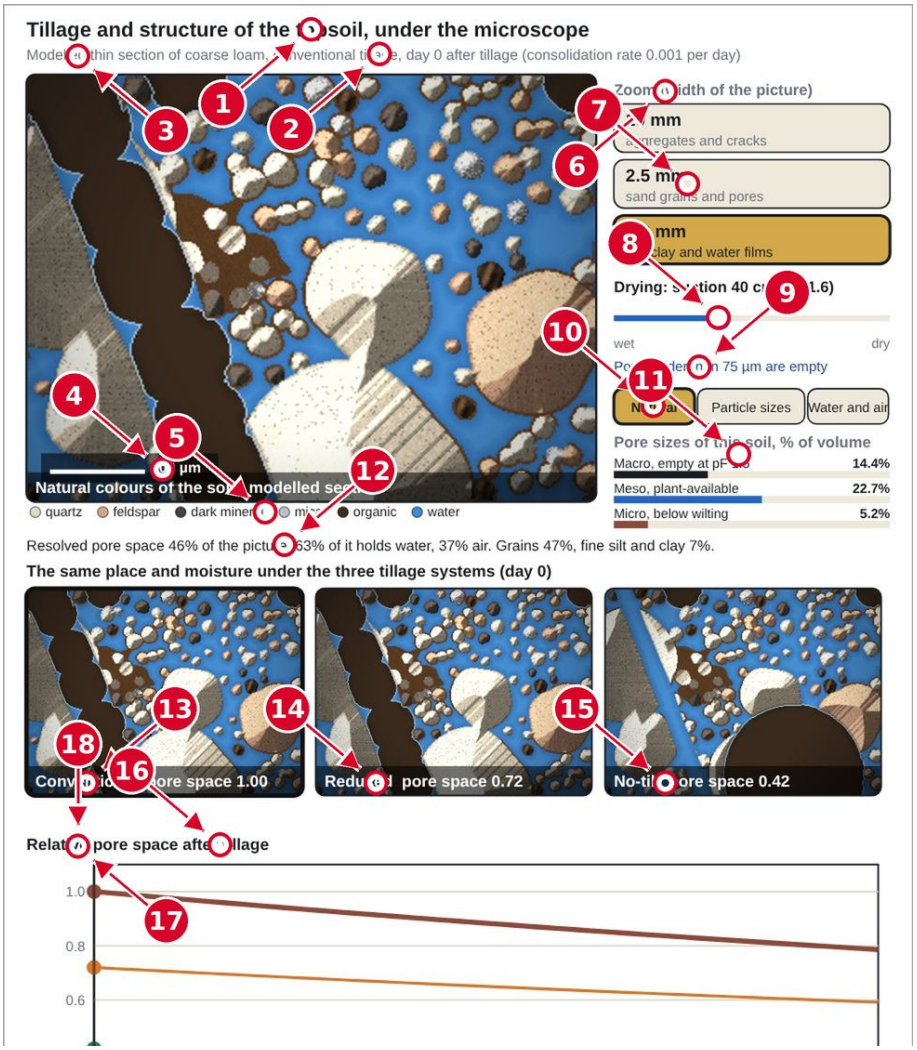


Figure 25.168: Tillage and structure, part 1. The 0.6 mm zoom chosen, so the picture shows silt, clay and the water films.

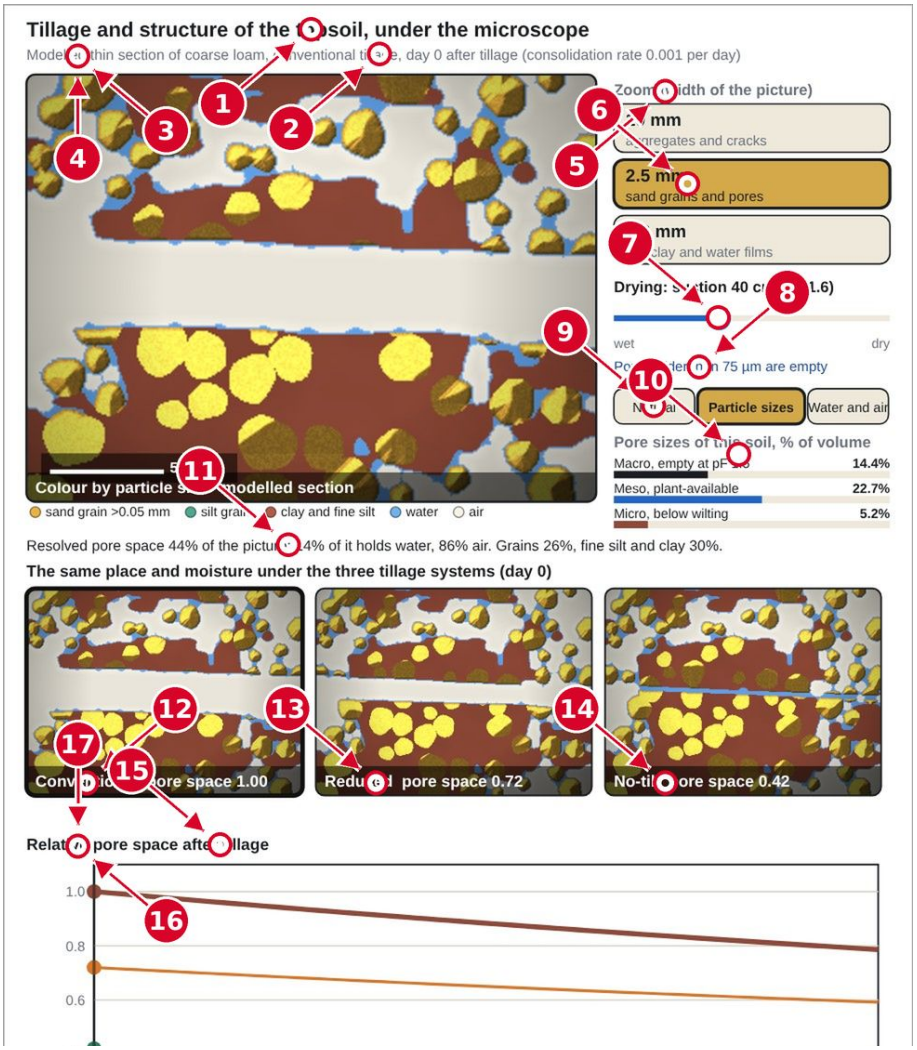


Figure 25.169: Tillage and structure, part 1. The Particle sizes view chosen, so the grains are coloured by size.

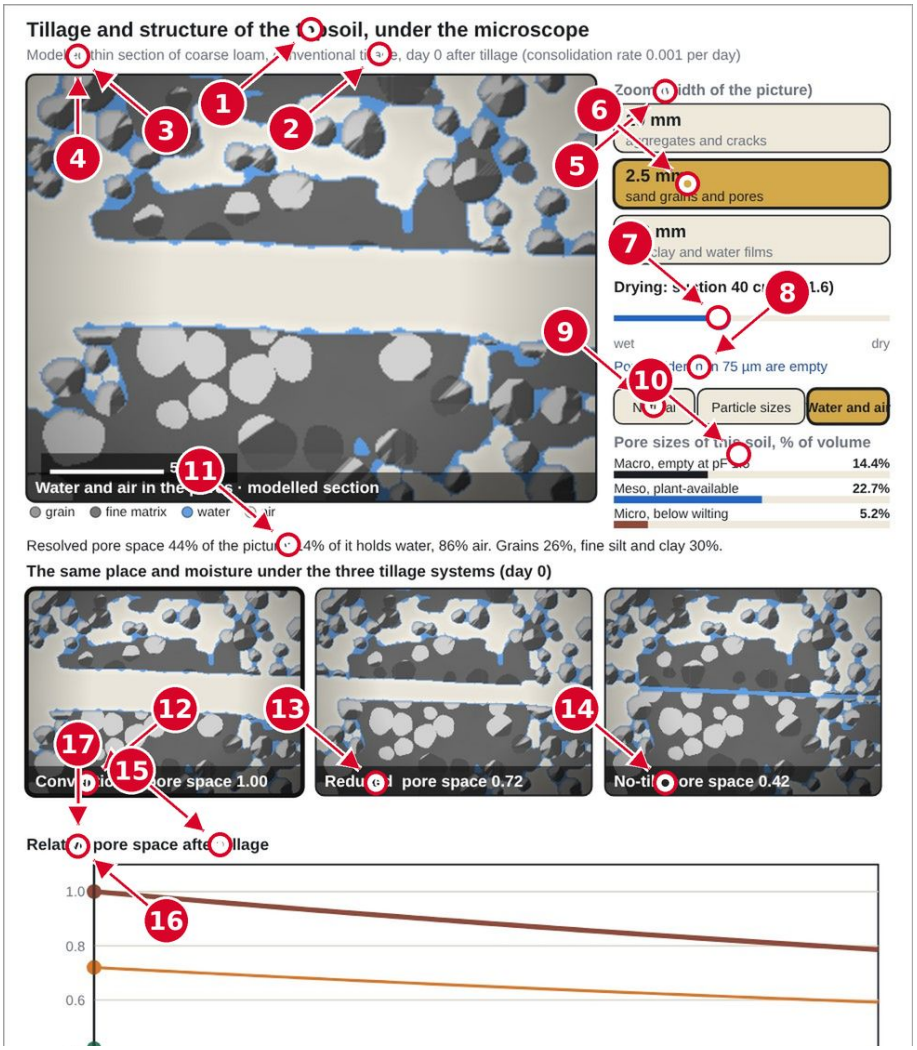


Figure 25.170: Tillage and structure, part 1. The Water and air view chosen, so the pores are coloured by whether they hold water or air.

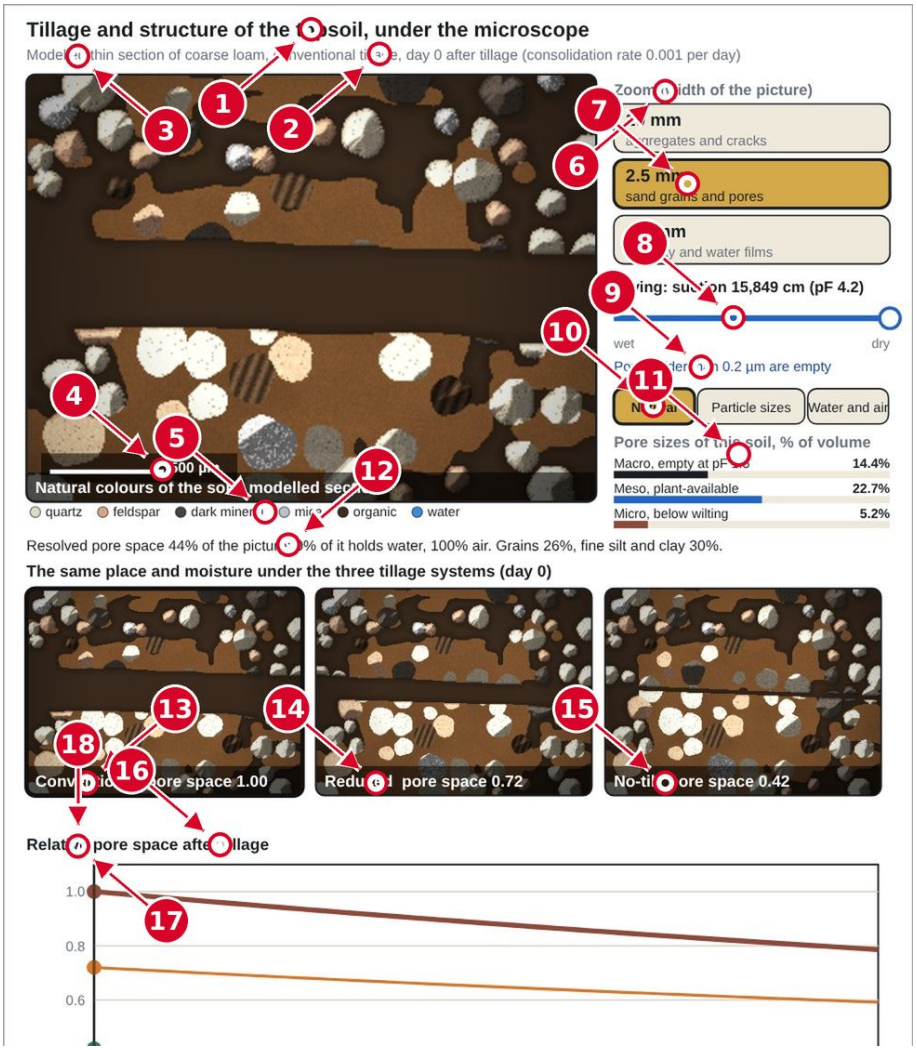


Figure 25.171: Tillage and structure, part 1. The drying value box dragged to the dry end, so most pores are empty.

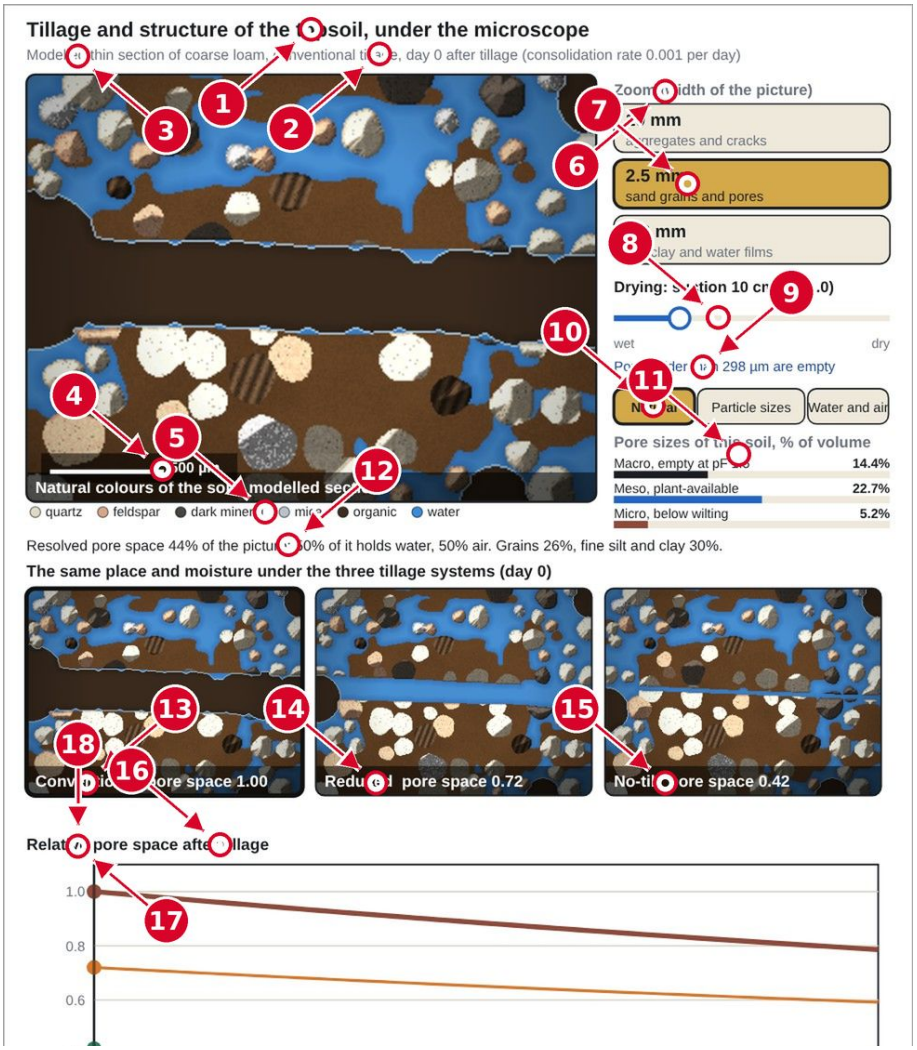


Figure 25.172: Tillage and structure, part 1. The drying value box dragged to the wet end, so most pores hold water.

25.11 Soil hydraulic drivers

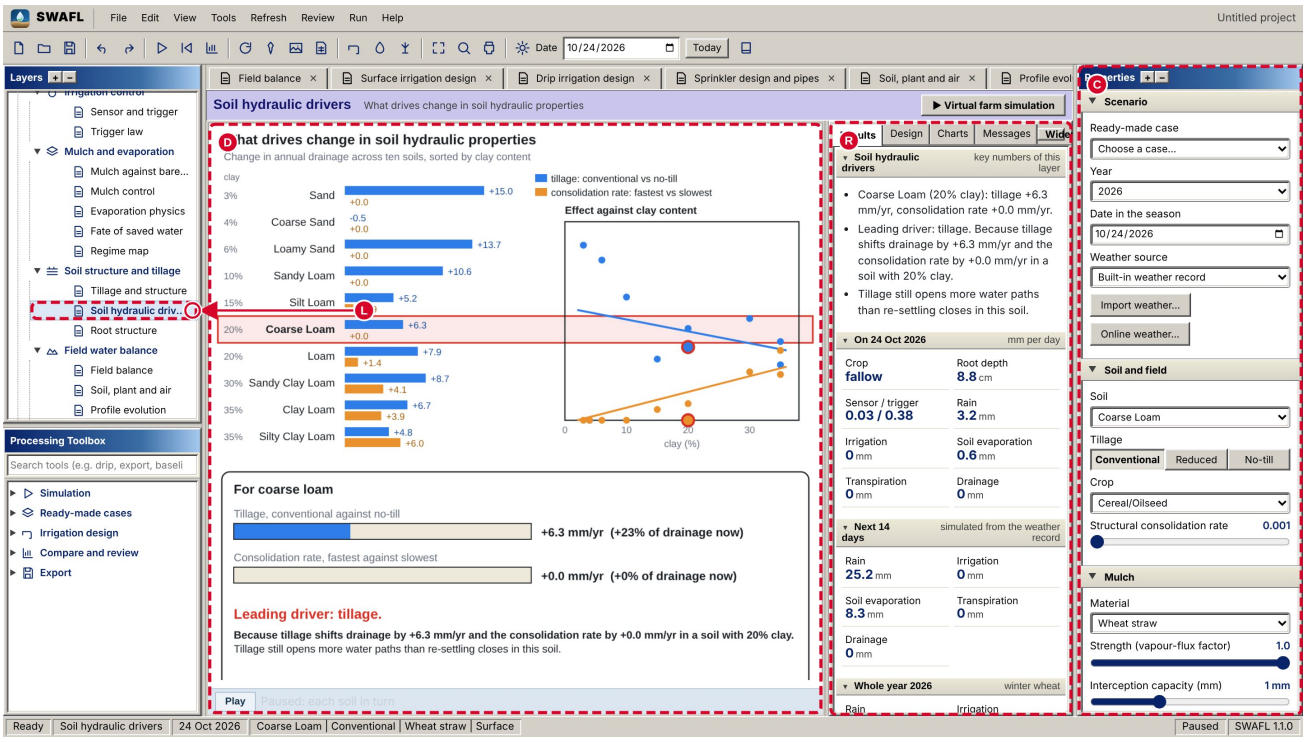


Figure 25.173: Soil hydraulic drivers: the whole workbench.

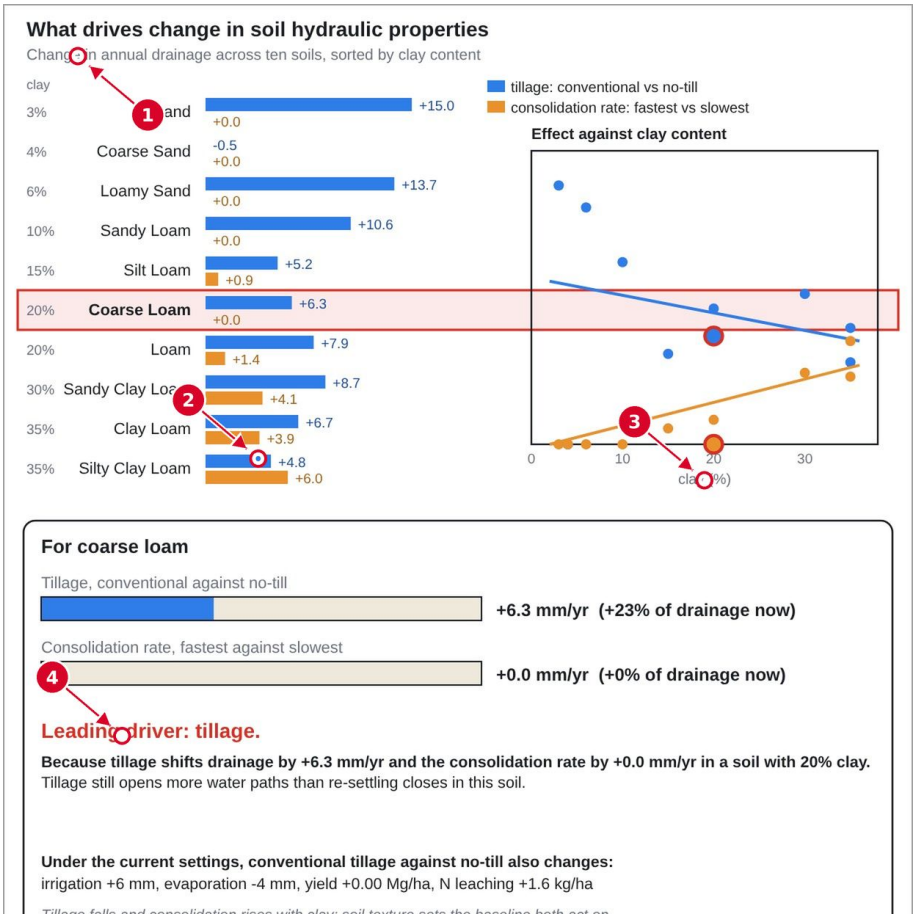


Figure 25.174: Soil hydraulic drivers: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Blue bar: effect of tillage on drainage, Delta_till = D(conventional) - D(no-till): Eq. 8.5.
- **2** Orange bar: effect of the consolidation rate on drainage, Delta_cons: Eq. 8.5.
- **3** Scatter of each effect against clay content with the least-squares fit: Eq. 8.6.
- **4** “Leading driver”: the larger of Delta_till and Delta_cons for the selected soil, Eq. 8.5.

Equations behind this screen. Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 8.1, Eq. 8.5, Eq. 8.6.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".

$$Q \leftarrow Q\,\text{clip}\Big(\frac{r_\kappa.Q}{r_{\kappa_0}.Q};\,0.05,\,6\Big)\;(Q \in \{E,T,ET,I,Y\}),\qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D),\quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N) \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".

$ET \in [\,P + I - D - 30,\; P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\; ET - T - E),\qquad \Delta S = P + I - ET - D,\qquad \text{IE} = \frac{100\,Y}{I}\;\;[\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 8.1. Where explained: chapter "Soil structure and tillage layers", section "Pore index and consolidation".

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t},\qquad p_0 = 1.00\;(\text{conventional}),\;0.72\;(\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 8.5. Where explained: chapter "Soil structure and tillage layers", section "Generation".

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*),\qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

Provenance (Eq. 8.5): **[Data-driven]**: differences of Daisy run-set E5 results.

Eq. 8.6. Where explained: chapter "Soil structure and tillage layers", section "Generation".

$$\hat{\Delta} = a + b\,c,\qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2},\quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

Provenance (Eq. 8.6): **[Data-driven]**: ordinary least squares applied to Daisy results.

25.11.1 The whole layer, part by part

The picture below is the complete drawing of the layer Soil hydraulic drivers, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

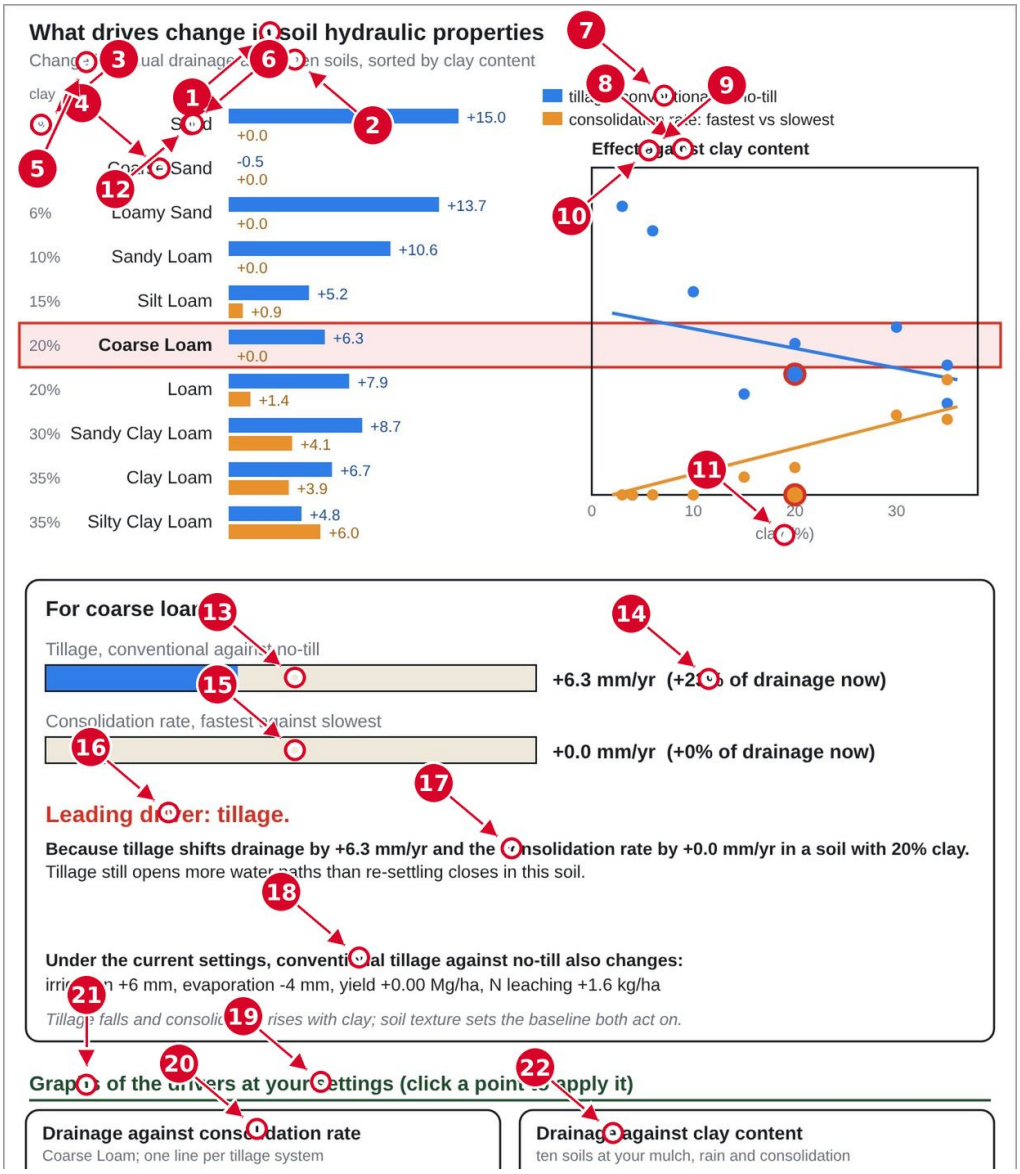


Figure 25.175: Soil hydraulic drivers, part 1 of 3.

What the arrows point to.

- **1** The title, what drives the change in the hydraulic properties of the soil.
- **2** The subtitle, the change in annual drainage across ten soils sorted by clay content.
- **3** The clay content of each soil, the sorting key.
- **4** The soil highlighted in the list, here sand, which every panel below refers to.
- **5** The blue bar, the change in drainage from tillage, conventional against no-till.
- **6** The orange bar, the change in drainage from the consolidation rate.
- **7** The legend of the two bars.
- **8** The scatter chart of the two effects against the clay content.
- **9** The highlighted soil on the scatter chart.

- **10** The blue trend line, the tillage effect falls with clay.
- **11** The orange trend line, the consolidation effect rises with clay.
- **12** The card for the selected soil, with the two effects in mm per year.
- **13** The bar of the tillage effect against its largest value.
- **14** The value of the tillage effect and its share of the drainage now.
- **15** The bar of the consolidation effect.
- **16** The leading driver, tillage or the consolidation rate.
- **17** The reasoning behind the leading driver for this clay content.
- **18** The other changes of the conventional tillage against no-till, irrigation, evaporation, yield and nitrogen leaching.
- **19** The graphs of the drivers; clicking a point applies it to the settings.
- **20** The graph of the drainage against the consolidation rate, one line per tillage system.
- **21** A point of the graph, which applies that value when it is clicked.
- **22** The graph of the drainage against the clay content of the ten soils.

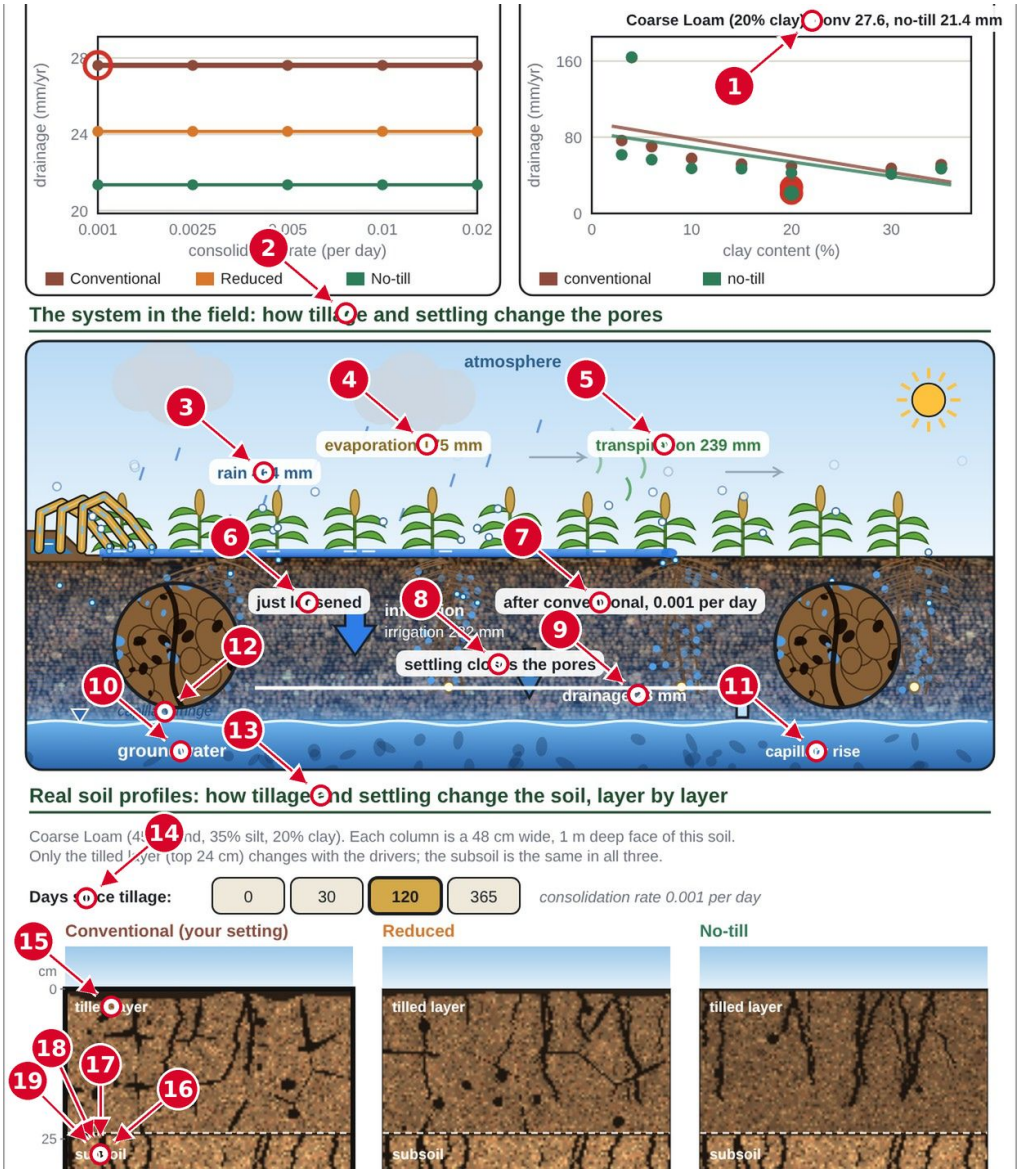


Figure 25.176: Soil hydraulic drivers, part 2 of 3.

What the arrows point to.

- **1** The readout of the soil selected, drainage of the conventional and no-till systems.
- **2** The system in the field, how tillage and settling change the pores.
- **3** The rain of the year.
- **4** The evaporation of the year.
- **5** The transpiration of the year.
- **6** The loosened layer just after the tillage.
- **7** The state of the layer after tillage at the consolidation rate in force.
- **8** The settling that closes the pores with time.
- **9** The drainage through the profile, from the Daisy simulation.
- **10** The groundwater below the profile.
- **11** The capillary rise from the groundwater.
- **12** The aggregate drawn inside the tilled layer.
- **13** The real soil profiles, the three tillage systems layer by layer.
- **14** The buttons of the days since tillage, 0, 30, 120 and 365.
- **15** The tilled layer, the top 24 cm, the only layer that changes.
- **16** The boundary of the tilled layer and the subsoil at 25 cm.
- **17** The conventional column, with the largest clods and cracks.
- **18** The reduced tillage column.
- **19** The no-till column, with fine cracks near the surface.

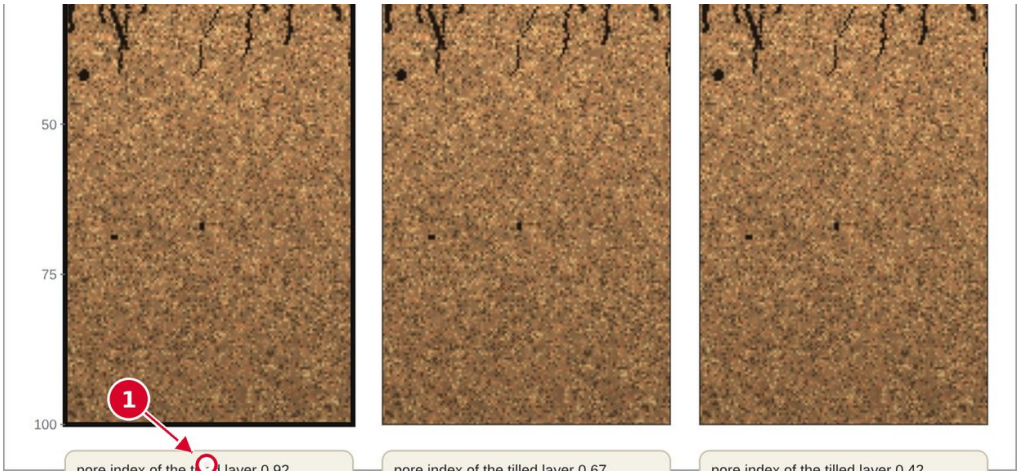


Figure 25.177: Soil hydraulic drivers, part 3 of 3.

What the arrows point to.

- **1** The pore index, the voids on the face and the share open to air.

25.11.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

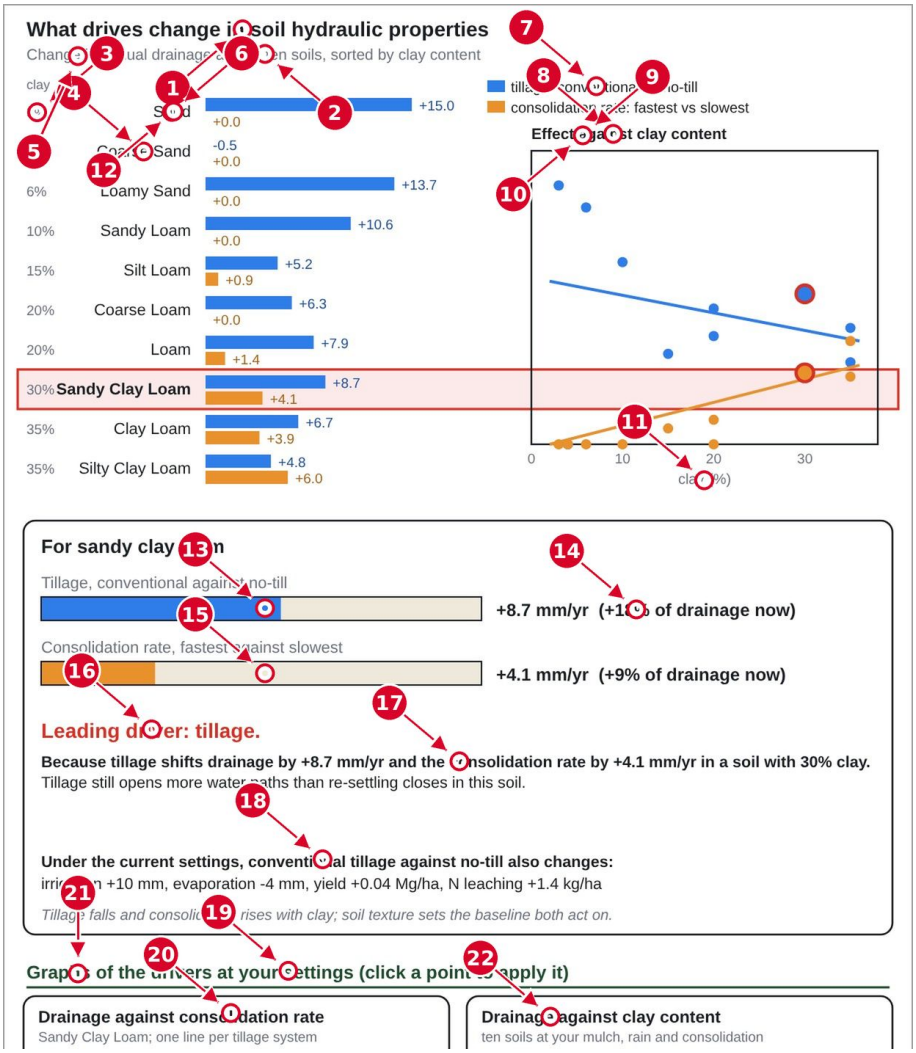


Figure 25.178: Soil hydraulic drivers, part 1. A different soil clicked in the list, so every panel refers to that soil.

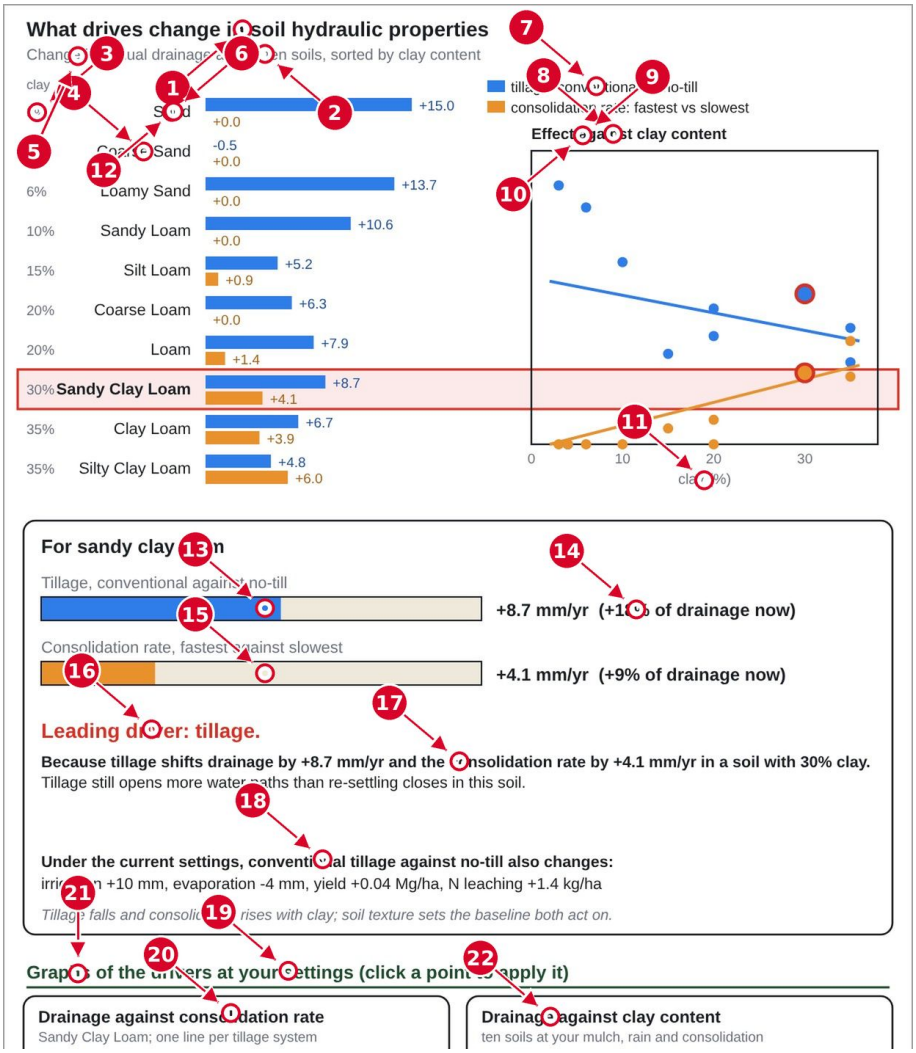


Figure 25.179: Soil hydraulic drivers, part 1. A different soil clicked in the list, so every panel refers to that soil.

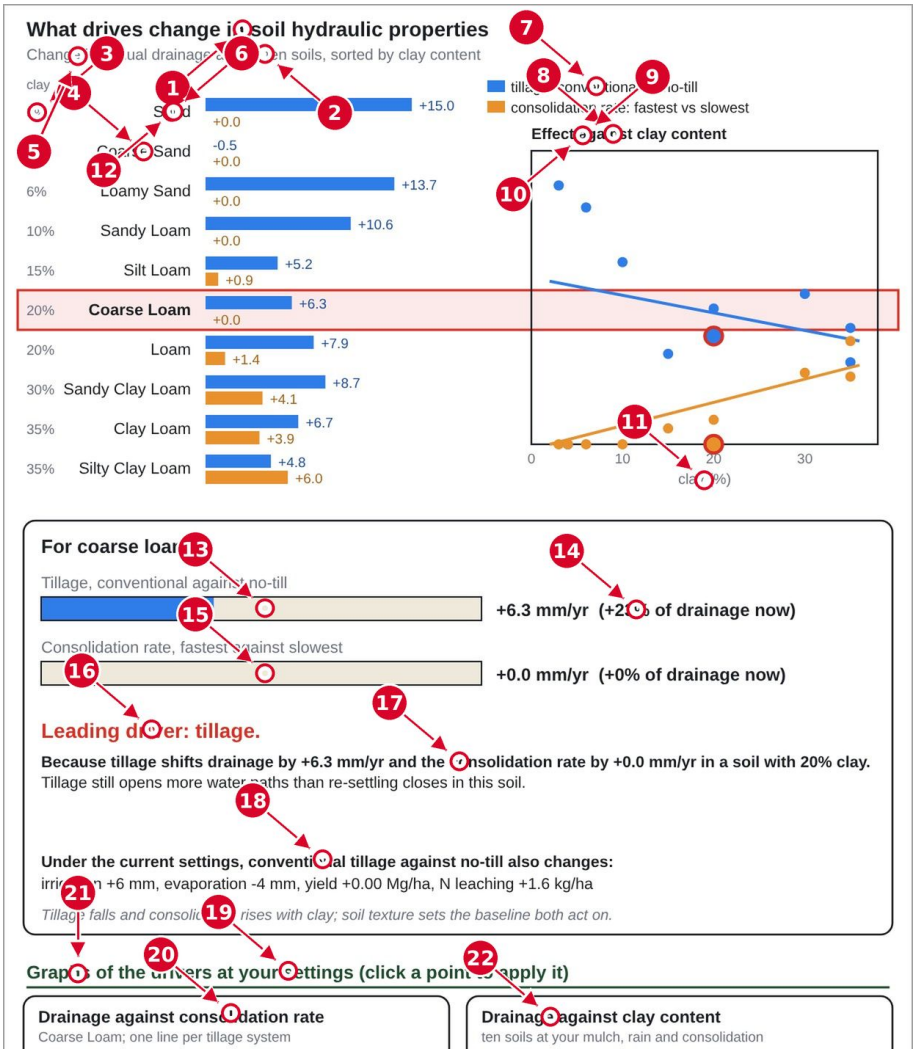


Figure 25.180: Soil hydraulic drivers, part 1. The 0 days since tillage button pressed, so the tilled layer is shown just after tillage.

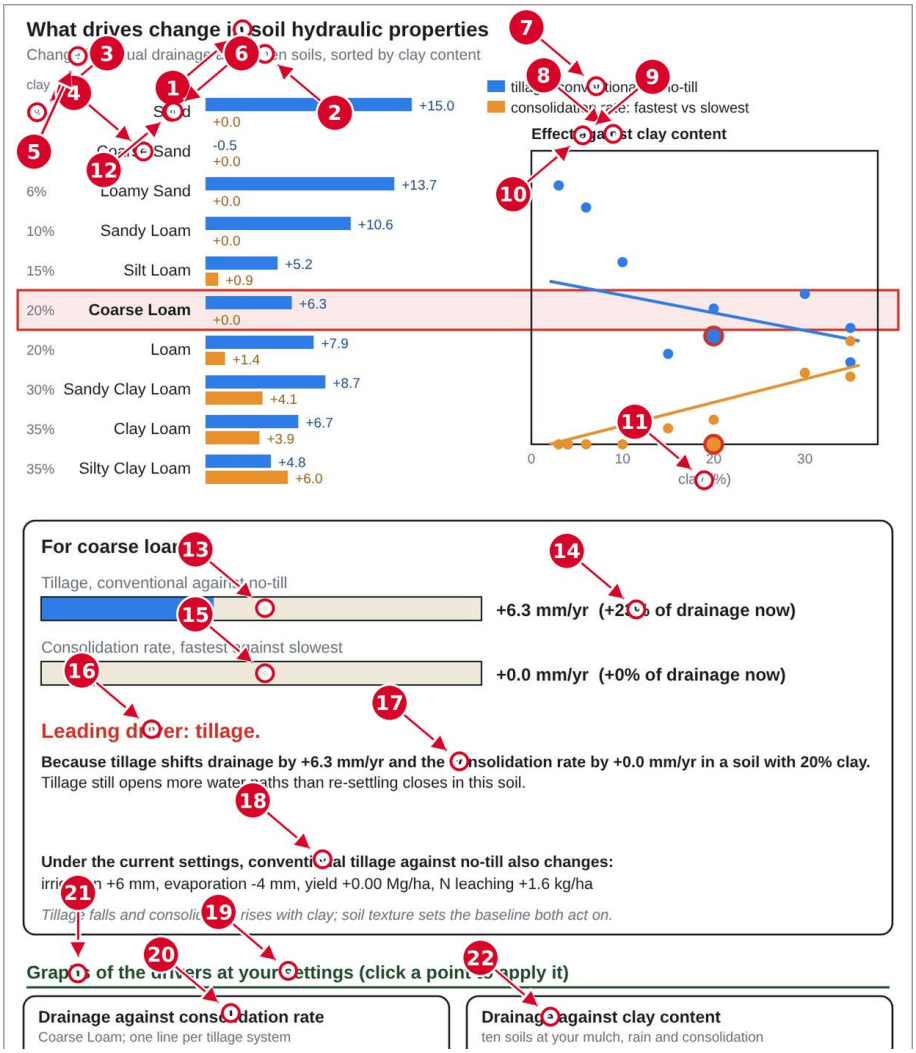


Figure 25.181: Soil hydraulic drivers, part 1. The 30 days since tillage button pressed.

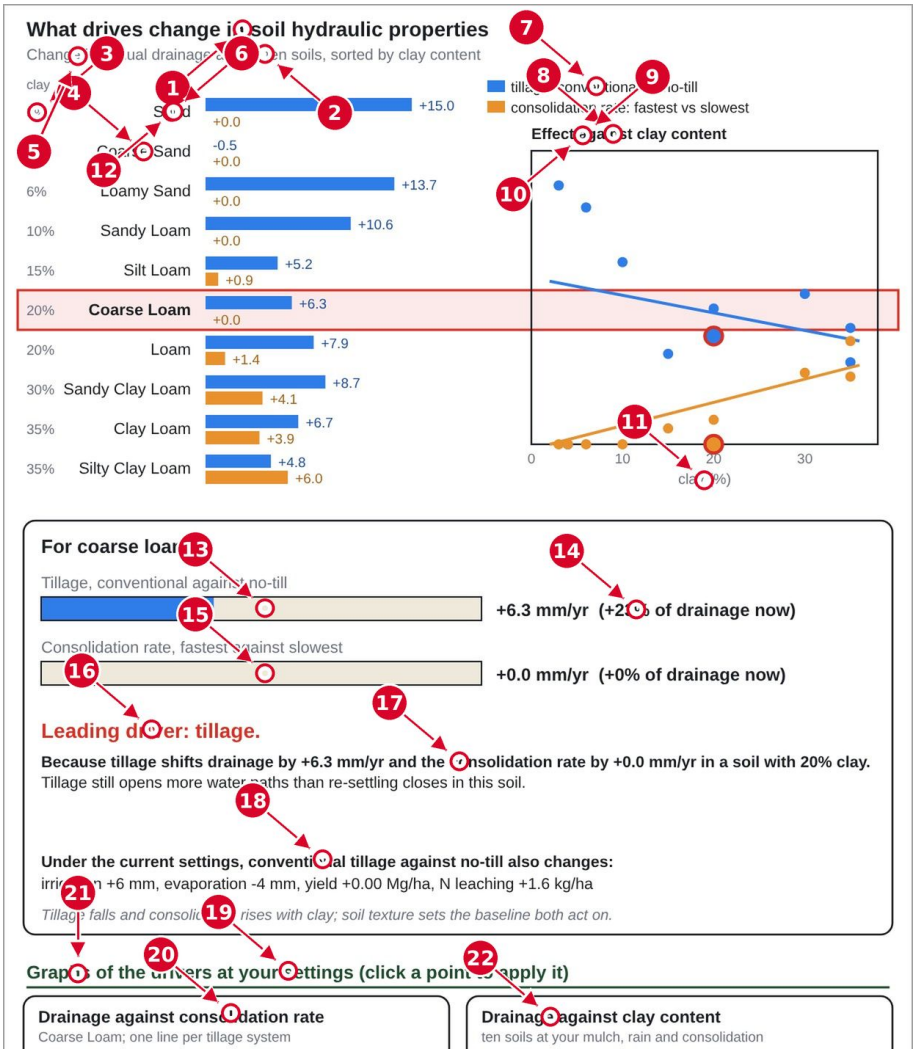


Figure 25.182: Soil hydraulic drivers, part 1. The 365 days since tillage button pressed, so the tilled layer has settled for a year.

25.12 Root structure

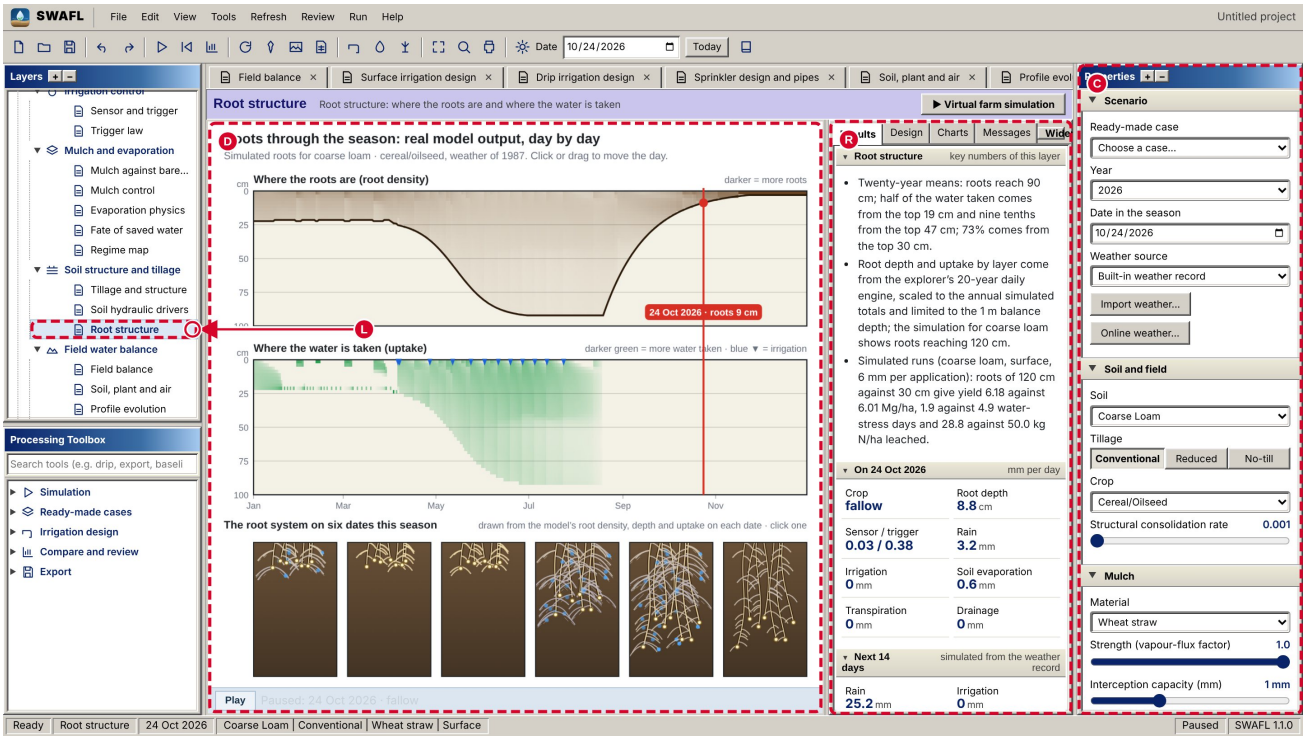


Figure 25.183: Root structure: the whole workbench.

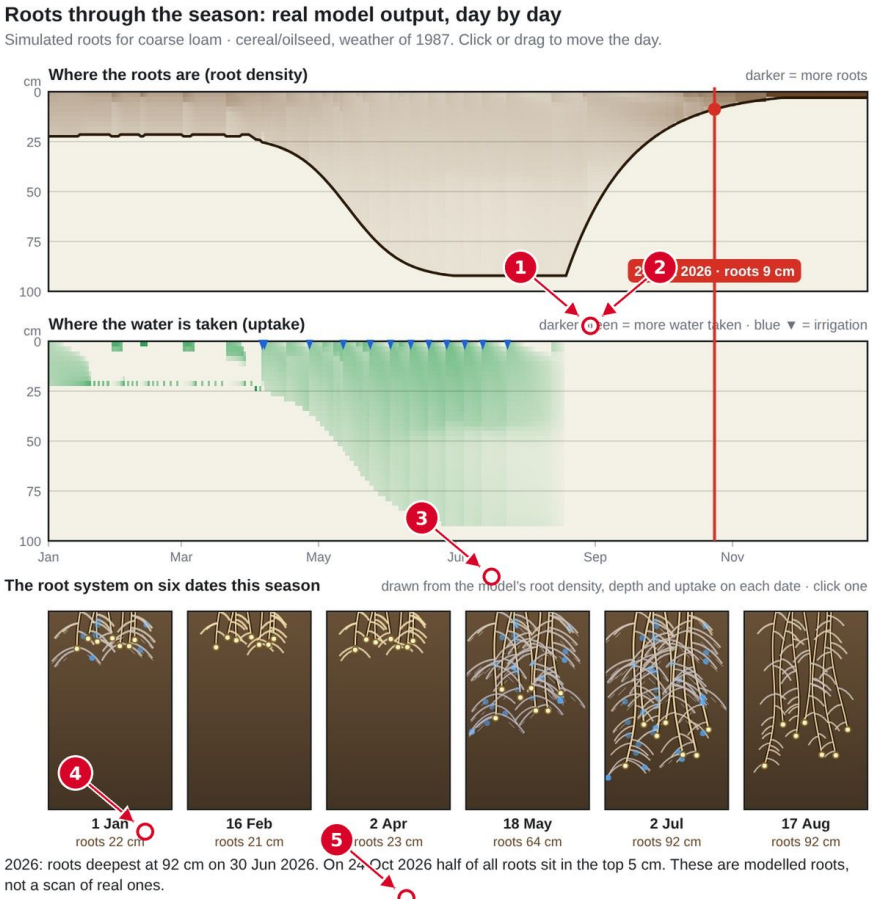


Figure 25.184: Root structure: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Heat map of relative root density by depth and day: Eq. 5.20.
- **2** Rooting depth through the season (black line): Eq. 5.18.
- **3** Heat map of water uptake per layer, u_i(d): Eqs. 5.21 to 5.23.
- **4** Root systems drawn on six dates from the same density: Eqs. 5.19 and 5.20.
- **5** Depth holding half of the uptake, z_p: Eq. 8.7.

Equations behind this screen. Eq. 5.18, Eq. 5.19, Eq. 5.20, Eq. 5.21, Eq. 5.22, Eq. 5.23, Eq. 8.7.

Eq. 5.18. Where explained: chapter "The two computing engines", section "Roots".

$$R_{\max} = \mathrm{clip}\Big(R_c\,(0.85{+}0.3\,\mathrm{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \mathrm{cm}, \quad (5.18)$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.19. Where explained: chapter "The two computing engines", section "Roots".

$$R_d \leftarrow \max\Big\{ R_d,\ \min\big\{ R_{\max}\,\mathrm{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35+\bar{s}_{\mathrm{wet}})\,\kappa \big\} \Big\}, \quad (5.19)$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.20. Where explained: chapter “The two computing engines”, section “Roots”.

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)\,(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.21. Where explained: chapter “The two computing engines”, section “Transpiration, evaporation”.

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65 \cdot \tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy} - 100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.22. Where explained: chapter “The two computing engines”, section “Transpiration, evaporation”.

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.23. Where explained: chapter “The two computing engines”, section “Transpiration, evaporation”.

$$q_i^T = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \text{clip}\Big(\frac{s_i - 0.15}{0.6};\,0,\,1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 8.7. Where explained: chapter “Soil structure and tillage layers”, section “Generation”.

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k :\ \frac{\sum_{i \le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d \in \text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

Provenance (Eq. 8.7): **[Engineered]**: diagnostic defined for the workbench.

25.12.1 The whole layer, part by part

The picture below is the complete drawing of the layer Root structure, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

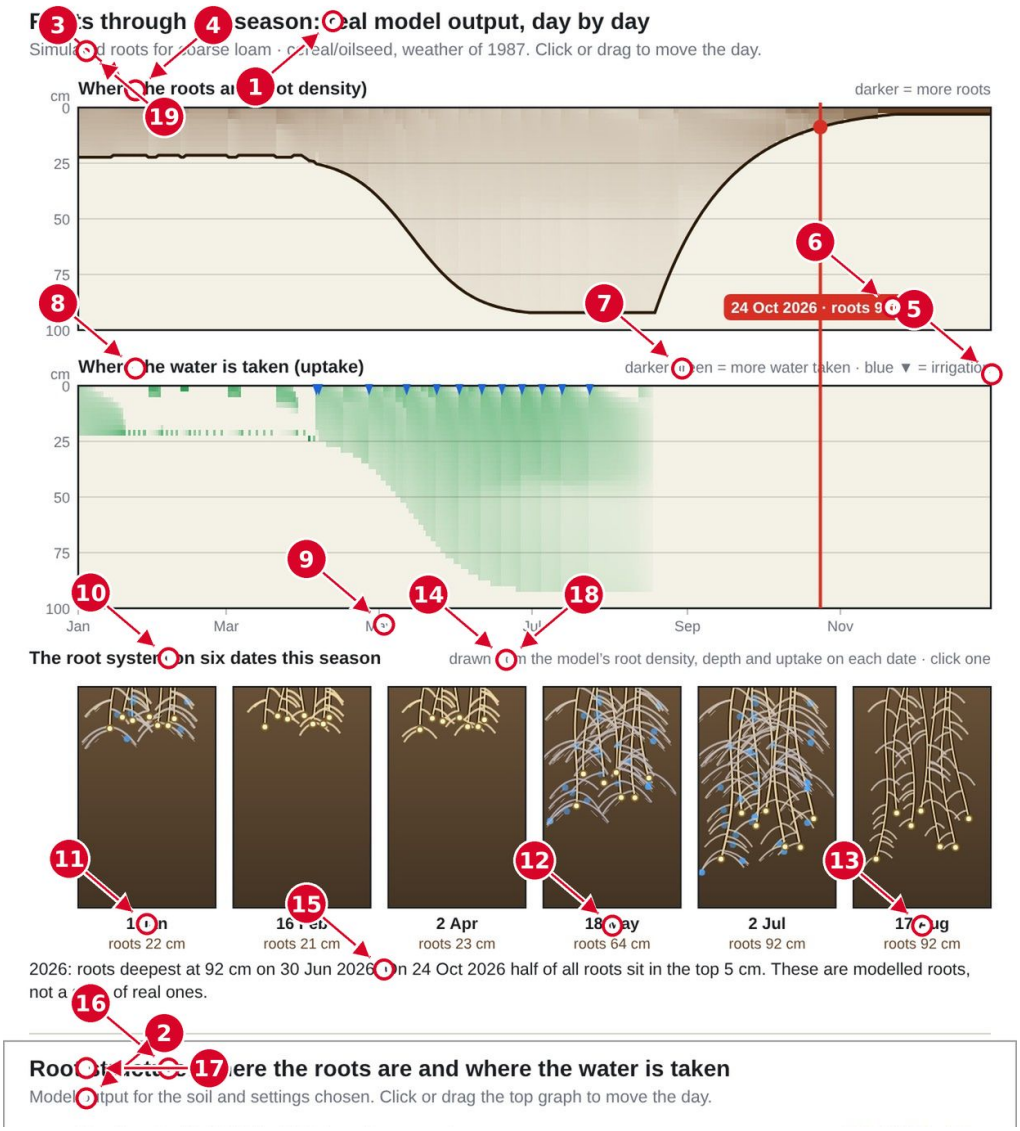


Figure 25.185: Root structure, part 1 of 4.

What the arrows point to.

- **1** The title, the roots through the season as real model output, day by day.
- **2** The instruction, the day is moved by clicking or dragging on the graphs.
- **3** The root density by depth and day, darker where there are more roots.
- **4** The black line is the rooting depth, which falls and grows with the season.
- **5** The red day cursor, the day of the clock.
- **6** The label of the cursor, the date and the rooting depth on that day.
- **7** The uptake of water by depth and day, darker green for more water taken.
- **8** The blue triangles mark the irrigations.
- **9** The month axis of the season.
- **10** The six dates on which the root system is drawn, each clickable.
- **11** The root system on the first date, still small.
- **12** The root system in the middle of the season, deep and branched.
- **13** The root system late in the season with the uptake in blue.
- **14** The date and the rooting depth under each drawing.
- **15** The sentence that reports the deepest roots and where half of all the roots sit.

- **16** The second section, the root structure where the roots are and where the water is taken.
- **17** The soil, roots and water of the day.
- **18** The relative root density by depth on the day.
- **19** The note that the runs exist for coarse loam only.

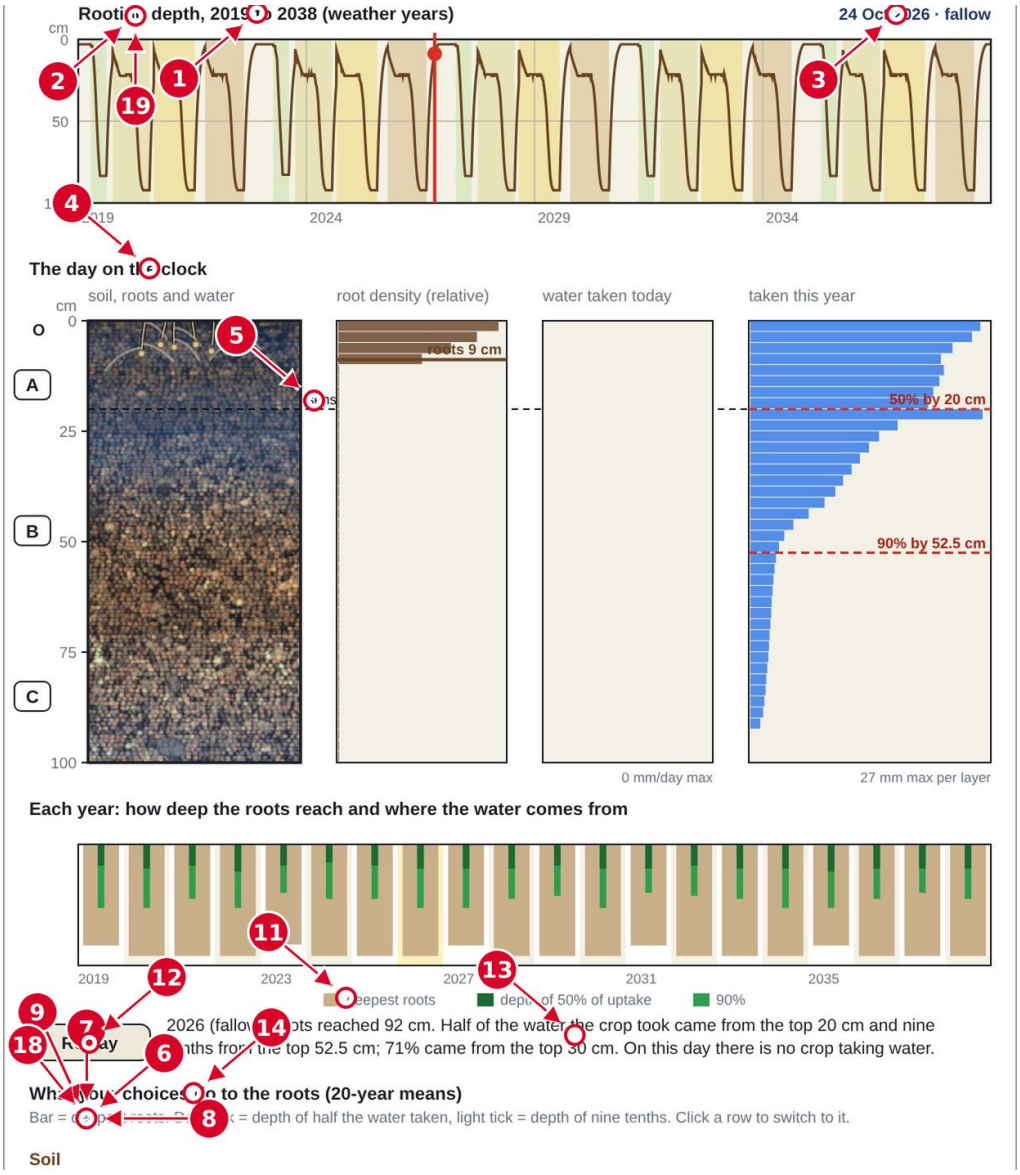


Figure 25.186: Root structure, part 2 of 4.

What the arrows point to.

- **1** The rooting depth for every year of the record, with the season colours.
- **2** The brown line is the rooting depth of each year.
- **3** The cursor of the selected day on the multi-year graph.
- **4** The day on the clock, four panels for the selected day.
- **5** The dashed line carries the sensor depth across the panels.
- **6** The water taken on the day by depth.
- **7** The water taken in the year by depth.
- **8** The depth by which half of the water was taken.
- **9** The depth by which nine tenths of the water was taken.

- **10** The deepest roots of each year and the depth of half and nine tenths of the uptake.
- **11** The legend of the deepest roots and of the two uptake depths.
- **12** The Replay button of the season.
- **13** The summary of the year, the depth of the roots and where the water came from.
- **14** What your choices do to the roots, as 20-year means; a row is clicked to switch to it.
- **15** The bar of the deepest roots of the soil.
- **16** The dark tick, the depth of half of the water taken.
- **17** The light tick, the depth of nine tenths of the water taken.
- **18** The green ticks, the depths of half and nine tenths of the uptake.
- **19** The red line, the rooting depth set for this run.

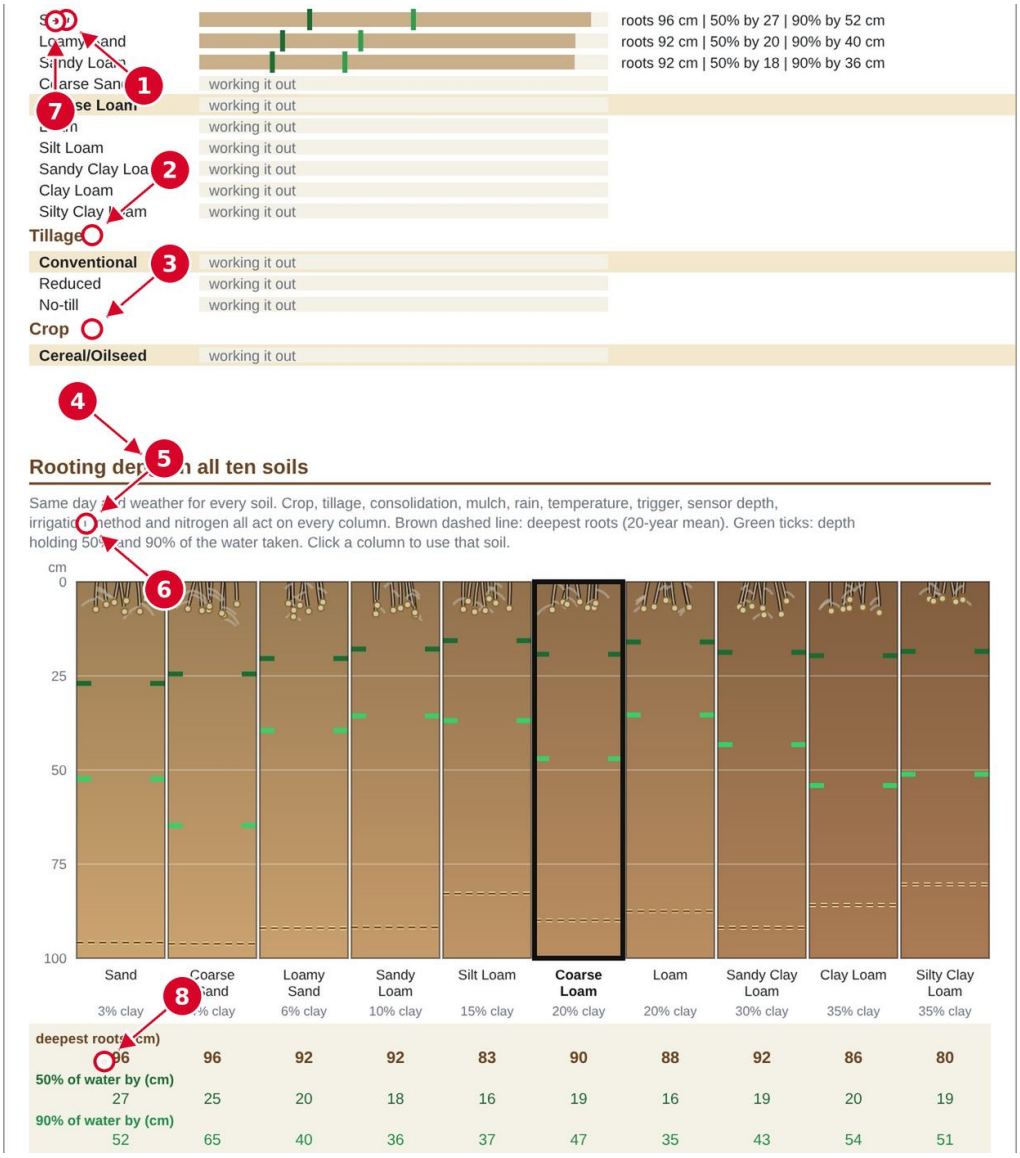


Figure 25.187: Root structure, part 3 of 4.

What the arrows point to.

- **1** The soil selected, highlighted in the list.
- **2** The tillage systems compared, conventional, reduced and no-till.
- **3** The crops compared, cereal, legume and vegetable.
- **4** The rooting depth in all ten soils, on the same day and weather.

- **5** The soil column of the selected soil.
- **6** The dashed line, the deepest roots of the 20-year mean.
- **7** The soil name and its clay content under the column.
- **8** The table of the deepest roots and of the uptake depths of the ten soils.

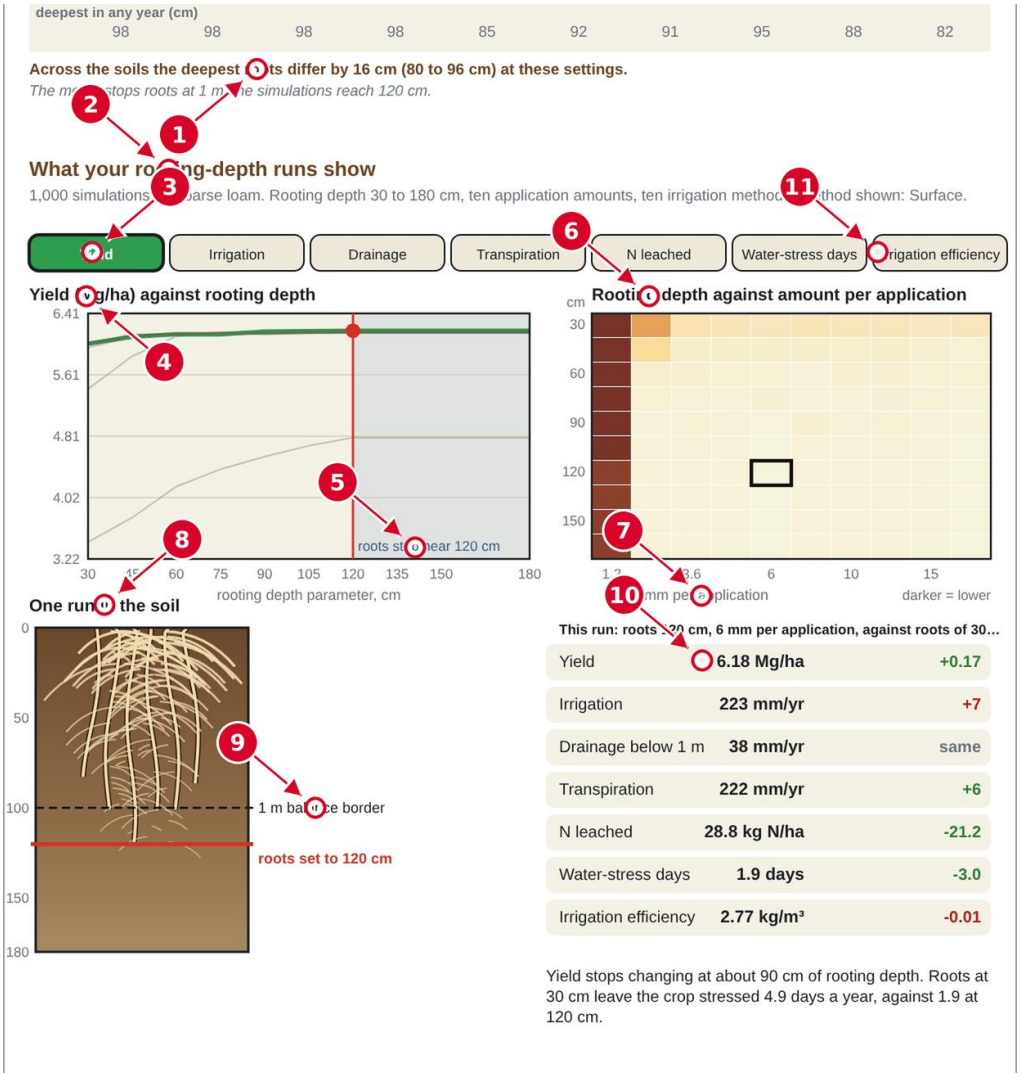


Figure 25.188: Root structure, part 4 of 4.

What the arrows point to.

- **1** The range of the deepest roots across the soils at these settings.
- **2** The rooting-depth runs, 1,000 Daisy runs on coarse loam.
- **3** The metric buttons, Yield, Irrigation, Drainage, Transpiration, N leached, Water-stress days and Irrigation efficiency.
- **4** The yield against the rooting depth for ten application amounts.
- **5** The red line, where the roots stop near 120 cm.
- **6** The map of the rooting depth against the amount per application.
- **7** The black square, the run selected.
- **8** One run in the soil, with the roots drawn to scale.
- **9** The dashed line at 1 m, the border of the balance.
- **10** The table of the run against the reference run with roots of 30 cm.
- **11** The water-stress days of the run and their change.

25.12.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

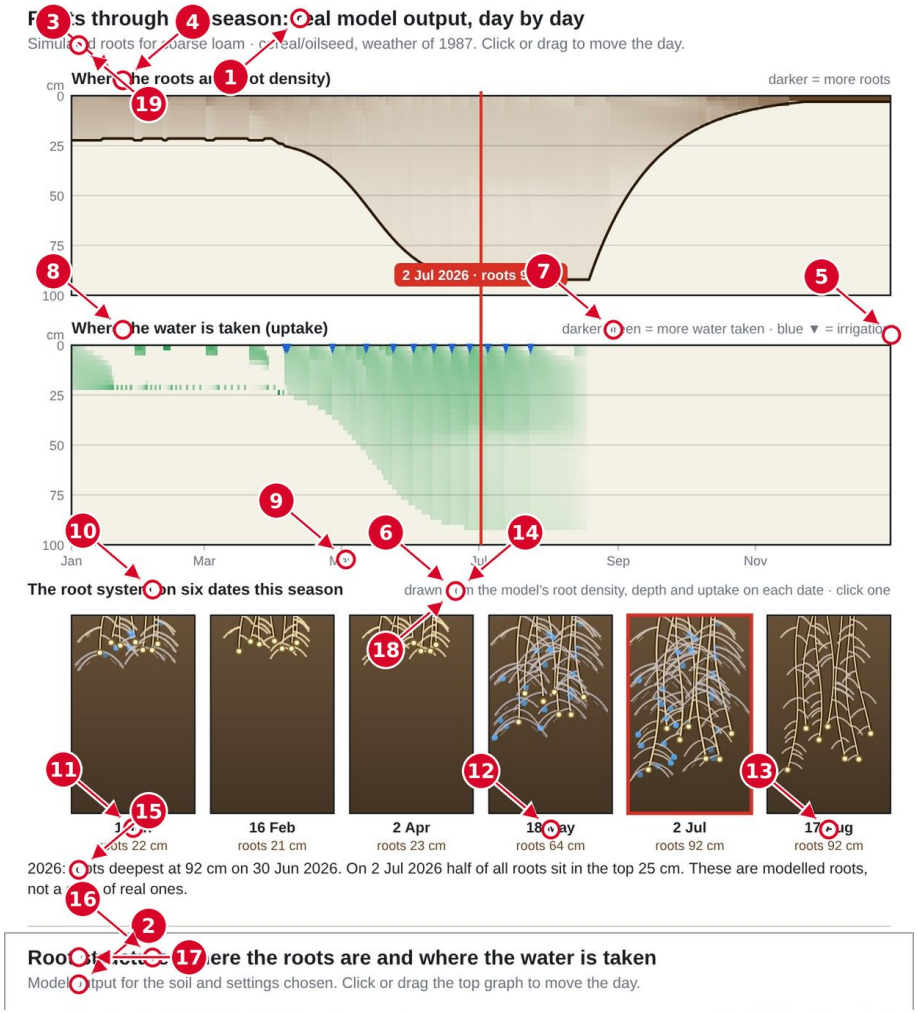


Figure 25.189: Root structure, part 1. The top graph clicked in the middle of the season, so the day on the clock moves there.

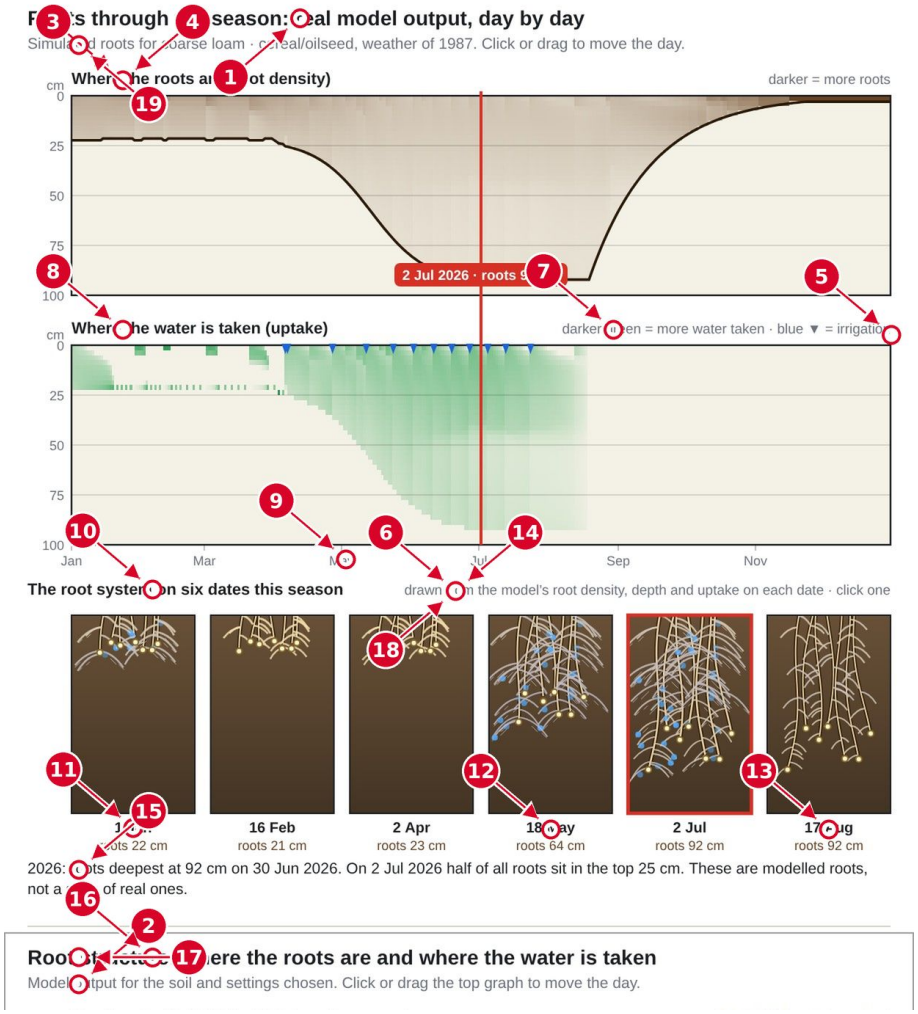


Figure 25.190: Root structure, part 1. The top graph clicked in the middle of the season, so the day on the clock moves there.

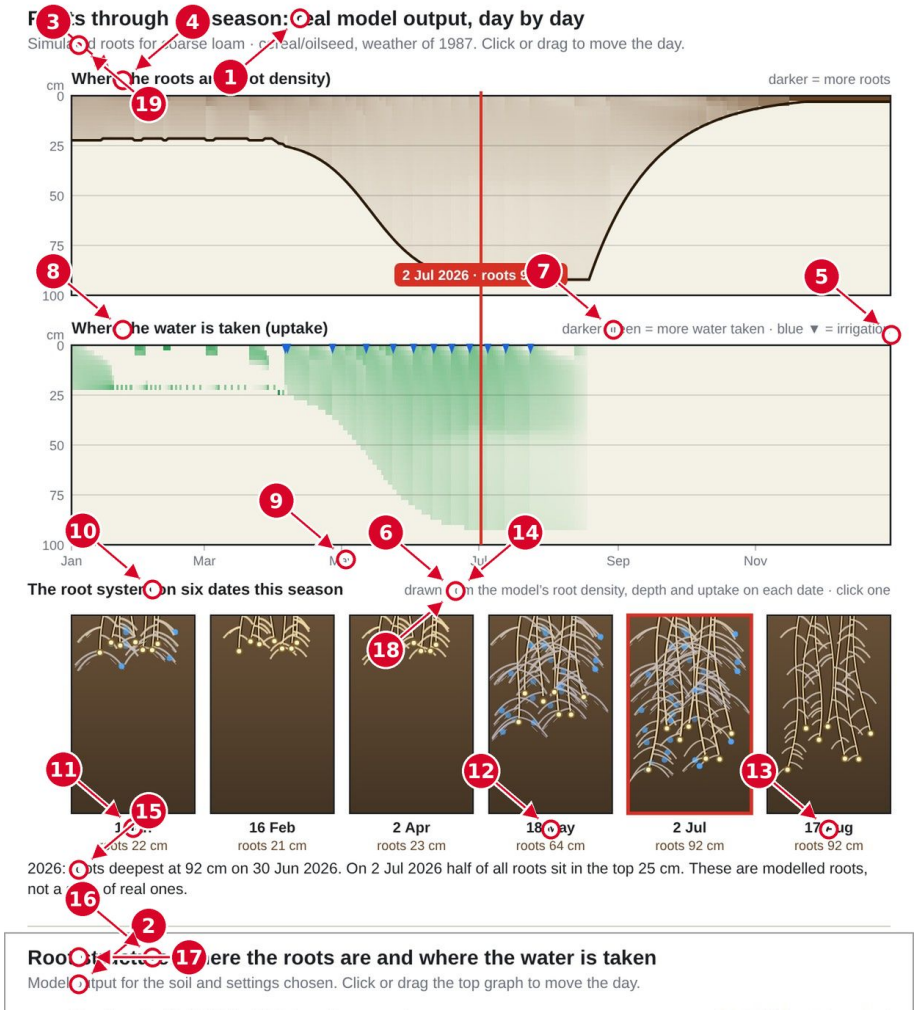


Figure 25.191: Root structure, part 1. The top graph clicked late in the season, so the day on the clock moves there.

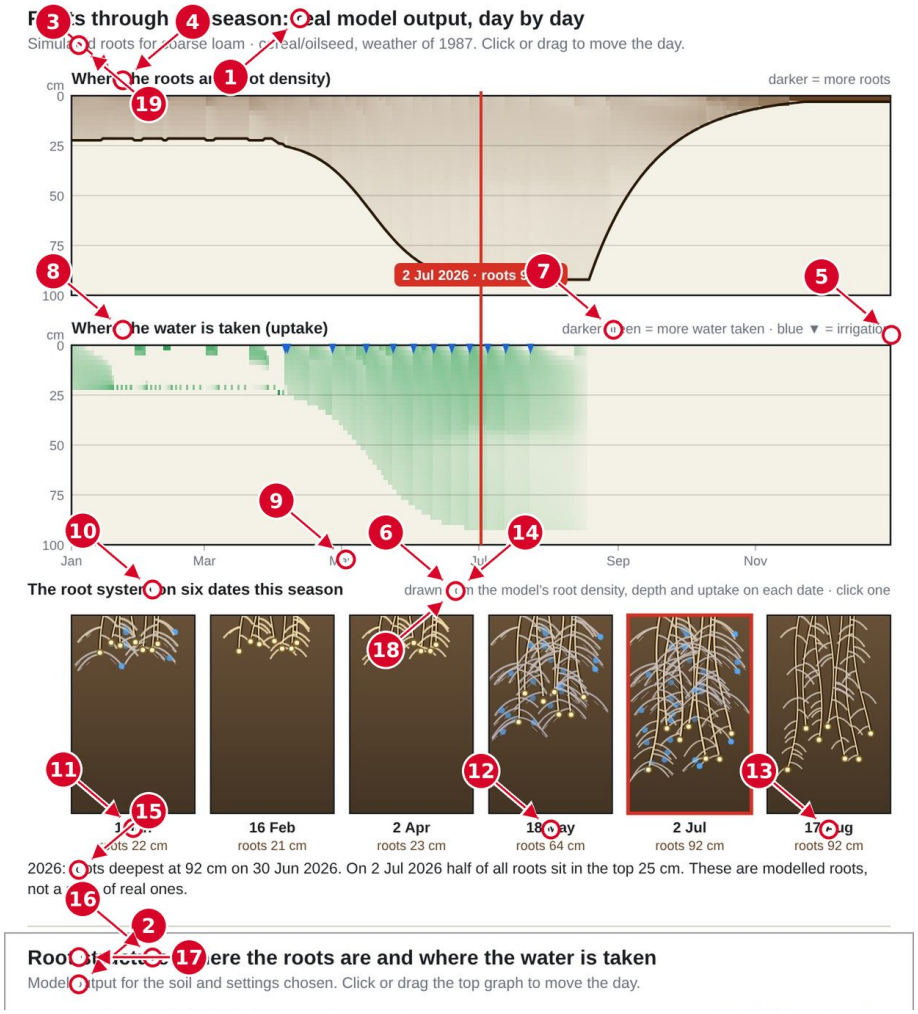


Figure 25.192: Root structure, part 1. The top graph clicked late in the season, so the day on the clock moves there.

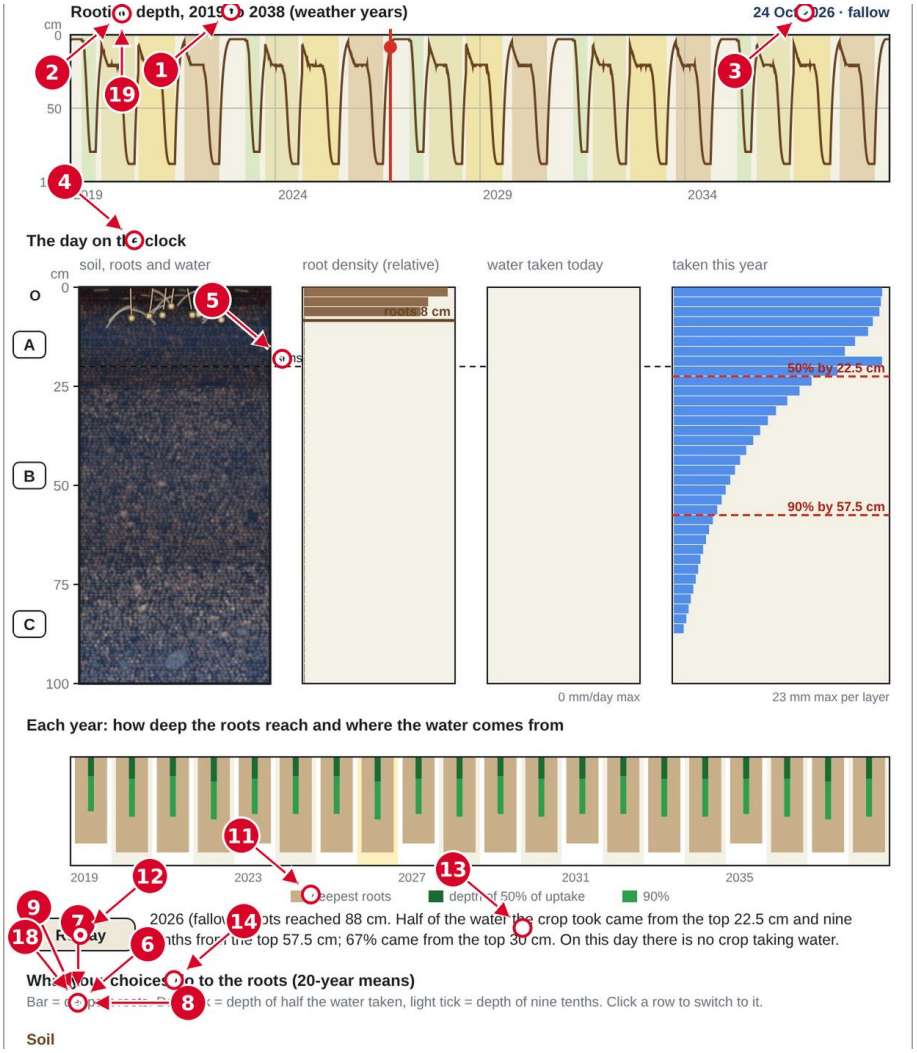


Figure 25.193: Root structure, part 2. A soil row of the list clicked, so the page switches to that soil.

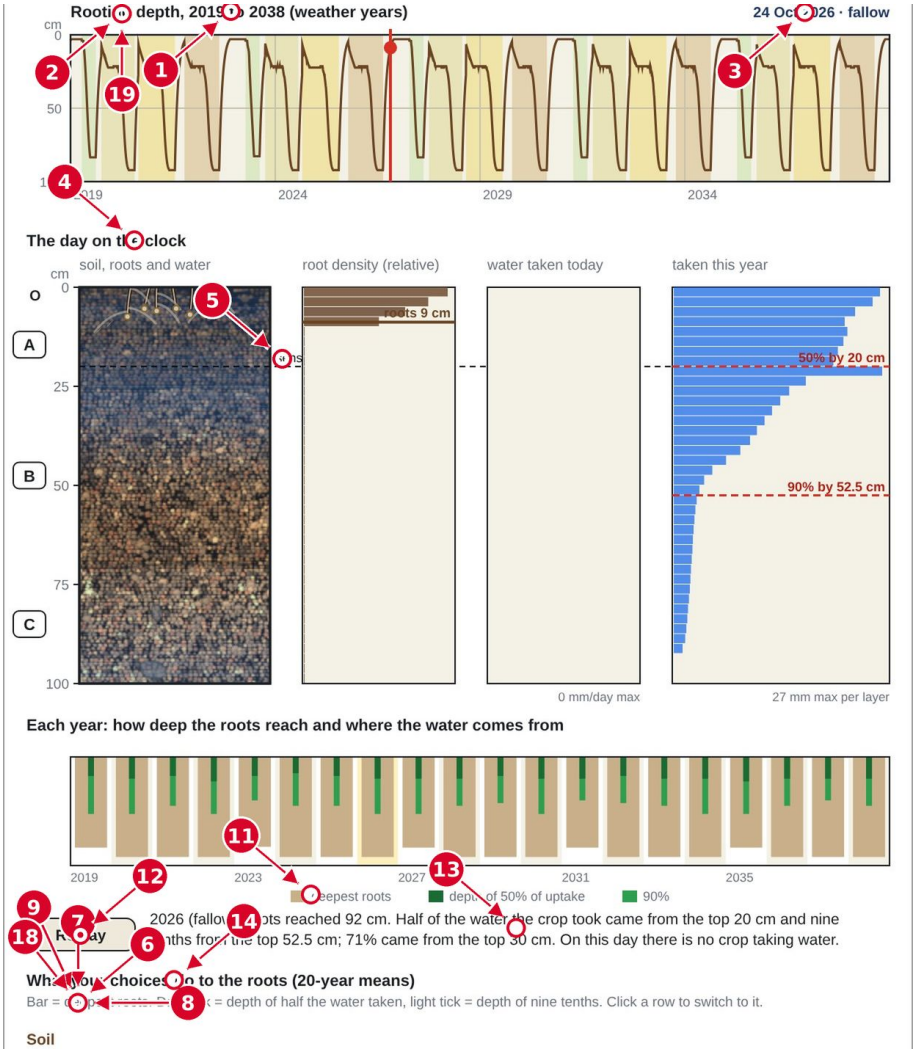


Figure 25.194: Root structure, part 2. The Irrigation button pressed, so the runs show the irrigation against the rooting depth.

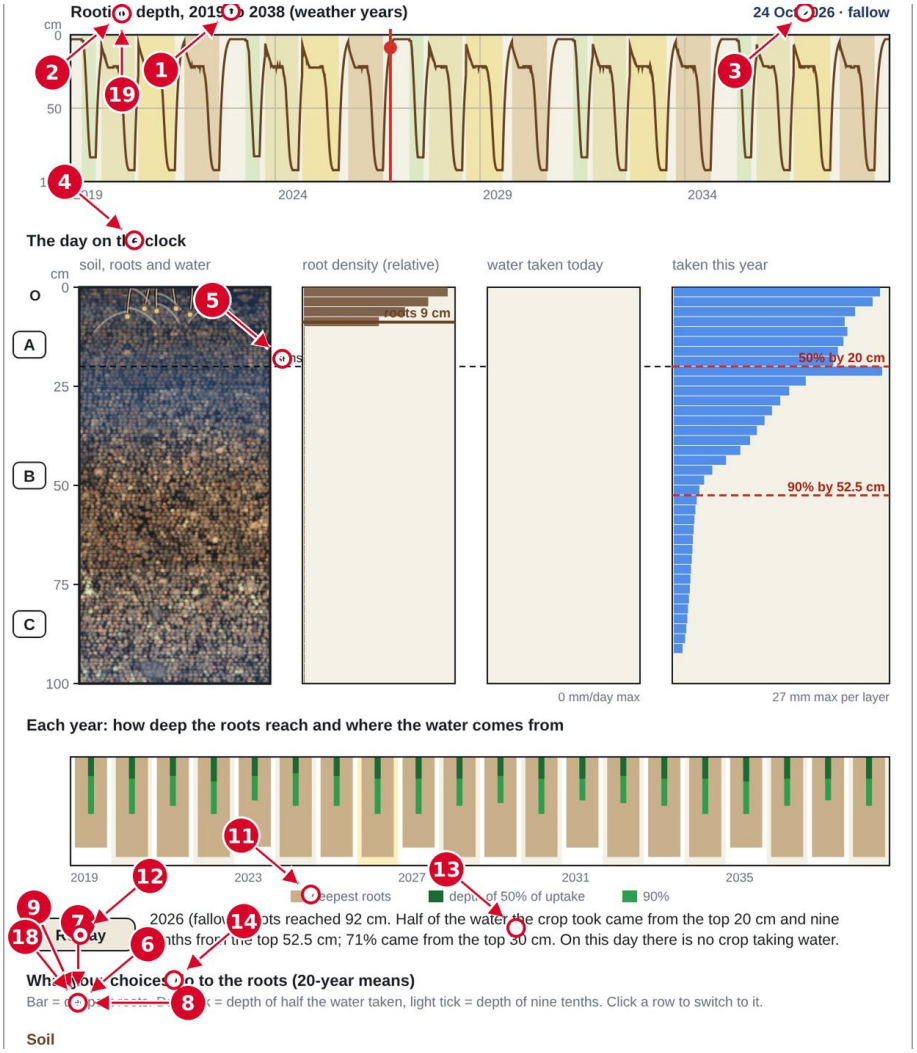


Figure 25.195: Root structure, part 2. The Drainage button pressed, so the runs show the drainage against the rooting depth.

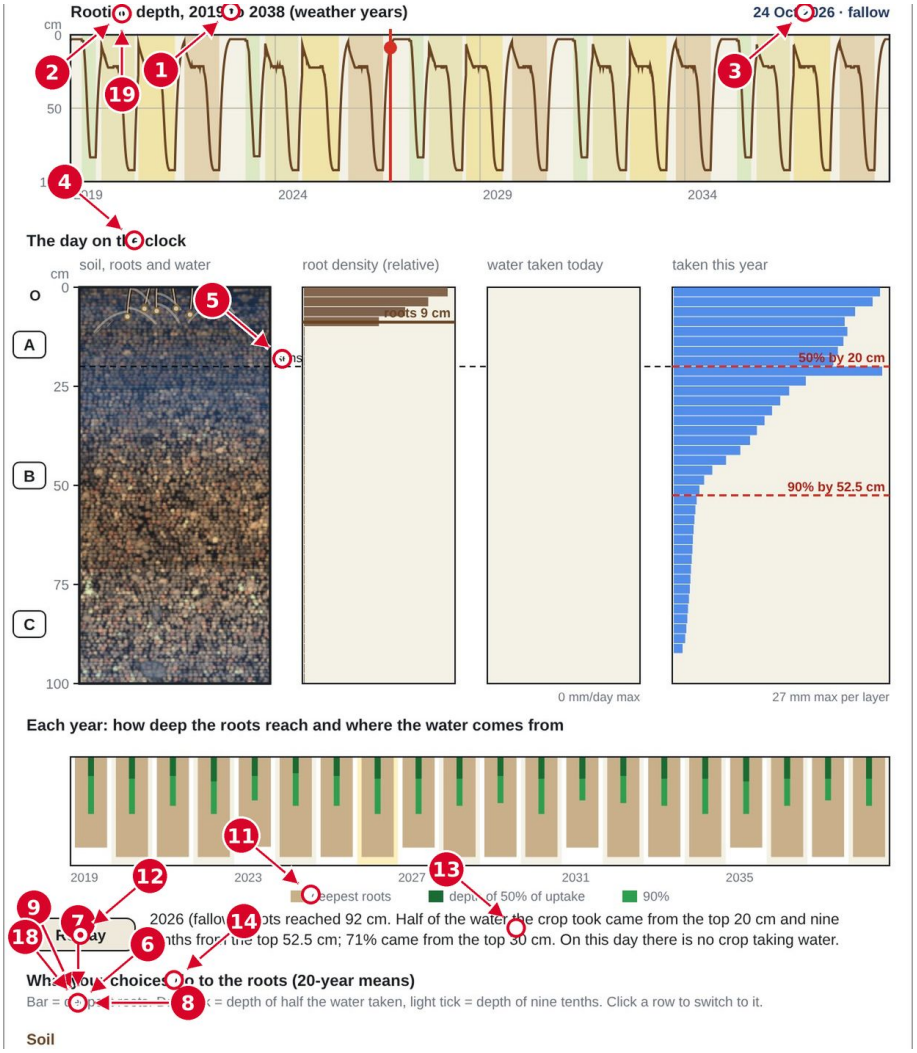


Figure 25.196: Root structure, part 2. The Transpiration button pressed.

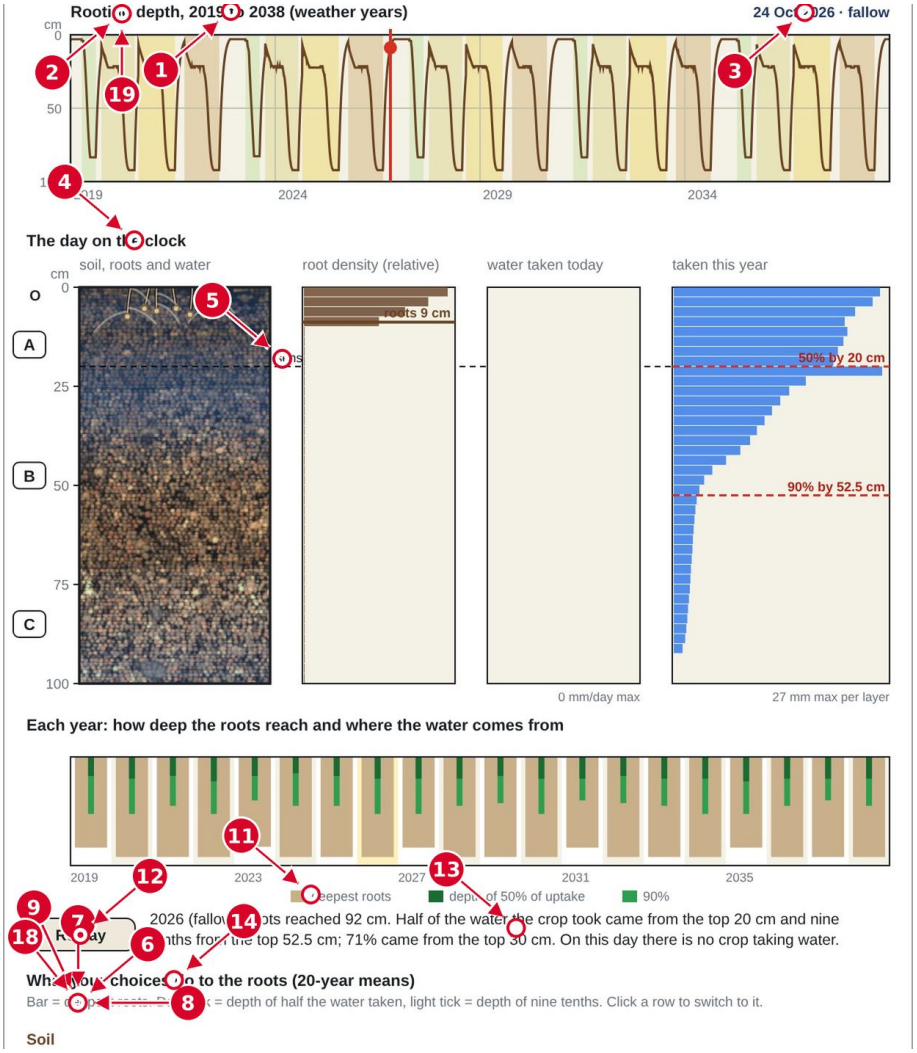


Figure 25.197: Root structure, part 2. The N leached button pressed.

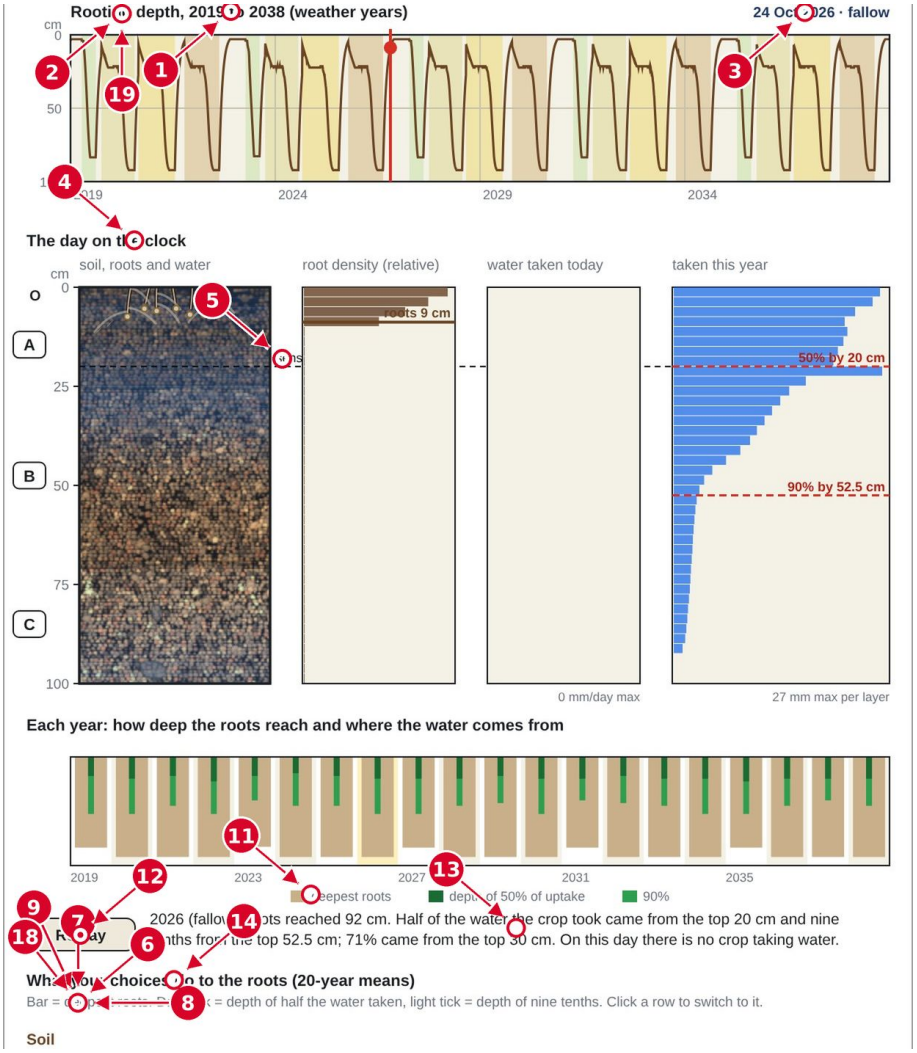


Figure 25.198: Root structure, part 2. The Water-stress days button pressed.

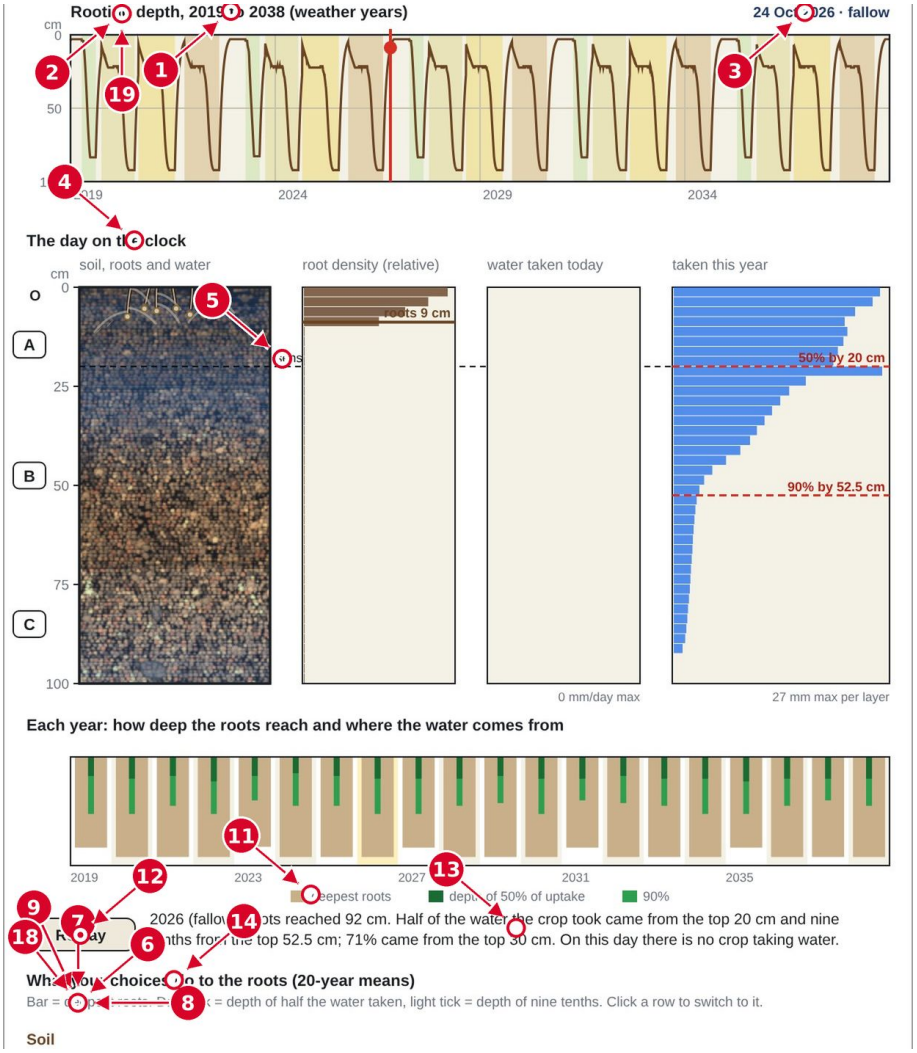


Figure 25.199: Root structure, part 2. The Irrigation efficiency button pressed.

25.13 Mulch against bare soil

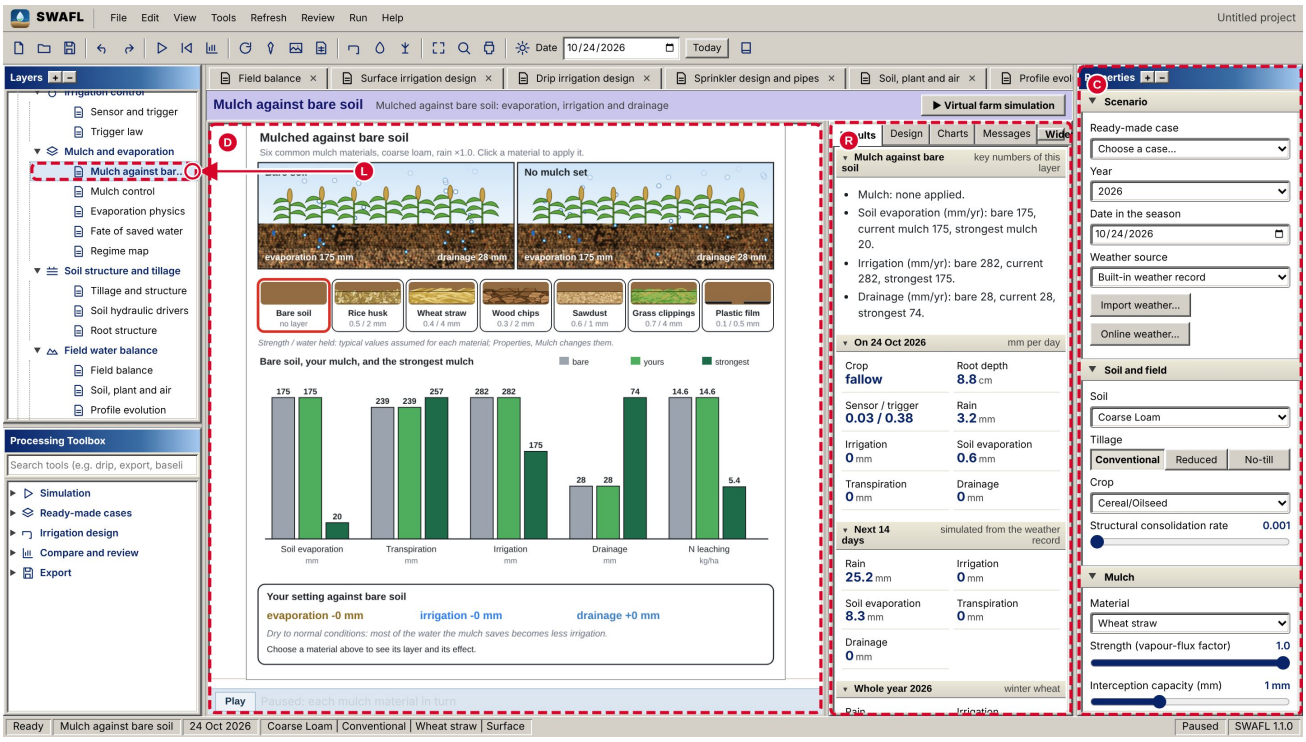


Figure 25.200: Mulch against bare soil: the whole workbench.

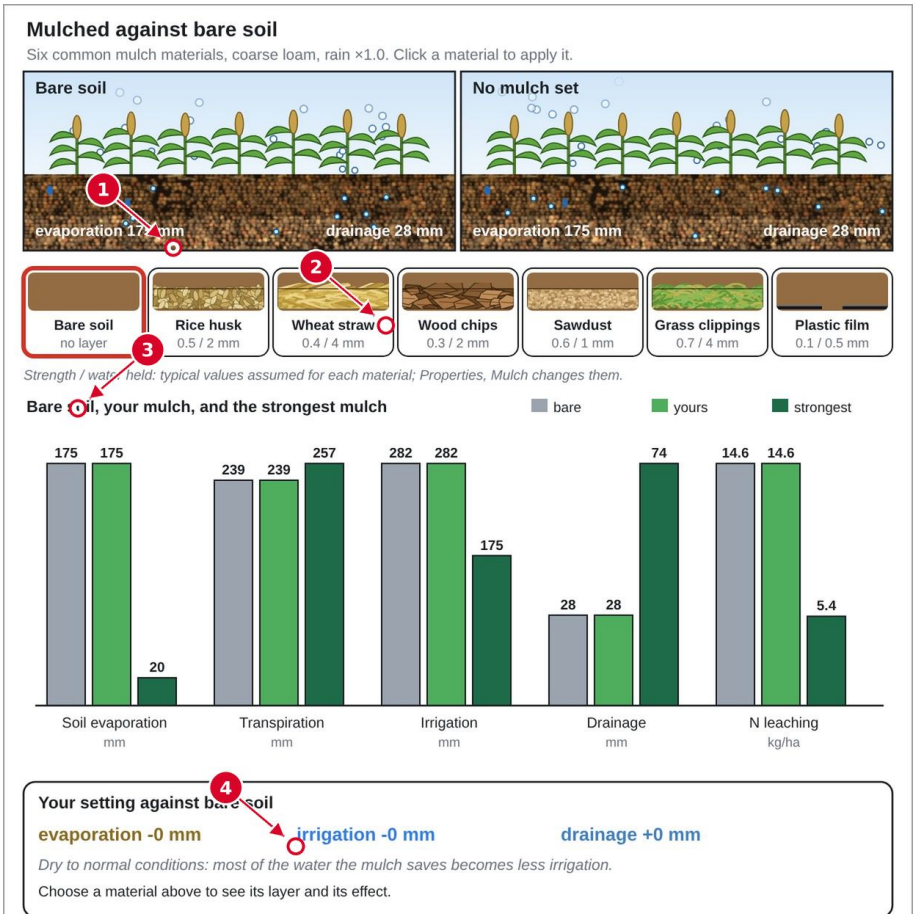


Figure 25.201: Mulch against bare soil: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Bare-soil reference: Engine 1 at mulch factor 1, Eq. 5.4.
- **2** Selected mulch (strength phi, capacity c): the input of Eq. 9.1.
- **3** Evaporation, transpiration, irrigation, drainage and nitrogen leaching of bare soil, your mulch and the strongest mulch: Eqs. 5.4, 5.7, 5.9 and 5.10.
- **4** Differences between bare soil and the selected mulch, Delta E, Delta I, Delta D: Eq. 9.1.

Equations behind this screen. Eq. 5.4, Eq. 5.7, Eq. 5.9, Eq. 9.1.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 9.1. Where explained: chapter “Mulch and evaporation layers”, section “Mulch against bare soil”.

$$\Delta E = E_{1.0} - E_{\varphi}, \qquad \Delta I = I_{1.0} - I_{\varphi}, \qquad \Delta D = D_{\varphi} - D_{1.0}, \tag{9.1}$$

Provenance (Eq. 9.1): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

25.13.1 The whole layer, part by part

The picture below is the complete drawing of the layer Mulch against bare soil, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

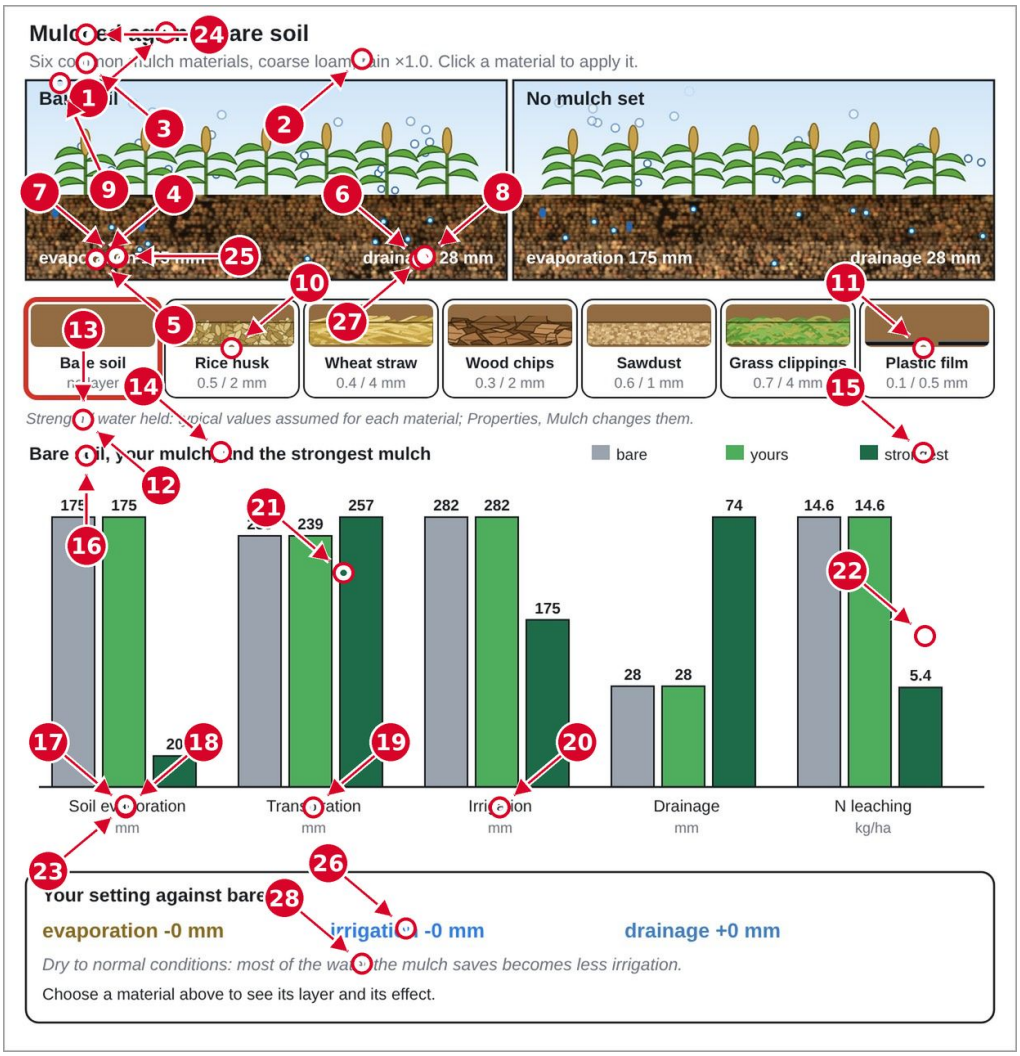


Figure 25.202: Mulch against bare soil, part 1 of 1.

What the arrows point to.

- **1** The title, the mulched soil against the bare soil.
- **2** The instruction, a material is clicked to apply it.
- **3** The bare soil panel, with its evaporation and drainage for the year.
- **4** The panel of the mulch selected, here rice husk.
- **5** The mulch layer drawn on top of the soil.

- **6** The close-up lens that shows the mulch material.
- **7** The evaporation of the bare soil in mm.
- **8** The drainage of the mulched soil in mm.
- **9** The bare soil card, no layer.
- **10** The material selected, highlighted in red.
- **11** The plastic film card, the last of the six materials.
- **12** The strength and the water held by the material, in mm of rain.
- **13** The note that the typical values are assumed and the value boxs change them.
- **14** The chart of the bare soil, your mulch and the strongest mulch.
- **15** The legend of the three bars.
- **16** The bar of the bare soil, grey.
- **17** The bar of the mulch selected, light green.
- **18** The bar of the strongest mulch, dark green.
- **19** The transpiration group, nearly the same for the three.
- **20** The irrigation group, lower under mulch.
- **21** The drainage group, higher under mulch.
- **22** The nitrogen leaching group, in kg per ha.
- **23** The unit of each group.
- **24** The result box, the mulch against the bare soil.
- **25** The change of the evaporation.
- **26** The change of the irrigation.
- **27** The change of the drainage.
- **28** The note that explains where the saved water goes.

25.13.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

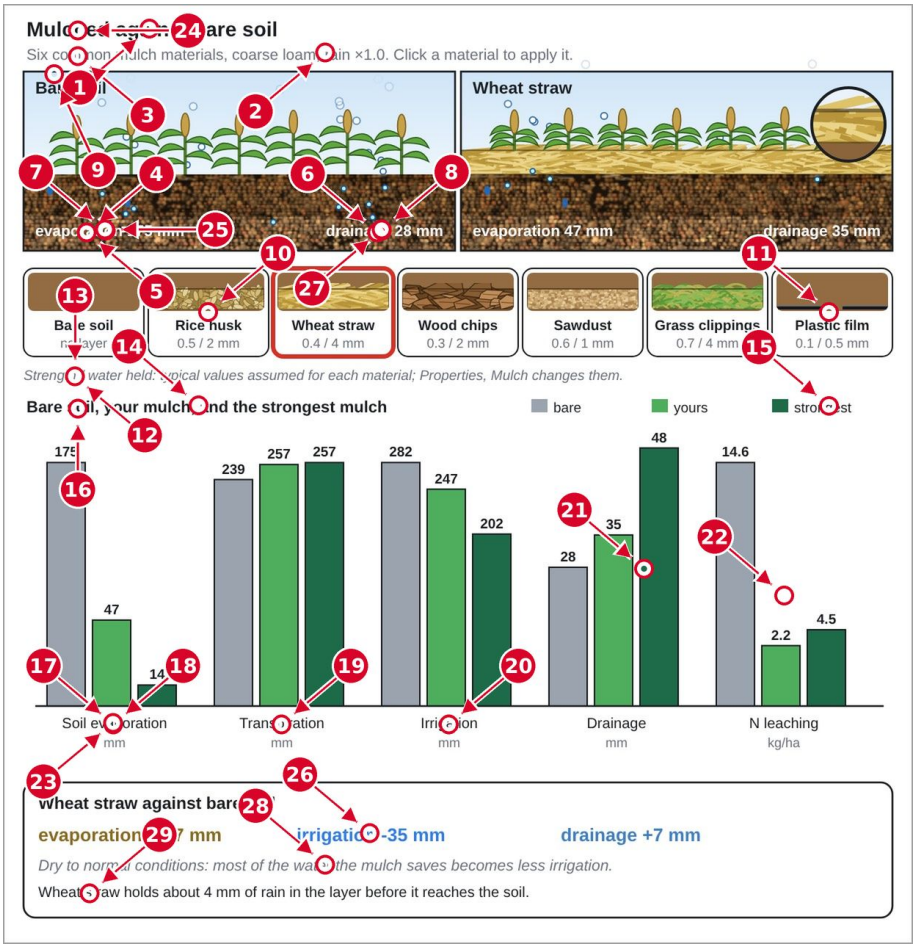


Figure 25.203: Mulch against bare soil, part 1. Wheat straw chosen as the material.

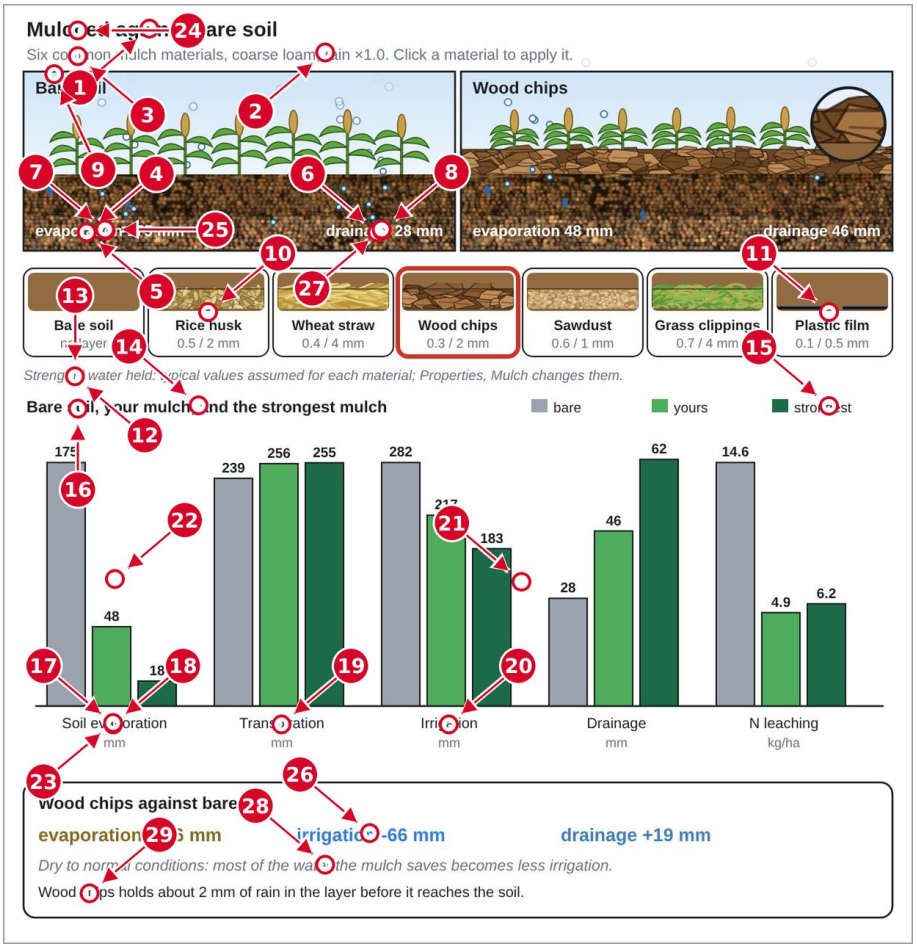


Figure 25.204: Mulch against bare soil, part 1. Wood chips chosen as the material.

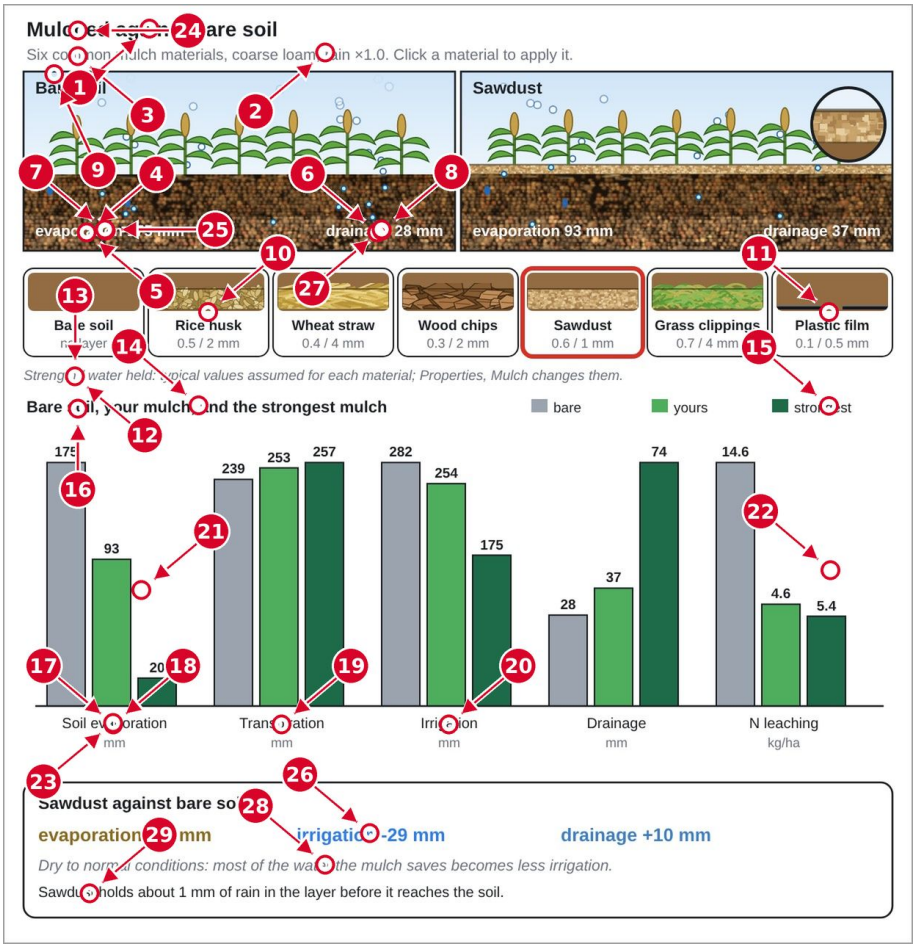


Figure 25.205: Mulch against bare soil, part 1. Sawdust chosen as the material.

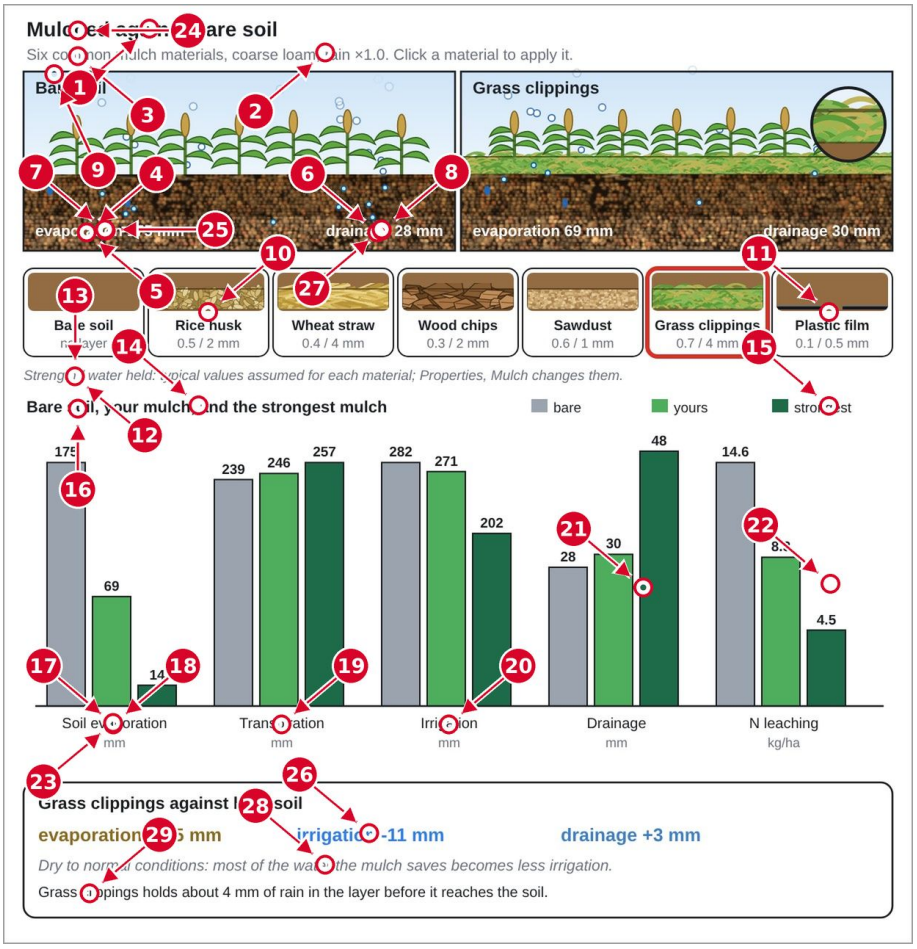


Figure 25.206: Mulch against bare soil, part 1. Grass clippings chosen as the material.

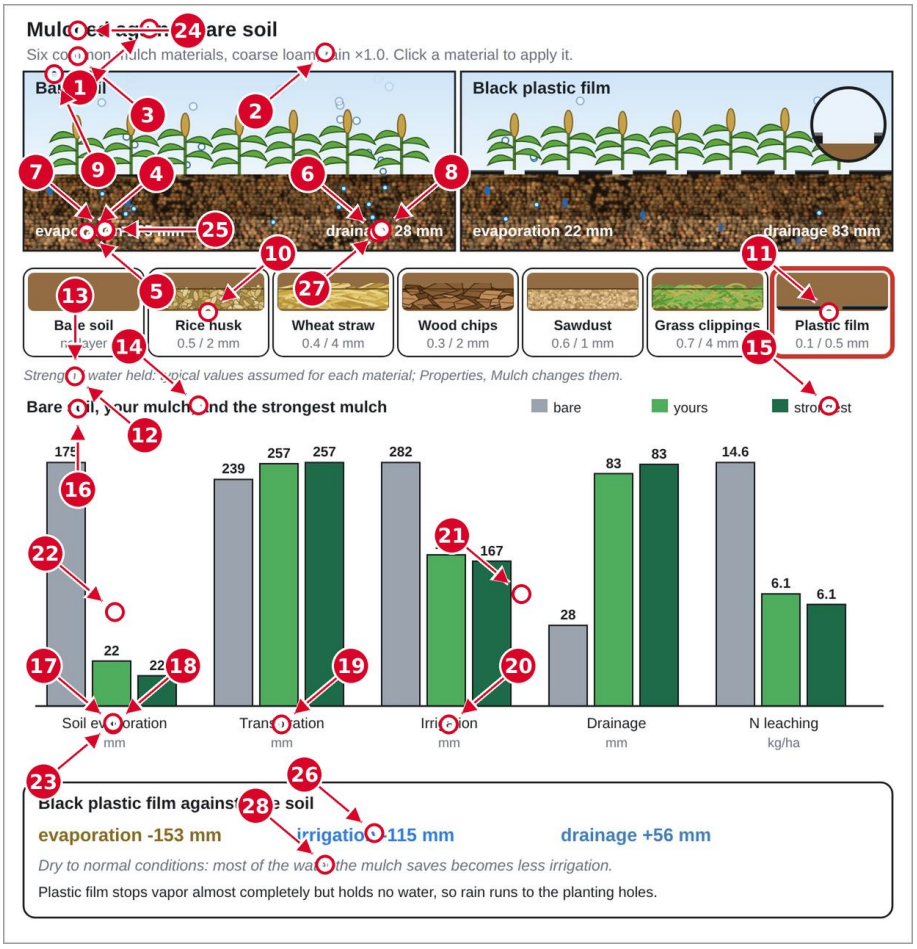


Figure 25.207: Mulch against bare soil, part 1. Plastic film chosen as the material.

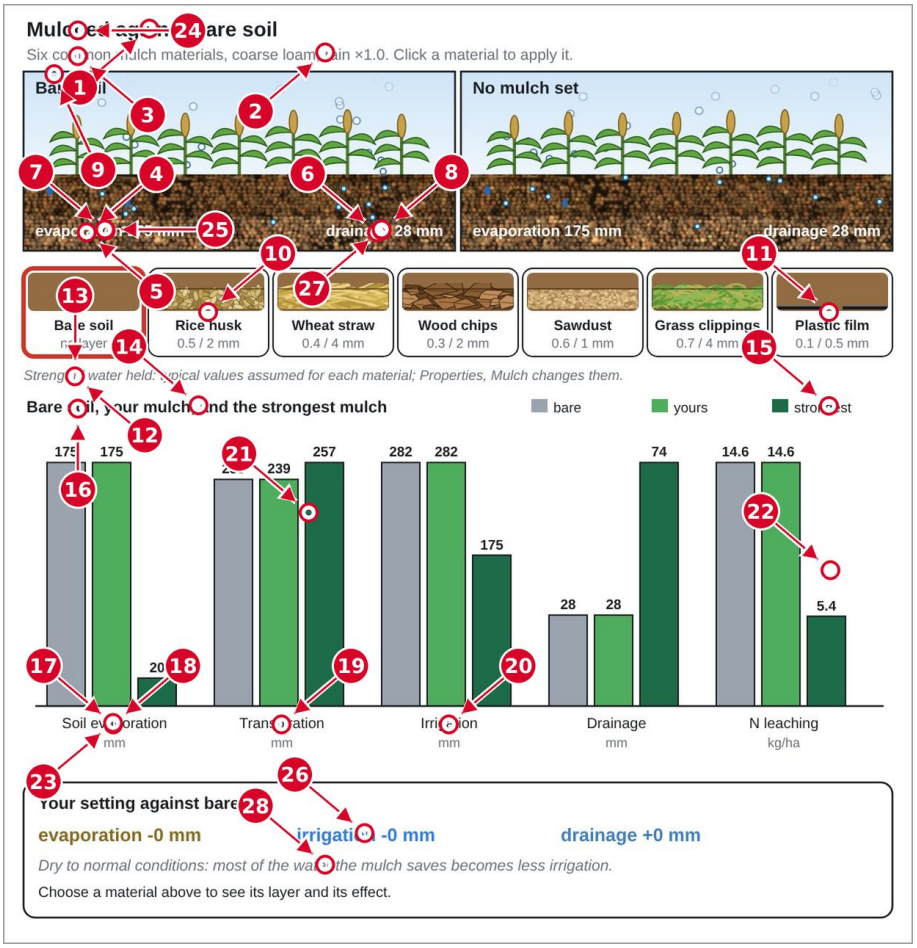


Figure 25.208: Mulch against bare soil, part 1. Bare soil chosen, so the comparison shows no change.

25.14 Mulch control

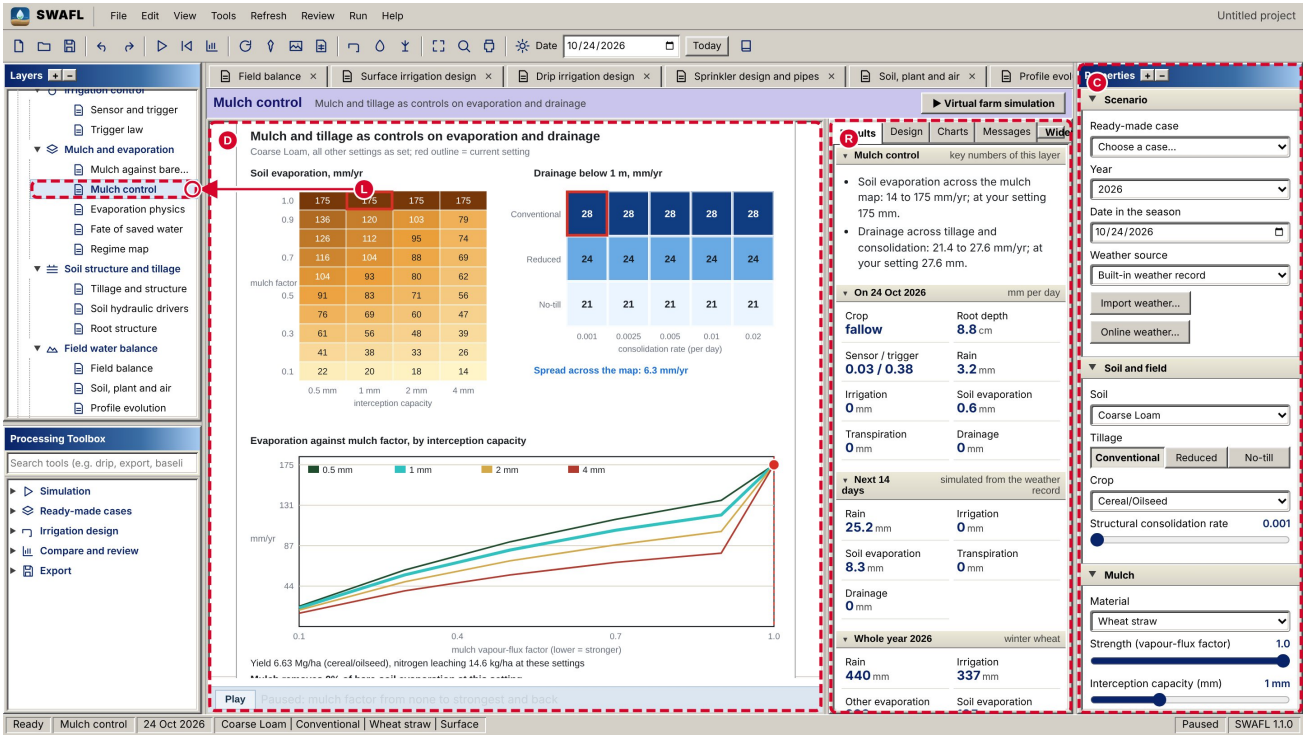


Figure 25.209: Mulch control: the whole workbench.

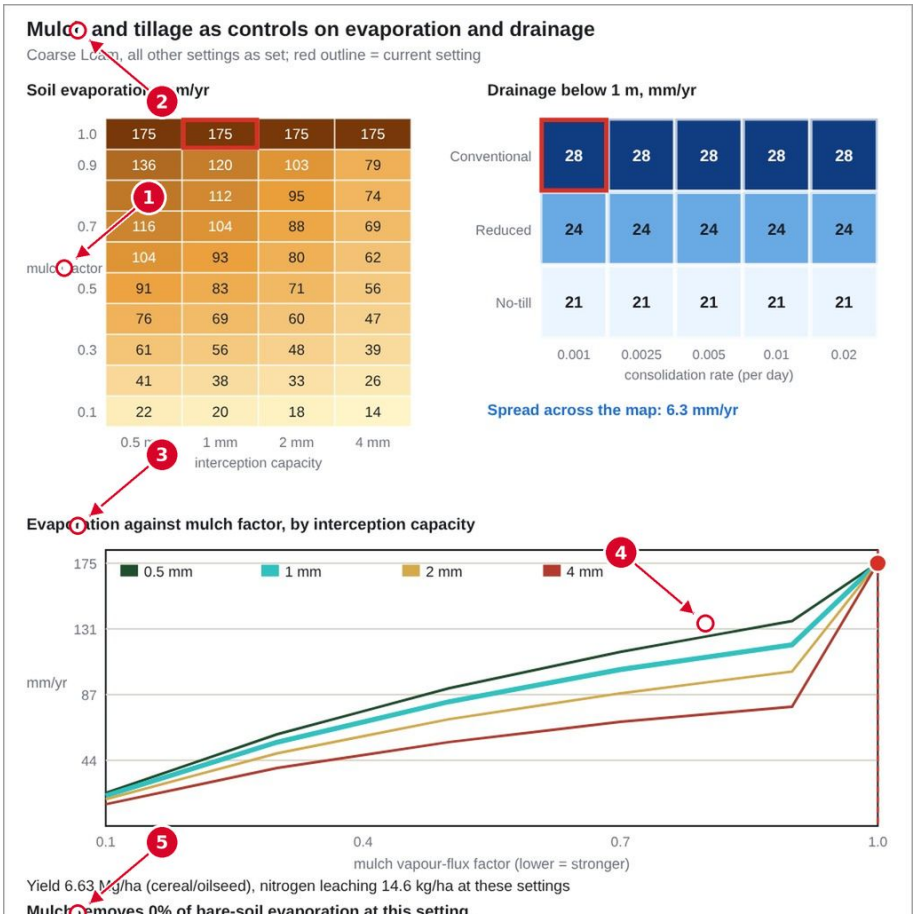


Figure 25.210: Mulch control: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Matrix of soil evaporation E(phi, c) for each mulch factor and interception capacity: Eq. 5.4.
- **2** Matrix of drainage D(tillage, consolidation rate): Eq. 5.9.
- **3** Evaporation against mulch factor for four capacities: Eq. 5.4.
- **4** Red dot: your current setting.
- **5** “Mulch removes 31 % of bare-soil evaporation”: Eq. 9.1.

Equations behind this screen. Eq. 5.4, Eq. 5.9.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

25.14.1 The whole layer, part by part

The picture below is the complete drawing of the layer Mulch control, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

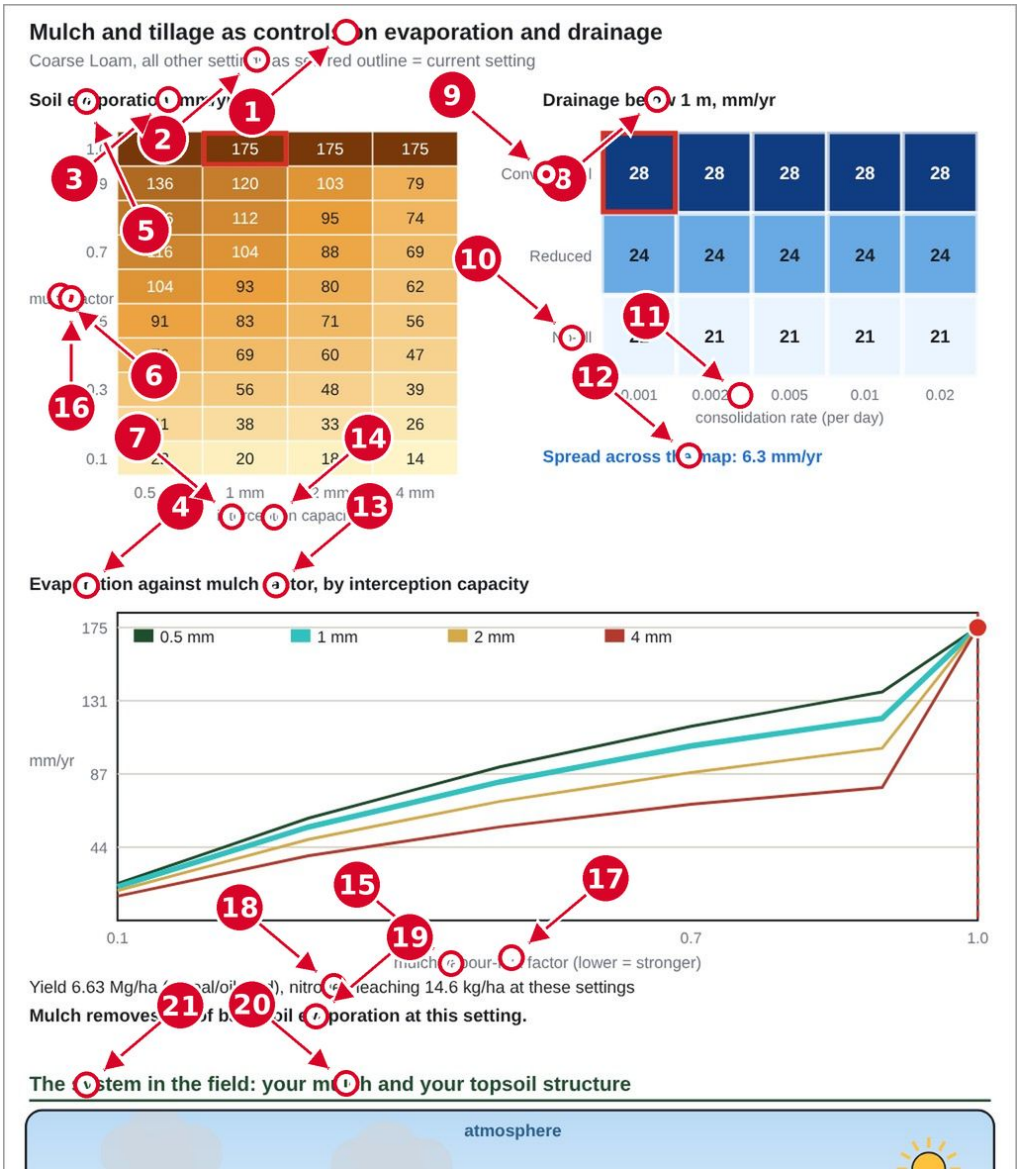


Figure 25.211: Mulch control, part 1 of 2.

What the arrows point to.

- **1** The title, mulch and tillage as controls on evaporation and drainage.
- **2** The note that the red outline marks the current setting.
- **3** The map of the soil evaporation in mm per year.
- **4** The red outlined cell, the evaporation at the current mulch factor and interception capacity.
- **5** The top row, no mulch effect, the bare soil evaporation.
- **6** The mulch factor axis, from 1.0 for no mulch down to 0.1 for the strongest.
- **7** The interception capacity axis, from 0.5 to 4 mm.
- **8** The map of the drainage below 1 m in mm per year.
- **9** The red outlined cell, the drainage under the current tillage and consolidation rate.
- **10** The no-till row, with the smallest drainage.
- **11** The consolidation rate axis, per day.
- **12** The spread of the drainage across the map.
- **13** The curves of the evaporation against the mulch factor.
- **14** The legend of the interception capacities.

- **15** The thick line, the curve of the capacity in force.
- **16** The red dashed line, the current mulch factor.
- **17** The mulch vapour-flux factor axis.
- **18** The yield and the nitrogen leaching at these settings.
- **19** The share of the bare-soil evaporation that the mulch removes.
- **20** The system in the field, your mulch and your topsoil structure.
- **21** The lens that shows the mulch material.

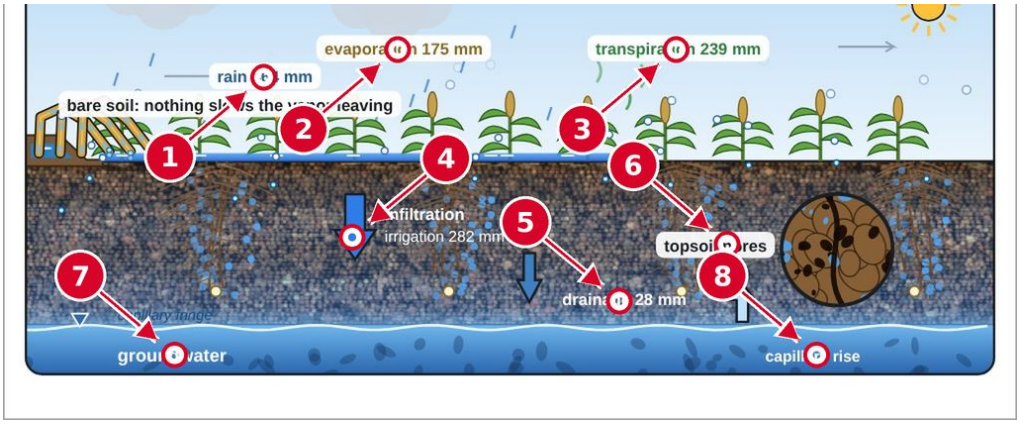


Figure 25.212: Mulch control, part 2 of 2.

What the arrows point to.

- **1** The rain of the year.
- **2** The evaporation of the year under the mulch.
- **3** The transpiration of the year.
- **4** The infiltration arrow with the irrigation applied.
- **5** The drainage arrow.
- **6** The topsoil pores, drawn as an aggregate.
- **7** The groundwater.
- **8** The capillary rise.

25.14.2 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

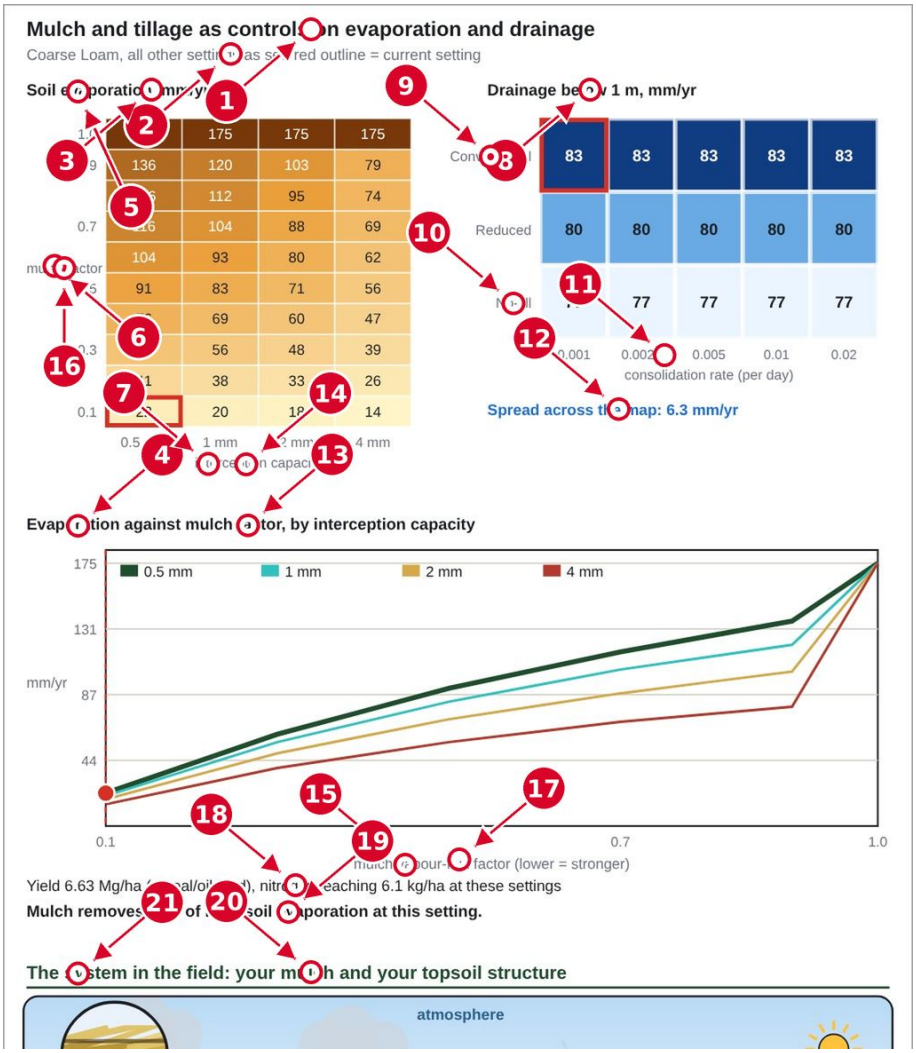


Figure 25.213: Mulch control, part 1. A cell of the evaporation map clicked, so that mulch factor and capacity become the current setting.

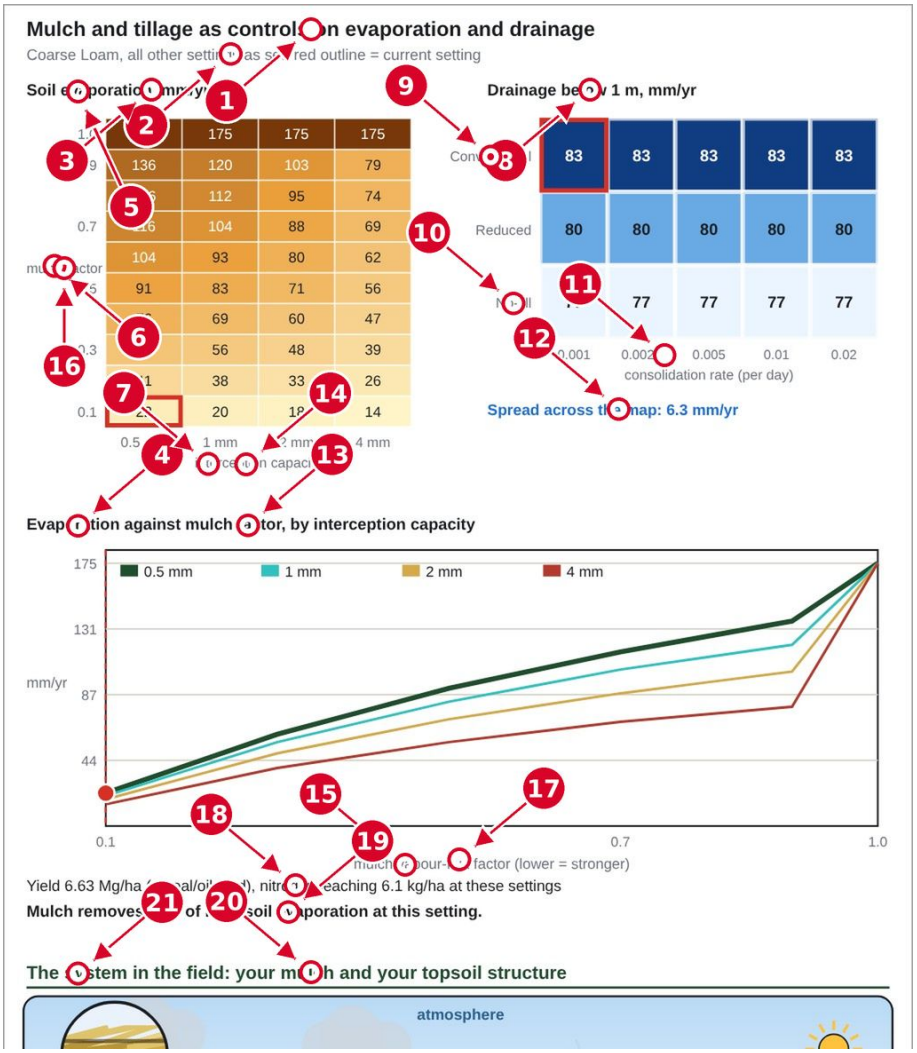


Figure 25.214: Mulch control, part 1. A cell of the drainage map clicked, so that tillage and consolidation rate become the current setting.

25.15 Evaporation physics

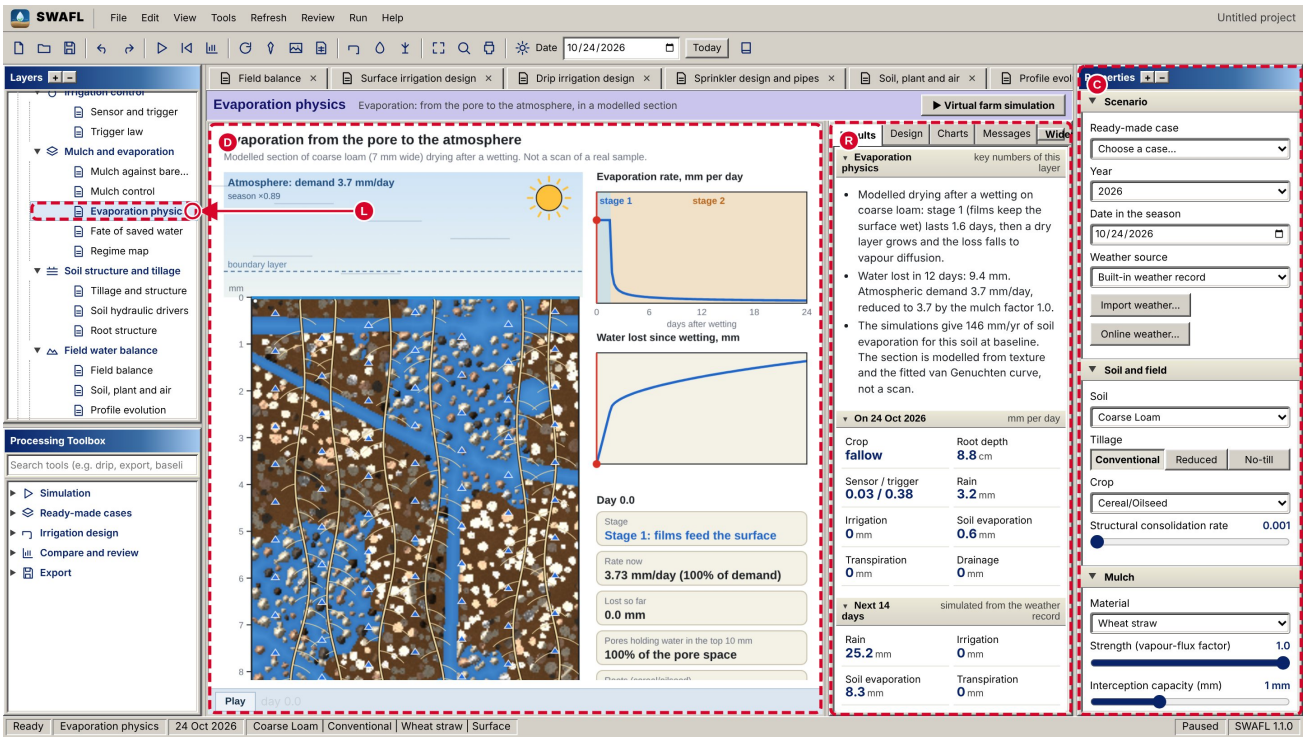


Figure 25.215: Evaporation physics: the whole workbench.

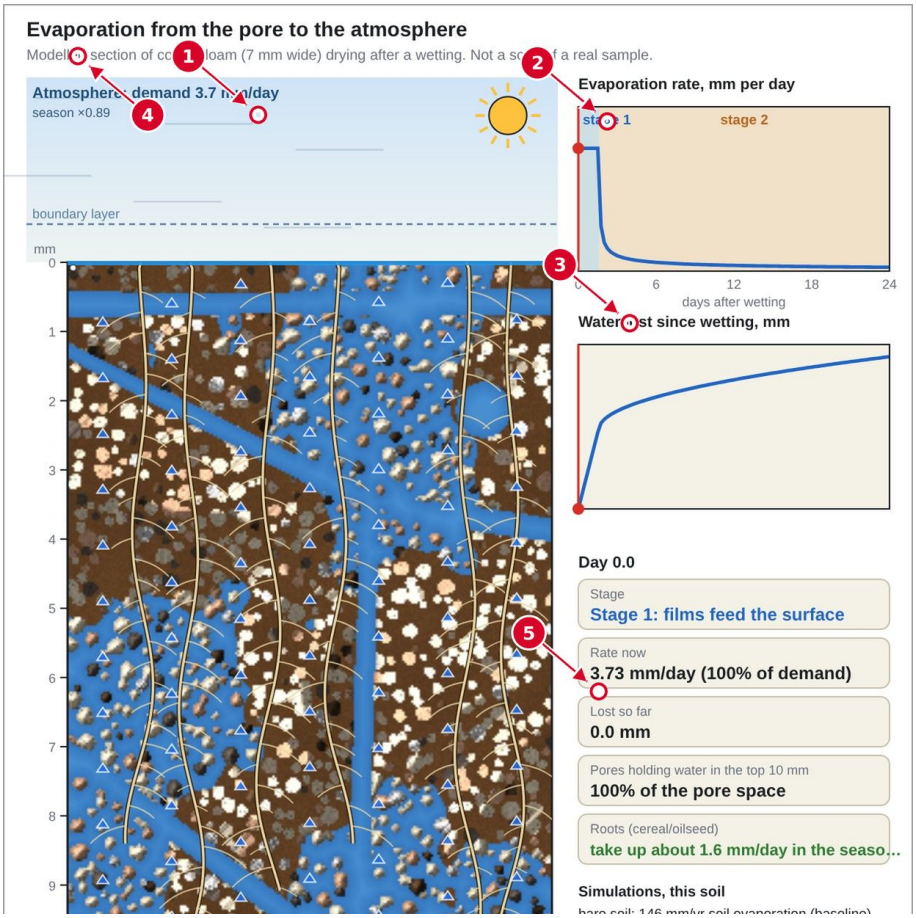


Figure 25.216: Evaporation physics: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Atmospheric demand under mulch and canopy, e_0 = e_0,b phi (1 - 0.4 c): Eq. 5.26.
- **2** Evaporation rate against time since wetting, stage 1 then stage 2: Eq. 9.2.
- **3** Cumulative water lost since wetting, Q(t): Eq. 9.3.
- **4** Section of soil: dry layer depth (Eq. 9.4) and water-filled pores from the capillary rule (Eq. 8.4) with the suction profile of Eq. 9.5.
- **5** Panel with the stage, the rate now and the water lost so far: Eqs. 9.2 and 9.3.

Equations behind this screen. Eq. 5.4, Eq. 5.26, Eq. 8.3, Eq. 8.4, Eq. 9.2, Eq. 9.3, Eq. 9.4, Eq. 9.5.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\,\left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\,\left[\frac{\text{now}.T}{\text{bare}.T}\right]\,\text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\,0.3,\,2.5\Big) \tag{5.5}$$

$$ET = ET_b\,\left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\,\text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\,0.5,\,2\Big) \tag{5.6}$$

$$I = I_b\,\left[\frac{\text{now}.I}{\text{bare}.I}\right]\,\text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\,0.4,\,2.5\Big) \tag{5.7}$$

$$Y = Y_b\,\left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.26. Where explained: chapter "The two computing engines", section "Evaporation under warming (used by the E1 layer)".

$$E_{0,\text{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65\cdot\tfrac{1}{2}[1+\sin(2\pi(\text{doy}-100)/365)]}{0.675}\,\text{clip}\big(1{-}0.12(\rho{-}1);\,0.85,\,1.12\big)\sqrt{\gamma_y}\ \text{m} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

Eq. 8.3. Where explained: chapter "Soil structure and tillage layers", section "How the section is generated".

$$\theta(h) = \theta_r + (\phi - \theta_r)\,S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

Eq. 8.4. Where explained: chapter "Soil structure and tillage layers", section "How the section is generated".

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\,g\,h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

Eq. 9.2. Where explained: chapter "Mulch and evaporation layers", section "Evaporation physics".

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

Provenance (Eq. 9.2): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.3. Where explained: chapter "Mulch and evaporation layers", section "Evaporation physics".

$$Q(t) = q_0 + \begin{cases} e_0\,t, & t \le t_1 \\ (Q_1 - q_0) + 2\,e_0\,\tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

Provenance (Eq. 9.3): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.4. Where explained: chapter “Mulch and evaporation layers”, section “Evaporation physics”.

$$\ell(t) = \min\!\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\}\ \text{mm} \quad (Q > Q_1), \tag{9.4}$$

Provenance (Eq. 9.4): **[Engineered]**: drying-front geometry chosen for drawing.

Eq. 9.5. Where explained: chapter “Mulch and evaporation layers”, section “Evaporation physics”.

$$\log_{10} h(z) = \log_{10} h_s + (\log_{10} h_i - \log_{10} h_s)(1 - e^{-(z-\ell)/\lambda}) \quad (z \ge \ell), \qquad h = 10^{5}\ \text{cm for } z < \ell. \tag{9.5}$$

Provenance (Eq. 9.5): **[Engineered]**: drying-front geometry chosen for drawing.

25.15.1 The whole layer, part by part

The picture below is the complete drawing of the layer Evaporation physics, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

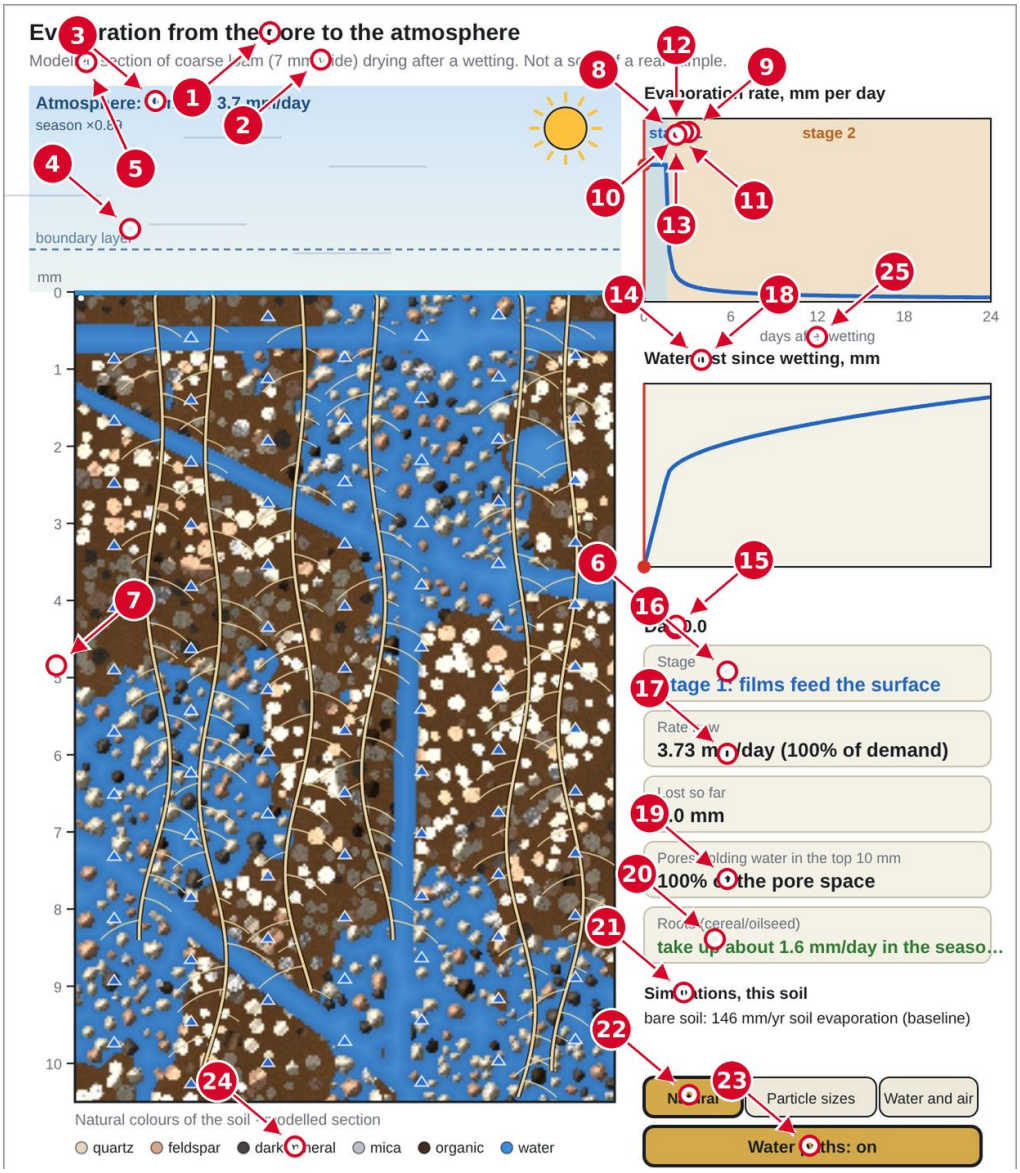


Figure 25.217: Evaporation physics, part 1 of 3.

What the arrows point to.

- **1** The title, the evaporation from the pore to the atmosphere.
- **2** The subtitle, a modelled section of the soil drying after a wetting.
- **3** The atmospheric demand in mm per day, with the season and weather scaling.
- **4** The boundary layer through which the vapour must pass.
- **5** The sun, the energy that drives the demand.
- **6** The modelled section of the soil, 7 mm wide, with grains, films and the water in blue.
- **7** The depth scale of the section in mm.
- **8** The chart of the evaporation rate in mm per day against the days after wetting.
- **9** The stage 1 band, the films feed the surface.
- **10** The stage 2 band, the dry layer limits the loss.
- **11** The blue line, the rate with the mulch.
- **12** The grey dashed line, the same soil without mulch.
- **13** The red dot, the day shown, which moves with the play bar.
- **14** The chart of the water lost since wetting in mm.

- **15** The day shown on the clock.
- **16** The stage card, whether the films feed the surface or the dry layer limits the loss.
- **17** The rate now in mm per day and its share of the demand.
- **18** The water lost so far and the water saved by the mulch.
- **19** The share of the pore space still holding water in the top 10 mm.
- **20** The roots card, the uptake of the crop.
- **21** The yearly figures from your Daisy simulations for this soil.
- **22** The colour buttons of the section, Natural, Particle sizes and Water and air.
- **23** The Water paths button, which draws the flow of the water and the vapour.
- **24** The legend of the minerals and the water.
- **25** The Day shown buttons, 0.5, 1, 1.75, 2.5, 4 and 7 days.

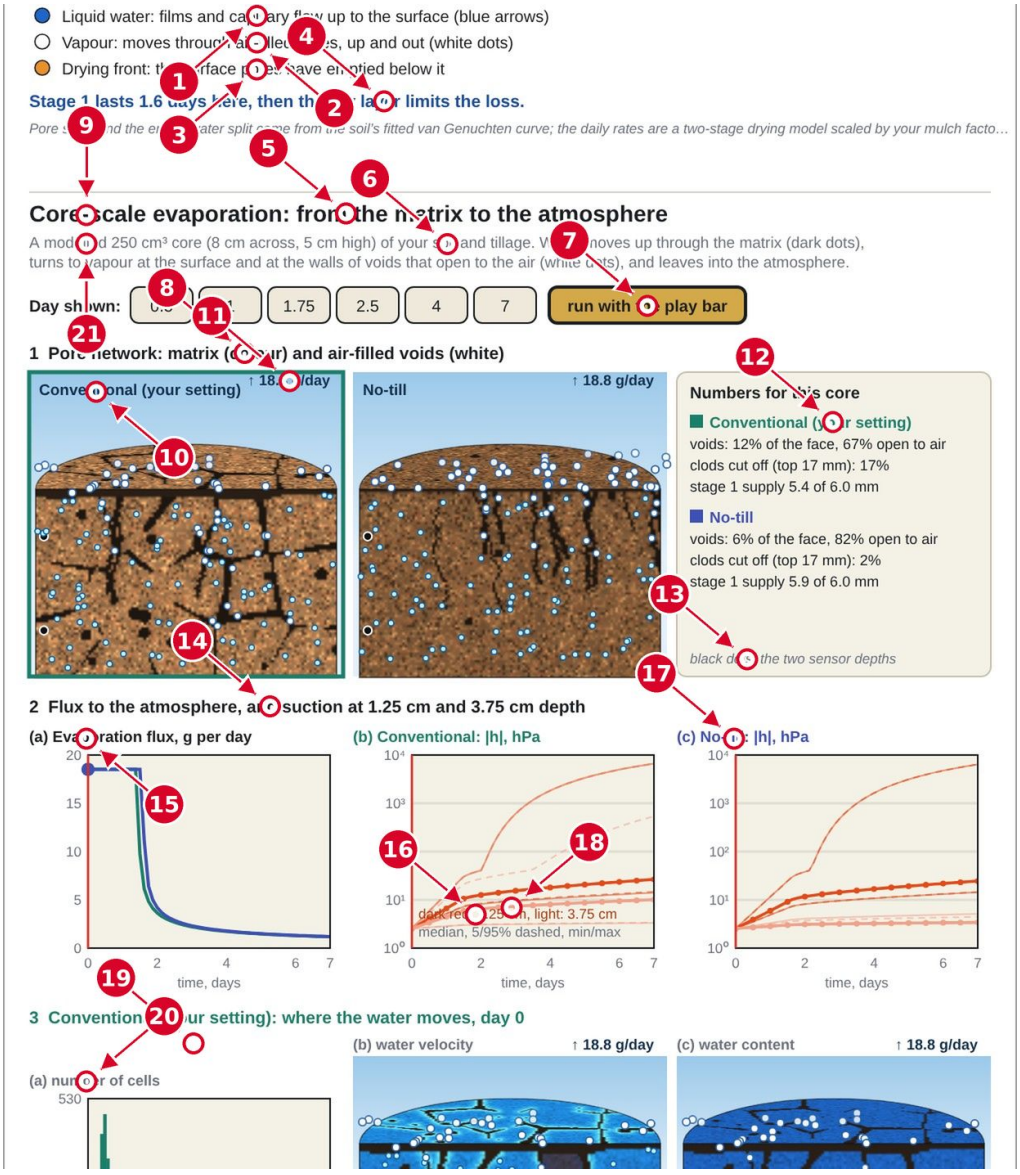


Figure 25.218: Evaporation physics, part 2 of 3.

What the arrows point to.

- **1** The blue arrows, the liquid water moving up by films and capillary flow.
- **2** The white dots, the vapour in the air-filled pores.

- **3** The drying front, below which the surface pores have emptied.
- **4** The duration of stage 1 with the mulch and without it.
- **5** The core-scale evaporation, from the matrix to the atmosphere.
- **6** The description of the modelled 250 cm³ core.
- **7** The button that runs the core with the play bar.
- **8** The pore network, the matrix in colour and the air-filled voids in white.
- **9** The flux of the core to the atmosphere in g per day.
- **10** The conventional core, with large clods and open cracks.
- **11** The no-till core, with fine open voids.
- **12** The numbers of the two cores, voids on the face, open to air and the stage 1 supply.
- **13** The black dots, the two sensor depths of the core.
- **14** The flux to the atmosphere and the suction at 1.25 and 3.75 cm depth.
- **15** The evaporation flux of the two cores against the days.
- **16** The suction of the conventional core at the two depths.
- **17** The suction of the no-till core at the two depths.
- **18** The dark red line, the median suction at 1.25 cm, with the light line at 3.75 cm.
- **19** The conventional core, where the water moves.
- **20** The histogram of the number of cells by water speed.
- **21** The histogram of the no-till core.

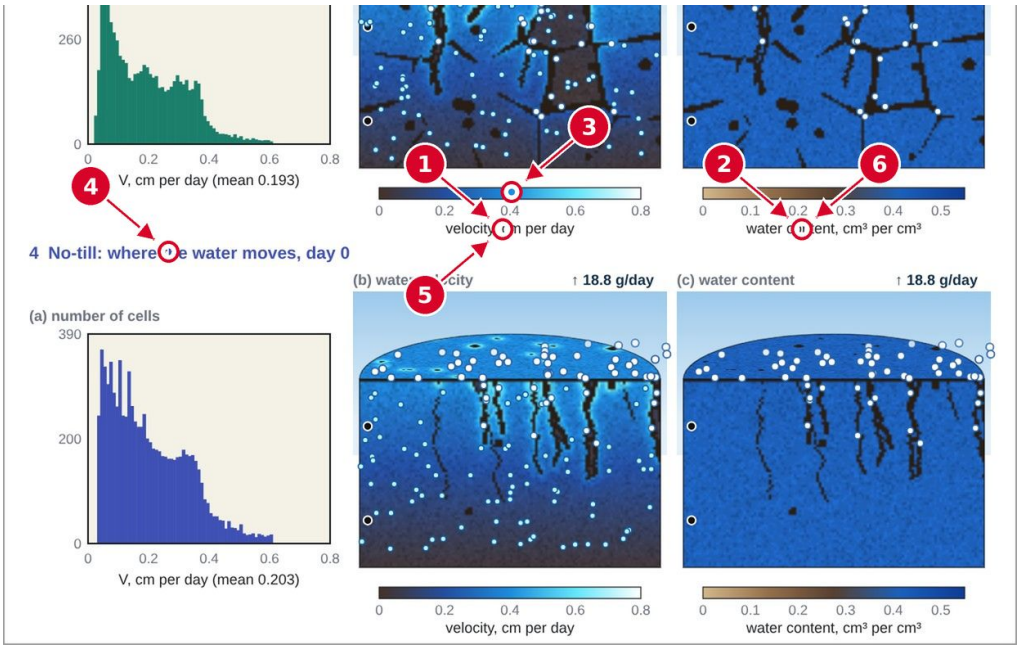


Figure 25.219: Evaporation physics, part 3 of 3.

What the arrows point to.

- **1** The map of the water velocity.
- **2** The map of the water content.
- **3** The colour scale of the velocity in cm per day.
- **4** The no-till core, where the water moves.
- **5** The velocity map of the no-till core.
- **6** The water content map of the no-till core.

25.15.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers

as in the corresponding part above.

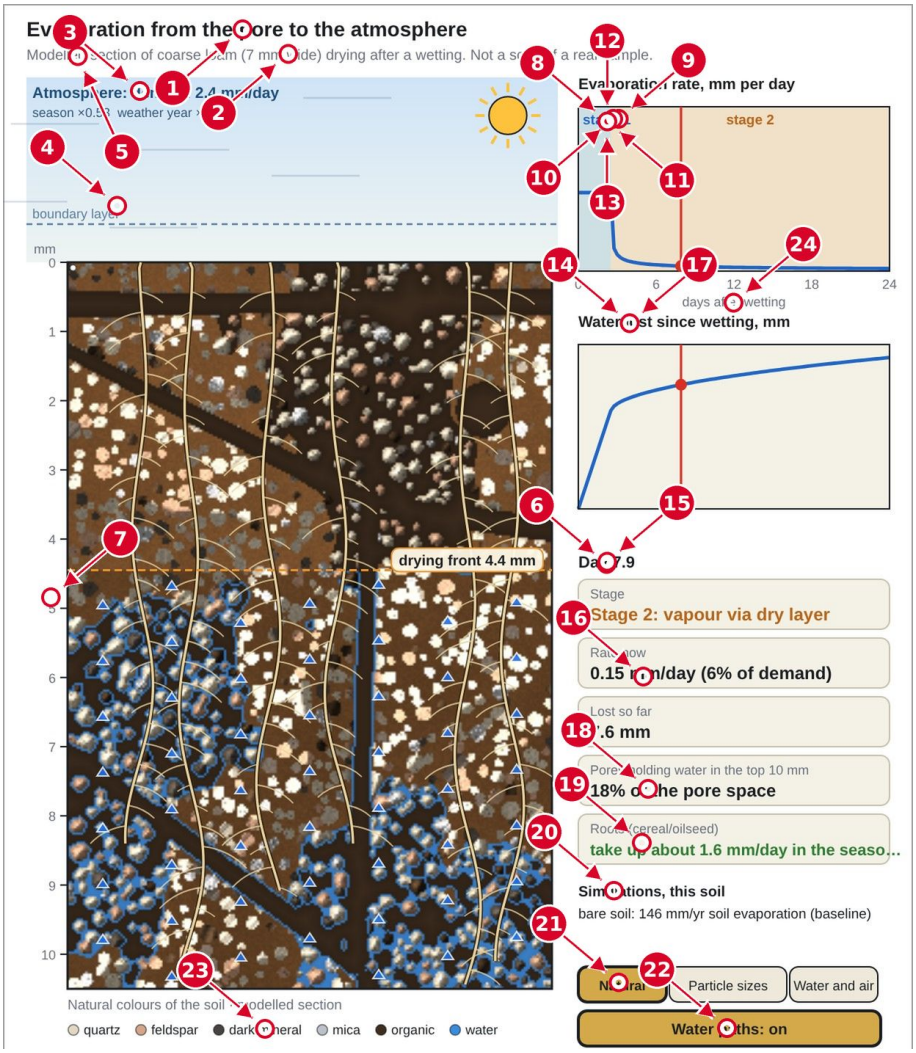


Figure 25.220: Evaporation physics, part 1, playback a third of the way (day 7.9).

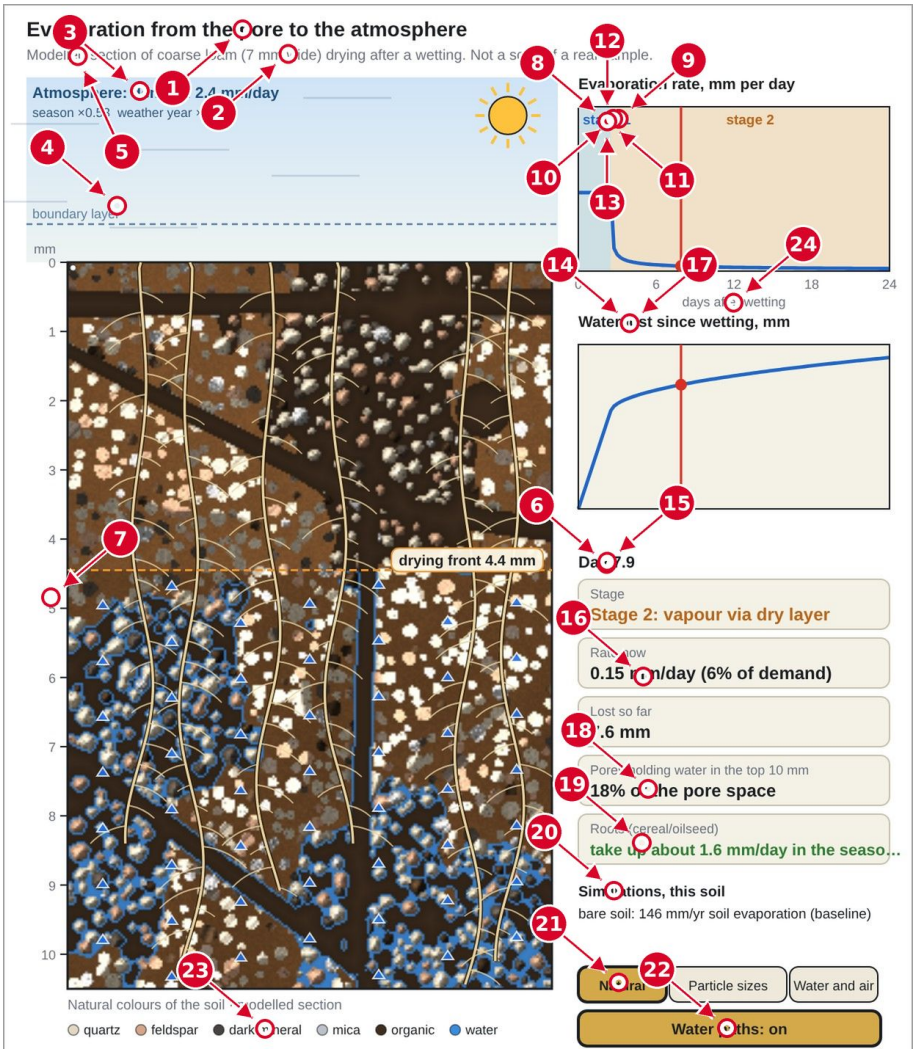


Figure 25.221: Evaporation physics, part 1, playback a third of the way (day 7.9).

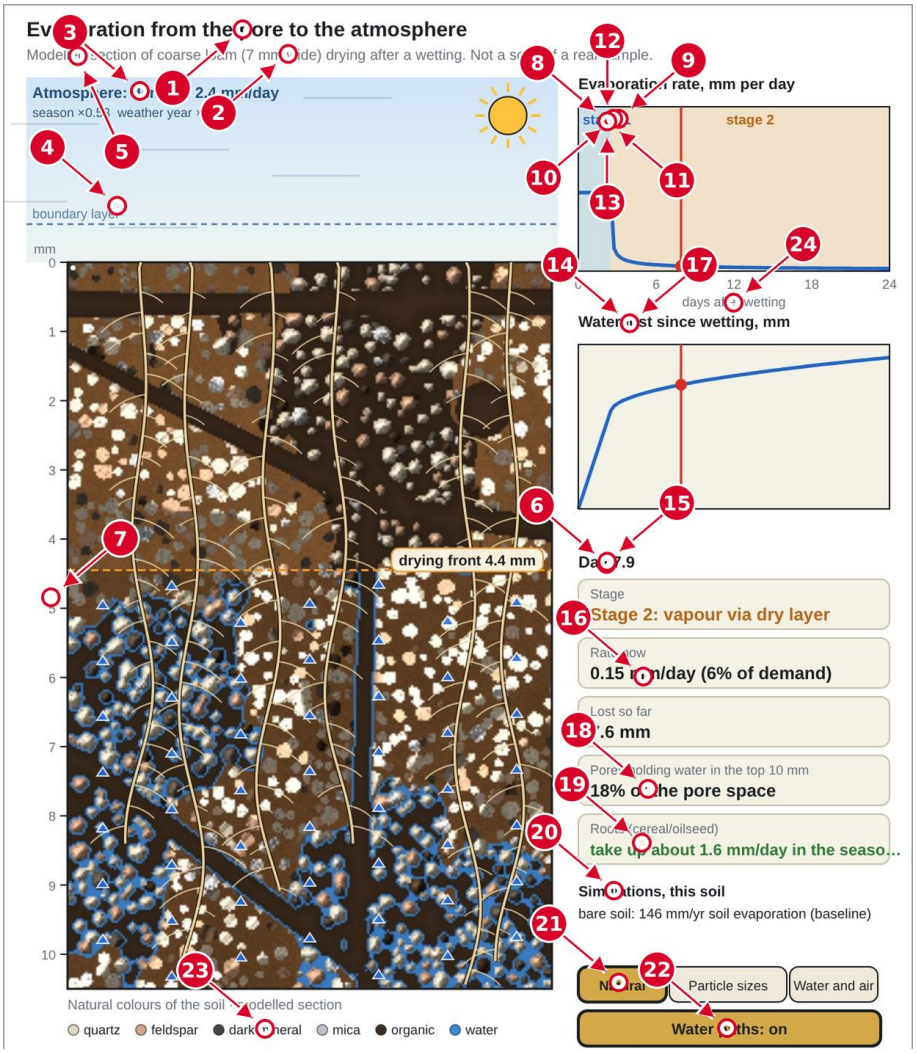


Figure 25.222: Evaporation physics, part 1, playback a third of the way (day 7.9).

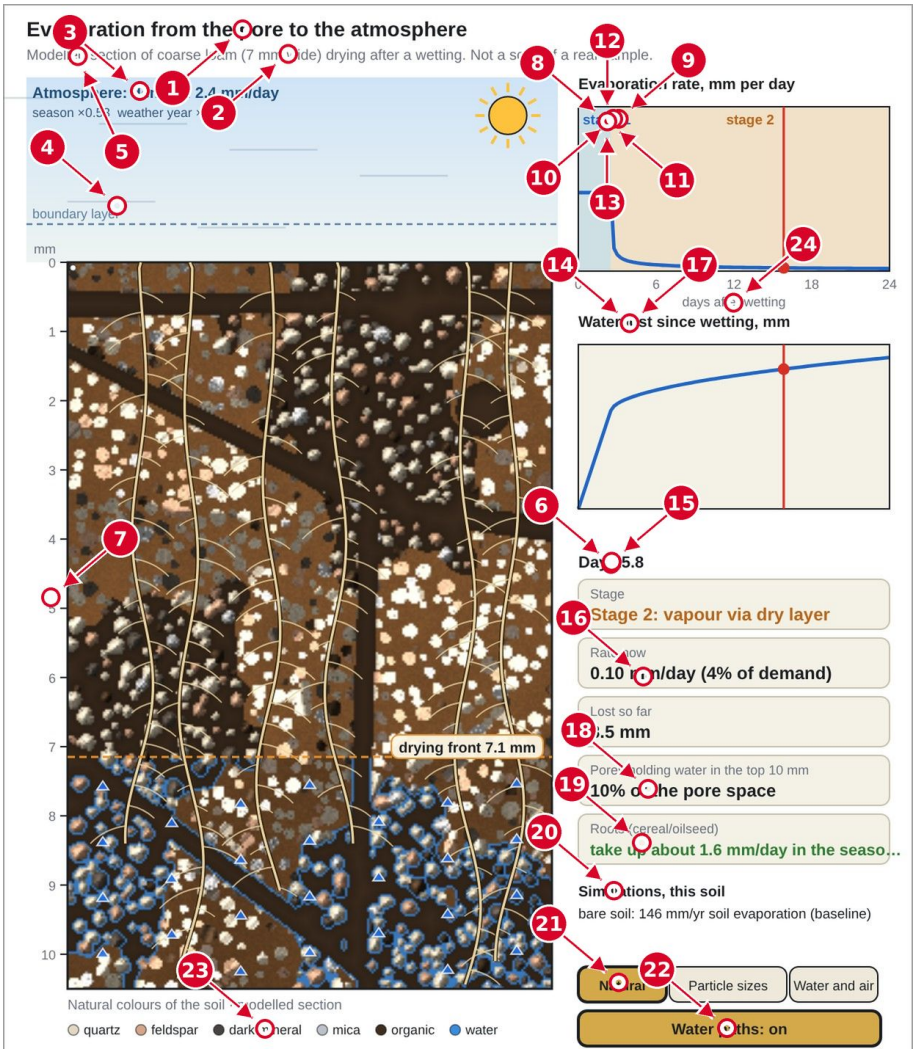


Figure 25.223: Evaporation physics, part 1, playback two thirds of the way (day 15.8).

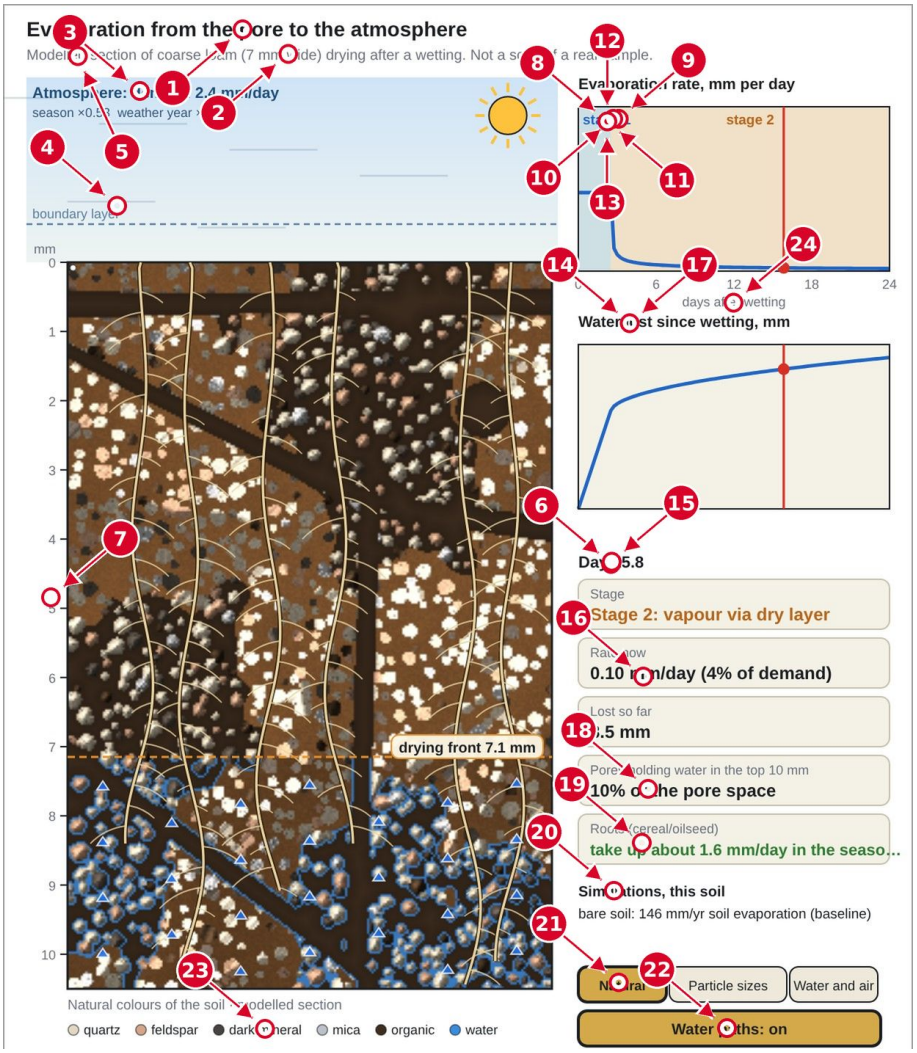


Figure 25.224: Evaporation physics, part 1, playback two thirds of the way (day 15.8).

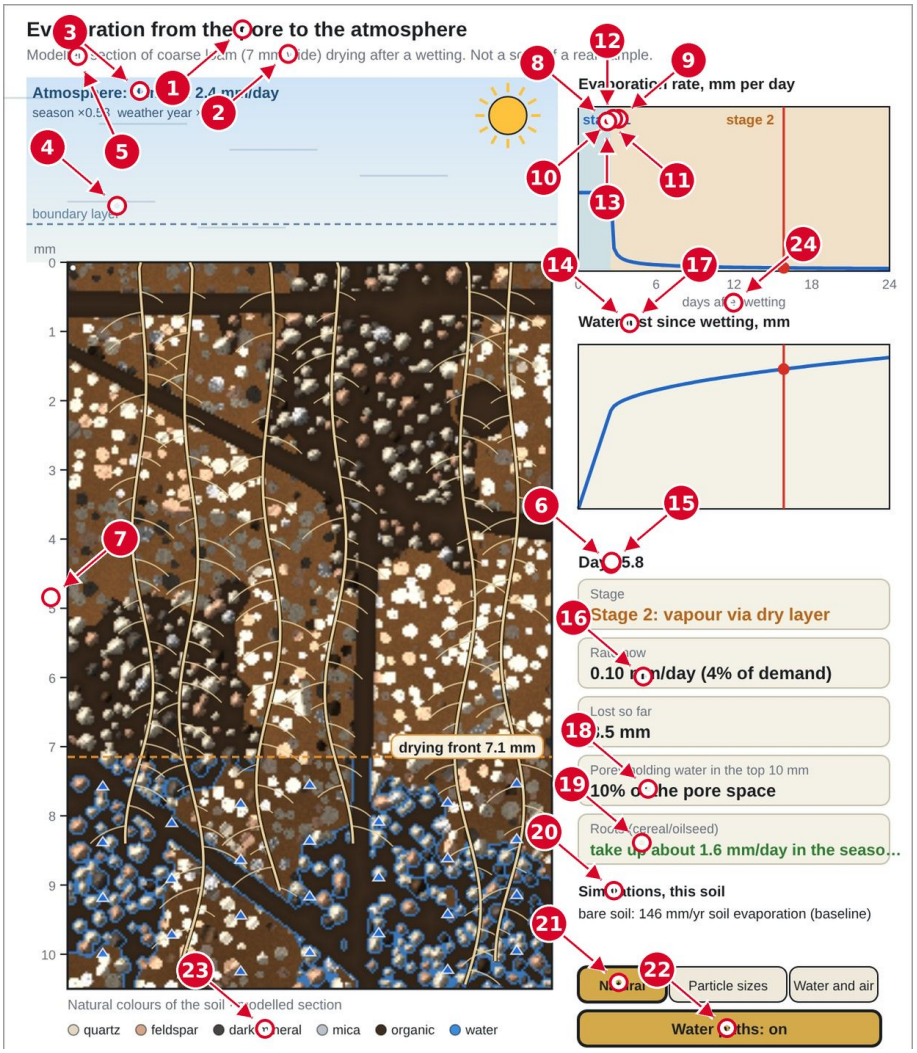


Figure 25.225: Evaporation physics, part 1, playback two thirds of the way (day 15.8).

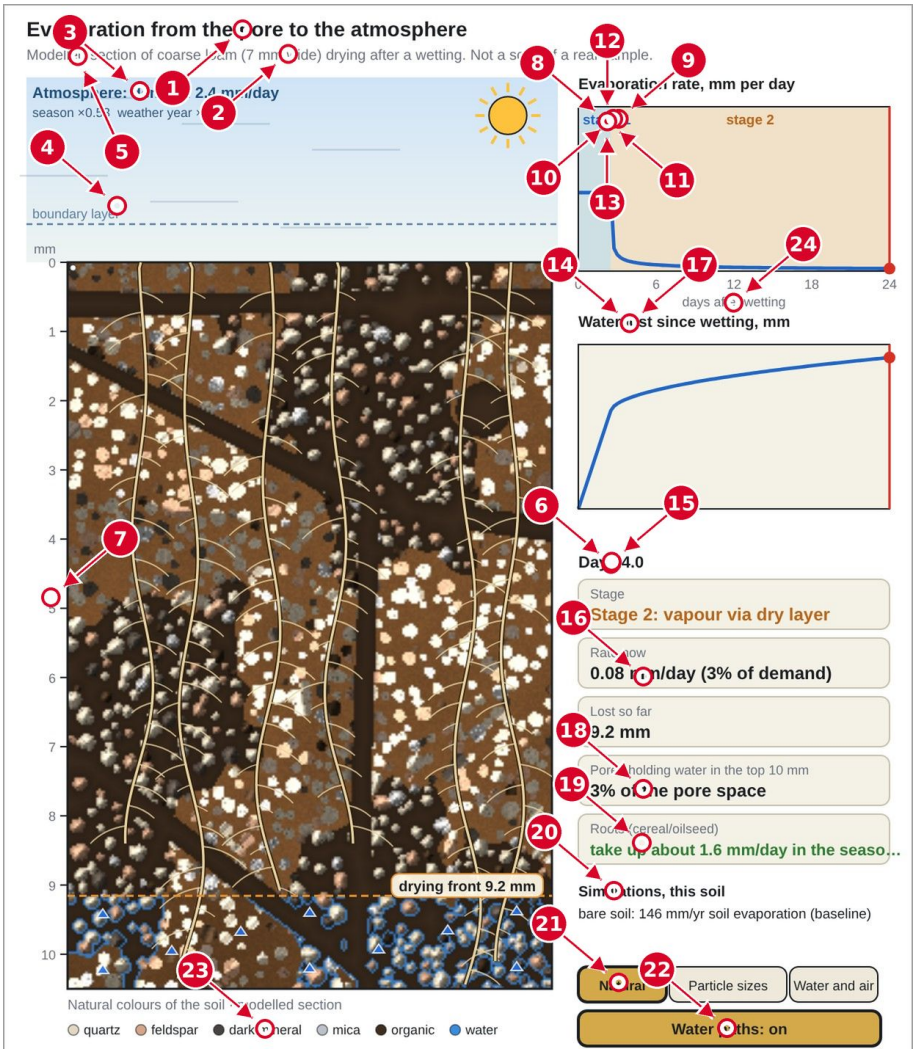


Figure 25.226: Evaporation physics, part 1, playback the end (day 24.0).

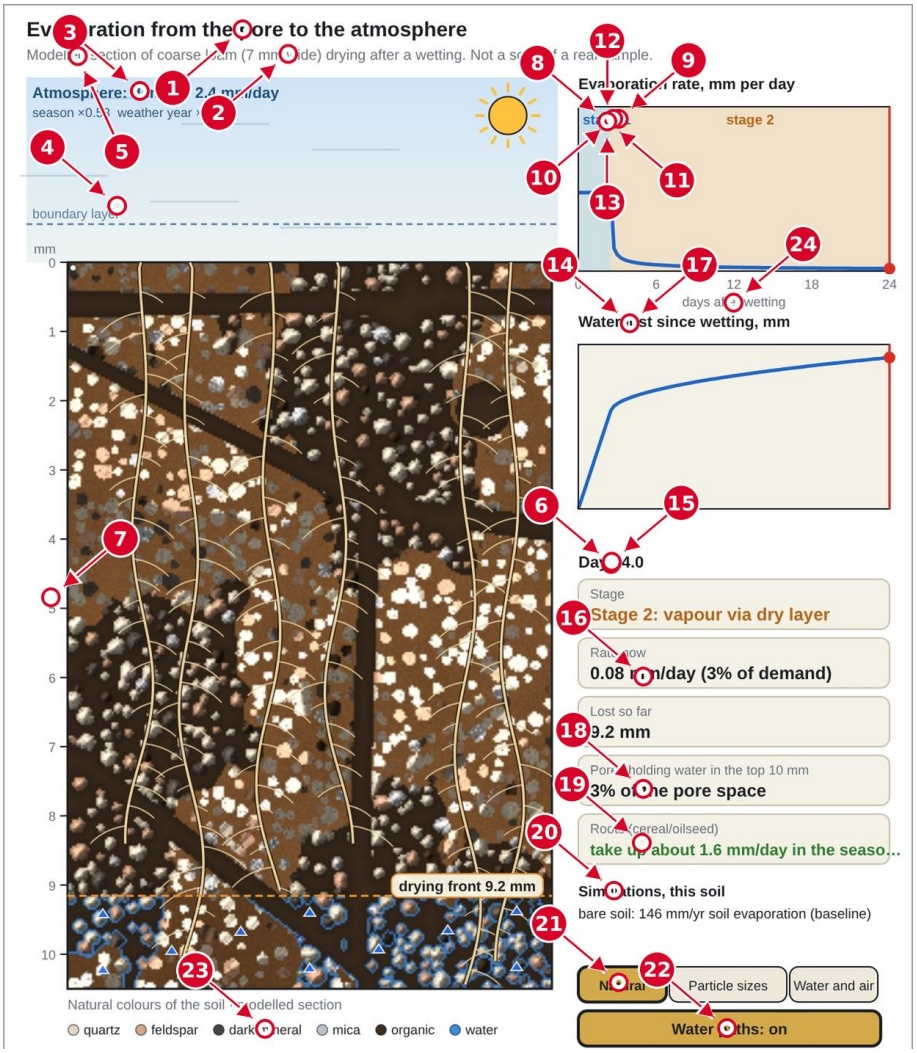


Figure 25.227: Evaporation physics, part 1, playback the end (day 24.0).

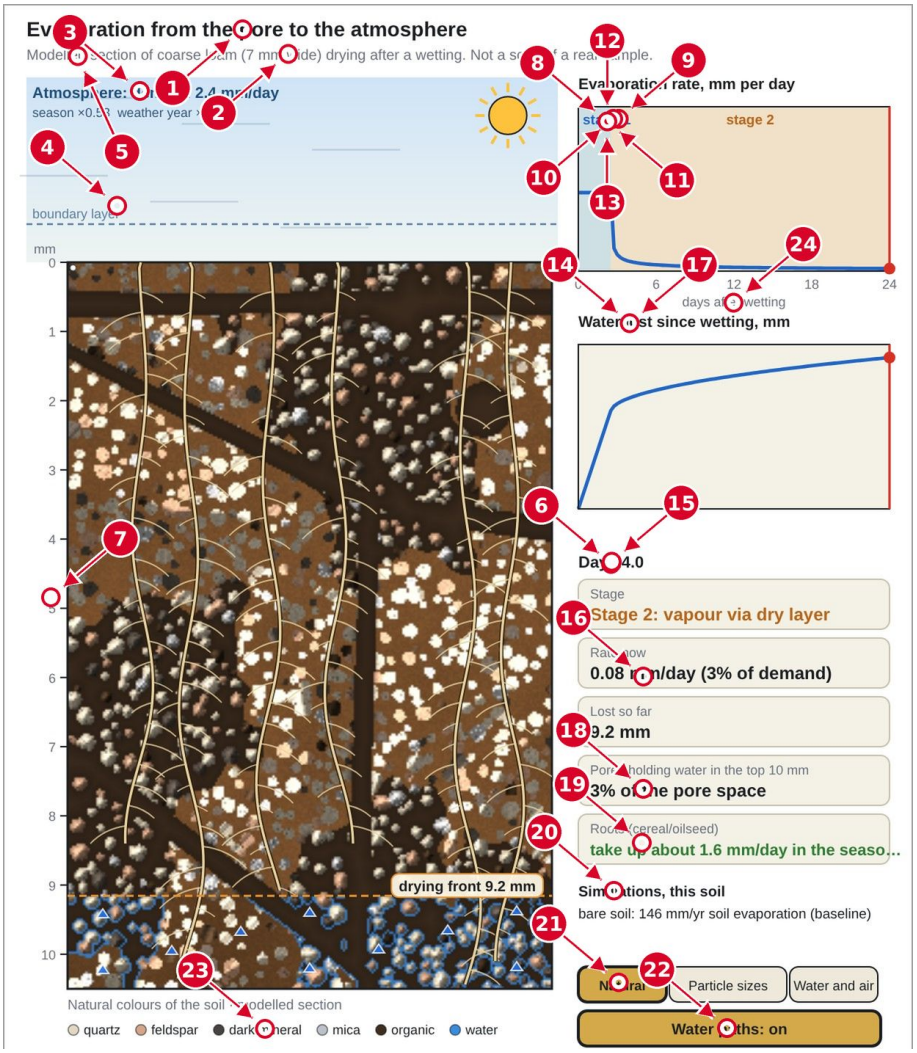


Figure 25.228: Evaporation physics, part 1, playback the end (day 24.0).

25.15.3 The controls of the layer, one state at a time

Each picture below shows the layer after one control has been pressed or dragged, with the same arrows and numbers as in the corresponding part of the layer above. The caption names the control and what it does.

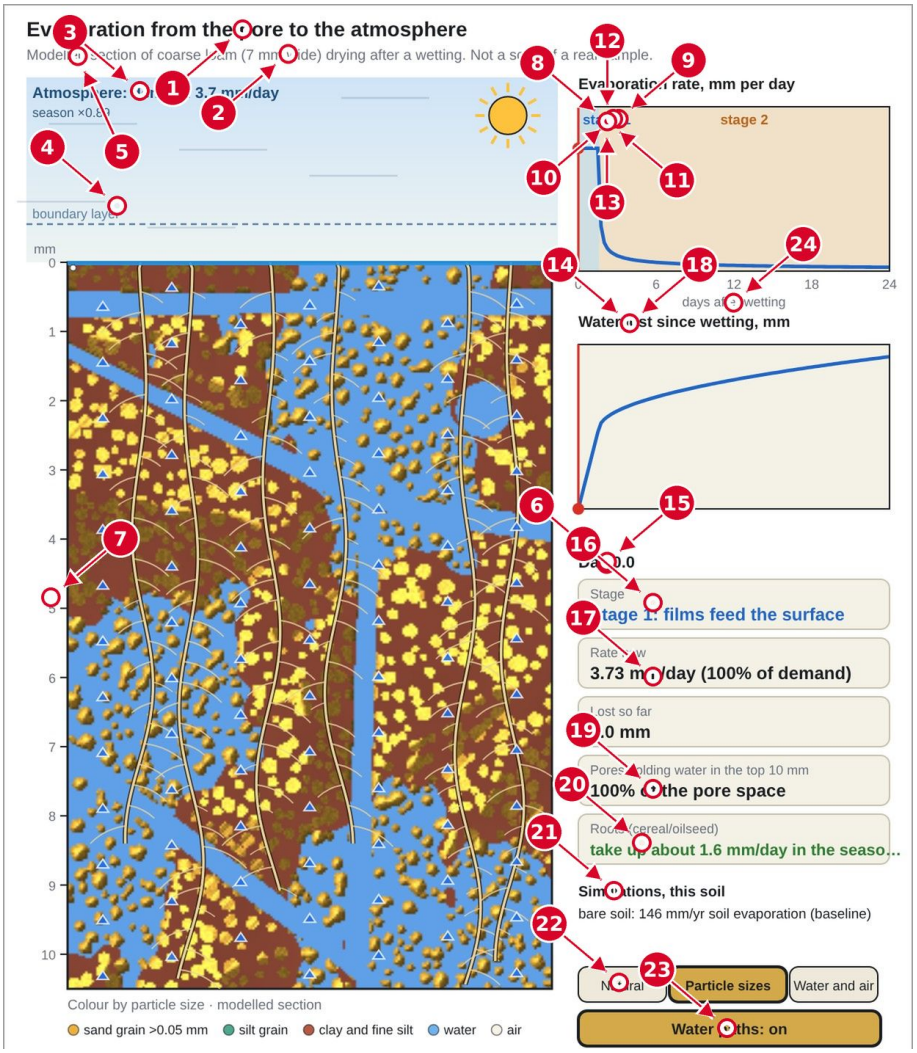


Figure 25.229: Evaporation physics, part 1. The Particle sizes view of the section chosen.

Figure 25.230: Evaporation physics, part 1. The Water and air view of the section chosen.

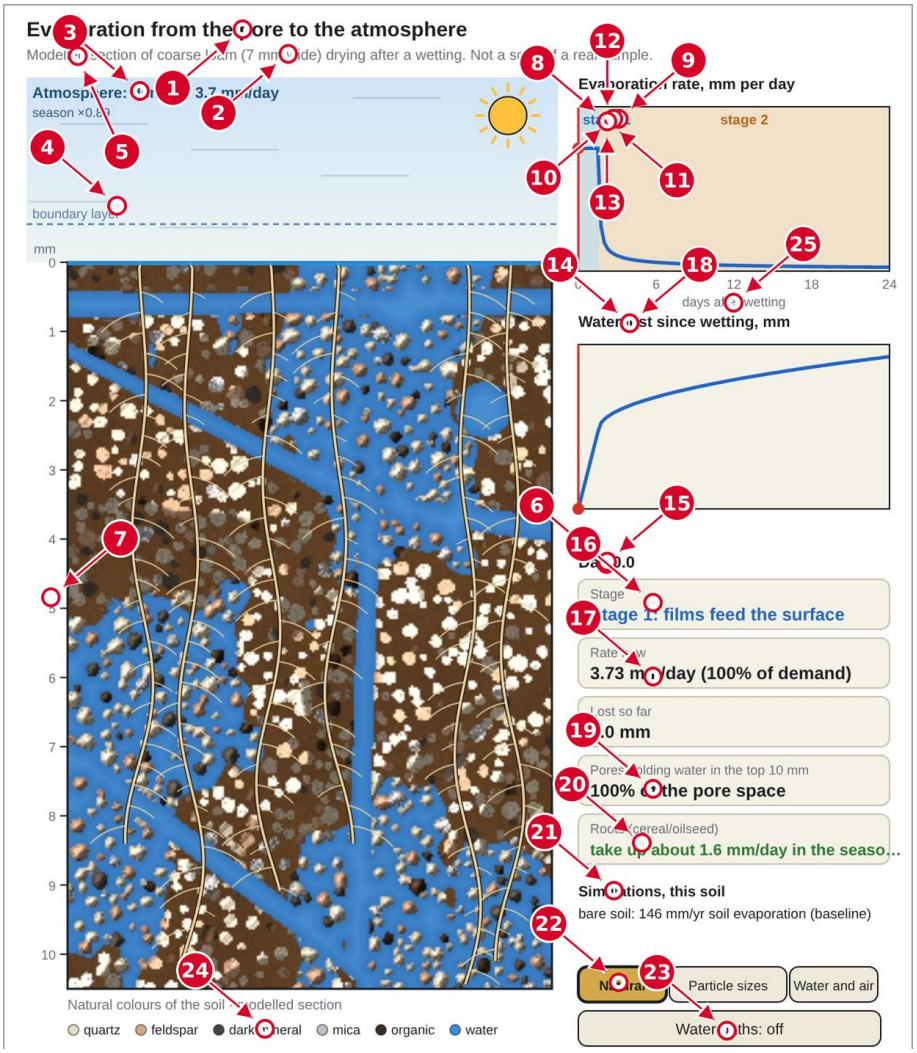


Figure 25.231: Evaporation physics, part 1. The Water paths button switched off, so the arrows and dots of the flow are hidden.

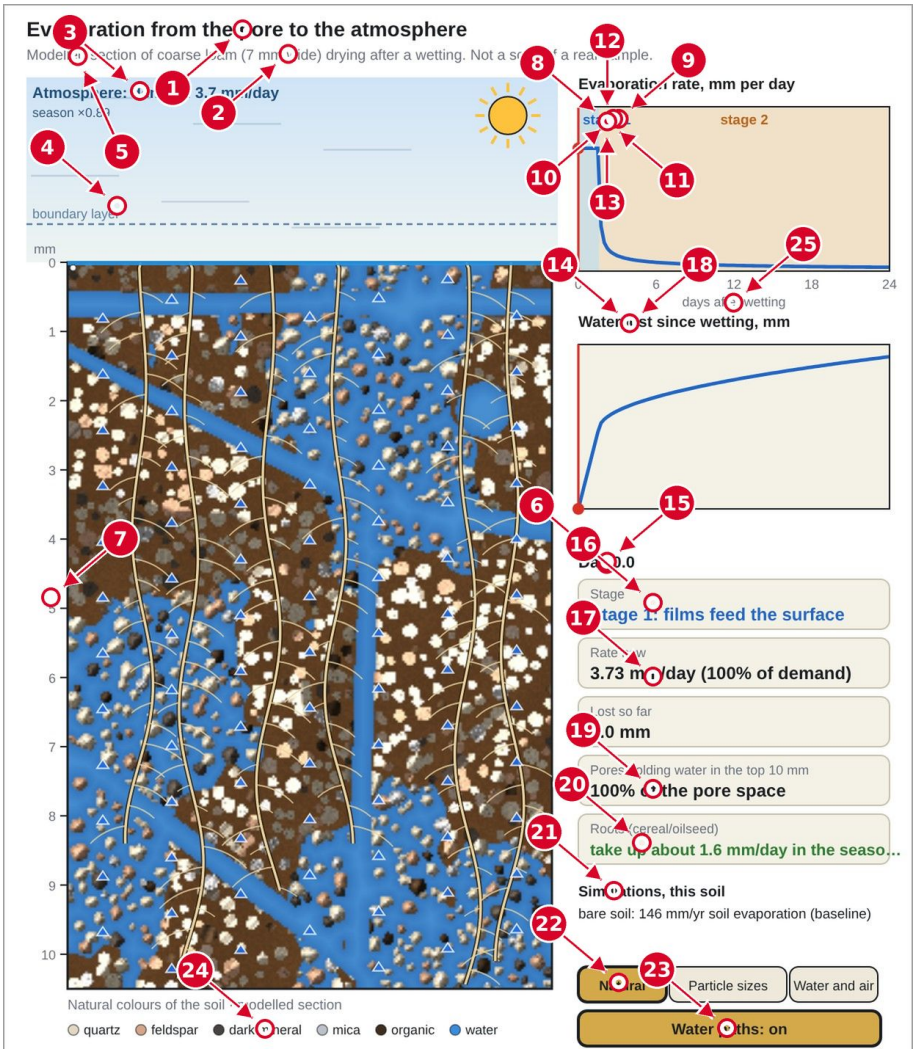


Figure 25.232: Evaporation physics, part 1. The day 0.5 button pressed, so the cores are shown at half a day.

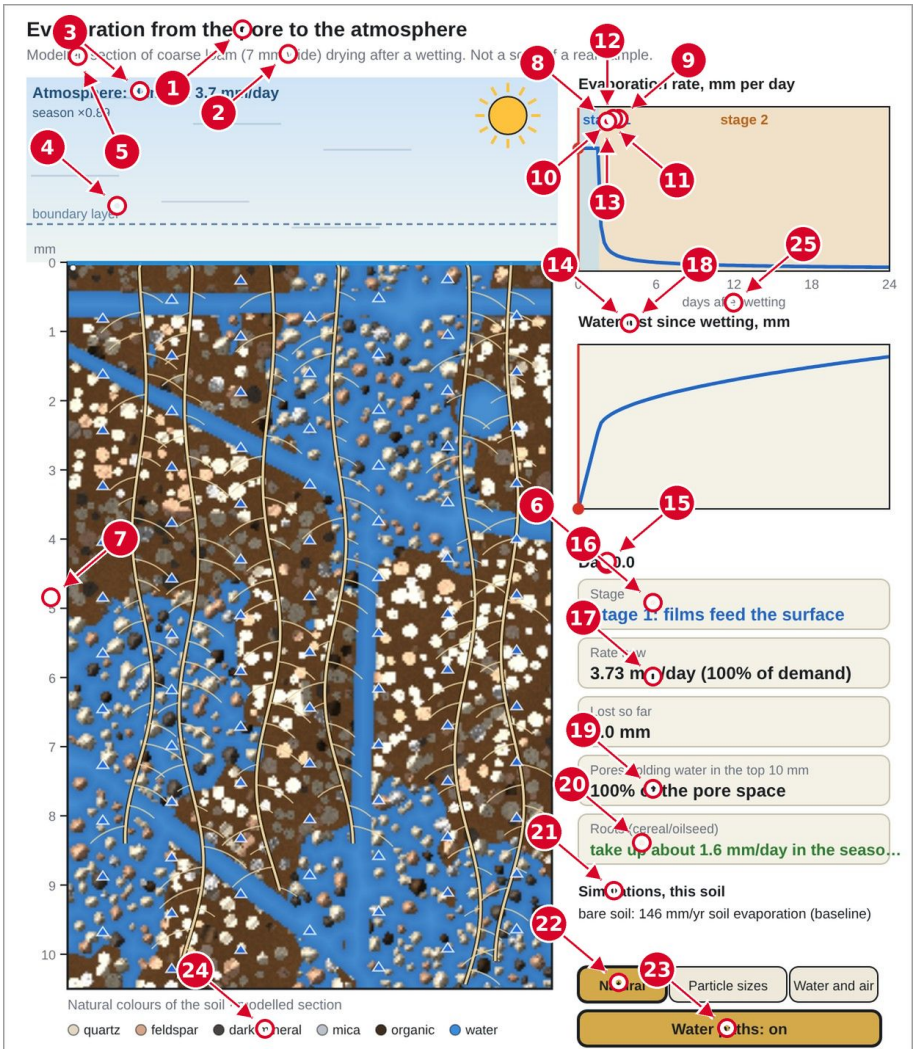


Figure 25.233: Evaporation physics, part 1. The day 0.5 button pressed, so the cores are shown at half a day.

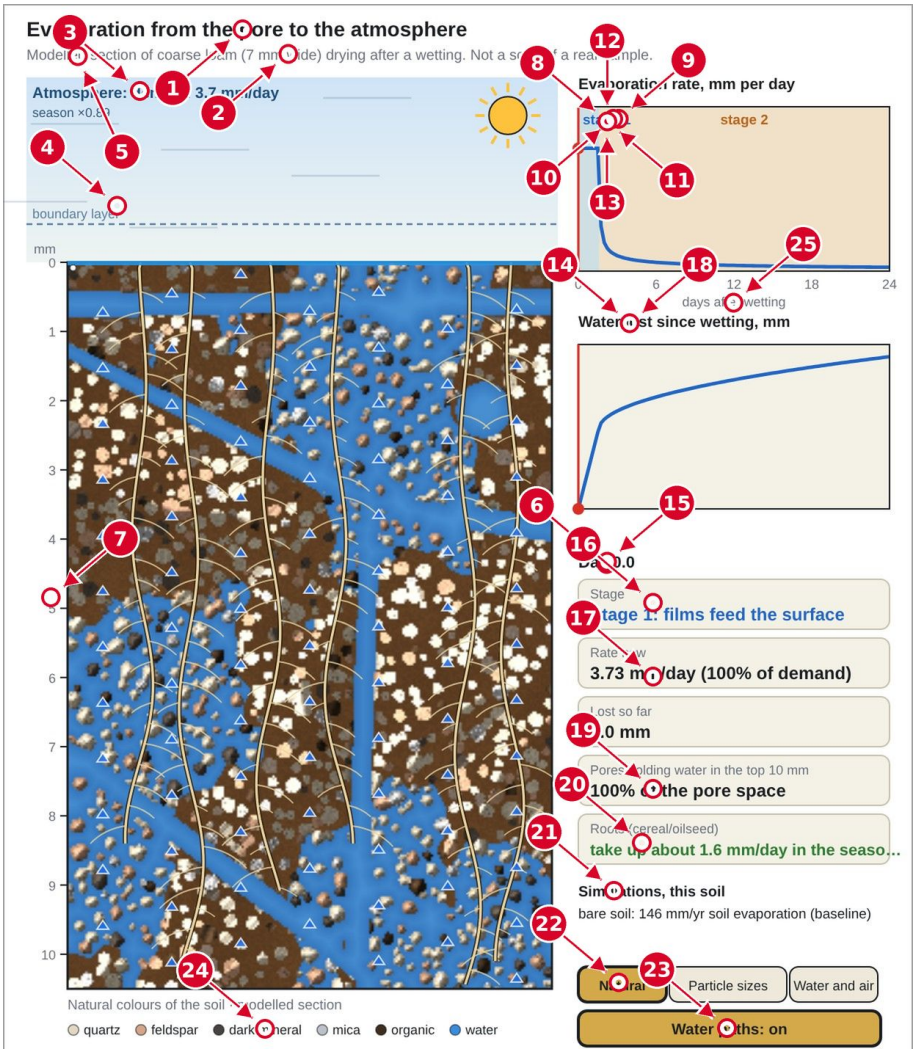


Figure 25.234: Evaporation physics, part 1. The day 1.75 button pressed.

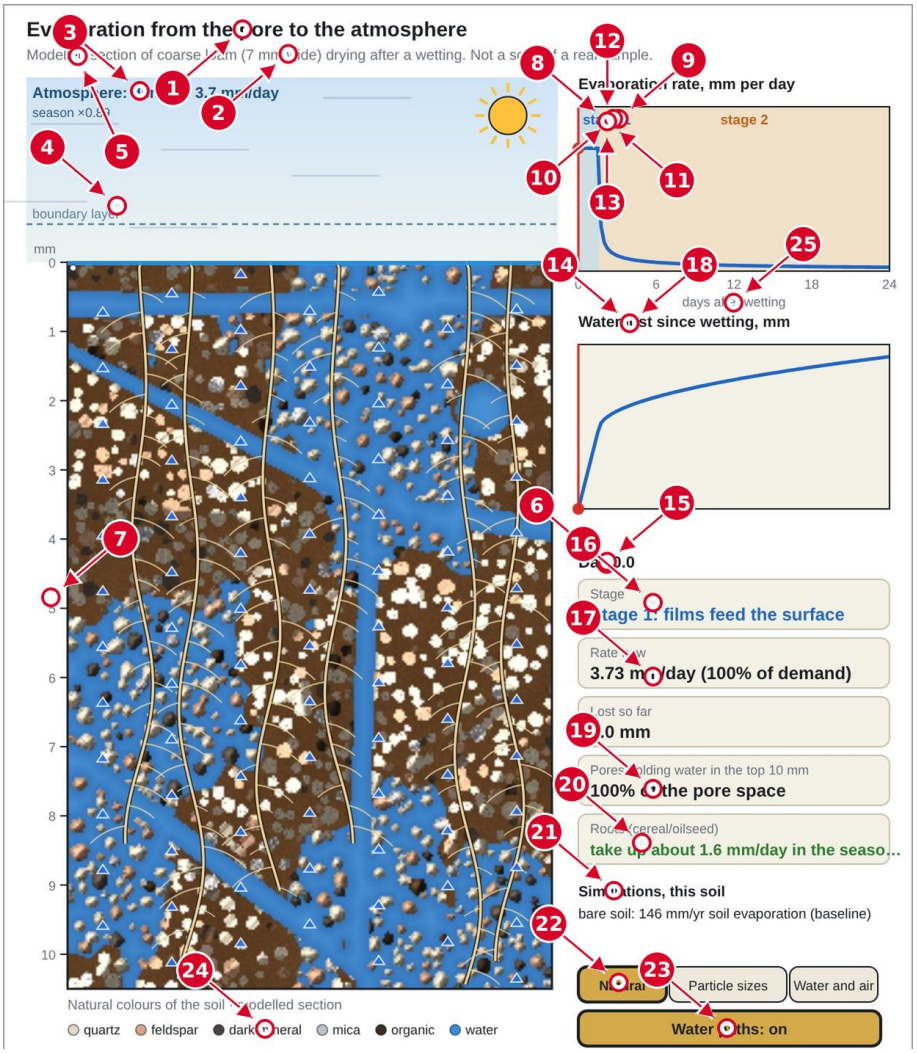


Figure 25.235: Evaporation physics, part 1. The day 1.75 button pressed.

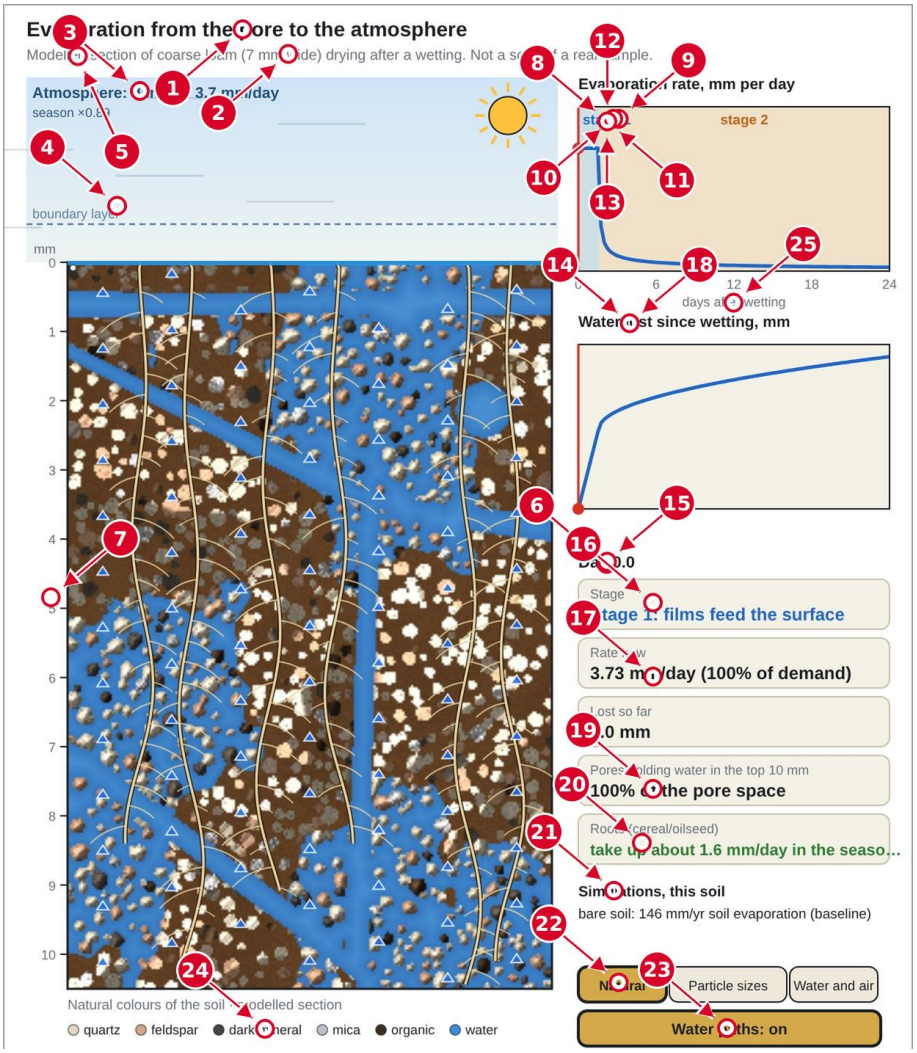


Figure 25.236: Evaporation physics, part 1. The day 4 button pressed.

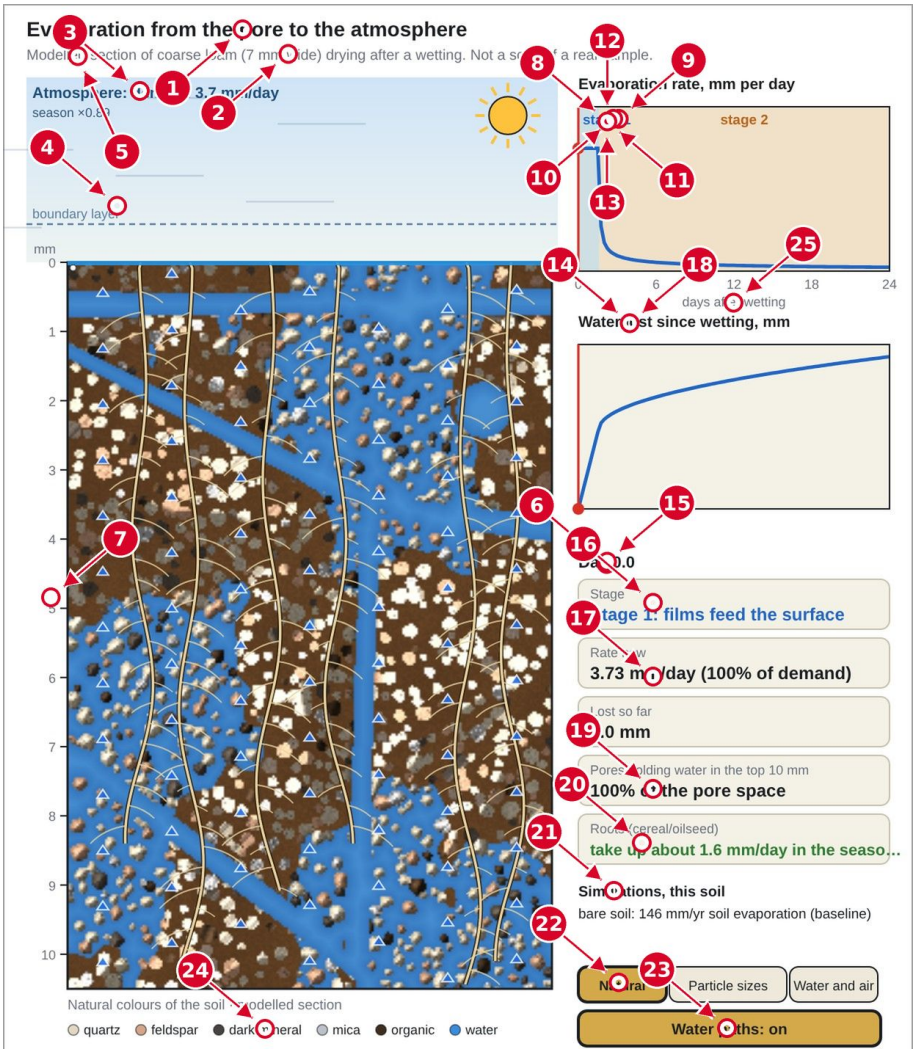


Figure 25.237: Evaporation physics, part 1. The day 4 button pressed.

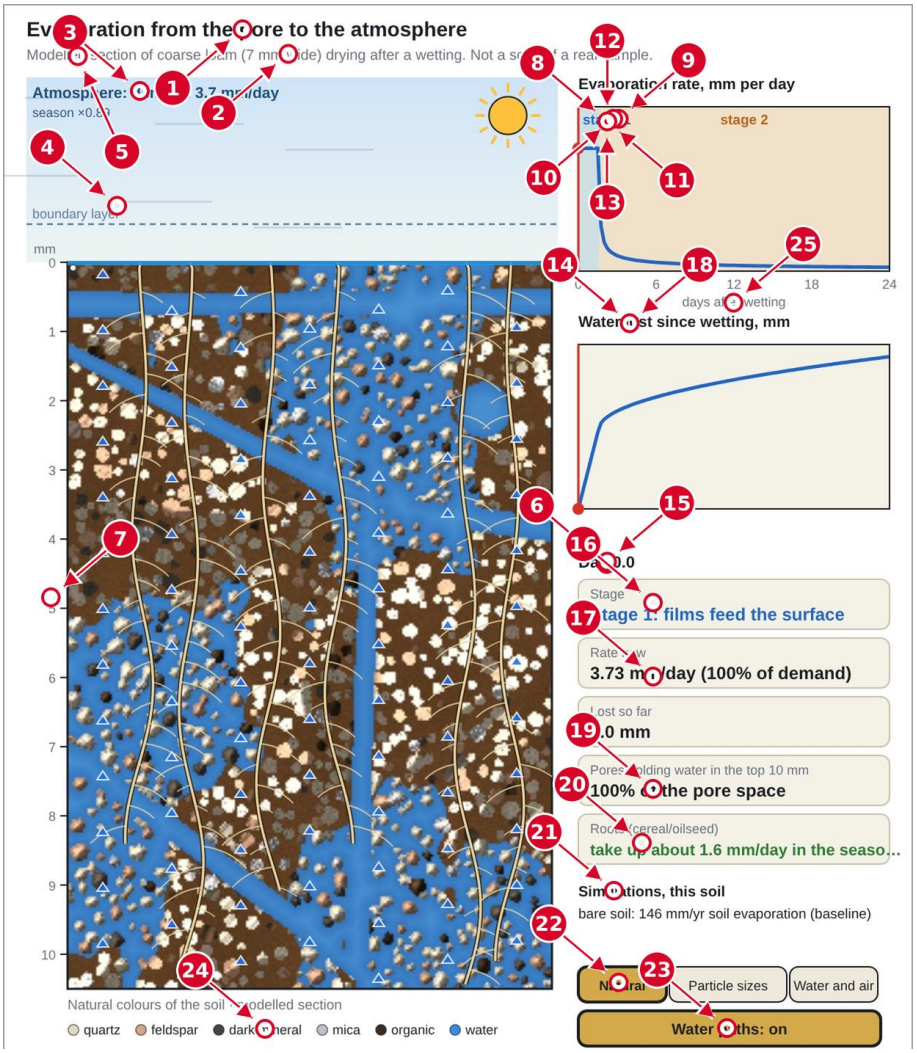


Figure 25.238: Evaporation physics, part 1. The day 7 button pressed, so the cores are shown at the end of stage 1.

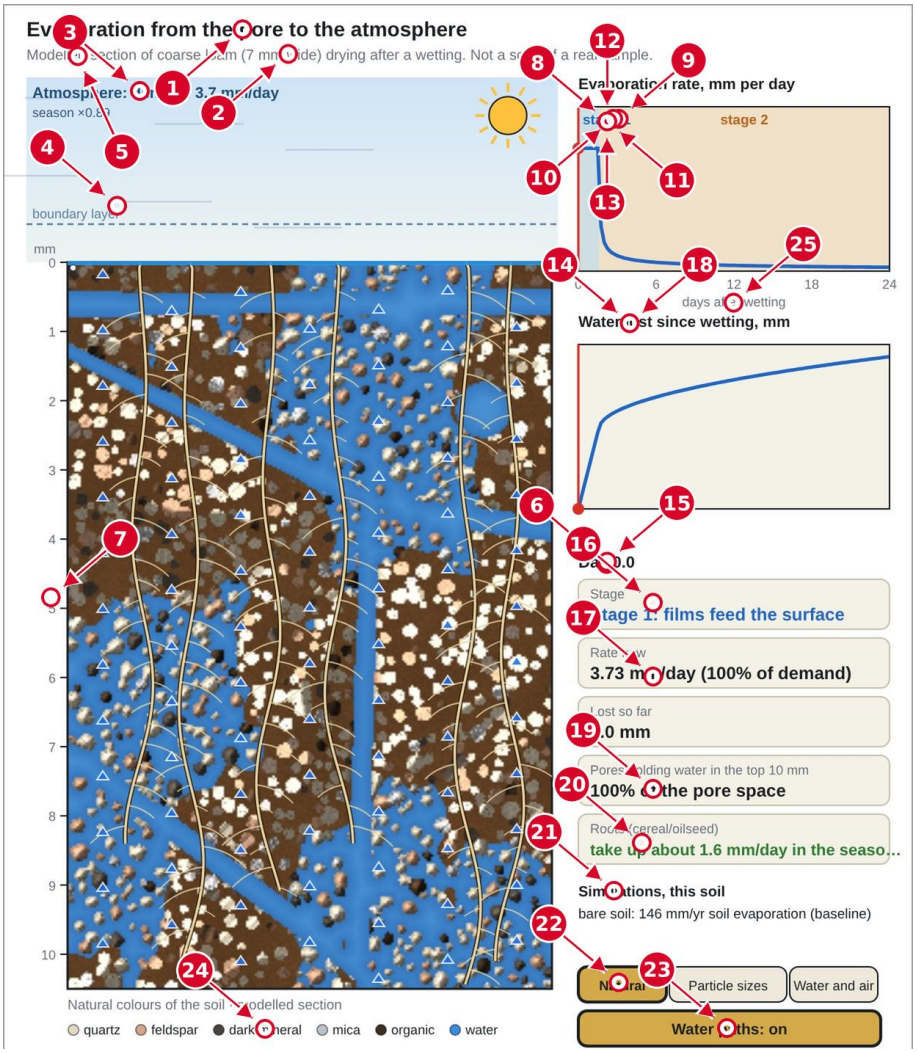


Figure 25.239: Evaporation physics, part 1. The day 7 button pressed, so the cores are shown at the end of stage 1.

25.16 Fate of saved water

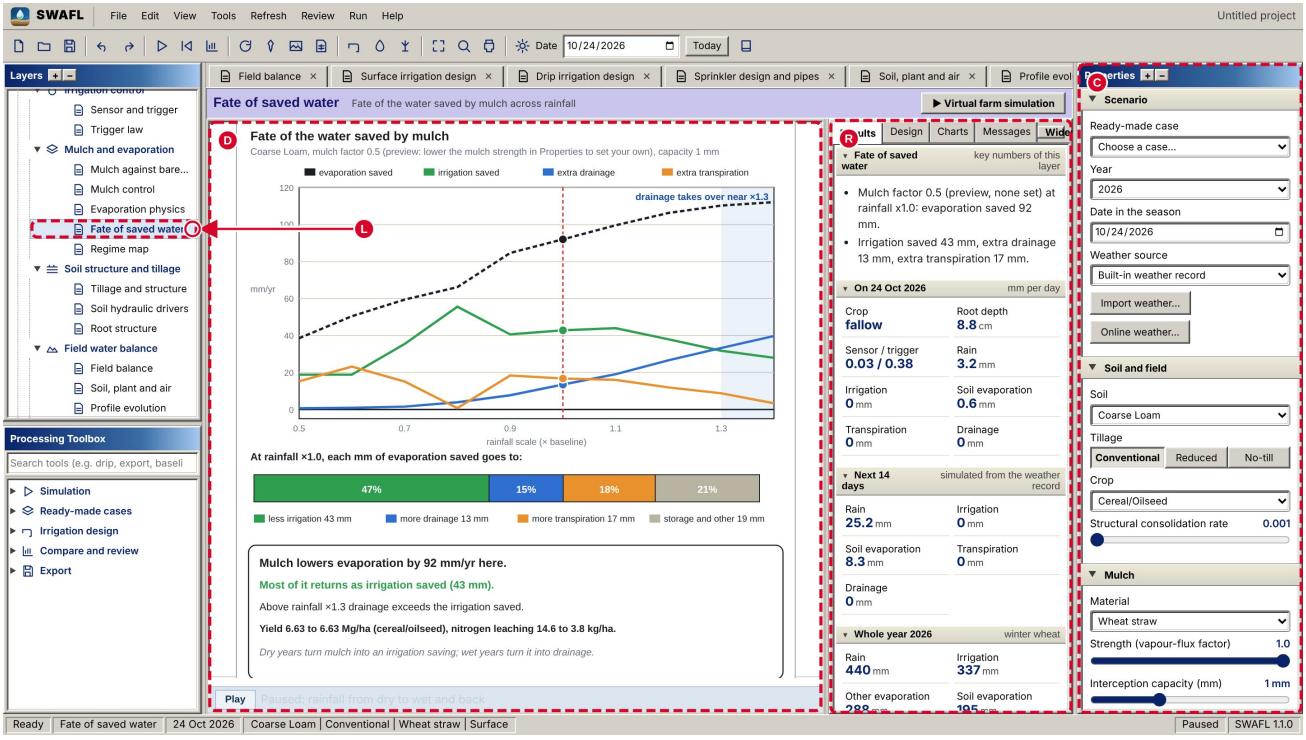


Figure 25.240: Fate of saved water: the whole workbench.

Figure 25.241: Fate of saved water: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Evaporation saved, irrigation saved, extra drainage and extra transpiration against rainfall scale: Eq. 9.6.
- **2** Rainfall at which drainage overtakes the irrigation saved: from the curves of Eq. 9.6.
- **3** Where each mm of evaporation saved goes (irrigation, drainage, transpiration, storage): Eq. 9.7.
- **4** The sentence under the picture is built from the shares of Eq. 9.7.

Equations behind this screen. Eq. 9.6, Eq. 9.7.

Eq. 9.6. Where explained: chapter "Mulch and evaporation layers", section "Fate of saved water".

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

Provenance (Eq. 9.6): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

Eq. 9.7. Where explained: chapter "Mulch and evaporation layers", section "Fate of saved water".

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\ f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\ f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\ f_{\text{rest}} = \max\{0,\ 1 - f_I - f_D - f_T\}. \tag{9.7}$$

Provenance (Eq. 9.7): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

25.16.1 The whole layer, part by part

The picture below is the complete drawing of the layer Fate of saved water, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

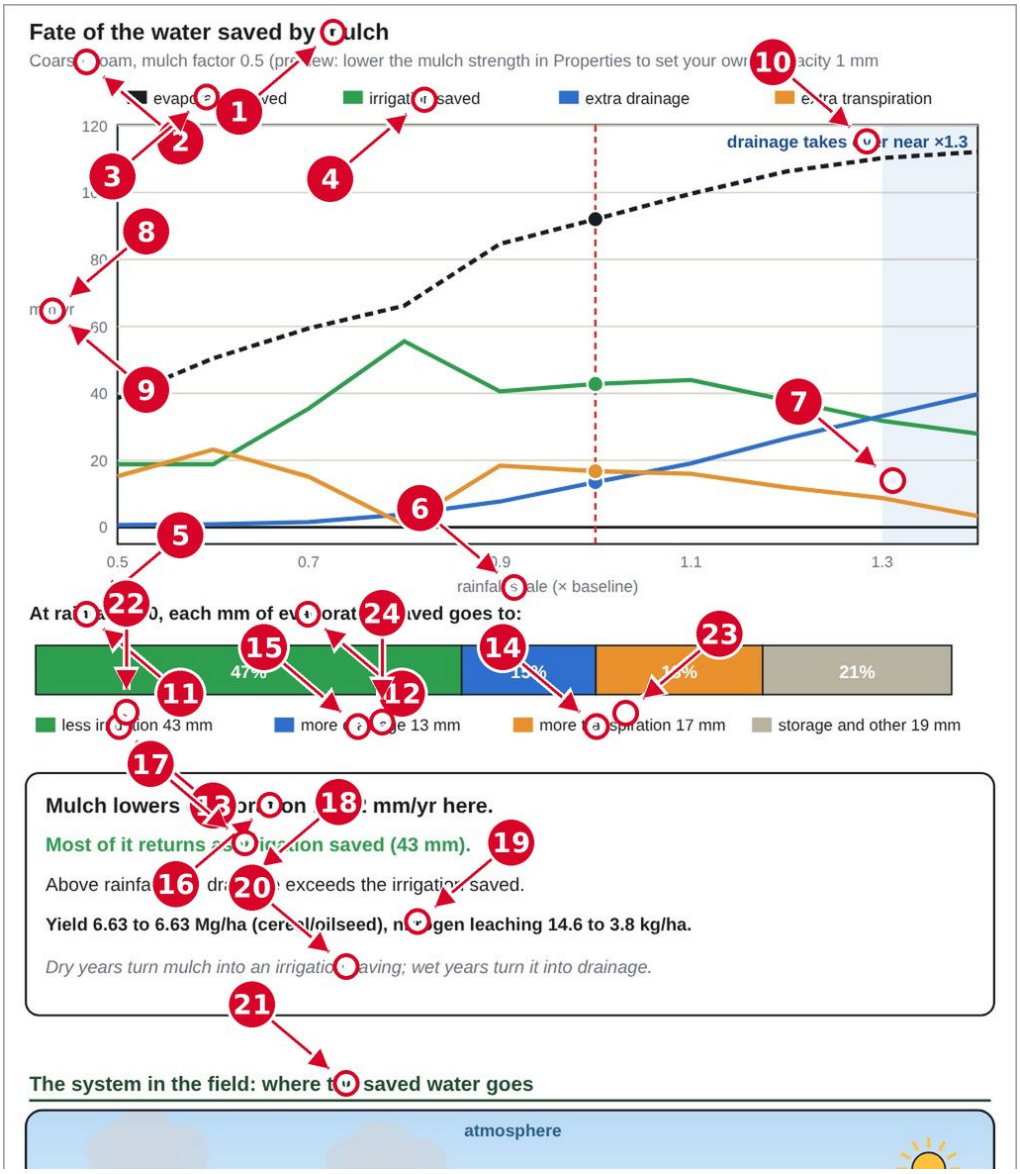


Figure 25.242: Fate of saved water, part 1 of 2.

What the arrows point to.

- **1** The title, the fate of the water saved by mulch.
- **2** The settings of the chart, the soil, the mulch factor and the capacity.
- **3** The legend of the four lines, evaporation saved, irrigation saved, extra drainage and extra transpiration.
- **4** The black dotted line, the evaporation saved by the mulch at each rainfall scale.

- **5** The orange line, the extra transpiration.
- **6** The green line, the irrigation saved.
- **7** The blue line, the extra drainage, which grows with the rainfall.
- **8** The red dashed line, the rainfall scale in force.
- **9** The point of the evaporation saved at that rainfall scale.
- **10** The rainfall scale above which the drainage takes over.
- **11** The rainfall scale axis, as a multiple of the baseline.
- **12** The partition of each mm of evaporation saved at the rainfall in force.
- **13** The share that returns as less irrigation, in green.
- **14** The share that stays as storage and other terms, in grey.
- **15** The legend of the partition with the values in mm.
- **16** The box with the conclusion for these settings.
- **17** The main destination of the saved water.
- **18** The rainfall above which the drainage exceeds the irrigation saved.
- **19** The yield and the nitrogen leaching before and after the mulch.
- **20** The message that dry years turn the mulch into irrigation saving and wet years into drainage.
- **21** The system in the field, where the saved water goes.
- **22** The irrigation saved.
- **23** The extra transpiration.
- **24** The extra drainage.

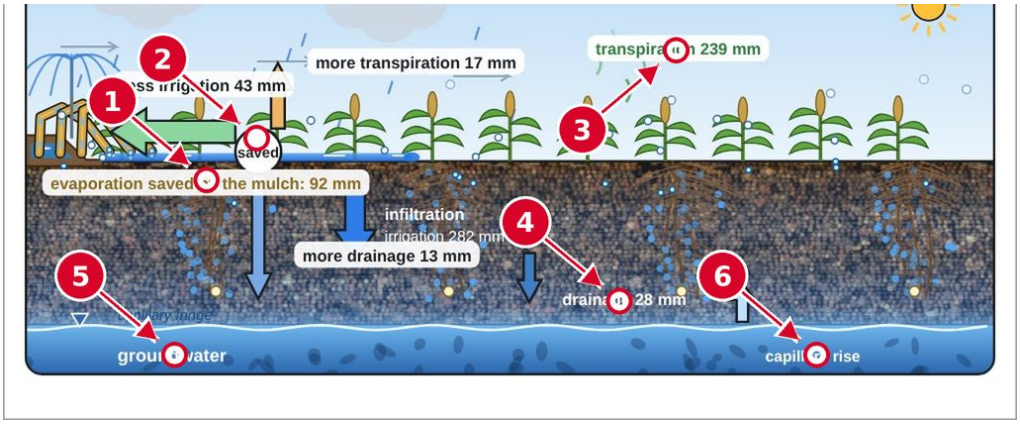


Figure 25.243: Fate of saved water, part 2 of 2.

What the arrows point to.

- **1** The tag of the evaporation saved by the mulch.
- **2** The saved water leaving the surface, drawn as the circle labelled saved.
- **3** The transpiration of the year.
- **4** The drainage arrow.
- **5** The groundwater.
- **6** The capillary rise.

25.17 Regime map

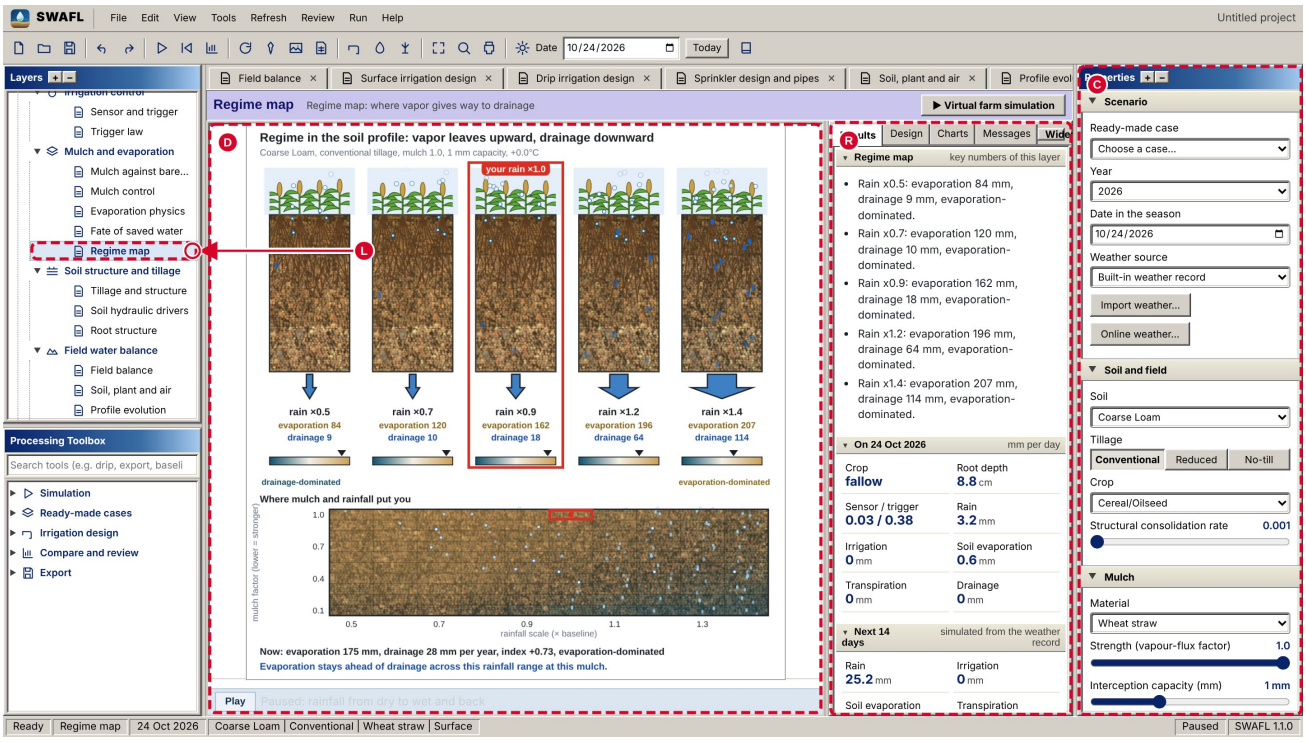


Figure 25.244: Regime map: the whole workbench.

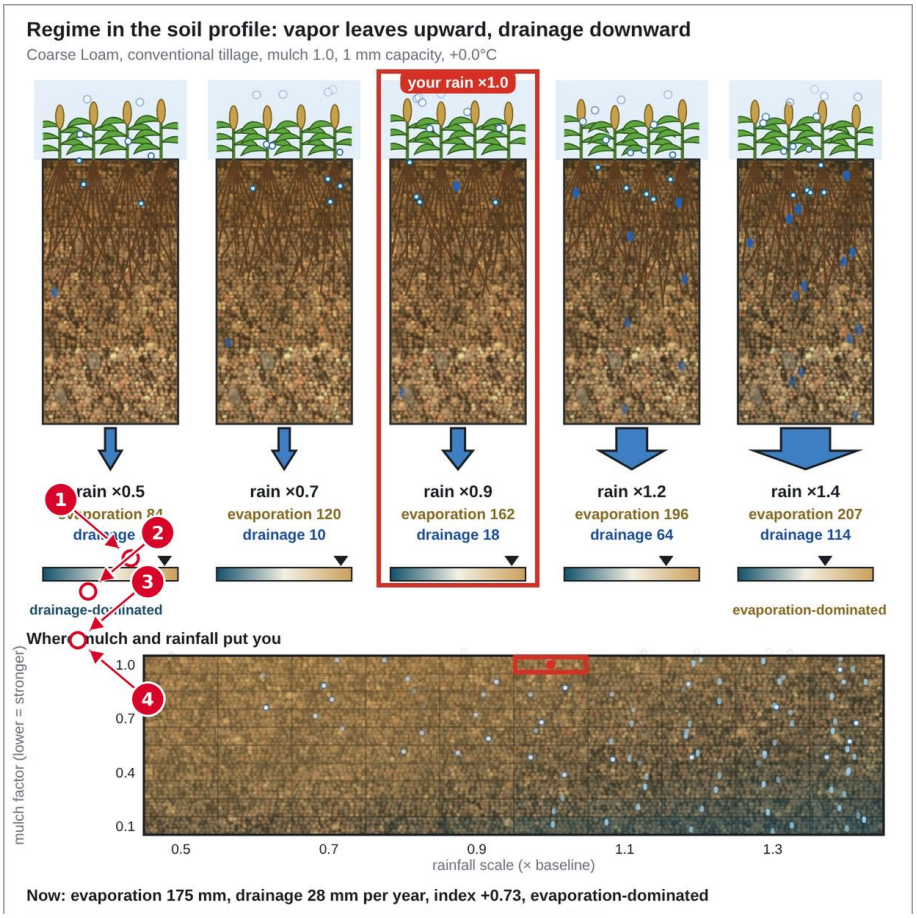


Figure 25.245: Regime map: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **1** Rainfall columns: evaporation and drainage of Engine 1 at each rainfall scale, Eqs. 5.4 and 5.9.
- **2** Colour bar under each column: regime index I = (E - D) / max(1, E + D): Eq. 9.8.
- **3** Regime map over rainfall scale and mulch factor: Eq. 9.8 in every cell.
- **4** Red box: your current setting on the map.

Equations behind this screen. Eq. 9.8.

Eq. 9.8. Where explained: chapter “Mulch and evaporation layers”, section “Regime map”.

$$\mathcal{I} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{I} > +0.05 \\ \text{drainage-dominated} & \mathcal{I} < -0.05 \\ \text{on the boundary} & |\mathcal{I}| \le 0.05 \end{cases} \tag{9.8}$$

Provenance (Eq. 9.8): **[Engineered]**: index defined for the workbench.

25.17.1 The whole layer, part by part

The picture below is the complete drawing of the layer Regime map, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

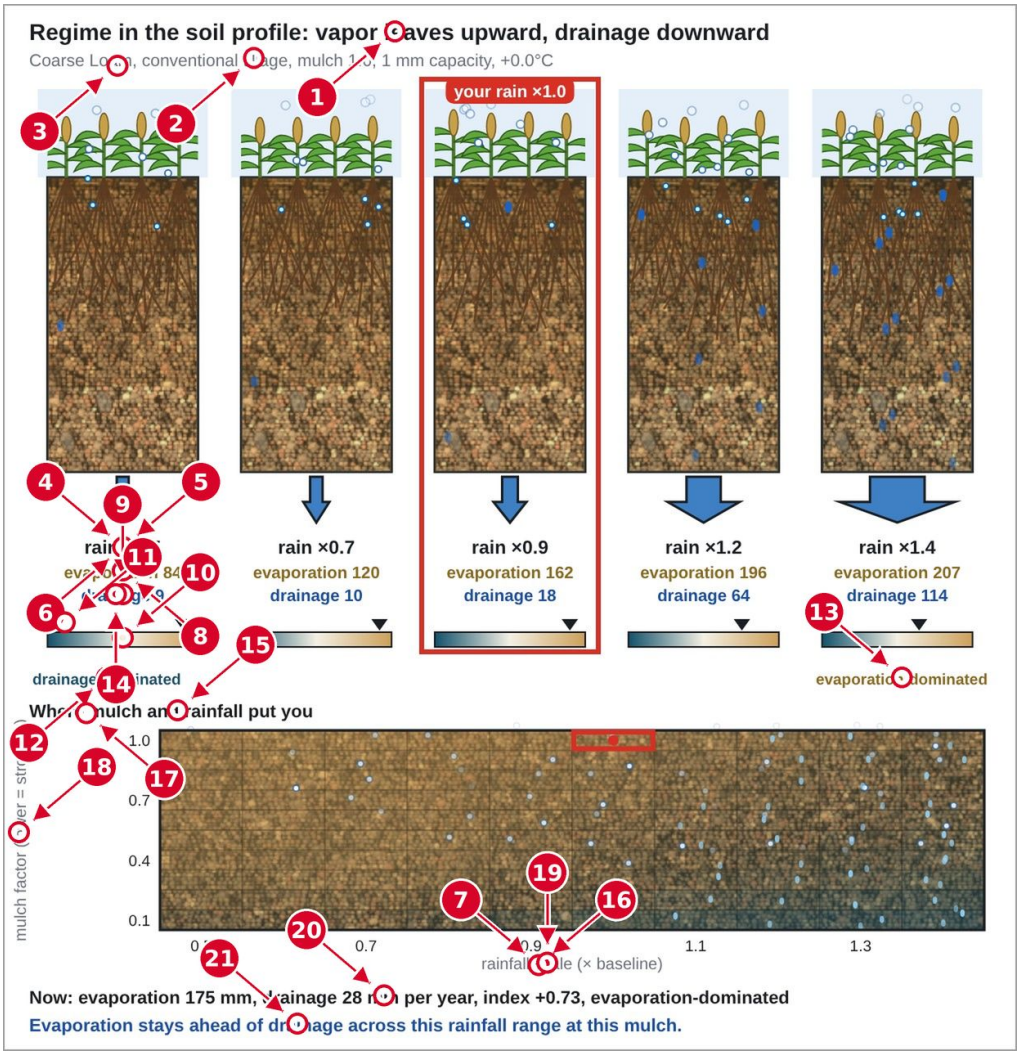


Figure 25.246: Regime map, part 1 of 1.

What the arrows point to.

- **1** The title, the regime of the soil profile, vapour upward and drainage downward.
- **2** The settings of the five profiles, the soil, tillage, mulch factor, capacity and warming.
- **3** The tag of the profile that matches your own rain scaling.
- **4** The profile outlined in red, the rainfall scale in force.
- **5** The next profile, a wetter rainfall scale.
- **6** The rain arrow, thicker for more rain.
- **7** The rainfall scale of the profile.
- **8** The evaporation of the year at that rainfall scale, in mm.
- **9** The drainage of the year at that rainfall scale, in mm.
- **10** The regime scale from drainage-dominated to evaporation-dominated.
- **11** The marker of the regime index on that scale.
- **12** The label of the drainage-dominated end.
- **13** The label of the evaporation-dominated end.
- **14** The drainage of the wettest profile, far above its evaporation.
- **15** The map of where the mulch and the rainfall put you.
- **16** The regime map over the rainfall scale and the mulch factor.
- **17** The red outlined cell, your own setting on the map.
- **18** The mulch factor axis, lower means stronger.

- **19** The rainfall scale axis, as a multiple of the baseline.
- **20** The evaporation, the drainage and the regime index of your setting.
- **21** The sentence on the rainfall at which the drainage overtakes the evaporation.

25.18 Feedback loops

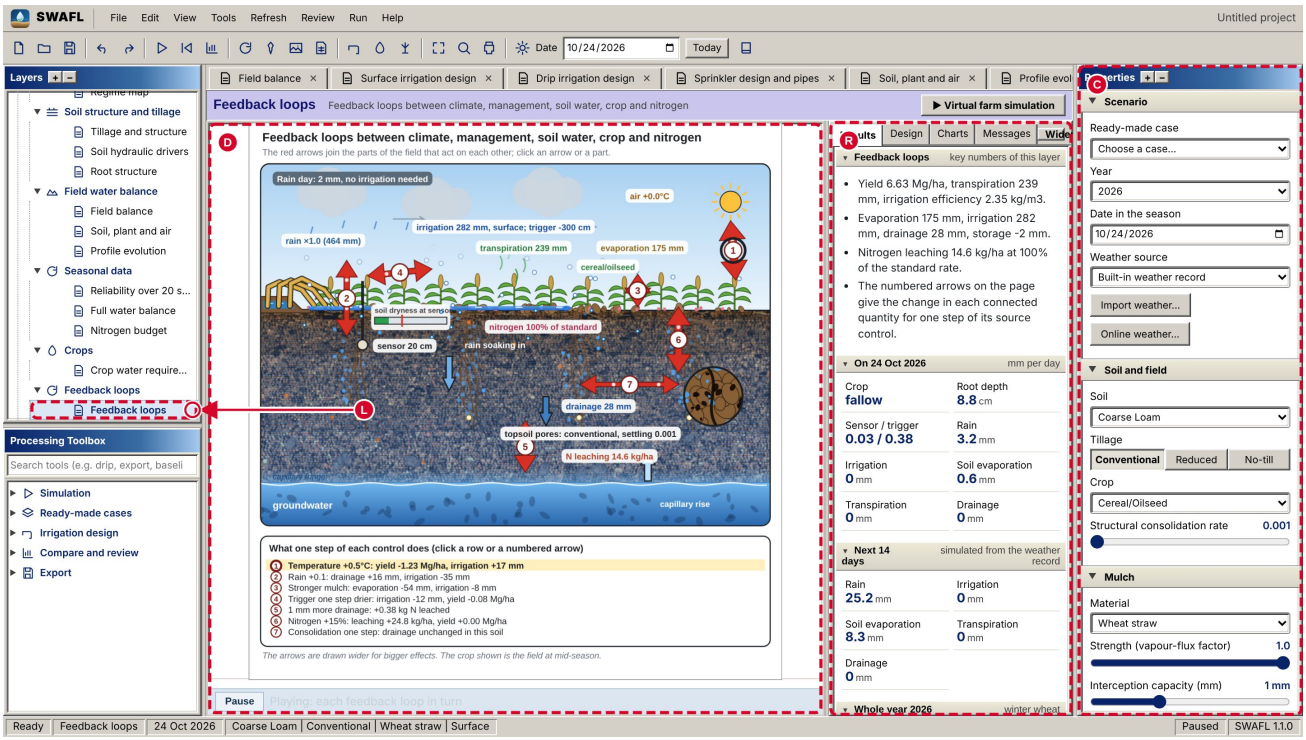


Figure 25.247: Feedback loops: the whole workbench.

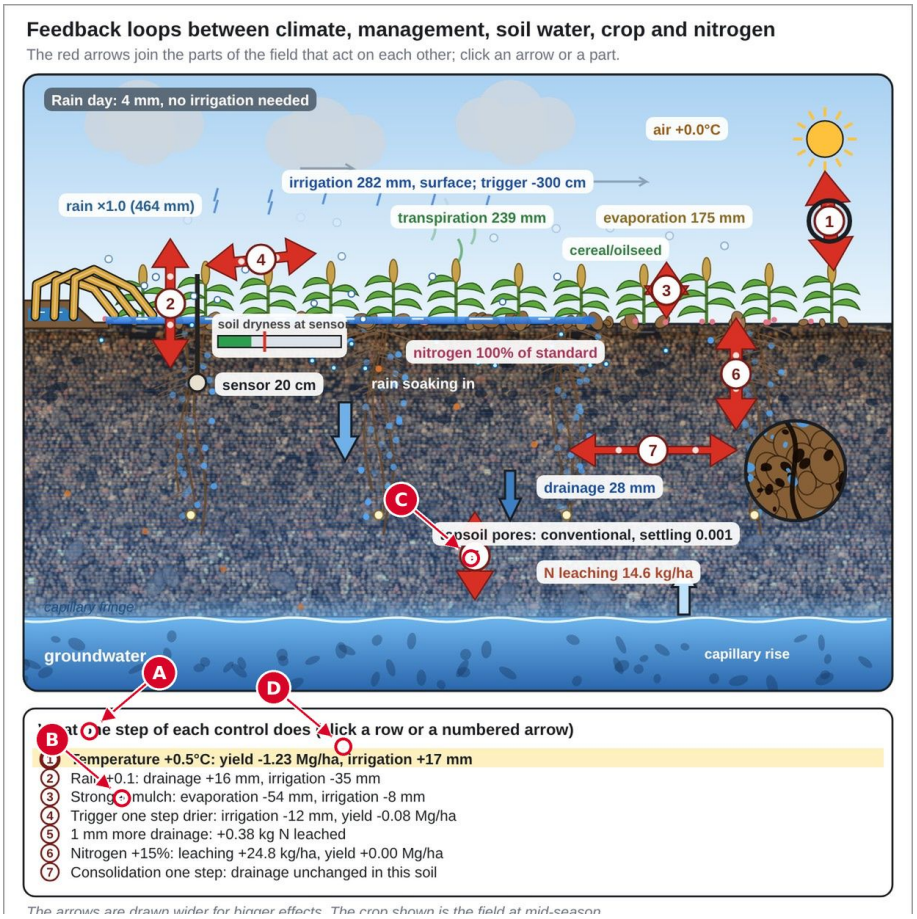


Figure 25.248: Feedback loops: the drawing area.

What the arrows point to. The dashed outlines in the whole-workbench picture mark the layer chosen in the Layers panel (**L**), the drawing (**D**), the Dashboard (**R**) and the controls in Properties (**C**). In the drawing, each numbered arrow ends on the item it names.

- **A** Arrow 1, temperature step: Engine 1 (Eqs. 5.4 to 5.12) re-evaluated; arrow width from Eq. 10.1.
- **B** Arrow 3, stronger mulch: change in evaporation and irrigation, Eqs. 5.4, 5.7 and 9.1.
- **C** Arrow 5, nitrate cost of drainage, kappa_N: Eq. 10.2.
- **D** List of the change produced by one step of each control; widths of the arrows: Eq. 10.1.

Equations behind this screen. Eq. 5.4, Eq. 5.5, Eq. 5.6, Eq. 5.7, Eq. 5.8, Eq. 5.9, Eq. 5.10, Eq. 5.11, Eq. 5.12, Eq. 10.1, Eq. 10.2.

Eq. 5.4. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10. Where explained: chapter “The two computing engines”, section “The meta-model”.

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11. Where explained: chapter "The two computing engines", section "The meta-model".

$$Q \leftarrow Q\,\text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12. Where explained: chapter "The two computing engines", section "The meta-model".

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 10.1. Where explained: chapter "Feedback loops", section "How the seven effects are computed".

$$w_k = 5 + \text{clip}(\mu_k;\ 0,\ 4). \tag{10.1}$$

Provenance (Eq. 10.1): **[Engineered]**: drawing rule.

Eq. 10.2. Where explained: chapter "Feedback loops", section "How the seven effects are computed".

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D}\ \ [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

Provenance (Eq. 10.2): **[Data-driven]**: one-step differences of Daisy results.

25.18.1 The whole layer, part by part

The picture below is the complete drawing of the layer Feedback loops, which in the workbench is scrolled inside the drawing area; here it is shown from top to bottom in several parts, with the playback bar at its start. Each numbered arrow ends on the item that its entry names.

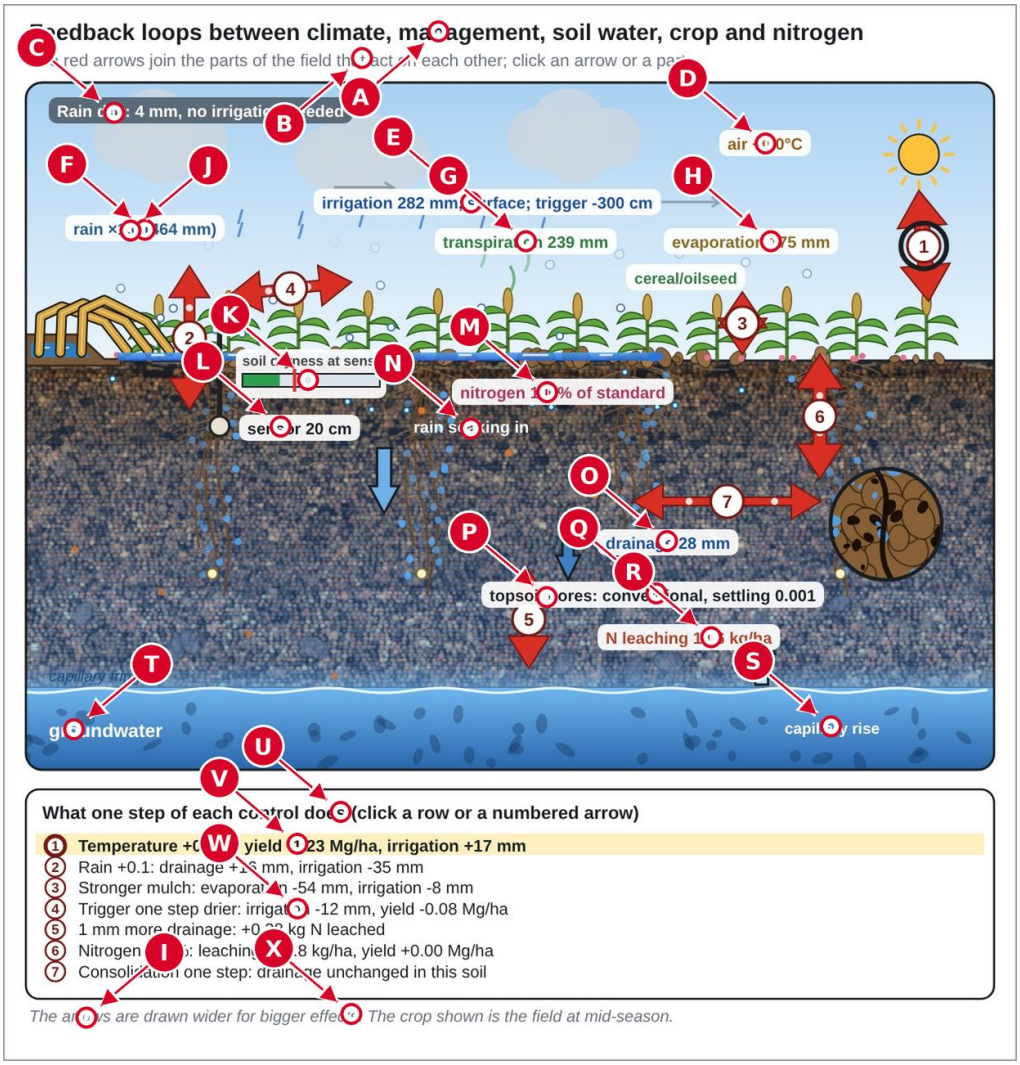


Figure 25.249: Feedback loops, part 1 of 1.

What the arrows point to.

- **A** The title, the feedback loops between climate, management, soil water, crop and nitrogen.
- **B** The instruction, the red arrows join the parts that act on each other, and an arrow or a part is clicked.
- **C** The banner of the rain on the day shown and whether irrigation is needed.
- **D** The air temperature change in force.
- **E** The irrigation of the year, its method and the trigger.
- **F** The rain scaling and the rain of the year.
- **G** The transpiration of the year.
- **H** The evaporation of the year and its change against bare soil.
- **I** The crop shown, the field at mid-season.
- **J** The water held by the mulch.
- **K** The gauge of the soil dryness at the sensor.
- **L** The sensor at 20 cm.
- **M** The nitrogen supply as a share of the standard.
- **N** The rain soaking into the soil.
- **O** The drainage of the year.
- **P** The aggregate that stands for the structure of the topsoil.
- **Q** The topsoil pores, the tillage and the settling rate.

- **R** The nitrogen leaching of the year.
- **S** The capillary rise from the groundwater.
- **T** The groundwater.
- **U** The table of what one step of each control does.
- **V** The row of the loop selected, highlighted and clickable.
- **W** The other rows, each loop with its effect on the yield, irrigation and drainage.
- **X** The note that the arrows are wider for bigger effects.

25.18.2 The layer through its playback

The playback bar runs the layer through time. The pictures below show the parts of the drawing that change, at a third of the run, at two thirds and at the end, with the same arrows and numbers as in the corresponding part above.

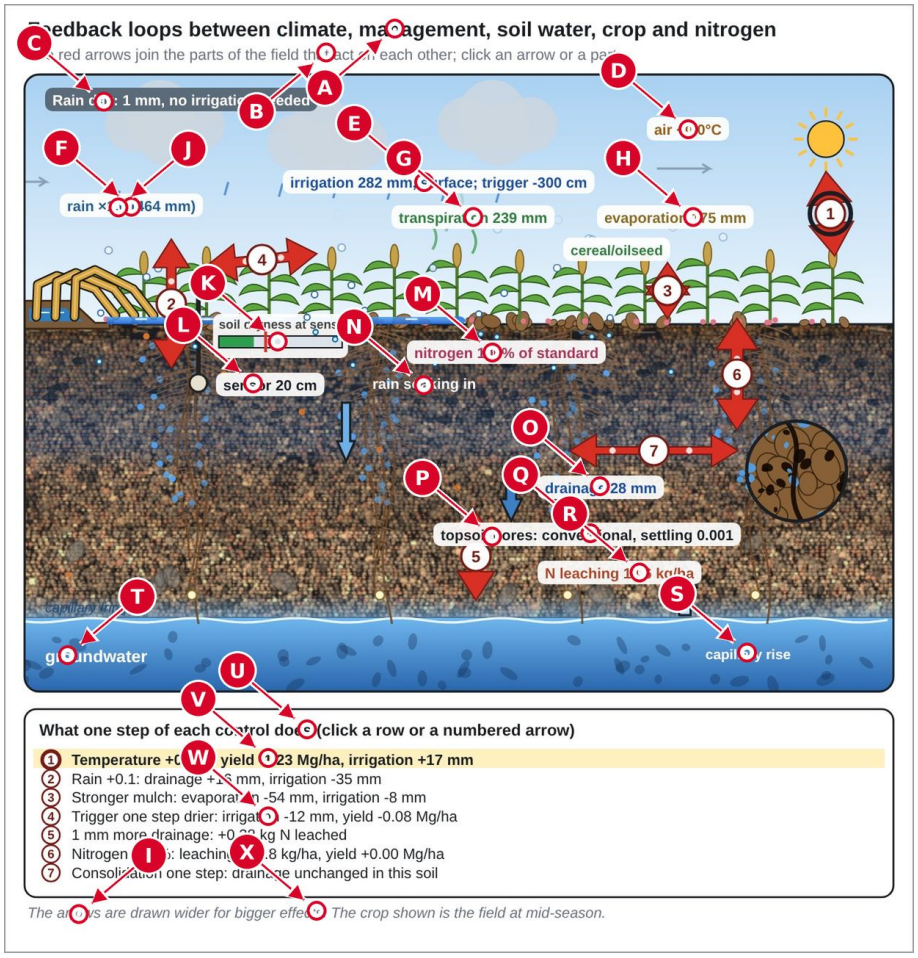


Figure 25.250: Feedback loops, part 1, playback a third of the way (Paused: each feedback loop in turn).

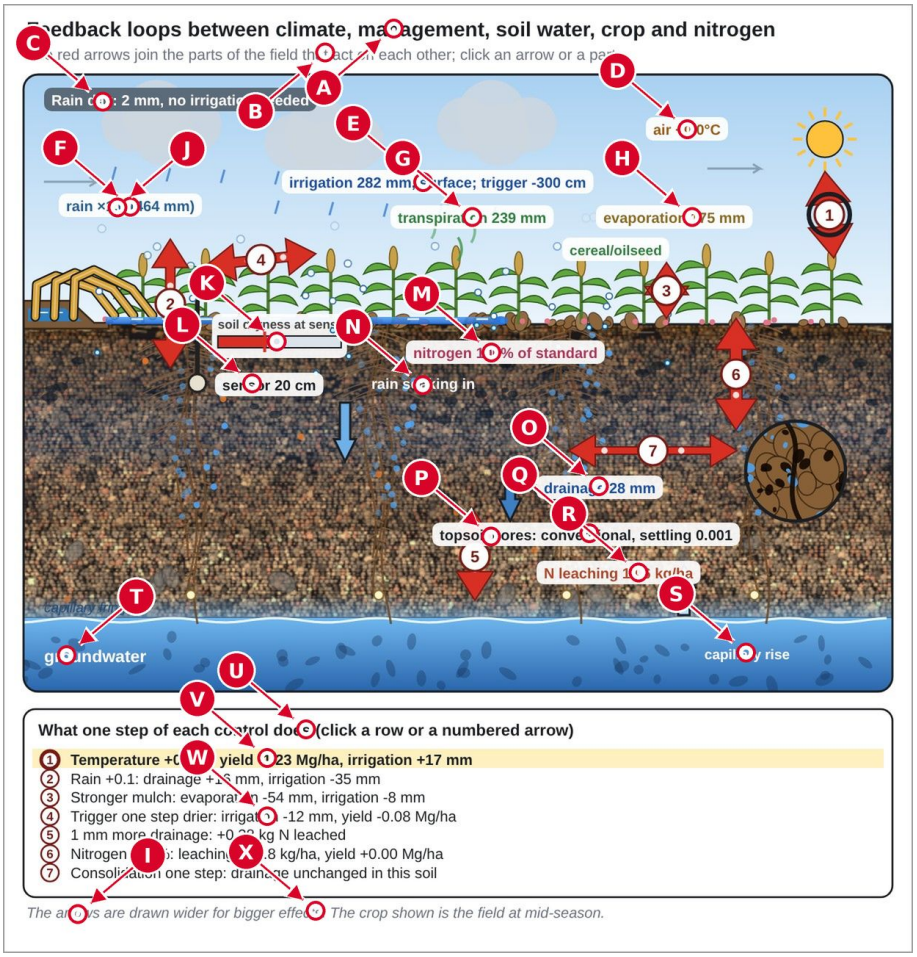


Figure 25.251: Feedback loops, part 1, playback two thirds of the way (Paused: each feedback loop in turn).

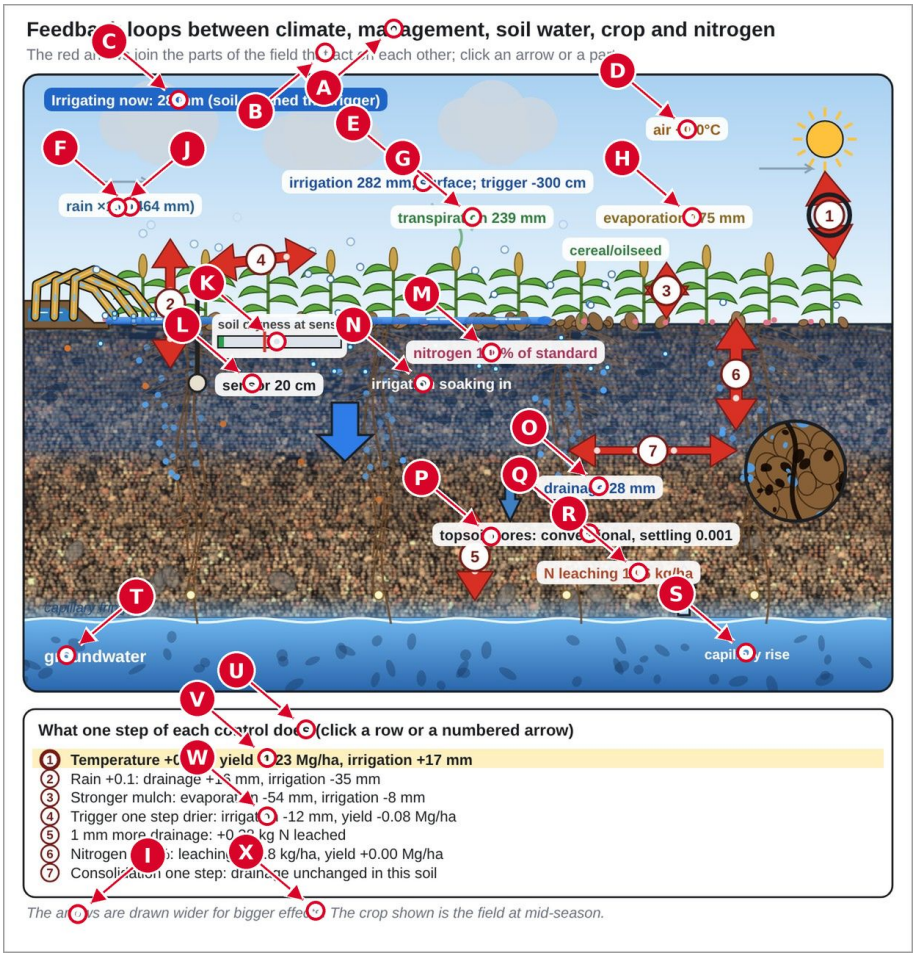


Figure 25.252: Feedback loops, part 1, playback the end (Paused: each feedback loop in turn).

25.19 Screens added in version 1.1

The screens below are new in version 1.1. The numbered arrows point to the headings of each part; the chapters named in each caption explain the screen and its equations.

25.19.1 A design before its run

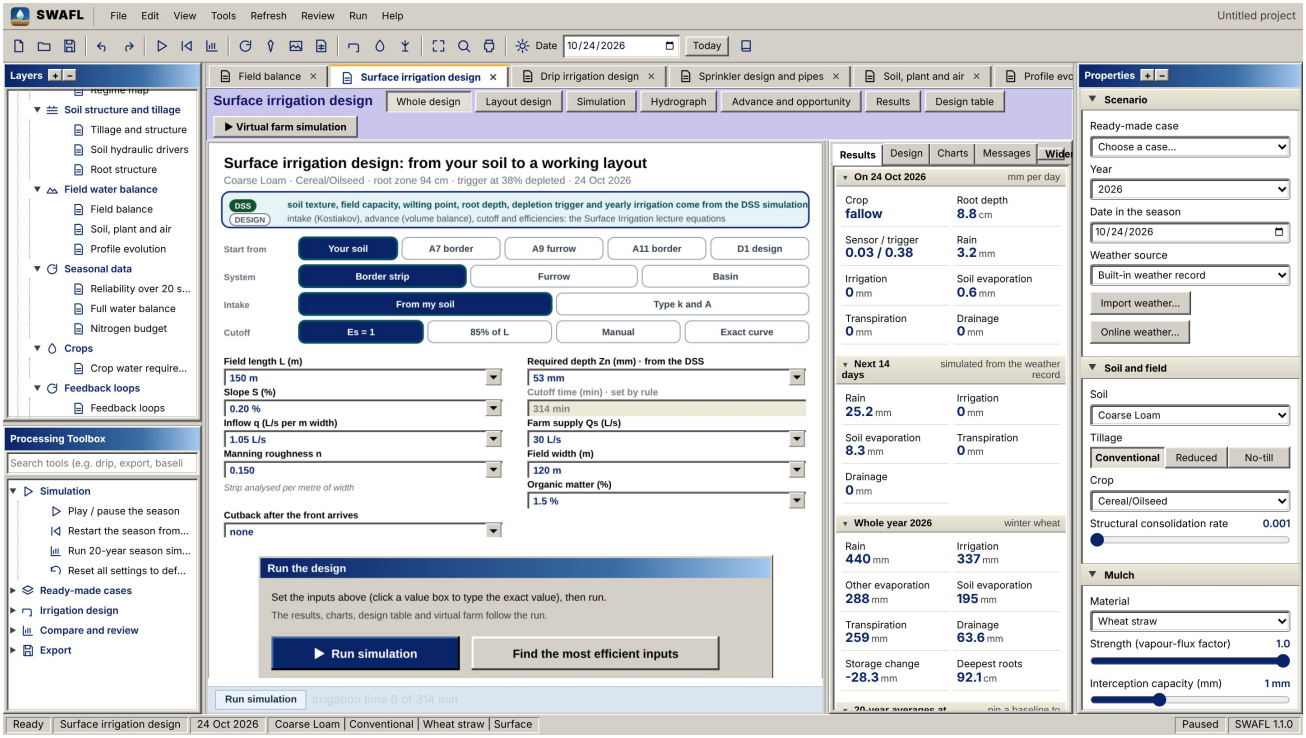


Figure 25.253: A design before its run (Chapter 18).

25.19.2 The value window of a design input

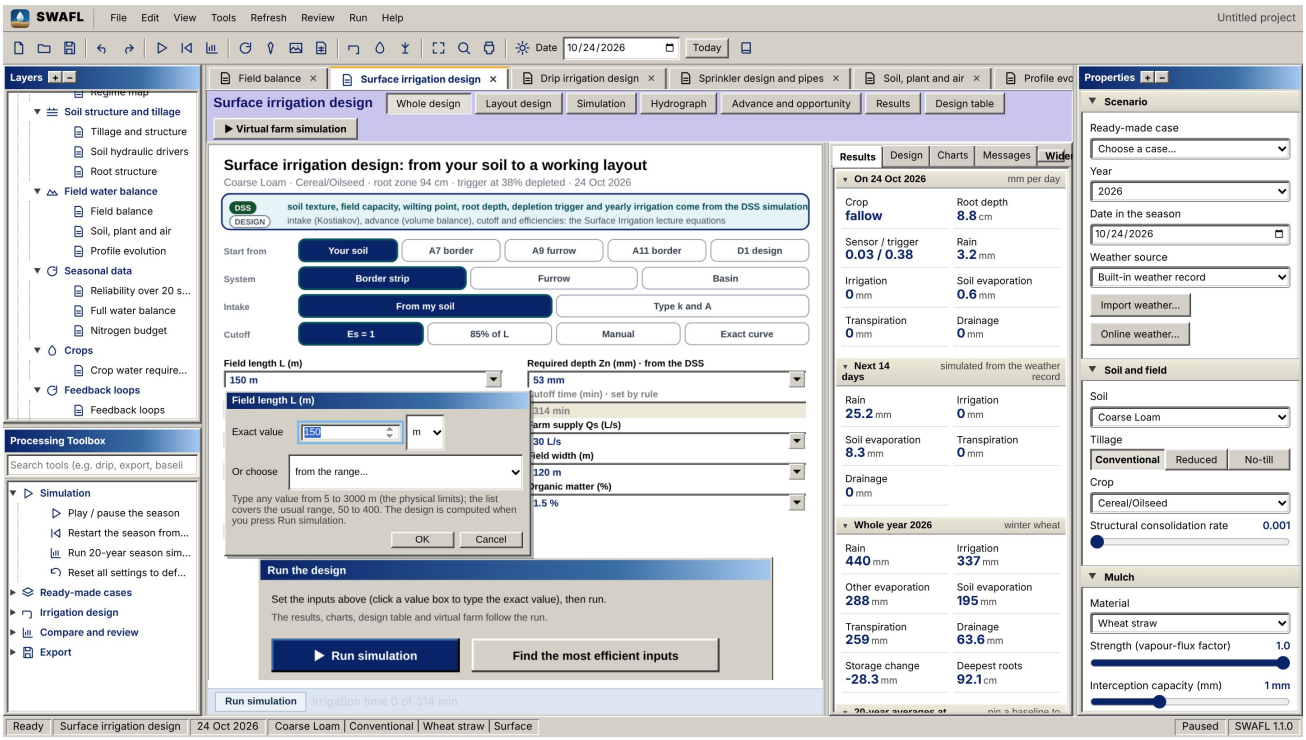


Figure 25.254: The value window of a design input (Chapter 18).

25.19.3 Import weather data

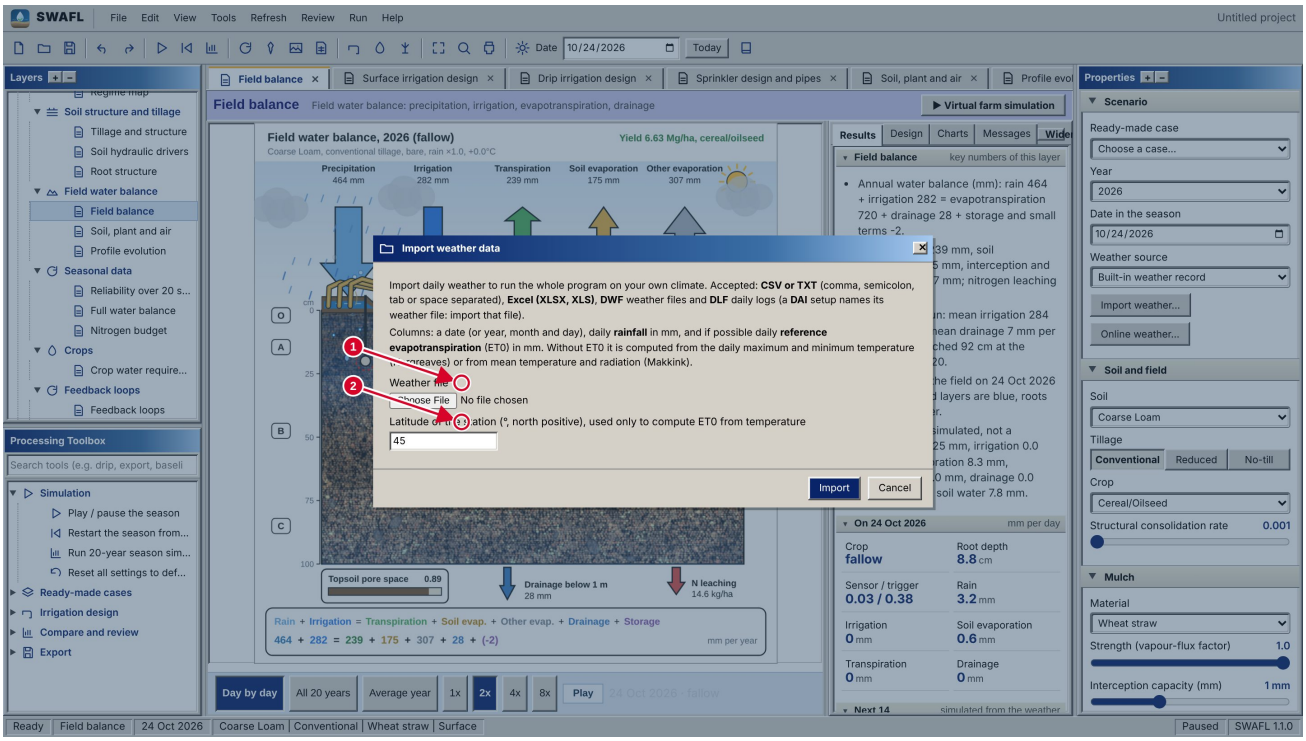


Figure 25.255: Import weather data (Chapter 19).

25.19.4 Online weather for a location

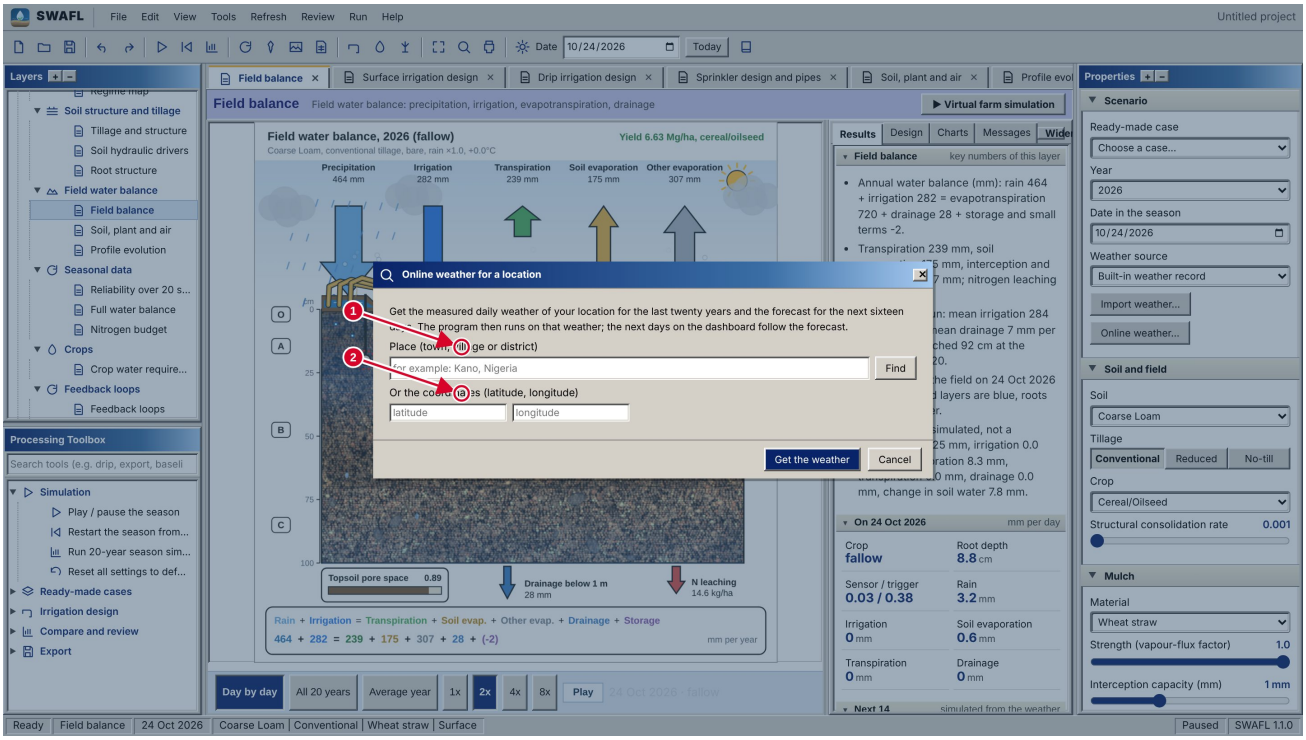


Figure 25.256: Online weather for a location (Chapter 19).

25.19.5 Save report as PDF: choose the layers

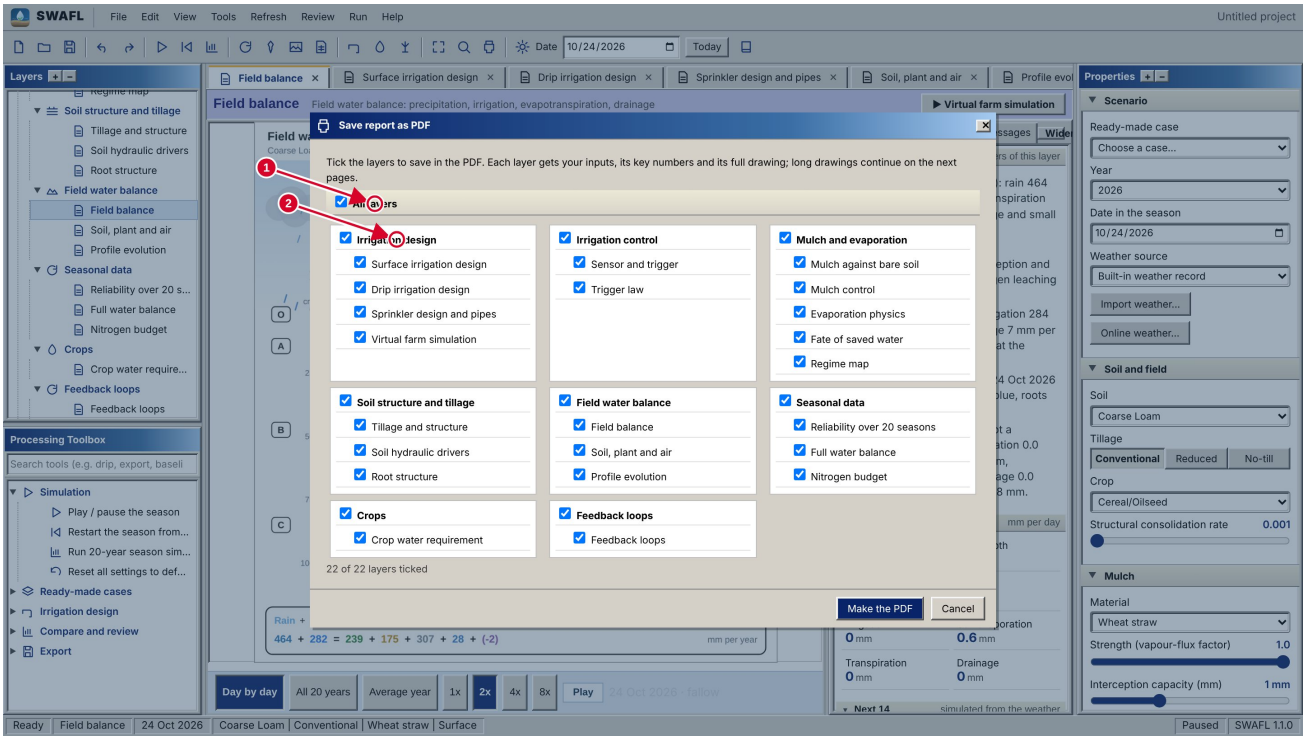


Figure 25.257: Save report as PDF: choose the layers (Chapter 23).

25.19.6 Field sketch to design

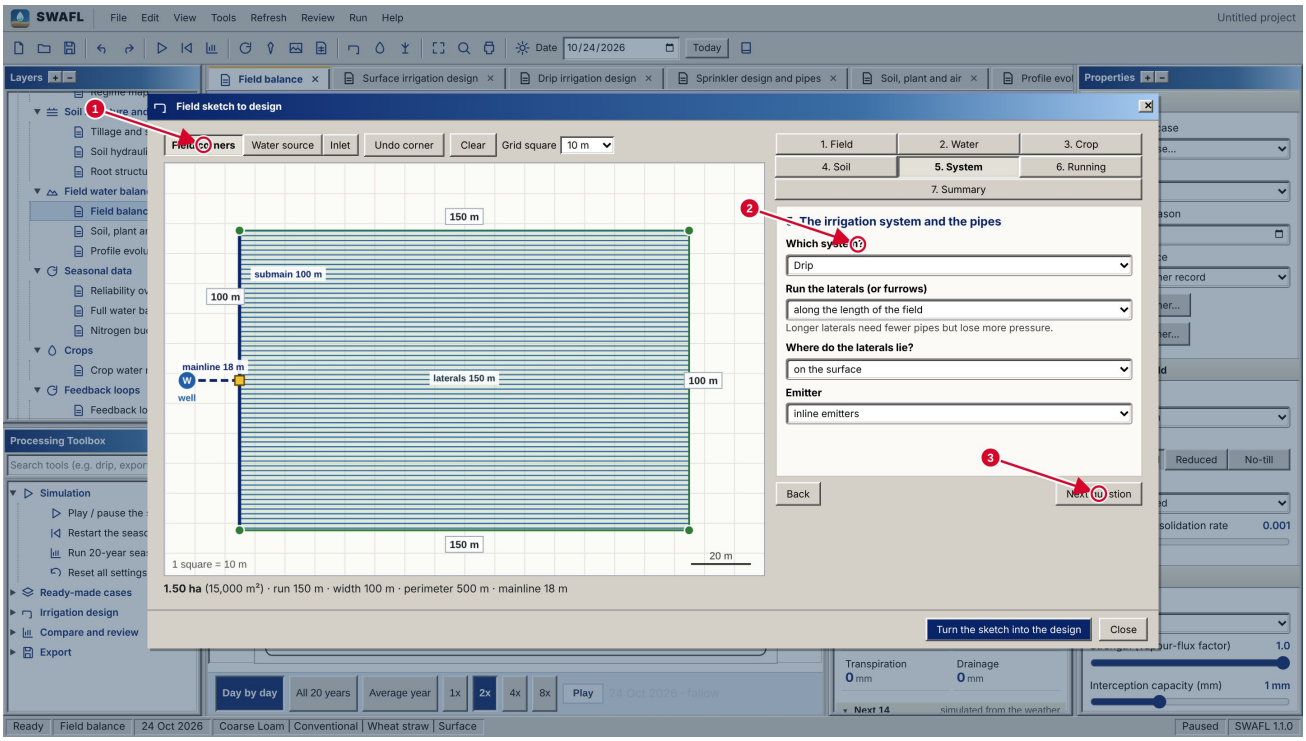


Figure 25.258: Field sketch to design (Chapter 18).

25.19.7 The irrigation systems compared on this field

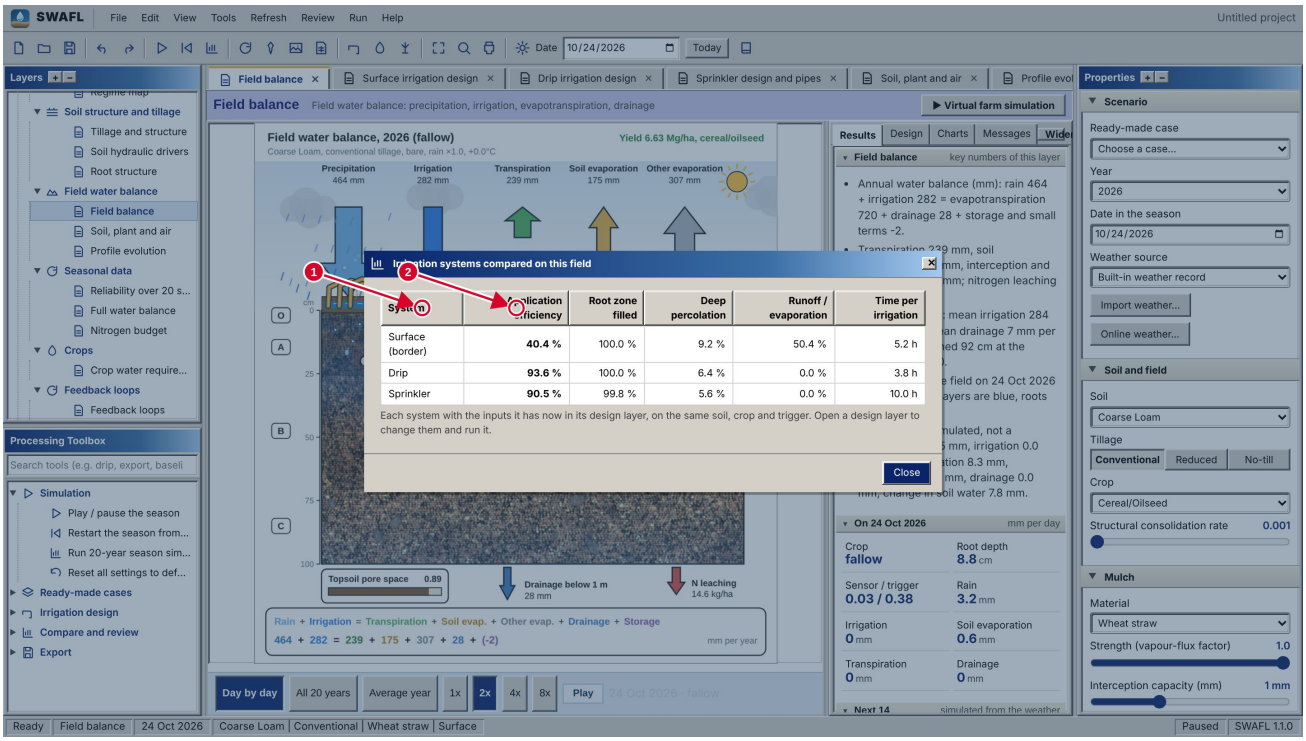


Figure 25.259: The irrigation systems compared on this field (Chapter 18).

25.19.8 The Charts tab of a design after its run

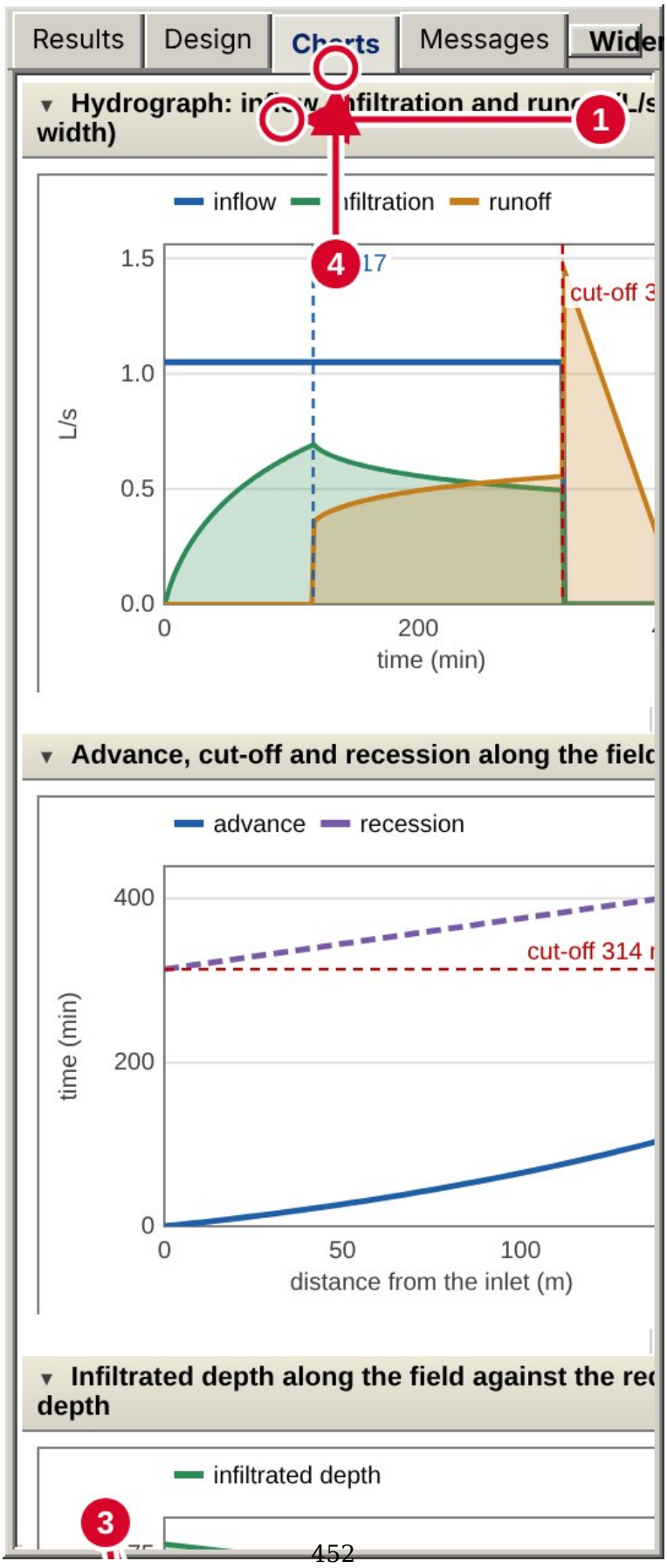


Figure 25.260: The Charts tab of a design after its run (Chapter 14).

25.19.9 Dashboard: the Results tab

The results at the working date and the 20-year averages at your inputs. Click a block header to fold or unfold it (Chapter 14).

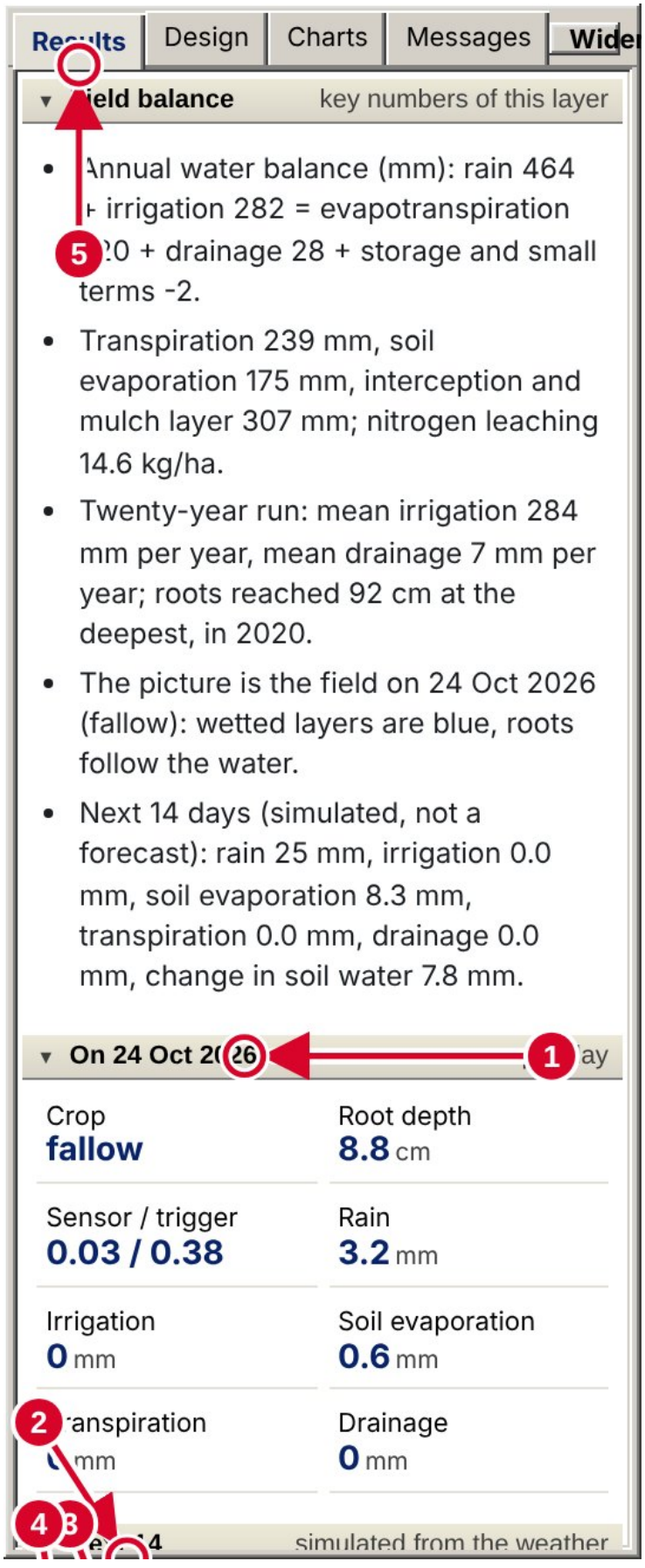


Figure 25.261: Dashboard: the Results tab.

25.19.10 Dashboard: the Design tab

The results of the last run of the open design: efficiencies, depths, times, the pump or the gate and the warnings of the run (Chapters 11 to 13 and 18).

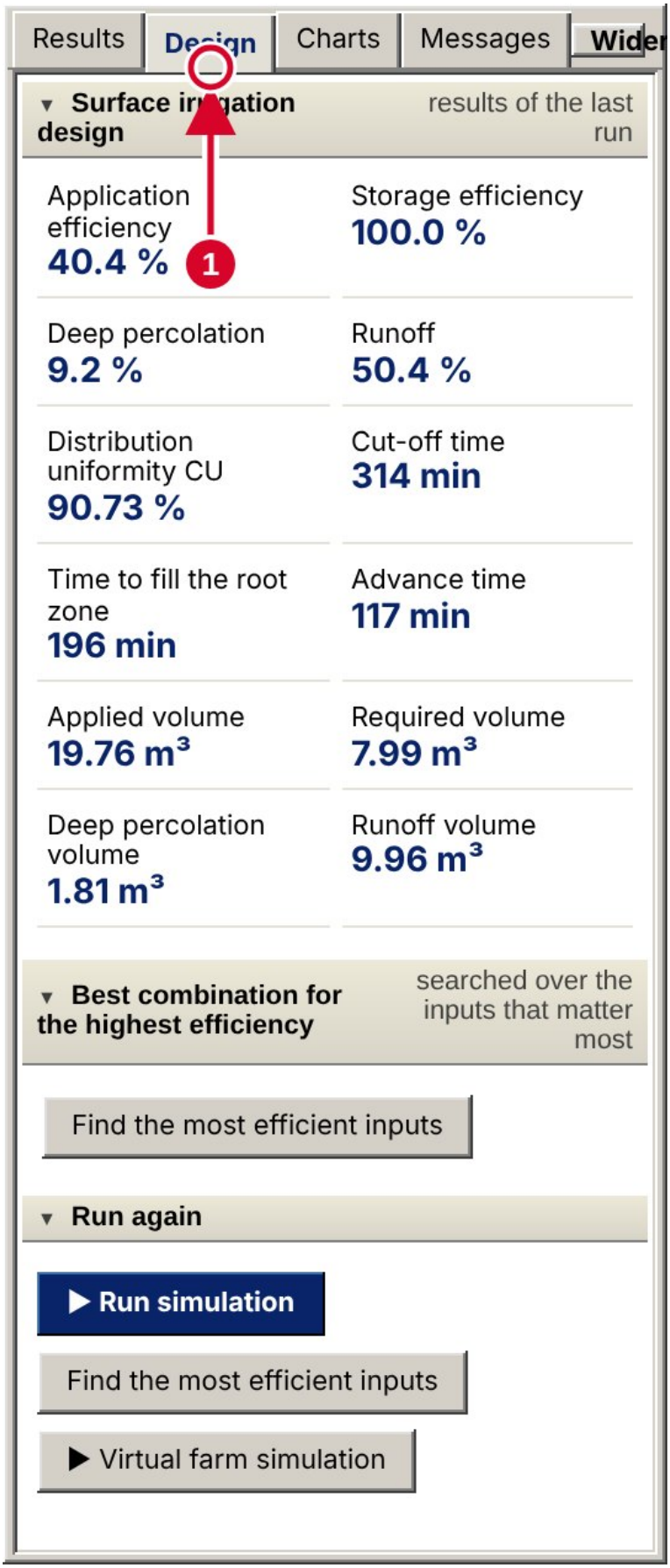


Figure 25.262: Dashboard: the Design tab.

25.19.11 Dashboard: the Charts tab of the Field balance

The yearly charts of the twenty seasons; every chart can be folded (Chapter 14).

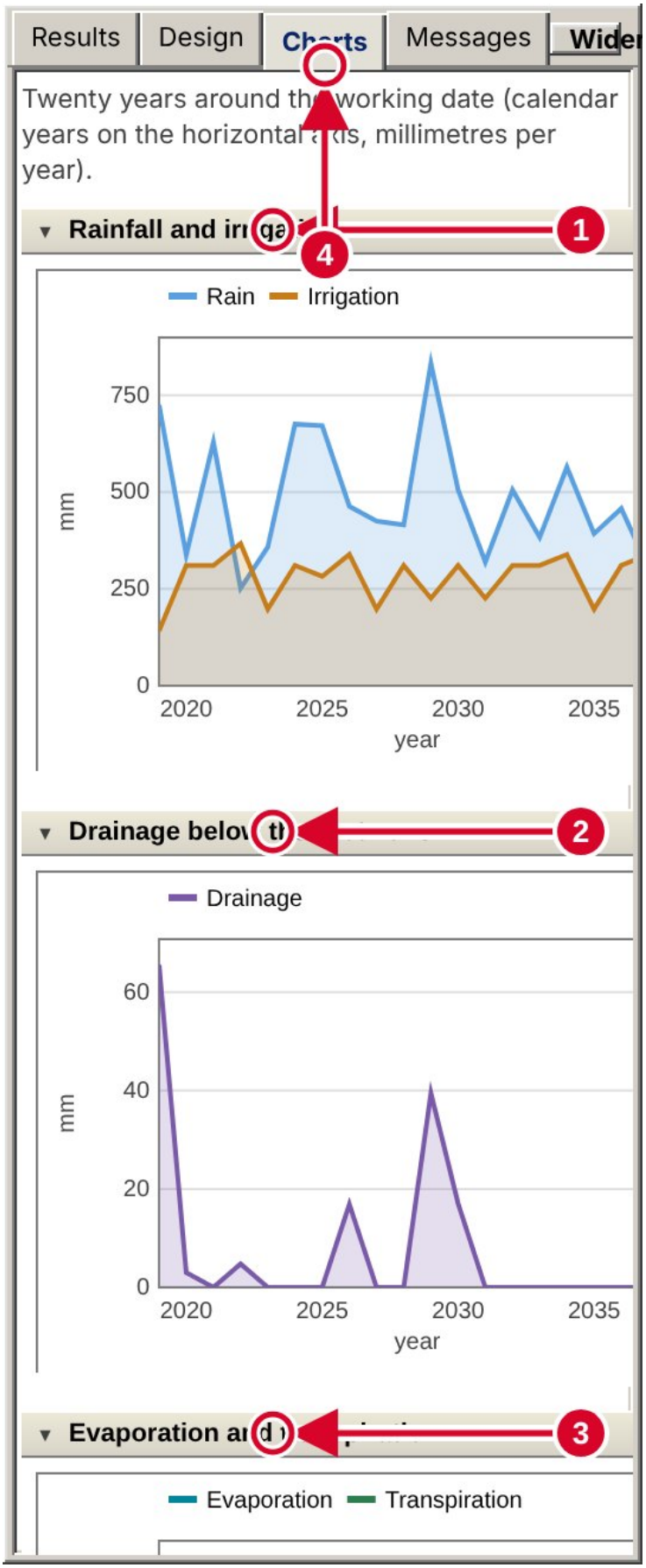


Figure 25.263: Dashboard: the Charts tab of the Field balance.

25.19.12 Dashboard: the Messages tab

The warnings first, then all messages of the session (Chapter 14).

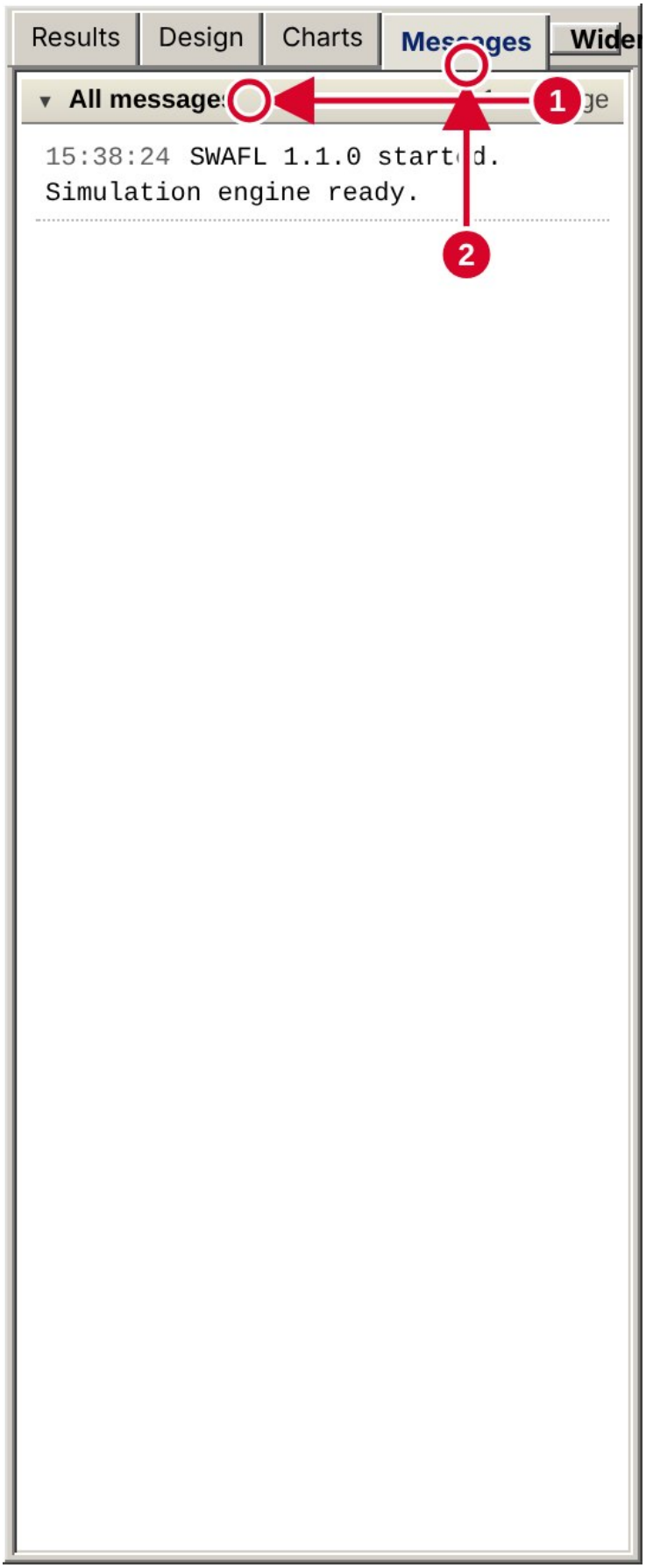


Figure 25.264: Dashboard: the Messages tab.

25.19.13 The surface irrigation design after its run

The whole workbench with the surface design open after Run simulation (Chapters 11 and 18).

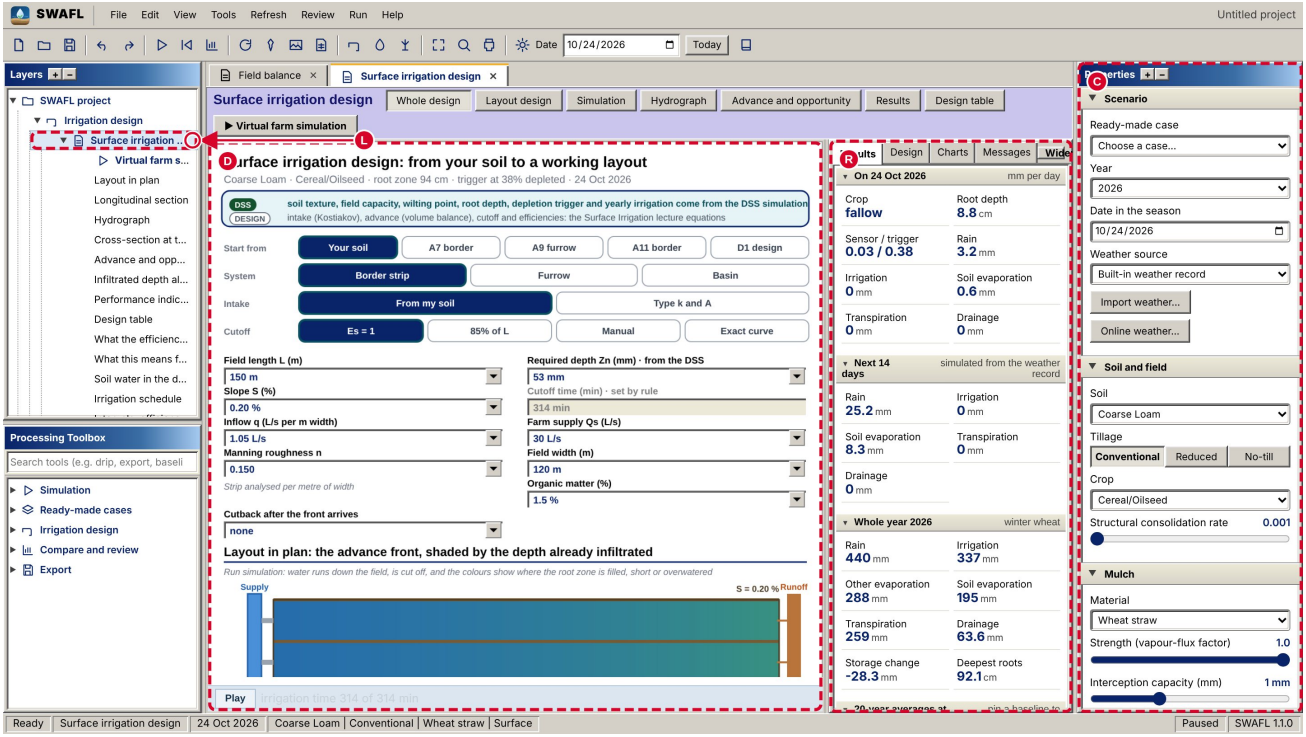


Figure 25.265: The surface irrigation design after its run.

25.19.14 The drip irrigation design after its run

The whole workbench with the drip design open after Run simulation (Chapters 12 and 18).

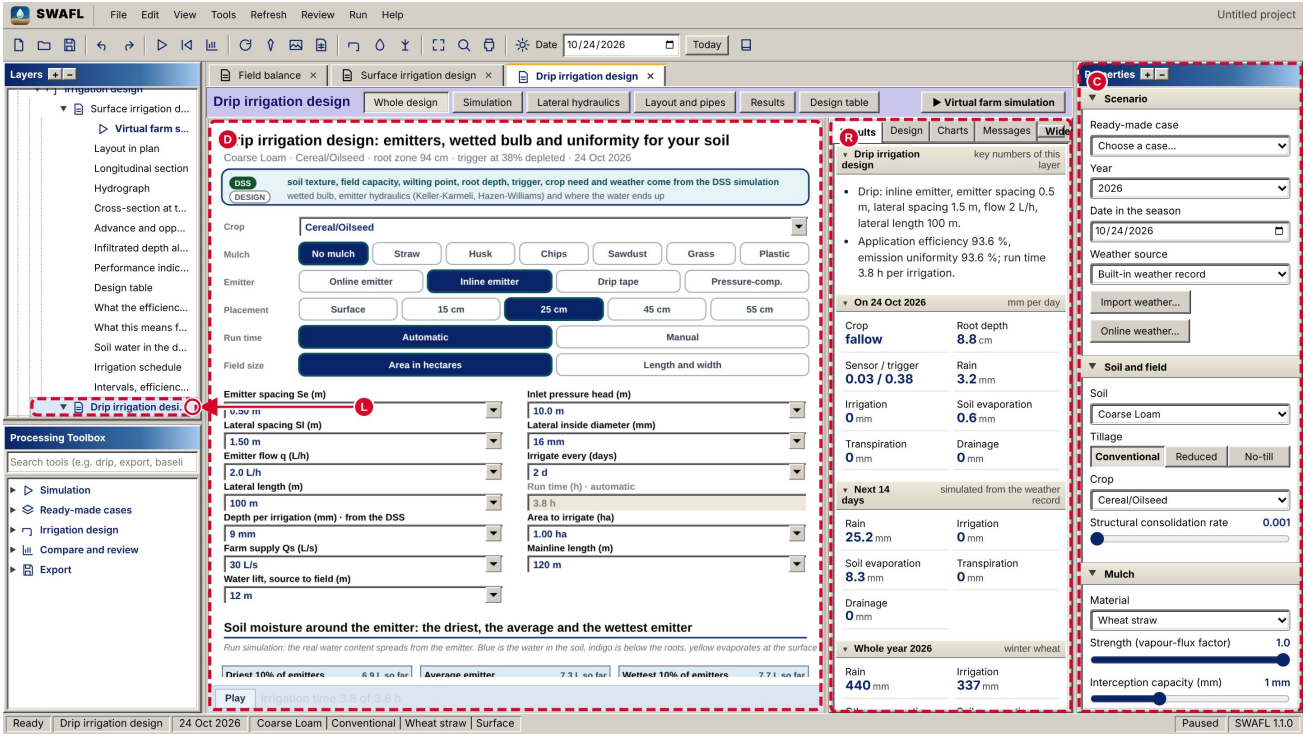


Figure 25.266: The drip irrigation design after its run.

25.19.15 The sprinkler design after its run

The whole workbench with the sprinkler design open after Run simulation (Chapters 13 and 18).

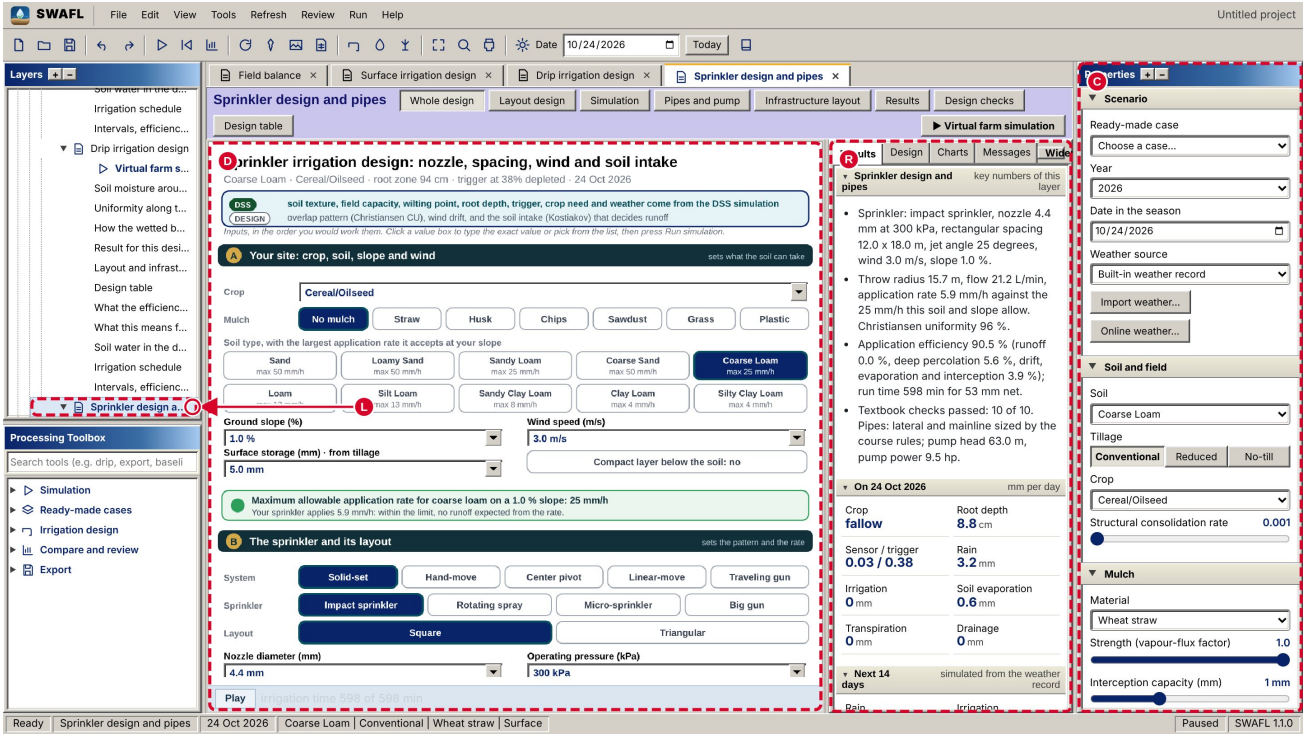


Figure 25.267: The sprinkler design after its run.

25.19.16 Crop water requirement

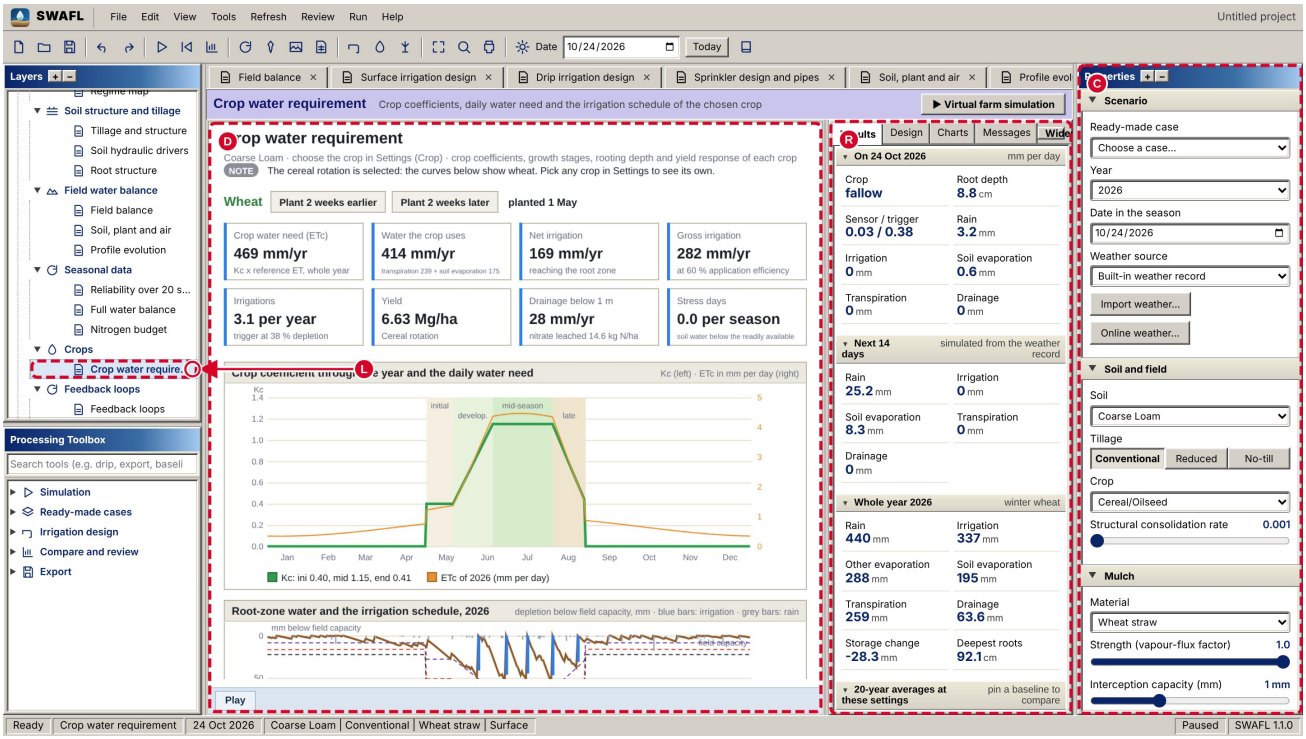


Figure 25.268: Crop water requirement: the whole workbench.

The drawing of the layer from top to bottom. It is explained in Chapter 21; its equations are Eqs. 21.1 to 21.9.

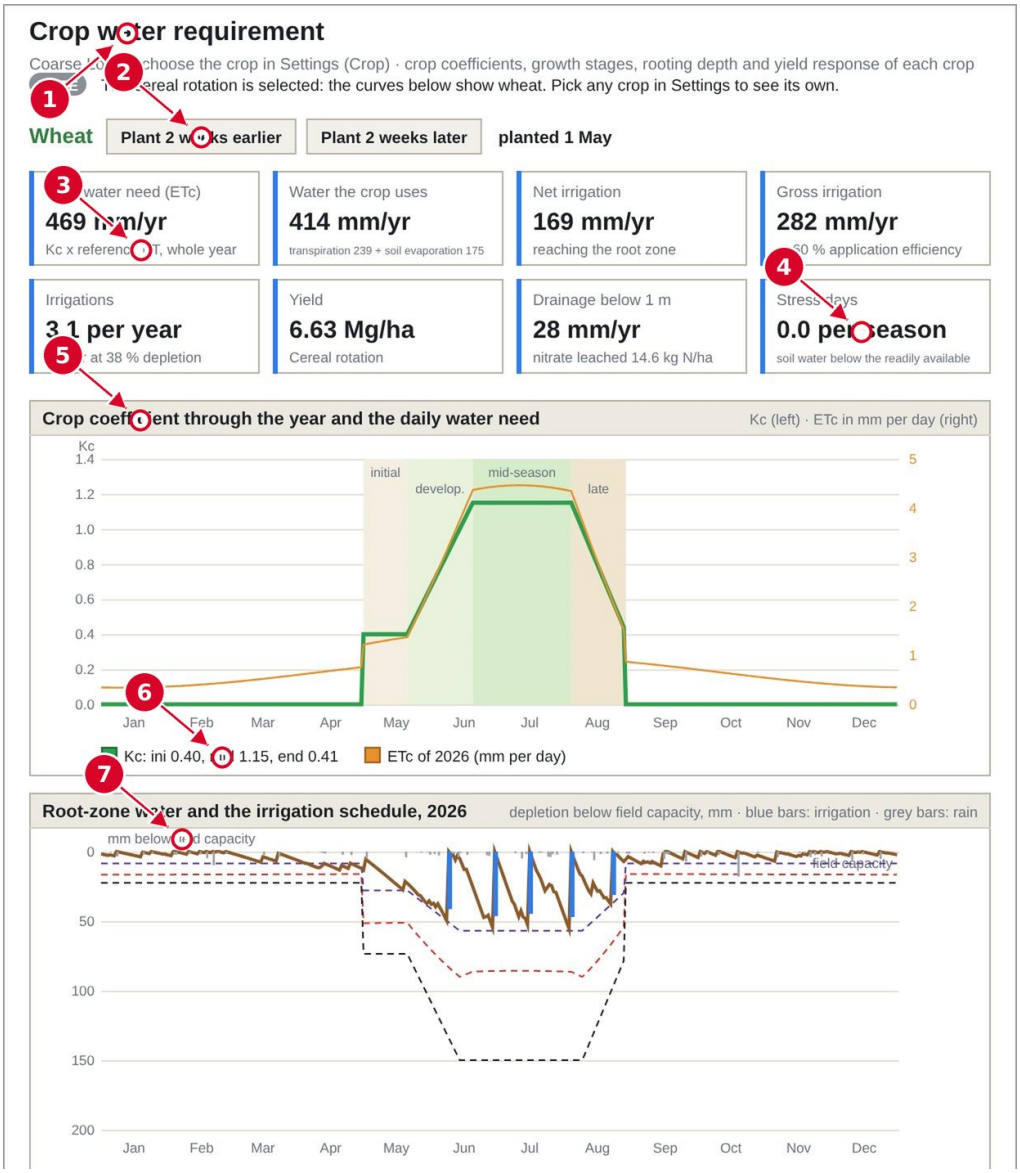
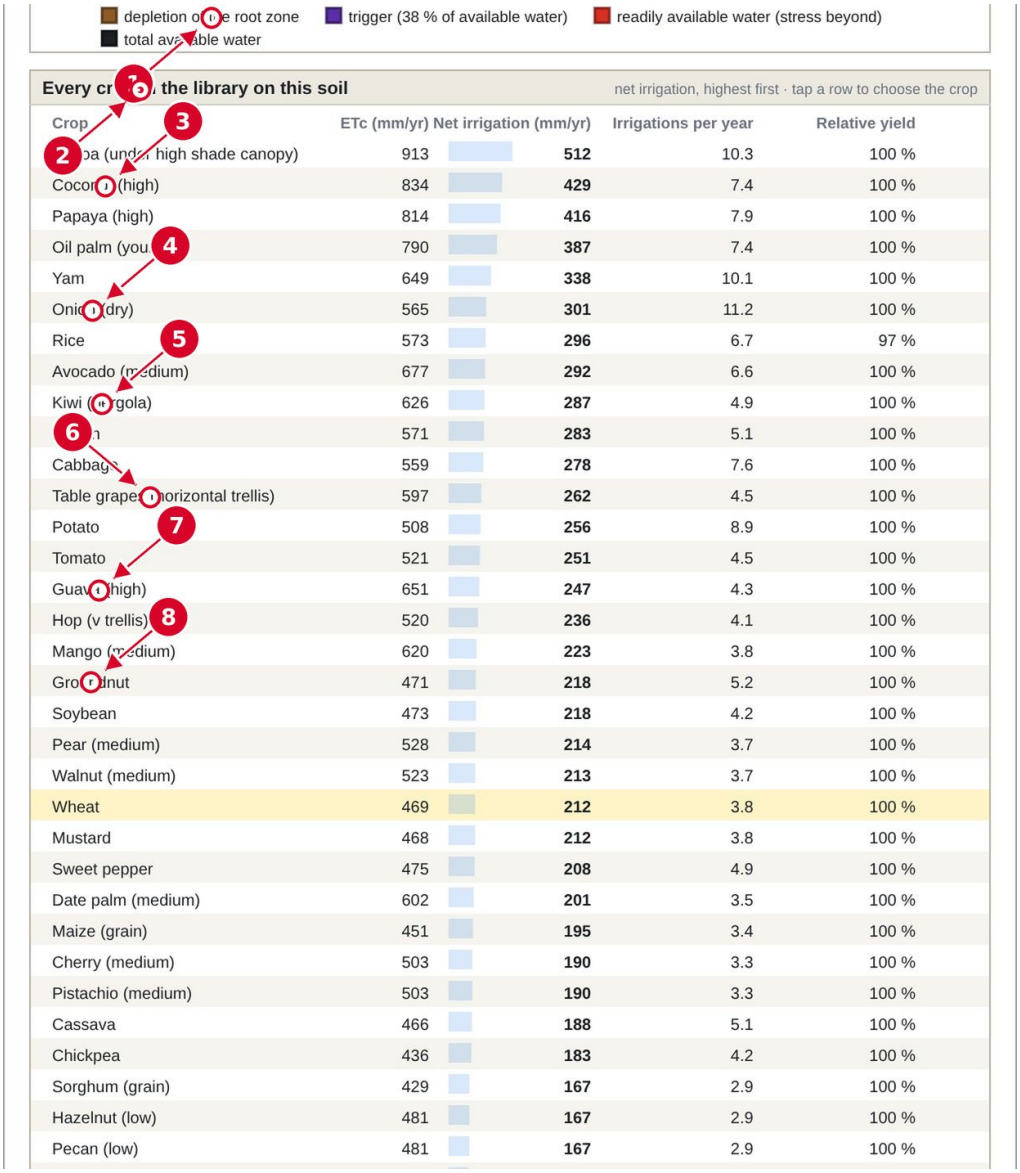


Figure 25.269: Crop water requirement, part 1 of 3.

What the arrows point to.

- **1** The heading or label *Crop water requirement*.
- **2** The heading or label *Plant 2 weeks earlier*.
- **3** The heading or label *Kc x reference ET, whole year*.
- **4** The heading or label *0.0 per season*.
- **5** The heading or label *Crop coefficient through the year and the daily water need*.
- **6** The heading or label *Kc: ini 0.40, mid 1.15, end 0.41*.
- **7** The heading or label *mm below field capacity*.



Crop	ETc (mm/yr)	Net irrigation (mm/yr)	Irrigations per year	Relative yield
[illegible] high shade canopy)	913	**512**	10.3	100 %
Coconut (high)	834	**429**	7.4	100 %
Papaya (high)	814	**416**	7.9	100 %
Oil palm (young)	790	**387**	7.4	100 %
Yam	649	**338**	10.1	100 %
Onion (dry)	565	**301**	11.2	100 %
Rice	573	**296**	6.7	97 %
Avocado (medium)	677	**292**	6.6	100 %
Kiwi (pergola)	626	**287**	4.9	100 %
[illegible]	571	**283**	5.1	100 %
Cabbage	559	**278**	7.6	100 %
Table grapes (horizontal trellis)	597	**262**	4.5	100 %
Potato	508	**256**	8.9	100 %
Tomato	521	**251**	4.5	100 %
Guava (high)	651	**247**	4.3	100 %
Hop (v trellis)	520	**236**	4.1	100 %
Mango (medium)	620	**223**	3.8	100 %
Groundnut	471	**218**	5.2	100 %
Soybean	473	**218**	4.2	100 %
Pear (medium)	528	**214**	3.7	100 %
Walnut (medium)	523	**213**	3.7	100 %
Wheat	469	**212**	3.8	100 %
Mustard	468	**212**	3.8	100 %
Sweet pepper	475	**208**	4.9	100 %
Date palm (medium)	602	**201**	3.5	100 %
Maize (grain)	451	**195**	3.4	100 %
Cherry (medium)	503	**190**	3.3	100 %
Pistachio (medium)	503	**190**	3.3	100 %
Cassava	466	**188**	5.1	100 %
Chickpea	436	**183**	4.2	100 %
Sorghum (grain)	429	**167**	2.9	100 %
Hazelnut (low)	481	**167**	2.9	100 %
Pecan (low)	481	**167**	2.9	100 %

Figure 25.270: Crop water requirement, part 2 of 3.

What the arrows point to.

- **1** The heading or label *depletion of the root zone*.
- **2** The heading or label *Every crop in the library on this soil*.
- **3** The heading or label *Coconut (high)*.
- **4** The heading or label *Onion (dry)*.
- **5** The heading or label *Kiwi (pergola)*.
- **6** The heading or label *Table grapes (horizontal trellis)*.
- **7** The heading or label *Guava (high)*.
- **8** The heading or label *Groundnut*.

Cherimoya (medium)	545		**161**	2.8	100 %
Citrus (medium [illegible]y, short)	552		**152**	2.6	100 %
[illegible]ple (medium)	459		**149**	2.6	100 %
Cowpea (beans)	398		**148**	3.4	100 %
[illegible]	395		**144**	2.6	100 %
Wine grapes (pergola)	457		**125**	2.2	100 %
Green gram (mung bean)	372		**123**	2.9	100 %
Peach (low)	435		**119**	2.1	100 %
Almond (medium)	435		**119**	2.1	100 %
Apricot (young)	380		**76**	1.3	100 %
[illegible]live (traditional, medium density)	446		**61**	1.1	100 %
Persimmon (young, low)	358		**58**	1.0	100 %
Pomegranate (low)	358		**58**	1.0	100 %
Highbush blueberry (low)	338		**55**	1.1	100 %
Fig tree (young, low)	327		**24**	0.4	100 %

Figure 25.271: Crop water requirement, part 3 of 3.

What the arrows point to.

- **1** The heading or label *Cherimoya (medium)*.
- **2** The heading or label *Cowpea (beans)*.
- **3** The heading or label *Green gram (mung bean)*.
- **4** The heading or label *Apricot (young)*.
- **5** The heading or label *Pomegranate (low)*.

25.19.17 Reliability over 20 seasons

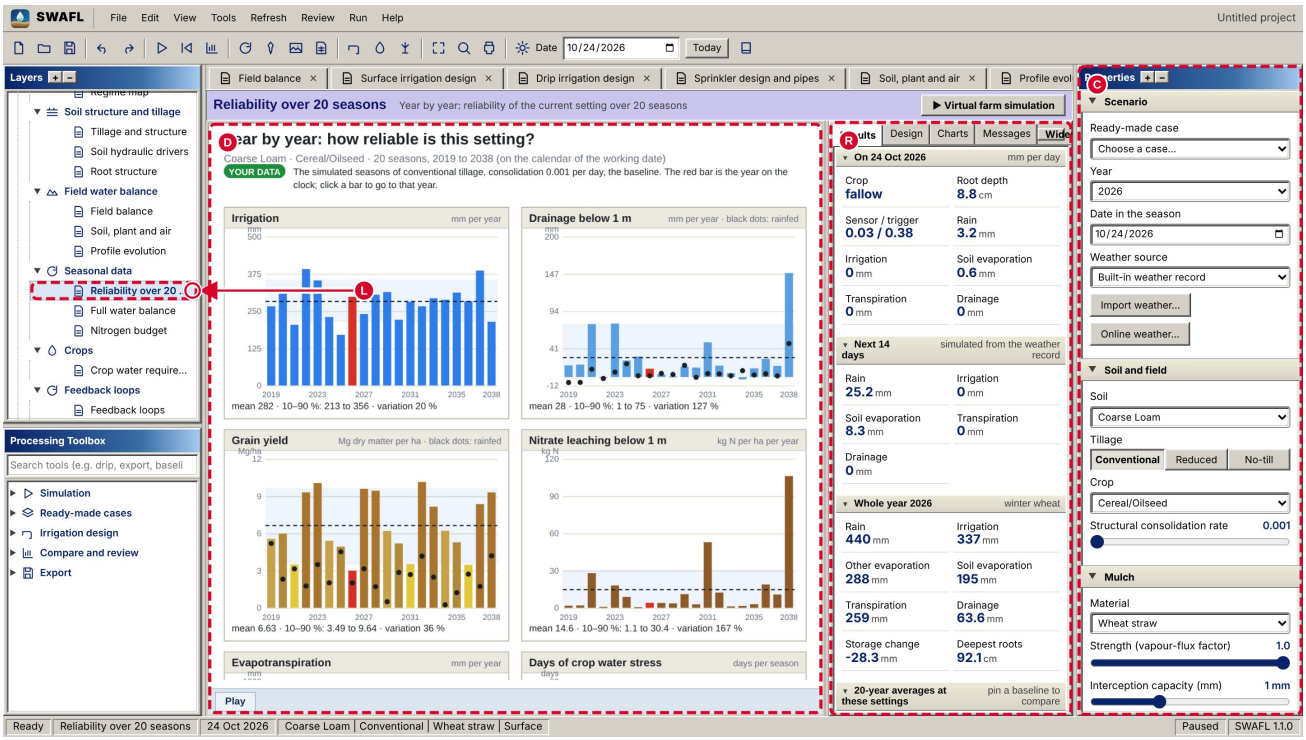


Figure 25.272: Reliability over 20 seasons: the whole workbench.

The drawing of the layer from top to bottom. It is explained in Chapter 22; its equations are Eqs. 22.1.

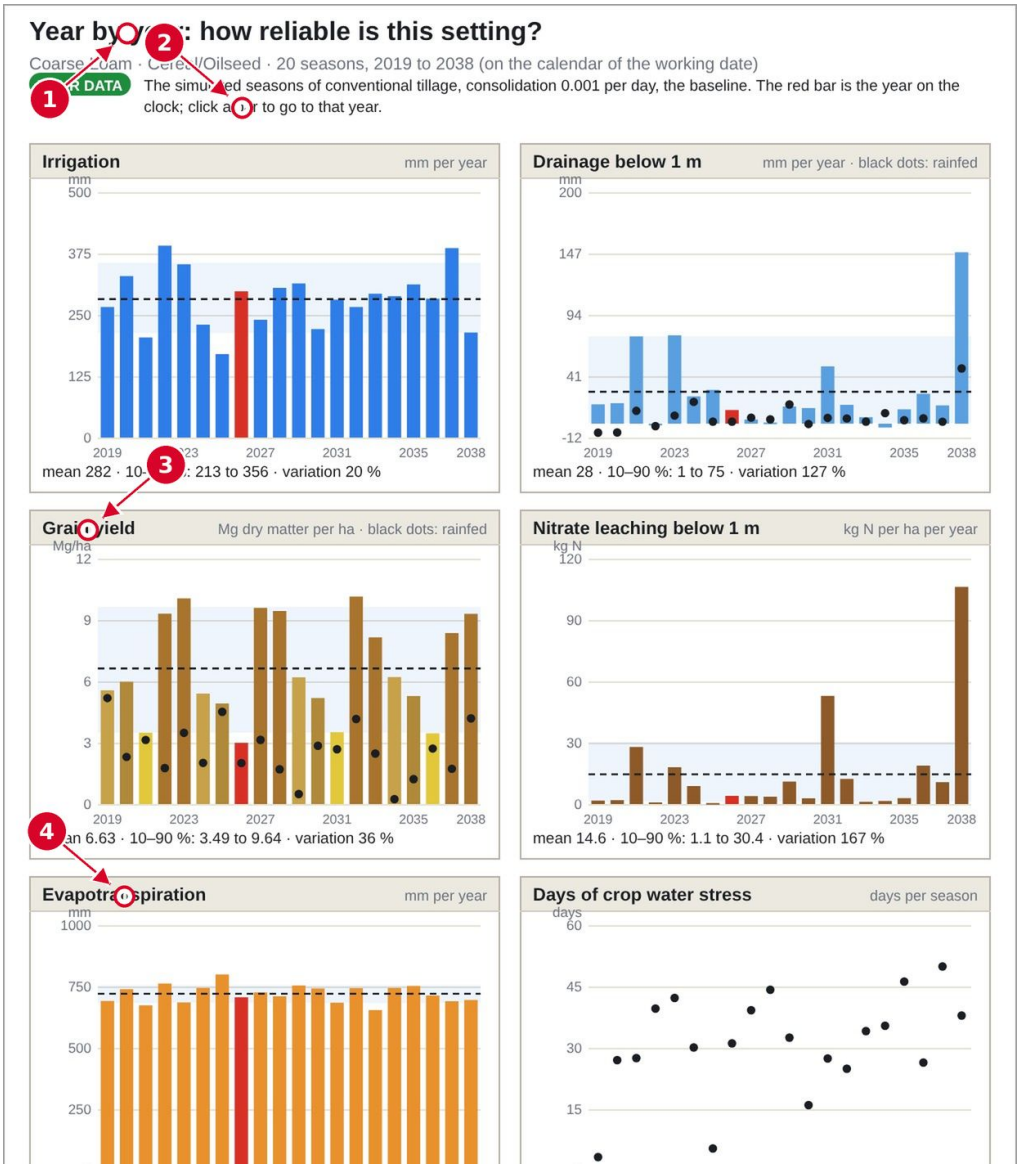


Figure 25.273: Reliability over 20 seasons, part 1 of 2.

What the arrows point to.

- **1** The heading or label *Year by year: how reliable is this setting?*.
- **2** The heading or label *clock; click a bar to go to that year.*.
- **3** The heading or label *Grain yield*.
- **4** The heading or label *Evapotranspiration*.

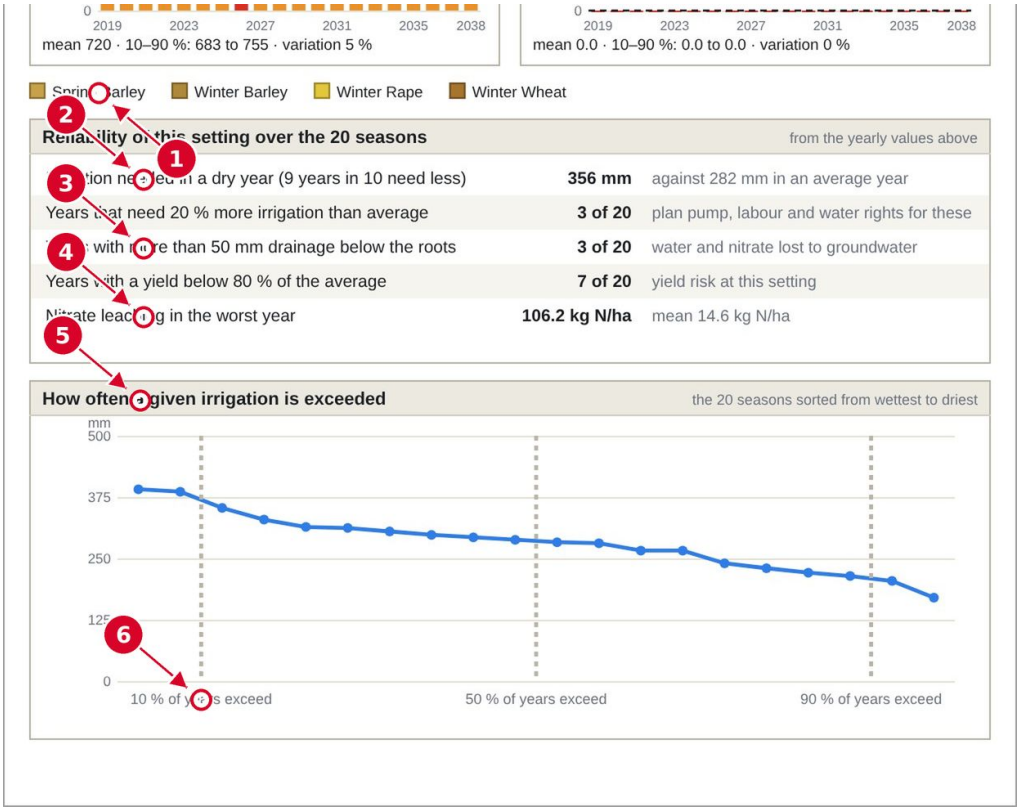


Figure 25.274: Reliability over 20 seasons, part 2 of 2.

What the arrows point to.

- **1** The heading or label *Spring Barley*.
- **2** The heading or label *Irrigation needed in a dry year (9 years in 10 need less)*.
- **3** The heading or label *Years with more than 50 mm drainage below the roots*.
- **4** The heading or label *Nitrate leaching in the worst year*.
- **5** The heading or label *How often a given irrigation is exceeded*.
- **6** The heading or label *10 % of years exceed*.

25.19.18 Full water balance

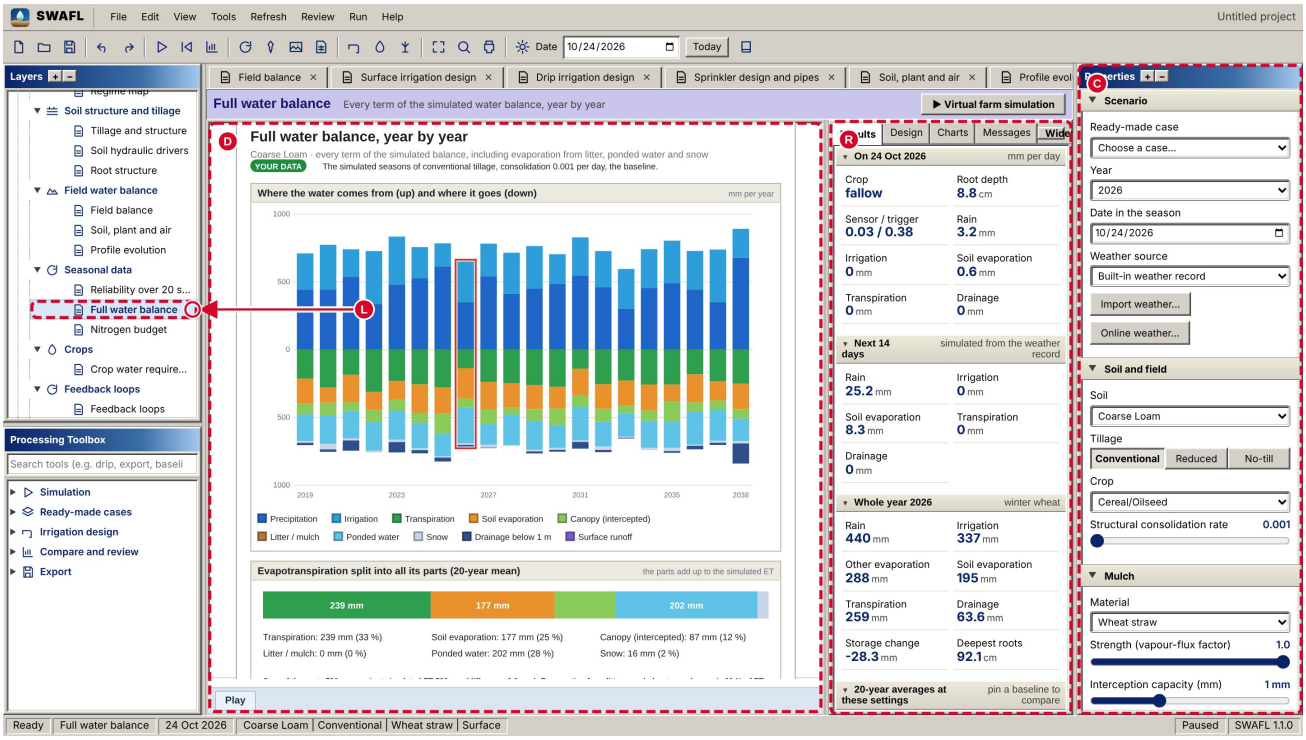


Figure 25.275: Full water balance: the whole workbench.

The drawing of the layer from top to bottom. It is explained in Chapter 22; its equations are Eqs. 22.1.

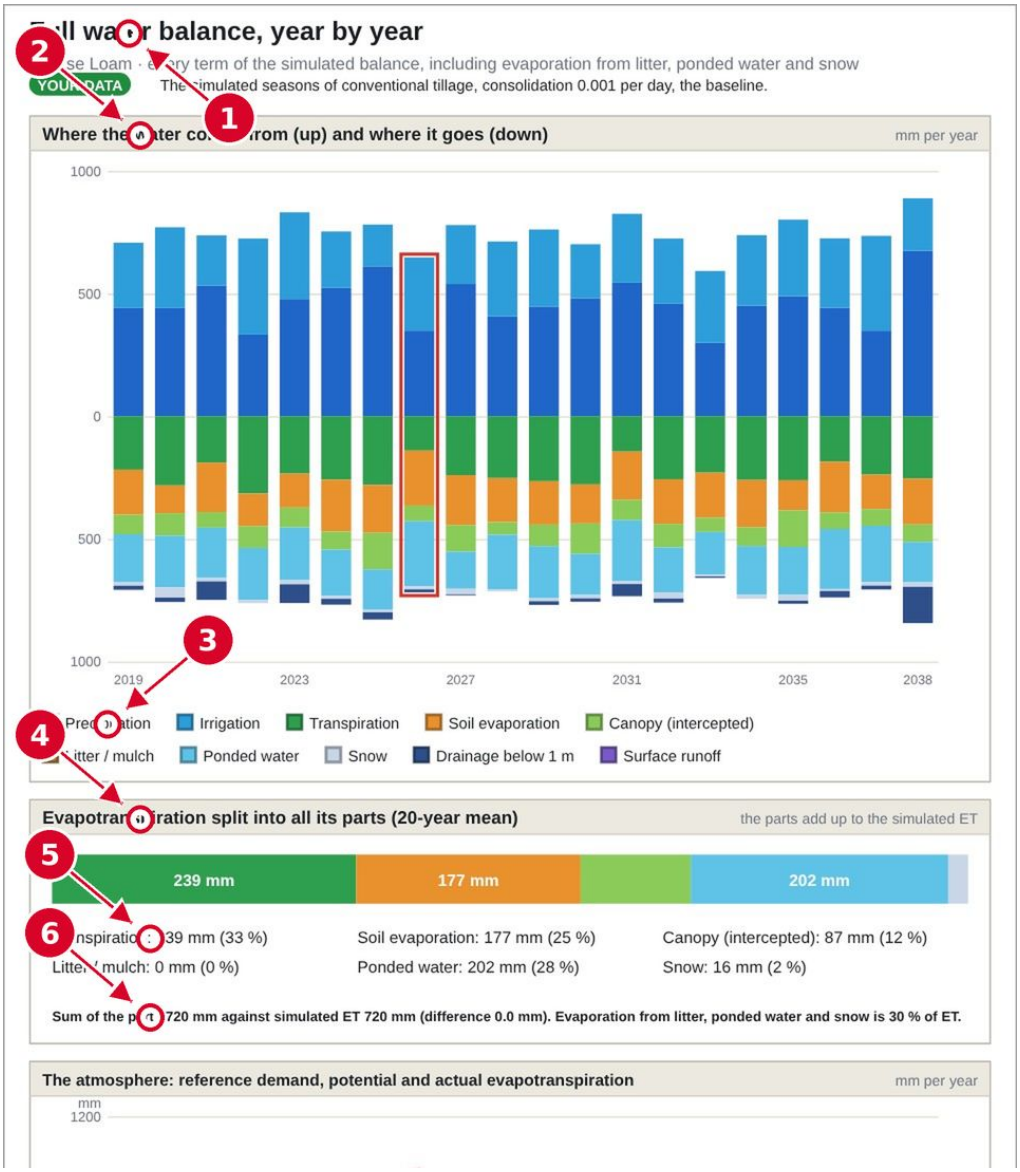


Figure 25.276: Full water balance, part 1 of 2.

What the arrows point to.

- **1** The heading or label *Full water balance, year by year*.
- **2** The heading or label *Where the water comes from (up) and where it goes (down)*.
- **3** The heading or label *Precipitation*.
- **4** The heading or label *Evapotranspiration split into all its parts (20-year mean)*.
- **5** The heading or label *Transpiration: 239 mm (33 %)*.
- **6** The heading or label *Sum of the parts 720 mm against simulated ET 720 mm (difference 0.0 mm). Evaporation from litter, ponded water and snow is 30 % of ET.*.

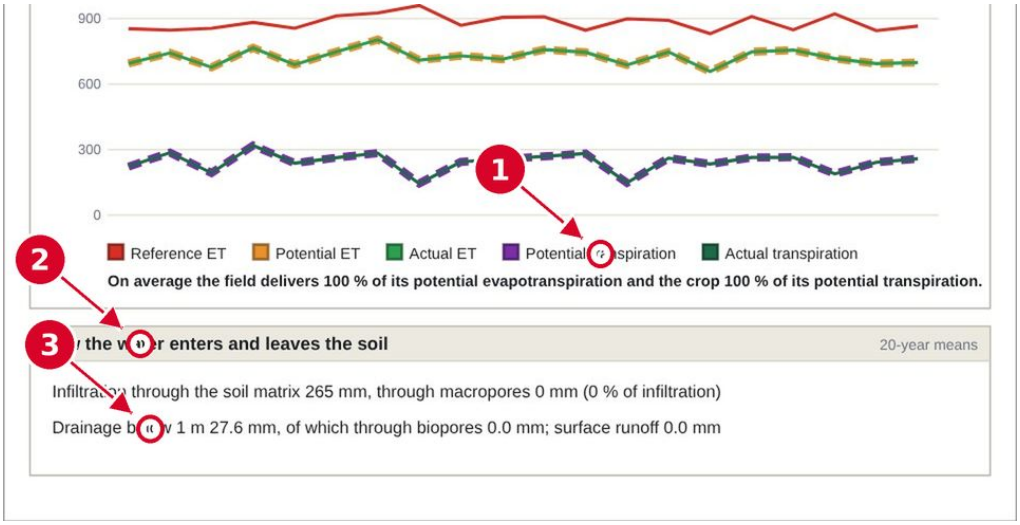


Figure 25.277: Full water balance, part 2 of 2.

What the arrows point to.

- **1** The heading or label *Potential transpiration*.
- **2** The heading or label *How the water enters and leaves the soil*.
- **3** The heading or label *Drainage below 1 m 27.6 mm, of which through biopores 0.0 mm; surface runoff 0.0 mm*.

25.19.19 Nitrogen budget

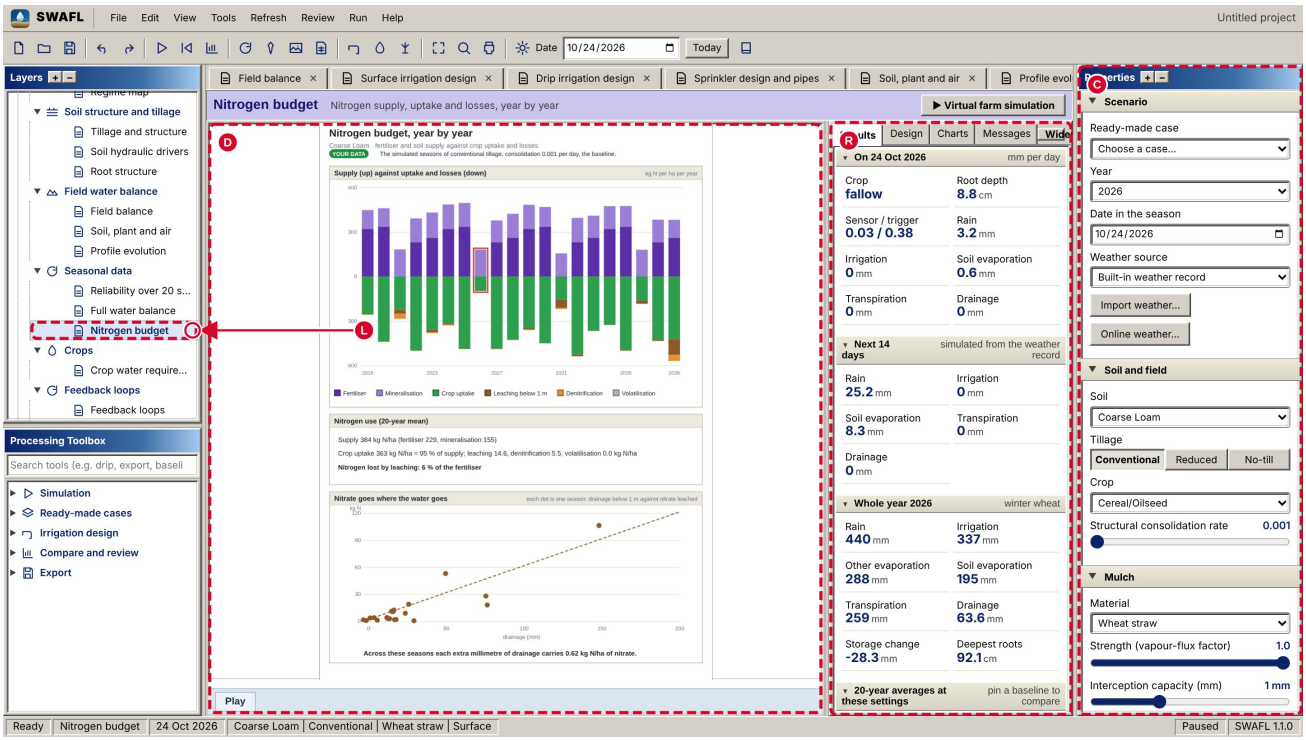


Figure 25.278: Nitrogen budget: the whole workbench.

The drawing of the layer from top to bottom. It is explained in Chapter 22; its equations are Eqs. 22.1.

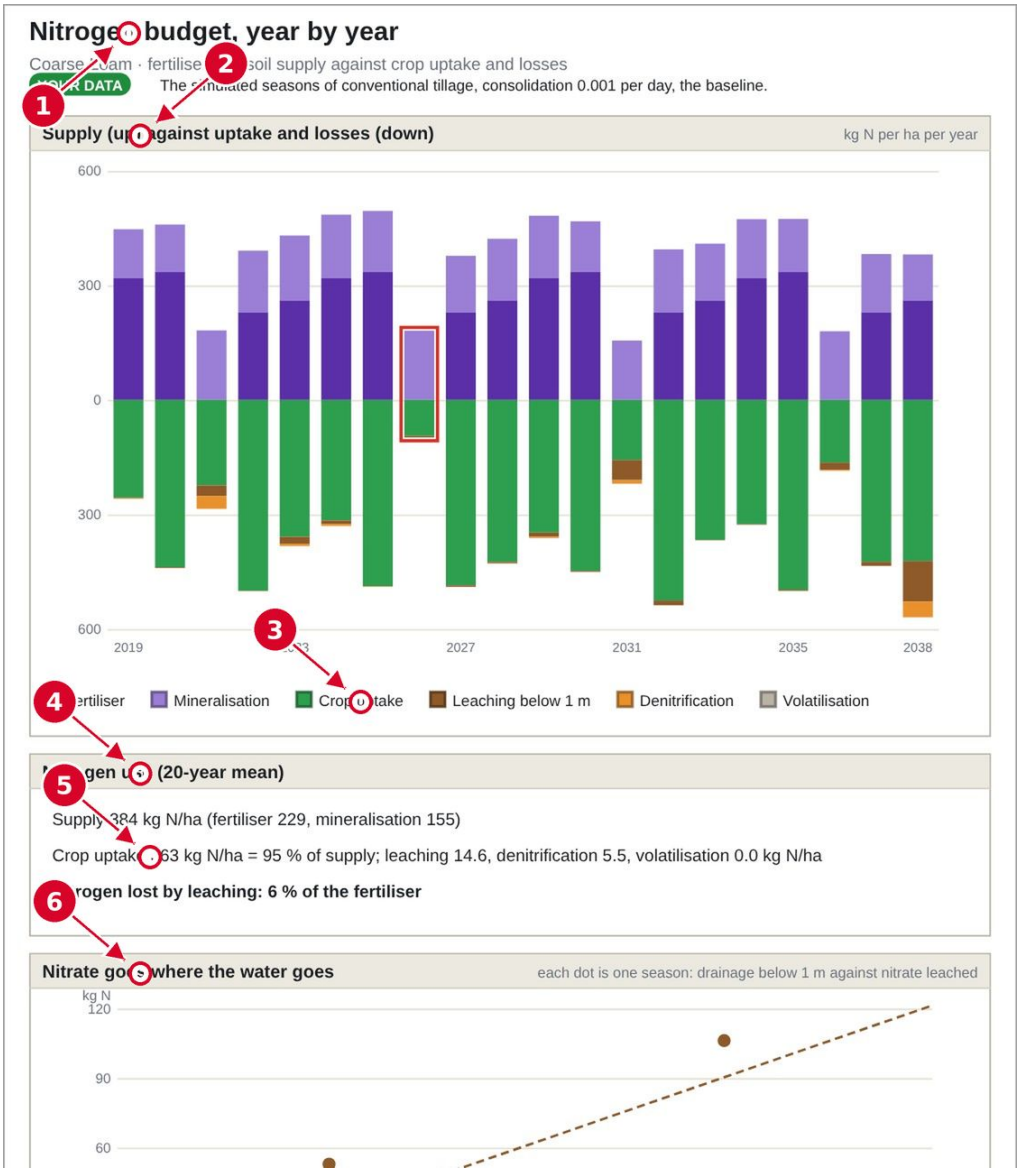


Figure 25.279: Nitrogen budget, part 1 of 2.

What the arrows point to.

- **1** The heading or label *Nitrogen budget, year by year*.
- **2** The heading or label *Supply (up) against uptake and losses (down)*.
- **3** The heading or label *Crop uptake*.
- **4** The heading or label *Nitrogen use (20-year mean)*.
- **5** The heading or label *Crop uptake 363 kg N/ha = 95 % of supply; leaching 14.6, denitrification 5.5, volatilisation 0.0 kg N/ha*.
- **6** The heading or label *Nitrate goes where the water goes*.

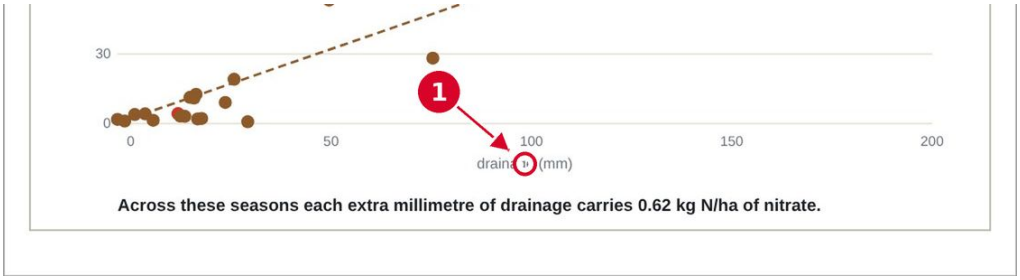


Figure 25.280: Nitrogen budget, part 2 of 2.

What the arrows point to.

- **1** The heading or label *drainage (mm)*.

25.20 Reverse index: where each equation appears on screen

Equation	Explained in chapter	Screens that use it

4.1 | Dates | The toolbar; The workbench |

4.2 | Dates | The toolbar; The workbench |

5.1 | The two computing engines | Soil, plant and air; Tillage and structure |

5.2 | The two computing engines | Soil, plant and air |

5.3 | The two computing engines | Soil, plant and air; Trigger law |

5.4 | The two computing engines | Evaporation physics; Feedback loops; Field balance; Mulch against bare soil; Mulch control; Sensor and trigger; Soil hydraulic drivers |

5.5 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.6 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.7 | The two computing engines | Feedback loops; Field balance; Mulch against bare soil; Sensor and trigger; Soil hydraulic drivers |

5.8 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.9 | The two computing engines | Feedback loops; Field balance; Mulch against bare soil; Mulch control; Sensor and trigger; Soil hydraulic drivers |

5.10 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.11 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.12 | The two computing engines | Feedback loops; Field balance; Sensor and trigger; Soil hydraulic drivers |

5.13 | The two computing engines | Field balance |

5.14 | The two computing engines | Field balance |

5.15 | The two computing engines | Field balance; Profile evolution |

5.16 | The two computing engines | Field balance |

5.17 | The two computing engines | Field balance; Profile evolution |

5.18 | The two computing engines | Field balance; Root structure; The Properties panel |

5.19 | The two computing engines | Field balance; Root structure |

5.20 | The two computing engines | Field balance; Root structure |

5.21 | The two computing engines | Field balance; Root structure |

5.22 | The two computing engines | Field balance; Root structure |

5.23 | The two computing engines | Field balance; Root structure |

5.24 | The two computing engines | Field balance |

5.25 | The two computing engines | Field balance; Soil, plant and air |

5.26 | The two computing engines | Evaporation physics; The Properties panel |

5.27 | The two computing engines | Surface irrigation design; Surface irrigation design / Cross-section at the inlet; Surface irrigation design / Layout in plan; Drip irrigation design; Sprinkler design and pipes; Trigger law |

6.1 | Irrigation control layers | Sensor and trigger; The Dashboard, Results tab |

6.2 | Irrigation control layers | Sensor and trigger |

6.3 | Irrigation control layers | Trigger law |

6.4 | Irrigation control layers | Trigger law |

6.5 | Irrigation control layers | Trigger law |

7.1 | Field water balance layers | Soil, plant and air |

7.2 | Field water balance layers | Soil, plant and air |

7.3 | Field water balance layers | Soil, plant and air |

7.4 | Field water balance layers | Soil, plant and air |

7.5 | Field water balance layers | Soil, plant and air |

7.6 | Field water balance layers | Profile evolution |

8.1 | Soil structure and tillage layers | Field balance; Soil hydraulic drivers; The Properties panel; Tillage and structure |

8.2 | Soil structure and tillage layers | Tillage and structure |

8.3 | Soil structure and tillage layers | Evaporation physics; Tillage and structure |

8.4 | Soil structure and tillage layers | Evaporation physics; Tillage and structure |

8.5 | Soil structure and tillage layers | Soil hydraulic drivers |

8.6 | Soil structure and tillage layers | Soil hydraulic drivers |

8.7 | Soil structure and tillage layers | Root structure |

9.1 | Mulch and evaporation layers | Mulch against bare soil |

9.2 | Mulch and evaporation layers | Evaporation physics |

9.3 | Mulch and evaporation layers | Evaporation physics |

9.4 | Mulch and evaporation layers | Evaporation physics |

9.5 | Mulch and evaporation layers | Evaporation physics |

9.6 | Mulch and evaporation layers | Fate of saved water |

9.7 | Mulch and evaporation layers | Fate of saved water |

9.8 | Mulch and evaporation layers | Regime map |

10.1 | Feedback loops | Feedback loops |

10.2 | Feedback loops | Feedback loops |

11.1 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.2 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.3 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet; Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

11.4 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.5 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.6 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Cross-section at the inlet; Surface irrigation design / Infiltrated depth along the field against the required depth Zn; Surface irrigation design / Longitudinal section; Sprinkler design and pipes; Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Soil intake against the application rate |

11.7 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Cross-section at the inlet |

11.8 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Layout in

plan |

11.9 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.10 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.11 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.12 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.13 | Surface irrigation design | Surface irrigation design; Surface irrigation design / Advance and opportunity time; Surface irrigation design / Layout in plan; Surface irrigation design / Longitudinal section |

11.15 | Surface irrigation design | Surface irrigation design / Advance and opportunity time; Surface irrigation design / Infiltrated depth along the field against the required depth Zn; Surface irrigation design / Longitudinal section |

11.16 | Surface irrigation design | Surface irrigation design / Advance and opportunity time; Surface irrigation design / Infiltrated depth along the field against the required depth Zn; Surface irrigation design / Longitudinal section |

11.17 | Surface irrigation design | Surface irrigation design / Advance and opportunity time; Surface irrigation design / Infiltrated depth along the field against the required depth Zn; Surface irrigation design / Longitudinal section |

11.18 | Surface irrigation design | Surface irrigation design / Hydrograph; Surface irrigation design / Performance indices and the water balance of one irrigation; The Dashboard, Design tab |

11.19 | Surface irrigation design | Surface irrigation design / Performance indices and the water balance of one irrigation; The Dashboard, Design tab |

11.20 | Surface irrigation design | Surface irrigation design / Hydrograph |

11.21 | Surface irrigation design | Surface irrigation design / Hydrograph |

11.22 | Surface irrigation design | Surface irrigation design / Design table |

11.23 | Surface irrigation design | Surface irrigation design / Design table |

11.24 | Surface irrigation design | Surface irrigation design / Intervals, efficiency and the advice in short; Surface irrigation design / What this means for the season and the farm |

12.1 | Drip irrigation design | Drip irrigation design |

12.2 | Drip irrigation design | Drip irrigation design |

12.3 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Layout and infrastructure; Drip irrigation design / Uniformity along the lateral; Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.4 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.5 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.6 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.7 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.8 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Uniformity along the lateral |

12.9 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.10 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Result for this design; Drip irrigation design / Soil moisture around the emitter; Drip irrigation design / Uniformity along the lateral |

12.11 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.12 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.13 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.14 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Soil moisture around the emitter |

12.15 | Drip irrigation design | Drip irrigation design; Drip irrigation design / How the wetted bulb grows with the run time; Drip irrigation design / Result for this design; Drip irrigation design / Soil moisture around the emitter; Drip irrigation design / Uniformity along the lateral |

12.16 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Result for this design; Drip irrigation design / Uniformity along the lateral |

12.17 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Result for this design; Drip irrigation design / Uniformity along the lateral |

12.18 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Result for this design; Drip irrigation design / Uniformity along the lateral |

12.19 | Drip irrigation design | Drip irrigation design; Drip irrigation design / Result for this design; Drip irrigation design / Uniformity along the lateral; The Dashboard, Design tab |

12.20 | Drip irrigation design | Drip irrigation design / Layout and infrastructure |

12.21 | Drip irrigation design | Drip irrigation design / Layout and infrastructure; Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.22 | Drip irrigation design | Drip irrigation design / Layout and infrastructure; Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.23 | Drip irrigation design | Drip irrigation design / Layout and infrastructure; Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

12.24 | Drip irrigation design | Drip irrigation design / Intervals, efficiency and the advice in short; Drip irrigation design / Irrigation schedule; Drip irrigation design / What this means for the season and the farm |

12.25 | Drip irrigation design | Drip irrigation design / Intervals, efficiency and the advice in short; Drip irrigation design / Irrigation schedule; Drip irrigation design / What this means for the season and the farm |

12.26 | Drip irrigation design | Drip irrigation design / Design table; Sprinkler design and pipes / Design table; Sprinkler design and pipes / What the efficiency is most sensitive to |

13.1 | Sprinkler design | Sprinkler design and pipes |

13.2 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / The jet; Sprinkler design and pipes / The sprinkler |

13.3 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / The jet; Sprinkler design and pipes / The sprinkler |

13.4 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Catch-can test;

Sprinkler design and pipes / Layout in plan; Sprinkler design and pipes / Layouts; Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.5 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Catch-can test; Sprinkler design and pipes / Layout in plan; Sprinkler design and pipes / Layouts; Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.6 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Catch-can test; Sprinkler design and pipes / Layout in plan; Sprinkler design and pipes / Layouts; Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.7 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Catch-can test; Sprinkler design and pipes / Layout in plan; Sprinkler design and pipes / Layouts; Sprinkler design and pipes / The water pattern of one sprinkler, and how overlap evens it out |

13.8 | Sprinkler design | Sprinkler design and pipes / The jet; Sprinkler design and pipes / What wind and soil allow |

13.9 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Soil intake against the application rate; The Properties panel |

13.10 | Sprinkler design | Sprinkler design and pipes; Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Soil intake against the application rate |

13.11 | Sprinkler design | Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Result for this design; Sprinkler design and pipes / Soil water as a reservoir; The Dashboard, Design tab |

13.12 | Sprinkler design | Sprinkler design and pipes / Along the lateral; Sprinkler design and pipes / Result for this design; Sprinkler design and pipes / Soil water as a reservoir |

13.13 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.14 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.15 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.16 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.17 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.18 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.19 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.20 | Sprinkler design | Sprinkler design and pipes / Does your design pass? The course checks |

13.21 | Sprinkler design | Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

13.22 | Sprinkler design | Sprinkler design and pipes / Layout and infrastructure; Sprinkler design and pipes / Pipes, pump and the hydraulic grade line (the course sizing) |

13.23 | Sprinkler design | Sprinkler design and pipes / Intervals, efficiency and the advice in short; Sprinkler design and pipes / Irrigation schedule; Sprinkler design and pipes / Soil water as a reservoir; Sprinkler design and pipes / What this means for the season and the farm |

14.1 | The Dashboard | The Dashboard, Results tab |

14.2 | The Dashboard | The Dashboard, Results tab |

14.3 | The Dashboard | The Dashboard, Results tab |

14.4 | The Dashboard | The Dashboard, Results tab |

14.5 | The Dashboard | The Dashboard, Charts tab |

Symbols and units

Table A.1. Symbols used in this manual, grouped by chapter of first use. Units are SI-derived as shown; $[-]$ is dimensionless.

Symbol	Meaning	Unit
θ, θ_r, θ_s	Volumetric water content; residual; saturated	$m^3\ m^{-3}$
θ_{fc}, θ_{wp}	Water content at field capacity; at wilting point	$m^3\ m^{-3}$
h, ψ	Soil-water suction (positive); matric potential	cm; kPa
α, n, m	van Genuchten parameters ($m = 1 - 1/n$)	cm^{-1}, [–], [–]
K_s, $K(h)$	Saturated and unsaturated hydraulic conductivity	$mm\ h^{-1}$
AWC	Plant-available water capacity	$mm\ cm^{-1}$
f, f_T	Relative depletion of the sensor layer; trigger depletion	[–]
$R_{\max}$, R_d	Maximum and present rooting depth	cm
P, I, E, T, D	Rain, irrigation, soil evaporation, transpiration, drainage	$mm\ d^{-1}$ or $mm\ a^{-1}$
E_0	Atmospheric evaporative demand	$mm\ d^{-1}$
$\mathrm{ET}_{\mathrm{pk}}$	Peak 5-day mean of $E + T$	$mm\ d^{-1}$
k_t	Calibrated transpiration coefficient of Engine 2	[–]
Z_n	Required (net) depth per irrigation	mm
$Z(\tau)$, $F(t)$	Cumulative infiltration (Kostiakov; Green–Ampt)	mm
k, A	Kostiakov intake constants	$mm\ min^{-A}$; [–]
ψ_f	Wetting-front suction	mm
L, S, w, n	Field length; slope; spacing or width; Manning roughness	m; %; m; $s\ m^{-1/3}$
q, Q	Unit and total inflow	$L\ s^{-1}\ m^{-1}$, $L\ s^{-1}$
t_x, t_L, t_{co}, τ	Advance time; time to end; cut-off time; opportunity time	min
σ_y, σ_z	Surface and subsurface shape factors	[–]
E_a, E_s	Application and storage efficiency	[–] or %
DP, RO, WD	Deep percolation, runoff, wind drift and interception fractions	[–]
CU, DU, E_u	Christiansen uniformity; distribution uniformity; emission uniformity	[–] or %
S_e, S_l, L_L, H_0, D	Emitter spacing; lateral spacing; lateral length; inlet head; inside diameter	m, m, m, m, mm
c_v, x	Emitter coefficient of variation; flow exponent	[–]
r, a, u	Bulb radius; aspect ratio; upper semi-axis ratio	mm, [–], [–]
d, P, h_s, R	Nozzle diameter; pressure; pressure head; wetted radius	mm, kPa, m, m

Symbol	Meaning	Unit
S_s, S_l	Sprinkler spacing along and between laterals	m
u	Wind speed	m s^{-1}
AR	Mean gross application rate	mm h^{-1}
C	Hazen–Williams roughness coefficient	[–]
J	Hydraulic gradient	m m^{-1}
TDH, H_{man}	Total dynamic head; manometric head	m
A_f, Q_s	Irrigated area; farm supply	ha; L s^{-1}

Control value lists

Table B.1. Index lists behind the Properties controls (defaults in bold). All other controls are described with their ranges in Chapter 3.

Control	Values
Trigger (sensor suction)	−50, −75, −100, −150, −200, −250, **−300**, −350, −400, −500, −600, −700, −800, −1000, −1200, −1500, −2000, −3000, −5000, −10000 cm
Sensor depth	5, 10, 15, …, 100 cm (20 values), default **20** cm
Nitrogen rate	10, 25, 40, 55, 70, 85, 100, 115, 130, 150
Mulch storage capacity (CAP)	0.5, 1, 2, 4 mm
Rainfall factor	0.5 to 1.4
Temperature offset	−1 to +3.5 °C
Mulch strength (vapour-flux factor)	0.1 to 1.0 (1.0 = bare soil)
Structural consolidation rate	0.001, 0.0025, 0.005, 0.01, 0.02 d^{-1}
Crop	the simulated cereal/oilseed rotation and the 48 library crops of Table 21.2

Table B.2. Mulch materials (thermal weight / storage capacity in mm): husk 0.5 / 2, straw 0.4 / 4, chips 0.3 / 2, sawdust 0.6 / 1, grass 0.7 / 4, plastic 0.1 / 0.5.

Simulation data sets

The annual meta-model (Engine 1) is built on the following Daisy 7.1 run sets of the study (Chapter 5, Table 5.1): E2, E3, E4, E5 (300 runs), E7, E11, E13 and E14. The study as a whole comprises 10,190 simulation runs in 17 experiments, available from the project repository (`Results/` and `Software/` folders). The 20-year weather record (1980–1999) is the baseline of the study; the projections to 2100 are analogue-year scenarios (Chapter 4).

Provenance ledger of the numbered equations

All 134 numbered equations are labelled in the text and listed here: 38 **[Literature]**, 18 **[Data-driven]** and 78 **[Engineered]**. A **[Literature]** label means the relation is the published one; it does not mean that SWAFL has been validated in the field. The surface, drip and sprinkler design relations marked in Paper 3 were tested against published worked examples; nothing has been tested against field measurements.

Eq.	Label	Basis
4.1	Engineered	calendar bookkeeping and choice of analogue year; no physical law
4.2	Engineered	calendar bookkeeping and choice of analogue year; no physical law
5.1	Literature	van Genuchten (1980)
5.2	Literature	Mualem (1976) with van Genuchten (1980)
5.3	Literature	standard definition of plant-available water between field capacity and wilting point
5.4	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.5	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.6	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.7	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.8	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.9	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.10	Data-driven	ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits
5.11	Data-driven	set-specific corrections from the Daisy run sets; clip bounds engineered
5.12	Engineered	closure constraint and bounds chosen for the workbench
5.13	Engineered	canopy curves, parameters chosen for the workbench; not a measured leaf-area record
5.14	Engineered	canopy curves, parameters chosen for the workbench; not a measured leaf-area record

Eq.	Label	Basis
5.15	Engineered	stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen
5.16	Engineered	stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen
5.17	Engineered	tipping-bucket redistribution with a chosen coefficient
5.18	Engineered	rooting rules with constants chosen for the workbench
5.19	Engineered	rooting rules with constants chosen for the workbench
5.20	Engineered	rooting rules with constants chosen for the workbench
5.21	Engineered	demand partitioning with chosen weights and a calibration factor
5.22	Engineered	demand partitioning with chosen weights and a calibration factor
5.23	Engineered	demand partitioning with chosen weights and a calibration factor
5.24	Data-driven	scaling of the daily series to the Engine 1 twenty-year means
5.25	Literature	water-balance identity
5.26	Engineered	atmosphere factor with chosen constants
5.27	Literature	readily available water as in FAO-56 practice, with an engineered clip
6.1	Data-driven	counts over the simulated daily series
6.2	Data-driven	counts over the simulated daily series
6.3	Literature	inverse of the van Genuchten curve, solved by bisection
6.4	Literature	same relation as Eq. (5.27) without the clip
6.5	Data-driven	ratio of Daisy run-set E2 yields
7.1	Literature	inverse of Eq. (5.3); the 30 cm floor is engineered
7.2	Engineered	series-resistance chain after van den Honert (1948); resistance values chosen for the workbench
7.3	Engineered	series-resistance chain after van den Honert (1948); resistance values chosen for the workbench
7.4	Engineered	series-resistance chain after van den Honert (1948); resistance values chosen for the workbench
7.5	Literature	Nobel (2009)
7.6	Engineered	indicator defined for the workbench
8.1	Engineered	exponential relaxation with chosen constants
8.2	Engineered	porosity relation with chosen coefficients
8.3	Literature	van Genuchten (1980)
8.4	Literature	Young–Laplace (Jurin) capillary law
8.5	Data-driven	differences of Daisy run-set E5 results
8.6	Data-driven	ordinary least squares applied to Daisy results
8.7	Engineered	diagnostic defined for the workbench
9.1	Data-driven	differences and ratios of Daisy run-set E14 and E11 results

Eq.	Label	Basis
9.2	Engineered	two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench
9.3	Engineered	two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench
9.4	Engineered	drying-front geometry chosen for drawing
9.5	Engineered	drying-front geometry chosen for drawing
9.6	Data-driven	differences and ratios of Daisy run-set E14 and E11 results
9.7	Data-driven	differences and ratios of Daisy run-set E14 and E11 results
9.8	Engineered	index defined for the workbench
10.1	Engineered	drawing rule
10.2	Data-driven	one-step differences of Daisy results
11.1	Literature	Saxton and Rawls (2006), implemented as written
11.2	Literature	Saxton and Rawls (2006), implemented as written
11.3	Engineered	rational-polynomial approximation of a Rawls-type suction
11.4	Engineered	initial and wetted water contents chosen for the workbench
11.5	Literature	Green and Ampt (1911)
11.6	Literature	Kostiakov (1932)
11.7	Literature	ordinary least squares on the log–log plot
11.8	Literature	Manning (1891)
11.9	Literature	volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3)
11.10	Literature	volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3)
11.11	Literature	volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3)
11.12	Engineered	alternative advance models set by worked examples and course notes
11.13	Engineered	alternative advance models set by worked examples and course notes
11.15	Literature	inversion of the Kostiakov law
11.16	Engineered	cut-off rules chosen as design options
11.17	Literature	standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997)
11.18	Literature	standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997)
11.19	Literature	standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997)
11.20	Engineered	hydrograph shapes chosen for drawing; volume conserving

Eq.	Label	Basis
11.21	Engineered	hydrograph shapes chosen for drawing; volume conserving
11.22	Engineered	search limit; the furrow stream limit follows Criddle et al. (1956)
11.23	Engineered	optimisation score and seasonal bookkeeping chosen for the workbench
11.24	Engineered	optimisation score and seasonal bookkeeping chosen for the workbench
12.1	Engineered	depth and demand rules chosen for the workbench
12.2	Engineered	depth and demand rules chosen for the workbench
12.3	Literature	Hazen and Williams (1905)
12.4	Literature	Keller and Karmeli (1974) emitter law
12.5	Engineered	pressure-compensating emitter model with chosen constants
12.6	Literature	definitions of mean and minimum discharge
12.7	Literature	Keller and Karmeli (1974)
12.8	Engineered	combined spread chosen to drive the bulb quantiles
12.9	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.10	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.11	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.12	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.13	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.14	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.15	Engineered	wetted-bulb geometry and loss partition; constants chosen for the workbench
12.16	Engineered	five-quantile delivery and run-time rules chosen for the workbench
12.17	Literature	definition of the requirement over the area served by one emitter
12.18	Engineered	five-quantile delivery and run-time rules chosen for the workbench
12.19	Engineered	five-quantile delivery and run-time rules chosen for the workbench
12.20	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.21	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.22	Engineered	Christiansen-type reduction with a chosen factor 0.4
12.23	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice

Eq.	Label	Basis
12.24	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.25	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
12.26	Engineered	layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice
13.1	Engineered	depression storage reduction with slope
13.2	Literature	Torricelli orifice equation
13.3	Engineered	range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen
13.4	Engineered	pattern and wind shapes chosen to reproduce published qualitative patterns
13.5	Engineered	pattern and wind shapes chosen to reproduce published qualitative patterns
13.6	Literature	overlap by superposition of single-sprinkler patterns
13.7	Literature	Christiansen (1942)
13.8	Engineered	loss and ponding rules chosen for the workbench
13.9	Engineered	loss and ponding rules chosen for the workbench
13.10	Engineered	loss and ponding rules chosen for the workbench
13.11	Literature	efficiency and distribution uniformity of Merriam and Keller (1978)
13.12	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.13	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.14	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.15	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.16	Engineered	run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified
13.17	Literature	Blasius friction law for smooth pipe
13.18	Literature	20 % rule of Keller and Bliesner (1990)
13.19	Engineered	head and sets bookkeeping
13.20	Literature	hydraulic power definition
13.21	Engineered	head and sets bookkeeping
13.22	Literature	Christiansen reduction factor with Hazen–Williams exponent
13.23	Engineered	head and sets bookkeeping

Eq.	Label	Basis
14.1	Engineered	bookkeeping of Engine 2 series
14.2	Engineered	bookkeeping of Engine 2 series
14.3	Literature	water-balance identity
14.4	Engineered	bookkeeping of Engine 2 series
14.5	Engineered	bookkeeping of Engine 2 series

References

Abrahamsen, P., and Hansen, S. (2000). Daisy: an open soil-crop-atmosphere system model. *Environmental Modelling & Software* 15(3), 313–330.

Allen, R. G., Pereira, L. S., Raes, D., and Smith, M. (1998). *Crop evapotranspiration: guidelines for computing crop water requirements*. FAO Irrigation and Drainage Paper 56. FAO, Rome.

Barnes, E. M., Clarke, T. R., Richards, S. E., et al. (2000). Coincident detection of crop water stress, nitrogen status and canopy density using ground-based multispectral data. *Proceedings of the 5th International Conference on Precision Agriculture*, Bloomington, MN.

Blasius, H. (1913). Das Ähnlichkeitsgesetz bei Reibungsvorgängen in Flüssigkeiten. *Forschungsarbeiten auf dem Gebiete des Ingenieurwesens* 131.

Brouwer, C., Prins, K., Kay, M., and Heibloem, M. (1988). *Irrigation water management: irrigation methods*. FAO Training Manual 9. FAO, Rome.

Burt, C. M., Clemmens, A. J., Strelkoff, T. S., Solomon, K. H., Bliesner, R. D., Hardy, L. A., Howell, T. A., and Eisenhauer, D. E. (1997). Irrigation performance measures: efficiency and uniformity. *Journal of Irrigation and Drainage Engineering* 123(6), 423–442.

Carsel, R. F., and Parrish, R. S. (1988). Developing joint probability distributions of soil water retention characteristics. *Water Resources Research* 24(5), 755–769.

Chamberlain, R. G., and Duquette, W. H. (2007). *Some algorithms for polygons on a sphere*. JPL Publication 07-03, Jet Propulsion Laboratory, Pasadena.

Christiansen, J. E. (1942). *Irrigation by sprinkling*. University of California Agricultural Experiment Station Bulletin 670.

Criddle, W. D., Davis, S., Pair, C. H., and Shockley, D. G. (1956). *Methods for evaluating irrigation systems*. USDA Soil Conservation Service, Agriculture Handbook 82.

Elliott, R. L., and Walker, W. R. (1982). Field evaluation of furrow infiltration and advance functions. *Transactions of the ASAE* 25(2), 396–400.

Gao, B.-C. (1996). NDWI: a normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sensing of Environment* 58, 257–266.

Gitelson, A. A., Kaufman, Y. J., Stark, R., and Rundquist, D. (2002). Novel algorithms for remote estimation of vegetation fraction. *Remote Sensing of Environment* 80, 76–87.

Green, W. H., and Ampt, G. A. (1911). Studies on soil physics: 1. The flow of air and water through soils. *Journal of Agricultural Science* 4(1), 1–24.

Hazen, A., and Williams, G. S. (1905). *Hydraulic tables*. John Wiley & Sons, New York.

Hillel, D. (1998). *Environmental soil physics*. Academic Press, San Diego.

Huete, A. R. (1988). A soil-adjusted vegetation index (SAVI). *Remote Sensing of Environment* 25, 295–309.

Huete, A., Didan, K., Miura, T., Rodriguez, E. P., Gao, X., and Ferreira, L. G. (2002). Overview of the radiometric and biophysical performance of the MODIS vegetation indices. *Remote Sensing of Environment* 83, 195–213.

Jurin, J. (1718). An account of some experiments shown before the Royal Society; with an enquiry into the cause of the ascent and suspension of water in capillary tubes. *Philosophical Transactions*

30(355), 739–747.

Keller, J., and Bliesner, R. D. (1990). *Sprinkle and trickle irrigation*. Van Nostrand Reinhold, New York.

Keller, J., and Karmeli, D. (1974). Trickle irrigation design parameters. *Transactions of the ASAE* 17(4), 678–684.

Kostiakov, A. N. (1932). On the dynamics of the coefficient of water-percolation in soils and on the necessity of studying it from a dynamic point of view for purposes of amelioration. *Transactions of the 6th Committee of the International Society of Soil Science*, Part A, 17–21.

Louhaichi, M., Borman, M. M., and Johnson, D. E. (2001). Spatially located platform and aerial photography for documentation of grazing impacts on wheat. *Geocarto International* 16, 65–70.

Manning, R. (1891). On the flow of water in open channels and pipes. *Transactions of the Institution of Civil Engineers of Ireland* 20, 161–207.

Merriam, J. L., and Keller, J. (1978). *Farm irrigation system evaluation: a guide for management*. Utah State University, Logan.

Mualem, Y. (1976). A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research* 12(3), 513–522.

Nobel, P. S. (2009). *Physicochemical and environmental plant physiology*, 4th ed. Academic Press, Amsterdam.

Philip, J. R. (1957). Evaporation, and moisture and heat fields in the soil. *Journal of Meteorology* 14(4), 354–366.

Poggio, L., de Sousa, L. M., Batjes, N. H., et al. (2021). SoilGrids 2.0: producing soil information for the globe with quantified spatial uncertainty. *SOIL* 7, 217–240.

Ritchie, J. T. (1972). Model for predicting evaporation from a row crop with incomplete cover. *Water Resources Research* 8(5), 1204–1213.

Rouse, J. W., Haas, R. H., Schell, J. A., and Deering, D. W. (1974). Monitoring vegetation systems in the Great Plains with ERTS. *Third ERTS Symposium*, NASA SP-351, 309–317.

Saxton, K. E., and Rawls, W. J. (2006). Soil water characteristic estimates by texture and organic matter for hydrologic solutions. *Soil Science Society of America Journal* 70(5), 1569–1578.

Tucker, C. J. (1979). Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment* 8, 127–150.

van den Honert, T. H. (1948). Water transport in plants as a catenary process. *Discussions of the Faraday Society* 3, 146–153.

van Genuchten, M. Th. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal* 44(5), 892–898.

Walker, W. R., and Skogerboe, G. V. (1987). *Surface irrigation: theory and practice*. Prentice-Hall, Englewood Cliffs, NJ.

Woebbecke, D. M., Meyer, G. E., Von Bargen, K., and Mortensen, D. A. (1995). Color indices for weed identification under various soil, residue, and lighting conditions. *Transactions of the ASAE* 38, 259–269.

Equation reference

This appendix prints every numbered equation of the program in full, in the order of the chapters, with its provenance and the layers of the program that draw on it. The equations added in version 1.1 (Eqs. 5.12a, 6.4a and those of Chapters 14, 18–23) are printed in their chapters.

Eq. 4.1 — dates

$$k_{\text{start}} = 365\,a + d_{\text{oy}}, \qquad \text{date}(k) = 1\ \text{Jan of year}\ \big(Y_0 + \lfloor k/365 \rfloor - a\big) + \big(k - 365\lfloor k/365 \rfloor\big)\ \text{days}, \tag{4.1}$$

Provenance (Eq. 4.1): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

Eq. 4.2 — dates

$$a = \begin{cases} \operatorname{argsort}_y \left(P_{\text{thr},y}\right)_{[10]} & \text{typical (median, 11th smallest of 20)} \\ \arg\min_y P_{\text{thr},y} & \text{dry} \\ \arg\max_y P_{\text{thr},y} & \text{wet} \\ Y_0 - 1980 & 1980 \le Y_0 \le 1999 \end{cases} \tag{4.2}$$

Provenance (Eq. 4.2): **[Engineered]**: calendar bookkeeping and choice of analogue year; no physical law.

Eq. 5.1 — engine

$$S_e(h) = \Big[1 + (\alpha|h|)^n\Big]^{-m}, \quad m = 1 - \tfrac{1}{n}, \qquad \theta(h) = \theta_r + (\theta_s - \theta_r)\, S_e(h) \tag{5.1}$$

Provenance (Eq. 5.1): **[Literature]**: van Genuchten (1980).

Eq. 5.2 — engine

$$K_r(S_e) = S_e^{1/2}\Big[1 - \big(1 - S_e^{1/m}\big)^m\Big]^2. \tag{5.2}$$

Provenance (Eq. 5.2): **[Literature]**: Mualem (1976) with van Genuchten (1980).

Eq. 5.3 — engine

$$f(h) \;=\; \frac{S_e(100) - S_e(|h|)}{S_e(100) - S_e(15\,849)}, \qquad f_{\text{used}} = \min\{1, \max\{0, f\}\}. \tag{5.3}$$

Provenance (Eq. 5.3): **[Literature]**: standard definition of plant-available water between field capacity and wilting point.

Eq. 5.4 — engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.5 — engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.6 — engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.7 — engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.8 — engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.9 — engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\frac{\text{now}.E}{\text{ref}.E}\right]\left[\frac{\text{tN}.E}{\text{tB}.E}\right]\left[\frac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\frac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\frac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\frac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\frac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\frac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\frac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\frac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\frac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.10 — engine

$$E = \bar{E}(\text{soil}, \varphi, c)\ \left[\tfrac{\text{now}.E}{\text{ref}.E}\right]\left[\tfrac{\text{tN}.E}{\text{tB}.E}\right]\left[\tfrac{E_b}{E_{b,\text{ref}}}\right] \tag{5.4}$$

$$T = T_b\ \left[\tfrac{\text{now}.T}{\text{bare}.T}\right]\ \text{clip}\Big(\tfrac{\text{tN}.T}{\text{tB}.T};\ 0.3,\ 2.5\Big) \tag{5.5}$$

$$ET = ET_b\ \left[\tfrac{\text{now}.ET}{\text{bare}.ET}\right]\ \text{clip}\Big(\tfrac{\text{tN}.ET}{\text{tB}.ET};\ 0.5,\ 2\Big) \tag{5.6}$$

$$I = I_b\ \left[\tfrac{\text{now}.I}{\text{bare}.I}\right]\ \text{clip}\Big(\tfrac{\text{tN}.I}{\text{tB}.I};\ 0.4,\ 2.5\Big) \tag{5.7}$$

$$Y = Y_b\ \left[\tfrac{\text{tN}.Y}{\text{tR}.Y}\right]\left[\tfrac{\text{now}.Y}{\text{nm}.Y}\right] \tag{5.8}$$

$$D = D_b + (\text{now}.D - \text{bare}.D) + (\text{tN}.D - \text{tB}.D) \tag{5.9}$$

$$N_L = N_b + (\text{now}.N - \text{bare}.N) + (\text{tN}.N - \text{tB}.N) \tag{5.10}$$

Provenance (Eq. 5.4–5.10): **[Data-driven]**: ratios of Daisy ensemble results (run sets E2–E14); the clip bounds are engineered limits.

Eq. 5.11 — engine

$$Q \leftarrow Q\,\text{clip}\Big(\tfrac{r_\kappa.Q}{r_{\kappa_0}.Q};\ 0.05,\ 6\Big)\ (Q \in \{E, T, ET, I, Y\}), \qquad D \leftarrow D + (r_\kappa.D - r_{\kappa_0}.D), \quad N_L \leftarrow N_L + (r_\kappa.N - r_{\kappa_0}.N \tag{5.11}$$

Provenance (Eq. 5.11): **[Data-driven]**: set-specific corrections from the Daisy run sets; clip bounds engineered.

Eq. 5.12 — engine

$ET \in [\,P + I - D - 30,\ P + I - D + 15\,]$ (if not, ET is clipped and E, T are scaled by the same factor),

$$E_{\text{other}} = \max(0,\ ET - T - E), \qquad \Delta S = P + I - ET - D, \qquad \text{IE} = \frac{100\,Y}{I}\ \ [\text{kg m}^{-3}]. \tag{5.12}$$

Provenance (Eq. 5.12): **[Engineered]**: closure constraint and bounds chosen for the workbench.

Eq. 5.13 — engine

$$c_{\text{spring}}(d) = \frac{1}{1 + \exp(-(d - d_{\text{mid}})/s)}, \quad s = 13\ (\text{vegetable } 8), \tag{5.13}$$

Provenance (Eq. 5.13): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.14 — engine

$$c_{\text{winter}}(d) = \begin{cases} 0.08 + 0.12\ \min\{1, (d - d_{\text{sow}})/60\} & d < d_{\text{pk}} - 90 \\ 0.2 + \dfrac{0.8}{1 + \exp(-(d - d_{\text{pk}} + 45)/12)} & \text{otherwise} \end{cases} \tag{5.14}$$

Provenance (Eq. 5.14): **[Engineered]**: canopy curves, parameters chosen for the workbench; not a measured leaf-area record.

Eq. 5.15 — engine

$$R = \min\{70,\ \mu_0\,\gamma_y\,(-\ln U)\}, \qquad U \sim \text{Uniform}(0, 1), \tag{5.15}$$

Provenance (Eq. 5.15): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.16 — engine

$$I_c = \mu_0 \ln\frac{1}{r}, \qquad r = \frac{\max\{40,\ P - P_{\text{other}}\}}{P}, \qquad T_f = \max\{0,\ R - I_c\}. \tag{5.16}$$

Provenance (Eq. 5.16): **[Engineered]**: stochastic rain and interception generator; the exponential wet-day depth is a standard weather-generator form, the constants are chosen.

Eq. 5.17 — engine

$$q_i = 0.08\,(w_{i+1} - w_i), \qquad w_i \leftarrow w_i + q_i,\ \ w_{i+1} \leftarrow w_{i+1} - q_i, \tag{5.17}$$

Provenance (Eq. 5.17): **[Engineered]**: tipping-bucket redistribution with a chosen coefficient.

Eq. 5.18 — engine

$$R_{\max} = \text{clip}\Big(R_c\,(0.85{+}0.3\,\text{sand})\big[1{+}0.12\log_{10}(|h_T|/300)\big]\big[1{-}0.18\,\tfrac{N-100}{100}\big];\ 25,\ 100\Big)\ \text{cm}, \tag{5.18}$$

Provenance (Eq. 5.18): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.19 — engine

$$R_d \leftarrow \max\Big\{R_d,\ \min\big\{R_{\max}\,\text{clip}\big(\tfrac{c-0.04}{0.96};0,1\big)^{0.8},\ R_{\max},\ R_d + 1.7\,(0.35 + \bar{s}_{\text{wet}})\,\kappa\big\}\Big\}, \tag{5.19}$$

Provenance (Eq. 5.19): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.20 — engine

$$\rho_i \propto \exp\!\Big(-\frac{z_i}{\lambda R_d + 8}\Big)(0.45 + 0.9\,s_i)\Big[1 + 3\,e^{-((z_i - z_d)/12)^2}\Big]_{\text{drip only}}, \quad z_i \le R_d, \qquad \sum_i \rho_i = 1, \tag{5.20}$$

Provenance (Eq. 5.20): **[Engineered]**: rooting rules with constants chosen for the workbench.

Eq. 5.21 — engine

$$E_d = E_{\text{yr}}\,\frac{w(\text{doy})}{\sum_{k=0}^{364} w(k)}\,(1 - 0.3\,c), \qquad w(\text{doy}) = 0.35 + 0.65\cdot\tfrac{1}{2}\Big[1 + \sin\tfrac{2\pi(\text{doy}-100)}{365}\Big], \tag{5.21}$$

Provenance (Eq. 5.21): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.22 — engine

$$T_d = k_t\,T_{\text{yr}}\,\frac{c(d)}{\bar{C}}, \qquad \bar{C} = \tfrac{1}{20}\sum_{20\ \text{yr}} c, \tag{5.22}$$

Provenance (Eq. 5.22): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.23 — engine

$$q_i^T = T_d\,\frac{\rho_i\,a_i}{\sum_j \rho_j a_j}, \qquad a_i = \text{clip}\Big(\frac{s_i - 0.15}{0.6};\ 0,\ 1\Big), \tag{5.23}$$

Provenance (Eq. 5.23): **[Engineered]**: demand partitioning with chosen weights and a calibration factor.

Eq. 5.24 — engine

$$S_P = \frac{P}{\overline{P}_{\text{sim}}}, \quad S_{\text{thr}} = \frac{P - P_{\text{other}}}{\overline{T}_{f,\text{sim}}}, \quad S_E = \frac{E}{\overline{E}_{\text{sim}}}, \quad S_T = \frac{T}{\overline{T}_{\text{sim}}}, \quad S_D = \frac{D}{\overline{D}_{\text{sim}}}, \tag{5.24}$$

Provenance (Eq. 5.24): **[Data-driven]**: scaling of the daily series to the Engine 1 twenty-year means.

Eq. 5.25 — engine

$$\Delta S_d = T_f + I - E - T - D. \tag{5.25}$$

Provenance (Eq. 5.25): **[Literature]**: water-balance identity.

Eq. 5.26 — engine

$$E_{0,\text{bare}} = 4.2\,(1{+}0.07\,\Delta T)\,\frac{0.35 + 0.65 \cdot \frac{1}{2}[1 + \sin(2\pi(\text{doy} - 100)/365)]}{0.675}\,\text{clip}(1{-}0.12(\rho{-}1);\,0.85,\,1.12)\sqrt{\gamma_y}\ \text{m} \tag{5.26}$$

Provenance (Eq. 5.26): **[Engineered]**: atmosphere factor with chosen constants.

Eq. 5.27 — engine

$$Z_n = \text{clip}\big(f_T \cdot \text{AWC} \cdot R_{\text{max}};\ 25,\ 150\big)\ \text{mm} \qquad \text{(surface; drip and sprinkler use their own limits)} \tag{5.27}$$

Provenance (Eq. 5.27): **[Literature]**: readily available water as in FAO-56 practice, with an engineered clip.

Eq. 6.1 — control

$$N_y(h_T) = \#\big\{\, d \in \text{growing season of year } y : h_d > h_T \,\big\} \quad (183 \text{ days from 1 April}), \tag{6.1}$$

Provenance (Eq. 6.1): **[Data-driven]**: counts over the simulated daily series.

Eq. 6.2 — control

$$X = 100\Big(1 - \frac{\bar{N}_{\text{irr}}}{\bar{N}_{\text{rainfed}}}\Big), \qquad \bar{N} = \tfrac{1}{20}\sum_{y} N_y. \tag{6.2}$$

Provenance (Eq. 6.2): **[Data-driven]**: counts over the simulated daily series.

Eq. 6.3 — control

$$h(f) = \Big\{\, h : f(h) = f \,\Big\}, \tag{6.3}$$

Provenance (Eq. 6.3): **[Literature]**: inverse of the van Genuchten curve, solved by bisection.

Eq. 6.4 — control

$$Z_n = f \cdot \text{AWC} \cdot R_{\text{max}} \quad [\text{mm}] \tag{6.4}$$

Provenance (Eq. 6.4): **[Literature]**: same relation as Eq. (5.27) without the clip.

Eq. 6.5 — control

$$Y_{\text{rel}} = \frac{Y(h_T)}{\max_h\, Y(h)}, \tag{6.5}$$

Provenance (Eq. 6.5): **[Data-driven]**: ratio of Daisy run-set E2 yields.

Eq. 7.1 — field

$$h_i = \max\{30,\ h(f_i)\}, \qquad f_i = 1 - s_i, \qquad \psi_{\text{soil},i} = -h_i \cdot 9.807 \times 10^{-5}\ \text{MPa cm}^{-1}. \tag{7.1}$$

Provenance (Eq. 7.1): **[Literature]**: inverse of Eq. (5.3); the 30 cm floor is engineered.

Eq. 7.2 — field

$$R_{\text{soil}} = 0.03\ \max\{K_r,\,0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.3 — field

$$R_{\text{soil}} = 0.03\ \max\{K_r,\,0.002\}^{-0.55}, \quad R_{\text{root}} = 0.10, \quad R_{\text{xyl}} = 0.05, \quad R_{\text{leaf}} = 0.08 \tag{7.2}$$

$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.4 — field

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$$\psi_{\text{root}} = \psi_{RZ} - a\,E\,R_{\text{soil}}, \qquad \psi_{\text{stem}} = \psi_{\text{root}} - a\,E\,R_{\text{root}}, \tag{7.3}$$

$$\psi_{\text{leaf}} = \text{clip}\big(\psi_{\text{stem}} - a\,E\,(R_{\text{xyl}} + R_{\text{leaf}});\ -3.2,\ 0\big), \tag{7.4}$$

Provenance (Eq. 7.2–7.4): **[Engineered]**: series-resistance chain after van den Honert (1948); resistance values chosen for the workbench.

Eq. 7.5 — field

$$\psi_{\text{air}} = \frac{R\,T}{V_w}\,\ln \text{RH} \ \approx\ 135.5\,\ln \text{RH}\ \ \text{MPa} \tag{7.5}$$

Provenance (Eq. 7.5): **[Literature]**: Nobel (2009).

Eq. 7.6 — field

$$\text{share} = \frac{\sum_{i:\ z_i \le 60\,\text{cm},\ h_i > h_T} \Delta z_i}{\sum_{i:\ z_i \le 60\,\text{cm}} \Delta z_i}, \tag{7.6}$$

Provenance (Eq. 7.6): **[Engineered]**: indicator defined for the workbench.

Eq. 8.1 — structure

$$p(t) = p_c + (p_0 - p_c)\,e^{-\tau_c t}, \qquad p_0 = 1.00\ (\text{conventional}),\ 0.72\ (\text{reduced}), \tag{8.1}$$

Provenance (Eq. 8.1): **[Engineered]**: exponential relaxation with chosen constants.

Eq. 8.2 — structure

$$\phi = \text{clip}\big(0.36 + 0.0016\,\text{clay} + 0.0009\,\text{silt};\ 0.34,\ 0.50\big), \qquad \theta_r = 0.01 + 0.0006\,\text{clay}, \tag{8.2}$$

Provenance (Eq. 8.2): **[Engineered]**: porosity relation with chosen coefficients.

Eq. 8.3 — structure

$$\theta(h) = \theta_r + (\phi - \theta_r)\, S_e(h). \tag{8.3}$$

Provenance (Eq. 8.3): **[Literature]**: van Genuchten (1980).

Eq. 8.4 — structure

$$r_c = \frac{2\,\sigma\cos\gamma}{\rho_w\, g\, h} \;\approx\; \frac{0.149\ \text{cm}^2}{h} \quad (\text{or } r_c[\text{mm}] \approx 1.49/h[\text{cm}]), \tag{8.4}$$

Provenance (Eq. 8.4): **[Literature]**: Young–Laplace (Jurin) capillary law.

Eq. 8.5 — structure

$$\Delta_{\text{till}}(s) = D_s(\text{conv}, \tau^*) - D_s(\text{notill}, \tau^*), \qquad \Delta_{\text{cons}}(s) = D_s(\text{till}^*, \tau_{\text{fast}}) - D_s(\text{till}^*, \tau_{\text{slow}}), \tag{8.5}$$

Provenance (Eq. 8.5): **[Data-driven]**: differences of Daisy run-set E5 results.

Eq. 8.6 — structure

$$\hat{\Delta} = a + b\,c, \qquad b = \frac{\sum_s (c_s - \bar{c})(\Delta_s - \bar{\Delta})}{\sum_s (c_s - \bar{c})^2}, \quad a = \bar{\Delta} - b\,\bar{c}. \tag{8.6}$$

Provenance (Eq. 8.6): **[Data-driven]**: ordinary least squares applied to Daisy results.

Eq. 8.7 — structure

$$z_p = k_p\,\Delta z, \qquad k_p = \min\Big\{k : \frac{\sum_{i\le k} U_i}{\sum_i U_i} \ge p\Big\}, \qquad U_i = \sum_{d\in\text{year}} u_i(d), \quad p = 0.5,\ 0.9, \tag{8.7}$$

Provenance (Eq. 8.7): **[Engineered]**: diagnostic defined for the workbench.

Eq. 9.1 — mulch

$$\Delta E = E_{1.0} - E_{\varphi}, \qquad \Delta I = I_{1.0} - I_{\varphi}, \qquad \Delta D = D_{\varphi} - D_{1.0}, \tag{9.1}$$

Provenance (Eq. 9.1): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

Eq. 9.2 — mulch

$$t_1 = \frac{Q_1 - q_0}{e_0}, \qquad e(t) = \begin{cases} e_0, & t \le t_1 \\ \dfrac{e_0}{\sqrt{1 + (t - t_1)/\tau}}, & t > t_1 \end{cases} \tag{9.2}$$

Provenance (Eq. 9.2): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.3 — mulch

$$Q(t) = q_0 + \begin{cases} e_0\, t, & t \le t_1 \\ (Q_1 - q_0) + 2\,e_0\,\tau\Big(\sqrt{1 + (t - t_1)/\tau} - 1\Big), & t > t_1 \end{cases} \tag{9.3}$$

Provenance (Eq. 9.3): **[Engineered]**: two-stage evaporation of the type described by Ritchie (1972) and Philip, with parameters chosen for the workbench.

Eq. 9.4 — mulch

$$\ell(t) = \min\Big\{60,\ \frac{Q(t) - Q_1}{\Delta\theta}\Big\}\ \text{mm} \quad (Q > Q_1), \tag{9.4}$$

Provenance (Eq. 9.4): **[Engineered]**: drying-front geometry chosen for drawing.

Eq. 9.5 — mulch

$$\log_{10} h(z) = \log_{10} h_s + (\log_{10} h_i - \log_{10} h_s)(1 - e^{-(z-\ell)/\lambda}) \quad (z \geq \ell), \qquad h = 10^5 \text{ cm for } z < \ell. \tag{9.5}$$

Provenance (Eq. 9.5): **[Engineered]**: drying-front geometry chosen for drawing.

Eq. 9.6 — mulch

$$\begin{aligned} \Delta E(\rho) &= E_{1.0}(\rho) - E_{\varphi^*}(\rho), \quad \Delta I(\rho) = I_{1.0}(\rho) - I_{\varphi^*}(\rho), \\ \Delta D(\rho) &= D_{\varphi^*}(\rho) - D_{1.0}(\rho), \quad \Delta T(\rho) = T_{\varphi^*}(\rho) - T_{1.0}(\rho), \end{aligned} \tag{9.6}$$

Provenance (Eq. 9.6): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

Eq. 9.7 — mulch

$$f_I = \frac{\max\{\Delta I, 0\}}{\max\{\Delta E, 0.1\}},\ f_D = \frac{\max\{\Delta D, 0\}}{\max\{\Delta E, 0.1\}},\ f_T = \frac{\max\{\Delta T, 0\}}{\max\{\Delta E, 0.1\}},\ f_{\text{rest}} = \max\{0,\ 1 - f_I - f_D - f_T\}. \tag{9.7}$$

Provenance (Eq. 9.7): **[Data-driven]**: differences and ratios of Daisy run-set E14 and E11 results.

Eq. 9.8 — mulch

$$\mathcal{J} = \frac{E - D}{\max\{1,\ E + D\}}, \qquad \text{regime} = \begin{cases} \text{evaporation-dominated} & \mathcal{J} > +0.05 \\ \text{drainage-dominated} & \mathcal{J} < -0.05 \\ \text{on the boundary} & |\mathcal{J}| \leq 0.05 \end{cases} \tag{9.8}$$

Provenance (Eq. 9.8): **[Engineered]**: index defined for the workbench.

Eq. 10.1 — loops

$$w_k = 5 + \text{clip}(\mu_k;\ 0,\ 4). \tag{10.1}$$

Provenance (Eq. 10.1): **[Engineered]**: drawing rule.

Eq. 10.2 — loops

$$\kappa_N = \frac{N_L^{(\text{rain step})} - N_L}{D^{(\text{rain step})} - D} \quad [\text{kg N ha}^{-1}\ \text{mm}^{-1}], \tag{10.2}$$

Provenance (Eq. 10.2): **[Data-driven]**: one-step differences of Daisy results.

Eq. 11.1 — surface

$$\begin{aligned} \theta_{1500,t} &= -0.024\,S + 0.487\,C + 0.006\,\text{OM} + 0.005\,S\,\text{OM} - 0.013\,C\,\text{OM} + 0.068\,S\,C + 0.031, \\ \theta_{1500} &= \theta_{1500,t} + (0.14\,\theta_{1500,t} - 0.02), \\ \theta_{33,t} &= -0.251\,S + 0.195\,C + 0.011\,\text{OM} + 0.006\,S\,\text{OM} - 0.027\,C\,\text{OM} + 0.452\,S\,C + 0.299, \\ \theta_{33} &= \theta_{33,t} + (1.283\,\theta_{33,t}^2 - 0.374\,\theta_{33,t} - 0.015), \end{aligned} \tag{11.1}$$

Provenance (Eq. 11.1): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.2 — surface

$$\begin{aligned}\theta_{(S-33),t} &= 0.278\,S + 0.034\,C + 0.022\,\mathrm{OM} - 0.018\,S\,\mathrm{OM} - 0.027\,C\,\mathrm{OM} - 0.584\,S\,C + 0.078,\\ \theta_{(S-33)} &= \theta_{(S-33),t} + \big(0.636\,\theta_{(S-33),t} - 0.107\big), \qquad \theta_s = \theta_{33} + \theta_{(S-33)} - 0.097\,S + 0.043,\\ B &= \frac{\ln 1500 - \ln 33}{\ln \theta_{33} - \ln \theta_{1500}}, \qquad \lambda = \frac{1}{B}, \qquad K_s = 1930\,(\theta_s - \theta_{33})^{\,3-\lambda}.\end{aligned} \tag{11.2}$$

Provenance (Eq. 11.2): **[Literature]**: Saxton and Rawls (2006), implemented as written.

Eq. 11.3 — surface

$$\psi_f = \frac{10}{\alpha}\,\frac{0.046\,m + 2.07\,m^2 + 19.5\,m^3}{1 + 4.7\,m + 16\,m^2}. \tag{11.3}$$

Provenance (Eq. 11.3): **[Engineered]**: rational-polynomial approximation of a Rawls-type suction.

Eq. 11.4 — surface

$$\theta_0 = \theta_{fc} - f_T\,(\theta_{fc} - \theta_{wp}), \qquad \Delta\theta = \max\{0.05,\ 0.95\,\theta_s - \theta_0\}. \tag{11.4}$$

Provenance (Eq. 11.4): **[Engineered]**: initial and wetted water contents chosen for the workbench.

Eq. 11.5 — surface

$$F - \psi_f\,\Delta\theta\,\ln\!\Big(1 + \frac{F}{\psi_f\,\Delta\theta}\Big) = K_s\,\frac{t}{60} \quad (t \text{ in min}), \tag{11.5}$$

Provenance (Eq. 11.5): **[Literature]**: Green and Ampt (1911).

Eq. 11.6 — surface

$$Z(\tau) = k\,\tau^{A} \qquad (Z \text{ in m},\ \tau \text{ in min}), \tag{11.6}$$

Provenance (Eq. 11.6): **[Literature]**: Kostiakov (1932).

Eq. 11.7 — surface

$$A = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}, \quad \ln k = \bar{y} - A\,\bar{x}, \qquad x_i = \ln \tau_i,\ y_i = \ln\!\big(F(\tau_i)/1000\big), \tag{11.7}$$

Provenance (Eq. 11.7): **[Literature]**: ordinary least squares on the log–log plot.

Eq. 11.8 — surface

$$Q = \frac{1}{n}\,A\,R_h^{2/3}\,S^{1/2}, \qquad R_h = \frac{A}{P}. \tag{11.8}$$

Provenance (Eq. 11.8): **[Literature]**: Manning (1891).

Eq. 11.9 — surface

$$Q\,t = \sigma_y\,A_0\,x + \sigma_z\,k\,t^{A}\,w\,x \implies x(t) = \frac{Q\,t}{\sigma_y A_0 + \sigma_z\,k\,t^{A}\,w}. \tag{11.9}$$

Provenance (Eq. 11.9): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.10 — surface

$$\sigma_z = \frac{A + r\,(1 - A) + 1}{(1 + A)(1 + r)}, \tag{11.10}$$

Provenance (Eq. 11.10): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.11 — surface

$$r = \frac{\ln 2}{\ln(t_L / t_{L/2})}, \tag{11.11}$$

Provenance (Eq. 11.11): **[Literature]**: volume balance of Walker and Skogerboe (1987) and Elliott and Walker (1982); tested against FAO-45 examples (Paper 3).

Eq. 11.12 — surface

$$t_x(x) = t_L \Big(\frac{x}{L}\Big)^{1/r} \quad \text{(given advance)}, \qquad t_x(x) = c\, q^{-q_e}\, x^{x_e} \quad \text{(field advance)}. \tag{11.12}$$

Provenance (Eq. 11.12): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.13 — surface

$$t_L = (1 + 0.15\,A)\,\frac{A_0 L}{Q} + \Big(\frac{\sigma_F\, k\, w\, L}{Q}\Big)^{\frac{1}{1-A}}, \qquad \sigma_F = \frac{\pi A(1 - A)}{\sin(\pi A)}, \qquad t_x(x) = t_V(x)\,\frac{t_L}{t_V(L)}, \tag{11.13}$$

Provenance (Eq. 11.13): **[Engineered]**: alternative advance models set by worked examples and course notes.

Eq. 11.15 — surface

$$t_{st} = \Big(\frac{Z_n}{k}\Big)^{1/A}. \tag{11.15}$$

Provenance (Eq. 11.15): **[Literature]**: inversion of the Kostiakov law.

Eq. 11.16 — surface

$$t_{co} = \begin{cases} t_L + t_{st} & \text{“}E_s = 1\text{”: the tail receives exactly } Z_n \\ t_x(0.85\,L) + t_{st} & \text{“85 \% of L”} \\ t_{co}^{\text{set}} & \text{“Manual”} \end{cases} \tag{11.16}$$

Provenance (Eq. 11.16): **[Engineered]**: cut-off rules chosen as design options.

Eq. 11.17 — surface

$$Z(x) = k\,\big[t_{co} - t_x(x)\big]^A, \quad Z_0 = k\,t_{co}^A \ \text{(inlet)}, \qquad Z_L = k\,(t_{co} - t_L)^A \ \text{(tail)}. \tag{11.17}$$

Provenance (Eq. 11.17): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.18 — surface

$$V_{\text{app}} = Q\,t_{co} \quad (\text{with cutback } c_b < 1:\ Q\,t_L + c_b\,Q\,(t_{co} - t_L)), \quad V_{\text{req}} = Z_n\,L\,w,$$
$$V_{\text{inf}} = w\int_0^L Z\,dx, \qquad V_{\text{st}} = w\int_0^L \min(Z, Z_n)\,dx,$$
$$V_{\text{dp}} = w\int_0^L \max(Z - Z_n, 0)\,dx, \qquad V_{\text{df}} = w\int_0^L \max(Z_n - Z, 0)\,dx,$$
$$V_{\text{ro}} = \max(V_{\text{app}} - V_{\text{inf}},\ 0). \tag{11.18}$$

Provenance (Eq. 11.18): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.19 — surface

$$E_a = \frac{V_{\text{st}}}{V_{\text{app}}}, \quad E_s = \frac{V_{\text{st}}}{V_{\text{req}}}, \quad \text{DP} = \frac{V_{\text{dp}}}{V_{\text{app}}}, \quad \text{RO} = \frac{V_{\text{ro}}}{V_{\text{app}}}, \quad \text{CU} = 100\Big(1 - \frac{\frac{1}{N+1}\sum_{i=0}^{N} |Z_i - \bar{Z}|}{\bar{Z}}\Big)\ \%. \tag{11.19}$$

Provenance (Eq. 11.19): **[Literature]**: standard opportunity-time and performance-index definitions (Walker and Skogerboe, 1987; Burt et al., 1997).

Eq. 11.20 — surface

$$V_c(t) = w\sum_i k\left[\min(t, t_{co}) - \bar{t}_{x,i}\right]_+^A \Delta x\ \cdot s_c, \qquad s_c = \frac{V_{\text{inf}}}{V_c(t_{co})}, \qquad I(t) = \frac{dV_c}{dt}, \tag{11.20}$$

Provenance (Eq. 11.20): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

Eq. 11.21 — surface

$$R(t) = \frac{2V_s}{t_r}\Big(1 - \frac{t - t_{co}}{t_r}\Big), \quad t_{co} < t < t_{co} + t_r. \tag{11.21}$$

Provenance (Eq. 11.21): **[Engineered]**: hydrograph shapes chosen for drawing; volume conserving.

Eq. 11.22 — surface

$$q_{\max} = \min\Big\{q_{\max}^{\text{sys}},\ \frac{0.04}{S}\cdot\frac{1000}{60}\Big\}\ \text{L s}^{-1} \quad (S \text{ in \%}), \tag{11.22}$$

Provenance (Eq. 11.22): **[Engineered]**: search limit; the furrow stream limit follows Criddle et al. (1956).

Eq. 11.23 — surface

$$\text{score} = \min(E_a, 1) - 3\max(0,\ 0.8 - E_s) - 3\,[c_b < 1]\ \max(0,\ 0.05 - \text{RO}) - 5\max(0,\ E_a - 1). \tag{11.23}$$

Provenance (Eq. 11.23): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

Eq. 11.24 — surface

$$n_{\text{ev}} = \sum I \,/\, Z_n, \qquad Z_g = \frac{V_{\text{app}}}{L\,w} \times 1000 \;\; (\text{mm}), \qquad Z_{\text{season}} = n_{\text{ev}}\, Z_g,$$
$$N_{0M} = \frac{Q_s}{q}, \qquad N_{\text{units}} = \frac{W_f}{w}, \quad N_{\text{sets}} = \max\!\Big(1, \frac{N_{\text{units}}}{N_{0M}}\Big), \qquad T_{\text{tot}} = \frac{N_{\text{sets}}\, t_{co}}{60} \;\; (\text{h}), \tag{11.24}$$

Provenance (Eq. 11.24): **[Engineered]**: optimisation score and seasonal bookkeeping chosen for the workbench.

Eq. 12.1 — drip

$$Z_n = \min\!\big\{ Z_n^{\text{DSS}},\; \max(4,\; \text{ET}_{\text{pk}}\, \Delta t) \big\} \;\; \text{mm}, \tag{12.1}$$

Provenance (Eq. 12.1): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.2 — drip

$$\text{ET}_{\text{pk}} = \text{clip}\Big\{ \max_{d} \tfrac{1}{5} \sum_{j=0}^{4} (E_{d+j} + T_{d+j});\; 2.5,\; 9 \Big\}. \tag{12.2}$$

Provenance (Eq. 12.2): **[Engineered]**: depth and demand rules chosen for the workbench.

Eq. 12.3 — drip

$$h_{f,j} = \frac{10.67\, S_e\, Q_j^{1.852}}{C^{1.852}\, D^{4.871}}, \qquad Q_j = \sum_{i \ge j+1} q_i \;\; (\text{m}^3\,\text{s}^{-1}), \quad D \text{ in m}. \tag{12.3}$$

Provenance (Eq. 12.3): **[Literature]**: Hazen and Williams (1905).

Eq. 12.4 — drip

$$q_j = q\Big(\frac{h_j}{H_0}\Big)^{x} \quad (\text{non-compensating};\; h_j/H_0 \ge 0.01), \tag{12.4}$$

Provenance (Eq. 12.4): **[Literature]**: Keller and Karmeli (1974) emitter law.

Eq. 12.5 — drip

$$q_j = \begin{cases} q\big[1 + 0.04\,(h_j - H_0)/H_0\big], & h_j \ge H_{\min}, \\ q\,\big(\max(h_j, 0.1)/H_{\min}\big)^{0.5}, & h_j < H_{\min}, \end{cases} \tag{12.5}$$

Provenance (Eq. 12.5): **[Engineered]**: pressure-compensating emitter model with chosen constants.

Eq. 12.6 — drip

$$\bar{q} = \frac{1}{N+1} \sum_{j=0}^{N} q_j, \qquad q_{\min} = \min_j q_j, \qquad Q_{\text{lat}} = \frac{N\,\bar{q}}{3600} \;\; (\text{L}\,\text{s}^{-1}), \tag{12.6}$$

Provenance (Eq. 12.6): **[Literature]**: definitions of mean and minimum discharge.

Eq. 12.7 — drip

$$E_u = \text{clip}\Big\{ \Big(1 - \frac{1.27\, c_v}{\sqrt{n_e}}\Big) \frac{q_{\min}}{\bar{q}};\; 0.3,\; 1 \Big\}. \tag{12.7}$$

Provenance (Eq. 12.7): **[Literature]**: Keller and Karmeli (1974).

Eq. 12.8 — drip

$$\sigma_q = \sqrt{c_v^2 + \sigma_P^2}, \qquad \sigma_P^2 = \frac{1}{N+1}\sum_{j=0}^{N}\Big(\frac{q_j}{\bar{q}} - 1\Big)^2. \tag{12.8}$$

Provenance (Eq. 12.8): **[Engineered]**: combined spread chosen to drive the bulb quantiles.

Eq. 12.9 — drip

$$\Delta\theta = \max\{0.03,\ f_T\,(\theta_{fc} - \theta_{wp})\}. \tag{12.9}$$

Provenance (Eq. 12.9): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.10 — drip

$$\rho(z) = r\,\sqrt{1-\zeta^2}, \qquad \zeta = \begin{cases} (z - z_d)/(a\,r), & z \ge z_d, \\ (z_d - z)/(u\,r), & z < z_d, \end{cases} \qquad \rho = 0 \text{ for } |\zeta| \ge 1. \tag{12.10}$$

Provenance (Eq. 12.10): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.11 — drip

$$a = \operatorname{clip}\Big\{1 + 0.6\,\log_{10}\Big(1 + \frac{K_s}{q\cdot 10^{6}/(2\pi\lambda^2)}\Big);\ 1,\ 1.9\Big\}, \qquad \lambda = \max(40, \psi_f), \tag{12.11}$$

Provenance (Eq. 12.11): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.12 — drip

$$V_w = \int \pi\,\rho(z)^2\,dz, \tag{12.12}$$

Provenance (Eq. 12.12): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.13 — drip

$$P_w = \operatorname{clip}\Big\{\frac{2r}{1000\,S_l};\ 0,\ 1\Big\}. \tag{12.13}$$

Provenance (Eq. 12.13): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.14 — drip

$$V_{\mathrm{dp}}^{(1)} = V\,\frac{\int_{z>R_{\max}} \pi\rho^2\,dz}{V_w}, \tag{12.14}$$

Provenance (Eq. 12.14): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.15 — drip

$$V_{\mathrm{ev}} = \min\{0.5\,V,\ \pi\rho(0)^2\cdot 10^{-6}\cdot 2E_0\} \quad (\mathrm{L},\ \text{area in m}^2,\ E_0 \text{ in mm d}^{-1}), \tag{12.15}$$

Provenance (Eq. 12.15): **[Engineered]**: wetted-bulb geometry and loss partition; constants chosen for the workbench.

Eq. 12.16 — drip

$$V_i = \max\{0.02\,\bar{V},\ \bar{V}\,(1+\sigma_q z_i)\}, \qquad \bar{V} = \bar{q}\,t, \tag{12.16}$$

Provenance (Eq. 12.16): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.17 — drip

$$V_{\text{req}} = Z_n\,S_e\,S_l\ \ (\text{L}), \qquad V_{\text{st},i} = \min\{V_{\text{root},i},\ V_{\text{req}}\}, \qquad V_{\text{ex},i} = V_{\text{root},i} - V_{\text{st},i}. \tag{12.17}$$

Provenance (Eq. 12.17): **[Literature]**: definition of the requirement over the area served by one emitter.

Eq. 12.18 — drip

$$t = \text{clip}\Big\{\frac{V_{\text{req}}}{\bar{q}\,E_u};\ 0.05,\ 96\Big\}\ \text{h}. \tag{12.18}$$

Provenance (Eq. 12.18): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.19 — drip

$$E_a = \frac{\langle V_{\text{st}}\rangle}{\langle V\rangle}, \quad E_s = \frac{\langle V_{\text{st}}\rangle}{V_{\text{req}}}, \quad \text{DP} = \frac{\langle V_{\text{dp}}^{(1)} + V_{\text{ex}}\rangle}{\langle V\rangle}, \quad \text{EV} = \frac{\langle V_{\text{ev}}\rangle}{\langle V\rangle}, \quad Z_{\text{gross}} = \frac{\bar{V}}{S_e S_l}, \tag{12.19}$$

Provenance (Eq. 12.19): **[Engineered]**: five-quantile delivery and run-time rules chosen for the workbench.

Eq. 12.20 — drip

$$W_f = \frac{10^4 A_f}{L_L}, \quad N_L = \Big\lfloor \frac{W_f}{S_l} \Big\rfloor, \quad n_{\text{zone}} = \Big\lceil \frac{N_L\,Q_{\text{lat}}}{Q_s} \Big\rceil, \quad n_l = \Big\lceil \frac{N_L}{n_{\text{zone}}} \Big\rceil, \quad Q_z = n_l\,Q_{\text{lat}}, \tag{12.20}$$

Provenance (Eq. 12.20): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.21 — drip

$$D \in \{32, 40, 50, 63, 75, 90, 110, 125, 160, 200, 250, 315\}\ \text{mm} \tag{12.21}$$

Provenance (Eq. 12.21): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.22 — drip

$$h_{f,sm} = 0.4\,\frac{10.67\,L_{sm}\,Q_z^{1.852}}{C^{1.852} D_{sm}^{4.871}}, \qquad h_{f,m} = \frac{10.67\,L_m\,Q_z^{1.852}}{C^{1.852} D_m^{4.871}}. \tag{12.22}$$

Provenance (Eq. 12.22): **[Engineered]**: Christiansen-type reduction with a chosen factor 0.4.

Eq. 12.23 — drip

$$\text{TDH} = z_{\text{lift}} + H_0 + h_{f,sm} + h_{f,m} + 9\ \text{m}, \qquad P = \frac{\rho g\,Q_z\,\text{TDH}}{\eta} = \frac{9.81\,(Q_z/1000)\,\text{TDH}}{0.65}\ \text{kW}. \tag{12.23}$$

Provenance (Eq. 12.23): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.24 — drip

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \quad Z_{\text{season}} = n_{\text{ev}}\, Z_{\text{gross}}, \quad q_{ha} = \frac{q}{3600}\,\frac{10^4}{S_e S_l}\ \ (\text{L s}^{-1}\,\text{ha}^{-1}), \tag{12.24}$$

Provenance (Eq. 12.24): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.25 — drip

$$n_{\text{set}} = \left\lceil \frac{q_{ha} A_f}{Q_s} \right\rceil, \qquad t_{\text{round}} = n_{\text{set}}\, t, \qquad H_{\text{pump}} = n_{\text{ev}}\, t_{\text{round}}, \qquad E_{\text{kWh}} = H_{\text{pump}}\, P. \tag{12.25}$$

Provenance (Eq. 12.25): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 12.26 — drip

$$\text{score} = \min(E_a, 1) - 2.5\,\max(0,\ 0.9 - E_s), \tag{12.26}$$

Provenance (Eq. 12.26): **[Engineered]**: layout, sizing and scoring rules; the 1.5 m/s velocity limit is a common practice.

Eq. 13.1 — sprinkler

$$S_d^{\text{eff}} = \frac{\max(0, S_d)}{1 + 0.25\,S} \qquad (S \text{ in \%}). \tag{13.1}$$

Provenance (Eq. 13.1): **[Engineered]**: depression storage reduction with slope.

Eq. 13.2 — sprinkler

$$q = 0.95\,A_n\sqrt{2g\,h} \times 6 \times 10^4\ \ (\text{L min}^{-1}), \qquad A_n = \pi\Big(\frac{d}{2000}\Big)^2\ \text{m}^2, \tag{13.2}$$

Provenance (Eq. 13.2): **[Literature]**: Torricelli orifice equation.

Eq. 13.3 — sprinkler

$$R = c_R\sqrt{d\,h}\ f_\alpha(\alpha), \qquad f_\alpha(\alpha) = \Big[\frac{\max\{0.05,\ \sin(3.1\,\alpha)\}}{0.9763}\Big]^{0.45}, \tag{13.3}$$

Provenance (Eq. 13.3): **[Engineered]**: range law of the Keller and Bliesner (1990) form; the impact coefficient 1.35 is the published value and is verified against their worked example (Paper 3); the coefficients of the other types and the angle function are chosen.

Eq. 13.4 — sprinkler

$$\hat{p}(x) = \begin{cases} 1 - x^m, & \text{impact } (m = 1.3),\ \text{gun } (m = 1), \\ (0.5 + 0.5x)(1 - x^6), & \text{rotating spray (ring)}, \\ (0.55 + 0.45x)(1 - x^5), & \text{micro-sprinkler (ring-2)}, \end{cases} \qquad \hat{p} = 0 \text{ for } x \geq 1. \tag{13.4}$$

Provenance (Eq. 13.4): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.5 — sprinkler

$$R_e(\theta) = R\,\mathrm{clip}\{1 + 0.022\,u\cos\theta;\ 0.7,\ 1.4\}, \qquad \text{shift} = 0.03\,u\,R, \tag{13.5}$$

Provenance (Eq. 13.5): **[Engineered]**: pattern and wind shapes chosen to reproduce published qualitative patterns.

Eq. 13.6 — sprinkler

$$a(x,y) = \sum_{i,j} p(x - x_{ij},\, y - y_{ij}) \quad (\mathrm{mm\,min^{-1}}), \qquad \bar{a} = \frac{1}{n}\sum_{k} a_k, \tag{13.6}$$

Provenance (Eq. 13.6): **[Literature]**: overlap by superposition of single-sprinkler patterns.

Eq. 13.7 — sprinkler

$$\mathrm{CU} = 1 - \frac{\sum_k |a_k - \bar{a}|}{n\,\bar{a}}. \tag{13.7}$$

Provenance (Eq. 13.7): **[Literature]**: Christiansen (1942).

Eq. 13.8 — sprinkler

$$w_d = \frac{1}{100}\,\mathrm{clip}\Big\{(1.2 + 0.5u + 0.04u^2)\,F\Big(\frac{P}{250}\Big)^{0.7}\,[1 + 0.025\,(T_a - 20)]\Big[0.6 + 0.4\,\frac{\sin\alpha}{0.4226}\Big];\ 0.5,\ 45\Big\}. \tag{13.8}$$

Provenance (Eq. 13.8): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.9 — sprinkler

$$t_p = \Big(\frac{a}{K}\Big)^{1/(A-1)}, \qquad F = \begin{cases} \sup, & T \le t_p \\ K\,T^A, & T > t_p \end{cases} \qquad R = \sup - F, \quad \mathrm{RO} = \max(0, R - S_d^{\mathrm{eff}}), \quad I = \sup - \mathrm{RO}. \tag{13.9}$$

Provenance (Eq. 13.9): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.10 — sprinkler

$$I_c = \min\{I,\ I_{\mathrm{int}}\}, \qquad I_{\mathrm{int}} = 0.6\,\mathrm{CAP}\,[\mathrm{VFF} < 0.95] + 1.2\,\mathrm{LAI}_{\mathrm{live}}, \tag{13.10}$$

Provenance (Eq. 13.10): **[Engineered]**: loss and ponding rules chosen for the workbench.

Eq. 13.11 — sprinkler

$$\begin{aligned} E_a &= \frac{\overline{\min(Z_k, Z_n)}}{\mathrm{gross}}, & E_s &= \frac{\overline{\min(Z_k, Z_n)}}{Z_n}, & \mathrm{DU} &= \frac{\overline{Z}_{25}}{\overline{Z}}, \\ \mathrm{DP} &= \frac{\overline{\max(Z_k - Z_n, 0)}}{\mathrm{gross}}, & \mathrm{RO} &= \frac{\overline{\mathrm{RO}_k}}{\mathrm{gross}}, & \mathrm{WD} &= w_d + \frac{\overline{I_c}}{\mathrm{gross}}, \end{aligned} \tag{13.11}$$

Provenance (Eq. 13.11): **[Literature]**: efficiency and distribution uniformity of Merriam and Keller (1978).

Eq. 13.12 — sprinkler

$$T = \min\{T :\ \overline{Z}_{25}(T) \ge Z_n\}, \qquad T \in [1,\ 2400]\ \mathrm{min}, \tag{13.12}$$

Provenance (Eq. 13.12): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.13 — sprinkler

$$k = \begin{cases} 1, & W \le 5.5 \\ 0.9, & 5.5 < W \le 13 \\ 0.8, & 13 < W \le 22 \\ 0.7, & W > 22. \end{cases} \tag{13.13}$$

Provenance (Eq. 13.13): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.14 — sprinkler

$$S_s \le k\,R_w, \qquad S_l \le 1.5\,R_w, \qquad L_y = \frac{10^4 A_f}{L_x}, \quad N = \lfloor L_x/S_s \rfloor, \quad N_m = \lfloor L_y/S_l \rfloor, \tag{13.14}$$

Provenance (Eq. 13.14): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.15 — sprinkler

$$t_{\text{set}} = \frac{Z_{\text{gross}}}{\text{AR}} \ \text{(h)}, \qquad n_{\text{sets/day}} = \left\lfloor \frac{h_{\text{op}}}{t_{\text{set}}} \right\rfloor, \qquad T_{\text{int}} = \max\!\left(1, \left\lfloor \frac{Z_n}{\text{ET}_{\text{pk}}} \right\rfloor\right) \text{ days}, \tag{13.15}$$

Provenance (Eq. 13.15): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.16 — sprinkler

$$N_d(n) = \left\lceil \frac{N_m}{n\, n_{\text{sets/day}}} \right\rceil, \qquad Q_{\text{in}}(n) = n\,N\,q_h, \tag{13.16}$$

Provenance (Eq. 13.16): **[Engineered]**: run-time, wind-spacing and set rules; the wind factors follow the practice of Keller and Bliesner (1990) but are not independently verified.

Eq. 13.17 — sprinkler

$$J = \frac{0.472\,Q^{1.75}}{D^{4.75}} \ \ (\text{m m}^{-1};\ Q \text{ in L h}^{-1},\ D \text{ in mm}), \tag{13.17}$$

Provenance (Eq. 13.17): **[Literature]**: Blasius friction law for smooth pipe.

Eq. 13.18 — sprinkler

$$h_{f,\text{lat}} = 1.1\,F\,J(Q_L, D)\,(L_x - S_s/2) \le 0.2\,h_s, \qquad Q_L = 1000\,N\,q_h, \tag{13.18}$$

Provenance (Eq. 13.18): **[Literature]**: 20 % rule of Keller and Bliesner (1990).

Eq. 13.19 — sprinkler

$$H_{\text{man}} = z_{\text{lift}} + J\,L_{\text{conv}} + J\,L_{\text{main}} + h_s + 0.75\,h_{f,\text{lat}}, \tag{13.19}$$

Provenance (Eq. 13.19): **[Engineered]**: head and sets bookkeeping.

Eq. 13.20 — sprinkler

$$P_{\text{CV}} = \frac{1000\,(Q_m/3.6\times10^6)\,H_{\text{man}}}{75\,\eta}, \quad \eta = 0.75, \qquad P_{\text{kW}} = 0.7355\,P_{\text{CV}}, \tag{13.20}$$

Provenance (Eq. 13.20): **[Literature]**: hydraulic power definition.

Eq. 13.21 — sprinkler

$$N_L = \Big\lfloor \frac{10^4 A_f / L_L}{S_l} \Big\rfloor, \quad n_l = \min\Big(N_L,\ \Big\lfloor \frac{Q_s}{Q_{\text{lat}}} \Big\rfloor\Big), \quad n_{\text{set}} = \Big\lceil \frac{N_L}{n_l} \Big\rceil, \quad Q_z = n_l Q_{\text{lat}}. \tag{13.21}$$

Provenance (Eq. 13.21): **[Engineered]**: head and sets bookkeeping.

Eq. 13.22 — sprinkler

$$F_m = \frac{1}{2.852} + \frac{1}{2N_s} + \frac{\sqrt{0.852}}{6N_s^2}, \qquad h_{f,\text{lat}} = F_m\, h_{HW}(Q_{\text{lat}}, D, L_L), \tag{13.22}$$

Provenance (Eq. 13.22): **[Literature]**: Christiansen reduction factor with Hazen–Williams exponent.

Eq. 13.23 — sprinkler

$$n_{\text{ev}} = \frac{y_I}{Z_n}, \qquad n_{\text{sp}} = \text{round}\Big(\frac{10^4 A_f}{S_s S_l}\Big), \qquad Q_{\text{tot}} = \frac{n_{\text{sp}}\, q}{60}, \qquad n_{\text{set}} = \Big\lceil \frac{Q_{\text{tot}}}{Q_s} \Big\rceil, \qquad t_{\text{round}} = \frac{n_{\text{set}}\, T}{60}\ \text{h}. \tag{13.23}$$

Provenance (Eq. 13.23): **[Engineered]**: head and sets bookkeeping.

Eq. 14.1 — dashboard

$$\text{Rain} = P_k, \quad \text{Irr} = I_k, \quad \text{Evap} = E_k, \quad \text{Transp} = T_k, \quad \text{Drain} = D_k \ \ (\text{mm}\,\text{d}^{-1}). \tag{14.1}$$

Provenance (Eq. 14.1): **[Engineered]**: bookkeeping of Engine 2 series.

Eq. 14.2 — dashboard

$$\Sigma X = \sum_{j=1}^{N} X_{k+j}, \qquad X \in \{P, I, E, T, D\}. \tag{14.2}$$

Provenance (Eq. 14.2): **[Engineered]**: bookkeeping of Engine 2 series.

Eq. 14.3 — dashboard

$$\Delta S = P + I - E_{\text{int}} - E - T - D, \tag{14.3}$$

Provenance (Eq. 14.3): **[Literature]**: water-balance identity.

Eq. 14.4 — dashboard

$$\Delta_i = x_i - x_i^{\text{pin}}, \tag{14.4}$$

Provenance (Eq. 14.4): **[Engineered]**: bookkeeping of Engine 2 series.

Eq. 14.5 — dashboard

$$Y^{(y)} = \sum_{d \in y} X_d, \qquad X \in \{P, I, D, E, T\}. \tag{14.5}$$

Provenance (Eq. 14.5): **[Engineered]**: bookkeeping of Engine 2 series.